TEXTBOOKS IN MATHEMATICS

ELEMENTS OF ALGEBRAIC TOPOLOGY

SECOND EDITION

James R. Munkres, Steven G. Krantz, and Harold R. Parks



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Elements of Algebraic Topology

This classic text appears here in a new edition for the first time in four decades. The new edition, with the aid of two new authors, brings it up to date for a new generation of mathematicians and mathematics students.

Elements of Algebraic Topology provides the most concrete approach to the subject. With coverage of homology and cohomology theory, universal coefficient theorems, Kunneth theorem, duality in manifolds, and applications to classical theorems of point-set topology, this book is perfect for communicating complex topics and the fun nature of algebraic topology for beginners.

This second edition retains the essential features of the original book. Most of the notation and terminology are the same. There are some useful additions. There is a new introduction to homotopy theory. A new Index of Notation is included. Many new exercises are added.

Algebraic topology is a cornerstone of modern mathematics. Every working mathematician should have at least an acquaintance with the subject. This book, which is based largely on the theory of triangulations, provides such an introduction. It should be accessible to a broad cross-section of the profession—both students and senior mathematicians. Students should have some familiarity with general topology.

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Elements of Algebraic Topology

Second Edition

James R. Munkres, Steven G. Krantz,
and Harold R. Parks

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For Barbara.

To the memory of Marvin J. Greenberg, who taught me topology.

To Susan.



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Preface

Algebraic topology is a cornerstone of modern mathematics. Every working mathematician should have at least an acquaintance with the subject. The book *Elements of Algebraic Topology* by James R. Munkres is a classic that provides such an introduction. It is now forty years old and needs to be brought up to date. This new edition is meant to fulfill that purpose.

Of course, we have retained the essential features of the original book. Most of the notation is the same and most of the terminology is the same. We have retained all of the key topics from the first edition. But there are some useful additions and changes.

Unlike the first edition, we do give an introduction to homotopy theory in a section on the fundamental group. The section on acyclic models has been completely revamped to make it more accessible to the reader. We have collected together at the end some very limited biographical information about each person mentioned in the text. There is also a short list of additional readings, mostly chosen from books published after the first edition. Another new feature is a very detailed Index of Notation. And the main Index for the book is now also very detailed. Finally, we have added quite a number of exercises. Most of these are in the nature of drill.

Organization of this book

The first two chapters are based on simplicial homology—the most concrete of homology theories—so they should be quite accessible to students. The third chapter introduces material that is more abstract, including the Eilenberg–Steenrod Axioms. The fourth chapter introduces the singular homology theory and many interesting applications. CW complexes appear as a computational tool. For a one-semester course, these first four chapters would be suitable.

Sections marked with asterisks may be considered optional for this one-semester course. Of course, the applications seem to the authors to be of great interest, so while optional, the choice may well be to cover some or all of them.

Cohomology theory is introduced in the fifth chapter, culminating with the cohomology ring. The sixth chapter introduces the tensor product and adds to the treatment of homology with arbitrary coefficients mentioned in an optional section in the first chapter. While likely too much for a

one-semester course, the first six chapters do provide a solid core that should be accessible to a broad cross-section of the profession—both students and senior mathematicians.

For a two-semester course, the entire book may be appropriate. Of course, the instructor will have to exercise some discretion in designing a suitable course. Sections that are optional for the one-semester course (for example, Categories and Functors) may be necessary for the two-semester course. The last two chapters are independent of each other. So if it is necessary to cut some material as the second semester winds down, it can be done.

Regarding the last two chapters, the seventh chapter deals with homological algebra, including the universal coefficient theorems and the Künneth theorems in various settings. The eighth chapter deals with manifolds—specifically, the duality theorems associated with the names of Poincaré, Lefschetz, Alexander, and Pontryagin. Čech cohomology is introduced to study Pontryagin duality.

If an instructor wishes to spend less time on the treatment of simplicial homology, then it is possible to omit Chapter 2. In that case, the topological invariance of the simplicial homology groups will not be shown directly, but instead must be obtained via the invariance of the singular homology groups and the isomorphism between simplicial and singular homology.

Prerequisites

We take it that the student has some familiarity with general topology. This includes compactness, connectedness, and quotient spaces. The Tietze extension theorem for normal spaces will be used, and quotient spaces are very important. Quotient spaces are reviewed at appropriate times.

We also would like the student to know some algebra. Notably some familiarity with groups, factor groups, and homomorphisms as well as rings, fields, and vector spaces. No particularly deep theorems will be needed. As with quotient spaces, basic algebra results are reviewed at appropriate times.

Acknowledgement

For our cover figure, we are grateful to Carlo H. Séquin of the University of California at Berkeley. The figure shows how two Möbius bands can fit together to form a Klein bottle.

We are grateful to our Editor Robert Ross for his support and encouragement. We look forward to feedback from our readership.

— Steven G. Krantz
— Harold R. Parks



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1

Homology Groups of a Simplicial Complex

1.1 Introduction

A fundamental problem of topology is that of determining, for two spaces, whether or not they are homeomorphic. To show two spaces *are* homeomorphic, one needs to construct a continuous bijective map, with a continuous inverse, mapping one space to the other. To show two spaces are not homeomorphic involves showing that such a map does not exist. To do that is often harder. The usual way of proceeding is to find some topological property [i.e., some property invariant under homeomorphisms] that is satisfied by one space but not the other. For example, the closed unit disc in $\mathbb{R}^2$ cannot be homeomorphic with the plane $\mathbb{R}^2$, because the closed disc is compact and the plane is not. Nor can the real line $\mathbb{R}$ be homeomorphic with $\mathbb{R}^2$, because deleting a point from $\mathbb{R}$ leaves a disconnected space remaining, while deleting a point from $\mathbb{R}^2$ does not.

Such elementary properties do not carry one very far in tackling homeomorphism problems. Classifying all compact surfaces up to homeomorphism, for instance, demands more sophisticated topological invariants than these. So does the problem of showing that, in general, $\mathbb{R}^n$ and $\mathbb{R}^m$ are not homeomorphic if $n \neq m$.

Algebraic topology originated in the attempts by such mathematicians as Henri Poincaré and Enrico Betti to construct such topological invariants. Poincaré introduced a certain group, called the *fundamental group* of a topological space; it is, by its definition, a topological invariant. One can show fairly readily that a number of familiar spaces, such as the sphere, torus, and Klein bottle, have fundamental groups that are different, so these spaces cannot be homeomorphic [see [Mun00], Section 60]. In fact, one can classify all compact surfaces using the fundamental group [see [Mas91], Chapter 4, Section 5].

The transition from focusing on the numerical invariants of a space to studying the homology groups of a space was hastened by the algebraist Emmy Noether. This happened because in the mid-1920s Paul Alexandroff and Heinz Hopf were visitors at Göttingen where Noether was a member of the department. She recognized that the betti numbers and torsion coefficients that Alexandroff and Hopf were discussing were in fact invariants associated with abelian groups: specifically the *homology groups*. Indeed, Hopf explicitly credited the influence of Noether in the introductory section of his 1928

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paper [Hop28] in which he generalized the Euler–Poincaré formula. Early on the homology groups presented the difficulty that they were not obviously a topological invariant. Once that was shown, then those groups could also be used to tackle homeomorphism problems.

We shall begin our discussion of algebraic topology by studying the homology groups. Later on, we shall deal with other topological invariants, such as the cohomology groups and the cohomology ring. We shall also give a brief treatment of the homotopy groups.

There are several different ways of defining homology groups, all of which lead to the same results for spaces that are sufficiently nice. The two that we shall consider in detail are the *simplicial* and the *singular* groups. We begin with the simplicial homology groups, which came first historically. Both conceptually and computationally, they are concrete and down-to-earth. They are defined, however, only for particularly nice spaces [i.e., polyhedra], and it is hard work to prove their topological invariance. After that, we shall treat the singular homology groups, which were introduced as a generalization of the simplicial groups. They are defined for arbitrary spaces, and it is immediate from their definition that they are topological invariants. Furthermore, they are much more convenient for theoretical purposes than the simplicial groups. They are not as readily computable as the simplicial groups, but they agree with the simplicial homology groups when both are defined.

A third way of defining homology groups for arbitrary spaces is due to Eduard Čech. The Čech homology theory is still not completely satisfactory, but Čech cohomology theory is both important and useful. It will appear near the end of this book.

1.2 Simplices

Before defining simplicial homology groups, we must discuss the class of spaces for which they are defined, which is the class of all polyhedra. A polyhedron is a space that can be built from such building blocks as line segments, triangles, tetrahedra, and their higher-dimensional analogues, by gluing them together along their faces. In this section, we shall discuss these basic building blocks, and in the next, we shall use them to construct polyhedra.

First, we need to study a bit of the analytic geometry of euclidean space. We denote the N-dimensional euclidean space by $\mathbb{R}^N$. The origin in euclidean space will be denoted by $\mathbf{0}$. For $x = (x_1, x_2, \ldots, x_N) \in \mathbb{R}^N$, the *norm* of x is defined by the equation

$$\|x\| = \left[\sum_{i=1}^{N} (x_i)^2\right]^{\frac{1}{2}}.$$

Given a set $\{a_0, a_1, \ldots, a_n\}$ of points of $\mathbb{R}^N$, this set is said to be *geometrically independent* if for any real scalars t_i, the equations

$$\sum_{i=0}^{n} t_i = 0 \quad \text{and} \quad \sum_{i=0}^{n} t_i a_i = \mathbf{0}$$

imply that $t_0 = t_1 = \cdots = t_n = 0$.

It is clear that a one-point set is always geometrically independent. Simple algebra shows that in general $\{a_0, a_1, \ldots, a_n\}$ is geometrically independent if and only if the vectors

$$a_1 - a_0, a_2 - a_0, \ldots, a_n - a_0$$

are linearly independent in the sense of linear algebra. Thus two distinct points in $\mathbb{R}^N$ form a geometrically independent set, as do three non-collinear points, four non-coplanar points, and so on.

Given a geometrically independent set of points $\{a_0, a_1, \ldots, a_n\}$, we define the *n-plane* P spanned by these points to consist of all points $x \in \mathbb{R}^N$ such that

$$x = \sum_{i=0}^{n} t_i a_i$$

for some scalars t_i with $\sum_i t_i = 1$. Since the a_i are geometrically independent, the t_i are uniquely determined by x. Note that each point a_i itself belongs to the plane P.

The plane P can also be described as the set of all points x such that

$$x = a_0 + \sum_{i=1}^{n} t_i (a_i - a_0)$$

for some scalars $t_1, t_2, \ldots, t_n$; in this form we say P is "the plane through a_0 parallel to the vectors $a_i - a_0$."

It is elementary to check that, if $\{a_0, a_1, \ldots, a_n\}$ is geometrically independent, and if w lies outside the plane that these points span, then $\{w, a_0, a_1, \ldots, a_n\}$ is geometrically independent.

An *affine transformation* T of $\mathbb{R}^N$ is a map that is a composition of translations [i.e., maps of the form $T(x) = x + p$ for fixed p], and non-singular linear transformations. If T is an affine transformation, it is immediate from the definitions that T preserves geometrically independent sets and that T carries the plane P spanned by $\{a_0, a_1, \ldots, a_n\}$ onto the plane spanned by $\{T(a_0), T(a_1), \ldots, T(a_n)\}$.

Now the translation $T(x) = x - a_0$ carries P onto the vector subspace of $\mathbb{R}^N$ having $a_1 - a_0, a_2 - a_0, \ldots, a_n - a_0$ as a basis. If we follow T by a linear transformation of $\mathbb{R}^N$ carrying $a_1 - a_0, a_2 - a_0, \ldots, a_n - a_0$ to the first n unit basis vectors $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$ in $\mathbb{R}^N$, we obtain an affine transformation S of $\mathbb{R}^N$ such that $S(s_0) = \mathbf{0}$ and $S(a_i) = \epsilon_i$ for $i > 0$. The map S carries P onto the plane $\mathbb{R}^n \times \mathbf{0}$ of the first n coordinates in $\mathbb{R}^N$. It is thus clear why we call P a "plane of dimension n" in $\mathbb{R}^N$.

Definition 1.2.1 Let $\{a_0, a_1, \ldots, a_n\}$ be a geometrically independent set in $\mathbb{R}^N$. We define the *n-simplex* σ spanned by $a_0, a_1, \ldots, a_n$ to be the set of all points $x \in \mathbb{R}^N$ such that

$$x = \sum_{i=0}^{n} t_i a_i, \quad \text{where} \quad \sum_{i=0}^{n} t_i = 1 \quad \text{and} \quad t_i \geq 0 \quad \text{for all} \quad i\,. \qquad (*)$$

The numbers t_i are uniquely determined by x and the condition $(*)$. They are called the *barycentric coordinates* of the point x with respect to $a_0, a_1, \ldots, a_n$.

EXAMPLE 1.2.2 In low dimensions, one can picture a simplex easily. A 0-simplex is a point, of course. The 1-simplex spanned by a_0 and a_1 consists of all points of the form

$$x = ta_0 + (1-t)a_1$$

with $0 \leq t \leq 1$, so it is simply the line segment joining a_0 and a_1. That line segment can be written a_0a_1.

Similarly, the 2-simplex σ spanned by a_0, a_1, a_2 equals the triangular region having these three points as vertices. This can be seen most easily as follows: Assume $x \neq a_0$. Then

$$x = \sum_{i=0}^{2} t_i a_i = t_0 a_0 + (1-t_0)[(t_1/\lambda)a_1 + (t_2/\lambda)a_2]$$

where $\lambda = 1 - t_0$. The expression in brackets represents a point p of the line segment joining a_1 and a_2, since $(t_1 + t_2)/\lambda = 1$ and $t_i/\lambda \geq 0$ for $i = 1, 2$.

Thus x is a point of the line segment joining a_0 and p. See Figure 1.2.1. Conversely, any point of such a line segment is in σ, as you can check. It follows that σ equals the union of all line segments joining a_0 to points of a_1a_2. That is, σ is a triangular region. That triangular region can be written $a_0a_1a_2$.

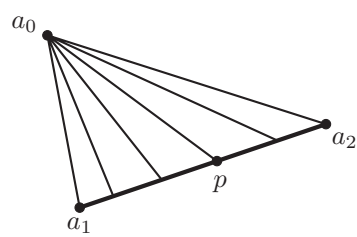


Figure 1.2.1

A similar proof shows that a 3-simplex is a tetrahedron.

Definition 1.2.3

(a) Let σ be the simplex spanned by the geometrically independent points $a_0, a_1, \ldots, a_n$. The points a_i are called the *vertices* of σ and the number n is called the *dimension* of σ. One can denote σ by $a_0 a_1 \ldots a_n$.

(b) Any simplex spanned by a subset of $\{a_0, a_1, \ldots, a_n\}$ is called a *face* of σ.

(c) Let s be the face of σ spanned by $a_0, a_1, \ldots, a_p$ [where $p < n$]; and let t be the face spanned by $a_{p+1}, a_{p+2}, \ldots, a_n$. Then t is called the face of σ *opposite* s. In particular, the face of σ spanned by $a_1, a_2, \ldots, a_n$ is called the face *opposite* a_0.

(d) The faces of σ different from σ itself are called the *proper faces* of σ.

(e) The union of the proper faces of σ is called the *boundary* of σ and is denoted Bd σ.

(f) The *interior* of σ, denoted Int σ is defined by the equation Int $\sigma = \sigma \setminus$ Bd σ. It is important to note that the interior of a 0-simplex is the 0-simplex itself—a fact that may be contrary to your intuition.

(g) The set Int σ is sometimes called an *open simplex*.

Let us recall some standard notation. The *unit n-ball*, $\mathbb{B}^n$, is the set of all points x of $\mathbb{R}^n$ for which $\|x\| \leq 1$, and the *unit sphere*, $\mathbb{S}^{n-1}$, is the set of points for which $\|x\| = 1$. The *upper hemisphere* E_+^{n-1} of $\mathbb{S}^{n-1}$ consists of all points x of $\mathbb{S}^{n-1}$ for which $x_n \geq 0$; while the *lower hemisphere* E_-^{n-1} of $\mathbb{S}^{n-1}$ consists of all points x of $\mathbb{S}^{n-1}$ for which $x_n \leq 0$.

With these definitions, $\mathbb{B}^0$ is a one-point space, $\mathbb{B}^1$ equals the interval $[-1, 1]$, and $\mathbb{S}^0$ is the two-point space $\{-1, 1\}$. The 2-ball is the unit disc in $\mathbb{R}^2$ centered at the origin, and $\mathbb{S}^1$ is the unit circle.

Definition 1.2.4 A subset A of $\mathbb{R}^N$ is said to be *convex* if, for each pair x, y of points in A, the line segment joining them lies in A.

There is also a weaker definition that is sometimes useful:

Definition 1.2.5 A subset A of $\mathbb{R}^N$ is said to be *star-convex* relative to a point $w \in A$ if, for each point x in A different from w, the line segment joining x to w lies in A.

We will now list some basic properties of simplices. The proofs are elementary and are largely left to the exercises.

Lemma 1.2.6 *Let P be the n-plane determined by the points of the geometrically independent set $\{a_0, a_1, \ldots, a_n\}$, and let σ be the n-simplex spanned by these points.*

(1) *The barycentric coordinates $t_i(x)$ of x with respect to $a_0, a_1, \ldots, a_n$ are continuous functions of x.*

(2) *The simplex σ equals the union of all line segments joining a_0 to points of the simplex s spanned by $a_1, \ldots, a_n$. Two such line segments intersect only in the point a_0.*

(3) *The simplex σ is a compact, convex set in $\mathbb{R}^N$, which equals the intersection of all convex sets in $\mathbb{R}^N$ containing $a_0, a_1, \ldots, a_n$.*

(4) *Given a simplex σ, there is one and only one geometrically independent sets of points spanning σ.*

(5) *The set* Int σ *is convex and is open in the plane P; its closure is σ. Furthermore,* Int σ *equals the union of all the open line segments joining a_0 to points of* Int s, *where s is the face of σ opposite a_0.*

(6) *There is a homeomorphism of σ with the unit ball $\mathbb{B}^n$ that carries* Bd σ *onto the unit sphere $\mathbb{S}^{n-1}$.*

Proof: We leave parts **(1)**–**(5)** as exercises. Part **(6)** is a consequence of the next lemma, which is a stronger result that will be useful to us later. □

For a subset A of a topological space, we will use the notation $\overline{A}$ to denote the closure of A, the notation Int A to denote the interior of A, and the notation Bd $A = \overline{A} \setminus$ Int A to denote the boundary of A.

Lemma 1.2.7 *Let U be a bounded, convex open set in $\mathbb{R}^n$; let $w \in U$.*

(a) *Each ray emanating from w intersects* Bd $U = \overline{U} \setminus U$ *in precisely one point.*

(b) *There is a homeomorphism of $\overline{U}$ with $\mathbb{B}^n$ carrying* Bd U *onto $\mathbb{S}^{n-1}$.*

Proof:
(a) Given a ray $\mathcal{R}$ emanating from w, its intersection with U is convex, bounded and open in $\mathcal{R}$. Hence it consists of all points of the form $w + tp$, where t ranges over a half-open interval $[0, a)$. Thus the point $x = w + ap$ is a point of $\mathcal{R}$ that is in $\overline{U} \setminus U$.

Suppose $\mathcal{R}$ intersects $\overline{U} \setminus U$ in another point, say y. Then we have $y = w + bp$ for some $b > a$. We can write

$$x = (1 - t)w + ty,$$

where $t = a/b$. We rewrite this last equation in the form

$$w = (x - ty)/(1 - t).$$

Then we choose a sequence of points $y_n \in U$ converging to y, and we define

$$w_n = (x - ty_n)/(1 - t).$$

See Figure 1.2.2. The sequence w_n converges to $w \in U$, so $w_n \in U$ for some n. But then, since $x = tw_n + (1-t)y_n$, the point x belongs to U, because U is convex. This fact contradicts our choice of x.

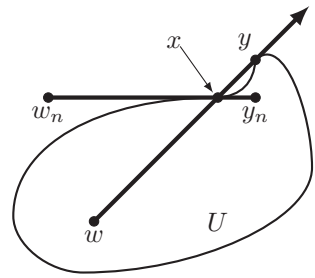


Figure 1.2.2

(b) Assume $w = \mathbf{0}$ for convenience. The equation $f(x) = x/\|x\|$ defines a continuous map f of $\mathbb{R}^n \setminus \mathbf{0}$ onto $\mathbb{S}^{n-1}$. By **(a)**, f restricts to a bijection of Bd U with $\mathbb{S}^{n-1}$. Since Bd U is compact, this restriction is a homeomorphism, and we let g be its inverse. Extend g to a bijection $G : \mathbb{B}^n \to \overline{U}$ by letting G map the line segment joining $\mathbf{0}$ to the point u of $\mathbb{S}^{n-1}$ linearly onto the line segment joining $\mathbf{0}$ to $g(u)$. Formally, we define

$$G(x) = \begin{cases} \|g(x/\|x\|)\|x & \text{if } x \neq \mathbf{0}, \\ \mathbf{0} & \text{if } x = \mathbf{0}. \end{cases}$$

Continuity of G for $x \neq \mathbf{0}$ is immediate. Continuity at $\mathbf{0}$ is easy: If M is a bound for $\|g(x)\|$, then whenever $\|x - \mathbf{0}\| < \delta$, we have $\|G(x) - G(\mathbf{0})\| < M\delta$. □

EXERCISES

1. Verify Lemma 1.2.6**(1)**–**(3)**. [**Hint:** If T is the affine transformation carrying a_0 to $\mathbf{0}$ and a_i to ϵ_i, then T carries the point

$$x = \sum_{i=0}^{n} t_i a_i$$

to the point $(t_1, t_2, \ldots, t_n, 0 \ldots, 0)$.]

2. Verify Lemma 1.2.6**(4)** as follows:

(a) Show that if $x \in \sigma$ and $x \neq a_0, a_1, \ldots, a_n$, then x lies in some open line segment contained in σ. [Assume $n > 0$.]

(b) Show that a_0 lies in no open line segment contained in σ, by showing that if $a_0 = tx + (1-t)y$, where $x, y \in \sigma$ and $0 < t < 1$, then $x = y = a_0$.

3. Verify Lemma 1.2.6**(5)**.

4. Generalize Lemma 1.2.6**(2)** as follows: Let σ be spanned by $a_0, a_1, \ldots, a_n$. Let s be the face of σ spanned by $a_0, a_1, \ldots, a_p$ [where $p < n$], and let t be the face opposite s.

(a) Show that σ is the union of all line segments joining points of s to points of t, and two of these line segments intersect in at most a common end point.

(b) Show that Int σ is the union of all open line segments joining points of Int s to points of Int t.

5. Let U be a bounded open set in $\mathbb{R}^n$. Suppose that U is star-convex relative to the origin:

(a) Show that a ray from $\mathbf{0}$ may intersect Bd U in more than one point.

*(b)** Show by example that $\overline{U}$ need not be homeomorphic to $\mathbb{B}^n$.

1.3 Simplicial Complexes and Simplicial Maps

Definition 1.3.1 A *simplicial complex* K in $\mathbb{R}^N$ is a collection of simplices in $\mathbb{R}^N$ such that

(1) Every face of a simplex of K is in K.

(2) The intersection of any two simplices of K is a face of each of them.

EXAMPLE 1.3.2 The collection K_1 pictured in Figure 1.3.1, of a 2-simplex and its faces, is a simplicial complex. The collection K_2, consisting of two 2-simplices with an edge in common, along with their faces, is a simplicial complex; while the collection K_3 is not. What about K_4?

The following lemma is sometimes useful in verifying that a collection of simplices is a simplicial complex:

Lemma 1.3.3 *A collection K of simplices is a simplicial complex if and only if the following hold:*

(1) *Every face of a simplex of K is in K.*

(2′) *Every pair of distinct simplices of K have disjoint interiors.*

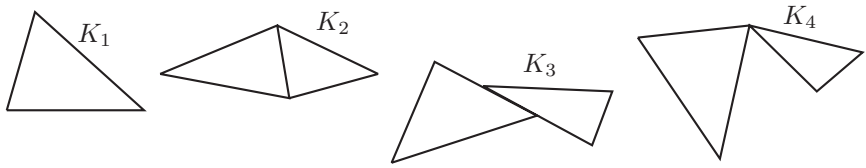


Figure 1.3.1

Proof: First, assume K is a simplicial complex. Given two simplices σ and τ of K we show that if there exists $x \in \text{Int } \sigma \cap \text{Int } \tau$, then $\sigma = \tau$. Let $s = \sigma \cap \tau$. If s were a proper face of σ, then x would belong to Bd σ, which it does not. Therefore, $s = \sigma$. Similarly, s is not a proper face of τ, so $s = \tau$. We conclude that $\sigma = s = \tau$.

To prove the converse, we assume **(1)** and **(2′)** hold. Let two simplices σ and τ of K be given. Suppose that $\sigma \cap \tau$ is non-empty. We will show that $\sigma \cap \tau$ is a face σ' of σ. Specifically, σ' is the face of σ spanned by the vertices $b_0, b_1, \ldots, b_m$ of σ that lie in τ. First, σ' is contained in $\sigma \cap \tau$ because $\sigma \cap \tau$ is convex and contains $b_0, b_1, \ldots, b_m$. To prove the reverse inclusion, suppose that $x \in \sigma \cap \tau$. Then $x \in \text{Int } s \cap \text{Int } t$, for some face s of σ and some face t of τ. It follows from **(2′)** that $s = t$. Hence the vertices of s lie in τ so that by definition they are elements of the set $\{b_0, b_1, \ldots, b_m\}$. Then s is a face of σ', as desired. Similarly, we can show that $\sigma \cap \tau$ is a face τ' of τ. Thus we have shown that **(2)** holds. □

It follows from that last lemma that if σ is a simplex, then the collection consisting of σ and its proper faces is a simplicial complex: Condition **(1)** is immediate, and condition **(2′)** holds because for each point $x \in \sigma$, there is exactly one face s of σ such that $x \in \text{Int } s$.

Definition 1.3.4 If L is a subcollection of K that contains all faces of its elements, then L is a simplicial complex in its own right, called a *subcomplex* of K. One subcomplex of K is the collection of all simplices of K of dimension at most p; it is called the *p-skeleton* of K and is denoted $K^{(p)}$. The points of the collection $K^{(0)}$ are called the *vertices* of K.

Definition 1.3.5 Let $|K|$ be the subset of $\mathbb{R}^N$ that is the union of the simplices of K. Giving each simplex its natural topology as a subspace of $\mathbb{R}^N$, we then topologize $|K|$ by declaring a subset A of $|K|$ to be closed in $|K|$ if and only if $A \cap \sigma$ is closed in σ, for each σ in K. It is easy to see that this defines a topology on $|K|$, for this collection of sets is closed under finite unions and arbitrary intersections. The space $|K|$ is called the *underlying space* of K, or the *polytope* of K.

A space that is the polytope of a simplicial complex will be called a *polyhedron*. [We note that some topologists reserve this term for the polytope of a *finite* simplicial complex.]

In general, the topology of $|K|$ is finer [i.e., larger] than the topology $|K|$ inherits as a subspace of $\mathbb{R}^N$. To see this, note that if A is closed in $|K|$ in the subspace topology, then $A = B \cap |K|$ for some closed set B in $\mathbb{R}^N$. Then $B \cap \sigma$ is closed in σ for each simplex σ of K, so $A \cap \sigma = B \cap |K| \cap \sigma = B \cap \sigma$ is closed in σ. By definition A is closed in the topology of $|K|$.

The two topologies on $|K|$, the subspace topology and the topology of Definition 1.3.5, are different in general. [See Examples 1.3.6 and 1.3.7.] Nonetheless, if K is finite, the two topologies are the same. For suppose K is finite and A is closed in $|K|$. Then $A \cap \sigma$ is closed in σ and hence closed in $\mathbb{R}^N$. Because A is the union of the finitely many closed sets $A \cap \sigma$, the set A also is closed in $\mathbb{R}^N$.

EXAMPLE 1.3.6 Let K be the collection of all 1-simplices in $\mathbb{R}$ of the form $[m, m+1]$, where $m \neq 0$ is an integer, along with all simplices of the form $[1/(n+1), 1/n]$ for n a positive integer, along with all faces of these simplices. Then K is a complex whose underlying space equals $\mathbb{R}$ as a set but not as a topological space. For instance, the set of points of the form $1/n$, $n > 0$, is closed in $|K|$ but not in $\mathbb{R}$.

EXAMPLE 1.3.7 Let K be the collection of 1-simplices $\sigma_1, \sigma_2, \ldots$ and their vertices, where σ_i is the 1-simplex in $\mathbb{R}^2$ having vertices $\mathbf{0}$ and $(1, 1/i)$. See Figure 1.3.2. Then K is a simplicial complex. The intersection of $|K|$ with the open parabolic arc $\{(x, x^2) : x > 0\}$ is closed in $|K|$, because its intersection with each simplex σ_i is a single point. It is not closed in the topology $|K|$ derives from $\mathbb{R}^2$, because in that topology it has the origin as a limit point.

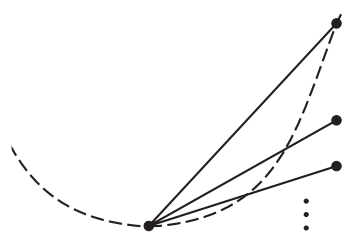


Figure 1.3.2

We prove some elementary topological properties of polyhedra.

Lemma 1.3.8 *If L is a subcomplex of K, then $|L|$ is a closed subspace of $|K|$. In particular, if $\sigma \in K$, then σ is a closed subspace of $|K|$.*

Proof: Suppose A is closed in $|L|$. If σ is a simplex of K, then $\sigma \cap |L|$ is the union of those faces s_i of σ that belong to L. Since A is closed in $|L|$, the set $A \cap s_i$ is closed in s_i and hence closed in σ. Since $A \cap \sigma$ is a finite union of the sets $A \cap s_i$, it is closed in σ. We conclude that A is closed in $|K|$.

Conversely, if B is closed in $|K|$, then $B \cap \sigma$ is closed in σ for each $\sigma \in K$, and in particular for each $\sigma \in L$. Hence $B \cap |L|$ is closed in $|L|$. □

Lemma 1.3.9 *A map $f : |K| \to X$ is continuous if and only if $f|\sigma$ is continuous for each $\sigma \in K$.*

Proof: If f is continuous, so is $f|\sigma$ since σ is a subspace of $|K|$. Conversely, suppose each map $f|\sigma$ is continuous. If C is a closed set of X, then $f^{-1}(C \cap \sigma) = (f|\sigma)^{-1}(C)$, which is closed in σ by the continuity of $f|\sigma$. Thus $f^{-1}(C)$ is closed in $|K|$ by definition. □

Definition 1.3.10 If X is a space and if $\mathcal{C}$ is a collection of subspaces of X whose union is X, then the topology of X is said to be *coherent* with the collection $\mathcal{C}$, provided a set A is closed in X if and only if $A \cap C$ is closed in C for each $C \in \mathcal{C}$. It is equivalent to require that U be open in X if and only if $U \cap C$ is open in C for each $C \in \mathcal{C}$.

In particular, the topology of $|K|$ is coherent with the collection of subspaces σ, for $\sigma \in K$.

The analogue of Lemma 1.3.9 holds for coherent topologies in general: a map $f : X \to Y$ is continuous if and only if $f|C$ is continuous for each $C \in \mathcal{C}$.

In the next definition, we extend the barycentric coordinates of Definition 1.2.1 to a family of functions defined on a polyhedron.

Definition 1.3.11 Let v be an arbitrary vertex of the simplicial complex K. The *barycentric coordinate* $t_v : |K| \to [0,1]$ is defined as follows: A point $x \in |K|$ is interior to precisely one simplex of K, whose vertices are, say, $a_0, a_1, \ldots, a_n$. Then there are positive numbers $t_0, t_1, \ldots, t_n$ such that

$$x = \sum_{i=0}^{n} t_i a_i \quad \text{and} \quad \sum_{i=0}^{n} t_i = 1\,.$$

If v is not one of the vertices a_i, set $t_v(x) = 0$. If v is one of the vertices a_i, say, $v = a_{i'}$, then set $t_v(x) = t_{i'}$.

For fixed v the function $t_v(x)$ is continuous when restricted to a fixed simplex σ of K, since either it is identically zero on σ or it equals the barycentric coordinate of x with respect to the vertex v of σ in the sense formerly defined. Therefore, $t_v(x)$ is continuous on $|K|$, by Lemma 1.3.9.

Lemma 1.3.12 *A polyhedron $|K|$ is Hausdorff.*

Proof: Given two points of $|K|$, $x_0 \neq x_1$, there is at least one vertex v such that $t_v(x_0) \neq t_v(x_1)$. Choose r between these two numbers. Then the sets $\{x : t_v(x) < r\}$ and $\{x : t_v(x) > r\}$ are the required disjoint open sets each containing one of the two points. □

Lemma 1.3.13 *If K is finite, then $|K|$ is compact. Conversely, if a subset A of $|K|$ is compact, then $A \subset |K_0|$ for some finite subcomplex K_0 of K.*

Proof: If K is finite, then $|K|$ is a finite union of compact subspaces σ, and hence is compact.

Now suppose that $A \subset |K|$ is compact and A does not lie in the polytope of any finite subcomplex of K. Choose a point $x_s \in A \cap \operatorname{Int} s$ whenever $A \cap \operatorname{Int} s$ is nonempty. Then the set $B = \{x_s\}$ is infinite. Furthermore, every subset of B is closed, since its intersection with any simplex σ is finite. Being closed and discrete, B has no limit point, contrary to the fact that every infinite subset of a compact space has a limit point. □

Three particular subspaces of $|K|$ are often useful when studying local properties of $|K|$. We mention them here.

Definition 1.3.14 If v is a vertex of K, the *star* of v in K, denoted by $\operatorname{St} v$, or sometimes by $\operatorname{St}(v, K)$, is the union of the interiors of those simplices of K that have v as a vertex. Its closure, denoted $\overline{\operatorname{St}}\ v$, is called the *closed star* of v in K. It is the union of all simplices of K having v as a vertex, and is the polytope of a subcomplex of K. The set $\overline{\operatorname{St}}\ v \setminus \operatorname{St} v$ is called the *link* of v in K and is denoted $\operatorname{Lk} v$. See Figure 1.3.3.

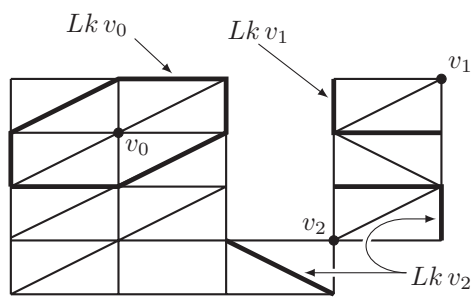


Figure 1.3.3

The set $\operatorname{St} v$ is open in $|K|$, since it consists of all points x of $|K|$ such that $t_v(x) > 0$. Its complement is the union of all simplices of K that do not

have v as a vertex; it is thus the polytope of a subcomplex of K. The set Lk v is also the polytope of a subcomplex of K; it is the intersection of $\overline{\text{St}}\ v$ and the complement of St v. The sets St v and $\overline{\text{St}}\ v$ are easily seen to be path connected, while the set Lk v need not be connected.

Definition 1.3.15 A simplicial complex K is said to be *locally finite* if each vertex of K belongs to only finitely many simplices of K. Said differently, a complex is locally finite if and only if each closed star $\overline{\text{St}}\ v$ is the polytope of a finite subcomplex of K.

Lemma 1.3.16 *The complex K is locally finite if and only if the space $|K|$ is locally compact.*

Proof: Suppose K is locally finite. Given $x \in |K|$, it lies in St v for some vertex v of K. Since $\overline{\text{St}}\ v$ is a compact set, $|K|$ is locally compact. We leave the converse as an exercise. □

Simplicial Maps

Now we introduce the notion of a "simplicial map" of one complex into another.

Lemma 1.3.17 *Let K and L be complexes, and let $f : K^{(0)} \to L^{(0)}$ be a map. Suppose that whenever the vertices $v_0, v_1, \ldots, v_n$ of K span a simplex of K, the points $f(v_0), f(v_1), \ldots, f(v_n)$ are vertices of a simplex of L. Then f can be extended to a continuous map $g : |K| \to |L|$ such that*

$$x = \sum_{i=0}^{n} t_i v_i \quad \Longrightarrow \quad g(x) = \sum_{i=0}^{n} t_i f(v_i)\,.$$

We call g the [linear] *simplicial map* induced by the vertex map f.

Proof: Note that although the vertices $f(v_0), f(v_1), \ldots, f(v_n)$ of L are not necessarily distinct, nonetheless they span a simplex τ of L, by hypothesis. When we "collect terms" in the expression for $g(x)$, it remains true that the coefficients are non-negative and that their sum is 1. To illustrate what we mean by collecting terms, suppose that $f(v_0) = f(v_1)$, while the remaining points $f(v_2), f(v_3), \ldots, f(v_n)$ are distinct. We have

$$g(x) = \sum_{i=0}^{n} t_i f(v_i) = (t_0 + t_1) f(v_1) + \sum_{i=3}^{n} t_i f(v_i)\,.$$

Thus τ is spanned by $f(v_1), f(v_2), \ldots, f(v_n)$ and $g(x)$ is the point of τ with barycentric coordinates $(t_0 + t_1), t_2, t_3, \ldots, t_n$. Hence g maps the n-simplex σ spanned by $v_0, v_1, \ldots, v_n$ continuously to the simplex τ whose vertex set is $\{f(v_0), f(v_1), \ldots, f(v_n)\}$.

The map g is continuous as a map of σ into τ, and hence as a map of σ into $|L|$. Then, by Lemma 1.3.9, g is continuous as a map of $|K|$ into $|L|$. □

The vertex map f from $K^{(0)}$ to $L^{(0)}$ determines the extension g from $|K|$ to $|L|$. Of course the restriction of g to the vertices $K^{(0)}$ brings us back to the original map f. Sometimes we will use only g [or only f] for both maps when no confusion will result.

Observe that, provided $v_0, v_1, \ldots, v_n$ span a simplex of K, the formula for g remains valid even if the points $v_0, v_1, \ldots, v_n$ are not distinct. It is essential that the t_i be non-negative and that they sum to 1. One sees that this is true, again by collecting terms.

The composition of simplicial maps $g : |K| \to |L|$ and $h : |L| \to |M|$ is again a simplicial map. We see that this is true as follows: Suppose σ is a simplex of K spanned by the vertices $v_0, v_1, \ldots, v_n$. By hypothesis, the vertices $g(v_0), g(v_1), \ldots, g(v_n)$ span a simplex of L whether they are distinct or not. Likewise, the vertices $h(g(v_0)), h(g(v_1)), \ldots, h(g(v_n))$ span a simplex of M, again whether they are distinct or not. For x in σ, we have $x = \sum t_i v_i$, where $t_i \geq 0$ and $\sum t_i = 1$, so we use the formula for g to obtain

$$h(g(x)) = h\left(\sum_{i=0}^{n} t_i g(v_i)\right) = \sum_{i=0}^{n} t_i h(g(v_i))\,.$$

Lemma 1.3.18 *Suppose that $f : K^{(0)} \to L^{(0)}$ is a bijective correspondence such that the vertices $v_0, v_1, \ldots, v_n$ of K span a simplex of K if and only if $f(v_0), f(v_1), \ldots, f(v_n)$ span a simplex of L. Then the induced simplicial map $g : |K| \to |L|$ is a homeomorphism.*

The map g is called a *simplicial homeomorphism*, or an *isomorphism*, of K with L.

Proof: Each simplex σ of K is mapped by g onto a simplex τ of the same dimension as σ. We need only show that the linear map $h : \tau \to \sigma$ induced by the vertex correspondence f^{-1} is the inverse of the map $g : \sigma \to \tau$. For that we note that if $x = \sum t_i v_i$, then $g(x) = \sum t_i f(v_i)$ by definition, so

$$h[\,g(x)\,] = h\Big[\,\sum t_i f(v_i)\,\Big] = \sum t_i f^{-1}[\,f(v_i)\,] = \sum t_i v_i = x. \qquad \square$$

Corollary 1.3.19 *Let Δ^N denote the complex consisting of an N-simplex and its faces. If K is a finite complex, then K is isomorphic to a subcomplex of Δ^N for some N.*

Proof: Let $v_0, v_1, \ldots, v_N$ be the vertices of K. Choose $a_0, a_1, \ldots, a_N$ to be geometrically independent points in $\mathbb{R}^N$, and let Δ^N consist of the N-simplex they span, along with all its faces. The vertex map $f(v_i) = a_i$ induces an isomorphism with a subcomplex of Δ^N. $\square$

General Simplicial Complexes

Our insistence that a simplicial complex K must lie in $\mathbb{R}^N$ for some N puts a limitation on the cardinality of K and on the dimension of the simplices of K. We now remove these restrictions.

Let J be an arbitrary index set, and let $\mathbb{R}^J$ denote the J-fold cartesian product of $\mathbb{R}$ with itself. An element of $\mathbb{R}^J$ is a function from J to $\mathbb{R}$, ordinarily denoted using the "tuple notation" $(x_\alpha)_{\alpha\in J}$. The product $\mathbb{R}^J$ is of course a vector space with the usual component-wise addition and multiplication by scalars.

Let $\mathbb{E}^J$ denote the subset of $\mathbb{R}^J$ consisting of all points $(x_\alpha)_{\alpha\in J}$ such that $x_\alpha = 0$ for all but finitely many values of α. Note that $\mathbb{E}^J$ is a vector subspace of $(x_\alpha)_{\alpha\in J}$. If $\epsilon_\alpha \in \mathbb{E}^J$ is the map of J into $\mathbb{R}$ whose value is 1 on the index α and is 0 on all other elements of J, then the set $\{\epsilon_\alpha : \alpha \in J\}$ is a basis for $\mathbb{E}^J$ [but not for $\mathbb{R}^J$, if J is infinite].

We call $\mathbb{E}^J$ *generalized euclidean space* and topologize it by the metric

$$|x - y| = \max_{\alpha\in J} |x_\alpha - y_\alpha| .$$

Everything we have done for complexes in $\mathbb{R}^N$ generalizes to complexes in $\mathbb{E}^J$. The space $\mathbb{E}^J$ is the union of certain of its finite-dimensional subspaces—namely, the subspaces spanned by finite subsets of the basis $\{\epsilon_\alpha : \alpha \in J\}$. Each such subspace is simply a copy of $\mathbb{R}^N$ for some N. Any finite set $\{a_0, a_1, \ldots, a_n\}$ of points of $\mathbb{E}^J$ lies in such a subspace. If the points are geometrically independent, then the simplex they span lies in the same subspace. Furthermore, the metric on $\mathbb{E}^J$ gives the usual topology on each such subspace. Therefore, any finite collection of simplices in $\mathbb{E}^J$ lies in a copy of $\mathbb{R}^N$ for some N. All that we are really doing is to allow ourselves to deal with complexes for which the total collection of simplices cannot be fitted entirely into any one $\mathbb{R}^N$. Conceptually it may seem more complicated to work in $\mathbb{E}^J$ than in $\mathbb{R}^N$, but in practice, it causes no difficulty at all.

We leave it to you to check that our results hold for complexes in $\mathbb{E}^J$. We shall use these results freely henceforth.

Let us make one further comment. If K is a complex in $\mathbb{R}^N$, then each simplex of K has dimension at most N. On the other hand, if K is a complex in $\mathbb{E}^J$, there need be no upper bound on the dimensions of the simplices of K. We define the *dimension* of K, denoted $\dim K$, to be the largest dimension of a simplex of K, if such a maximum exists; otherwise, we say that K has *infinite dimension*.

EXERCISES

1. Let K be a simplicial complex; let $\sigma \in K$. When is $\operatorname{Int}\sigma$ open in $|K|$? When is σ open in $|K|$?

2. Show that in general, $\operatorname{St} v$ and $\overline{\operatorname{St}}\, v$ are path connected.

3. **(a)** Show directly that the polyhedron of Example 1.3.7 is not locally compact.

(b) Show that, in general, if a complex K is not locally finite, then the space $|K|$ is not locally compact.

4. Show that the polyhedron of Example 1.3.7 is not metrizable. [**Hint:** Show that the first countability axiom fails.]

5. If $g : |K| \to |L|$ is a simplicial map carrying the vertices of σ onto the vertices of τ, show that g maps some face of σ homeomorphically onto τ.

6. Check that the proofs of all eight lemmas of this section apply without change to complexes in $\mathbb{E}^J$, if one simply replaces $\mathbb{R}^N$ by $\mathbb{E}^J$ throughout.

7. Let K be a complex. Show that $|K|$ is metrizable if and only if K is locally finite. [**Hint:** The function

$$d(x, y) = \sup |t_v(x) - t_v(y)|$$

is a metric for the topology of each finite subcomplex of K.]

8. Let K be a complex. Show that $|K|$ is normal. [**Hint:** If A is closed in $|K|$ and if $f : A \to [0, 1]$ is continuous, extend f step-by-step to $A \cup |K^{(p)}|$, using the Tietze extension theorem.]

9. Let K be a complex in $\mathbb{R}^N$. Show that $|K|$ is a subspace of $\mathbb{R}^N$ if and only if each point x of $|K|$ lies in an open set of $\mathbb{R}^N$ that intersects only finitely many simplices of K. Generalize to $\mathbb{E}^J$.

10. Show that the collection of all simplices in $\mathbb{R}$ of the form $[1/(n + 1), 1/n]$ for n a positive integer, along with their vertices, is a complex whose polytope is the subspace $(0, 1]$ of $\mathbb{R}$.

1.4 Abstract Simplicial Complexes

In practice, specifying a polyhedron X by giving a collection of simplices whose union is X is not a very convenient way of dealing with specific polyhedra. One quickly gets immersed in details of analytic geometry; it is messy to specify all the simplices and to make sure they intersect only when they should. It is much easier to specify X by means of an "abstract simplicial complex," a notion we now introduce.

Definition 1.4.1 An *abstract simplicial complex* is a collection $\mathcal{S}$ of finite nonempty sets, such that if A is an element of $\mathcal{S}$, then so is every nonempty subset of A.

The element A of $\mathcal{S}$ is called a *simplex* of $\mathcal{S}$; its *dimension* is one less than the number of its elements. Each nonempty subset of A is called a *face* of A.

The *dimension* of $\mathcal{S}$ is the largest dimension of one of its simplices or is infinite if there is no such largest dimension. The *vertex set* V of $\mathcal{S}$ is the union of the one-point elements of $\mathcal{S}$; we shall make no distinction between the vertex $v \in V$ and the 0-simplex $\{v\} \in \mathcal{S}$. A subcollection of $\mathcal{S}$ that is itself a complex is called a *subcomplex* of $\mathcal{S}$.

Two abstract complexes $\mathcal{S}$ and $\mathcal{T}$ are said to be *isomorphic* if there is a bijective correspondence f mapping the vertex set of $\mathcal{S}$ to the vertex set of $\mathcal{T}$ such that $\{a_0, a_1, \ldots, a_n\} \in \mathcal{S}$ if and only if $\{f(a_0), f(a_1), \ldots, f(a_n)\} \in \mathcal{T}$.

Definition 1.4.2 If K is a simplicial complex, let V be the vertex set of K. Let $\mathcal{K}$ be the collection of all subsets $\{a_0, a_1, \ldots, a_n\}$ of V such that the vertices $a_0, a_1, \ldots, a_n$ span a simplex of K. The collection $\mathcal{K}$ is called the *vertex scheme* of K.

The collection $\mathcal{K}$ is a particular example of an abstract simplicial complex. It is in fact the crucial example, as the following theorem shows.

Theorem 1.4.3

> **(a)** *Every abstract complex $\mathcal{S}$ is isomorphic to the vertex scheme of some simplicial complex K.*
>
> **(b)** *Two simplicial complexes are linearly isomorphic if and only if their vertex schemes are isomorphic as abstract simplicial complexes.*

Proof: Part **(b)** follows at once from Lemma 1.3.18. To prove **(a)**, we proceed as follows: Given an index set J, let Δ^J be the collection of all simplices in $\mathbb{E}^J$ spanned by finite subsets of the standard basis $\{\epsilon_\alpha\}_{\alpha \in J}$. It is easy to see that Δ^J is a simplicial complex. If σ and τ are two simplices of Δ^J, then their combined vertex set is geometrically independent and spans a simplex of Δ^J. If J is an infinite set, then Δ^J is an infinite dimensional simplex.

Now let $\mathcal{S}$ be an abstract simplicial complex with vertex set V. Choose an index set J large enough that there is an injective function $f : V \to \{\epsilon_\alpha\}_{\alpha \in J}$. [For instance, one could choose $J = V$.] We specify a subcomplex K of Δ^J by the condition that for each [abstract] simplex $\{a_0, a_1, \ldots, a_n\} \in \mathcal{S}$, the [geometric] simplex spanned by $f(a_0), f(a_1), \ldots, f(a_n)$ is to be in K. It is immediate that K is a simplicial complex and $\mathcal{S}$ is isomorphic to the vertex scheme of K; f is the required correspondence between their vertex sets. □

Definition 1.4.4 If the abstract simplicial complex $\mathcal{S}$ is isomorphic to the vertex scheme of the simplicial complex K, we call K a *geometric realization* of $\mathcal{S}$. A geometric realization is uniquely determined up to a linear isomorphism.

Let us illustrate how abstract complexes can be used to specify particular simplicial complexes.

EXAMPLE 1.4.5 Suppose we wish to indicate a simplicial complex K whose underlying space is homeomorphic to the cylinder $\mathbb{S}^1 \times I$. [Here I denotes the closed unit interval $[0, 1]$.] One way of doing so is to draw the picture

in Figure 1.4.1, which specifies K as a collection consisting of six 2-simplices and their faces. Another way of picturing this same complex K is to draw the diagram in Figure 1.4.2. This diagram consists of two things: first, a complex L whose underlying space is a rectangle, and second, a particular labeling of the vertices of L [some vertices being given the same label]. We shall consider this diagram to be a short-hand way of denoting the abstract complex $\mathcal{S}$ whose vertex set consists of the letters a, b, c, d, e, and f, and whose simplices are the sets $\{a, f, d\}$, $\{a, b, d\}$, $\{b, c, d\}$, $\{c, d, e\}$, $\{a, c, e\}$, $\{a, e, f\}$, along with their nonempty subsets. Of course, this abstract complex is isomorphic to the vertex scheme of the complex K pictured earlier, so it specifies precisely the same complex [up to linear isomorphism]. That is, the complex K of Figure 1.4.1 is a geometric realization of $\mathcal{S}$.

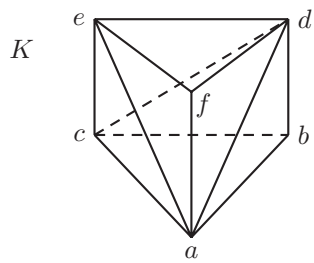


Figure 1.4.1

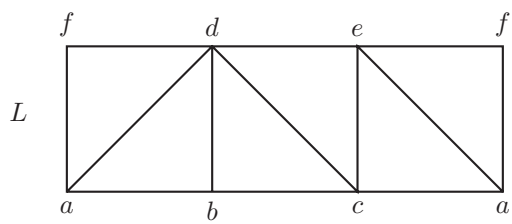


Figure 1.4.2

Let $f : L^{(0)} \to K^{(0)}$ be the map that assigns to each vertex of L the corresponding labeled vertex of K. Then f extends to a simplicial map $g : |L| \to |K|$. Because the spaces are compact Hausdorff, g is a quotient map, or "pasting map." It identifies the right edge of $|L|$ linearly with the left edge of L. And of course this is the usual way one forms a cylinder from a rectangular piece of paper—one bends it around and pastes the right edge to the left edge!

EXAMPLE 1.4.6 Now suppose we *begin* with a complex L and a labeling of its vertices. Consider for instance the same complex L with a different labeling of the vertices, as in Figure 1.4.3. Just as before, this diagram indicates a certain abstract complex $\mathcal{S}$, whose simplices one can list. Let K be a geometric realization of $\mathcal{S}$. As before, the vertices of K correspond to the letters

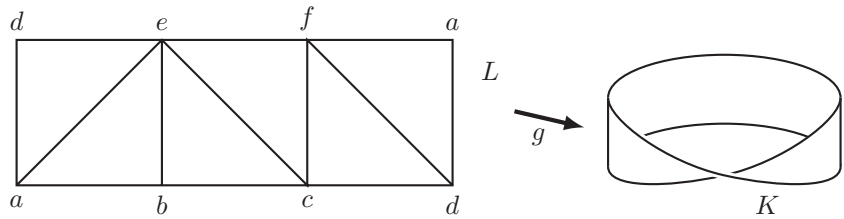


Figure 1.4.3

$a, b, \ldots, f$. We consider the linear simplicial map $g : |L| \to |K|$ that assigns to each vertex of L the correspondingly labeled vertex of K. Again, g is a quotient map; in this case, it identifies the left edge of $|L|$ linearly with the right edge of $|L|$, *but with a twist.* The space $|K|$ is the one we call the *Möbius band.* The Möbius band can be pictured in $\mathbb{R}^3$ as the familiar space indicated in Figure 1.4.3.

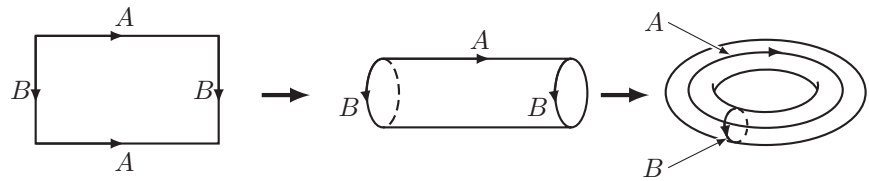
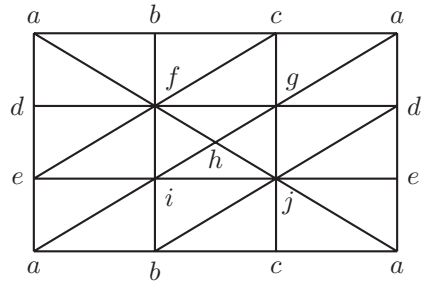


Figure 1.4.4

EXAMPLE 1.4.7 The *torus* is often defined as the quotient space obtained from a rectangle by making the identifications pictured in Figure 1.4.4. If we wish to construct a complex whose underlying space is homeomorphic to the torus, we can thus obtain it by using the diagram in Figure 1.4.5. You can check that the resulting quotient map of L onto the geometric realization of this diagram carries out precisely the identifications needed to form the torus.

EXAMPLE 1.4.8 Some care is required in general to make sure that the quotient map g does not carry out more identifications than one wishes. For instance, you may think that the diagram in Figure 1.4.6 determines the torus, but it does not. The quotient map in this case does more than paste opposite edges together, as you will see if you examine the diagram more closely.

EXAMPLE 1.4.9 The diagram in Figure 1.4.7 indicates an abstract complex whose underlying space is called the *Klein bottle.* It is the quotient space



L

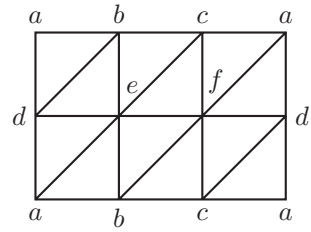


Figure 1.4.5

Figure 1.4.6

obtained from the rectangle by pasting the edges together as indicated in Figure 1.4.8. The resulting space cannot be imbedded in $\mathbb{R}^3$, but one can picture it in $\mathbb{R}^3$ by letting it "pass through itself."

EXAMPLE 1.4.10 The *projective plane* $\mathbb{P}^2$ is defined as the space obtained from the 2-sphere $\mathbb{S}^2$ by identifying x with $-x$ for each $x \in \mathbb{S}^2$. Exercise 2(b) asks you to show that Figure 1.4.9 determines a complex whose space is homeomorphic to $\mathbb{P}^2$.

We now describe more carefully the process indicated in the preceding examples: Given a finite complex L, a *labeling* of the vertices of L is a surjective function f mapping the vertex set of L to a set called the *set of labels*. Corresponding to this labeling is an abstract complex $\mathcal{S}$ whose vertices are the labels and whose simplices consist of all sets of the form $\{f(v_0), f(v_1), \ldots, f(v_n)\}$, where $v_0, v_1, \ldots, v_n$ span a simplex of L. Let K be a geometric realization of $\mathcal{S}$. Then the vertex map of $L^{(0)}$ onto $K^{(0)}$ derived from f extends to a surjective simplicial map $g : |L| \to |K|$. We say that K is the *complex derived from the labeled complex* and we call g the *associated pasting map*.

Because $|L|$ is compact and $|K|$ is Hausdorff, g maps closed sets to closed sets. Thus g is a closed quotient map. Of course, g can in general map a simplex of L onto a simplex of K of lower dimension. For instance, g might collapse all of L to a single point. We are more interested in cases where this

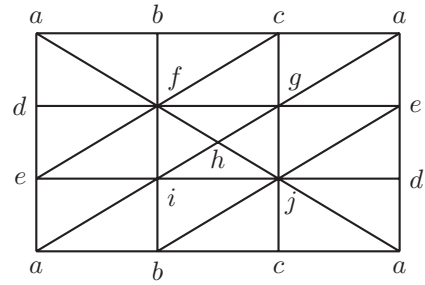


Figure 1.4.7

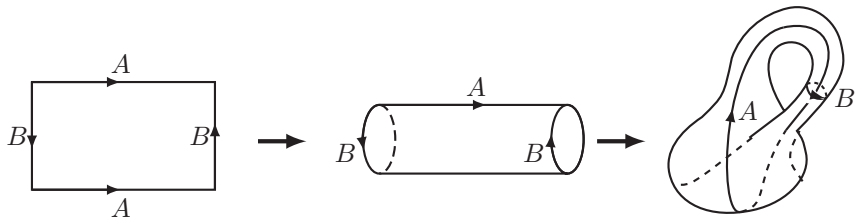


Figure 1.4.8

does not occur. We are particularly interested in situations similar to those of the preceding examples. We now state a lemma giving conditions under which the general "pasting map" g behaves like those in our examples. First, we need a definition.

Definition 1.4.11 If L is a complex, a subcomplex L_0 of L is said to be a *full subcomplex* of L provided each simplex of L whose vertices belong to L_0 belongs to L_0 itself.

For example, the boundary of the rectangle L pictured in Figure 1.4.5 is a full subcomplex of L, but the boundary of the rectangle pictured in Figure 1.4.6 is not.

Lemma 1.4.12 *Let L be a complex; let f be a labeling of its vertices; let $g : |L| \to |K|$ be the associated pasting map. Let L_0 be a full subcomplex of L. Suppose that whenever v and w are vertices of L having the same label:*

(1) *v and w belong to L_0.*

(2) *$\overline{\mathrm{St}}\, v$ and $\overline{\mathrm{St}}\, w$ are disjoint.*

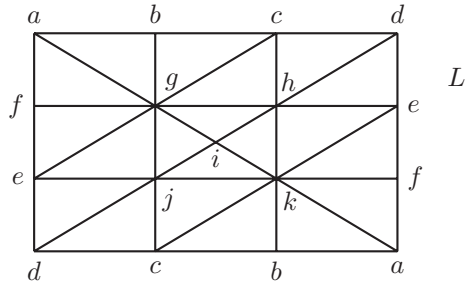


Figure 1.4.9

Then $\dim g(\sigma) = \dim \sigma$ *for all* $\sigma \in L$. *Furthermore, if* $g(\sigma_1) = g(\sigma_2)$, *then* σ_1 *and* σ_2 *must be disjoint simplices belonging to* L_0.

The proof is easy and is left as an exercise. In the usual applications of this lemma, $|L|$ is a polyhedral region in the plane or $\mathbb{R}^N$, and $|L_0|$ is the boundary of the region.

EXERCISES

1. **(a)** Show that $\mathbb{P}^2$ is homeomorphic to the space obtained from $\mathbb{B}^2$ by identifying x with $-x$ for each $x \in \mathbb{S}^1$.

(b) Show that the labeled complex L of Figure 1.4.9 determines a complex K whose space is homeomorphic to $\mathbb{P}^2$.

(c) Describe the space determined by the labeled complex of Figure 1.4.10.

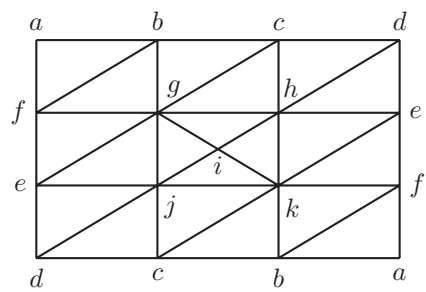


Figure 1.4.10

2. Describe the spaces determined by the labeled complexes in Figure 1.4.11–14.

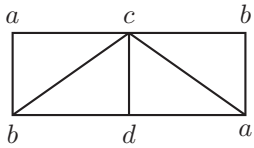


Figure 1.4.11

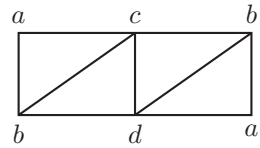


Figure 1.4.12

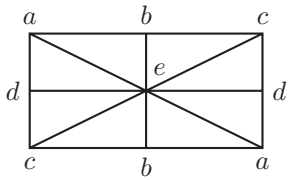


Figure 1.4.13

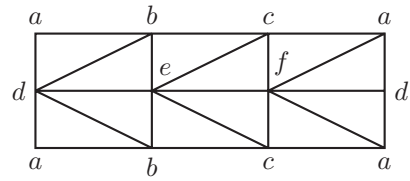


Figure 1.4.14

3. Prove Lemma 1.4.12.

4. Let S be a set with a partial order relation $\leq$. A standard technique in combinatorics is to associate with S the abstract complex $\mathcal{S}$ whose vertices are the elements of S and whose simplices are the finite simply-ordered subsets of S. Suppose one is given the partial order on $\{a_1, a_2, \ldots, a_8\}$ generated by the following relations:

$$a_1 \leq a_3 \leq a_7 \leq a_8; \quad a_1 \leq a_5 \leq a_7;$$
$$a_2 \leq a_6 \leq a_8; \quad a_2 \leq a_5.$$

Describe a geometric realization of $\mathcal{S}$.

1.5 Review of Abelian Groups

In this section we review some results from algebra that we shall be using—specifically, facts about abelian groups.

We write abelian groups with additive notation, so 0 denotes the identity element and $-g$ denotes the inverse of g. If n is a positive integer, then ng denotes the n-fold sum $g + g + \cdots + g$ and $(-n)g$ denotes $n(-g)$.

We denote the group of integers by $\mathbb{Z}$, the rationals by $\mathbb{Q}$, and the complex numbers by $\mathbb{C}$.

Homomorphisms

If $f : G \to H$ is a homomorphism, the *kernel* of f is the subgroup $f^{-1}(0)$ of G, the *image* of f is the subgroup $f(G)$ of H, and the *cokernel* of f is the quotient group $H/f(G)$. We denote these groups by $\ker f$, $\text{im } f$, and $\text{cok } f$, respectively. The map f is a *monomorphism* if and only its kernel contains only the identity—often more colorfully expressed by saying the kernel "vanishes." The map f is an *epimorphism* if and only if the cokernel of f vanishes; in this case, f induces an isomorphism $G/\ker f \cong H$.

Free Abelian Groups

An abelian group G is *free* if it has a *basis*—that is, if there is a family $\{g_\alpha\}_{\alpha \in J}$ of elements of G such that each $g \in G$ can be written uniquely as a finite sum

$$g = \sum n_\alpha g_\alpha,$$

with each n_α an integer. Uniqueness implies that each element g_α has infinite order, that is, g_α generates an infinite cyclic subgroup of G.

More generally, if each $g \in G$ can be written as a finite sum $g = \sum n_\alpha g_\alpha$, but not necessarily uniquely, then we say that the family $\{g_\alpha\}_{\alpha \in J}$ *generates* G. In particular, if the set $\{g_\alpha\}_{\alpha \in J}$ is finite, then we say that G is *finitely generated*.

If G is free and has a basis consisting of n elements, say $g_1, g_2, \ldots, g_n$, then every basis for G consists of precisely n elements. To see this we will show how the number of elements of the group $G/2G$ is related to the number of basis elements for G: Given the basis, we see that the group $G/2G$ consists of all cosets of the form

$$\sum \epsilon_i g_i + 2G,$$

where $\epsilon_i = 0$ or 1; this fact implies that the group $G/2G$ consists of precisely 2^n elements. The number of elements in a basis for G is called the *rank* of G.

It is true more generally that if G has an infinite basis, any two bases for G have the same cardinality. We will not use this fact.

A crucial property of free abelian groups is the following: if G has a basis $\{g_\alpha\}_{\alpha \in J}$, then any function f from the set $\{g_\alpha\}_{\alpha \in J}$ to an abelian group H extends uniquely to a homomorphism of G into H.

One specific way of constructing free abelian groups is the following: Given a set S, we define the *free abelian group G generated by S* to be the set of all functions $\phi : S \to \mathbb{Z}$ such that $\phi(x) \neq 0$ for only finitely many values of x. We add two such functions by adding their values. Given $x \in S$, there is a characteristic function ϕ_x for x defined by setting

$$\phi_x(y) = \begin{cases} 0 & \text{if } y \neq x, \\ 1 & \text{if } y = x. \end{cases}$$

The functions $\{\phi_x : x \in S\}$ form a basis for G, for each function $\phi \in G$ can be written uniquely as a finite sum

$$\sum n_x \phi_x,$$

where $n_x = \phi(x)$ and the summation extends over all x for which $\phi(x) \neq 0$. We often abuse notation and identify the element $x \in S$ with its characteristic function ϕ_x. With this notation, the general element of G can be written uniquely as a finite "formal linear combination"

$$\sum n_\alpha x_\alpha,$$

of elements of the set S.

If G is an abelian group, an element g of G has *finite order* if $ng = 0$ for some positive integer n. The set of all elements of finite order in G is a subgroup T of G, called the *torsion subgroup*. If T vanishes, we say G is *torsion-free*. A free abelian group is necessarily torsion-free, but the converse is not true.

If T consists of only finitely many elements, then the number of elements in T is called the *order* of T. If T has finite order, then each element of T has finite order, but again the converse is not true.

Internal Direct Sums

Suppose G is an abelian group, and suppose $\{G_\alpha\}_{\alpha \in J}$ is a collection of subgroups of G, indexed bijectively by some index set J. Suppose that each g in G can be written uniquely as a finite sum $g = \sum g_\alpha$, where $g_\alpha \in G_\alpha$ for each α. Then G is said to be the *internal direct sum* of the groups G_α, and we write

$$G = \bigoplus_{\alpha \in J} G_\alpha \,.$$

If the collection $\{G_\alpha\}$ is finite, say $\{G_\alpha\} = \{G_1, G_2, \ldots, G_n\}$, we also write this direct sum in the form $G = G_1 \oplus G_2 \oplus \cdots \oplus G_n$.

If each g in G can be written as a finite sum $g = \sum g_\alpha$, but not necessarily uniquely, we say simply that G is the *sum* of the groups $\{G_\alpha\}$, and we write $G = \sum G_\alpha$, or, in the finite case, $G = G_1 + G_2 + \cdots + G_n$. In this situation, we also say that the groups $\{G_\alpha\}$ *generate* G.

If $G = \sum G_\alpha$, then this sum is direct if and only if the equation $0 = \sum g_\alpha$ implies that $g_\alpha = 0$ for each α. This in turn occurs only if for each fixed index α_0, one has

$$G_{\alpha_0} \cap \left(\sum_{\alpha \neq \alpha_0} G_\alpha \right) = \{0\} \,.$$

In particular, if $G = G_1 + G_2$, then this sum is direct if and only if $G_1 \cap G_2 = \{0\}$.

The resemblance to free abelian groups is strong. Indeed, if G is free with basis $\{g_\alpha\}$, then G is the direct sum of the subgroups $\{G_\alpha\}$, where G_α is

the infinite cyclic group generated by g_α. Conversely, if G is a direct sum of infinite cyclic subgroups, then G is a free abelian group.

If G is the direct sum of subgroups $\{G_\alpha\}$, and if for each α, one has a homomorphism f_α of G_α into the abelian group H, then the homomorphisms $\{f_\alpha\}$ extend uniquely to a homomorphism of G into H.

Here is a useful criterion for showing that G is a direct sum:

Lemma 1.5.1 *Let G be an abelian group. If G is the direct sum of the subgroups $\{G_\alpha\}$, then there are homomorphisms*

$$j_\beta : G_\beta \to G \quad \text{and} \quad \pi_\beta : G \to G_\beta$$

such that $\pi_\beta \circ j_\alpha$ is the zero homomorphism if $\alpha \neq \beta$, and the identity homomorphism if $\alpha = \beta$.

Conversely, suppose $\{G_\alpha\}$ is a family of abelian groups, and there are homomorphisms j_β and π_β as above. Then j_β is a monomorphism. Furthermore, if the groups $j_\alpha(G_\alpha)$ generate G, then G is their direct sum.

Proof: Suppose $G = \oplus_{\alpha\in J} G_\alpha$. We define j_β to be the inclusion homomorphism. To define π_β, write $g = \sum g_\alpha$, where $g_\alpha \in G_\alpha$ for each α, and let $\pi_\beta(g) = g_\beta$. Uniqueness of the representation of g shows π_β is a well-defined homomorphism.

Next we consider the converse. Because $\pi_\alpha \circ j_\alpha$ is the identity on G_α, j_α is injective [and π_α is surjective]. If the groups $j_\alpha(G_\alpha)$ generate G, then every element of G can be written as a finite sum $\sum j_\alpha(g_\alpha)$. To show this representation is unique, suppose that

$$\sum j_\alpha(g_\alpha) = \sum j_\alpha(g'_\alpha)\,.$$

Applying π_β to both sides of this last equation, we see that $g_\beta = g'_\beta$. □

Direct Products and External Direct Sums

Let $\{G_\alpha\}_{\alpha\in J}$ be an indexed family of abelian groups. Their *direct product* $\prod_{\alpha\in J} G_\alpha$ is the group whose underlying set is the cartesian product of the sets G_α and whose group operation is component-wise addition. Their *external direct sum* G is the subgroup of the direct product consisting of all tuples $(g_\alpha)_{\alpha\in J}$ such that $g_\alpha = 0_\alpha$ for all but finitely many values of α. [Here 0_α is the zero element of G_α.] The group G is sometimes also called the *weak direct product* of the groups G_α.

The relation between internal and external direct sums is described as follows: Suppose that G is the external direct sum of the groups $\{G_\alpha\}$. Then for each β, we define $\pi_\beta : G \to G_\beta$ to be the projection onto the βth factor. We then define $j_\beta : G_\beta \to G$ by letting it carry the element $g \in G_\beta$ to the tuple $(g_\alpha)_{\alpha\in J}$ for which $g_\alpha = 0_\alpha$ for α different from β and $g_\beta = g$. Then $\pi_\beta \circ j_\alpha = 0$ for $\alpha \neq \beta$, and $\pi_\alpha \circ j_\alpha$ is the identity. It follows that G equals the *internal* direct sum of the groups $G'_\alpha = j_\alpha(G_\alpha)$, where G'_α is isomorphic to G_α.

Thus the notions of internal and external direct sums are closely related. The difference is mainly one of notation. For this reason, we customarily use the notations

$$G = G_1 \oplus G_2 \oplus \cdots \oplus G_n \qquad \text{and} \qquad G = \bigoplus_{\alpha \in J} G_\alpha$$

to denote either internal or external direct sums, relying on the context to make clear which is meant [if indeed, it is important]. With this notation, one can for instance express the fact that G is free abelian of rank 3 by writing $G \cong \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$.

If G_1 is a subgroup of G, we say that G_1 is a *direct summand* in G if there is a subgroup G_2 of G such that $G = G_1 \oplus G_2$. In this case, if H_i is a subgroup of G_i, for $i = 1, 2$, then the sum $H_1 + H_2$ is direct, and furthermore,

$$\frac{G}{H_1 \oplus H_2} \cong \frac{G_1}{H_1} \oplus \frac{G_2}{H_2}\,.$$

In particular, if $G = G_1 \oplus G_2$, then $G/G_1 \cong G_2$.

Of course, one can have $G/G_1 \cong G_2$ without its following that $G = G_1 \oplus G_2$, that is, G_1 may be a subgroup of G without being a direct summand in G. For instance, the subgroup $n\mathbb{Z}$ of the integers is not a direct summand in $\mathbb{Z}$, for that would mean that would mean that

$$\mathbb{Z} \cong n\mathbb{Z} \oplus G_2$$

for some subgroup G_2 of $\mathbb{Z}$. But then G_2 is isomorphic to $\mathbb{Z}/n\mathbb{Z}$, which is a group of finite order, while no subgroup of $\mathbb{Z}$ has finite order.

Incidentally, we shall denote the group $\mathbb{Z}/n\mathbb{Z}$ of integers modulo n simply by $\mathbb{Z}/n$, in accordance with current usage.

The Fundamental Theorem of Finitely Generated Abelian Groups

There are actually two theorems that are important to us. The first is a theorem about subgroups of free abelian groups. We state it here, and give a proof in Section 1.12:

Theorem 1.5.2 *Let F be a free abelian group. If R is a subgroup of F, then R is also a free abelian group. If F has rank n, then R has rank $r \leq n$; furthermore, there is a basis $e_1, e_2, \ldots, e_n$ for F and integers $t_1, t_2, \ldots, t_k$ with $t_i > 1$ such that*

(1) *$t_1 e_1, t_2 e_2, \ldots, t_k e_k,\ e_{k+1}, e_{k+2}, \ldots, e_r$ is a basis for R,*

(2) *$t_1 \mid t_2 \mid \cdots \mid t_k$, that is, t_i divides t_{i+1} for all i.*

The integers $t_1, t_2, \ldots, t_k$ are uniquely determined by F and R, although the basis $e_1, e_2, \ldots, e_n$ is not.

An immediate corollary of this theorem is the following:

Theorem 1.5.3 (The fundamental theorem of finitely generated abelian groups) *Let G be a finitely generated abelian group. Let T be its torsion subgroup.*

> **(a)** *There is a free abelian subgroup H of G having finite rank β such that $G = H \oplus T$.*
>
> **(b)** *There are finite cyclic groups $T_1, T_2, \ldots, T_k$, where T_i has order $t_i > 1$, such that $t_1 \mid t_2 \mid \cdots \mid t_k$ and*
>
> $$T = T_1 \oplus T_2 \oplus \cdots \oplus T_k \,.$$
>
> **(c)** *The numbers β and $t_1, t_2, \ldots, t_k$ are uniquely determined by G.*

The number β is called the *betti number* of G. The numbers $t_1, t_2, \ldots, t_k$ are called the *torsion coefficients* of G. Note that β is the rank of the free abelian group $G/T \cong H$. The rank of the subgroup H and the orders of the subgroups T_i are uniquely determined, but the subgroups themselves are not.

Proof: Let S be a finite set of generators $\{g_i\}$ for G, and let F be the free abelian group on the set S. The map carrying each g_i to itself extends to a homomorphism carrying F *onto* G. Let R be the kernel of this homomorphism. Then $F/R \cong G$. Choose bases for F and R as in Theorem 1.5.2. Then we have

$$F = F_1 \oplus F_2 \oplus \cdots \oplus F_n \,,$$

where F_i is infinite cyclic with generator e_i, and

$$R = t_1 F_1 \oplus t_2 F_2 \oplus \cdots \oplus t_k F_k \oplus F_{k+1} \oplus F_{k+2} \oplus \cdots \oplus F_r \,.$$

We compute the quotient group as follows:

$$F/R = (F_1/t_1 F_1 \oplus F/t_2 F_2 \oplus \cdots \oplus F/t_k F_k) \oplus (F_{r+1} \oplus F_{r+2} \oplus \cdots \oplus F_n) \,.$$

Thus there is an isomorphism

$$f : G \to (\mathbb{Z}/t_1 \oplus \mathbb{Z}/t_2 \oplus \cdots \mathbb{Z}/t_k) \oplus (\mathbb{Z} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}) \,.$$

The torsion subgroup T of G must be mapped to the subgroup

$$\mathbb{Z}/t_1 \oplus \mathbb{Z}/t_2 \oplus \cdots \oplus \mathbb{Z}/t_k$$

by f, since any isomorphism preserves torsion subgroups. Parts **(a)** and **(b)** of the theorem follow. Part **(c)** is left for the exercises. □

This theorem shows that any finitely generated abelian group G can be written as a finite direct sum of cyclic groups, that is,

$$G \cong (\mathbb{Z} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}) \oplus \mathbb{Z}/t_1 \oplus \mathbb{Z}/t_2 \oplus \cdots \mathbb{Z}/t_k \,.$$

with $t_i > 1$ and $t_1 \mid t_2 \mid \cdots \mid t_k$. This representation is in some sense a "canonical form" for G. There is another such canonical form, derived as follows:

Recall first the fact that if m and n are relatively prime positive integers, then

$$\mathbb{Z}/m \oplus \mathbb{Z}/n \cong \mathbb{Z}/mn\,.$$

It follows that any finite cyclic group can be written as a direct sum of cyclic groups whose orders are powers of primes. Theorem 1.5.3 then implies that for any finitely generated group G,

$$G \cong (\mathbb{Z} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}) \oplus \mathbb{Z}/a_1 \oplus \mathbb{Z}/a_2 \oplus \cdots \mathbb{Z}/a_s\,,$$

where each a_i is a power of a prime. This is another canonical form for G, since the numbers a_i are uniquely determined by G up to rearrangement, as we shall see.

EXERCISES

1. Show that if G is a finitely generated abelian group, then every subgroup of G is finitely generated. [This result does not hold for non-abelian groups.]

2. **(a)** Show that if G is free, then G is torsion-free.

(b) Show that if G is finitely generated and torsion-free, then G is free.

(c) Show that the additive group of rationals $\mathbb{Q}$ is torsion free but not free. [**Hint:** If $\{g_\alpha\}$ is a basis for $\mathbb{Q}$, let β be fixed and express $g_\beta/2$ in terms of this basis.]

3. **(a)** Show that if m and n are relatively prime, then $\mathbb{Z}/m \oplus \mathbb{Z}/n$ is cyclic of order mn.

(b) If $G \cong \mathbb{Z}/18 \oplus \mathbb{Z}/36$, then express G as a direct sum of cyclic groups of prime power order.

(c) If $G \cong \mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/9$, then find the torsion coefficients of G.

(d) If $G \cong \mathbb{Z}/15 \oplus \mathbb{Z}/20 \oplus \mathbb{Z}/18$, then find the torsion coefficients of G.

4. **(a)** Let p be a prime; let $b_1, b_2, \ldots, b_k$ be non-negative integers. Show that if

$$G \cong (\mathbb{Z}/p)^{b_1} \oplus (\mathbb{Z}/p^2)^{b_2} \oplus \cdots \oplus (\mathbb{Z}/p^k)^{b_k}\,,$$

then the integers b_i are uniquely determined by G. [**Hint:** Consider the kernel of the homomorphism $f_i : G \to G$ that is multiplication by p^i. Show that f_1 and f_2 determine b_1. Proceed similarly.]

(b) Let $p_1, p_2, \ldots, p_N$ be a sequence of distinct primes. Generalize **(a)** to a finite direct sum of terms of the form $(\mathbb{Z}/(p_i)^k)^{b_{ik}}$, where $b_{ik} \geq 0$.

(c) Verify **(c)** of Theorem 1.5.3. That is, show that the betti number and torsion coefficients of a finitely generated abelian group G are uniquely determined by G.

(d) Show that the numbers t_i appearing in the conclusion of Theorem 1.5.2 are uniquely determined by F and R.

1.6 Homology Groups

Now we are ready to define the homology groups. First we must discuss the notion of "orientation."

Definition 1.6.1 Let σ be a simplex [either geometric or abstract]. Define two orderings of its vertex set to be equivalent if they differ from one another by an even permutation. If $\dim \sigma > 0$, then the orderings of the vertices of σ fall into two equivalence classes. Each of these classes is called an *orientation* of σ. [If σ is a 0-simplex, then there is only one class and hence only one orientation of σ.] An *oriented simplex* is a simplex σ together with an orientation of σ.

If the points $v_0, v_1, \ldots, v_p$ are independent, we will use the symbol

$$v_0 v_1 \ldots v_p$$

to denote the simplex they span, and we shall use the symbol

$$[v_0, v_1, \ldots, v_p]$$

to denote the oriented simplex consisting of the simplex $v_0 v_1 \ldots v_p$ and the equivalence class of the particular ordering $(v_0, v_1, \ldots, v_p)$.

Occasionally, when the context makes the meaning clear, we may use a single letter such as σ to denote either a simplex of an oriented simplex.

EXAMPLE 1.6.2 We often picture an orientation of a 1-simplex by drawing an arrow on it. The oriented simplex $[v_0, v_1]$ is pictured in Figure 1.6.1; one draws an arrow pointing from v_0 to v_1. An orientation of a 2-simplex is pictured by a circular arrow. The oriented simplex $[v_0, v_1, v_2]$ is indicated in the

figure by drawing an arrow in the direction from v_0 to v_1 to v_2. You can check that $[v_1, v_2, v_0]$ and $[v_2, v_0, v_1]$ are indicated by the same clockwise arrow. An arrow in the counterclockwise direction would indicate the oppositely oriented simplex.

Similarly, the oriented simplex $[v_0, v_1, v_2, v_3]$ is pictured by drawing a spiral arrow, as in the figure. The arrow in this picture is called a "right-hand screw." If one curls the fingers of the right hand in the direction from v_0 to v_1 to v_2, then the thumb points toward v_3. You can check that $[v_0, v_2, v_3, v_1]$, and each of the other ten orderings equivalent to these two, also give rise to right-hand screws. A "left-hand screw" is used to picture the opposite orientation.

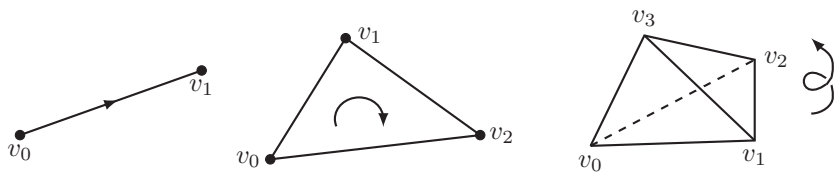


Figure 1.6.1

Definition 1.6.3 Let K be a simplicial complex. A *p-chain* on K is a function c from the set of oriented p-simplices of K to the integers such that:

(1) $c(\sigma) = -c(\sigma')$ if σ and σ' are opposite orientations of the same simplex,

(2) $c(\sigma) = 0$ for all but finitely many oriented simplices of K.

We add p-chains by adding their values; the resulting group is denoted $C_p(K)$ and is called the *group of oriented p-chains* of K. If $p < 0$ or $p > \dim K$, then we let $C_p(K)$ denote the trivial group.

If σ is an oriented simplex, then the *elementary chain* corresponding to σ is the function c defined by

$$\begin{aligned} &c(\sigma) = 1, \\ &c(\sigma') = -1 \quad \text{if } \sigma' \text{ is the opposite orientation of } \sigma, \\ &c(\tau) = 0 \qquad \text{for all other oriented simplices } \tau. \end{aligned}$$

By abuse of notation, we often use the symbol σ to denote not only a simplex, or an oriented simplex, but also to denote the elementary p-chain corresponding to the oriented simplex σ. With this convention, if σ and σ' are opposite orientations of the same simplex, then we write $\sigma' = -\sigma$, because this equation holds when σ and σ' are interpreted as elementary chains.

Lemma 1.6.4 *The group $C_p(K)$ is free abelian. A basis for $C_p(K)$ can be obtained by orienting each p simplex and using the corresponding elementary chains as a basis.*

Proof: The proof is straightforward. Once all the p-simplices of K are oriented—each arbitrarily with no need to reference any other p-simplex—then each p-chain, c, can be written uniquely as a finite linear combination

$$c = \sum n_i \sigma_i$$

of the corresponding elementary chains σ_i. The chain c assigns the value n_i to the oriented p-simplex σ_i, the value $-n_i$ to the opposite orientation of σ_i, and the value 0 to all the oriented p-simplices not appearing in the summation. □

The group $C_0(K)$ differs from the others in that it has a natural basis, because each 0-simplex has only one orientation. The group $C_p(K)$ has no "natural" basis if $p > 0$, since one must orient the p-simplices of K in some arbitrary fashion in order to obtain a basis.

Corollary 1.6.5 *Any function f from the oriented p-simplices of K to an abelian group G, extends uniquely to a homomorphism $C_p(K) \to G$, provided that $f(-\sigma) = -f(\sigma)$ for all oriented p-simplices.*

Definition 1.6.6 We now define a homomorphism

$$\partial_p : C_p(K) \to C_{p-1}(K)$$

called the *boundary operator*. If $\sigma = [v_0, v_1, \ldots, v_p]$ is an oriented simplex with $p > 0$, then we define

$$\partial_p \sigma = \partial_p [v_0, v_1, \ldots, v_p] = \sum_{i=0}^{p} (-1)^p \, [v_0, v_1, \ldots, \widehat{v_i}, \ldots, v_p] \,, \qquad (*)$$

where the symbol $\widehat{v_i}$ means that the vertex is to be deleted from the array. Since $C_p(K)$ is the trivial group for $p < 0$, the operator ∂_p is the trivial homomorphism for $p \leq 0$.

We must check that ∂_p is well-defined. Specifically, we must show that $\partial_p(-\sigma) = -\partial_p(\sigma)$. For this purpose, it suffices to show that the right-hand side of $(*)$ changes sign if we exchange two adjacent vertices in the array $[v_0, v_1, \ldots, v_p]$. So let us compare

$$\partial_p \sigma = \partial_p [v_0, v_1, \ldots, v_j, v_{j+1}, \ldots, v_p]$$

and

$$\partial_p (-\sigma) = \partial_p [v_0, v_1, \ldots, v_{j+1}, v_j, \ldots, v_p] \,.$$

We will compute both in detail and compare the results:

$$\begin{aligned}
\partial_p\sigma \;=\; & \sum_{i<j}(-1)^i[v_0,\ldots,\widehat{v_i},\ldots,v_{j-1},v_j,v_{j+1},v_{j+2},\ldots,v_p] && (a)\\
+\; & \sum_{j+1<i}(-1)^i[v_0,\ldots,v_{j-1},v_j,v_{j+1},v_{j+2},\ldots,\widehat{v_i},\ldots,v_p] && (b)\\
& +(-1)^j[v_0,\ldots,v_{j-1},v_{j+1},v_{j+2},\ldots,v_p] && (c)\\
& +(-1)^{j+1}[v_0,\ldots,v_{j-1},v_j,v_{j+2},\ldots,v_p]\,. && (d)
\end{aligned}$$

$$\begin{aligned}
\partial_p(-\sigma) \;=\; & \sum_{i<j}(-1)^i[v_0,\ldots,\widehat{v_i},\ldots,v_{j-1},v_{j+1},v_j,v_{j+2},\ldots,v_p] && (a')\\
+\; & \sum_{j+1<i}(-1)^i[v_0,\ldots,v_{j-1},v_{j+1},v_j,v_{j+2},\ldots,\widehat{v_i},\ldots,v_p] && (b')\\
& +(-1)^j[v_0,\ldots,v_{j-1},v_j,v_{j+2},\ldots,v_p] && (c')\\
& +(-1)^{j+1}[v_0,\ldots,v_{j-1},v_{j+1},v_{j+2},\ldots,v_p]\,. && (d')
\end{aligned}$$

The right-hand side of line (a') is the negative of the right-hand side of line (a), because v_j and v_{j+1} are interchanged, while all other vertices are in their same positions. Likewise, line (b') is the negative of line (b). Line (c') is the negative of line (d), because the exponent on -1 differs by 1, while all vertices are in their same positions. Likewise (d') is the negative of line (c). Thus we have confirmed that $\partial_p(-\sigma) = -\partial_p(\sigma)$.

EXAMPLE 1.6.7 For a 1-simplex, we compute $\partial_1[v_0, v_1] = v_1 - v_0$. For a 2-simplex, one has

$$\partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]\,.$$

For a 3-simplex, one has the formula

$$\partial_3[v_0, v_1, v_2, v_3] = [v_1, v_2, v_3] - [v_0, v_2, v_3] + [v_0, v_1, v_3] - [v_0, v_1, v_2]\,.$$

The geometric content of these formulas is pictured in Figure 1.6.2. If you remember the versions of Green's, Stokes', and Gauss' theorem that you studied in calculus, these pictures should look rather familiar.

EXAMPLE 1.6.8 Consider the 1 chain $\partial_2[v_0, v_1, v_2]$ pictured in Figure 1.6.2. If you apply the operator ∂_1 to this 1-chain, you get zero; everything cancels out because each vertex appears as the initial point of one edge and as the end point of another edge. You can check that a similar cancellation occurs when you compute $\partial_2\partial_3[v_0, v_1, v_2, v_3]$.

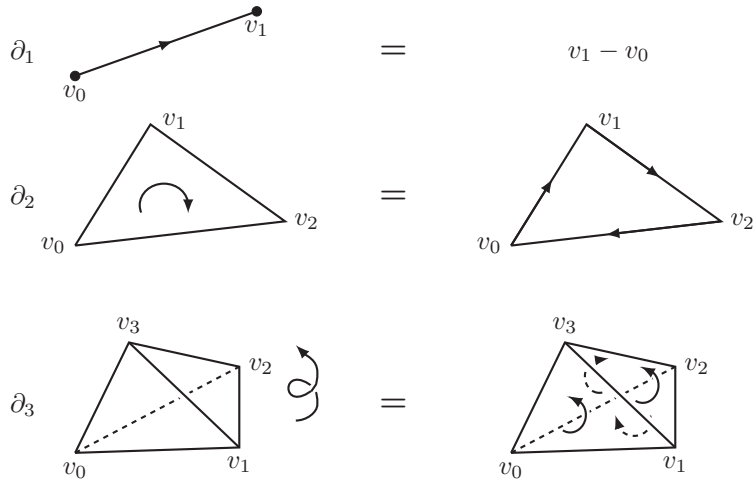


Figure 1.6.2

The computation discussed in Example 1.6.8 illustrates a general fact:

Lemma 1.6.9 *For each integer p, it holds that $\partial_{p-1} \circ \partial_p = 0$.*

Proof: Assume $0 < p \leq \dim K$, since otherwise the result is trivial. Let $[v_0, v_1, \ldots, v_p]$ be an oriented simplex in K. We compute

$$\partial_{p-1}\ \partial_p[v_0, v_1, \ldots, v_p] = \partial_{p-1}\ \left(\sum_{i=0}^{p}(-1)^i\ [v_0, v_1, \ldots, \widehat{v}_i, \ldots, v_p]\right)$$

$$= \sum_{i=0}^{p}(-1)^i\ \partial_{p-1}[v_0, v_1, \ldots, \widehat{v}_i, \ldots, v_p]$$

$$= \sum_{i=0}^{p}\sum_{j<i}(-1)^i\ (-1)^j\ [v_0, v_1, \ldots, \widehat{v}_j, \ldots, \widehat{v}_i, \ldots, v_p]$$

$$+ \sum_{i=0}^{p}\sum_{j>i}(-1)^i\ (-1)^{j-1}\ [v_0, v_1, \ldots, \widehat{v}_i, \ldots, \widehat{v}_j, \ldots, v_p]\,.$$

The terms of these two summations cancel in pairs. □

Definition 1.6.10 The kernel of $\partial_p : C_p(K) \to C_{p-1}(K)$ is called the group of *p-cycles* and is denoted $Z_p(K)$. [The "Z" comes from the German word "Zyklus,"—the letter "C" has already been used for chains.] The image of $\partial_{p+1} : C_{p+1}(K) \to C_p(K)$ is called the group of *p-boundaries* and is denoted

$B_p(K)$. By the preceding lemma, the boundary of a $(p+1)$-chain is automatically a p-cycle, that is, $B_p(K) \subset Z_p(K)$. We define

$$H_p(K) = Z_p(K)/B_p(K)\,,$$

and call this quotient group the pth *homology group* of K.

Let us compute a few examples.

EXAMPLE 1.6.11 Consider the complex K of Figure 1.6.3, whose underlying space is the boundary of a square with edges $e_1,\ e_2,\ e_3,\ e_4$. The group $C_1(K)$ is free abelian of rank 4; the general 1-chain c is of the form $c = \sum n_i e_i$. Computing $\partial_1 c$, we see that its value on the vertex v is $n_1 - n_2$. A similar argument, applied to the other vertices, shows that c is a cycle if and only if $n_1 = n_2 = n_3 = n_4$. One concludes that $Z_1(K)$ is infinite cyclic, and is generated by the chain $e_1 + e_2 + e_3 + e_4$. Since there are no 2-simplices in K, $B_1(K)$ is trivial. Therefore, we have

$$H_1(K) = Z_1(K)/B_1(K) \cong \mathbb{Z}\,.$$

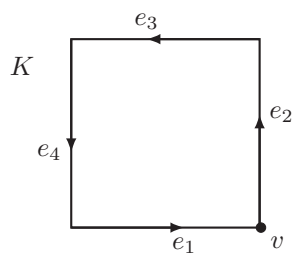


Figure 1.6.3

EXAMPLE 1.6.12 Consider the complex L of Figure 1.6.4, whose underlying space is a square region [not just the boundary of the square, but the interior also]. The general 1-chain is of the form $c = \sum n_i e_i$. One reasons as in the preceding example to conclude that this 1-chain is a cycle if and only if $n_1 = n_2$, $n_3 = n_4$, and $n_5 = n_3 - n_2$. One can assign values to n_2 and n_3 arbitrarily, and then the others are determined. Therefore, $Z_1(L)$ is free abelian of rank 2. One possible basis for $Z_1(L)$ consists of the chain $e_1 + e_2 - e_5$, obtained by taking $n_2 = 1$ and $n_3 = 0$, and the chain $e_3 + e_4 + e_5$, obtained by taking $n_2 = 0$ and $n_3 = 1$. This choice of basis is convenient, because $\partial_2\sigma_1 = e_1 + e_2 - e_5$ and $\partial\sigma_2 = e_3 + e_4 + e_5$. Therefore we have $Z_1(L) = B_1(L)$ and

$$H_1(K) = Z_1(L)/B_1(L) = 0\,.$$

We also have $H_2(L) = 0$. This last equation holds because the general 2-chain in L is $m_1\sigma_1 + m_2\sigma_2$ which is a cycle if and only if $m_1 = m_2 = 0$.

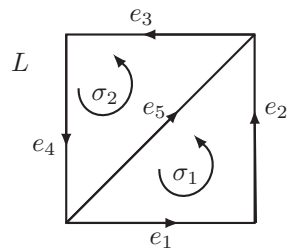


Figure 1.6.4

These examples may begin to give you a feeling for what the homology groups mean geometrically. Only by computing many more examples can one begin to "see" what a homology class is. Our hope is that after you get a feeling for what homology means geometrically, you will begin to believe what is at the moment far from clear—that the homology groups of a complex K actually depend only on the underlying space $|K|$.

Now let us consider another example. It involves a complex having more simplices than those in the preceding examples. In general, as the number of simplices increases, calculating the group of cycles Z_p and the group of boundaries B_p becomes more tedious. We can short-cut some of these calculations by avoiding calculating these groups and proceeding directly to a calculation of the homology groups H_p.

We deal here only with the groups $H_p(K)$ for $p > 0$, postponing discussion of the group $H_0(K)$ to Section 1.8.

We need some terminology. We shall say that a chain c is *carried by* a subcomplex L of K if c has value 0 on every simplex that is not in L. We say that two p-chains c and c' are *homologous* if $c - c' = \partial_{p+1} d$ for some $(p+1)$-chain d. In particular, if $c = \partial_{p+1} d$, we say that c is *homologous to zero*, or simply that c *bounds*.

EXAMPLE 1.6.13 Consider the complex M indicated in Figure 1.6.5, whose underlying space is a square region. Instead of computing the group of 1-cycles specifically, we reason as follows: Given a 1-chain c, let a be the value of c on e_1. Then by direct computation, the chain

$$c_1 = c + \partial_2(a\sigma_1)$$

has value 0 on the oriented simplex e_1. Intuitively speaking, by modifying c by a boundary, we have "pushed it off e_1." We then push c_1 off e_2 in a similar manner, as follows: Let b be the value of c_1 on e_2, then the chain

$$c_2 = c_1 + \partial_2(b\sigma_2)$$

has value 0 on the oriented simplex e_2. It also still has value 0 on e_1, since e_1 does not appear in the expression for $\partial_2\sigma_2$. Now letting d denote the value of c_2 on e_3, one sees that

$$c_3 = c_2 + \partial_2(d\sigma_3)$$

has value 0 on e_3 and on e_2 and on e_1. We have thus pushed c off all of e_1, e_2, and e_3. Said differently, we have proved the following result:

Given a 1-chain c, it is homologous to a chain c_3 that is carried by the subcomplex of M that is pictured in Figure 1.6.5.

Now if c happens to be a *cycle*, then c_3 is also a cycle. It follows that the value of c_3 on the simplex e_4 must be 0, because otherwise $\partial_1 c_3$ would have a non-zero value on the center vertex v. Thus every 1-cycle of M is homologous to a 1-cycle carried by the boundary of the square. By the same argument as used before in Example 1.6.11, such a cycle must be some multiple of the cycle $e_5 + e_6 + e_7 + e_8$. This last cycle bounds; indeed, it clearly equals $\partial_2(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4)$. Thus we have $H_1(M) = 0$, as expected.

The fact that $H_2(M) = 0$ is easy: One sees readily [as before] that $\sum m_i\sigma_i$ is a cycle if and only if $m_i = 0$ for all i.

Note that the homology groups of M are the same as the homology groups of the complex L of Example 1.6.12. This fact lends some plausibility to our remark [yet to be proved] that the homology groups of a complex depend only on its underlying space.

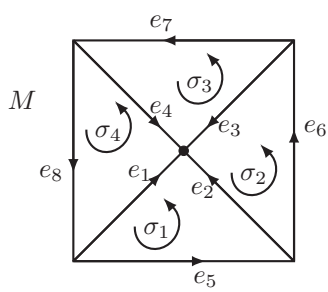


Figure 1.6.5

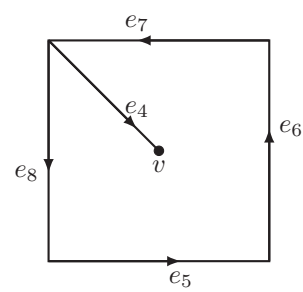


Figure 1.6.6

Definition 1.6.14 Let K be a complex. An *infinite p-chain* on K is a function c from the oriented p-simplices of K to the integers such that $c(\sigma) = -c(\sigma')$ if σ and σ' are opposite orientations of the same simplex. We do not require that $c(\sigma) = 0$ for all but finitely many oriented simplices. Let $C_p^\infty(K)$ denote the *group of infinite p-chains*. It is abelian, but it will not in general be free.

After one has defined an appropriate boundary operator [see Exercise 6] $\partial_p^\infty : C_p^\infty(K) \to C_{p-1}^\infty(K)$, the resulting groups

$$H_p^\infty(K) = \ker \partial_p^\infty / \text{im } \partial_{p+1}^\infty$$

are called the *homology groups based on infinite chains.*

The next definition is used in Exercise 3.

Definition 1.6.15 A 1-dimensional complex if called a *tree* if its 1-dimensional homology vanishes.

EXERCISES

1. Let $\mathcal{S}$ be the abstract complex consisting of the 1-simplices $\{v_0, v_1\}$, $\{v_1, v_2\}$, ..., $\{v_{n-1}, v_n\}$, $\{v_n, v_0\}$, and their vertices. If K is a geometric realization of $\mathcal{S}$, compute $H_i(K)$.

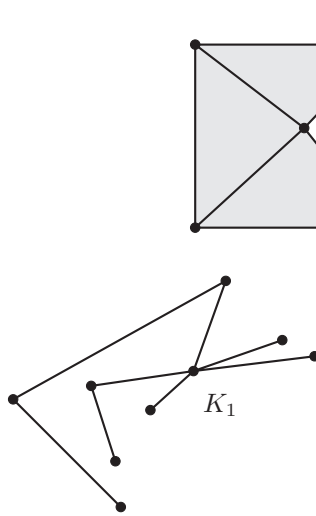


Figure 1.6.7

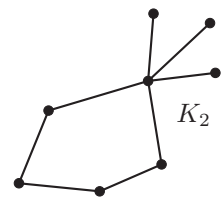


Figure 1.6.8

2. Consider the complex M pictured in Figure 1.6.7; it is the union of three triangles and a line segment. Compute the homology groups $H_1(M)$ and $H_2(M)$.

3. Is either of the complexes pictured in Figure 1.6.8 a tree?

4. Let K be the complex consisting of the proper faces of a 3-simplex. Compute $H_1(K)$ and $H_2(K)$.

5. For what values of i is it true that

$$H_i(K^{(p)}) \cong H_i(K)\,?$$

6. **(a)** Show that if K is locally finite, then one can define a boundary operator

$$\partial_p^\infty : C_p^\infty(K) \to C_{p-1}^\infty(K)$$

by the formula used earlier, and Lemma 1.6.9 holds.

(b) Let K be the complex whose space is $\mathbb{R}$ and whose vertices are the integers. Show that

$$H_1(K) = 0 \quad \text{and} \quad H_1^\infty(K) \cong \mathbb{Z}\,.$$

1.7 Homology Groups of Surfaces

If K is a finite complex, then the chain group $C_p(K)$ has finite rank, so the cycle group $Z_p(K)$ also has finite rank. Then $H_p(K)$ is finitely generated, so that the fundamental theorem of abelian groups applies. The betti number and torsion coefficients of $H_p(K)$ are called, classically, the betti number of K and the torsion coefficients of K in dimension p. The fact that these numbers are topological invariants of $|K|$ will be proved in Chapter 2.

In former times, much attention was paid to the *numerical* invariants, not only in topology, but also in algebra and other parts of mathematics. Nowadays, mathematicians are likely to feel that homology *groups* are the more important notion, and one is more likely to study properties of these groups than to compute their numerical invariants. Nevertheless, it is still important in many situations to compute the homology groups specifically—that is, to find the betti numbers and torsion coefficients for a given space.

One of the greatest virtues of the *simplicial* homology groups is that it is in fact possible to do precisely this. In Section 1.12 we shall prove a theorem to the effect that, for a finite complex K, the homology groups are effectively computable. This means that there is a specific algorithm for finding the betti numbers and torsion coefficients of K.

In the present section, we shall compute the betti numbers and torsion coefficients of the compact surfaces. The techniques we shall use may seem a bit awkward and ad hoc in nature. But, in fact, they are effective on a large class of spaces. In a later section, when we study CW complexes, we shall return to these techniques and show they are part of a systematic approach to computing homology groups.

For the record, we define what we mean by a surface.

Definition 1.7.1 By a *surface* we mean a nonempty, connected Hausdorff space, X, with a countable basis, such that each point of X has a neighborhood homeomorphic with an open subset of $\mathbb{R}^2$.

In Section 4.8 we will define the notion of an m-manifold. According to that definition, every surface is a 2-manifold, but because we require a surface to be connected and have a countable basis, not every 2-manifold is a surface.

Convention. *For convenience in notation, we shall henceforth delete the dimensional subscript p on the boundary operator ∂_p, and rely on the context to make clear which of these operators is intended.*

We shall compute the homology of the torus, the Klein bottle, and several other surfaces that can be constructed from a rectangle L by identifying its edges appropriately. Thus we begin by proving certain facts about L itself.

Lemma 1.7.2 *Let L be the complex of Figure 1.7.1, whose underlying space is a rectangle. Let* Bd L *denote the complex whose space is the boundary of the rectangle. Orient each 2-simplex σ_i of L by a counterclockwise arrow. Orient the 1-simplices arbitrarily. Then:*

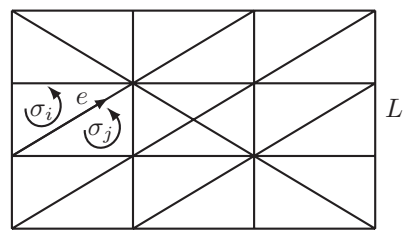


Figure 1.7.1

(1) *Every 1-cycle of L is homologous to a 1-cycle carried by* Bd L.

(2) *If d is a 2-chain of L and if ∂d is carried by* Bd L, *then d is a multiple of the chain $\sum \sigma_i$.*

Proof: The proof of **(2)** is easy. If σ_i and σ_j have an edge e in common, then ∂d must have value 0 on e. It follows that d must have the same value on σ_i as it does on σ_j. Continuing this process, we see that d has the same value on every oriented 2-simplex.

To prove **(1)**, we proceed as in Example 1.6.13. Given a 1-chain c of L, one pushes it off the 1-simplices, one step at a time. First, one shows that c is homologous to a 1-chain c_1 carried by the subcomplex pictured in Figure 1.7.2.

Then one shows that c_1 is in turn homologous to a 1-chain c_2 carried by the subcomplex of Figure 1.7.3. Finally, one notes that in the case where the original chain c is a cycle, then the chain c_2 is also a cycle. It follows that c_2 must be carried by Bd L, for otherwise c_2 would have a non-zero coefficient on one or more of the vertices v_1, v_2, v_3, v_4, v_5. □

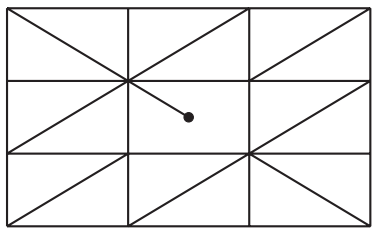


Figure 1.7.2

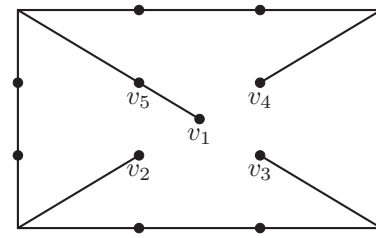


Figure 1.7.3

Theorem 1.7.3 *Let T denote the complex represented by the labeled rectangle L of Figure 1.7.4; its underlying space is the torus. Then*

$$H_1(T) \cong \mathbb{Z} \oplus \mathbb{Z} \quad \text{and} \quad H_2(T) \cong \mathbb{Z}\,.$$

Orient each 2 simplex of L counterclockwise; use the induced orientation of the 2-simplices of T; let γ denote their sum. Let

$$\begin{aligned} w_1 &= [a,b] + [b,c] + [c,a] \\ z_1 &= [a,d] + [d,e] + [e,a]\,. \end{aligned}$$

Then γ generates $H_2(T)$ and w_1 and z_1 represent a basis for $H_1(T)$.

Proof: Let $g : |L| \to |T|$ be the pasting map; let $A = g(|\text{Bd } L|)$. Then A is homeomorphic to a space that is the union of two circles with a point in common. [Such a space is called the *wedge of two circles*.] Orient the 1-simplices of T arbitrarily. Because g makes identifications only among simplices of Bd L, the arguments we gave earlier in proving Lemma 1.7.2 apply verbatim to prove the following:

(1) *Every 1-cycle of T is homologous to a 1-cycle carried by A.*

(2) *If d is a 2-chain of T and if ∂d is carried by A, then d is a multiple of γ.*

We show that, in the complex T, two further results hold:

(3) *If c is a 1-cycle of T carried by A, then c is of the form $nw_1 + mz_1$.*

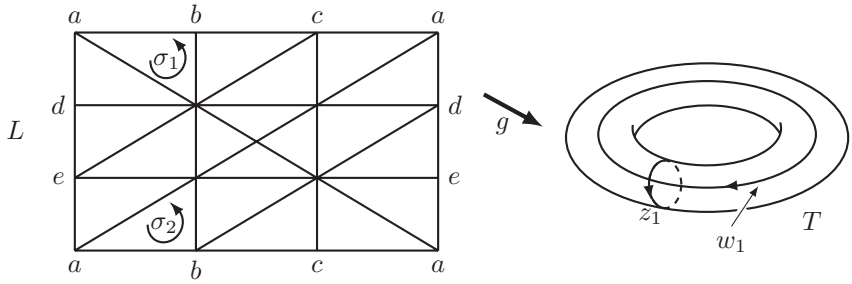


Figure 1.7.4

(4) $\partial\gamma = 0$.

The proof of **(3)** is easy, given the fact that A is simply the 1-dimensional complex pictured in Figure 1.7.5. The proof of **(4)** is similarly direct: It is clear that $\partial\gamma$ has value 0 on every 1-simplex of T not in A. One checks directly that it also has value 0 on each 1-simplex in A: For instance, the elementary chain $[a, b]$ appears in the expression for $\partial\sigma_1$ with value -1 and in the expression for $\partial\sigma_2$ with value $+1$, so that $\partial\gamma$ has value 0 on $[a, b]$. [See Figure 1.7.4.] You can check that similar conclusions apply to the other oriented 1-simplices in w_1 and z_1.

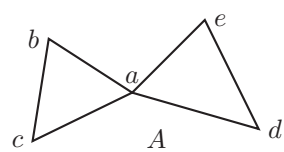


Figure 1.7.5

Using the results of **(1)**–**(4)**, we can compute the homology of T. Every 1-cycle of T is homologous to a 1-cycle of the form $nw_1 + mz_1$, by **(1)** and **(3)**. Such a cycle bounds only if it is trivial: For if $c = \partial d$ for some d, then **(2)** applies to show that $d = q\gamma$. Since $\partial\gamma = 0$ by **(4)**, we have $c = \partial d = 0$. We conclude that

$$H_1(T) \cong \mathbb{Z} \oplus \mathbb{Z}\,,$$

and the (cosets of the) 1-cycles w_1 and z_1 form a basis for the 1-dimensional homology.

To compute $H_2(T)$, note that by **(2)** any 2-cycle d of T must be of the form $q\gamma$ for some q. Each such 2-chain is in fact a cycle, by **(4)**, and there are

no 3-chains for it to bound. We conclude that

$$H_2(T) \cong \mathbb{Z}\,,$$

and this group has as generator the 2-cycle γ. □

Theorem 1.7.4 *Let S denote the complex represented by the labeled rectangle of Figure 1.7.6; its underlying space is the Klein bottle. Then*

$$H_1(S) \cong \mathbb{Z} \oplus \mathbb{Z}/2 \quad \text{and} \quad H_2(S) = 0\,.$$

The torsion element of $H_1(S)$ is represented by the chain z_1, and a generator for the group $H_1(S)$ modulo torsion is represented by w_1, where

$$\begin{aligned} w_1 &= [a,b] + [b,c] + [c,a] \\ z_1 &= [a,d] + [d,e] + [e,a]\,. \end{aligned}$$

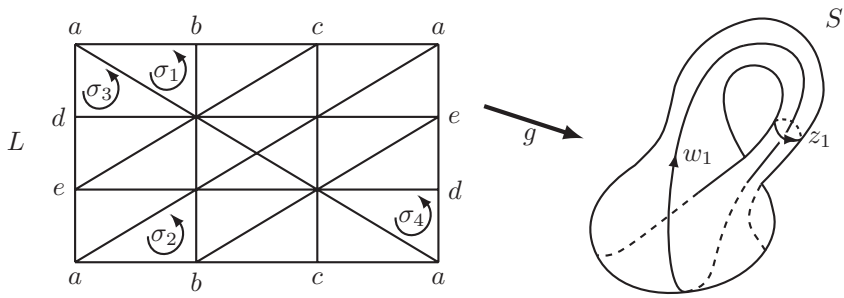


Figure 1.7.6

Proof: As in the proof of Theorem 1.7.3, let $g : |L| \to |S|$ be the pasting map, and let $A = g(|\text{Bd } L|)$. As before, A is the wedge of two circles. Orient the 2-simplices of S counterclockwise, and let γ be their sum. Orient the 1-simplices of S arbitrarily. The statements **(1)** and **(2)** of the proof of Theorem 1.7.3 hold in this case too, since neither involves the particular identification on the boundary. Because A is the wedge of two circles, **(3)** holds as well. The final condition **(4)** is different; instead one has

(4) $\partial\gamma = 2z_1$.

To see that **(4)** holds, one does a direct computation. For example, $[a,b]$ appears in $\partial\sigma_1$ with coefficient -1 and in $\partial\sigma_2$ with coefficient $+1$. On the other hand, $[a,d]$ appears in both $\partial\sigma_3$ and $\partial\sigma_4$ with coefficient $+1$. You can check that similar conclusions apply to the other oriented 1-simplices in w_1 and z_1.

Putting these facts together, we compute the homology of S as follows: Every 1-cycle of S is homologous to a 1-cycle of the form $nw_1 + mz_1$, by **(1)** and **(3)**. If $c = \partial d$ for some d, then $d = p\gamma$ by **(2)**, so **(4)** gives us $\partial d = 2pz_1$. Thus $nw_1 + mz_1$ bounds if and only if m is even and n is zero. We conclude that

$$H_1(S) = \mathbb{Z} \oplus \mathbb{Z}/2\,.$$

The cycle z_1 represents the torsion element, and w_1 represents a generator of the infinite cyclic group $H_1(S)/T_1(S)$.

To compute $H_2(S)$, note that any 2-cycle d of S must be of the form $p\gamma$ by **(2)**. By **(4)**, $p\gamma$ is not a cycle, so we have $H_2(S) = 0$. □

Theorem 1.7.5 *Let $\mathbb{P}^2$ be the complex indicated by the labeled rectangle of Figure 1.7.7. [Recall from Example 1.4.10 that its underlying space is called the projective plane.] Then*

$$H_1(\mathbb{P}^2) \cong \mathbb{Z}/2 \quad \text{and} \quad H_2(\mathbb{P}^2) = 0\,.$$

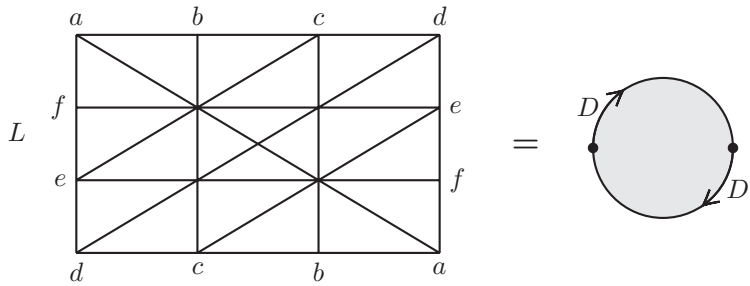


Figure 1.7.7

Proof: Let $g : |L| \to |\mathbb{P}^2|$ be the pasting map. Let $A = g(\text{Bd } L)$, so A is homeomorphic to a circle. Let γ be as in the proofs of the preceding two theorems. Let

$$z_1 = [a, b] + [b, c] + [c, d] + [d, e] + [e, f] + [f, a]\,.$$

The statements **(1)** and **(2)** of the proof of Theorem 1.7.3 hold as before. The additional results, which are easy to verify, are the following:

(3) *Every 1-cycle carried by A is a multiple of z_1.*

(4) $\partial\gamma = -2z_1$.

From these facts, we conclude that

$$H_1(\mathbb{P}^2) \cong \mathbb{Z}/2 \quad \text{and} \quad H_2(\mathbb{P}^2) = 0\,.$$

The non-zero element of H_1 is represented by the cycle z_1. □

Definition 1.7.6 The *connected sum* of two surfaces, S_1 and S_2, is defined to be the surface obtained by deleting a small open disc from each, and pasting together the pieces that remain, along their free edges. We denote this space by $S_1 \# S_2$.

Let T be the torus. The space $T \# T$, the *double torus* is illustrated in Figure 1.7.8.

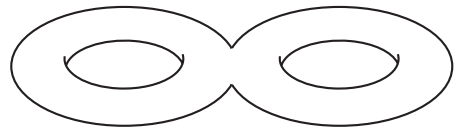


Figure 1.7.8

The space $\mathbb{P}^2 \# \mathbb{P}^2$ can be represented as a quotient space of a rectangle obtained by pasting its edges together in the manner indicated in Figure 1.7.9. [Note that if you cut the rectangle along the dotted line C, you have two projective planes with an open disc removed from each, as indicated in the figure.]

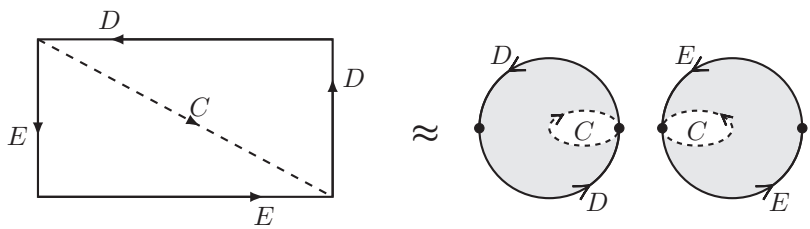


Figure 1.7.9

Theorem 1.7.7 *Let $\mathbb{P}^2 \# \mathbb{P}^2$ be the connected sum of two projective planes. Then*

$$H_1(\mathbb{P}^2 \# \mathbb{P}^2) \cong \mathbb{Z} \oplus \mathbb{Z}/2 \quad \text{and} \quad H_2(\mathbb{P}^2 \# \mathbb{P}^2) = 0\,.$$

Proof: We represent $\mathbb{P}^2 \# \mathbb{P}^2$ by the same rectangle L as before, labeling the vertices so as to be consistent with Figure 1.7.8. In this case, the complex $A = g(|\text{Bd } L|)$ is again the wedge of two circles. Let w_1 be the 1-cycle running right-to-left on the top edge of the rectangle as in Figure 1.7.9, and let z_1 be the 1-cycle running top-to-bottom on the left edge of the rectangle as in the same figure. Statements **(1)** and **(2)** of the proof of Theorem 1.7.3 hold as before. We also have

(3) *Every* 1*-cycle carried by* A *is of the form* $nw_1 + mz_1$.

(4) $\partial\gamma = 2w_1 + 2z_1$.

It is then clear that $H_2(\mathbb{P}^2 \# \mathbb{P}^2) = 0$, but computing H_1 requires additional work.

To determine $H_1(\mathbb{P}^2 \# \mathbb{P}^2)$ we need to compute the quotient of the group G having w_1 and z_1 as a basis by the subgroup H generated by $2(w_1 + z_1)$. For this purpose, we need to choose bases for the two groups that are "compatible," as in our basic theorem on subgroups of free abelian groups [Theorem 1.5.2]. In this case, it is easy to see what is needed: We need to choose $w_1 + z_1$ to be one of the basis elements for G. Can we do this? Of course, we can: For $\{w_1, w_1 + z_1\}$ will serve as a basis for G just as well as $\{w_1, z_1\}$ does. [One can express each basis in terms of the other, since $z_1 = -(w_1) + (w_1 + z_1)$.] If we use the basis $\{w_1, w_1 + z_1\}$ for G, then it is clear that

$$H_1(\mathbb{P}^2 \# \mathbb{P}^2) \cong \mathbb{Z} \oplus \mathbb{Z}/2\,,$$

the torsion element is represented by $w_1 + z_1$, and w_1 represents a generator of H_1/T_1. □

We remark that w_1 is not the only cycle representing a generator of H_1/T_1. The cycle z_1 does just as well, because $\{z_1, w_1 + z_1\}$ is a basis for G. Similarly, $\{2w_1 + 3z_1, w_1 + z_1\}$ is a basis for G that allows us to conclude that $2w_1 + 3z_1$ represents a generator of H_1/T_1. Indeed, many other cycles represent a generator of H_1/T_1.

The astute reader might notice that the homology groups here are the same as those for the Klein bottle [Theorem 1.7.4]. This is no accident, for the two spaces are in fact homeomorphic. Figure 1.7.10 indicates the proof.

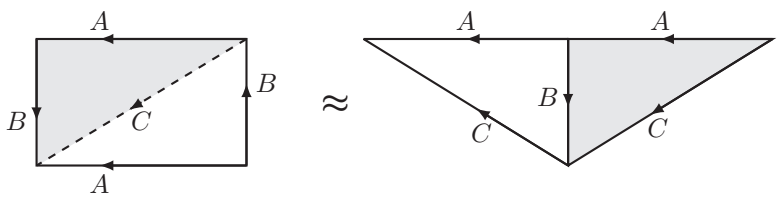


Figure 1.7.10

Definition 1.7.8 The quotient space indicated in Figure 1.7.11 is called the *dunce cap*.

The dunce cap will be generalized in Exercise 6.

The next theorem was a major accomplishment of nineteenth-century mathematics.

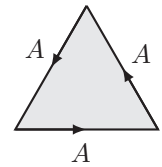


Figure 1.7.11

Theorem 1.7.9 (Classification theorem for compact surfaces) *Every compact surface is homeomorphic to a sphere, a connected sum of n tori, or a connected sum of m projective planes.*

We will not prove this theorem here, but the topic is covered in Chapter 12 of [Mun00].

In light of the classification theorem and the examples we have worked, you should be able to compute the homology groups of the general compact surface. We leave this computation to the exercises.

EXERCISES

1. Let w_1 and z_1 be the cycles on the Klein bottle pictured in Figure 1.7.6. Show that $w_1 + z_1$ represents a generator of the infinite cyclic group $H_1(S)/T_1(S)$.

2. The connected sum $T \# T$ of two tori can be represented as a quotient space of an octagonal region in the plane by making identifications on the boundary as indicated in Figure 1.7.12. [Splitting this octagon along the dotted line gives two tori with open discs deleted.]

(a) Construct a complex K whose underlying space is $T \# T$ by an appropriate vertex-labeling of the complex L pictured in Figure 1.7.13.

(b) Compute the homology of $T \# T$ by following the pattern of Theorem 1.7.3. Specifically, let A be the image of Bd L under the quotient map; then A is a wedge of four circles. Orient each 2-simplex of L counterclockwise; let γ be the sum of the correspondingly oriented simplices of K. Show that every 1-cycle of K is homologous to one carried by A. Then show that every 2-chain of K whose boundary is carried by A is a multiple of γ. Complete the computation by analyzing the 1-cycles carried by A, and by computing $\partial\gamma$.

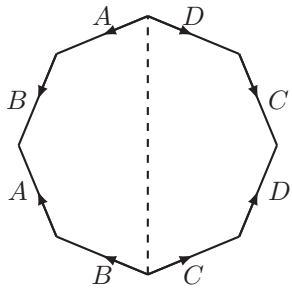


Figure 1.7.12

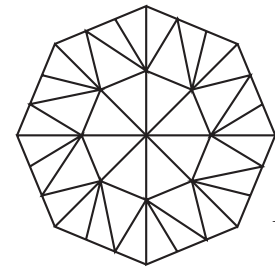


Figure 1.7.13

3. Represent the 4-fold connected sum $\mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$ by an appropriate labeling of the complex L of Figure 1.7.12. Compute its homology in dimensions 1 and 2.

4. **(a)** Define the n-fold connected sum $X_n = T \# T \# \cdots \# T$ of tori, and compute its homology in dimensions 1 and 2.

(b) Define the n-fold connected sum $Y_n = \mathbb{P}^2 \# \mathbb{P}^2 \# \cdots \# \mathbb{P}^2$ of projective planes, and compute its homology in dimensions 1 and 2.

5. Compute the homology of $T \# \mathbb{P}^2$. To which of the surfaces listed in Exercise 4 must it be homeomorphic? Can you construct the homeomorphism?

6. **(a)** Compute the homology in dimensions 1 and 2 of the quotient space indicated in Figure 1.7.14. We call this space the "5-fold dunce cap."

(b) Define analogously the "k-fold dunce cap" and compute its homology.

7. Compute the homology of the space indicated in Figure 1.7.15.

8. Given finitely generated abelian groups G_1 and G_2 with G_2 free, show there is a finite 2-dimensional complex K such that $|K|$ is connected, $H_1(K) \cong G_1$, and $H_2(K) \cong G_2$.

1.8 Zero-Dimensional Homology

We have not yet computed any zero-dimensional homology group. In this section, we shall show that this group has a simple topological interpretation that makes its computation trivial.

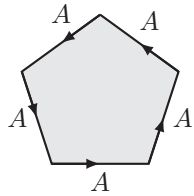


Figure 1.7.14

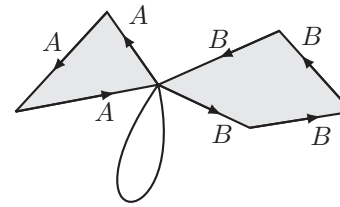


Figure 1.7.15

We prove the following theorem:

Theorem 1.8.1 *Let K be a complex. Then the group $H_0(K)$ is free abelian. If $\{v_\alpha\}$ is a collection consisting of one vertex from each component of $|K|$, then the homology classes of the chains v_α form a basis for $H_0(K)$.*

Proof: Step 1. If v and w are vertices of K, let us define $v \sim w$ if there is a sequence

$$a_0, a_1, \ldots, a_n$$

of vertices of K such that $v = a_0$, $w = a_n$, and $a_i a_{i+1}$ is a 1-simplex of K for each i. This relation is clearly an equivalence relation. Given v, define

$$C_v = \cup\{\text{St } w : w \sim v\}\,.$$

We will show that the sets C_v are the components of $|K|$.

Note first that C_v is open because it is a union of open sets. Furthermore, $C_v = C_{v'}$ holds if $v \sim v'$.

Second, we show that C_v is connected, in fact, path connected. Given v, let $w \sim v$ and let x be a point of St w. Choose a sequence $a_0, a_1, \ldots, a_n$ of vertices of K, as before. Then the path of line segments with successive vertices $a_0, a_1, \ldots, a_n, x$ lies in C_v: We see this because $a_i \sim v$ by definition, so that St $a_i \subset C_v$, and in particular, the line segment $a_i a_{i+1}$ lies in C_v. Similarly, the line segment $a_n x$ lies in St a_n, which is contained in C_v. Hence C_v is path connected.

Third, we show that distinct sets C_v and C'_v are disjoint. Suppose x is a point of $C_v \cap C'_v$. Then $x \in$ St w for some w equivalent to v, and $x \in$ St w' for some w' equivalent to v'. Since x has positive barycentric coordinates with respect to both w and w', some simplex of K has w and w' as vertices. Then ww' must be a 1-simplex of K, so $w \sim w'$. It follows that $v \sim v'$, so the two sets C_v and $C_{v'}$ are the same.

Being connected, open, and disjoint, the sets C_v are necessarily the components of $|K|$. Note that each is the space of a subcomplex of K; each simplex of K [being connected] lies entirely in one component of $|K|$.

Step 2. Now we prove the theorem. Let $\{v_\alpha\}$ be a collection of vertices containing one vertex v_α from each component C_α of $|K|$. Given a vertex w of K, it belongs to some component of $|K|$, say C_α. By hypothesis, $w \sim v_\alpha$, so there is a sequence $a_0, a_1, \ldots, a_n$ of vertices of K, as before, leading from v_α to w. The 1-chain

$$[a_0, a_1] + [a_1, a_2] + \cdots + [a_{n-1}, a_n]$$

has as its boundary the 0-chain $a_n - a_0 = w - v_\alpha$. Thus the 0-chain w is homologous to the 0-chain v_α. We conclude that every 0-chain in K is homologous to a linear combination of the elementary 0-chains v_α.

To finish the proof, we show that no non-trivial 0-chain of the form $c = \sum n_\alpha v_\alpha$ bounds. Suppose $c = \partial d$ for some 1-chain d. Since each 1-simplex of K lies in a unique component of $|K|$, we can write $d = \sum d_\alpha$, where d_α consists of those terms of d that are carried by C_α. Since $\partial d = \sum \partial d_\alpha$ and ∂d_α is carried by C_α, we conclude that $\partial d_\alpha = n_\alpha v_\alpha$. It follows that $n_\alpha = 0$ for each α. For let $\epsilon : C_0(K) \to \mathbb{Z}$ be the homomorphism defined by setting $\epsilon(v) = 1$ for each vertex v of K. Then $\epsilon(\partial[v, w]) = \epsilon(w - v) = 1 - 1 = 0$ for any elementary 1-chain $[v, w]$. As a result, $\epsilon(\partial e) = 0$ for every 1-chain e. In particular, $0 = \epsilon(\partial d_\alpha) = \epsilon(n_\alpha v_\alpha) = n_\alpha$. □

For some purposes, it is convenient to consider another version of 0-dimensional homology. We consider that situation now.

Definition 1.8.2 Let $\epsilon : C_0(K) \to \mathbb{Z}$ be the surjective homomorphism defined by $\epsilon(v) = 1$ for each vertex v of K. Then, if c is a 0-chain, $\epsilon(c)$ equals the sum of the values of c on the vertices of K. The map ϵ is called an *augmentation map* for $C_0(K)$. We have noted in the last proof that $\epsilon(\partial d) = 0$ if d is a 1-chain. We define the *reduced homology group* of K in dimension 0, denoted $\widetilde{H}_0(K)$, by the equation

$$\widetilde{H}_0(K) = \ker \epsilon / \text{im } \partial_1 .$$

[If $p > 0$, we let $\widetilde{H}_p(K)$ denote the usual group $H_p(K)$.]

The relation between reduced and ordinary homology is as follows:

Theorem 1.8.3 *The group $\widetilde{H}_0(K)$ is free abelian, and*

$$\widetilde{H}_0(K) \oplus \mathbb{Z} \cong H_0(K) . \qquad (*)$$

Thus $\widetilde{H}_0(K)$ vanishes if $|K|$ is connected. If $|K|$ is not connected, let $\{v_\alpha\}$ consist of one vertex from each component of $|K|$; let α_0 be a fixed index. Then the homology classes of the chains $v_\alpha - v_{\alpha_0}$, for $\alpha \neq \alpha_0$, form a basis for $\widetilde{H}_0(K)$.

Proof: As above, let $\{v_\alpha\}$ consist of one vertex from each component of $|K|$. Given a 0-chain c, it is homologous to a 0-chain of the form $c' = \sum n_\alpha v_\alpha$; and

the chain bounds only if $n_\alpha = 0$ for all α. On the other hand, if $c \in \ker \epsilon$, then $\epsilon(c) = \epsilon(c') = \epsilon(\sum n_\alpha v_\alpha) = \sum n_\alpha = 0$.

If $|K|$ has one component, then $c \in \ker \epsilon$ implies $c' = 0$. We conclude that $\widetilde{H}_0(K) = 0$. Theorem 1.8.1 tells us that $H_0(K) \cong \mathbb{Z}$. Thus, $(*)$ is verified.

If $|K|$ has more than one component, then $c \in \ker \epsilon$ implies that c' is a linear combination of the 0-chains $v_\alpha - v_{\alpha_0}$, $\alpha \neq \alpha_0$. Again, Theorem 1.8.1 verifies $(*)$. □

EXERCISE

1. (a) Let G be an abelian group and let $\phi : G \to \mathbb{Z}$ be an epimorphism. Show that G has an infinite cyclic subgroup H such that

$$G = (\ker \phi) \oplus H\,.$$

[**Hint:** Define a homomorphism $\psi : \mathbb{Z} \to G$ such that $\phi \circ \psi$ is the identity; let $H = \text{im } \psi$.]

(b) Show that if $\phi : C_0 \to \mathbb{Z}$ is any epimorphism such that $\phi \circ \partial_1 = 0$, then

$$H_0(K) \cong (\ker \phi)/(\text{im } \partial_1) \oplus \mathbb{Z}\,.$$

1.9 The Homology of a Cone

Now we compute the homology of the n-simplex and its boundary. A convenient way of doing this is to introduce the notion of a *cone*.

Definition 1.9.1 Suppose that K is a complex in $\mathbb{E}^J$, and w is a point of $\mathbb{E}^J$ such that each ray emanating from w intersects $|K|$ in at most one point. We define the *cone on K with vertex w* to be the collection of all simplices of the form $wa_0a_1 \dots a_p$, where $a_0a_1 \dots a_p$ is a simplex of K, along with all faces of such simplices. We denote this collection by $w * K$. See Figure 1.9.1.

We show that $w * K$ is a well-defined complex, and it contains K as a subcomplex; K is often called the *base* of the cone.

First we show that the set $\{wa_0a_1 \dots a_p\}$ is geometrically independent: If w were in the plane P determined by $a_0, a_1, \dots, a_p$, we could consider the line segment joining w to an interior point x of $\sigma = a_0a_1 \dots a_p$. The set Int σ, being open in P, would contain an interval of points on this line segment. But the ray from w through x intersects $|K|$ in only one point, by hypothesis.

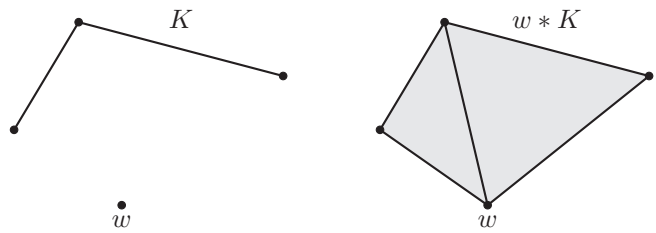


Figure 1.9.1

We now show that $w * K$ is a complex. The simplices of $w * K$ are of three types: simplices $a_0a_1 \dots a_p$ of K, simplices of the form $wa_0a_1 \dots a_p$, and the 0-simplex w. A pair of simplices of the first type have disjoint interiors because K is a complex. The open simplex Int $(wa_0a_1 \dots a_p)$ is the union of all open line segments joining w to points of Int $(a_0a_1 \dots a_p)$; no two such open simplices can intersect because no ray from w contains more than one point of $|K|$. For the same reason, simplices of the first and second types have disjoint interiors.

EXAMPLE 1.9.2 If K_σ is the complex consisting of the simplex $\sigma = v_0v_1 \dots v_n$ and its faces, then $K_\sigma = v_0 * K_s$, where s is the face of σ opposite v_0. Thus every simplex of positive dimension is a cone.

EXAMPLE 1.9.3 If K is the complex in $\mathbb{R}^2$ consisting of the intervals $[n, n+1] \times 0$ on the x-axis and their vertices, and if w is a point on the y-axis different from the origin, then $w * K$ is the complex illustrated in Figure 1.9.2. Although $|K|$ is a subspace of $\mathbb{R}^2$, $|w * K|$ is not a subspace of $\mathbb{R}^2$. [See Exercise 9 of Section 1.3.]

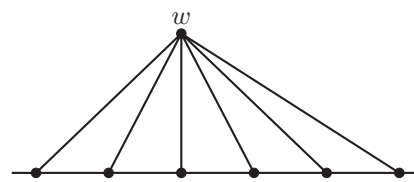


Figure 1.9.2

One particularly useful consequence of the cone construction is the following:

Lemma 1.9.4 *Let U be a bounded convex open set in $\mathbb{R}^n$, and let $w \in U$. If K is a finite complex such that $|K| = \overline{U} \setminus U$, then $w * K$ is a finite complex such that $|w * K| = \overline{U}$.*

Proof: It follows at once from Lemma 1.2.7 that each ray emanating from w intersects $|K|$ in precisely one point, and that $\overline{U}$ is the union of all line segments joining w to points of $|K|$. □

Given K, we note that any two cones $w*K$ and $z*K$ over K are isomorphic. The vertex map that carries each vertex of K to itself and carries w to z induces an isomorphism of $w * K$ with $z * K$.

Note also that for a complex K in $\mathbb{R}^N$, there may be no point w in $\mathbb{R}^N$ such that the cone complex $w * K$ can be formed. Nonetheless, we can always consider K as a complex in $\mathbb{R}^N \times 0 \subset \mathbb{R}^{N+1}$; then the point $w = (0, 0, \ldots, 0, 1) \in \mathbb{R}^{N+1}$ will do. A similar remark applies to a complex in $\mathbb{E}^J$.

Now we compute the homology of a cone and show that it vanishes in positive dimensions. For this purpose, we shall introduce a certain bracket operation that will also be useful later.

Definition 1.9.5 Let $w * K$ be a cone. If $\sigma = [a_0, a_1, \ldots, a_p]$ is an oriented simplex of K, let $[w, \sigma]$ denote the oriented simplex $[w, a_0, a_1, \ldots, a_p]$ of $w * K$. This operation is well-defined; exchanging two vertices in the array $[a_0, a_1, \ldots, a_p]$ results in exchanging two vertices in the array $[w, a_0, a_1, \ldots, a_p]$. More generally, if

$$c_p = \sum n_i \sigma_i$$

is a p-chain of K, we define

$$[w, c_p] = \sum n_i [w, \sigma_i] \,.$$

This *bracket operation* is a homomorphism carrying $C_p(K)$ into $C_{p+1}(K)$,

We compute readily from the boundary formula:

$$\partial[w, \sigma] = \begin{cases} \sigma - w & \text{if } \dim \sigma = 0, \\ \sigma - [w, \partial\sigma] & \text{if } \dim \sigma > 0. \end{cases}$$

This leads to the following more general formulas:

$$\partial[w, c_0] = c_0 - \epsilon(c_0) w,$$
$$\partial[w, c_p] = c_p - [w, \partial c_p] \quad \text{if } p > 0.$$

Theorem 1.9.6 *If $w * K$ is a cone, then for all p,*

$$\widetilde{H}_p(w * K) = 0 \,.$$

In general, a complex whose reduced homology vanishes in all dimensions is said to be *acyclic*.

Proof: The reduced homology of $w * K$ vanishes in dimension 0, because $|w * K|$ is connected. Consider the case $p > 0$. Let z_p be a p-cycle of $w * K$; we show that z_p bounds. Let us write

$$z_p = c_p + [w, d_{p-1}]\,,$$

where c_p consists of those terms of z_p that are carried by K, and d_{p-1} is a chain of K. We will show that

$$z_p - \partial[w, c_p] = 0\,;$$

then our result is proved. By direct computation,

$$\begin{aligned} z_p - \partial[w, c_p] &= c_p + [w, d_{p-1}] - c_p + [w, \partial c_p] \\ &= [w, e_{p-1}]\,, \end{aligned}$$

where $e_p = d_{p-1} + \partial c_p$ is a chain of K. Now since z_p is a cycle,

$$0 = \begin{cases} e_{p-1} - \epsilon(e_{p-1})w & \text{if } p = 1, \\ e_{p-1} - [w, \partial e_{p-1}] & \text{if } p > 1. \end{cases}$$

Now the only portion of this chain carried by K is e_{p-1}, therefore, $e_{p-1} = 0$. We conclude that

$$z_p - \partial[w, c_p] = [w, e_{p-1}] = 0\,,$$

as desired. □

Theorem 1.9.7 *Let σ be an n-simplex. The complex K_σ consisting of σ and its faces is acyclic. If $n > 0$, let Σ^{n-1} denote the complex whose polytope is* Bd σ*. Orient σ. Then $\widetilde{H}_{n-1}(\Sigma^{n-1})$ is infinite cyclic and is generated by the chain $\partial\sigma$; furthermore, $\widetilde{H}_i(\Sigma^{n-1}) = 0$ for $i \neq n-1$.*

Recall from Lemma 1.2.6 that Bd σ is homeomorphic to $\mathbb{S}^{n-1}$, so the theorem tells us the homology of the sphere.

Proof: Because K_σ is a cone, it is acyclic. [Recall Example 1.9.2.] Let us compare the chain groups of K_σ and Σ^{n-1}; they are equal except in dimension n. [In the diagram below, we keep track of the dimension of the boundary operators and we distinguish between the boundary operators on the chain groups of K_σ and on the chain groups of Σ^{n-1}.] Now

$$\begin{array}{ccccccccccc} C_n(K_\sigma) & \xrightarrow{\partial_n} & C_{n-1}(K_\sigma) & \xrightarrow{\partial_{n-1}} & \cdots & \xrightarrow{\partial_1} & C_0(K_\sigma) & \xrightarrow{\epsilon} & \mathbb{Z}\,. \\ & & \| & & & & \| & & \| \\ 0 & \xrightarrow{\partial'_n} & C_{n-1}(\Sigma^{n-1}) & \xrightarrow{\partial'_{n-1}} & \cdots & \xrightarrow{\partial'_1} & C_0(\Sigma^{n-1}) & \longrightarrow & \mathbb{Z}\,. \end{array}$$

It follows at once that $\widetilde{H}_i(\Sigma^{n-1}) = \widetilde{H}_i(K_\sigma) = 0$ for $i \neq n-1$. Let us compute

the homology in dimension $n-1$. First, take the case $n > 1$. Because $C_n(\Sigma^{n-1})$ is the trivial group, there are no boundaries in $C_{n-1}(\Sigma^{n-1})$. One has

$$\begin{aligned} H_{n-1}(\Sigma^{n-1}) &= Z_{n-1}(\Sigma^{n-1}) \\ &= \ker \partial'_{n-1} = \ker \partial_{n-1} \\ &= \text{im } \partial_n\,, \text{ because } H_{n-1}(K_\sigma) = 0\,. \end{aligned}$$

Now $C_n(K_\sigma)$ is infinite cyclic and is generated by σ. Hence im ∂_n is cyclic and is generated by $\partial_n\sigma$; it is infinite because $C_{n-1}(K_\sigma)$ has no torsion.

The argument for $n = 1$ is similar, except that ∂_{n-1} is replaced by ϵ throughout. □

The next definition will be needed in the exercises, but the notion of a "suspension" will be seen again in later sections.

Definition 1.9.8 Let K be a complex; let $w_0 * K$ and $w_1 * K$ be two cones on K whose polytopes intersect only in $|K|$. Exercise 1(a) asks you to show that $(w_0 * K) \cup (w_1 * K)$ is a complex. It is called a *suspension* of K and is denoted $S(K)$. See Figure 1.9.3.

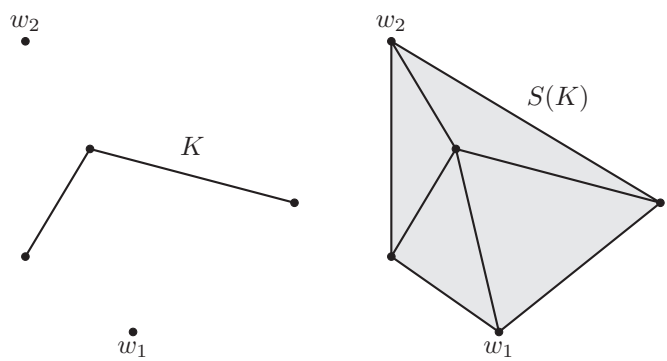


Figure 1.9.3

EXERCISE

1. Let K be a complex; let $w_0 * K$ and $w_1 * K$ be two cones on K whose polytopes intersect only in $|K|$.

(a) Show that $(w_0 * K) \cup (w_1 * K)$ is a complex.

(b) Using the bracket notation, define $\phi : C_p(K) \to C_{p+1}(S(K))$ by the equation

$$\phi(c_p) = [w_0, c_p] - [w_1, c_p] .$$

Show that ϕ induces a homomorphism

$$\phi_* : \widetilde{H}_p(K) \to \widetilde{H}_{p+1}(S(K)) .$$

(c) Show that ϕ_* is an isomorphism when K consists of the proper faces of a 2-simplex.

We will see later that ϕ_* is an isomorphism in general [see Section 3.4].

1.10 Relative Homology

Suppose that K_0 is a subcomplex of the complex K. In many of the applications of topology, it is convenient to consider what are called the *relative* homology groups of K modulo K_0. We introduce them briefly here and compute some examples, postponing a more complete discussion to Chapter 3.

If K_0 is a subcomplex of the complex K, then the chain group $C_p(K_0)$ can be considered to be a subgroup of the chain group $C_p(K)$ in the natural way. Formally, if c_p is a chain on K_0 [that is, a function on the oriented simplices of K_0], one extends it to a chain on K by letting its value be zero on each oriented p-simplex of K not in K_0. When we write c_p as a linear combination of oriented p-simplices of K_0, we need merely consider these simplices as belonging to K in order to consider c_p as a chain of K.

Definition 1.10.1 If K_0 is a subcomplex of K, then we call (K, K_0) a *simplicial pair* or simply a *pair* if the context makes the meaning clear. The quotient group

$$C_p(K)/C_p(K_0)$$

is called the group of *relative chains of* K *modulo* K_0, and is denoted by $C_p(K, K_0)$.

Note that the group $C_p(K, K_0)$ is free abelian. Indeed, if we orient the p-simplices of K so as to obtain a basis for $C_p(K)$, then the subcollection consisting of the oriented p-simplices of K_0 is a basis for $C_p(K_0)$. Then the quotient $C_p(K)/C_p(K_0)$ is free, for it has as basis all cosets of the form

$$\{\sigma_i\} = \sigma_i + C_p(K_0) ,$$

where σ_i is a p-simplex of K that is *not* in K_0.

The boundary operator $\partial : C_p(K_0) \to C_{p-1}(K_0)$ is simply the restriction of the boundary operator on $C_p(K)$. We use the same symbol to denote both these homomorphisms, when no confusion will result. This homomorphism induces a homomorphism

$$C_p(K, K_0) \to C_{p-1}(K, K_0)$$

of the relative chain groups, which we also denote by ∂. As before, it satisfies the equation $\partial \circ \partial = 0$. We let

$$\begin{aligned} Z_p(K, K_0) &= \ker \partial : C_p(K, K_0) \to C_{p-1}(K, K_0)\,, \\ B_p(K, K_0) &= \text{im } \partial : C_{p+1}(K, K_0) \to C_p(K, K_0)\,, \\ H_p(K, K_0) &= Z_p(K, K_0)/B_p(K, K_0)\,. \end{aligned}$$

These groups are called, respectively, the group of *relative p-cycles*, the group of *relative p-boundaries*, and the *relative homology group* in dimension p, of K modulo K_0.

Note that a relative p-chain, which is a coset $c_p + C_p(K_0)$, is a relative cycle if and only if ∂c_p is carried by K_0. Furthermore, it is a relative boundary if and only if there is a $p+1$ chain d_{p+1} of K such that $c_p - \partial d_{p+1}$ is carried by K_0.

EXAMPLE 1.10.2 Let K consist of an n-simplex and its faces; let K_0 be the proper faces of K. Then the group $C_p(K, K_0)$ vanishes except when $p = n$, in which case it is infinite cyclic. It follows that

$$\begin{aligned} H_i(K, K_0) &= 0 \quad \text{for} \quad i \neq n\,, \\ H_n(K, K_0) &\cong \mathbb{Z}\,. \end{aligned}$$

EXAMPLE 1.10.3 Let K be a complex and let K_0 consist of a single vertex v of K. Using the results of Section 1.8, one sees readily that $H_0(K, v)$ is free abelian; one obtains a basis for $H_0(K, v)$ by choosing one vertex from each component of $|K|$ other than the component containing v. Then $H_0(K, v) \cong \widetilde{H}_0(K)$.

It is not hard to show that $\widetilde{H}_p(K, v) \cong H_p(K)$ for $p > 0$; see the exercises.

EXAMPLE 1.10.4 Let K be the complex indicated in Figure 1.10.1, whose underlying space is a square. Let K_0 be the subcomplex whose space is the boundary of the square. It is easy to see that the 2-chain $\sum m_i \sigma_i$ represents a relative cycle of K modulo K_0 if and only if $m_1 = m_2 = m_3 = m_4$. Since there are no boundaries in dimension 2,

$$H_2(K, K_0) \cong \mathbb{Z}$$

and the chain $\gamma = \sum \sigma_i$ represents a generator.

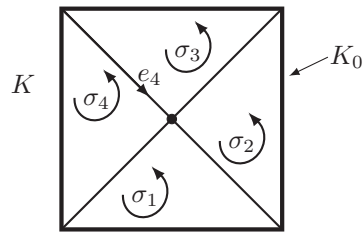


Figure 1.10.1

We showed in Example 1.6.13 that any 1-chain c of K is homologous to a 1-chain c_3 carried by $K_0 \cup e_4$. Now if c represents a relative 1-cycle [so that ∂c is carried by K_0], then $\partial c_3 = \partial c$ is also carried by K_0. It follows that the value of c_3 on e_4 must be zero, whence c_3 is actually carried by K_0. We conclude that

$$H_1(K, K_0) = 0\,.$$

Comparing this computation with that of Example 1.10.2 lends some plausibility to the statement [yet to be proved] that the relative homology groups are topological invariants.

EXAMPLE 1.10.5 Let K be the complex indicated in Figure 1.10.2. Its underlying space is called an *annulus*. Let K_0 denote the 1-dimensional complex whose space is the union of the inner and outer edges of K. We compute the homology of K modulo K_0.

First, $H_0(K, K_0) = 0$ because the relative chain group itself vanishes in dimension 0. To compute H_1 and H_2, one first verifies three facts:

(i) If c is a 1-chain of K, then c is homologous to a 1-chain of K that is carried by the subcomplex M pictured in Figure 1.10.3.

(ii) Orient each 2-simplex of K counterclockwise. If d is any 2-chain of K such that ∂d is carried by M, then $d = m\gamma$, where γ is the sum of all the oriented 2-simplices of K.

(iii) $\partial\gamma$ is carried by K_0.

The computation then proceeds as follows: Let e_0 be the oriented 1-simplex pictured in Figure 1.10.2; then e_0 represents a relative 1-cycle of K modulo K_0, because ∂e_0 lies in K_0. It follows from **(i)** that *any* relative 1-cycle is relatively homologous to a multiple of $\{e_0\}$. Furthermore, no such relative cycle bounds. For suppose $ne_0 - \partial d$ is carried by K_0, for some 2-chain d of K. Then ∂d is carried by M, whence by **(ii)**, $d = m\gamma$ for some integer m. But $\partial d = m\partial\gamma$ is carried by K_0, by **(iii)**, so that $n = 0$. We conclude that

$$H_1(K, K_0) \cong \mathbb{Z}\,,$$

and the relative cycle e_0 represents a generator.

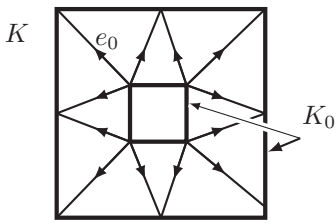


Figure 1.10.2

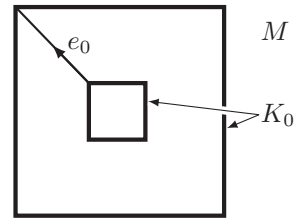


Figure 1.10.3

A similar argument, using **(ii)** and **(iii)**, shows that

$$H_2(K, K_0) \cong \mathbb{Z}\,,$$

and the relative cycle γ represents a generator.

Students often visualize the relative homology group $H_p(K, K_0)$ as representing the homology of the quotient space $X = |K|/|K_0|$ obtained by collapsing $|K_0|$ to a point p, modulo that point. Assuming X is homeomorphic to a polyhedron [so its simplicial homology is defined] this is in fact correct, but the proof is not easy. [See Lemma 8.10.1 and the exercises of Section 4.13.]

Roughly speaking, the relative homology group $H_p(K, K_0)$ depends only on the part of K lying outside of K_0 or on the boundary of K_0; it effectively ignores the part of K lying inside K_0. We express this fact formally in the following theorem:

Theorem 1.10.6 (Excision theorem) *Let K be a complex; let K_0 be a subcomplex. Let U be an open set contained in $|K_0|$, such that $|K| \setminus U$ is the polytope of a subcomplex L of K. Let L_0 be the subcomplex of K whose polytope is $|K_0| \setminus U$. Then inclusion induces an isomorphism*

$$H_p(L, L_0) \cong H_p(K, K_0)\,.$$

We think of $(|L|, |L_0|)$ as having been formed by excising away the open set U from $|K|$ and $|K_0|$. See Figure 1.10.4.

Proof: Consider the composite map ϕ given by

$$C_p(L) \to C_p(K) \to C_p(K)/C_p(K_0)\,,$$

which is inclusion followed by projection. Then ϕ is surjective, because a basis for $C_p(K)/C_p(K_0)$ consists of all cosets $\{\sigma_i\}$ for σ_i not in K_0, and L contains all such simplices σ_i. The kernel of ϕ is precisely $C_p(L_0)$. Thus ϕ induces an isomorphism

$$C_p(L)/C_p(L_0) \cong C_p(K)/C_p(K_0)\,,$$

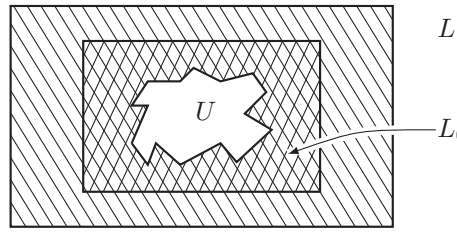


Figure 1.10.4

for all p. Since the boundary operator is preserved under this isomorphism, it follows that $H_p(L, L_0) \cong H_p(K, K_0)$. □

This elementary fact has some useful consequences which we shall consider later.

EXERCISES

1. Let K be the complex pictured in Figure 1.10.2; let K_1 be its outer edge. Compute $H_i(K)$ and $H_i(K, K_1)$.

2. Let K be a complex whose underlying space is the Möbius band; let K_0 be its edge. Compute $H_i(K)$ and $H_i(K, K_1)$. [**Hint:** See Example 1.4.6. Here the edge consists of the line segments ab, bc, cd, de, ef, and fa.]

3. Show that if K is a complex and v is a vertex of K, then $H_i(K, v) \cong \widetilde{H}_i(K)$ for all i. [**Hint:** Care is needed when $i = 1$.]

4. Describe $H_0(K, K_0)$ in general.

5. Let $|K|$ be the torus, represented by a labeled rectangle in the usual way; let K_0 be the subcomplex represented by the top edge of the rectangle. Compute $H_i(K, K_0)$. [See Figure 1.7.4; $|K_0|$ is the union of the line segments ab, bc, and ca.]

6. Let K be a 2-dimensional complex; let σ be a 2-simplex of K; let K_0 be the subcomplex whose space is $|K| \setminus \operatorname{Int} \sigma$. Compute $H_i(K, K_0)$.

1.11 *Homology with Arbitrary Coefficients

There is one further version of homology that we shall mention here, although we shall not study it in detail until Chapter 6. It arises when one introduces an arbitrary abelian group as the coefficient group.

Let G be an abelian group. Let K be a simplicial complex. A *p-chain of K with coefficients in G* is a function c_p from the oriented p-simplices of K to G that vanishes on all but finitely many p-simplices and is such that

$$c_p(\sigma') = -c_p(\sigma)$$

if σ' and σ are opposite orientations of the same simplex. Two chains are added by adding their values. The resulting group is denoted $C_p(K;G)$.

If σ is an oriented simplex and if $g \in G$, we use $g\sigma$ to denote the elementary chain whose value is g on σ, $-g$ on the opposite orientation of σ, and 0 on all other oriented simplices. In this notation, $g(-\sigma) = (-g)\sigma$, where $-\sigma$ as usual denotes σ with the opposite orientation. If one orients all the p-simplices of K, then each chain c_p can be written uniquely as a finite sum

$$c_p = \sum g_i \sigma_i$$

of elementary chains. Thus $C_p(K;G)$ is the direct sum of subgroups isomorphic to G, one for each p-simplex of K.

The boundary operator $\partial : C_p(K;G) \to C_{p-1}(K;G)$ is defined easily by the formula

$$\partial(g\sigma) = g(\partial\sigma)$$

where $\partial\sigma$ is the ordinary boundary, defined earlier. As before, $\partial \circ \partial = 0$ holds. We define $Z_p(K;G)$ to be the kernel of the homomorphism

$$\partial : C_p(K;G) \to C_{p-1}(K;G)\,,$$

$B_{p-1}(K;G)$ to be its image, and

$$H_p(K;G) = Z_p(K;G)/B_p(K;G)\,.$$

These groups are called, respectively, the *cycles*, *boundaries*, and *homology of K with coefficients in G*.

Of course, one can also study *relative homology with coefficients in G*. The details are clear. The groups in question are denoted by $H_p(K, K_0; G)$.

We are not going to do much concerning homology with general coefficients for some time to come. But you should be aware of its existence at an early stage, for it is sometimes useful.

EXAMPLE 1.11.1 One group that is frequently used as a coefficient group is the group $\mathbb{Z}/2$ of integers mod 2. Let us calculate the homology of the torus

and the Klein bottle using these coefficients. The argument for the torus given in Section 1.7 goes through essentially unchanged to show that

$$\begin{aligned} H_1(T;\mathbb{Z}/2) &\cong \mathbb{Z}/2 \oplus \mathbb{Z}/2\,, \\ H_2(T;\mathbb{Z}/2) &\cong \mathbb{Z}/2\,. \end{aligned}$$

For the Klein bottle S, the argument goes through, but with some changes. One has the results

$$\begin{aligned} H_1(S;\mathbb{Z}/2) &\cong \mathbb{Z}/2 \oplus \mathbb{Z}/2\,, \\ H_2(S;\mathbb{Z}/2) &\cong \mathbb{Z}/2\,. \end{aligned}$$

For $\mathbb{Z}/2$ coefficients, the basic 2-chain γ, which is the sum of the 2-simplices of S, has boundary *zero*. [In the group $\mathbb{Z}/2$, one has $2 = 1 + 1 = 0$.]

Note that homology with $\mathbb{Z}/2$ coefficients is inadequate to distinguish between T and S.

EXAMPLE 1.11.2 Let us compute the homology of the torus and Klein bottle using the rational numbers $\mathbb{Q}$ as coefficients. The same arguments as before apply, but the end results are different. For the torus, one has

$$H_1(T;\mathbb{Q}) \cong \mathbb{Q} \oplus \mathbb{Q}\,, \qquad H_2(T;\mathbb{Q}) \cong \mathbb{Q}\,.$$

For the Klein bottle, one has

$$H_1(S;\mathbb{Q}) \cong \mathbb{Q}\,, \qquad H_2(S;\mathbb{Q}) = 0\,.$$

For $\mathbb{Q}$ coefficients, the cycle z_1 bounds the chain $1/2\,\gamma$.

EXERCISES

1. Compute the homology of $\mathbb{P}^2$ with $\mathbb{Z}/2$ coefficients and with $\mathbb{Q}$ coefficients.
2. Show that homology with $\mathbb{Q}$ coefficients suffices to distinguish among $\mathbb{S}^2$ and the connected sums $\mathbb{P}^2 \# \mathbb{P}^2 \# \cdots \# \mathbb{P}^2$ and $T \# T \# \cdots \# T$.
3. Compute the homology of the Klein bottle S with $\mathbb{Z}/3$ coefficients and with $\mathbb{Z}/4$ coefficients.
4. Compute the homology of the k-fold dunce cap with $\mathbb{Z}/n$ coefficients and with $\mathbb{Q}$ coefficients. [See Exercise 6 of Section 1.7.]
5. Let (K, K_0) be the pair consisting of the Möbius band and its edge. Compute $H_i(K, K_0; \mathbb{Z}/2)$ and $H_i(K, K_0; \mathbb{Q})$. [See Exercise 2 of Section 1.10.]

1.12 *The Computability of Homology Groups

We have computed the homology groups of some familiar spaces, such as the sphere and the torus and the Klein bottle. Now we ask the question whether one can in fact compute homology groups in general. For finite complexes, the answer is affirmative. In this section, we present an explicit algorithm for carrying out the computation.

First, we prove a basic theorem giving a "normal form" for homomorphisms of finitely generated free abelian groups. The proof is constructive in nature. One corollary is the theorem about subgroups of free abelian groups that we stated earlier as Theorem 1.5.2. A second corollary is a theorem concerning standard bases for free chain complexes. A third corollary gives our desired algorithm for computing the homology groups of a finite complex.

First, we need two lemmas with which you might already be familiar.

Lemma 1.12.1 *Let A be a free abelian group of rank n. If B is a subgroup of A, then B is free abelian of rank $r \leq n$.*

Proof: We may without loss of generality assume that B is a subgroup of the n-fold direct product $\mathbb{Z}^n = \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$. We construct a basis for B as follows:

Let $\pi_i : \mathbb{Z}^n \to \mathbb{Z}$ denote projection on the ith coordinate. For each $m \leq n$, let B_m be the subgroup of B defined by the equation

$$B_m = B \cap (\mathbb{Z}^m \times \mathbf{0}) .$$

That is, B_m consists of all $\mathbf{x} \in B$ such that $\pi_i(\mathbf{x}) = 0$ for $i > m$. In particular, $B_n = B$. Now the homomorphism

$$\pi_m : B_m \to \mathbb{Z}$$

carries B_m onto a subgroup of $\mathbb{Z}$. If this subgroup is trivial let $\mathbf{x}_m = \mathbf{0}$; otherwise, choose $\mathbf{x}_m \in B_m$ so that its image $\pi_m(\mathbf{x}_m)$ generates this subgroup. We assert that the non-zero elements of the set $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ form a basis for B.

First, we show that, for each m, the elements $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$ generate B_m. [Then, in particular, the elements $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ generate B.] We use induction on m. It is trivial that $\mathbf{x}_1$ generates B_1; indeed, if d is the integer $\pi_1(\mathbf{x}_1)$, then

$$\mathbf{x}_1 = (d, 0, 0, \ldots, 0)$$

and B_1 consists of all multiples of this element. [Even if $\mathbf{x}_1 = \mathbf{0}$ and $d = 0$, B_1 still consists of all multiples of $\mathbf{x}_1$.]

For the induction step, assume that $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{m-1}$ generate B_{m-1}, and let $\mathbf{x} \in B_m$. Now $\pi_m(\mathbf{x}) = k\pi_m(x_m)$ for some integer k. It follows that

$$\pi_m(\mathbf{x} - k\mathbf{x}_m) = 0 ,$$

so $\mathbf{x} - k\mathbf{x}_m$ belongs to B_{m-1}. Then, by the induction hypothesis, we have

$$\mathbf{x} - k\mathbf{x}_m = k_1\mathbf{x}_1 + k_2\mathbf{x}_2 + \cdots + k_{m-1}\mathbf{x}_{m-1}\,.$$

Hence, $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m$ generate B_m.

Second, for each m, we show the independence of the non-zero elements in the set $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m\}$. Again we use induction. The result is trivial when $m = 1$. Suppose it true for $m - 1$. Then we show that if

$$\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 + \cdots + \lambda_m\mathbf{x}_m = \mathbf{0}\,,$$

then it follows that for each i, $\lambda_i = 0$ whenever $\mathbf{x}_i \neq \mathbf{0}$; once we have done that, independence follows.

Applying the map π_m, we obtain

$$\lambda_m\pi_m(\mathbf{x}_m) = 0\,.$$

From this equation, it follows that either $\lambda_m = 0$ or $\mathbf{x}_m = \mathbf{0}$. For if $\lambda_m \neq 0$, then $\pi_m(\mathbf{x}_m) = 0$ whence the subgroup $\pi_m(B_m)$ is trivial and $\mathbf{x}_m = \mathbf{0}$ by definition. We conclude two things:

$$\lambda_m = 0 \;\text{ if }\; \mathbf{x}_m \neq \mathbf{0}\,,$$

$$\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 + \cdots + \lambda_{m-1}\mathbf{x}_{m-1} = \mathbf{0}\,.$$

The induction hypothesis now applies to show that, for $i < m$,

$$\lambda_i = 0 \;\text{ if }\; \mathbf{x}_i \neq \mathbf{0}\,. \qquad \square$$

For later use, we generalize this result to arbitrary free abelian groups:

Lemma 1.12.2 *If A is a free abelian group, any subgroup B of A is free.*

Proof: The proof given above for the finite case generalizes to the situation in which A has an infinite basis, as we now show. Using the well-ordering theorem [which is equivalent to the axiom of choice], we index the basis for A by a well-ordered set J having a largest element.

We begin by assuming [without loss of generality] that A equals a direct sum of copies of $\mathbb{Z}$; that is, A equals the subgroup of $\mathbb{Z}^J$ consisting of all tuples $(n_\alpha)_{\alpha\in J}$ such that $n_\alpha = 0$ for all but finitely many α. Then we proceed as before. Since the process may be unfamiliar, we will give the details:

Let B be a subgroup of A. For $\beta \in J$, let B_β consist of those elements $\mathbf{x}$ of B such that $\pi_\alpha(\mathbf{x}) = 0$ for $\alpha > \beta$. Consider the subgroup $\pi_\beta(B_\beta)$ of $\mathbb{Z}$; if it is trivial then define $\mathbf{x}_\beta = \mathbf{0}$, otherwise choose $\mathbf{x}_\beta \in B_\beta$ so the $\pi_\beta(\mathbf{x}_\beta)$ generates the subgroup.

We show first that the set $\{x_\alpha : \alpha \leq \beta\}$ generates B_β. This fact is trivial if β is the smallest element of J. We prove it in general by transfinite induction. Given $\mathbf{x} \in B_\beta$, we have

$$\pi_\beta(\mathbf{x}) = k\pi_\beta(\mathbf{x}_\beta)$$

for some integer k. Hence $\pi_\beta(\mathbf{x} - k\mathbf{x}_\beta) = 0$. Consider the set of those indices α for which $\pi_\alpha(\mathbf{x} - k\mathbf{x}_\beta) \neq 0$. [If there are none, then $\mathbf{x} = k\mathbf{x}_\beta$ and we are through.] All of these indices are less than β, because $\mathbf{x}$ and $\mathbf{x}_\beta$ belong to B_β. Furthermore, this set of indices if *finite*, so it has a largest element γ, which is less than β. This means that $\mathbf{x} - k\mathbf{x}_\beta$ belongs to B_γ, whence by the induction hypothesis, $\mathbf{x} - k\mathbf{x}_\beta$ can be written as a linear combination of elements $\mathbf{x}_\alpha$ with each $\alpha \leq \gamma$.

Second, we show that the non-zero elements in the set $\{\mathbf{x}_\alpha : \alpha \leq \beta\}$ are independent. Again, this fact is trivial if β is the smallest element of J. In general, suppose that

$$\lambda_{\alpha_1}\mathbf{x}_{\alpha_1} + \lambda_{\alpha_2}\mathbf{x}_{\alpha_2} + \cdots + \lambda_{\alpha_k}\mathbf{x}_{\alpha_k} + \lambda_\beta\mathbf{x}_\beta = \mathbf{0}\,,$$

where $\alpha_i < \beta$. Applying π_β, we see that

$$\lambda_\beta\pi_\beta(\mathbf{x}_\beta) = 0\,.$$

As before, it follows that either $\lambda_\beta = 0$ or $\mathbf{x}_\beta = \mathbf{0}$. We conclude that

$$\lambda_\beta = 0 \quad \text{if} \quad \mathbf{x}_\beta \neq \mathbf{0}\,,$$

$$\lambda_{\alpha_1}\mathbf{x}_{\alpha_1} + \lambda_{\alpha_2}\mathbf{x}_{\alpha_2} + \cdots + \lambda_{\alpha_k}\mathbf{x}_{\alpha_k} = \mathbf{0}\,.$$

The induction hypothesis now implies that $\lambda_{\alpha_i} = 0$ whenever $\mathbf{x}_{\alpha_i} \neq \mathbf{0}$. □

We now prove our basic theorem. First we need a definition.

Definition 1.12.3 Let G and G' be free abelian groups with bases $a_1, a_2, \ldots, a_n$ and $a'_1, a'_2, \ldots, a'_m$, respectively. If $f : G \to G'$ is a homomorphism, then

$$f(a_j) = \sum_{i=1}^{m} \lambda_{ij} a'_i$$

for unique integers λ_{ij}. The matrix (λ_{ij}) is called the *matrix of* f relative to the given bases for G and G'.

Theorem 1.12.4 *Let G and G' be free abelian groups of ranks n and m, respectively; let $f : G \to G'$ be a homomorphism. Then there are bases for G and G' such that, relative to these bases, the matrix of f has the form*

$$B = \left[\begin{array}{ccc|c} b_1 & & 0 & \\ & \ddots & & 0 \\ 0 & & b_\ell & \\ \hline & 0 & & 0 \end{array}\right]$$

where $b_i \geq 1$ and $b_1 \mid b_2 \mid \cdots \mid b_\ell$.

This matrix is in fact uniquely determined by f [although the bases involved are not]. It is called the *normal form* for the matrix of f.

Proof: We begin by choosing bases in G and G' arbitrarily. Let A be the matrix of f relative to these bases. We shall give shortly a procedure for modifying these bases so as to bring the matrix into the normal form described in the theorem. The theorem will then follow. □

Definition 1.12.5 The *elementary row operations* on an integer matrix A are the following:

(1) Exchange row i and row k.

(2) Multiply row i by -1.

(3) Replace row i by (row i) + q (row k), where q is an integer and $k \neq i$.

Each of these row operations corresponds to a change of basis in G'. The first corresponds to an exchange of a'_i and a'_k. The second corresponds to replacing a'_i by $-a'_i$. The third corresponds to replacing a'_k by $a'_k - qa'_i$, as you can readily check.

There are three similar *elementary column operations* on A that correspond to changes of basis in G.

We will show how to apply these six elementary operations to an arbitrary integer matrix A so as to reduce it to our desired normal form. We assume A is not the zero matrix, since in that case the result is trivial.

Before we begin the description of the reduction algorithm, we note the following fact:

If c is an integer that divides every entry of the matrix A, and if B is obtained from A by applying any one of the elementary operations, then c also divides every entry of B.

The Reduction Algorithm

The reduction algorithm starts with a given matrix $A = (a_{ij})$ of integers, not all zero, and uses elementary operations to produce a matrix in reduced form as in Theorem 1.12.4.

Definition 1.12.6 Given the matrix $A = (a_{ij})$ of integers, not all zero, we let $\alpha(A)$ denote the smallest non-zero element of the set of numbers $|a_{ij}|$. We call a_{ij} a *minimal entry* of A if $|a_{ij}| = \alpha(A)$.

Lemma 1.12.7 *Let A be a non-zero integer matrix. The number $\alpha(A)$ fails to divide some entry of A if and only if it is possible to decrease the value of α by applying elementary operations to A.*

Proof: Suppose that a_{ij} is a minimal entry of A that fails to divide some entry of A. If the entry a_{ij} fails to divide some entry a_{kj} in its column, then we use Euclid's division lemma to write

$$a_{kj} = q\ a_{ij} + r$$

where the remainder r satisfies $0 < |r| < |a_{ij}|$. Signs do not matter here; q and r may be either positive or negative. We then replace (row k) of A by (row i) $-\, q\,$(row k). The result is to replace the entry a_{kj} in the kth row and jth column of A by $a_{kj} - qa_{ij} = r$. The value of α for this new matrix is at most $|r|$, which is less than $\alpha(A)$.

A similar argument applies if a_{ij} fails to divide some entry in its *row*.

Finally, suppose a_{ij} divides each entry in its row and each entry in its column, but fails to divide the entry a_{st}, where $s \neq i$ and $t \neq j$. Consider the following four entries of A:

$$\begin{matrix} a_{ij} & \dots & a_{it} \\ \vdots & & \vdots \\ a_{sj} & \dots & a_{st} \end{matrix}$$

Because a_{ij} divides a_{sj}, we can use an elementary row operation to bring the matrix to the form where the entries in these four places are as follows:

$$\begin{matrix} a_{ij} & \dots & a_{it} \\ \vdots & & \vdots \\ 0 & \dots & a_{st} + \ell a_{it} \end{matrix}$$

We then replace (row i) of this matrix by (row i) + (row s), to obtain

$$\begin{matrix} a_{ij} & \dots & a_{st} + (\ell+1)a_{it} \\ \vdots & & \vdots \\ 0 & \dots & a_{st} + \ell a_{it} \end{matrix}$$

Since a_{ij} divides a_{it}, it does *not* divide $a_{st} + (\ell+1)a_{it}$, and we are back in the situation where a_{ij} fails to divide some entry in its row.

To prove the other direction, suppose that $\alpha(A)$ divides every entry of A. Any matrix B obtained by applying elementary operations to A will be non-zero and $\alpha(A)$ will also divide each entry of B. Choose any non-zero entry b_{ij} of B. Then $\alpha(A)$ divides b_{ij}, so $|b_{ij}| \geq \alpha(A)$. □

Main Step of the Algorithm. Suppose we are given a matrix

$$\left[\begin{array}{ccc|c} b_1 & & 0 & \\ & \ddots & & 0 \\ 0 & & b_k & \\ \hline & 0 & & A \end{array}\right] \qquad (*)$$

where the following hold:

(1) The [possibly empty] set $\{b_1, b_2, \ldots, b_k\}$ of positive integers is such that $b_1 \,|\, b_2 \,|\, \cdots \,|\, b_k$.

(2) The matrix $A = (a_{ij})$ has integer entries, not all zero, such that b_k divides every entry of A.

Note that applying an elementary operation to the matrix A is equivalent to applying that elementary operation to the corresponding row or column of the larger matrix $(*)$.

Set $A_0 = A$. Lemma 1.12.7 tells us that if $\alpha(A_0)$ does not divide every entry of A_0, then elementary operations can be applied to A_0 to produce a matrix A_1 with $\alpha(A_1) < \alpha(A_0)$. Repeating this process [no more than $\alpha(A_0)$ times] will yield a matrix A_r for which $\alpha(A_r)$ divides every element of A_r.

We will apply further elementary operations to A_r, but for convenience of notation, we will continue to denote it by A_r.

Apply elementary operations to bring a minimal entry of A_r to the upper left corner of the matrix and to make it positive. Because the minimal entry divides all entries in its row and column, we can apply elementary operations to make all the other entries in its row and column into zeros. Note that at the end of this process, the entry in the upper left corner of A_r divides all entries of the matrix. In particular, it divides all the entries of the submatrix A' obtained by omitting the first row and first column from A_r. Let b_{k+1} be the entry in the upper left corner of A_r.

Thus, we see that by applying elementary operations to A, or equivalently to the matrix $(*)$, we have brought the matrix to the following form:

$$\left[\begin{array}{ccc|c} b_1 & & 0 & 0 & \\ & \ddots & & & 0 \\ 0 & & b_k & 0 & \\ 0 & & 0 & b_{k+1} & \\ \hline & 0 & & & A' \end{array}\right]$$

where $b_i \geq 1$ and $b_1 \,|\, b_2 \,|\, \cdots \,|\, b_k \,|\, b_{k+1}$, b_{k+1} divides every entry of A', and A' has one less row and one less column than has A.

Termination Condition. If A' contains only zeros or is empty, then we have reached the desired normal form. Otherwise, we repeat the Main Step.

Completion of the Proof of Theorem 1.12.4: It now follows immediately from Exercise 4 of Section 1.5 that the numbers $b_1, b_2, \ldots, b_\ell$ are uniquely determined by the homomorphism f. For the number ℓ of non-zero entries in

the matrix is simply the rank of the free abelian group $f(G) \subset G'$. And those numbers b_i that are greater than 1 are the torsion coefficients $t_1, t_2, \ldots, t_k$ of the torsion group $G'/f(G)$. □

Applications of the Reduction Algorithm

Now we prove the basic theorem concerning subgroups of free abelian groups that we stated in Section 1.5.

Proof of Theorem 1.5.2: Given a free abelian group F of rank n, we know from Lemma 1.12.1 that any subgroup R is free of rank $r \leq n$. Consider the *inclusion* homomorphism $j : R \to F$, and choose bases $a_1, a_2, \ldots, a_r$ for R and $e_1, e_2, \ldots, e_n$ for F relative to which the matrix of j is in the normal form of the preceding Theorem 1.12.4. Because j is a monomorphism, this normal form has no zero columns. Thus, $j(a_i) = b_i e_i$ for $i = 1, 2, \ldots, r$, where $b_i \geq 1$ and $b_1 \,|\, b_2 \,|\, \ldots \,|\, b_r$. Since $j(a_i) = a_i$, it follows that $b_1 e_1, b_2 e_2, \ldots, b_r e_r$ is a basis for R □

Now we introduce the notion of a chain complex and prove the standard basis theorem for free chain complexes.

Definition 1.12.8 A *chain complex* $\mathcal{C}$ is a sequence

$$\ldots \longrightarrow C_{p+1} \xrightarrow{\partial_{p+1}} C_p \xrightarrow{\partial_p} C_{p-1} \longrightarrow \ldots$$

of abelian groups C_i and homomorphisms ∂_i, indexed by the integers, such that $\partial_p \circ \partial_{p+1} = 0$ for all p. The pth *homology group* of $\mathcal{C}$ is defined by

$$H_p(\mathcal{C}) = \ker \partial_p / \text{im } \partial_{p+1} .$$

If $H_p(\mathcal{C})$ is finitely generated, its betti number and torsion coefficients are called the betti number and torsion coefficients of $\mathcal{C}$ in dimension p.

Theorem 1.12.9 (Standard bases for free chain complexes) *Let $\{C_p, \partial_p\}$ be a chain complex; suppose each group C_p is free of finite rank. Then for each p there are subgroups U_p, V_p, and W_p of C_p such that*

$$C_p = U_p \oplus V_p \oplus W_p ,$$

where $\partial_p(U_p) \subset W_{p-1}$, $\partial_p(V_p) = 0$, and $\partial_p(W_p) = 0$. Furthermore, there are bases for U_p and W_{p-1} relative to which $\partial_p : U_p \to W_{p-1}$ has a matrix of the form

$$B = \begin{bmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_\ell \end{bmatrix} ,$$

where $b_i \geq 1$ and $b_1 \,|\, b_2 \,|\, \cdots \,|\, b_\ell$.

Proof: Step 1. Let

$$Z_p = \ker \partial_p \quad \text{and} \quad B_p = \text{im } \partial_{p+1}\,.$$

Let W_p consist of all elements c_p of C_p such that some non-zero multiple of c_p belongs to B_p. This is a subgroup of C_p called the group of *weak boundaries*. We have

$$B_p \subset W_p \subset Z_p \subset C_p\,.$$

The second inclusion holds because C_p is torsion free, so that the equation $mc_p = \partial_{p+1} d_{p+1}$ implies $\partial_p c_p = 0$.

Consider the natural projection

$$Z_p \to H_p(\mathcal{C}) \to H_p(\mathcal{C})/T_p(\mathcal{C})\,,$$

where $T_p(\mathcal{C})$ is the torsion subgroup of $H_p(\mathcal{C})$. The kernel of this projection is W_p; therefore $Z_p/W_p \cong H_p/T_p$. The latter group is finitely generated and torsion-free, so it is free. If $c_1 + W_p, c_2 + W_p, \ldots, c_k + W_p$ is a basis for Z_p/W_p, and $d_1, d_2, \ldots, d_\ell$ is a basis for W_p, then it is straightforward to check that $c_1, c_2, \ldots, c_k, d_1, d_2, \ldots, d_\ell$ is a basis for Z_p. Then $Z_p = V_p \oplus W_p$, where V_p is the group with basis $c_1, c_2, \ldots, c_k$.

Step 2. Suppose we choose bases $e_1, e_2, \ldots, e_n$ for C_p, and $e'_1, e'_2, \ldots, e'_m$ for C_{p-1}, relative to which the matrix of $\partial_p : C_p \to C_{p-1}$ has the normal form

$$\begin{array}{c} \\ e'_1 \\ \vdots \\ e'_\ell \\ e'_{\ell+1} \\ \vdots \\ e'_m \end{array}
\begin{array}{c}
\begin{array}{cccc} e_1 & \cdots & e_\ell & e_{\ell+1} \cdots e_n \end{array} \\
\left[\begin{array}{ccc|c}
b_1 & & 0 & \\
& \ddots & & 0 \\
0 & & b_\ell & \\
\hline
\multicolumn{3}{c|}{0} & 0
\end{array}\right]
\end{array}$$

where $b_i \geq 1$ and $b_1 \,|\, b_2 \,|\, \cdots \,|\, b_\ell$. Then the following hold:

- **(1)** $e_{\ell+1}, e_{\ell+2}, \ldots, e_n$ is a basis for Z_p.
- **(2)** $e'_1, e'_2, \ldots, e'_\ell$ is a basis for W_{p-1}.
- **(3)** $b_1 e'_1, b_2 e'_2, \ldots, b_\ell e'_\ell$ is a basis for B_{p-1}.

We prove these results as follows: Let c_p be the general p-chain. We compute its boundary. If

$$c_p = \sum_{i=1}^{n} a_i e_i\,, \quad \text{then} \quad \partial_p c_p = \sum_{i=1}^{\ell} a_i b_i e'_i\,.$$

To prove **(1)**, we note that since $b_i \neq 0$, the p-chain c_p is a cycle if and only if $a_i = 0$ for $i = 1, 2, \ldots, \ell$. To prove **(3)**, we note that any $p-1$ boundary $\partial_p c_p$ lies in the group generated by $b_1 e'_1, b_2 e'_2, \ldots, b_\ell e'_\ell$; since $b_i \neq 0$, these elements are independent. Finally, we prove **(2)**. Note first that each of $e'_1, e'_2, \ldots, e'_\ell$ belongs to W_{p-1}, since $b_i e'_i = \partial e_i$. Conversely, let

$$c_{p-1} = \sum_{i=1}^{m} d_i e'_i$$

be a $p-1$ chain and suppose $c_{p-1} \in W_{p-1}$. Then c_{p-1} satisfies an equation of the form

$$\lambda c_{p-1} = \partial_p c_p = \sum_{i=1}^{\ell} a_i b_i e'_i$$

for some $\lambda \neq 0$. Equating coefficients, we see that $\lambda d_i = 0$ for $i > \ell$, whence $d_i = 0$ for $i > \ell$. Thus $e'_1, e'_2, \ldots, e'_\ell$ is a basis for W_{-1}.

Step 3. We prove the theorem. Choose bases for C_p and C_{p-1} as in Step 2. Define U_p to be the group spanned by $e_1, e_2, \ldots, e_\ell$; then

$$C_p = U_p \oplus Z_p \,.$$

Using Step 1, choose V_p so that $Z_p = V_p \oplus W_p$. Then we have a decomposition of C_p such that $\partial_p(V_p) = 0$ and $\partial_p(W_p) = 0$. The existence of the desired bases for U_p and W_p follows from Step 2. □

Note that W_p and $Z_p = V_p \oplus W_p$ are uniquely determined subgroups of C_p. On the other hand, the subgroups U_p and V_p are not uniquely determined.

Theorem 1.12.10 *The homology groups of a finite complex K are effectively computable.*

Proof: By the preceding theorem, there is a decomposition

$$C_p(K) = U_p \oplus V_p \oplus W_p$$

where $Z_p = V_p \oplus W_p$ is the group of p-cycles and W_p is the group of weak p-boundaries. Now we have

$$H_p(K) = Z_p / B_p \cong V_p \oplus (W_p / B_p) \cong (Z_p / W_p) \oplus (W_p / B_p) \,.$$

The group Z_p / W_p is free and the group W_p / B_p is a torsion group; computing $H_p(K)$ thus reduces to computing these two groups.

Let us choose bases for the chain groups $C_p(K)$ by orienting the simplices of K, once and for all. Then consider the matrix of the boundary homomorphism $\partial_p : C_p(K) \to C_{p-1}(K)$ relative to this choice of bases; the entries of this matrix will in fact have values in the set $\{-1, 0, 1\}$. Using the reduction algorithm described earlier, we reduce this matrix to normal form. Examining Step 2 of the preceding proof, we conclude from the results proved there the following facts about this normal form:

(1) The rank of Z_p equals the number of zero columns.

(2) The rank of W_{p-1} equals the number of non-zero rows.

(3) There is an isomorphism

$$W_{p-1}/B_{p-1} \cong \mathbb{Z}/b_1 \oplus \mathbb{Z}/b_2 \oplus \cdots \oplus \mathbb{Z}/b_\ell \,.$$

Thus the normal form for the matrix of $\partial_p : C_p(K) \to C_{p-1}(K)$ gives us the torsion coefficients of K in dimension $p-1$; they are the entries of the matrix that are greater than 1. This normal form also gives us the rank of Z_p. On the other hand, the normal form for $\partial_{p+1} : C_{p+1}(K) \to C_p(K)$ gives us the rank of W_p. The difference of these numbers is the rank of Z_p/W_p—that is, the betti number of K in dimension p. □

EXERCISES

1. Show that the reduction algorithm is not needed if one wishes merely to compute the betti numbers of a finite complex K; instead all that is needed is an algorithm for determining the rank of a matrix. Specifically, show that if A_p is the matrix of $\partial_p : C_p(K) \to C_{p-1}(K)$ relative to some choice of basis, then

$$\beta_p(K) = \text{rank } C_p(K) - \text{rank } A_p - \text{rank } A_{p+1} \,.$$

2. Compute the homology groups of the quotient space indicated in Figure 1.12.1. [**Hint:** First check whether all the vertices are identified.]

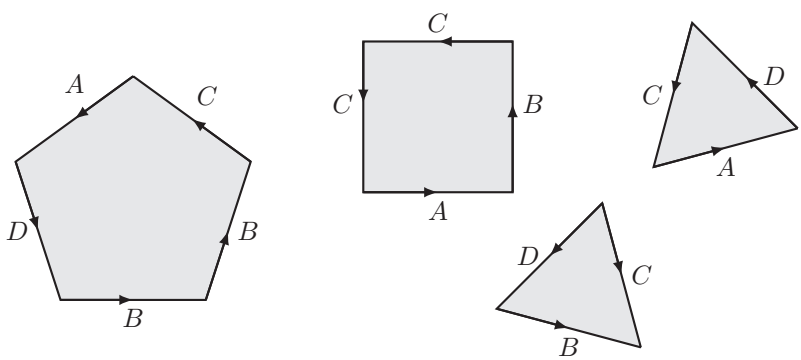


Figure 1.12.1

3. Reduce the following matrix to normal form

$$\begin{bmatrix} 2 & 6 & 4 \\ 4 & -7 & 4 \\ 4 & 8 & 4 \end{bmatrix}.$$

1.13 Homomorphisms Induced by Simplicial Maps

If f is a simplicial map of $|K|$ into $|L|$, then f maps each p-simplex σ_i of K onto a simplex τ_i of L of the same or lower dimension. We shall define a homomorphism of p-chains that carries a formal sum $\sum m_i\sigma_i$ of oriented p-simplices of K onto a formal sum $\sum m_i\tau_i$ of their images. [We delete from the latter sum those simplices τ_i whose dimension is less than p.] This map in turn induces a homomorphism of homology groups, as we shall see.

As a general convention, we shall use the phrase

$$f : K \to L \text{ is a simplicial map}$$

to mean that f is a continuous map of $|K|$ into $|L|$ that maps each simplex of K linearly onto a simplex of L. Thus f maps each vertex of K to a vertex of L, and it equals the simplicial map induced by this vertex map, as defined in Section 1.3.

Definition 1.13.1 Let $f : K \to L$ be a simplicial map. If $v_0v_1 \dots v_p$ is a simplex of K, then the points $f(v_0), f(v_1), \dots, f(v_p)$ span a simplex of L. We define a homomorphism $f_\# : C_p(K) \to C_p(L)$ by defining it on oriented simplices as follows:

$$f_\#([v_0, v_1, \dots, v_p]) = \begin{cases} [f(v_0), f(v_1), \dots, f(v_p)] & \text{if } f(v_0), f(v_1), \dots, f(v_p) \text{ are distinct,} \\ 0 & \text{otherwise.} \end{cases}$$

This map is clearly well-defined; exchanging two vertices in the expression $[v_0, v_1, \dots, v_p]$ changes the sign of the right side of the equation. The family of homomorphisms $\{f_\#\}$, one in each dimension, is called the *chain map induced by the simplicial map* f.

Properly speaking, one should use dimensional subscripts to distinguish these homomorphisms, denoting the map in dimension p by $(f_\#)_p : C_p(K) \to C_p(L)$. Normally, we shall omit the subscript, relying on the context to make the situation clear, just as we do with the boundary operator.

In a similar vein, we shall use the symbol ∂ to denote the boundary operators in both K and L, in order to keep the notation from becoming cumbersome. If it is necessary to distinguish them, we can use the notations ∂_K and ∂_L.

Lemma 1.13.2 *The homomorphism $f_\#$ commutes with ∂; therefore $f_\#$ induces a homomorphism $f_* : H_p(K) \to H_p(L)$.*

Proof: We need to show that

$$\partial f_\#([v_0, v_1, \ldots, v_p]) = f_\#(\partial[v_0, v_1, \ldots, v_p])\,. \qquad (*)$$

Let τ be the simplex of L spanned by $f(v_0), f(v_1), \ldots, f(v_p)$. We consider various cases.

Case 1. $\dim \tau = p$. In this case, the vertices $f(v_0), f(v_1), \ldots, f(v_p)$ are distinct and the result follows at once from the definitions of $f_\#$ and ∂.

Case 2. $\dim \tau \leq p-2$. In this case, the left-hand side of $(*)$ vanishes because $f(v_0), f(v_1), \ldots, f(v_p)$ are not distinct, and the right-hand side vanishes because for each i because at least two of the points $f(v_0), f(v_1), \ldots f(v_{i-1})$, $f(v_{i+1}), \ldots, f(v_p)$ are the same.

Case 3. $\dim \tau = p-1$. In this case, we may assume the vertices are ordered so that $f(v_0) = f(v_1)$, while $f(v_1), f(v_2), \ldots, f(v_p)$ are distinct. Then the left-hand side of $(*)$ vanishes because $f(v_0), f(v_1), \ldots, f(v_p)$ are not all distinct. Many of the terms on the right-hand side of $(*)$ are of the form

$$(-1)^i\,[f(v_0), f(v_1), \ldots f(v_{i-1}), f(v_{i+1}), \ldots, f(v_p)]\,,$$

in which the simplex is of dimension $p-2$. The only two non-zero terms on the right-hand side are

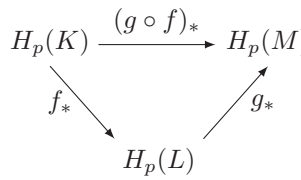
$$[f(v_1), f(v_2), \ldots, f(v_p)] - [f(v_0), f(v_2), \ldots, f(v_p)]\,,$$

and they cancel each other, as desired.

The homomorphism $f_\#$ carries cycles to cycles, since the equation $\partial c_p = 0$ implies that $\partial f_\#(c_p) = f_\#(\partial c_p) = 0$. Also, $f_\#$ carries boundaries to boundaries, since the equation $c_p = \partial d_{p+1}$ implies that $f_\#(c_p) = f_\#(\partial d_{p+1}) = \partial f_\#(d_{p+1})$. Thus $f_\#$ induces a homomorphism $f_* : H_p(K) \to H_p(L)$ of homology groups. □

Theorem 1.13.3 **(a)** *Let $i : K \to K$ be the identity simplicial map. Then $i_* : H_p(K) \to H_p(K)$ is the identity homomorphism.*

(b) *Let $f : K \to L$ and $g : L \to M$ be simplicial maps. Then $(g \circ f)_* = g_* \circ f_*$; that is, the following diagram commutes:*



Proof: It is immediate from the definition that i_* is the identity and that $(g \circ f)_* = g_* \circ f_*$, as you can check. □

This last theorem expresses what are called the "functorial properties" of the induced homomorphism. This phrase will be defined formally later when we discuss categories and functors. For the present, we point out simply that the operator H_p assigns to each simplicial complex an abelian group, and the operator $*$ assigns to each simplicial map of one complex into another a homomorphism of the corresponding abelian groups. Because **(a)** and **(b)** hold, we say that $(H_p, *)$ is a "functor" from the "category" of simplicial complexes and simplicial maps to the "category" of abelian groups and homomorphisms.

Lemma 1.13.4 *The chain map $f_\#$ preserves the augmentation map ϵ; therefore it induces a homomorphism f_* of reduced homology groups.*

Proof: Let $f : K \to L$ be simplicial. Then $\epsilon f_\#(v) = 1$ and $\epsilon(v) = 1$ for each vertex v of K. Thus $\epsilon \circ f_\# = \epsilon$. This equation implies that $f_\#$ carries the kernel of $\epsilon_K : C_0(K) \to \mathbb{Z}$ into the kernel of $\epsilon_L : C_0(L) \to \mathbb{Z}$, and thus induces a homomorphism $f_* : \widetilde{H}_0(K) \to \widetilde{H}_0(L)$. □

EXAMPLE 1.13.5 Consider the complexes K and T shown in Figure 1.13.1. Their underlying spaces are the circle and the torus, respectively. We have $H_1(K) \cong \mathbb{Z}$; let us use the cycle z indicated in the figure as a generator. Similarly, we have $H_1(T) \cong \mathbb{Z} \oplus \mathbb{Z}$. As a basis, let us use the cosets of the cycles

$$w_1 = [A, B] + [B, C] + [C, A] \quad \text{and} \quad z_1 = [A, D] + [D, E] + [E, A]\,.$$

Now consider the simplicial maps:

$f : a \to A$	$g : a \to A$	$h : a \to A$
$b \to P$	$b \to B$	$b \to I$
$c \to D$	$c \to C$	$c \to J$
$d \to D$	$d \to A$	$d \to G$
$e \to P$	$e \to E$	$e \to G$
$p \to E$	$p \to D$	$p \to A$

You can check that $f_\#(z)$ is homologous to z_1, that $g_\#(z)$ equals $w_1 - z_1$, and that $h_\#(z)$ is homologous to $w_1 - z_1$. Thus g_* and h_* are equal as homomorphisms of 1-dimensional homology. They are also equal on 0-dimensional homology.

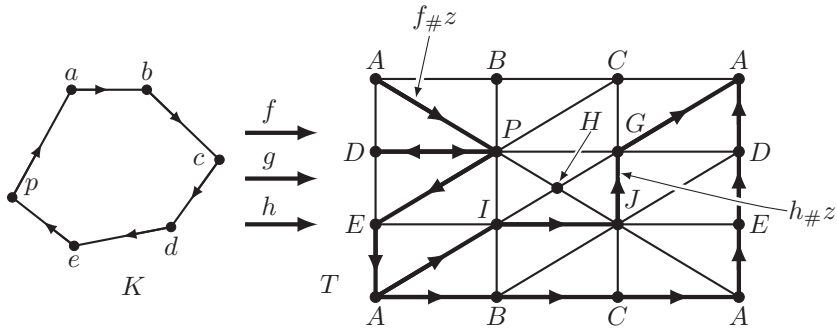


Figure 1.13.1

In general, a given homomorphism can be induced by quite different simplicial maps, as the preceding example shows. This fact leads us to consider the general question: Under what conditions do two simplicial maps induce the same homomorphism of homology groups? Answering this question involves an important technique, one we shall use many times. So we begin by explaining the underlying motivation.

Given simplicial maps $f, g : K \to L$, we wish to find conditions under which $f_\#(z)$ and $g_\#(z)$ are homologous for each cycle $z \in Z_p(K)$. Said differently, we want to find conditions under which there is a function [commonly called D] that assigns to each p-cycle z of K, a $p+1$ chain Dz of L, such that

$$\partial Dz = g_\#(z) - f_\#(z)\,.$$

Let us consider an example in which this is possible. Suppose that K is the boundary of a triangular region, and L consists of the sides of a triangular prism, as pictured in Figure 1.13.2. Suppose f and g are the simplicial maps that carry K onto the two ends of the prism, respectively. If z is the 1-cycle generating $H_1(K)$, as indicated in the figure, it is quite easy to find a 2-chain Dz whose boundary is $g_\#(z) - f_\#(z)$; we let Dz equal the sum of all the 2-simplices of L, oriented appropriately. You can check that this 2-chain does the job.

In general, it is awkward to define D only for cycles, since that requires us to compute the group of cycles. What proves more satisfactory is to define D as a homomorphism on the entire chain group, because then one can define D one simplex at a time. How might that procedure work in the present example? Let us illustrate the process. Given a vertex v of K, we define Dv to be the edge of L that leads from $f(v)$ to $g(v)$, as indicated in Figure 1.13.3. And given the oriented 1-simplex σ of K, we define $D\sigma$ to be the sum of the two oriented simplices that are heavily shaded in the figure, which form one side of the prism. We proceed similarly for the other simplices of K. Since z is the

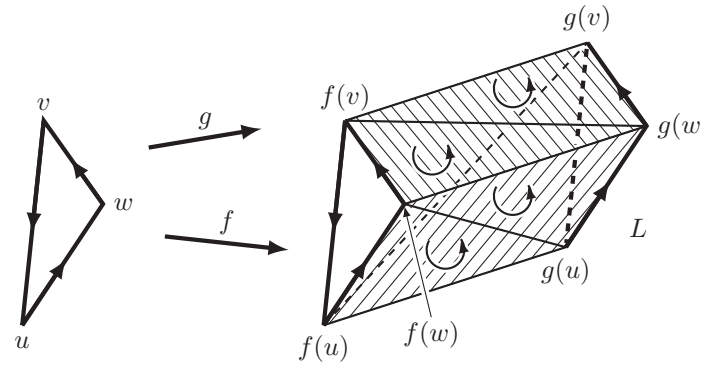


Figure 1.13.2

sum of the oriented edges of K, the chain Dz will be the sum of the oriented 2-simplices of L, just as before, and $\partial Dz = g_{\#}(z) - f_{\#}(z)$, as desired.

One can ask what sort of formula holds for ∂Dc when c is an arbitrary chain. The answer is clear from the definition. Given the oriented simplex σ, we compute $\partial D\sigma = g_{\#}(\sigma) - Dv - f_{\#}(\sigma) + Dw$; see Figure 1.13.3 for verification. That is,

$$\partial(D\sigma) = g_{\#}(\sigma) - f_{\#}(\sigma) - D(\partial\sigma)\,. \qquad (**)$$

This same formula holds for the other simplices of K, as you can check.

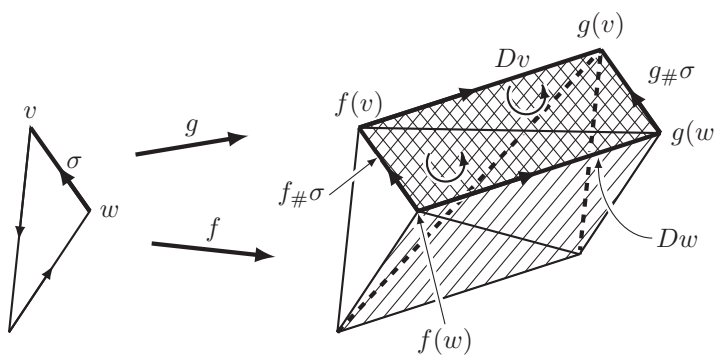


Figure 1.13.3

This formula expresses the crucial algebraic property of the homomorphism D. It is the formula at which we were aiming; we make it part of the following definition.

Definition 1.13.6 Let $f, g : K \to L$ be simplicial maps. Suppose that, for each p, one has a homomorphism

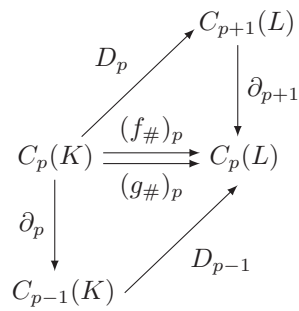
$$D : C_p(K) \to C_{p+1}(L)$$

satisfying the equation

$$\partial D + D\partial = g_\# - f_\# \,.$$

Then D is said to be *chain homotopy* between $f_\#$ and $g_\#$.

We have omitted dimensional subscripts in the above formula; the following diagram may make the maps involved more clear:



With subscripts, the formula becomes

$$\partial_{p+1} D_p + D_{p-1}\partial_p = (g_\#)_p - (f_\#)_p \,.$$

This formula is more precise, but messier. We shall customarily omit dimensional subscripts.

The importance of chain homotopies comes from the following theorem:

Theorem 1.13.7 *If there is a chain homotopy between $f_\#$ and $g_\#$, then the induced homomorphisms f_* and g_*, for both reduced and ordinary homology, are equal.*

Proof: Let z be a p-cycle of K. Then

$$g_\#(z) - f_\#(z) = \partial Dz + D\partial z = \partial Dz + 0 \,,$$

so $g_\#(z)$ and $f_\#(z)$ are in the same homology class. Thus $g_*(\{z\}) = f_*(\{z\})$, as desired. □

We still want to find conditions on two simplicial maps f and g under which the induced homomorphisms f_* and g_* are equal. We have reduced this problem to the problem of finding conditions under which one can construct a chain homotopy between $f_\#$ and $g_\#$. Here is one set of conditions under which this is possible:

Definition 1.13.8 Given two simplicial maps $f, g : K \to L$, these maps are said to be *contiguous* if for each simplex $v_0 v_1 \dots v_p$ of K, the points

$$f(v_0), f(v_1), \dots, f(v_p), g(v_0), g(v_1), \dots, g(v_p)$$

span a simplex τ of L. [The simplex τ may be of any dimension from 0 to $2p+1$, depending on how many of these points are distinct.]

Roughly speaking, this condition says that f and g are fairly close, in the sense that one can move a simplex $f(\sigma)$ to the simplex $g(\sigma)$ by crossing some possibly larger simplex τ of which both are faces.

Theorem 1.13.9 *If $f, g : K \to L$ are contiguous maps, then there is a chain homotopy between $f_\#$ and $g_\#$.*

Proof: The argument we give here is a standard one; you should master it.

For each simplex $\sigma = v_0 v_1 \dots v_p$ of K, let $L(\sigma)$ denote the subcomplex of L consisting of the simplex whose vertex set is

$$\{f(v_0), f(v_1), \dots, f(v_p), g(v_0), g(v_1), \dots, g(v_p)\},$$

and its faces. We note the following facts:

- **(1)** $L(\sigma)$ is nonempty, and $\widetilde{H}_i(L(\sigma)) = 0$ for all i. (Recall Theorem 1.9.7.)
- **(2)** If s is a face of σ, then $L(s) \subset L(\sigma)$.
- **(3)** For each oriented simplex σ, the chains $f_\#(\sigma)$ and $g_\#(\sigma)$ are carried by $L(\sigma)$.

Applying these facts, we shall use induction on p to construct the required chain homotopy $D : C_p(K) \to C_{p+1}(L)$. For each σ, the chain $D\sigma$ will be carried by $L(\sigma)$.

Let $p = 0$; let v be a vertex of K. Because $f_\#$ and $g_\#$ preserve augmentation, $\epsilon(g_\#(v) - f_\#(v)) = 1 - 1 = 0$. Thus $g_\#(v) - f_\#(v)$ represents an element of the reduced homology group $\widetilde{H}_0(L(v))$. Because this group vanishes, we can choose a 1-chain Dv of L carried by the subcomplex $L(v)$ such that

$$\partial(Dv) = g_\#(v) - f_\#(v).$$

Then $\partial Dv + D\partial v = \partial Dv + 0 = g_\#(v) - f_\#(v)$, as desired. Define D in this way for each vertex of K.

Now suppose D is defined, in dimensions less than p, such that for each oriented simplex s of dimension less than p, the chain Ds is carried by $L(s)$, and satisfies

$$\partial Ds + D\partial s = g_\#(s) - f_\#(s).$$

Let σ be an oriented simplex of dimension p. We wish to define $D\sigma$ so that $\partial(D\sigma)$ equals the chain

$$c = g_\#(\sigma) - f_\#(\sigma) - D\partial\sigma.$$

Note that c is a well-defined chain, since $D\partial\sigma$ is defined because $\partial\sigma$ has dimension $p-1$. Furthermore, c is a cycle as we now check: Applying the induction hypothesis to the $p-1$ chain $\partial\sigma$ and using the fact that $\partial\circ\partial = 0$, we compute

$$\begin{aligned}\partial c &= \partial g_{\#}(\sigma) - \partial f_{\#}(\sigma) - \partial D(\partial\sigma)\\ &= \partial g_{\#}(\sigma) - \partial f_{\#}(\sigma) - \Big[g_{\#}(\partial\sigma) - \partial f_{\#}(\partial\sigma) - D\partial(\partial\sigma)\Big]\\ &= \partial g_{\#}(\sigma) - \partial f_{\#}(\sigma) - g_{\#}(\partial\sigma) + \partial f_{\#}(\partial\sigma) = 0\,.\end{aligned}$$

Finally, we check that c is carried by $L(\sigma)$: Both $g_{\#}(\sigma)$ and $f_{\#}(\sigma)$ are carried by $L(\sigma)$, by **(3)**. To show that $D\partial\sigma$ is carried by $L(\sigma)$, note that the chain $\partial\sigma$ is a sum of oriented faces of σ. For each such face s, the chain Ds is carried by $L(s)$, and $L(s)\subset L(\sigma)$ by **(2)**. Thus $D\partial\sigma$ is carried by $L(\sigma)$.

Since c is a p-cycle carried by $L(\sigma)$, and since $H_p(L(\sigma)) = 0$, we can choose a $p+1$ chain $D\sigma$ carried by $L(\sigma)$ such that

$$\partial D\sigma = c = g_{\#}(\sigma) - f_{\#}(\sigma) - D\partial\sigma\,.$$

We then define $D(-\sigma) = -D(\sigma)$. We repeat this process for each p-simplex σ of K; then we have the required chain homotopy D in dimension p. The theorem follows. □

Some students are bothered by the fact that constructing the chain homotopy D involves arbitrary choices. They would be happier if there were a definite formula for D. Unfortunately, there is no such neat formula. Basically, this is because there are many possible chain homotopies between $f_{\#}$ and $g_{\#}$, and there is no reason for preferring one over the other.

To illustrate this fact, let us consider the preceding proof in a particular case. At the first step of the proof, when v is a vertex, the chain Dv is in fact uniquely determined. Since Dv is to be a chain carried by $L(v)$, then necessarily

$$\begin{aligned} Dv &= 0 && \text{if } f(v) = g(v),\\ Dv &= [f(v), g(v)] && \text{if } f(v)\neq g(v).\end{aligned}$$

In the first case, $L(v)$ is a vertex; and in the second case, $L(v)$ consists of a 1-simplex and its faces.

But at the very next step of the proof, choices can arise. Let $\sigma = vw$ be a 1-simplex, and suppose the points $f(v)$, $f(w)$, $g(v)$, $g(w)$ are all distinct, so they span a 3-simplex τ, as indicated in Figure 1.13.4. The chains Dv and Dw are indicated in the figure. How shall we define $D\sigma$? There are two obvious choices for a 2-chain $D\sigma$ whose boundary is the 1-chain $g_{\#}(\sigma) - f_{\#}(\sigma) - D\partial\sigma$ pictured in Figure 1.13.5. One choice would consist of the front two faces of τ, oriented as indicated; the other choice would consist of the back two faces, appropriately oriented. There is no reason to prefer one choice to the other; we must simply choose one arbitrarily.

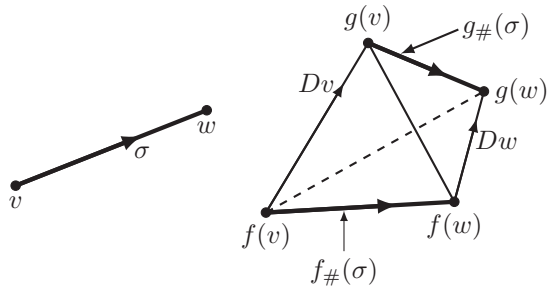


Figure 1.13.4

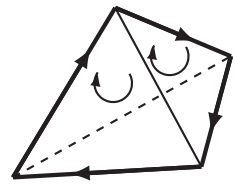


Figure 1.13.5

Application to Relative Homology

Let K_0 be a subcomplex of K, and let L_0 be a subcomplex of L. Let $f : K \to L$ be a simplicial map that carries each simplex of K_0 into a simplex of L_0. We often express this by the phrase

$$\text{“}f : (K, K_0) \to (L, L_0) \text{ is a simplicial map.”}$$

In this case, it is immediate that $f_\#$ maps $C_p(K_0)$ into $C_p(L_0)$, so that one has an induced map [also denoted by $f_\#$]

$$f_\# : C_p(K, K_0) \to C_p(L, L_0)\,.$$

This map commutes with ∂ and thus induces a homomorphism

$$f_* : H_p(K, K_0) \to H_p(L, L_0)\,.$$

The functorial properties stated in Theorem 1.13.3 carry over immediately to relative homology.

Definition 1.13.10 Let $f, g : (K, K_0) \to (L, L_0)$ be two simplicial maps. We say f and g are *contiguous as maps of pairs* if for each simplex $\sigma = v_0 v_1 \dots v_p$ of K, the points

$$f(v_0), f(v_1), \dots, f(v_p), g(v_0), g(v_1), \dots, g(v_p)$$

span a simplex of L, and if $\sigma \in K_0$, they span a simplex of L_0.

Theorem 1.13.11 *Let $f, g : (K, K_0) \to (L, L_0)$ be contiguous as maps of pairs. Then for all p there is a homomorphism*

$$D : C_p(K, K_0) \to C_{p+1}(L, L_0)$$

such that $\partial D + D\partial = g_\# - f_\#$. It follows that f_* and g_* are equal as maps of relative homology groups.

Proof: The chain homotopy D constructed in the preceding proof of Theorem 1.13.9 automatically maps $C_p(K_0)$ into $C_{p+1}(L_0)$. For if $\sigma \in K_0$, then the complex $L(\sigma)$ is by definition a subcomplex of L_0. Given σ, the chain $D\sigma$ is carried by $L(\sigma)$; therefore, D maps $C_p(K_0)$ into $C_{p+1}(L_0)$. Then D induces the required homomorphism of the relative chain groups. □

EXERCISES

1. Check the assertions made in Example 1.13.5.

2. Consider the complex K indicated by the vertex labeling in Figure 1.13.6, whose underlying space is the torus. Although this complex is slightly different from the one considered earlier, the computation of the homology of the torus remains the same. Let γ be the sum of all the 2-simplices of K, oriented counterclockwise; then γ generates $H_2(K)$. The cycles

$$w_1 = [a, b] + [b, c] + [c, a] \quad \text{and} \quad z_1 = [a, d] + [d, e] + [e, a]$$

generate $H_1(K)$.

(a) Define the simplicial maps f, g, h, k of K with itself such that

$f : a \to a$	$g : a \to a$	$h : a \to d$	$k : a \to p$
$b \to d$	$b \to c$	$b \to p$	$b \to q$
$c \to e$	$c \to b$	$c \to q$	$c \to d$
$d \to c$	$d \to d$	$d \to d$	$d \to u$
$e \to b$	$e \to e$	$e \to d$	$e \to b$

(b) Compute the values of f_*, g_*, h_*, k_* on the homology classes $\{w_1\}$, $\{z_1\}$, and $\{\gamma\}$.

3. Use the complex in Figure 1.13.7 to represent the Klein bottle S. Let $w_1 = [a, b] + [b, c] + [c, a]$ and $z_1 = [a, d] + [d, e] + [e, a]$, as before.

(a) Define a simplicial map $f : S \to S$ such that $f_*(\{w_1\}) = \{z_1\}$.

(b) Show there is no simplicial map $g : S \to S$ such that $g_*(\{z_1\}) = \{w_1\}$.

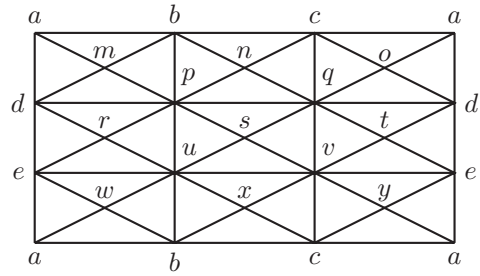


Figure 1.13.6

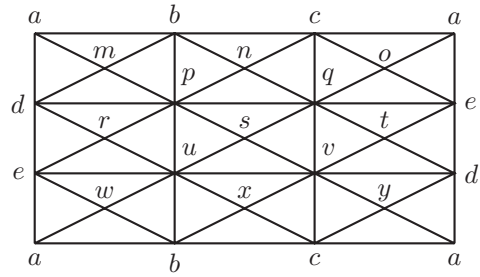


Figure 1.13.7

4. Let K be the torus, as in Exercise 2; let γ be the 2-cycle indicated there. Let L be the complex consisting of the proper faces of the 3-simplex having vertices A, B, C, D; and let γ' be the 2-cycle $\partial[A, B, C, D]$. [See Figure 1.13.8.]

(a) Show that any map of the vertices of K to the vertices of L induces a simplicial map of K to L.

(b) Let f be the simplicial map carrying m and r to A, p to B , b and u to C, and all other vertices to D. Compute the induced homomorphism $f_* : H_2(K) \to H_2(L)$, in terms of γ and γ'.

(c) Let g be the simplicial map agreeing with f on the vertices of K except that $g(r) = C$. Compute g_*.

(d) Let h agree with g on the vertices of K except that $h(u) = A$. Compute h_*.

5. Let $f, g : (K, K_0) \to (L, L_0)$ be simplicial maps. Show that if f and g are contiguous as maps of K into L, and if L_0 is a full subcomplex of L, then f and g are contiguous as maps of pairs. [Recall Definition 1.4.11.]

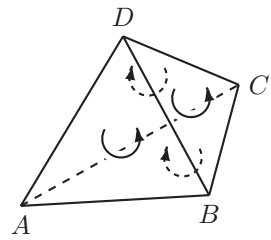


Figure 1.13.8

1.14 Chain Complexes and Acyclic Carriers

Many of the definitions and constructions we have made within the context of simplicial complexes also occur in more general situations. We digress at this point to discuss this more general context. We shall use these results many times later on. Proofs are left as exercises.

First we define the algebraic analogues to our chain groups. [We mentioned them earlier in Section 1.12.]

Definition 1.14.1 A *chain complex* $\mathcal{C}$ is a family $\{C_p, \partial_p\}$ of abelian groups C_p and homomorphisms

$$\partial_p : C_p \to C_{p-1}\,,$$

indexed by the integers, such that $\partial_p \circ \partial_{p+1} = 0$ for all p.

If $C_p = 0$ for $p < 0$, then $\mathcal{C}$ is said to be a *non-negative* chain complex. If C_p is a free abelian group for each p, then $\mathcal{C}$ is called a *free* chain complex. The group

$$H_p(\mathcal{C}) = \ker \partial_p / \text{im } \partial_{p+1}$$

is called the pth *homology group* of the chain complex $\mathcal{C}$.

If $\mathcal{C}$ is a non-negative chain complex, an *augmentation* for $\mathcal{C}$ is an epimorphism $\epsilon : C_0 \to \mathbb{Z}$ such that $\epsilon \circ \partial_1 = 0$. The *augmented chain complex* $\{\mathcal{C}, \epsilon\}$ is the chain complex obtained from $\mathcal{C}$ by adjoining the group $\mathbb{Z}$ in dimension -1 and using ϵ as the boundary operator in dimension 0. The homology groups of the augmented chain complex are called the *reduced homology groups* of the original chain complex $\mathcal{C}$, relative to the augmentation ϵ. They are denoted either by $H_i(\{\mathcal{C}, \epsilon\})$ or by $\widetilde{H}_i(\mathcal{C})$. It follows readily that $\widetilde{H}_p(\mathcal{C}) = H_p(\mathcal{C})$ for $p \neq 0$, and

$$H_0(\mathcal{C}) \cong \widetilde{H}_0(\mathcal{C}) \oplus \mathbb{Z}\,.$$

[See the exercises.]

An arbitrary chain complex $\mathcal{D}$ is said to be *acyclic* if $H_i(\mathcal{D}) = 0$ for all i. In particular, the augmented chain complex $\{\mathcal{C}, \epsilon\}$ is acyclic if $H_i(\{\mathcal{C}, \epsilon\}) = \widetilde{H}_i(\mathcal{C}) = 0$ for all i or equivalently if $H_0(\mathcal{C}) \cong \mathbb{Z}$ and $H_i(\mathcal{C}) = 0$ for $i \neq 0$.

Definition 1.14.2 Let $\mathcal{C} = \{C_p, \partial_p\}$ and $\mathcal{C}' = \{C'_p, \partial'_p\}$ be chain complexes. A *chain map* $\phi : \mathcal{C} \to \mathcal{C}'$ is a family of homomorphisms

$$\phi_p : C_p \to C'_p$$

such that $\partial'_p \circ \phi_p = \phi_{p-1} \circ \partial_p$ for all p.

A chain map $\phi : \mathcal{C} \to \mathcal{C}'$ gives rise to an *induced homology homomorphism*

$$(\phi_*)_p : H_p(\mathcal{C}) \to H_p(\mathcal{C}') .$$

Furthermore, the following hold:

(a) The identity map i of $\mathcal{C}$ is a chain map, and $(i_*)_p$ is the identity map of $H_p(\mathcal{C})$.

(b) If $\phi : \mathcal{C} \to \mathcal{C}'$ and $\psi : \mathcal{C}' \to \mathcal{C}''$ are chain maps, then $\psi \circ \phi$ is a chain map, and $(\psi \circ \phi)_* = \psi_* \circ \phi_*$.

If $\{\mathcal{C}, \epsilon\}$ and $\{\mathcal{C}', \epsilon'\}$ are augmented chain complexes, then the chain map $\phi : \mathcal{C} \to \mathcal{C}'$ is said to be *augmentation-preserving* if $\epsilon' \circ \phi_0 = \epsilon$. If we extend ϕ to the (-1)-dimensional groups by letting it equal the identity map of $\mathbb{Z}$, then ϕ is called a *chain map of augmented chain complexes*. It follows that an augmentation-preserving chain map ϕ induces a homomorphism $\phi_* : \widetilde{H}_p(\mathcal{C}) \to \widetilde{H}_p(\mathcal{C}')$ of reduced homology groups.

EXAMPLE 1.14.3 The chain complex

$$\mathcal{C}(K, K_0) = \{C_p(K, K_0), \partial_p\}$$

defined in Section 1.10 is called the *oriented chain complex of the simplicial pair* (K, K_0). It is both free and non-negative, but in general, it does not have an augmentation. [For example, in Example 1.10.5 the entire group $C_0(K, K_0)$ vanishes, so there can be no surjective map $C_0(K, K_0) \to \mathbb{Z}$.] If K_0 is empty, then the complex $\mathcal{C}(K, \emptyset) = \mathcal{C}(K)$ has a standard augmentation, defined by $\epsilon(v) = 1$ for each vertex of K, as we have seen.

EXAMPLE 1.14.4 If $f : K \to L$ is a simplicial map of simplicial complexes K and L, then $f_\#$ is an augmentation-preserving chain map of $\mathcal{C}(K)$ into $\mathcal{C}(L)$. In general, there exist augmentation-preserving chain maps $\phi : \mathcal{C}(K) \to \mathcal{C}(L)$ that are not induced by simplicial maps.

Definition 1.14.5 If $\phi, \psi : \mathcal{C} \to \mathcal{C}'$ are chain maps, then a *chain homotopy* of ϕ to ψ is a family of homomorphisms

$$D_p : C_p \to C'_{p+1}$$

such that

$$\partial'_{p+1} D_p + D_{p-1} \partial_p = \psi_p - \phi_p$$

for all p. We say ϕ and ψ are *chain homotopic*.

Henceforth we shall normally omit subscripts on boundary operators and chain maps and chain homotopies. The preceding formula, for instance, then assumes the more familiar form $\partial' D + D\partial = \psi - \phi$.

Definition 1.14.6 A chain map $\phi : \mathcal{C} \to \mathcal{C}'$ is called a *chain equivalence* if there is a chain map $\phi' : \mathcal{C}' \to \mathcal{C}$ such that $\phi' \circ \phi$ and $\phi \circ \phi'$ are chain homotopic to the identity maps of $\mathcal{C}$ and $\mathcal{C}'$, respectively. We call ϕ' a *chain-homotopy inverse* to ϕ.

We list several properties of chain homotopies; proofs are left to the exercises.

(1) *Chain homotopy is an equivalence relation on the set of chain maps from $\mathcal{C}$ to $\mathcal{C}'$.*

(2) *Composion of chain maps induces a well-defined composition operation of chain-homotopy classes.*

(3) *If ϕ and ψ are chain homotopic, then they induce the same homomorphism in homology.*

(4) *If ϕ is a chain equivalence, with chain-homotopy inverse ϕ', then ϕ_* and $(\phi')_*$ are homology isomorphisms that are inverse to each other.*

(5) *If $\phi : \mathcal{C} \to \mathcal{C}'$ and $\psi : \mathcal{C}' \to \mathcal{C}''$ are chain equivalences, then $\psi \circ \phi$ is a chain equivalence.*

Now we investigate what all this means in the special case of augmented chain complexes.

Lemma 1.14.7 *Let $\mathcal{C}$ and $\mathcal{C}'$ be non-negative chain complexes. Let $\phi, \psi : \mathcal{C} \to \mathcal{C}'$ be chain maps; let D be a chain homotopy between them. Suppose $\mathcal{C}$ and $\mathcal{C}'$ are augmented by ϵ and ϵ', respectively. If ϕ preserves augmentation, then so does ψ. If we extend ϕ and ψ to be the identity in dimension -1, and extend D to be zero in dimension -1, then D is a chain homotopy between the extended chain maps.*

Proof: If $c_0 \in C_0$, then we have

$$\partial D c_0 = \phi(c_0) - \psi(c_0)$$

because $\partial c_0 = 0$. Then

$$0 = \epsilon'(\partial D c_0) = \epsilon' \phi(c_0) - \epsilon' \psi(c_0) = \epsilon(c_0) - \epsilon' \psi(c_0)\,,$$

as desired. Furthermore, we have in dimension 0 the equation

$$D\epsilon(c_0) + \partial D(c_0) = \phi(c_0) - \psi(c_0)\,,$$

because $D\epsilon(c_0) = 0$, and we have in dimension -1 the equation

$$\epsilon'(D(1)) = \phi(1) - \psi(1)$$

because both sides vanish. Thus D is a chain homotopy between the extended chain maps. □

Lemma 1.14.8 *Let $\mathcal{C}$ and $\mathcal{C}'$ be non-negative chain complexes. Let $\phi : \mathcal{C} \to \mathcal{C}'$ be a chain equivalence with chain-homotopy inverse ϕ'. Suppose that $\mathcal{C}$ and $\mathcal{C}'$ are augmented by ϵ and ϵ', respectively. If ϕ preserves augmentation, so does ϕ'. Furthermore, ϕ and ϕ' are chain-homotopy inverses as maps of augmented complexes. Therefore they induce inverse isomorphisms in reduced homology.*

Proof: If D' is a chain homotopy between $\phi \circ \phi'$ and the identity, then in dimension 0,

$$\partial' D' c'_0 = \phi\, \phi'(c'_0) - c'_0\,,$$

so

$$0 = \epsilon'(\partial' D' c'_0) = \epsilon'(\phi\, \phi'(c'_0)) - \epsilon'(c_0) = \epsilon(\phi'(c'_0)) - \epsilon'(c'_0)\,.$$

Thus ϕ' preserves augmentation. The remainder of the statement follows from Lemma 1.14.7. □

These definitions set up the general algebraic framework into which the oriented simplicial chain groups fit as a special case. Now we seek to put into this general context the method by which in the preceding section we constructed a chain homotopy. As motivation, we first consider the case where the maps ϕ and ψ are general chain maps, but the chain groups are the familiar oriented simplicial chain groups. Later, we consider general chain complexes and general chain maps.

Definition 1.14.9 Let K and L be simplicial complexes. An *acyclic carrier* from K to L is a function Φ that assigns to each simplex σ of K a subcomplex $\Phi(\sigma)$ of L such that:

(1) $\Phi(\sigma)$ is nonempty and acyclic.

(2) If s is a face of σ, then $\Phi(s) \subset \Phi(\sigma)$.

If $f : C_p(K) \to C_q(L)$ is a homomorphism, we say that f is *carried* by Φ if for each oriented p-simplex σ of K, the chain $f(\sigma)$ is carried by the subcomplex $\Phi(\sigma)$ of L.

Theorem 1.14.10 (Acyclic carrier theorem, geometric version) *Let Φ be an acyclic carrier from K to L.*

> **(a)** *If ϕ and ψ are two augmentation-preserving chain maps from $\mathcal{C}(K)$ to $\mathcal{C}(L)$ that are carried by Φ, then there exists a chain homotopy D of ϕ to ψ that is also carried by Φ.*

(b) *There exists an augmentation-preserving chain map from* $\mathcal{C}(K)$ *to* $\mathcal{C}(L)$ *that is carried by* Φ.

Proof: Part **(a)** of the theorem is proved by copying the proof of Theorem 1.13.9; one simply replaces the subcomplex $L(\sigma)$ by the subcomplex $\Phi(\sigma)$ throughout.

To prove **(b)**, we proceed as follows: For each vertex v of K, define $\phi(v)$ to be a 0-chain c of $\Phi(v)$ such that $\epsilon(c) = 1$. This we can do because $\Phi(v)$ is nonempty, so $\epsilon : \Phi(v) \to \mathbb{Z}$ is surjective. [In fact, we can simply choose $\phi(v)$ to be a vertex of $\Phi(v)$.] Then ϕ preserves augmentation, and

$$\partial\phi(v) = 0 = \phi(\partial v)\,.$$

Let $\sigma = [v, w]$ be an oriented 1-simplex of K. The chain $c = \phi(\partial\sigma)$ is well-defined because $\partial\sigma$ is a 0-chain and ϕ has been defined in dimension 0. Furthermore, c is carried by $\Phi(\sigma)$; for $\phi(\partial\sigma)$ is carried by $\Phi(v)$ and $\Phi(w)$, both of which are contained in $\Phi(\sigma)$ by **(2)**. Finally,

$$\epsilon(c) = \epsilon\phi(\partial\sigma) = \epsilon(\partial\sigma) = 0\,,$$

because ϕ preserves augmentation. Hence c represents an element of $\widetilde{H}_0(\Phi(\sigma))$. Because $\Phi(\sigma)$ is acyclic [by **(1)**], we can choose a 1-chain carried by $\Phi(\sigma)$ whose boundary is c. We denote this 1-chain by $\phi(\sigma)$; then

$$\partial\phi(\sigma) = c = \phi(\partial\sigma)\,.$$

For the induction step, let $p > 1$. Assume that if $\dim s < p$, then $\phi(s)$ is defined and $\partial\phi(s) - \phi\partial(s)$. Let σ be an oriented simplex of dimension p; we seek to define $\phi(\sigma)$. The chain $c = \phi(\partial\sigma)$ is a well-defined $p-1$ chain. It is carried by $\Phi(\sigma)$, since $\Phi(\partial\sigma)$ is carried by the union of the complexes $\Phi(s_i)$, where s_i ranges over the $p-1$ faces of σ, and each of these complexes is contained in $\Phi(\sigma)$. Furthermore, c is a cycle since

$$\partial c = \partial\phi(\partial\sigma) = \phi\partial(\partial\sigma) = 0\,.$$

Here we apply the induction hypothesis to the $p-1$ chain $\partial\sigma$. Since $\Phi(\sigma)$ is acyclic, we can choose $\phi(\sigma)$ to be a p-chain carried by $\Phi(\sigma)$ such that $\partial\phi(\sigma) = c$. Then $\partial\phi(\sigma) = \phi\partial(\sigma)$ as desired.

The theorem follows by induction. □

With this proof as motivation, we now formulate an even more general version of this theorem. In this form, all geometry disappears and only algebra remains!

Definition 1.14.11 Let $\mathcal{C} = \{C_p, \partial_p\}$ be a chain complex. A *subchain complex* $\mathcal{D}$ of $\mathcal{C}$ is a chain complex whose pth chain group is a subgroup of C_p, and whose boundary operator in each dimension p is the restriction of ∂_p.

Definition 1.14.12 Let $\{\mathcal{C}, \epsilon\} = \{C_p, \partial_p, \epsilon\}$ be an augmented chain complex. Suppose that $\mathcal{C}$ is free; let $\{\sigma_p^\alpha\}$ be a basis for C_p, as α ranges over some index set J_p. Let $\{\mathcal{C}', \epsilon'\} = \{C'_p, \partial'_p, \epsilon'\}$ be an arbitrary augmented chain complex. An *acyclic carrier* from $\mathcal{C}$ to $\mathcal{C}'$, relative to the given bases, is a function Φ that assigns to each basis element σ_p^α a subchain complex $\Phi(\sigma_p^\alpha)$ of $\mathcal{C}'$, satisfying the following conditions:

(1) The chain complex $\Phi(\sigma_p^\alpha)$ is augmented by ϵ' and is acyclic.

(2) If σ_{p-1}^β appears in the expression of $\partial_p \sigma_p^\alpha$ in terms of a given basis for C_{p-1}, then $\Phi(\sigma_{p-1}^\beta)$ is a subchain complex of $\Phi(\sigma_p^\alpha)$.

A homomorphism $f : C_p \to C'_q$ is said to be *carried* by Φ if $f(\sigma_p^\alpha)$ belongs to the q-dimensional group of the subchain complex $\Phi(\sigma_p^\alpha)$ of $\mathcal{C}'$, for each α.

Theorem 1.14.13 (Acyclic carrier theorem, algebraic version) *Let $\mathcal{C}$ and $\mathcal{C}'$ be augmented chain complexes; let $\mathcal{C}$ be free. Let Φ be an acyclic carrier from $\mathcal{C}$ to $\mathcal{C}'$, relative to some set of preferred bases for $\mathcal{C}$. Then there is an augmentation-preserving chain map $\phi : \mathcal{C} \to \mathcal{C}'$ carried by Φ, and any two such are chain homotopic, and the chain homotopy is also carried by Φ.*

Proof: The proof of this theorem is simply an enhanced version of the preceding proof of Theorem 1.14.10. The requirement that the restriction of ϵ' give an augmentation for $\Phi(\sigma_p^\alpha)$ means that ϵ' must map the 0-dimensional group of this chain complex *onto* $\mathbb{Z}$; this corresponds to the requirement in the earlier version that $\Phi(\sigma)$ be nonempty for all σ. □

Application: Ordered Simplicial Homology

The algebraic version of the acyclic carrier theorem may seem unnecessarily abstract to you. But it is indeed useful. Let us give one application now; it is a theorem that we will use later when we study singular homology. This theorem involves a new way of defining the simplicial homology groups, using *ordered* simplices rather than *oriented* simplices.

Let K be a simplicial complex. An *ordered p-simplex* of K is a $p+1$ tuple $(v_0, v_1, \ldots, v_p)$ of vertices of K, where the v_i are vertices of a simplex of K *but they need not be distinct.* [For example, if vw is a 1-simplex of K, then (v, w, w, v) is an ordered 3-simplex of K.]

Let $C'_p(K)$ be the free abelian group generated by the ordered p-simplices of K; it is called the group of *ordered p-chains* of K. As usual, we shall identify the ordered p-simplex $(v_0, v_1, \ldots, v_p)$ with the elementary p-chain whose value is 1 on this ordered simplex and 0 on all other ordered simplices. Then every element of $C'_p(K)$ can be written uniquely as a finite linear combination, with integral coefficients, of ordered p-simplices. We define $\partial'_p : C'_p(K) \to C'_{p-1}(K)$ by the formula

$$\partial'_p(v_0, v_1, \ldots v_p) = \sum_{i=0}^{p} (-1)^i (v_0, v_1, \ldots, \widehat{v_i}, \ldots, v_p)\,.$$

The hat over a term indicates that it is omitted. Then ∂'_p is a well-defined homomorphism; and one checks as before that $\partial'_p \circ \partial'_{p+1} = 0$.

The chain complex $\mathcal{C}'(K) = \{C'_p(K), \partial'_p\}$ is called the *ordered chain complex* of K. The resulting homology groups, $H_i(\mathcal{C}'(K))$, give the *ordered homology* of K. There is also a relative version, $H_i(\mathcal{C}'(K, K_0))$. The ordered chain complex is augmented by defining $\epsilon'(v) = 1$ for every vertex v of K. The reduced groups are of course denoted by $\widetilde{H}_i(\mathcal{C}'(K))$.

Although the ordered chain complex is much too huge to be useful for computational purposes, it is sometimes quite convenient for theoretical purposes. Its homology is, surprisingly enough, isomorphic to the simplicial homology of K, in a natural way. We outline a proof of this fact, leaving the details to you.

Lemma 1.14.14 *If $w * K$ is a cone over the complex K, then $w * K$ is acyclic in ordered homology.*

Proof: Define

$$D : C'_p(w * K) \to C'_{p+1}(w * K)$$

for $p \geq 0$ by the equation

$$D((v_0, v_1, \ldots, v_p)) = (w, v_0, v_1, \ldots, v_p)\,.$$

Note that it is irrelevant here whether any of the v_i are equal to w. Let $c_p \in C'_p(w * K)$. We compute

$$\begin{aligned} \partial'_1 D c_0 &= c_0 - \epsilon'(c_0) w\,, \\ \partial'_{p+1} D c_p &= c_p - D\partial'_p c_p \quad \text{if} \quad p > 0\,. \end{aligned}$$

The lemma follows: If c_p is a *cycle* and $p > 0$, then $c_p = \partial'_{p+1} D c_p$; if c_0 is a 0-chain lying in $\ker \epsilon'$, then $c_0 = \partial'_1 D c_0$. □

Lemma 1.14.15 *Let K be a complex. Then there exists a partial ordering of the vertices of K that induces a linear ordering on the vertices of each simplex of K.*

We leave the proof of the lemma to the exercises.

Theorem 1.14.16 *Choose a partial ordering of the vertices of K that induces a linear ordering on the vertices of each simplex of K. Define $\phi : C_p(K) \to C'_p(K)$ by letting*

$$\phi([v_0, v_1, \ldots, v_p]) = (v_0, v_1, \ldots, v_p)$$

if $v_0 < v_1 < \cdots < v_p$ in the given ordering. Define $\psi : C'_p(K) \to C_p(K)$ by the equation

$$\psi((w_0, w_1, \ldots, w_p)) = \begin{cases} [w_0, w_1, \ldots, w_p] & \text{if the } w_i \text{ are distinct,} \\ 0 & \text{otherwise.} \end{cases}$$

Then ϕ and ψ are augmentation-preserving chain maps that are chain-homotopy inverses.

If K_0 is a subcomplex of K, then ϕ and ψ induce chain maps of the relative chain complexes that are chain-homotopy inverses.

The proof is an application of the acyclic carrier theorem; it is left as an exercise.

It follows from this theorem that oriented and ordered homology are isomorphic in a rather "natural" way. To explain what we mean by "naturality," we need to consider how a simplicial map acts in ordered homology.

Definition 1.14.17 *Let $f : K \to L$ be a simplicial map. Define $f'_\# : \mathcal{C}'(K) \to \mathcal{C}'(L)$ by the rule*

$$f'_\#((v_0, v_1, \ldots, v_p)) = (f(v_0), f(v_1), \ldots, f(v_p)) .$$

It is easy to check that $f'_\#$ is a chain map, easier in fact than in the oriented case, for we need not worry whether or not vertices are distinct. Clearly $f'_\#$ preserves augmentation. If f maps K_0 to L_0, then $f'_\#$ induces a chain map

$$f'_\# : \mathcal{C}'(K, K_0) \to \mathcal{C}'(L, L_0) ,$$

and a corresponding homomorphism in homology.

Theorem 1.14.18 *Let $f : (K, K_0) \to (L, L_0)$ be a simplicial map. Let ϕ and ψ be as in the preceding Theorem 1.14.16. Then the following diagram commutes:*

$$\begin{array}{ccc} H_i(K, K_0) & \xrightarrow{f_*} & H_i(L, L_0) \\ \uparrow \psi_* & & \uparrow \psi_* \\ H_i(\mathcal{C}'(K, K_0)) & \xrightarrow{f'_*} & H_i(\mathcal{C}'(L, L_0)) . \end{array}$$

Similarly, $\phi_* \circ f_* = f'_* \circ \phi_*$ holds.

Proof: One checks directly from the definition that $f_\# \circ \psi = \psi \circ f'_\#$. Thus the diagram already commutes on the chain level, so it commutes on the homology level as well. It is not true that $\phi \circ f_\# = f'_\# \circ \phi$, because ϕ depends on a particular ordering of vertices. Nonetheless, because ϕ_* is the inverse of ψ_*, it is true that $\phi_* \circ f_* = f'_* \circ \phi_*$. □

EXERCISES

1. If $\{\mathcal{C}, \epsilon\}$ is an augmented chain complex, show that $\widetilde{H}_{-1}(\mathcal{C}) = 0$ and

$$\widetilde{H}_0(\mathcal{C}) \oplus \mathbb{Z} \cong H_0(\mathcal{C}) .$$

[See the exercises of Section 1.8.] Conclude that $\{\mathcal{C}, \epsilon\}$ is acyclic if and only if $H_p(\mathcal{C})$ is infinite cyclic for $p = 0$ and vanishes for $p \neq 0$.

2. Check the properties **(1)**–**(5)** of chain homotopies. Only **(2)** and **(5)** require care.

3. Prove **(a)** of Theorem 1.14.10.

4. Consider Example 1.13.5. Show that although the maps g and h are not contiguous, there is nevertheless a chain homotopy between $g_\#$ and $h_\#$, as follows:

- **(a)** Define an acyclic carrier Φ from K to L carrying both $g_\#$ and $h_\#$.
- **(b)** Define a specific chain homotopy between $g_\#$ and $h_\#$ that is carried by Φ.
- **(c)** Define a chain map $\phi : \mathcal{C}(K) \to \mathcal{C}(L)$ carried by Φ that is not induced by a simplicial map; define a chain homotopy between ϕ and $h_\#$.

5. Check the details of the proof of Theorem 1.14.13.

6. Prove Lemma 1.14.15. [**Hint:** Use induction on the number of vertices in the complex.]

7. Prove Theorem 1.14.16 as follows:

- **(a)** Show ϕ and ψ are augmentation-preserving chain maps, and show that $\psi \circ \phi$ equals the identity map of $\mathcal{C}(K)$
- **(b)** Define an acyclic carrier from $\mathcal{C}'(K)$ to $\mathcal{C}'(K)$ that carries both $\phi \circ \psi$ and the identity map.

2

Topological Invariance of the Homology Groups

2.1 Introduction

In Chapter 1, we defined a function which assigns to each simplicial complex K a sequence of abelian groups called *the homology groups of* K. Our purpose now is to show that these groups are determined by the topology of K.

We shall attack this problem by studying continuous maps from one polyhedron to another; we are particularly interested in what such a map does to the homology groups. We will in fact prove that a continuous map $h : |K| \to |L|$ of polyhedra induces, in a canonical fashion, a homomorphism $h_* : H_p(K) \to H_p(L)$ of the homology groups of the corresponding simplicial complexes. Constructing this homomorphism will entail a fair bit of work.

We will see that when h is a homeomorphism of topological spaces, then h_* is an isomorphism of groups. The topological invariance of the simplicial homology groups is an immediate consequence.

We can see intuitively why there should be an induced homomorphism of homology groups. If we think of a homology class as a geometric object, then it is plausible that its image under h should be a well-defined homology class. For instance, if X is a topological space, then a closed loop on the surface T is mapped by $h : T \to X$ into a closed loop in X. To make this intuition rigorous will be our task for some time.

We know that a simplicial map $f : |K| \to |L|$ induces a homomorphism of homology groups. The current chapter is dedicated to the task of showing that an arbitrary continuous map h can be suitably approximated by a simplicial map f, and that the resulting induced homomorphism depends only on the map h, and not on the particular approximation chosen.

2.2 Simplicial Approximations

Here we talk about what it means for an arbitrary continuous map to be "approximated" by a simplicial map.

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Definition 2.2.1 Let $h : |K| \to |L|$ be a continuous map. We say that h satisfies the *star condition* with respect to K and L if, for each vertex v of K, there is a vertex w of L such that

$$h(\text{St } v) \subset \text{St } w\,.$$

Lemma 2.2.2 *Let $h : |K| \to |L|$ satisfy the star condition with respect to K and L. Choose $f : K^{(0)} \to L^{(0)}$ so that, for each vertex v of K,*

$$h(\text{St } v) \subset \text{St } f(v)\,.$$

Then

(a) *Given $\sigma \in K$, choose $x \in \text{Int } \sigma$ and choose τ so that $h(x) \in \text{Int } \tau$. Then f maps each vertex of σ to a vertex of τ.*

(b) *The function f may be extended to a simplicial map of K into L. We also denote this extended map by f.*

(c) *If $g : K \to L$ is another simplicial map such that $h(\text{St } v) \subset \text{St } g(v)$ for each vertex v of K, then f and g are contiguous.*

Proof: (a) Let $\sigma = v_0 v_1 \dots v_p$. Then $x \in \text{St } v_i$ for each i hence

$$h(x) \subset h(\text{St } v_i) \subset \text{St } f(v_i)\,.$$

This means that $h(x)$ has a positive barycentric coordinate with respect to each of the vertices $f(v_i)$, $i = 0, 1, \dots, p$. These vertices must therefore form a subset of the vertex set of τ.

(b) Because f carries the vertices of σ to vertices of a simplex of L, it may be extended to a simplicial map $f : K \to L$.

(c) Let σ, x, and τ be as above. Since

$$h(x) \in h(\text{St } v_i) \subset \text{St } g(v_i)$$

for $i = 0, 1, \dots, p$, the vertices $g(v_i)$ must also be vertices of τ. Therefore $f(v_0), f(v_1), \dots, f(v_p)$ and $g(v_0), g(v_1), \dots, g(v_p)$ span a face of τ. Thus f and g are contiguous. □

Definition 2.2.3 Let $h : |K| \to |L|$ be a continuous map. If $f : K \to L$ is a simplicial map such that

$$h(\text{St } v) \subset \text{St } f(v)$$

for all vertices v of K, then f is called a *simplicial approximation* to h.

In some sense the simplicial approximation f is "close" to the map h. More precisely, if $x \in |K|$, then there is a simplex L that contains both $h(x)$ and $f(x)$.

Lemma 2.2.4 *Let $f : K \to L$ be a simplicial approximation to $h : |K| \to |L|$. If $x \in |K|$, then there is a simplex τ of L such that $h(x) \in \text{Int } \tau$ and $f(x) \in \tau$.*

Proof: This result is immediate from part **(a)** of Lemma 2.2.2. □

Theorem 2.2.5 *Let $h : |K| \to |L|$ and $k : |L| \to |M|$ have simplicial approximations $f : K \to L$ and $g : L \to M$ respectively. Then $g \circ f$ is a simplicial approximation to $k \circ h$.*

Proof: Certainly $g \circ f$ is a simplicial map. If v is a vertex of K, then

$$h(\text{St } v) \subset \text{St } f(v)$$

just because f is a simplicial approximation to h. Therefore

$$k(h(\text{St } v)) \subset k(\text{St } f(v)) \subset \text{St } g(f(v))$$

because g is a simplicial approximation to k. □

EXAMPLE 2.2.6 Let K and L be the complexes shown in Figure 2.2.1.

Note that the underlying spaces are homeomorphic to the circle and the annulus respectively. Let K' be the complex obtained from K by inserting vertices as shown. Let h be the indicated continuous map, where we denote $h(a)$ by A, and similarly for the other vertices.

Now h does not satisfy the star condition relative to K and L, but it does satisfy the star condition relative to K' and L. Thus h has a simplicial approximation $f : K' \to L$. One such is shown in the figure. We denote $f(a)$ by A' and similarly for the other vertices.

If $h : |K| \to |L|$ satisfies the star condition relative to K and L, then there is a well-defined homomorphism

$$h_* : H_p(K) \to H_p(L)$$

obtained by setting $h_* = f_*$, where f is any simplicial approximation to h. It is straightforward to see that the "functorial properties" are satisfied.

In general an arbitrary continuous map $h : |K| \to |L|$ will not satisfy the star condition relative to K and L. Hence we cannot obtain an induced homomorphism h_* as above. What can we do instead? Two ideas are needed.

First we show that, given $h : |K| \to |L|$, it is possible to "subdivide" K, forming a new complex K', with the same underlying space as K, so that h does satisfy the star condition relative to K' and L. [This is what we did in the preceding Example 2.2.6.] This step is geometric in nature, and is carried out in Sections 2.3 and 2.4.

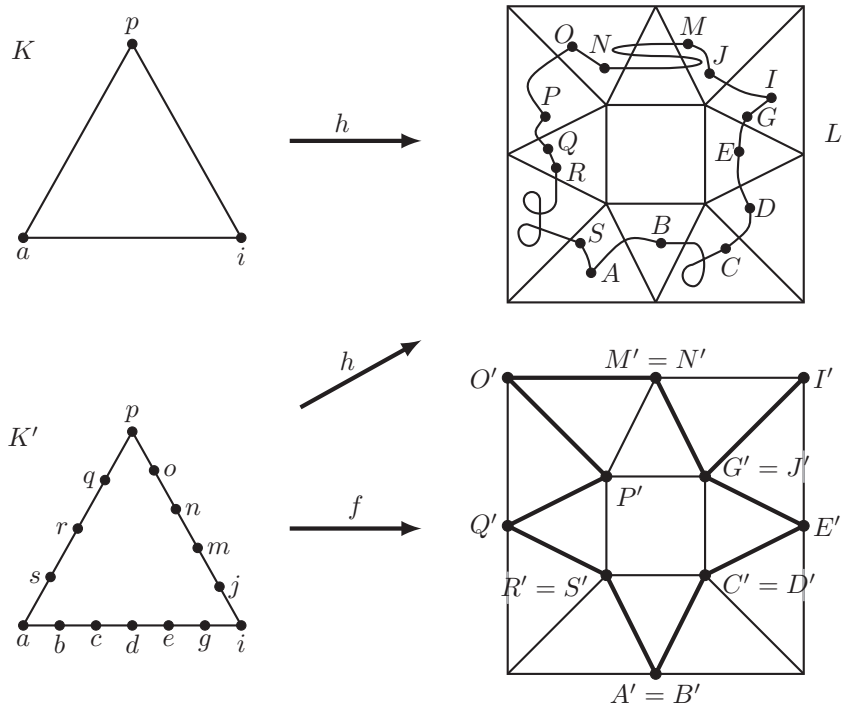


Figure 2.2.1

Next we show that the identity map $i : |K'| \to |K|$ has a simplicial approximation $g : K' \to K$ which induces a homology isomorphism g_*. This step is algebraic in nature and is carried out in Section 2.5.

The homomorphism $h_* : H_p(K) \to H_p(L)$ induced by h is then defined by the equation $h_* = f_* \circ g_*^{-1}$. We will see that the "functorial properties" also hold for this induced homomorphism. See Section 2.6.

Application to Relative Homology

The preceding results about simplicial approximation generalize to relative homology in a straightforward manner:

Lemma 2.2.7 *Let $h : |K| \to |L|$ satisfy the star condition relative to K and L. Suppose that h maps $|K_0|$ to $|L_0|$, where K_0 and L_0 are subcomplexes. Then*

> **(a)** *Any simplicial approximation $f : K \to L$ to the map h also maps $|K_0|$ to $|L_0|$. Also the restriction of f to K_0 is a simplicial approximation to the restriction of h to $|K_0|$.*

(b) *Any two simplicial approximations f, g to h are contiguous as maps of pairs.*

Proof: Let f, g be simplicial approximations to h. Given $\sigma \in K_0$, choose $x \in \operatorname{Int}\sigma$ and let τ be the simplex of L such that $h(x) \in \operatorname{Int}\tau$. Because h maps $|K_0|$ to $|L_0|$, the simplex τ must belong to L_0. Since both f and g map σ onto faces of τ, they map K_0 into L_0 and are contiguous as maps of pairs.

We now show that $f|K_0$ is a simplicial approximation to the restriction of h to $|K_0|$. Let v be a vertex of K_0. Then $\operatorname{St}(v, K_0) = \operatorname{St}(v, K) \cap |K_0|$. In conclusion, we have

$$\begin{aligned} h(\operatorname{St}(v, K_0)) &\subset h(\operatorname{St}(v, K)) \cap h(|K_0|) \\ &\subset \operatorname{St}(f(v), L) \cap |L_0| \\ &= \operatorname{St}(f(v), L_0)\,. \end{aligned}$$

This is the desired result. □

The next definition is used in Exercise 2. It will be seen again in Section 2.7.

Definition 2.2.8 A *homotopy* between two maps $f : X \to Y$ and $h : X \to Y$ is a continuous map $F : X \times I \to Y$, where $I = [0, 1]$, such that $F(x, 0) = f(x)$ and $F(x, 1) = h(x)$ for all $x \in X$.

The particular homotopy defined in equation $(*)$ below is called the *straight-line homotopy* between f and h.

EXERCISES

1. Consider the map $h : |K| \to |L|$ in Example 2.2.6. Determine how many different simplicial approximations $f : K' \to L$ to the map h there are. Let f and g be two such approximations. Choose a cycle g generating $H_1(K')$. Find a chain d of L such that $\partial d = f_\#(z) - g_\#(z)$.

2. **(a)** For two maps f and h mapping X into $\mathbb{R}^N$, show that the formula

$$F(x, t) = (1 - t)f(x) + th(x) \qquad (*)$$

defines a homotopy between them. This is the *straight-line homotopy*.

(b) Let K and L be finite complexes in $\mathbb{R}^N$. Let $f : K \to L$ be a simplicial approximation to $h : |K| \to |L|$. Show that $(*)$ defines a homotopy between f and h.

*__(c)__ Discuss part **(b)** in the case where K and L are not finite.

3. Let f, h be maps from $\mathbb{R}$ to $\mathbb{R}^2$ defined by $f(x) = (x, x^3)$ and $h(x) = (x, e^x)$. Prove that f and h are homotopic.

2.3 Barycentric Subdivision

We show here that a finite complex may be "subdivided" into simplices that are as small as we wish. We will use this result in our consideration of simplicial homology, and later on in our discussion of singular homology in Chapter 4.

Definition 2.3.1 Let K be a geometric complex in $\mathbb{E}^J$. A complex K' is said to be a *subdivision* of K if

(1) Each simplex of K' is contained in a simplex of K.

(2) Each simplex of K is the union of finitely many simplices of K'.

These conditions imply that the union of the simplices of K' equals the union of the simplices of K. That is, $|K'|$ and $|K|$ are equal as sets. The finiteness part of condition **(2)** guarantees that $|K'|$ and $|K|$ are equal as topological spaces. This is easy to check.

Note that if K'' is a subdivision of K' and if K' is a subdivision of K, then K'' is a subdivision of K.

Observe that if K' is a subdivision of K, and if K_0 is a subcomplex of K, then the collection of all simplices of K' that lie in $|K_0|$ is automatically a subdivision of K_0. We in fact call it the subdivision of K_0 *induced by* K'.

The following result will be useful in the sequel.

Lemma 2.3.2 *Let K' be a subdivision of K. Then for each vertex w of K' there is a vertex v of K such that*

$$\mathrm{St}(w, K') \subset \mathrm{St}(v, K)\,. \qquad (*)$$

In fact, if σ is the simplex of K such that $w \in \mathrm{Int}\ \sigma$, then this inclusion holds precisely when v is a vertex of σ.

Proof: If the inclusion $(*)$ holds, then since w belongs to $\mathrm{St}(w, K')$, so w must lie in some open simplex of K that has v as a vertex.

Conversely, suppose that $w \in \mathrm{Int}\ \sigma$ and v is a vertex of σ. It suffices to show that

$$|K| \setminus \mathrm{St}(v, K) \subset |K| \setminus \mathrm{St}(w, K')\,.$$

The set on the left-hand side of this last inclusion is the union of all simplices of K that do not have v as a vertex. Hence it is also a union of simplices τ of K'. No such simplex τ can have w as a vertex, just because $w \in \mathrm{Int}\ \sigma \subset \mathrm{St}(v, K)$. Thus any such simplex lies in $|K| \setminus \mathrm{St}(w, K')$. □

EXAMPLE 2.3.3 Let K consist of the 1-simplex $[0, 1]$ and it vertices. Let L consist of the 1-simplices $[1/(n+1), 1/n]$, $n = 1, 2, \ldots$ and their vertices, along with the vertex 0. Then $|L| = |K|$ as sets but not as topological spaces. In fact L satisfies all the conditions for a subdivision except the finiteness part of condition **(2)**.

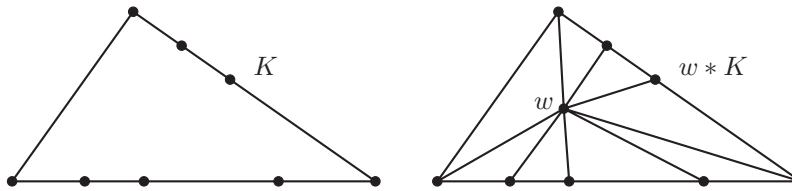


Figure 2.3.1

EXAMPLE 2.3.4 Let Σ be the complex consisting of a 2-simplex σ and its faces. The subdivision K of Bd σ indicated in Figure 2.3.1 can be extended to a subdivision Σ' of Σ by forming the cone $w * K$, where w is an interior point of Σ. The subdivision Σ is said to be obtained by "starring K from w." This method of subdividing complexes will be useful in what follows.

Next we describe the starring method for subdividing complexes in general. We shall need the following lemma, whose proof is left as an exercise.

Lemma 2.3.5 *If K is a complex, then the intersection of any collection of subcomplexes of K is also a subcomplex of K. Conversely, if $\{K_\alpha\}$ is a collection of complexes in $\mathbb{E}^J$ and if the intersection of every pair $|K_\alpha| \cap |K_\beta|$ is the polytope of a complex that is a subcomplex of both K_α and K_β, then the union $\cup_\alpha K_\alpha$ is a complex.*

Our method for constructing subdivisions is step-by-step. We now describe one step in the process.

Definition 2.3.6 Let K be a complex. Suppose that L_p is a subdivision of the p-skeleton of K. Let σ be a $p+1$ simplex of K. The set Bd σ is the polytope of a subcomplex of the p-skeleton of K, and hence of a subcomplex of L_p. We denote the latter subcomplex by L_σ. If w_σ is an interior point of σ, then the cone $w_\sigma * L_\sigma$ is a complex whose underlying space is σ. We define L_{p+1} to be the union of L_p and the complexes $w_\sigma * L_\sigma$ as σ ranges over all $(p+1)$-simplices of K. We show that L_{p+1} is a complex. It is said to be the *subdivision of K^{p+1} obtained by starring L_p from the points w_σ.*

To verify that L_{p+1} is a complex, we note that

$$|W_p * L_\sigma| \cap |L_p| = \text{Bd } \sigma\,,$$

which is the polytope of the subcomplex L_p of both $W_\sigma * L_\sigma$ and L_p. Similarly, if τ is another $(p+1)$-simplex of K, then the space $|w_\sigma * L_\sigma|$ and $|w_\tau * L_\tau|$ intersect in the simplex $\sigma \cap \tau$ of K, which is the polytope of a subcomplex of L_p and hence of both L_σ and L_τ. It follows from the preceding lemma that L_{p+1} is a complex.

Now the complex L_{p+1} depends on the choice of the point w_σ. Often it is convenient to choose a canonical interior point of σ for starring purposes. The usual choice is the following:

Definition 2.3.7 If $\sigma = \{v_0, v_1, \ldots, v_p\}$ then the *barycenter* of σ is defined to be the point

$$\widehat{\sigma} = \frac{1}{p+1} \sum_{i=0}^{p} v_i \, .$$

It is the point of Int σ all of whose barycentric coordinates with respect to the vertices of σ are equal.

If σ is a 1-simplex, then $\widehat{\sigma}$ is its midpoint. If σ is a 0-simplex, then $\widehat{\sigma} = \sigma$. In general $\widehat{\sigma}$ is the centroid of σ, but that fact will have no significance for us.

Now we describe the general method for constructing subdivisions.

Definition 2.3.8 Let K be a complex. We define a sequence of subdivisions of the skeletons of K as follows: Let $L_0 = K^{(0)}$, the 0-skeleton of K. In general, if L_p is a subdivision of the p-skeleton of K, let L_{p+1} be the subdivision of the $(p+1)$-skeleton obtained by starring L_p from the barycenters of the $(p+1)$ simplices of K. By Lemma 2.3.5, the union of the complexes L_p is a subdivision of K. It is called the *first barycentric subdivision* of K, and denoted by sd K.

Having formed a complex sd K, we can now construct its first barycentric subdivision sd(sd K), which we denote by sd^2 K. This complex is called the *second barycentric subdivision* of K. Similarly one defines sd^n K for n a positive integer.

Sometimes it is useful to have an explicit description of the simplices of the first barycentric subdivision. Now we give such a description. Let us use the notation $\sigma_1 \succ \sigma_2$ to mean "σ_2 is a proper face of σ_1."

Lemma 2.3.9 *The complex* sd K *equals the collection of all simplices of the form*

$$\widehat{\sigma}_1 \widehat{\sigma}_2 \ldots \widehat{\sigma}_n \, ,$$

where $\sigma_1 \succ \sigma_2 \succ \cdots \succ \sigma_n$.

Proof: The proof is by induction. It is immediate that the simplices of sd K lying in the subdivision of $K^{(0)}$ are of this form. [Each such simplex is a vertex of K, and $\widehat{v} = v$ for a vertex.]

Suppose now that each simplex of sd K lying in $|K^{(p)}|$ is of this form. Let τ be a simplex of sd K lying in $|K^{(p+1)}|$ and not in $|K^{(p)}|$. Then τ belongs to one of the complexes $\widehat{\sigma} * L_\sigma$, where σ is a $(p+1)$-simplex of K and L_σ is the first barycentric subdivision of the complex consisting of the proper faces of σ. By the induction hypothesis, each simplex of L_σ is of the form $\widehat{\sigma}_1 \widehat{\sigma}_2 \ldots \widehat{\sigma}_n$, where $\sigma_1 \succ \sigma_2 \succ \cdots \succ \sigma_n$ and σ_i is a proper face of σ. Then τ must be of the form

$$\widehat{\sigma} \widehat{\sigma}_1 \widehat{\sigma}_2 \ldots \widehat{\sigma}_n$$

which is the desired form. □

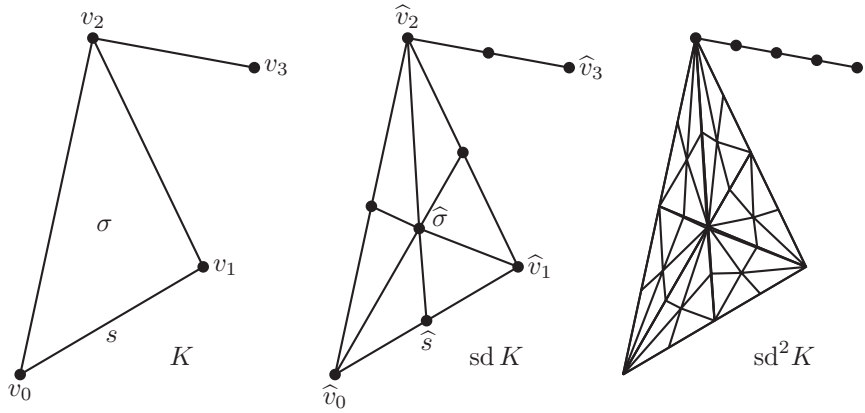


Figure 2.3.2

EXAMPLE 2.3.10 Consider the complex K shown in Figure 2.3.2. Its first and second barycentric subdivisions are displayed. Note that each of the simplices of sd K is of the form described in Lemma 2.3.9. Note also how rapidly the simplices of sd$^n K$ decrease in size as n increases. This is a general fact, that we next prove.

Theorem 2.3.11 *Given a finite complex K, given a metric for $|K|$, and given $\epsilon > 0$, there is a positive integer N such that each simplex of* sd$^N K$ *has diameter less than ϵ.*

Proof: Because K is finite, $|K|$ is a subspace of the Euclidean space $\mathbb{R}^J$ in which it lies. Because $|K|$ is compact, it is irrelevant which metric we use for $|K|$. [For if d_1 and d_2 are two metrics for $|K|$, then the identity map of $(|K|, d_1)$ to $(|K|, d_2)$ is uniformly continuous. Thus, given $\epsilon > 0$, there is a $\delta > 0$ such that any set with d_1 diameter less then δ has d_2 diameter less then ϵ.] Therefore we may as well use the metric of $\mathbb{R}^J$, which is

$$|x - y| = \max_{\alpha} |x_\alpha - y_\alpha| .$$

Step 1: We show that if $\sigma = v_0 v_1 \dots v_p$ is a simplex, then the diameter of σ equals the number $\ell = \max |v_i - v_j|$, which is the maximum distance between the vertices of σ. Because $v_i, v_j \in \sigma$, we know that $\operatorname{diam} \sigma \geq \ell$. Now we want to prove the reverse inequaity.

We first show that $|x - v_i| \leq \ell$ for every $x \in \sigma$. Consider the closed neighborhood of v_i of radius ℓ in $\mathbb{R}^J$ defined by the equations

$$C(v_i; \ell) = \{x : |x - v_i| \leq \ell\} .$$

One can check that this set is convex. Therefore, since it contains all the vertices of σ, it must contain σ. Then $|x - v_i| \leq \ell$ for $x \in \sigma$.

Next we show that $|x - z| \leq \ell$ for all $x, z \in \sigma$ so that $\operatorname{diam} \sigma \leq \ell$. Given x, consider the closed neighborhood $C(x; \ell)$. This set contains all the vertices of σ, by the result of the preceding paragraph. Being convex, it contains σ. Thus $|x - z| \leq \ell$ for all $x, z \in \sigma$.

Step 2: We show that if σ has dimension p, then for every $z \in \sigma$,

$$|\widehat{\sigma} - z| \leq \frac{p}{p+1} \operatorname{diam} \sigma \, .$$

For this purpose, we compute

$$\begin{aligned} |v_0 - \widehat{\sigma}| &= \left| v_0 - \sum_{i=0}^{p} \frac{1}{p+1} v_i \right| \\ &\leq \sum_{i=0}^{p} \frac{1}{p+1} |v_0 - v_i| \\ &\leq \frac{p}{p+1} \max |v_0 - v_i| \leq \frac{p}{p+1} \operatorname{diam} \sigma \, . \end{aligned}$$

A similar computation holds for $|v_j - \widehat{\sigma}|$. Therefore the closed neighborhood of $\widehat{\sigma}$ of radius $p/(p+1)\operatorname{diam} \sigma$ contains all vertices of σ. Being convex, it contains σ itself.

Step 3: We show that if σ is a p-simplex and τ is a simplex in the first barycentric subdivision of σ, then

$$\operatorname{diam} \tau \leq \frac{p}{p+1} \operatorname{diam} \sigma \, .$$

We proceed by induction. The result is trivial for $p = 0$. Suppose now that it is true in dimensions less than p. Let σ be a p-simplex. In view of Lemma 2.3.9 and of **Step 1** above, it suffices to show that if s and s' are faces of σ such that $s \succ s'$, then

$$|\widehat{s} - \widehat{s}'| \leq \frac{p}{p+1} \operatorname{diam} \sigma \, .$$

If s equals σ itself, then this inequality follows from **Step 2**. If instead s is a proper face of σ of dimension q, then

$$|\widehat{s} - \widehat{s}'| \leq \frac{q}{q+1} \operatorname{diam} s \leq \frac{p}{p+1} \operatorname{diam} \sigma \, .$$

The first inequality follows by the induction hypothesis, and the second from the fact that $f(x) = x/(x+1)$ is increasing for $x > 0$.

Step 4: Let K have dimension n: let d be the maximum diameter of a simplex of K. The maximum diameter of a simplex in the Nth barycentric subdivision of K is $(n/(n+1))^N d$. If N is sufficiently large then this number is less than ϵ. □

The next definition is used in Exercises 2 and 3.

Definition 2.3.12

(1) If $\mathcal{A}$ and $\mathcal{B}$ are collections of sets, we say that $\mathcal{B}$ *refines* $\mathcal{A}$ if for each $B \in \mathcal{B}$ there is an $A \in \mathcal{A}$ such that $B \subset A$.

(2) A space X is said to have *finite covering dimension* if there is an integer m satisfying the following condition: For every open covering $\mathcal{A}$ of X there is an open covering $\mathcal{B}$ of X that refines $\mathcal{A}$ and such that no point of X lies in more than $m+1$ elements of $\mathcal{B}$.

(3) The *covering dimension* of such a space X is the least integer m for which the condition in **(2)** holds.

Exercise 3 refers to the Cantor ternary set. For completeness we give a definition next. The resulting set is the usual "middle-thirds removed" set.

Definition 2.3.13 For each positive integer m, set

$$C_m = \bigcup_{k=0}^{(3^m-1)/2} \left[\frac{2k}{3^m}, \frac{2k+1}{3^m} \right].$$

The *Cantor ternary set*, C, is given by

$$C = \bigcap_{m=1}^{\infty} C_m .$$

EXERCISES

1. Let K be a complex and let $x_0 \in |K|$.

(a) Show that there is a subdivision of K whose vertex set contains x_0.

*__(b)__ Show that there is a subdivision of K whose vertex set consists of x_0 and the vertices of K.

2. **(a)** Show that a discrete set has covering dimension 0.

(b) Show that $[0,1]$ has covering dimension 1.

(c) Show that if X has covering dimension m, then any closed subspace A of X has covering dimension at most m.

(d) Show that if K is a finite complex of dimension m, then the covering dimension of $|K|$ exists and is at most m. [We will see later that it is precisely m.]

3. Calculate the covering dimension of the Cantor ternary set.

2.4 The Simplicial Approximation Theorem

We now show that if $h : |K| \to |L|$ is a continuous map, then there is a subdivision K' of K such that h has a simplicial approximation $f : K' \to L$. The proof when K is finite follows easily from the results of the preceding section; barycentric subdivision will suffice. The general case requires a slightly more sophisticated technique of subdivision, called "generalized barycentric subdivision," that we will describe below in Definition 2.4.5.

We will have need of the following lemma:

Lemma 2.4.1 (Lebesgue's covering lemma) *For every open cover, $\mathcal{A}$, of a compact metric space, there exists a positive real number λ such that every subset of diameter less than λ is contained in some element of $\mathcal{A}$.*

The positive number λ is called a *Lebesgue number* for $\mathcal{A}$.

Proof: If there were no such λ, then one could choose a sequence C_n of sets, where C_n has diameter less than $1/n$ but does not lie in any element of $\mathcal{A}$. Choose $x_n \in C_n$; by compactness, some subsequence x_{n_i} converges to some element x. Now $x \in A$ for some $A \in \mathcal{A}$. Because A is open, it contains C_{n_i} for i sufficiently large, contrary to our construction. □

Theorem 2.4.2 (The finite simplicial approximation theorem) *Let K and L be complexes, and assume that K is finite. Given a continuous map $h : |K| \to |L|$, there is an N such that h has a simplicial approximation $f : \mathrm{sd}^N K \to L$.*

Proof: Cover $|K|$ by the open sets $h^{-1}(\mathrm{St}\ w)$ as w ranges over the vertices of L. Now given this open covering $\mathcal{A}$ of the compact metric space $|K|$, we let λ be a Lebesgue number for $\mathcal{A}$.

Choose N so that each simplex in $\mathrm{sd}^N K$ has diameter less than $\lambda/2$. Then each star of a vertex in $\mathrm{sd}^N k$ has diameter less than λ, so it lies in one of the sets $h^{-1}(\mathrm{St}\ w)$. Then $h : |K| \to |L|$ satisfies the star condition relative to $\mathrm{sd}^N K$ and L, and the desired simplicial approximation exists. □

As a preliminary step toward our generalized version of barycentric subdivision, we show how to subdivide a complex K in a way that leaves a given subcomplex K_0 unchanged.

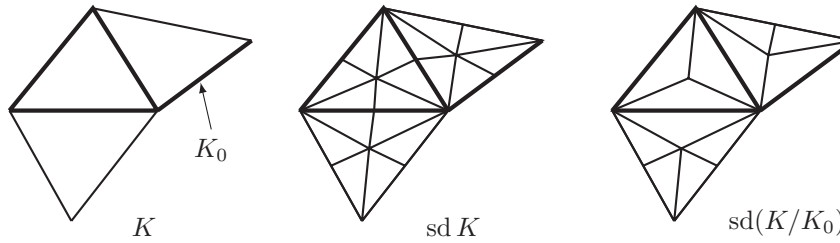


Figure 2.4.1

Definition 2.4.3 Let K be a complex and let K_0 be a subcomplex. We define a sequence of subdivisions of the skeleton of K as follows: Let $J_0 = K^{(0)}$. In general, suppose that J_p is a subdivision of the p-skeleton of K and each simplex of K_0 of dimension at most p belongs to J_p. Define J_{p+1} to be the union of the complex J_p, all $(p+1)$-simplices σ belonging to K_0, and the cones $\widehat{\sigma} * J_\sigma$, as σ ranges over all $(p+1)$-simplices of K not in K_0. [Here J_σ is the subcomplex of J_p whose polytope is Bd σ.] The union of the complexes J_p is a subdivision of K, denoted sd(K/K_0) and called the *first barycentric subdivision of K, holding K_0 fixed.*

As with barycentric subdivisions, this process can now be repeated. The complex sd(sd$(K/K_0)/K_0$) will be called the *second barycentric subdivision of K, holding K_0 fixed,* and will be denoted by sd$^2(K/K_0)$ and so forth.

EXAMPLE 2.4.4 Figure 2.4.1 illustrates the first barycentric subdivision sd K of a complex K and the first barycentric subdivision sd(K/K_0) holding a complex K_0 fixed.

To prove the general simplicial approximation theorem, iterated barycentric subdivisions will not suffice, just because the Lebesgue number argument used in the proof of Theorem 2.4.2 requires the space $|K|$ to be compact. For a general complex K, the number λ that measures how finely a simplex must be subdivided may vary from one simplex to another. Thus we must generalize our notion of barycentric subdivision to allow for this possibility.

Definition 2.4.5 Let K be a complex. Let N be a function assigning to each positive-dimensional simplex σ of K a non-negative integer $N(\sigma)$. We construct a subdivision of K as follows. Let $L_0 = K^{(0)}$. In general suppose that L_p is a subdivision of the p-skeleton of K. For each $(p+1)$-simplex σ of K, let L_σ be the subcomplex of L_p whose polytope is Bd σ. Form the cone $\widehat{\sigma} * L_\sigma$. Then subdivide this cone barycentrically $N(\sigma)$ times, holding L_σ fixed. Define L_{p+1} to be the union of L_p and the complexes

$$\text{sd}^{N(\sigma)}(\widehat{\sigma} * L_\sigma / L_\sigma)$$

as σ ranges over all $(p+1)$-simplices of K. Then L_{p+1} is a subdivision of the $(p+1)$-skeleton of K. The union of the complexes L_p is a subdivision of K. It is called the *generalized barycentric subdivision of* K corresponding to the function $N(\sigma)$.

The remainder of this section will show that this generalized barycentric subdivision is adequate to prove the general simplicial approximation theorem.

Keep Figure 2.4.1 in mind as we compare the complex sd (K/K_0) with the complex sd K in general.

Lemma 2.4.6 *Let K_0 be a subcomplex of K.*

(a) *If τ is a simplex of* sd(K/K_0)*, then τ is of the form*

$$\tau = \widehat{\sigma}_1\widehat{\sigma}_2\cdots\widehat{\sigma}_q v_0 v_1\cdots v_p\,,$$

where $s = v_0v_1\cdots v_p$ is a simplex of K_0 and $\widehat{\sigma}_1, \widehat{\sigma}_2, \cdots, \widehat{\sigma}_q$ are simplices of K not in K_0 and $\sigma_1 \succ \sigma_2 \succ \cdots \succ \sigma_q \succ s$.

(b) *Either $v_0v_1\cdots v_p$ or $\widehat{\sigma}_1\widehat{\sigma}_2\cdots\widehat{\sigma}_q$ may be missing from this expression. The simplex τ is disjoint from $|K_0|$ if and only if $v_0v_1\cdots v_p$ is missing. In this case τ is a simplex of* sd K*.*

Proof: (a) The result is true if τ is in J_0. In general, let τ be a simplex of J_{p+1}, not in J_p. Then either τ belongs to K_0, in which case τ is of the form $v_0v_1\cdots v_r$, or τ belongs to one of the cones $\widehat{\sigma} * J_\sigma$. Now each simplex of J_σ has the form $\widehat{\sigma}_1\widehat{\sigma}_2\cdot\widehat{\sigma}_q v_0v_2\cdots v_r$ by the induction hypothesis, where $\sigma \succ \sigma_1$. Then τ has the form $\widehat{\sigma}\widehat{\sigma}_1\widehat{\sigma}_2\cdots\widehat{\sigma}_q v_0v_2\cdots v_r$ as desired.

(b) Let $\tau = \widehat{\sigma}_1\widehat{\sigma}_2\cdots\widehat{\sigma}_q v_0v_2\cdots v_r$. If $v_0v_1\cdots v_p$ is not missing from this expression, then τ intersects $|K_0|$ in $v_0v_1\cdots v_p$ at least. Conversely, if the set $\tau \cap |K_0|$ is nonempty, then it contains a face of τ and hence a vertex of τ. Since none of the points $\widehat{\sigma}_1, \widehat{\sigma}_2, \ldots, \widehat{\sigma}_q$ is in $|K_0|$, the term $v_0v_1\cdots v_p$ cannot be missing. □

To prove the general simplicial approximation theorem, we need to show that, given an arbitrary continuous map $h : |K| \to |L|$, there is a subdivision K' of K such that h satisfies the star condition relative to K' and L. This is equivalent to the statement that if $\mathcal{A}$ is the open covering of K defined by

$$\mathcal{A} = \{h^{-1}(\mathrm{St}(w, L)) : w \text{ is a vertex of } L\}\,,$$

then there is a subdivision of K' of K such that the collection of open stars

$$\mathcal{B} = \{\mathrm{St}(v, K') : v \text{ is a vertex of } K'\}$$

refines $\mathcal{A}$. [Recall that a collection $\mathcal{B}$ *refines* a collection $\mathcal{A}$ if, for each element $B \in \mathcal{B}$, there is an element $A \in \mathcal{A}$ that contains B.]

To make the proof work, we actually have to prove something slightly stronger than this. We shall construct a division K' of K fine enough that the collection $\{\overline{\mathrm{St}}(v, K')\}$ of *closed stars* in K' refines the collection $\mathcal{A}$.

The following lemma gives the crux of the argument. It will enable us to carry out the inductive step of the proof.

Lemma 2.4.7 *Let $K = p * B$ be a cone over the finite complex B. Let $\mathcal{A}$ be an open covering of $|K|$. Suppose that there is a function assigning to each vertex v of the complex B an element A_v of $\mathcal{A}$ such that*

$$\overline{\mathrm{St}}(v, B) \subset A_v\,.$$

Then there is an N such that the collection of closed stars of the subdivision $\mathrm{sd}^N(K/B)$ refines $\mathcal{A}$, and furthermore such that for each vertex v of B,

$$\overline{\mathrm{St}}(v, \mathrm{sd}^N(K/B)) \subset A_v\,.$$

Proof: We assume that $|B|$ lies in $\mathbb{R}^m \times \{0\}$ for some m. Let $n = \dim K$. Set $p = (0, 0, \ldots, 0, 1) \in \mathbb{R}^m \times \mathbb{R}$.

Step 1: In general, as N increases, the maximum diameter of the simplices of $\mathrm{sd}^N(K/B)$ does not go to 0. For if σ has a positive-dimensional face in B, that face never gets subdivided. However, it is true that, as N increases, the simplices that intersect the plane $\mathbb{R}^m \times \{0\}$ lie closer and closer to this plane. More generally, we show that if K' is any subdivision of K that keeps B fixed, and if the simplices of K' that intersect $\mathbb{R}^m \times \{0\}$ lie in the strip $\mathbb{R}^m \times [0, \epsilon]$, then any simplex τ of $\mathrm{sd}(K'/B)$ that intersects $\mathbb{R}^m \times \{0\}$ lies in the strip $\mathbb{R}^m \times [0, n\epsilon/(n+1)]$.

The simplex τ is of the form $\widehat{\sigma}_1 \widehat{\sigma}_2 \cdots \widehat{\sigma}_q v_0 v_1 \cdots v_p$ as in Lemma 2.5.5. Assuming that τ intersects $\mathbb{R}^m \times \{0\}$ but does not lie in it, neither $\widehat{\sigma}_1 \widehat{\sigma}_2 \cdots \widehat{\sigma}_q$ nor $v_0 v_1 \cdots v_p$ is missing from this expression. Each vertex v_i lies in $\mathbb{R}^m \times \{0\}$.

Consider the vertex $\widehat{\sigma}_j$ of τ. The simplex σ_j of K' intersects $\mathbb{R}^m \times \{0\}$ because σ_j has $v_0 v_1 \cdots v_p$ as a face. Therefore $\sigma_j \in \mathbb{R}^m \times [0, \epsilon]$. Let $w_0, w_1, \ldots, w_k$ be the vertices of σ_j, and let $\pi : \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}$ be the projection on the last coordinate. Then $|\pi(w_j)| \le \epsilon$ for $i = 0, \ldots, k$. And $\pi(w_i) = 0$ for at least one index i. We compute

$$\pi(\widehat{\sigma}_j) = \sum_{i=0}^{k} \frac{1}{k+1} \pi(w_i) \le \left(\frac{k}{k+1}\right) \epsilon\,.$$

Thus each vertex of τ lies in the set $\mathbb{R}^m \times [0, (n/(n+1))\epsilon]$. Because this set is convex, all of τ lies in it.

Step 2: For convenience, let K_N denote the complex $\mathrm{sd}^N(K/B)$. We show that there is an integer N_0 such that if $N \ge N_0$ then, for each vertex v of B,

$$\overline{\mathrm{St}}(v, K_N) \subset A_v\,. \qquad (*)$$

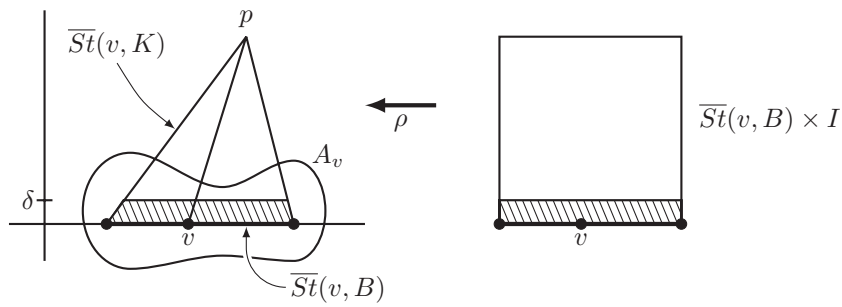


Figure 2.4.2

This is part of what we want to prove.

We are given that $\overline{\mathrm{St}}(v,B) \subset A_v$. We claim that we can choose $\delta > 0$ such that

$$\overline{\mathrm{St}}(v,K) \cap (\mathbb{R}^m \times [0,\delta]) \subset A_v$$

for each $v \in B$. See Figure 2.4.2.

To prove this assertion, consider the continuous map $\rho : |B| \times I \to |K|$ defined by $\rho(x,t) = (1-t)x + tp$. The map ρ carries $\overline{\mathrm{St}}(v,B) \times I$ onto $\overline{\mathrm{St}}(v,K)$ because K is a cone over B. Furthermore, ρ preserves the last coordinate since

$$\begin{aligned} \pi\rho(x,t) &= (1-t)\pi(x) + t\pi(p) \\ &= (1-t)\cdot 0 + t\cdot 1 = \pi(x,t)\,. \end{aligned}$$

Now $\overline{\mathrm{St}}(v,B)$ is a compact set. The "tube lemma" of point-set topology [see for instance Section 26 of [Mun00]] enables us to choose $\delta > 0$ so that

$$\overline{\mathrm{St}}(v,B) \times [0,\delta] \subset \rho^{-1}(A_v)\,.$$

[More directly, one can cover $\overline{\mathrm{St}}(v,B) \times \{0\}$ with finitely many sets of the form $U_i \times [0,\delta_i]$, each lying in $\rho^{-1}(A_v)$. Then one can choose $\delta = \min_i \delta_i$.] It follows that the set

$$\rho(\overline{\mathrm{St}}(v,b) \times [0,\delta]) = \overline{\mathrm{St}}(v,K) \cap (\mathbb{R}^m \times [0,\delta])$$

lies in A_v as desired.

Now we can choose N_0. Applying **Step 1**, we choose N_0 so that when $N \geq N_0$, each simplex of K_N that intersects $\mathbb{R}^m \times \{0\}$ lies in $\mathbb{R}^m \times [0,\delta]$. Then, if v is a vertex of B, the set $\overline{\mathrm{St}}(v,K_N)$ lies in $\mathbb{R}^m \times [0,\delta]$. Since this set also lies in $\overline{\mathrm{St}}(v,K)$, it lies in A_v as desired.

Step 3: The integer N_0 is now fixed. Consider the complex K_{N_0+1}. Let Q be the union of all simplices of K_{N_0+1} that do not intersect B. See Figure 2.4.3.

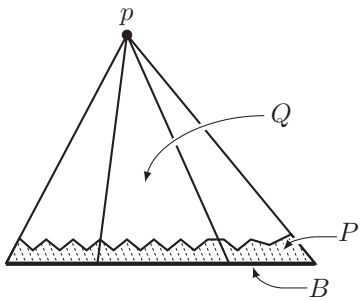


Figure 2.4.3

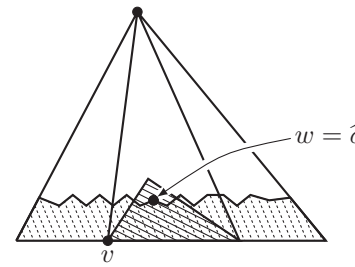


Figure 2.4.4

We prove the following: If $N \geq N_0 + 1$, then for each vertex w of K_N lying in P but not in $|B|$, there is an element A of $\mathcal{A}$ such that

$$\overline{\mathrm{St}}(w, K_N) \subset A\,. \qquad (**)$$

We prove $(**)$ first in the case $N = N_0 + 1$. Now P is the polytope of a subcomplex of K_{N_0+1} by definition. If w is a vertex of K_{N_0+1} lying in P but not in B, then $w = \widehat{\sigma}$ for some simplex σ of K_{N_0} that intersects B but does not lie in B, by Lemma 2.4.6. See Figure 2.4.4. Let v be a vertex of σ lying in B. Because w lies in $\operatorname{Int}\sigma$, we have

$$\mathrm{St}(w, K_{N_0+1}) \subset \mathrm{St}(v, K_{N_0})$$

by Lemma 2.3.2. Then

$$\overline{\mathrm{St}}(w, K_{N_0+1}) \subset \overline{\mathrm{St}}(v, K_{N_0}) \subset A_v$$

by $(*)$ of **Step 2**.

Now we prove $(**)$ in the case $N > N_0 + 1$. If w' is a vertex of K_N lying in P, then $w' \in \mathrm{St}(w, K_{N_0+1})$ for some vertex w of K_{N_0+1} lying in P. Hence $\mathrm{St}(w', K_N) \subset \mathrm{St}(w, K_{N_0+1})$ by Lemma 2.3.2. Therefore $(**)$ follows from the result of the preceding paragraph.

Step 4: We now complete the proof. Let λ be a Lebesgue number for the open covering $\mathcal{A}$ of $|K|$. Consider the space Q. It is the polytope of a subcomplex J of K_{N_0+1}. In forming the subdivision K_{N_0+2}, each simplex of J is subdivided barycentrically—by the preceding lemma. Thus K_{N_0+2} has $\mathrm{sd}\, J$ as a subcomplex. Repeating the argument, we see that in general K_{N_0+1+M} has $\mathrm{sd}^M\, J$ as a subcomplex.

Choose M large enough that each simplex of $\mathrm{sd}^M\, J$ has diameter less than $\lambda/2$. Then if $N \geq N_0 + 1 + M$ and if w is a vertex of K_N not in P, we show that there is an element A of $\mathcal{A}$ such that

$$\overline{\mathrm{St}}(w, K_N) \subset A\,. \qquad (***)$$

For since w is not in P, each simplex of K_N having w as a vertex must lie in Q and hence must be a simplex pf $\mathrm{sd}^M\ J$. Therefore $\overline{\mathrm{St}}(w, K_N)$ has diameter less than λ, so it lies in an element of $\mathcal{A}$.

The combination of $(*)$, $(**)$, and $(***)$ proves the lemma. □

Theorem 2.4.8 *Let K be a complex. Let $\mathcal{A}$ be an open covering of $|K|$. There exists a generalized barycentric subdivision K' of K such that the collection of closed stars $\{\overline{\mathrm{St}}(w, K')\}$, for w a vertex of K', refines $\mathcal{A}$.*

Proof: We proceed step by step. Initially we let $L_0 = K^{(0)}$ and for each point v of $K^{(0)}$ we let A_v denote an element of $\mathcal{A}$ that contains v.

In general we assume that a subdivision L_p of $K^{(p)}$ is given and that a function f_p is given assigning to each vertex v of L_p an element A_v of $\mathcal{A}$ such that

$$\overline{\mathrm{St}}(v, L_p) \subset A_v\,.$$

We extend L_p to a subdivision L_{p+1} of the $(p+1)$-skeleton of K and we extend f_p to a function f_{p+1} as we now describe.

For each $(p+1)$-simplex σ of K, the space Bd σ is the polytope of a subcomplex L_σ of L_p. Consider the cone $\widehat{\sigma} * L_\sigma$. By the preceding lemma, there is an integer $N(\sigma)$ such that if we set

$$C(\sigma) = \mathrm{sd}^{N(\sigma)}\ (\widehat{\sigma} * L_\sigma / L_\sigma)\,,$$

then the following two conditions hold:

For each vertex v of $C(\sigma)$ belonging to L_σ,

$$\overline{\mathrm{St}}(v, C(\sigma)) \subset A_v\,.$$

For each vertex v of $C(\sigma)$ not in L_σ there exists an element A of $\mathcal{A}$ such that

$$\overline{\mathrm{St}}(w, C(\sigma)) \subset A\,.$$

We define L_{p+1} to be the union of L_p and the complexes $C(\sigma)$ as σ ranges over the $(p+1)$-simplices of K.

If v is a vertex of L_p, then $\overline{\mathrm{St}}(v, L_{p+1})$ is the union of the sets $\overline{\mathrm{St}}(v, L_p)$ and $\overline{\mathrm{St}}(v, C(\sigma))$ as σ ranges over the $(p+1)$-simplices of K containg v. Each of these sets lies in A_v by construction.

On the other hand, if w is a vertex of L_{p+1} not in L_p, then w lies interior to some $(p+1)$-simplex σ of K so that

$$\overline{\mathrm{St}}(w, L_{p+1}) = \overline{\mathrm{St}}(w, C(\sigma))\,.$$

The latter set is contained in some element A of $\mathcal{A}$. We define $f_{p+1}(w)$ to be such an element A_w of $\mathcal{A}$. The inductive step is therefore complete.

The theorem follows. The complex K' is defined to be the union of the complexes L_p. The function $f(v) = A_v$ from the vertices of K' to $\mathcal{A}$ is defined to be the union of the functions f_p. The function f satisfies the requirements of the theorem. For let v be a vertex of K'. Then v is a vertex of L_p for some p and

$$\overline{\mathrm{St}}(v, L_{p+k}) \subset f_{p+k}(v) = f(v) = A_v$$

for all $k \geq 0$. From this it follows that $\overline{\mathrm{St}}(v, K') \subset A_v$. □

Theorem 2.4.9 (The general simplicial approximation theorem) *Let K and L be complexes. Let $h : |K| \to |L|$ be a continuous map. There exists a subdivision K' of K such that h has a simplicial approximation $f : K' \to L$.*

Proof: Let $\mathcal{A}$ be the covering of $|K|$ by the open sets $h^{-1}(\mathrm{St}(w, L))$ as w ranges over the vertices of L. Choose a subdivision K' of K whose closed stars refine $\mathcal{A}$. Then h satisfies the star condition relative to K' and L. □

EXERCISES

1. **(a)** Using Theorem 2.4.2 show that if K and L are finite and $\dim K = m$, then any continuous map $h : |K| \to |L|$ is homotopic to a map carrying K into $L^{(m)}$, where $L^{(m)}$ is the m-skeleton of L. [**Hint:** See Exercise 2 of Section 2.2.] We shall consider the non-finite case later.

(b) Show that if $h : \mathbb{S}^m \to \mathbb{S}^n$ and $m < n$, then h is homotopic to a constant map. [**Hint:** Any map $f : X \to \mathbb{S}^n \setminus \{p\}$ is homotopic to a constant.]

2. Let $h : |K| \to |L|$ be a continuous map. Given $\epsilon > 0$, show that there are subdivisions K' and L' of K and L respectively and a simplicial map $f : K' \to L'$ such that $|f(x) - h(x)| < \epsilon$ for all $x \in |K|$.

3. Show that if K is a complex of dimension m, then $|K|$ has covering dimension at most m. [See the exercises of Section 2.3.]

4. Provide examples to show that the result of Exercise 3 is sharp.

2.5 The Algebra of Subdivision

Now we explore some of the algebraic consequences of subdivision, determining what subdivision does to the homology groups. We prove that if K is any

complex and K' is any subdivision of K, then there is a uniquely defined chain map $\lambda : C_p(K) \to C_p(K')$ called the *subdivision operator* that induces an isomorphism of homology groups. Furthermore, if $g : K' \to K$ is any simplicial approximation to the identity of $|K|$, then λ and $g_\#$ are chain-homotopy inverse to each other, so $g_\#$ is also an isomorphism.

Lemma 2.5.1 *Let K' be a subdivision of K. Then the identity map $i : |K| \to |K|$ has a simplicial approximation*

$$g : K' \to K\,.$$

Let τ be a simplex of K' and let σ be a simplex of K. If $\tau \subset \sigma$ then $g(\tau) \subset \sigma$.

Proof: By Lemma 2.3.2, the map i has a simplicial approximation g. Given $\tau \subset \sigma$, let w be a vertex of τ. Then w lies interior to σ or to a face of σ. Then g maps w to a vertex of σ by Lemma 2.2.2. □

Definition 2.5.2 Let K' be a subdivision of K. If σ is a simplex of K, let $K(\sigma)$ denote the subcomplex of K consisting of σ and its faces. Let $K'(\sigma)$ denote the subcomplex of K' whose polytope is σ.

Theorem 2.5.3 (The algebraic subdivision theorem) *Let K' be a subdivision of K. There is a unique augmentation-preserving chain map*

$$\lambda : \mathcal{C}(K) \to \mathcal{C}(K')$$

such that $\lambda(\sigma)$ is carried by $K'(\sigma)$ for each σ. If $g : K' \to K$ is a simplicial approximation to the identity, then λ and $g_\#$ are chain-homotopy inverses. Hence λ_* and g_* are isomorphisms.

We call λ the *subdivision operator*.

Proof: Step 1: We show first that if K' satisfies the condition that for each $\sigma \in K$ the induced subdivision $K'(\sigma)$ of σ is acyclic: We shall use Theorem 1.14.10, the geometric version of the acyclic carrier theorem. We define acyclic carriers as follows:

$$\Psi \circlearrowright K \underset{\Theta}{\overset{\Lambda}{\rightleftarrows}} K' \circlearrowleft \Phi$$

The carriers Ψ and Λ are easy to define. We set

$$\Psi(\sigma) = K(\sigma)\,,$$

$$\Lambda(\sigma) = K'(\sigma)$$

for each $\sigma \in K$. The complex $K(\sigma)$ is acyclic because it consists of a simplex and its faces. The complex $K'(\sigma)$ is acyclic by hypothesis. The inclusion condition for an acyclic carrier is immediate. If $\sigma \succ s$ then $K(s) \subset K(\sigma)$ and $K'(s) \subset K'(\sigma)$.

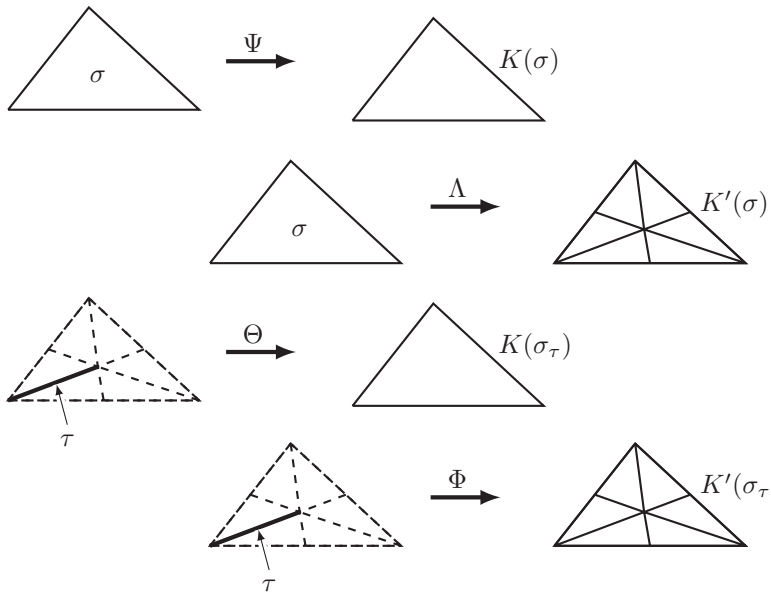


Figure 2.5.1

To define Θ and Φ we proceed as follows. For each simplex $\tau \in K'$, let σ_τ be the simplex of K of least dimension such that $\tau \subset \sigma_\tau$. Then, if t is a face of τ, we have $\sigma_t \subset \sigma_\tau$. For since σ_t and σ_τ both contain t, their intersection also contains t. Since σ_t has minimal dimension, it must equal this intersection. We define

$$\Theta(\tau) = K(\sigma_\tau)\,,$$
$$\Phi(\tau) = K'(\sigma_\tau)\,.$$

Both complexes are acyclic. The inclusion condition follows from the fact that if $\tau \succ t$ then $\sigma_t \subset \sigma_\tau$. See Figure 2.5.1, which illustrates these carriers in the case that $K' = \text{sd } K$.

By Theorem 1.14.10, there exist chain maps λ and θ with

$$C_p(K) \underset{\theta}{\overset{\lambda}{\rightleftarrows}} C_p(K')$$

preserving augmentation and carried by Λ and Θ respectively.

Now the identity map $C_p(K) \to C_p(K)$ is carried by Ψ (trivially). We show that $\theta \circ \lambda$ is also carried by Ψ. It follows then that $\theta \circ \lambda$ is chain homotopic to the identity. If σ is a simplex of K, then $\lambda(\sigma)$ is a chain of $K'(\sigma)$. Now each simplex τ in the subdivision $K'(\sigma)$ of σ is contained in σ, hence σ_τ equals

either σ or a face of σ. In any case, if τ appears in the chain $\lambda(\sigma)$, then $\theta(\tau)$ is carried by $K(\sigma_\tau) \subset K(\sigma)$. Thus $\theta(\lambda(\sigma))$ is a chain of $K(\sigma)$, so $\theta \circ \lambda$ is carried by Ψ.

The identity map $C_p(K') \to C_p(K')$ is carried by Φ. For τ is contained in σ by definition, so that τ is a simplex of $K'(\sigma_\tau)$. We show that $\lambda \circ \theta$ is also carried by Φ, hence $\lambda \circ \theta$ is chain homotopic to the identity. If $\tau \in K'$, then $\theta(\tau)$ is carried by the complex $K(\sigma_\tau)$ consisting of σ_τ and its faces, so that it equals a sum of oriented faces of σ_τ. Now if s is any face of σ_τ, then $\lambda(s)$ is carried by $K'(s) \subset K'(\sigma_\tau)$. It follows that $\lambda\theta(\tau)$ is carried by $K'(\sigma_\tau) = \Phi(\tau)$, as desired.

The preceding discussion is independent of the choice of the particular chain maps θ and λ. One choice for θ is the chain map $g_\#$. It follows from the preceding lemma that $g_\#$ is carried by θ. Therefore $g_\#$ and λ are chain-homotopy inverses.

We now show that λ is unique. Suppose that λ' is another augmentation-preserving chain map carried by λ. Then, by Theorem 1.14.10, there is a chain homotopy D, also carried by Λ, between λ and λ'. Note that if σ is a p-simplex, then $\Lambda(\sigma) = K'(\sigma)$ is a complex of dimension p. Since $D(\sigma)$ is a $(p+1)$-chain carried by $K'(\sigma)$, it must be zero. Thus D is identically zero. The equation $\partial D + D\partial = \lambda - \lambda'$ now implies that $\lambda = \lambda'$.

Step 2: We show the theorem holds if $K' = \text{sd } K$: For in this case, given $\sigma \in K$, the complex $K'(\sigma)$ is a cone. In fact $K'(\sigma) = \widehat{\sigma} * J$, where J is the first barycenetric subdivision of Bd σ. And we know from Theorem 1.9.6 that cones are acyclic. [*This is the place where barycentric subdivisions are essential for the proof.*]

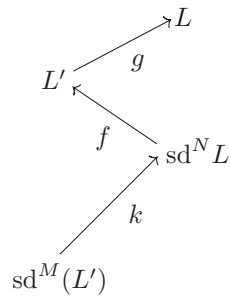
Step 3: We show the theorem is true if $K' = \text{sd}^N K$: In view of **Step 1**, it suffices to prove that for any simplex σ of K, the complex $\text{sd}^N K(\sigma)$ is acyclic. This follows from **Step 2**, which implies that for any complex L,

$$\widetilde{H}_i(L) \cong \widetilde{H}_i(\text{sd } L) \cong \widetilde{H}_i(\text{sd}^2 L) \cong \cdots .$$

In particular, if L is acyclic, then so is $\text{sd}^N L$.

Step 4: We show the theorem is true in general: In view of **Step 1**, it suffices to show that if $\sigma \in K$ and if K' is any subdivision of K, then $K'(\sigma)$ is acyclic.

Let $L = K(\sigma)$ and $L' = K'(\sigma)$. Then L is acyclic, and we wish to prove that L' is acyclic. Let $g : L' \to L$ be a simplicial approximation to the identity. Choose N so that the identity map of $|L|$ to itself has a simplicial approximation $f : \text{sd}^N\ L \to L'$. For this we need the (finite) simplicial approximation theorem. Proceeding similarly, choose M so that the identity has a simplicial approximation $k : \text{sd}^M(L') \to \text{sd}^N L$.



Now we note that $g \circ f$ is a simplicial approximation to the identity, so that by **Step 3**, $(g \circ f)_* = g_* \circ f_*$ is an isomorphism. For the same reason, $(f \circ k)_* = f_* \circ k_*$ is an isomorphism. The first fact implies that f_* is injective and the second implies that f_* is surjective. Thus f_* is an isomorphism. Finally, because $g_* \circ f_*$ and f_* are isomorphisms, then so is g_*. Thus L' is acyclic. □

Definition 2.5.4 In the special case when K' is the first barycentric subdivision of K, we denote the subdivision operator λ by

$$\mathrm{sd} : C_p(K) \to C_p(\mathrm{sd}\ K)\,.$$

We call it the *barycentric subdivision operator*. [Here we abuse notation, letting sd denote both the algebraic and the geometric subdivision operators.]

There is an inductive formula for the operator sd. It is the following.

$$\begin{aligned} \mathrm{sd}(v) &= v\,, \\ \mathrm{sd}(\sigma) &= [\widehat{\sigma}, \mathrm{sd}(\partial\sigma)]\,. \end{aligned}$$

Here the bracket notation has the same meaning that we gave it in Section 1.9. It is an exercise for the reader to check this last formula.

Application to Relative Homology

We can generalize the preceding result to relative homology with no trouble.

Theorem 2.5.5 *Let K_0 be a subcomplex of K. Given the subdivision K' of K, let K_0' denote the induced subdivision of K_0. The subdivision operator λ induces a chain map*

$$\lambda : C_p(K, K_0) \to C_p(K', K_0')\,.$$

If $g : (K', K_0') \to (K, K_0)$ is any simplicial approximation to the identity, then λ and $g_\#$ are chain-homotopy inverse to each other.

Proof: We check that each of the acyclic carriers defined in **Step 1** of the preceding proof preserves the subcomplexes involved. Certainly if $\sigma \in K_0$ then $\Psi(\sigma) = K(\sigma)$ and $\Lambda(\sigma) = K'(\sigma)$ are subcomplexes of K_0 and K_0', respectively. On the other hand, if $\tau \in K_0'$, then τ is contained in some simplex σ of K_0. It follows that the simplex σ_τ of K of least dimension containing τ must belong to K_0. Thus if $\tau \in K_0'$, then $\Theta(\tau) = K(\sigma_\tau)$ and $\Phi(\tau) = K'(\sigma_\tau)$ are subcomplexes of K_0 and K_0' respectively.

It follows that the chain maps λ and θ carried by Λ and Θ, respectively, induce chain maps on the relative level, and that the chain homotopies of $\theta \circ \lambda$ and $\lambda \circ \theta$ to the respecitve identity maps induce chain homotopies on the relative level as well. Since one choice for θ is the map $g_\#$, it follows that $g_\#$ and λ are chain-homotopy inverses on the relative level. □

EXERCISES

1. Let K' be a subdivision of K. Let $\lambda : C_p(K) \to C_p(K')$ be the subdivision operator. Let $g : K' \to K$ be a simplicial approximation to the identity. Show that $g_\# \circ \lambda$ equals the identity map of $C_p(K)$.

2. Let K and K' be the complexes shown in Figure 2.5.2. Their common underlying space is a square region.

(a) Find a formula for the subdivision operator $\lambda : C_p(K) \to C_p(K')$.

(b) Find two different simplicial approximations $g, g' : K' \to K$ to the identity. Conclude that the chain equivalence θ constructed in the proof of the algebraic subdivision theorem is not unique.

(c) Check that $\lambda \circ g_\#$ does not equal the identity on the chains of K'.

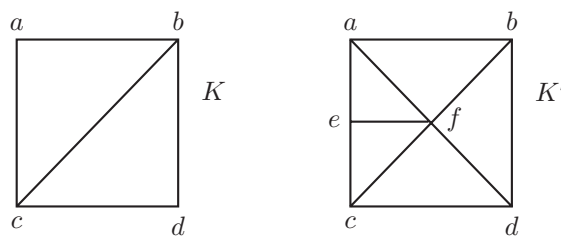


Figure 2.5.2

3. **(a)** Show that the inductive formula for the barycentric subdivision operator sd defines an augmentation-preserving chain map that is carried by Λ.

(b) Compute sd σ for the case of a 1-simplex and the case of a 2-simplex.

4. Let K be a pentagonal region square [the 1-dimensional figure] and let K' be the same pentagonal region split up by two diagonals. Answer 2(a), 2(b), and 2(c) for these two complexes.

2.6 Topological Invariance of the Homology Groups

In this section we achieve the main goal of this chapter: to prove the topological invariance of the simplicial homology groups.

Definition 2.6.1 Let K and L be simplicial complexes. Let $h : |K| \to |L|$ be a continuous map. Choose a subdivision K' of K such that h has a simplicial approximation $f : K' \to L$. Let $\lambda : \mathcal{C}(K) \to \mathcal{C}(K')$ be the subdivision operator. We define the *homomorphism induced by* h,

$$h_* : H_p(K) \to H_p(L)\,,$$

by the equation $h_* = f_* \circ \lambda_*$.

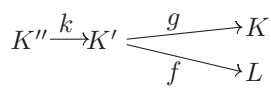
Note that, once K' has been chosen, the homomorphism h_* is independent of the particular choice of the simplicial approximation $f : K' \to L$ to h. For any two such simplicial approximations are contiguous.

Observe also that if $g : K' \to K$ is a simplicial approximation to the identity map $i_{|K|}$ of $|K|$ with itself, then λ_* and g_* are inverse to each other. Therefore one could just as well define

$$h_* = f_* \circ (g_*)^{-1}\,.$$

We use this fact to show that h_* is independent of the choice of the subdivision K'. Suppose that K'' is another subdivision of K such that h has a simplicial approximation mapping of K'' into L. We show that if h_* is defined using the subdivision K'', the result is the same as if one uses the subdivision K'.

The proof is especially easy in the case where the identity map of $|K|$ has a simplicial approximation $k : K'' \to K'$ as in the following diagram.



Then, since $g \circ k$ and $f \circ k$ are simplicial approximations to the identity and to h, respectively, the homomorphism h_*, defined using the subdivision K'', equals the composite

$$(f \circ k)_* \circ (g \circ k)_*^{-1} = (f_* \circ k_*) \circ (g_* \circ k_*)^{-1} = f_* \circ g_*^{-1} .$$

The result is thus the same as when h_* is defined using the subdivision K'.

The general case is proved by choosing a subdivision K''' of K such that the identity map has the simplicial approximations

$$k_1 : K''' \to K' \qquad \text{and} \qquad k_2 : K''' \to K'' .$$

Then using K''' to define h_* gives the same result as using K' or K''.

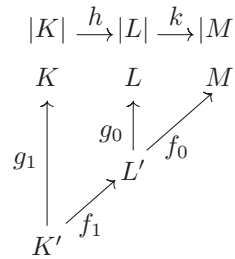
We should remark that the homomorphism h_* depends not only on the spaces $X = |K|$ and $Y = |L|$ and the continuous map $h : X \to Y$ but also on the particular complexes K and L whose polytopes are X and Y respectively. If M and N are other complexes whose polytopes are X and Y, respectively, then h also induces a homomorphism

$$h_* : H_p(M) \to H_p(N) .$$

One should really use a notation like $(h_{K,L})_*$ or $(h_{M,N})_*$ to distinguish between these homomorphisms. We shall abuse terminology and use the simple notation h_*, relying on the context to make the meaning clear. We give further discussion of this point later, when we define the homology of a triangulable space (Section 3.6).

Theorem 2.6.2 (Functorial properties) *The identity map $i : |K| \to |K|$ induces the identity homomorphism $i_* : H_p(K) \to H_p(K)$. If $h : |K| \to |L|$ and $k : |L| \to |M|$ are continuous maps, then $(k \circ h)_* = k_* \circ h_*$. The same result holds for reduced homology.*

Proof: That i_* is the identity is immediate from the definition. To check the second statement, choose $f_0 : L' \to M$ and $g_0 : L' \to L$ as simplicial approximations to k and $i_{|L|}$ respectively. Then choose $f_1 : K' \to L'$ and $g_1 : K' \to K$ as simplicial approximations to h and $i_{|K|}$ respectively. We have the following diagram of continuous maps and simplicial maps:



Now $f_0 \circ f_1$ is a simplicial approximation to $k \circ h$. Therefore

$$(k \circ h)_* = (f_0 \circ f_1)_* \circ (g_1)_*^{-1}$$

by definition. Since $g_0 \circ f_1$ is a simplicial approximation to h, we have

$$h_* = (g_0 \circ f_1)_* \circ (g_1)_*^{-1} \qquad \text{and} \qquad k_* = (f_0)_* \circ (g_0)_*^{-1}$$

again by definition. Combining these equations and applying Theorem 1.13.3, we obtain the desired result:

$$(k \circ h)_* = k_* \circ h_* \,. \qquad \square$$

Corollary 2.6.3 (The topological invariance of homology groups) *If $h : |K| \to |L|$ is a homeomorphism, then $h_* : H_p(K) \to H_p(L)$ is an isomorphism, and the same result holds for reduced homology.*

Proof: Let $k : |L| \to |K|$ be the inverse of h. Then $h_* \circ k_*$ equals $(i_{|L|})_*$ and $k_* \circ h_*$ equals $(i_{|K|})_*$. Thus $h_* \circ k_*$ and $k_* \circ h_*$ are isomorphisms, so h_* is an isomorphism. $\square$

Application to Relative Homology

We have proved the topological invariance of the (absolute) homology groups. We can also do the same for the relative homology groups. Everything we have done so far in this section goes through for relative homology. One simply replaces each occurrence of a complex by the appropriate pair consisting of a complex and a subcomplex and applies Theorem 1.13.11, Lemma 2.2.7, and Theorem 2.5.5. We restate Theorem 2.6.2 and Corollary 2.6.3 in the current context:

Theorem 2.6.4

(a) *The identity map i of $(|K|, |K_0|)$ with itself induces the identity homomorphism in relative homology.*

(b) *If*

$$(|K|, |K_0|) \xrightarrow{h} (|L|, |L_0|) \xrightarrow{k} (|M|, |M_0|)$$

are continuous maps, then $(k \circ h)_* = k_* \circ h_*$ in relative homology.

(c) *If h is a homeomorphism of $|K|$ with $|L|$ that maps $|K_0|$ onto $|L_0|$, then h_* is an isomorphism in relative homology.*

The next definition isolates the often used notion of a retraction. It is the focus of the exercises. The idea will be strengthened in the next section in Definition 2.7.10.

Definition 2.6.5 If $A \subset X$, then a *retraction* $r : X \to A$ is a continuous map such that $r(a) = a$ for all $a \in A$. If there is a retraction of X onto A, then we say that A is a *retract* of X.

EXERCISES

1. **(a)** If $r : X \to A$ is a retraction and X is Hausdorff, then show that A is closed in X.

(b) Let $r : |K| \to |K_0|$ be a retraction, where K_0 is a subcomplex of K. Show that $r_* : H_p(K) \to H_p(K_0)$ is surjective and the homomorphism j_* induced by the inclusion $j : |K_0| \to |K|$ is injective.

(c) Show that there is no retraction $r : \mathbb{B}^n \to \mathbb{S}^{n-1}$.

2. **(a)** Show that there is a retraction of the Klein bottle S onto the embedded circle A pictured in Figure 2.6.1. But there is no retraction of S onto the circle C.

(b) Show that there is no retraction of the projective plane onto the embedded circle C pictured in Figure 2.6.2.

3. Determine whether there are retractions of the torus onto the tube A depicted in Figure 2.6.3. Also onto the disc B and the circle C.

4. Determine all the spaces X such that there is a retraction from the circle C onto X.

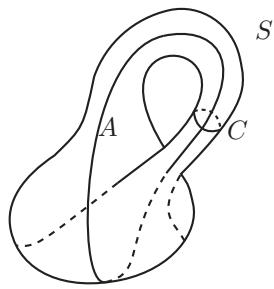


Figure 2.6.1

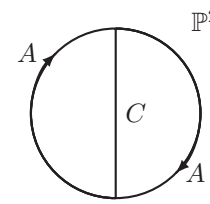


Figure 2.6.2

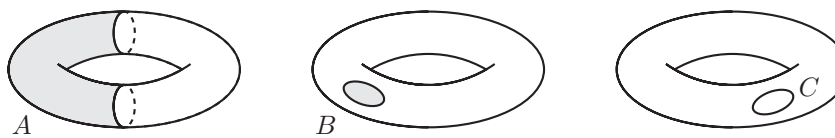


Figure 2.6.3

2.7 Homomorphisms Induced by Homotopic Maps

We now study the important concept of homotopy. Throughout this discussion, I will denote the closed unit interval $[0, 1]$.

Definition 2.7.1 Let X and Y be topological spaces. Two continuous maps $h : X \to Y$ and $k : X \to Y$ are said to be *homotopic* if there is a continuous map

$$f : X \times I \to Y$$

such that $F(x, 0) = h(x)$ and $F(x, 1) = k(x)$ for all $x \in X$. If h and k are homotopic then we write $h \simeq k$. The map F is called a *homotopy* of h with k.

We think of F as a means of "deforming" h continuously to k as t varies from 0 to 1.

We shall prove that if $h, k : |K| \to |L|$ are homotopic then the homology homomorphisms h_* and k_* that they induce are the same. This leads us to the important result that the homology groups are invariants of the homotopy type of a space.

EXAMPLE 2.7.2 Let $X = \mathbb{S}^1$. Then $H_1(X)$ is infinite cyclic. Choose the cycle z generating $H_1(X)$ indicated by the arrow in Figure 2.7.1.

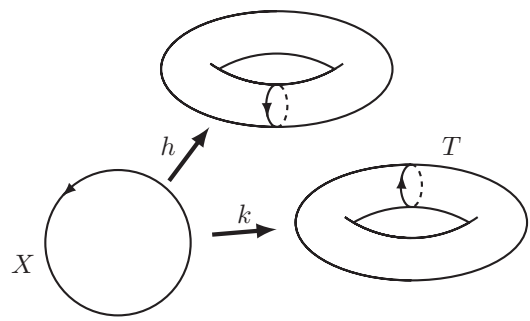


Figure 2.7.1

Let T be the torus. Let $h, k : X \to T$ be the maps shown in Figure 2.7.1. These maps are clearly homotopic, for one can "push h around the ring" until it coincides with k. It is also geometrically clear that the cycles $h_\#(z)$ and $k_\#(z)$ are homologous. Indeed, the 2-chain d obtained by chopping the right half of the torus into triangles and orienting them appropriately satisfies the equation $\partial d = h_\#(z) - k_\#(z)$. Since z represents a generator of $H_1(X)$, it follows that in this case $h_* = k_*$.

We now prove in general that if $h \simeq k$, then $h_* = k_*$. We need two preliminary results. The first is a basic fact about the topology of $|K| \times I$, which we shall prove in the next section [see Corollary 2.8.7].

The topology of the product space $|K| \times I$ is coherent with the subspaces $\sigma \times I$ for $\sigma \in K$.

The second concerns the fact that $|K| \times I$ is a polyhedron.

Lemma 2.7.3 *If K is a complex, then $|K| \times I$ is the polytope of a complex M such that each set $\sigma \times I$ is the polytope of a subcomplex of M, and the sets $\sigma \times 0$ and $\sigma \times 1$ are simplices of M, for each simplex σ of K.*

Proof: We have $|K| \subset \mathbb{E}^J$ for some J. Then $|K| \times I \subset \mathbb{E}^J \times \mathbb{R}$. We shall subdivide $|K| \times I$ into simplices by a variant of the starring procedure used in defining barycentric subdivision.

For $p \geq 0$ let us define

$$X_p = (|K| \times 0) \cup (|K| \times 1) \cup (|K^{(p)}| \times I)\,.$$

We proceed inductively to subdivide X_p into simplices. Consider first the case $p = 0$. The space $(|K| \times 0) \cup (|K| \times 1)$ is the polytope of a complex consisting of all simplices of the form $\sigma \times 0$ or $\sigma \times 1$, for $\sigma \in K$. The space $|K^{(0)}| \times I$ is the polytope of the complex consisting of all 1-simplices of the form $v \times I$ for $v \in K^{(0)}$ and their vertices. Their union is a complex M_0 whose polytope is X_0.

In general, suppose that M_{p-1} is a complex whose polytope is X_{p-1} such that each set $s \times I$, for s a simplex of K of dimension less than p, is the polytope of a subcomplex of M_{p-1}. Let $\dim \sigma = p$ and consider the set $\sigma \times I$. Now let

$$\begin{aligned} \mathrm{Bd}(\sigma \times I) &= (\sigma \times I) - (\mathrm{Int}\,\sigma \times \mathrm{Int}\,I) \\ &= ((\mathrm{Bd}\,\sigma) \times I) \cup (\sigma \times 0) \cup (\sigma \times 1)\,. \end{aligned}$$

Since Bd σ is the union of simplices s of K of dimension $p-1$, $\mathrm{Bd}(\sigma \times I)$ is the polytope of a subcomplex M_σ of M_{p-1}. It is finite because $\mathrm{Bd}(\sigma \times I)$ is compact. Let w_σ denote the point $(\widehat{\sigma}, 1/2) \in \sigma \times I$. Then the cone $w_\sigma * M_\sigma$ is a complex whose polytope is $\sigma \times I$. The intersection of $|w_\sigma * M_\sigma|$ and $|M_{p-1}|$ is the polytope of a subcomplex of each of them.

Define M_p to be the union of M_{p-1} and the cones $w_\sigma * M_\sigma$ as σ ranges over all p-simplices of K. Finally, define M to be the union of the complexes M_p for all p.

Now M is a complex whose underlying space consists precisely of the points of the space $|K| \times I$. However, it is not at all obvious that the spaces $|M|$ and $|K| \times I$ are equal as topological spaces. To prove that assertion, we need the fact about the topology of $|K| \times I$ that was just quoted.

We know that the topology of $|K| \times I$ is coherent with the subspaces $\sigma \times I$ for $\sigma \in K$. On the other hand, the topology of $|M|$ is coherent with the subspaces s for $s \in M$. Now if C is closed in $|K| \times I$, then $C \cap (\sigma \times I)$ is closed in $\sigma \times I$. If s is a simplex of M lying in $\sigma \times I$, then s is a subspace of $\sigma \times I$

(both are subspaces of $\mathbb{E}^J \times \mathbb{R}$, being compact). Hence $C \cap s$ is closed in s. It follows that C is closed in $|M|$.

Conversely, if C is closed in $|M|$, then $C \cap s$ is closed in s for each $s \in M$. Because $\sigma \times I$ is a finite union of simplices s of M, the set $C \cap (\sigma \times I)$ is closed in $\sigma \times I$. Thus σ is closed in $|K| \times I$. □

EXAMPLE 2.7.4 If K is the complex consisting of a 1-simplex and its faces, then $|K| \times I$ is by the procedure of the preceding lemma subdivided into the complex depicted in Figure 2.7.2. In the case of a complex consisting of a 2-simplex and its faces, the procedure is illustrated in Figure 2.7.3. The top-left part of that figure shows $|K| \times I$ and the bottom-left part shows its subdivision. On the right-hand side of the figure we have shown the subdivided complex pushed apart from the point $(\widehat{\sigma}, \frac{1}{2})$ in the directions of the five faces of $|K| \times I$.

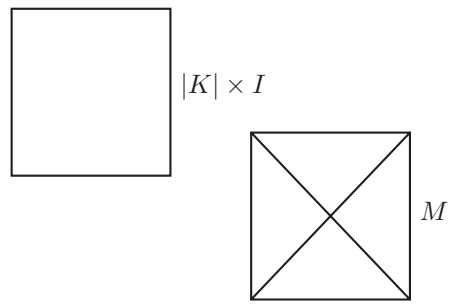


Figure 2.7.2

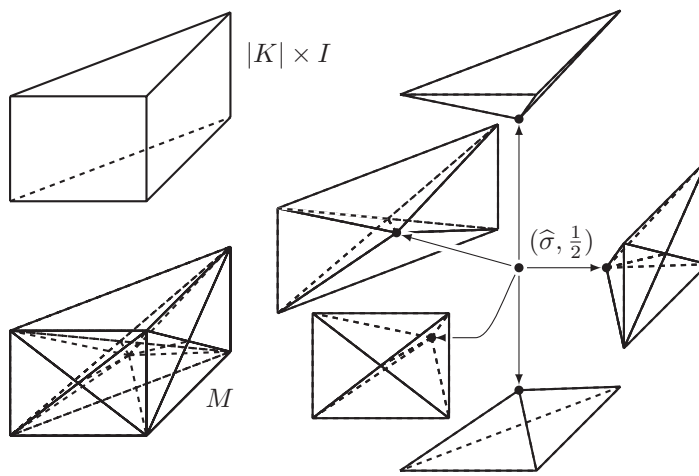


Figure 2.7.3

Theorem 2.7.5 *If the maps $h, k : |K| \to |L|$ are homotopic, then the homomorphisms $h_*, k_* : H_p(K) \to H_p(L)$ are equal. The same holds for reduced homology.*

Proof: Let K be a complex. Let M be a complex whose underlying space is $|K| \times I$, such that for each $\sigma \in K$, both $\sigma \times 0$ and $\sigma \times 1$ are simplices of M. Also suppose that $\sigma \times I$ is the polytope of a subcomplex of M.

Let $F : |K| \times I \to |L|$ be the homotopy of h to k. Let $i, j : |K| \to |K| \times I$ be the maps $i(x) = (x, 0)$ and $j(x) = (x, 1)$, as shown in Figure 2.7.4. Then i and j are simplicial maps of K into the complex M. Furthermore,

$$F \circ i = h \qquad \text{and} \qquad F \circ j = k\,.$$

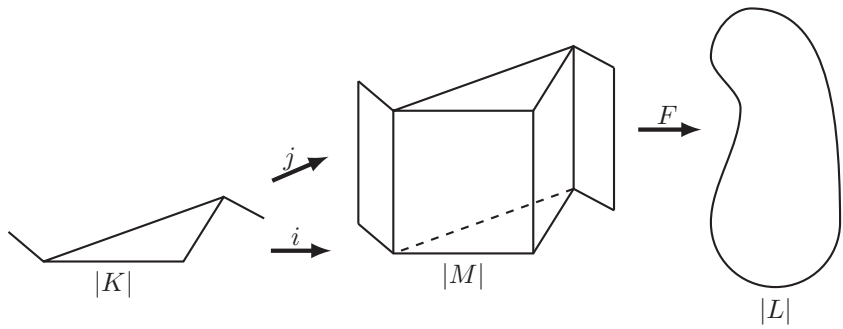


Figure 2.7.4

We claim that the chain maps $i_\#$ and $j_\#$ induced by i and j are chain homotopies. Consider the function Φ assigning to each simplex σ of K the subcomplex of M whose polytope is $\sigma \times I$. Now the space $\sigma \times I$ is acyclic because it is homeomorphic to a closed ball. And if $\sigma \succ s$, then $s \times I \subset \sigma \times I$ so $\Phi(s)$ is a subcomplex of $\Phi(\sigma)$. Hence Φ is an acyclic carrier from K to M. Furthermore, it carries both $i_\#$ and $j_\#$ because both $i(\sigma) = \sigma \times 0$ and $j(\sigma) = \sigma \times 1$ belong to $\Phi(\sigma)$. It follows from Theorem 1.14.10 that $i_\#$ and $j_\#$ are chain homotopic. We conclude that $i_* = j_*$. Thus

$$h_* = F_* \circ i_* = F_* \circ j_* = k_*$$

as desired. □

This result carries over directly to the relative homology groups. Given maps $h, k : (|K|, |K_0|) \to (|L|, |L_0|)$, we say that they are homotopic (as maps of pairs of spaces) if there is a homotopy $H : |K| \times I \to |L|$ of h to k such that H maps $|K_0| \times I$ into $|L_0|$. Now we have the following theorem:

Theorem 2.7.6 *If h and k are homotopic as maps of pairs of spaces, then $h_* = k_*$ as maps of relative homology groups.*

Proof: The proof of Theorem 2.7.5 goes through directly. Both i and j carry $|K_0|$ into $|K_0| \times I$ and so does the chain homotopy connecting $i_\#$ to $j_\#$. Then $i_* = j_*$ as maps of relative homology, and the proof proceeds as before. □

Here is another result which follows from our knowledge of the topology of $|K| \times I$. We shall make use of it in what follows.

Theorem 2.7.7 *If $f : K \to L$ is a simplicial approximation to the continuous map $h : |K| \to |L|$, then f is homotopic to h.*

Proof: For each $x \in |K|$ we know from Lemma 2.2.4 that $f(x)$ and $h(x)$ lie in a single simplex of L. Therefore the straight line homotopy given by

$$F(x,t) = (1-t)f(x) + th(x)$$

maps $|K| \times I$ into $|L|$. If L is finite then F is automatically continuous because it is continuous as a map into Euclidean space, of which $|L|$ is a subspace. To show that F is continuous in general, we show that its restriction to $\sigma \times I$ is continuous for each $\sigma \in K$. Since the topology of $|K| \times I$ is coherent with the subspaces of $\sigma \times I$, this will suffice.

For each $x \in \sigma$, let τ_x denote the simplex of L whose interior contains $h(x)$. Because $h(\sigma)$ is compact, the collection of simplices τ_x for $x \in \sigma$ is finite. Let L_σ be the subcomplex of L consisting of these simplices and their faces. By Lemma 2.2.4, the point $f(x)$ lies in τ_x. Therefore F carries the set $x \times I$ into τ_x. Thus F maps $\sigma \times I$ into $|L_\sigma|$. Since L_σ is a finite complex, its space is a subspace of Euclidean space. Hence $F : \sigma \times I \to |L_\sigma|$ is continuous. Because the inclusion $|L_\sigma| \to |L|$ is continuous, the map $F : \sigma \times I \to |L|$ is also continuous, as desired. □

We know that if two spaces are homeomorphic then they have isomorphic homology groups. There is in fact a weaker relation than homeomorphism that implies the same result. It is the relation of "homotopy equivalence," which we now introduce.

Definition 2.7.8 Two spaces X and Y are said to be *homotopy equivalent*, or said to have the same *homotopy type*, if there are maps

$$f : X \to Y \qquad \text{and} \qquad g : Y \to X$$

such that $g \circ f \simeq i_X$ and $f \circ g \simeq i_Y$, where i_X and i_Y are the identity maps on the spaces X and Y, respectively. The maps f and g are often called *homotopy equivalences* and g is called the *homotopy inverse* of f.

Symmetry and reflexivity of this relation are trivial. Transitivity is left as an exercise. So homotopy equivalence is an equivalence relation.

If X has the homotopy type of a single point then X is said to be *contractible*. This is equivalent to the statement that the identity map $i_X : X \to X$ is homotopic to a constant map. For example, the unit ball is contractible because the map $F(x,t) = (1-t)x$ is a homotopy between the identity and a constant.

Theorem 2.7.9 *If $f : |K| \to |L|$ is a homotopy equivalence, then f_* is an isomorphism. In particular, if $|K|$ is contractible, then K is acyclic.*

Proof: The proof is immediate. If g is an inverse for f, then g_* is an inverse for f_*. □

Homotopy equivalences are difficult to visualize in general. There is a special type of homotopy equivalence that is geometrically easier to understand.

Definition 2.7.10 A *deformation retraction* of X onto A is a continuous map $F : X \times I \to X$ such that

$$\begin{aligned} F(x,0) &= x \quad \text{for } x \in X; \\ F(x,1) &\in A \quad \text{for } x \in X; \\ F(a,t) &= a \quad \text{for } a \in A\,. \end{aligned}$$

If such an F exists, then A is called a *deformation retract* of X.

If F is a deformation retraction of X onto A, then the map $r(x) = F(x,1)$ is a retraction of X onto A. The latter fact is equivalent to the statement that the composite

$$A \xrightarrow{j} X \xrightarrow{r} A$$

(where j is inclusion) is the identity map i_A. On the other hand, the map F is a homotopy between the map i_X and the composite

$$X \xrightarrow{r} A \xrightarrow{j} X\,.$$

In fact each point of A remains fixed during the homotopy. It follows that r and j are homotopy inverse to each other.

One can visualize a deformation retraction as a gradual collapsing of the space X onto the subspace A, such that each point of A remains fixed during the process of collapse. This type of homotopy equivalence can thus be visualized geometrically. It is intuitively clear that if A is a deformation retract of X and B is a deformation retract of A, then B is a deformation retract of X. This is also easy to prove.

Now we look at some special cases:

Theorem 2.7.11 *The unit sphere $\mathbb{S}^{n-1}$ is a deformation retract of the punctured euclidean space $\mathbb{R}^n \setminus \{\mathbf{0}\}$.*

Proof: Let $X = \mathbb{R}^n \setminus \{\mathbf{0}\}$. We define $F : X \times I \to X$ by

$$F(x,t) = (1-t)x + t \cdot \frac{x}{\|x\|}\,.$$

Clearly the map F shrinks each open ray from the origin to the point that it intersects on the unit sphere. Thus it does the job. □

Corollary 2.7.12 *The Euclidean spaces $\mathbb{R}^n$ and $\mathbb{R}^m$ are not homeomorphic if $n \neq m$.*

Proof: Suppose to the contrary that h is a homeomorphism of $\mathbb{R}^n$ with $\mathbb{R}^m$. Then, by restriction, h is a homeomorphism of $\mathbb{R}^n \setminus \{\mathbf{0}\}$ with $\mathbb{R}^m \setminus \{p\}$ for $p = h(\mathbf{0}) \in \mathbb{R}^m$. The latter space is of course homeomorphic with $\mathbb{R}^m \setminus \{\mathbf{0}\}$. It now follows from the preceding theorem that $\mathbb{S}^{n-1}$ and $\mathbb{S}^{m-1}$ are homotopy equivalent. This cannot be true if $n \neq m$ because $\widetilde{H}_{n-1}(\mathbb{S}^{n-1}) = \mathbb{Z}$ while $\widetilde{H}_{n-1}(\mathbb{S}^{m-1}) = 0$. □

EXAMPLE 2.7.13 The wedge of two circles has the same homotopy type as the letter θ. Let X be a doubly punctured elliptical region in the plane as shown in Figure 2.7.5.

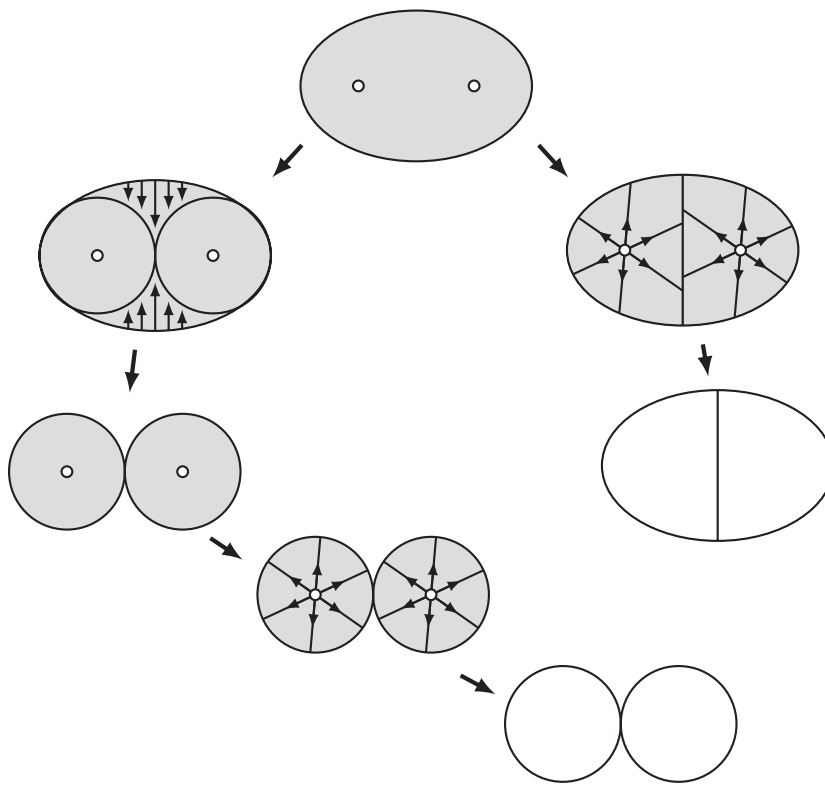


Figure 2.7.5

The sequence of arrows on the left side of this figure indicates how one can collapse X to the wedge of two circles. The arrows on the right side show how one can collapse X to the letter θ. Since each of these spaces is homotopy equivalent to X, they are therefore homotopy equivalent to each other.

The situation in the last example is more general than is immediately apparent. It is an interesting fact that every homotopy equivalence can be expressed in terms of deformation retractions. Specifically, there is a theorem to the effect that two spaces X and Y have the same homotopy type if and only if there is a space Z and embeddings $h : X \to Z$ and $k : Y \to Z$ such that both $h(X)$ and $k(Y)$ are homotopy retracts of Z. See [Whi78], [Fuc71]. This fact helps in visualizing what the notion of homotopy equivalence means geometrically.

Homotopy equivalences are a powerful tool for computing homology groups. Given a complex K, it is often much easier to show that its space is homotopy equivalent to a space whose homology is known than it is to calculate the homology of K directly. The exercises below will illustrate this idea.

EXERCISES

1. Show that if $\mathcal{A}$ is a collection of spaces then homotopy equivalence is an equivalence relation on $\mathcal{A}$.

2. Show that if A is a deformation retract of X and B is a deformation retract of A then B is a deformation retract of X.

3. Group the following spaces into homotopy equivalence classes. Assuming that they are all polytopes of complexes, calculate their homology groups.

- **(a)** The Möbius band.
- **(b)** The torus.
- **(c)** The solid torus $\mathbb{B}^1 \times \mathbb{S}^1$.
- **(d)** The torus minus one point.
- **(e)** The torus minus two points.
- **(f)** The Klein bottle minus a point.
- **(g)** $\mathbb{R}^3$ with the z-axis removed.
- **(h)** $\mathbb{R}^3$ with the circle $\{x^2 + y^2 = 1, z = 0\}$ removed.
- **(i)** The intersection of the spaces in **(g)** and **(h)**.
- **(j)** $\mathbb{S}^3$ with two linked circles removed.
- **(k)** $\mathbb{S}^3$ with two unlinked circles removed.

4. Prove that if K is a complex of dimension n, then $|K|$ has covering dimension at least n.

5. Prove Theorem 2.7.5 without using the fact that $|K| \times I$ is the space of a complex, as follows:

(a) Let K and L be finite; let $h : |K| \to |L|$. Show there is $\epsilon > 0$ such that if $k : |K| \to |L|$ and $|h(x) - k(x)| < \epsilon$ for all x, then $h_* = k_*$. [*Hint:* Choose K' so that $h(\overline{\mathrm{St}}(v, K')) \subset \mathrm{St}(w, L)$ for some w. If ϵ is small, this same inclusion holds with h replaced by k.]

(b) Prove the theorem when K and L are finite.

(c) Prove the theorem in general. [*Hint:* Each cycle of K is carried by a finite subcomplex of K.]

2.8 Review of Quotient Spaces

In this section we review some of the basic ideas concerning quotient spaces.

A surjective map $p : X \to Y$ is called a *quotient map* provided that a subset $U \subset Y$ is open if and only if $p^{-1}(U) \subset X$ is open. This condition is equivalent to requiring that $E \subset Y$ be closed if and only if $p^{-1}(E) \subset X$ is closed.

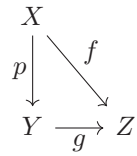
A subset C of X is *saturated* with respect to p if it equals the complete inverse image set $p^{-1}(A)$ for some subset $A \subset Y$. To say that p is a *quotient map* is to say that p is continuous and p maps saturated open subsets of X to open subsets of Y (or, equivalently, saturated closed subsets of X to closed subsets of Y).

Let $p : X \to Y$ be a surjective, continuous map. If p is either an open map or a closed map (that is, p either maps open sets to open sets or closed sets to closed sets), then p is a quotient map. In particular, if X is compact and Y is Hausdorff, then p is a closed map and hence a quotient map.

First we list some elementary facts about quotient maps. Their proofs are straightforward and we leave them to the reader.

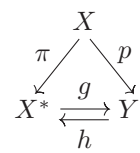
- A one-to-one quotient map is a homeomorphism.
- The composition of quotient maps is a quotient map.
- The restriction of a quotient map can *sometimes* be a quotient map. If $p : X \to Y$ is a quotient map, and if $A \subset X$ is a saturated subspace that is either open or closed, then p restricted to A is a quotient map.

Continuous maps and quotient maps are related as follows: If $p : X \to Y$ is a quotient map and $f : X \to Z$ is a continuous map that is constant on each set $p^{-1}(y)$, then there is a unique continuous map $g : Y \to Z$ such that $g \circ p = f$. We say that g is *induced* by f. See the diagram.



We define the notion of a *quotient space* as follows: Let X^* be a partition of X into disjoint subsets whose union is X. Let $\pi : X \to X^*$ map each point to the subset that contains it. Let us topologize X^* by declaring a set $U \subset X^*$ to be open if $\pi^{-1}(U) \subset X$ is open. Then π is a quotient map and we say that X^* is a *quotient space* of X. We often say that this quotient is obtained by identifying each element of the partition to a point.

If $p : X \to Y$ is a quotient map, then we can consider Y to be a quotient space of X in the following manner: Given p, let X^* be the partition of X into the disjoint sets $p^{-1}(y)$ for $y \in Y$. Then X^* is homeomorphic to Y. We now apply the preceding remark twice to obtain continuous maps $g : X^* \to Y$ and $h : Y \to X^*$ as shown in the next diagram:



These maps g and h are readily seen to be inverse to each other.

EXAMPLE 2.8.1 Let X be the subspace of $\mathbb{R}^3$ obtained by rotating the unit circle in the x-z plane with center $(2, 0, 0)$ about the z-axis. Using cylindrical coordinates (r, θ, z) in $\mathbb{R}^3$, we can express X as the set of points satisfying the equation $(r-2)^2 + z^2 = 1$. Letting $I^2 = I \times I$, we define $p : I^2 \to X$ by setting, for $(s, t) \in I^2$,

$$r - 2 = \cos 2\pi t \quad , \quad z = \sin 2\pi t \quad , \quad \theta = 2\pi s \, .$$

You can check that p maps I^2 onto X, and is a closed quotient map.

By use of this map, we can consider X to be the quotient space obtained from I^2 by identifying $(s, 0)$ with $(s, 1)$ and $(0, t)$ with $(1, t)$ for $s, t \in I$. Refer to Figure 2.8.1. This is the topological version of the method by which we constructed the torus in Section 1.4 by pasting together the edges of a rectangle.

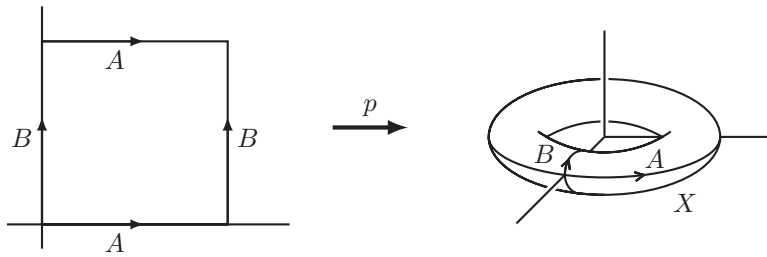


Figure 2.8.1

The separation axioms are not necessarily valid for quotient spaces. For example, a quotient space of a Hausdorff space need not be Hausdorff. All that one can say in general is the following: If $p : X \to Y$ is a quotient map, and if each $p^{-1}(y)$ is closed in X, then Y is T_1. That is to say, if X is Hausdorff, then one-point sets in Y are closed. So if X^* is a partition of X into closed sets, then the quotient space X^* is a T_1-space.

We will treat separation axioms for quotient spaces in more detail later in the book. We now treat one final question: Under what circumstances is the cartesian product of two quotient maps a quotient map? We have two results that address this question.

Theorem 2.8.2 *Let $p : X \to Y$ be a quotient map. If C is a locally compact Hausdorff space, then*

$$p \times i_C : X \times C \to Y \times C$$

is a quotient map.

Proof: Let $\pi = p \times i_C$. Let A be a subset of $Y \times C$ such that $\pi^{-1}(A)$ is open in $X \times C$. We show that A is open in $Y \times C$. That is, given $(y_0, c_0) \in A$, we will find an open set containing (y_0, c_0) and lying in A. Choose $x_0 \in X$ so that $p(x_0) = y_0$. Then $\pi(x_0, c_0) = (y_0, c_0)$. Since $\pi^{-1}(A)$ is open, we can choose neighborhoods U_1 of x_0 and W of c_0 such that $U_1 \times W \subset \pi^{-1}(A)$. Because C is locally compact Hausdorff, we can choose a neighborhood V of c_0 so that $\overline{V}$ is compact and $\overline{V} \subset W$. Then $U_1 \times V$ is a neighborhood of (x_0, c_0) such that $\overline{V}$ is compact and

$$U_1 \times \overline{V} \subset \pi^{-1}(A)\,.$$

In general, suppose that U_i is a neighborhood of x_0 such that $U_i \times \overline{V} \subset \pi^{-1}(A)$. Now $p^{-1}p(U_i)$ is not necessarily open in X, but it contains U_i. We construct an open set $U_{i+1} \subset X$ such that

$$p^{-1}p(U_i) \times \overline{V} \subset U_{i+1} \times \overline{V} \subset \pi^{-1}(A)$$

as follows: For each point x of $p^{-1}p(U_i)$, the space $\{x\} \times \overline{V}$ lies in $\pi^{-1}(A)$. Using the compactness of $\overline{V}$, we choose a neighborhood W_x of x such that $W_x \times \overline{V} \subset \pi^{-1}(A)$. Let U_{i+1} be the union of the open sets W_x. Then U_{i+1} is the desired open subset of X. See Figure 2.8.2.

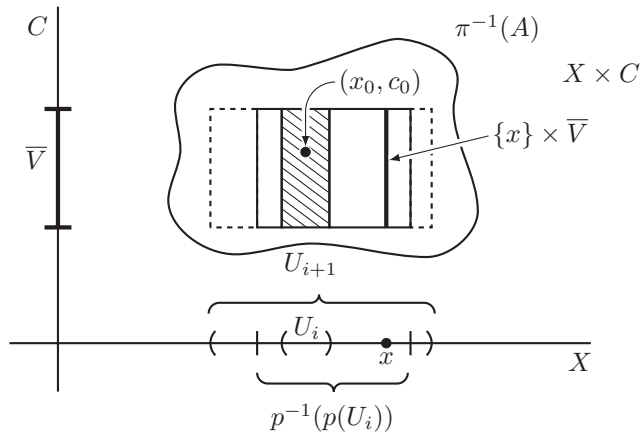


Figure 2.8.2

Finally let U be the union of the open sets $U_1 \subset U_2 \subset \cdots$. Then $U \times V$ is a neighborhood of (x_0, c_0) and $U \times \overline{V} \subset \pi^{-1}(A)$. In addition, U is saturated with respect to p, for

$$U \subset p^{-1}p(U) = \bigcup_{i=1}^{\infty} p^{-1}p(U_i) = \bigcup_{i=1}^{\infty} U_{i+1} = U\,.$$

Hence $p(U)$ is open in Y. Thus

$$p(U) \times V = \pi(U \times V) \subset A$$

is a neighborhood of (y_0, c_0) lying in A as desired. □

Corollary 2.8.3 *If $p : A \to B$ and $q : C \to D$ are quotient maps, and if the domain of p and the range of q are locally compact Hausdorff spaces, then*

$$p \times q : A \times C \to B \times D$$

is a quotient map.

Proof: We can write $p \times q$ as the composition

$$A \times C \overset{i_A \times q}{\longrightarrow} A \times D \overset{p \times i_D}{\longrightarrow} B \times D\,.$$

Since each of these maps is a quotient map, then so is $p \times q$. □

There is a close relationship between coherent topologies and quotient maps, which can be described as follows.

Definition 2.8.4 Suppose that E is a space that is the disjoint union of disjoint subspaces E_α, each of which is open (and closed) in E. Then we say that E is the *topological sum* of the E_α and we write $E = \sum E_\alpha$. A set U is open in E if and only if $U \cap E_\alpha$ is open in E_α for each α.

More generally, let $\{X_\alpha\}_{\alpha \in J}$ be a family of topological spaces, which may or may not be disjoint. Let E be the set that is the union of the disjoint spaces

$$E_\alpha = X_\alpha \times \{\alpha\}$$

for $\alpha \in J$. We topologize E by declaring U to be open in E if and only if $U \cap E_\alpha$ is open in E_α for each α. Then E is the topological sum of the disjoint spaces E_α. We have a natural map $p : E \to \cup X_\alpha$ which projects $X_\alpha \times \{\alpha\}$ onto X_α for each α. [We sometimes abuse terminology and speak of E as the topological sum of the spaces X_α in this situation.]

In these circumstances, we have the following result. The proof is immediate.

Lemma 2.8.5 *Let X be a space which is the union of certain of its subspaces X_α. Let E be the topological sum of the X_α. Let $p : E \to X$ be the natural projection. Then the topology of X is coherent with the subspaces X_α if and only if p is a quotient map.*

In this circumstance we often say that X is *coherent union* of the spaces X_α.

Theorem 2.8.6 *If the topology of X is coherent with the subspaces X_α and if Y is a locally compact Hausdorff space, then $X \times Y$ is coherent with the subspaces $X_\alpha \times Y$.*

Proof: Define $E = \sum(X_\alpha \times \{\alpha\})$. Let $p : E \to X$ be the projection map. Because Y is locally compact Hausdorff, the map

$$p \times i_Y : E \times Y \to X \times Y$$

is also a quotient map. Because E is the topological sum of the subspaces $E_\alpha = X_\alpha \times \{\alpha\}$, we see that $E \times Y$ is the disjoint union of its subspaces $E_\alpha \times Y$, each of which is open in $E \times Y$. Therefore $E \times Y$ is the topological sum of the spaces $E_\alpha \times Y = X_\alpha \times \{\alpha\} \times Y$. Since $p \times i_Y$ is a quotient map, the topology of $X \times Y$ is coherent with the subspaces $X_\alpha \times Y$. □

Corollary 2.8.7 *The topology of $|K| \times I$ is coherent with the subspaces $\sigma \times I$ for $\sigma \in K$.*

Proof: By definition, the topology of $|K|$ is coherent with the subspaces σ, for $\sigma \in K$. Since I is locally compact Hausdorff (in fact compact Hausdorff), the preceding theorem applies. □

Corollary 2.8.8 *Suppose $w * K$ is a cone over the complex K. The map $\pi : |K| \times I \to |w * K|$ defined by*

$$\pi(x, t) = (1 - t)x + tw$$

is a quotient map. It collapses $|K| \times I$ to the point w and is otherwise one-to-one.

Proof: If $\sigma = v_o \cdots v_n$ is a simplex of K, let $w*\sigma$ denote the simplex $wv_0 \cdots v_n$ of $w * K$. A set B is closed in $|w * K|$ if its intersection with each simplex $w * \sigma$ is closed in that simplex. A set A is closed in $|K| \times I$ if its intersection with each set $\sigma \times I$ is closed in $\sigma \times I$. Therefore, for π to be a quotient map, it suffices to show that the map

$$\pi' : \sigma \times I \to w * \sigma\,,$$

obtained by restricting π, is a quotient map. But that fact is immediate since π' is continuous and surjective, and the spaces involved are compact Hausdorff. □

The preceding corollary suggests a way to define a cone over an arbitrary topological space.

Definition 2.8.9 Let X be a space. We define the *cone over* X to be the quotient space obtained from $X \times I$ by identifying the subset $X \times 1$ to a point. This point is called the *vertex* of the cone. The cone itself is denoted by $C(X)$. Formally, we define $C(X)$ by partitioning $X \times I$ into the one-point sets $\{(x, t)\}$ for $t < 1$ and the set $X \times 1$, and passing to the resulting quotient space.

The next definition is used in Exercise 2.

Definition 2.8.10 The space obtained from two copies of $\mathbb{R}^2$, say $\mathbb{R}^2 \times \{a\}$ and $\mathbb{R}^2 \times \{b\}$, by identifying (x, a) and (x, b) whenever $x \neq 0$ is called the *plane with two origins.*

EXERCISES

1. Verify the results about quotient spaces that were stated without proof in this section.
2. **(a)** Show that each point of the plane with two origins has a neighborhood homeomorphic to an open subset of $\mathbb{R}^2$.
   (b) Show that the plane with two origins is not Hausdorff.

3. **(a)** Show that if $p : X \to Y$ is an open quotient map and A is open in X, then $p|A : A \to P(A)$ is an open quotient map.
(b) Repeat **(a)** with "open" replaced by "closed."

4. Show that if $p : A \to B$ and $q : C \to D$ are open quotient maps, then so is $p \times q$.

5. Let X be the coherent union of the subspaces $\{X_\alpha\}$. Show that if Y is a subspace of X that is either open or closed in X, then Y is the coherent union of its subspaces $\{X_\alpha \cap Y\}$.

6. Let K and L be complexes. Show that if K is locally finite, then the topology of $|K| \times |L|$ is coherent with the subspaces $\sigma \times \tau$, for $\sigma \in K$ and $\tau \in L$.

7. Show that the direct sum of quotient maps is a quotient map.

2.9 *Application: Maps of Spheres

In this section we present several applications of homology theory to classical problems of geometry and topology. The theorems that we prove here will be generalized in the next section. That is when we will treat the celebrated Lefschetz fixed point theorem.

Definition 2.9.1 Let $n \geq 1$. Let $f : \mathbb{S}^n \to \mathbb{S}^n$ be a continuous map. If α is one of the two generators of the infinite cyclic group $H_n(\mathbb{S}^n)$, then $f_*(\alpha) = d\alpha$ for some integer d. The integer d is independent of the choice of generator since $f_*(-\alpha) = d(-\alpha)$. It is called the *degree* of the map f and is denoted $\deg f$

Lemma 2.9.2 *The degree has the following properties:*

(1) *If $f \simeq g$, then* $\deg f = \deg g$.

(2) *If f extends to a continuous map $\widetilde{f} : \mathbb{B}^{n+1} \to \mathbb{S}^n$, then* $\deg f = 0$.

(3) *The identity map has degree 1.*

(4) $\deg(f \circ g) = (\deg f) \cdot (\deg g)$.

Proof: Property **(1)** follows from Theorem 2.7.5. Property **(2)** is true because $f_* : H_n(\mathbb{S}^n) \to H_n(\mathbb{S}^n)$ equals the composition

$$H_n(\mathbb{S}^n) \xrightarrow{j_*} H_n(\mathbb{B}^{n+1}) \xrightarrow{h_*} H_n(\mathbb{S}^n)\,,$$

where j is inclusion. Since $\mathbb{B}^{n+1}$ is acyclic, this composition is the zero homomorphism. Properties **(3)** and **(4)** are immediate consequences of Theorem 2.6.2. □

Theorem 2.9.3 *There is no retraction $r : \mathbb{B}^{n+1} \to \mathbb{S}^n$.*

Proof: Property **(3)** of Lemma 2.9.2 tells us that the identity map $i : \mathbb{S}^n \to \mathbb{S}^n$ has degree 1. If a retraction $r : \mathbb{B}^{n+1} \to \mathbb{S}^n$ existed, then it would be an extension of the identity map to $\mathbb{B}^{n+1}$. Property **(2)** of Lemma 2.9.2 would then tell us that the degree of i is 0, a contradiction. □

Theorem 2.9.4 (Brouwer fixed point theorem) *If $\phi : \mathbb{B}^n \to \mathbb{B}^n$ is a continuous map, then ϕ has a fixed point, that is, there exists $x \in \mathbb{B}^n$ such that $\phi(x) = x$.*

Proof: If ϕ has no fixed point, then we can define a map $h : \mathbb{B}^n \to \mathbb{S}^{n-1}$ by

$$h(x) = \frac{x - \phi(x)}{\|x - \phi(x)\|}$$

because $x - \phi(x) \neq 0$. Let $f : \mathbb{S}^{n-1} \to \mathbb{S}^{n-1}$ be the restriction of h to $\mathbb{S}^{n-1}$. Then by **(2)** of Lemma 2.9.2 f has degree 0.

On the other hand, we will now show that f has degree 1. That gives a contradiction. Define a homotopy $H : \mathbb{S}^{n-1} \times I \to \mathbb{S}^{n-1}$ by

$$H(u, t) = \frac{u - t\phi(u)}{\|u - t\phi(u)\|}\,.$$

The denominator is non-zero for $t = 1$ since $u \neq \phi(u)$. It is also non-zero for $t < 1$ because $\|u\| = 1$ while $\|t\phi(u)\| = t\|\phi(u)\| \leq t < 1$. The map H is a homotopy between the identity map of $\mathbb{S}^{n-1}$ and the map f. Therefore $\deg f = 1$. □

Definition 2.9.5 The *antipodal map* $a : \mathbb{S}^n \to \mathbb{S}^n$ is the map $a(x) = -x$.

For our further applications, we need to show that the degree of the antipodal map from $\mathbb{S}^n$ to $\mathbb{S}^n$ equals $(-1)^{n+1}$. Below we shall give a direct proof of the result. In the next section we give an alternative proof. Yet another proof is given in Section 4.4.

The key result that we need here is the following lemma:

Lemma 2.9.6 *The reflection map*

$$\rho(x_1, x_2, \ldots, x_n, x_{n+1}) = (x_1, x_2, \ldots, x_n, -x_{n+1})$$

from $\mathbb{S}^n$ to $\mathbb{S}^n$ has degree -1.

It may be intuitively clear that the reflection map reverses the orientation of the tangent plane at each point of the sphere. Nonetheless, we will need to give a proof in the context of Definition 2.9.1 that $\deg \rho = -1$.

In the proof of the lemma, we shall need the notion of a "triangulation." We give that definition next.

Definition 2.9.7 A *triangulation* of a space X is a complex L and a homeomorphism $h : |L| \to X$. If such a triangulation exists, then X is said to be *triangulable*.

Proof: Step 1: We shall construct a triangulation of $\mathbb{S}^n$ by an n-dimensional complex such that the reflection map ρ induces a simplicial map of this complex to itself.

For K a finite complex in $\mathbb{R}^N \times \{0\} \subset \mathbb{R}^{N+1}$, let $w_0 = (0, 0, \ldots, 0, 1)$ and $w_1 = (0, 0, \ldots, 0, -1)$ in $\mathbb{R}^{N+1}$ and let

$$S(K) = (w_0 * K) \cup (w_1 * K) .$$

Then $S(K)$ is called a *suspension* of K. Let $r : S(K) \to S(K)$ be the simplicial map that exchanges w_0 and w_1 and maps each vertex of K to itself. We show that there exists a complex K of dimension $n-1$ and a triangulation

$$k : |S(K)| \to \mathbb{S}^n$$

such that the following diagram commutes:

$$\begin{array}{ccc} S(K) & \xrightarrow[k]{} & \mathbb{S}^n \\ \Big\downarrow{\scriptstyle r} & & \Big\downarrow{\scriptstyle \rho} \\ S(K) & \xrightarrow[k]{} & \mathbb{S}^n \end{array} \tag{2.1}$$

Once this is done, **Step 1** will have been accomplished.

Let $h : |K| \to \mathbb{S}^{n-1}$ be any triangulation of $\mathbb{S}^{n-1}$ by a complex of dimension $n-1$. Let $y \in |S(K)|$. If $y = (1-t)x + tw_0$ for some $x \in |K|$, define

$$k(y) = (\sqrt{1-t^2}h(x), t) .$$

If $y = (1-t)x + tw_1$, define

$$k(y) = (\sqrt{1-t^2}h(x), -t) .$$

See Figure 2.9.1.

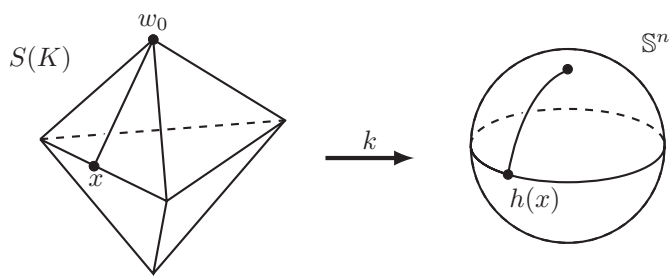


Figure 2.9.1

It is easy to check that k carries $S(K)$ homeomorphically onto $\mathbb{S}^n$. The fact that $\rho \circ k = k \circ r$ is immediate since

$$r((1-t)x + tw_0) = (1-t)x + tw_1 \,.$$

Step 2: In view of **Step 1**, it suffices to show that $\deg r = -1$ to see that our lemma is true.

Let z be an n-cycle of $S(K)$. Then z is a chain of the form

$$z = [w_0, c_m] + [w_1, d_m] \,,$$

where c_m, d_m are chains of K and $m = n-1$. Here we are using the bracket notation of Section 1.9.

Assume that $n > 1$. Since z is a cycle,

$$0 = \partial z = c_m - [w_0, \partial c_m] + d_m - [w_1, \partial d_m] \,.$$

Restricting this chain to K, we obtain the equation $c_m + d_m = 0$. Hence

$$z = [w_0, c_m] - [w_1, c_m] \,.$$

Since r exchanges w_0 and w_1, we have

$$r_\#(z) = [w_1, c_m] - [w_0, c_m] = -z$$

as desired. A similar computation holds if $n = 1$. □

Theorem 2.9.8 *For $n \geq 1$, the degree of the antipodal map is $(-1)^{n+1}$.*

Proof: We claim that any reflection map

$$\rho_i(x_1, x_2, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_{n+1}) = (x_1, x_2, \ldots, x_{i-1}, -x_i, x_{i+1}, \ldots, x_{n+1})$$

has degree -1. To see this set $\rho_i = h^{-1} \circ \rho \circ h$ where h is the homeomorphism of $\mathbb{R}^{n+1}$ that exchanges x_i and x_{n+1}. Using Lemma 2.9.6 and **(3)** and **(4)** of Lemma 2.9.2, we see that

$$\deg \rho_i = (\deg h^{-1})(\deg \rho)(\deg h) = \deg(h^{-1} \circ h) \deg \rho = \deg \rho \,.$$

Next we observe that the antipodal map a equals $\rho_1 \circ \rho_2 \circ \cdot \circ \rho_{n+1}$, so we see that $\deg a = (-1)^{n+1}$. □

Theorem 2.9.9 *If $h : \mathbb{S}^n \to \mathbb{S}^n$ has degree different from $(-1)^{n+1}$, then h has a fixed point.*

Proof: We assume to the contrary that h has no fixed point. Below we will show that $h \simeq a$ from which Theorem 2.9.8 and **(1)** of Lemma 2.9.2 allow us to conclude that $\deg h = (-1)^{n+1}$.

Intuitively, we construct the homotopy by simply moving the point $h(x)$ to the point $-x$ along the shorter great circular arc joining these two points. Because $h(x)$ and $-x$ are not antipodal, there is a unique such arc. So the

homotopy is well defined. Formally, we define the homotopy $H : \mathbb{S}^n \times I \to \mathbb{S}^n$ by

$$H(x,t) = \frac{(1-t)h(x) + t(-x)}{\|(1-t)h(x) + t(-x)\|} .$$

The proof is complete once we show that the denominator does not vanish. If $(1-t)h(x) = tx$ for some t and x, then taking norms of both sides, we conclude that $1 - t = t$, so $t = 1/2$. From this it follows that $h(x) = x$, contrary to hypothesis. □

Theorem 2.9.10 *If $h : \mathbb{S}^n \to \mathbb{S}^n$ has degree different from 1, then h takes some point to its antipode.*

Proof: If a is the antipodal map as usual, then $a \circ h$ has degree different from $(-1)^{n+1}$, so it has a fixed point x. Thus $a(h(x)) = x$ or $-h(x) = x$ as desired. □

Corollary 2.9.11 *The sphere $\mathbb{S}^n$ has a non-vanishing tangent vector field if and only if n is odd.*

Proof: First, suppose n is odd. We simply construct an appropriate vector field as follows: Set $n = 2k - 1$. For $x \in \mathbb{S}^n \subset \mathbb{R}^{2k}$, we define

$$v(x) = (-x_2, x_1, -x_4, x_3, \ldots, -x_{2k}, x_{2k-1}) .$$

We see that $v(x)$ is perpendicular to x, hence tangent to the sphere at x. Since $\|v(x)\| = \|x\| = 1$, v never vanishes.

For the converse, suppose that $w(x)$ is a tangent vector field to $\mathbb{S}^n$ at x with $w(x) \neq \mathbf{0}$ for all $x \in \mathbb{S}^n$. Then

$$h(x) = \frac{w(x)}{\|w(x)\|}$$

is a map from $\mathbb{S}^n$ into $\mathbb{S}^n$. Since $w(x)$ is perpendicular to x for all x, we cannot have either $h(x) = x$ or $h(x) = -x$ for any x. Thus h has no fixed point, and Theorem 2.9.9 tells us that $\deg(h) = (-1)^{n+1}$. Also h maps no point to its antipode, so Theorem 2.9.10 tells us that $\deg(h) = 1$. We conclude that $(-1)^{n+1} = 1$. Thus n must be odd. □

Exercise 2 asks you to prove the fundamental theorem of algebra. For completeness, we state that theorem here.

Theorem 2.9.12 (The Fundamental Theorem of Algebra) *Every polynomial*

$$z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$$

with real or complex coefficients and $n \geq 1$ has a root in the complex plane.

EXERCISES

1. **(a)** Let K be the complex shown in Figure 2.9.2. Its space is the boundary of a square.

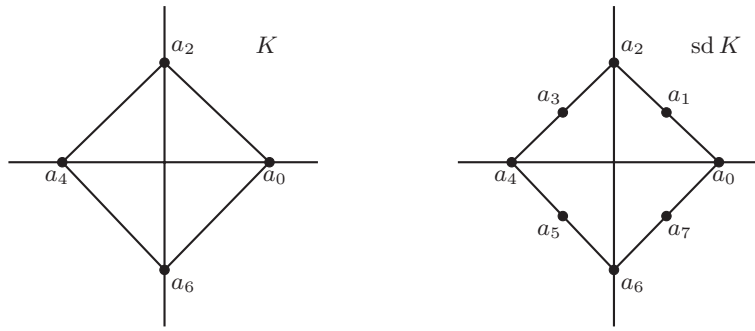


Figure 2.9.2

Let sd K be its first barycentric subdivision. Let $f : \text{sd } K \to K$ be the simplicial map specified by $F(a_i) = a_{2i}$, where the index $2i$ is reduced modulo 8 if necessary. Show that the map f has degree 2.

(b) Consider $\mathbb{S}^1$ as the set of complex numbers of unit modulus. Show that the map $h : \mathbb{S}^1 \to \mathbb{S}^1$ given by $h(z) = z^2$ has degree 2. [**Hint:** By radial projection, h induces a map of $|K|$ to itself that is homotopic to the simplicial map f of part **(a)**.]

(c) If n is any positive integer, show that the map $k : \mathbb{S}^1 \to \mathbb{S}^1$ given by $k(z) = z^n$ has degree n.

2. Use the result of Exercise 1 to prove the fundamental theorem of algebra by the following steps:

Let S_c be the circle $|z| = c$ with center 0 and radius c in the complex plane. Suppose that the given polynomial has no root in the disc $\{z : |z| \leq c\}$. Let

$$h : S_c \to \mathbb{C} \setminus \{0\}$$

be given by

$$h(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0 \,.$$

(a) Show that h_* is trivial.

(b) Show that if c is sufficiently large, then h is homotopic to the map $k : S_c \to \mathbb{C} \setminus \{0\}$ given by $k(z) = z^n$. [**Hint:** Set $F(z,t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$.]

(c) Derive a contradiction.

3. Show that the map k defined in **Step 1** of the proof of Theorem 2.9.8 is a homeomorphism by following steps:

(a) Show that the maps

$$\begin{aligned} p : |K| \times [-1,1] &\rightarrow |S(K)| \\ q : |K| \times [-1,1] &\rightarrow \mathbb{S}^n \,, \end{aligned}$$

given by the equations

$$p(x,t) = \begin{cases} (1-t)x + tw_0 & \text{if} \quad t \geq 0\,, \\ (1+t)x - tw_1 & \text{if} \quad t < 0 \end{cases}$$

and

$$q(x,t) = (\sqrt{1-t^2}h(x), t)$$

are quotient maps.

(b) Show that the quotient maps p, q induce the homeomorphism k.

4. Refer to Exercise 1**(c)**. Show that the result is true for integers $n \leq 0$.

5. Refer to Exercise 1**(b)**. Let m be any complex constant. Show that the function $f(z) = (z+m)^2$ has degree 2. [**Hint:** Use a homotopy argument to answer this question.]

2.10 *The Lefschetz Fixed Point Theorem

The fixed point theorems proved earlier concern maps of spheres and balls to themselves. There is a far-reaching generalization due to Solomon Lefschetz that applies to more arbitrary topological spaces. We shall prove it in this section.

We begin by reviewing some ideas from algebra.

If $A = (a_{ij})$ is an $n \times n$ square matrix, then the *trace* of A, denoted $\operatorname{tr} A$, is defined by

$$\operatorname{tr} A = \sum_{j=1}^{n} a_{jj}\,.$$

If A and B are $n \times n$ matrices, then

$$\operatorname{tr} AB = \sum_{i,j} a_{ij} b_{ji} = \operatorname{tr} BA\,.$$

If G is a free abelian group with basis $e_1, e_2, \ldots, e_n$, and if $\phi : G \to G$ is a homomorphism, then we define the trace of ϕ to be the number $\operatorname{tr} A$, where A is the matrix of ϕ relative to the given basis. This number is independent of the choice of basis, since the matrix of ϕ relative to another basis equals $B^{-1}AB$ for some square matrix B, and $\operatorname{tr}(B^{-1}(AB)) = \operatorname{tr}((AB)B^{-1}) = \operatorname{tr}(A)$. The same argument shows that if $i : G \to G$ is an isomorphism, then $\operatorname{tr}(i \circ \phi \circ i^{-1}) = \operatorname{tr}(\phi)$.

If K is a finite complex, and if $\phi : C_p(K) \to C_p(K)$ is a chain map, then since $C_p(K)$ is free abelian with finite rank, the trace of ϕ is defined. We denote it by $\operatorname{tr}(\phi, C_p(K))$. The group $H_p(K)$ is not necessarily free abelian, but if $T_p(K)$ is its torsion subgroup, then $H_p(K)/T_p(K)$ is free abelian. Furthermore, ϕ_* induces a homomorphism of this group with itself. We use the notation $\operatorname{tr}(\phi_*, H_p(K)/T_p(K))$ to denote the trace of the induced homomorphism.

There is no obvious relation between the two numbers $\operatorname{tr}(\phi, C_p(K))$ and $\operatorname{tr}(\phi_*, H_p(K)/T_p(K))$. As a result, the following formula is rather striking.

Theorem 2.10.1 (Hopf trace theorem) *Let K be a finite complex. Let $\phi : C_p(K) \to C_p(K)$ be a chain map. Then*

$$\sum_p (-1)^p \operatorname{tr}(\phi, C_p(K)) = \sum_p (-1)^p \operatorname{tr}(\phi_*, H_p(K)/T_p(K)) .$$

Proof: Step 1: Let G be free abelian of finite rank, let H be a subgroup (also automatically free abelian), and suppose that G/H is free abelian. Let $\phi : G \to G$ be a homomorphism that carries H into itself. We show that

$$\operatorname{tr}(\phi, G) = \operatorname{tr}(\phi', G/H) + \operatorname{tr}(\phi'', H) ,$$

where ϕ', ϕ'' are the homomorphisms induced by ϕ.

Let

$$\{\alpha_1 + H, \alpha_2 + H, \ldots, \alpha_n + H\}$$

be a basis for G/H and let $\beta_1, \beta_2, \ldots, \beta_p$ be a basis for H. If A and B are the matrices of ϕ' and ϕ'' relative to these respective bases, then

$$\phi'(\alpha_j + H) = \sum_i a_{ij}(\alpha_i + H) ,$$

$$\phi''(\beta_j) = \sum_i b_{ij}\beta_i .$$

Now it is easy to check that $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_p$ is a basis for G. Furthermore, it follows from the preceding equations that

$$\phi(\alpha_j) = \sum_i a_{ij}\alpha_i + (\text{something in } H) ,$$

$$\phi(\beta_j) = \sum_i b_{ij}\beta_i .$$

Therefore the matrix of ϕ relative to this basis for G has the form

$$C = \begin{bmatrix} A & 0 \\ * & B \end{bmatrix}.$$

Obviously, $\operatorname{tr} C = \operatorname{tr} A + \operatorname{tr} B$ and our result follows.

Step 2: As usual let C_p denote the chain group $C_p(K)$. Let Z_p denote the p-cycles, B_p the p-boundaries. Let W_p denote the group of *weak p-boundaries* which are those p-chains some multiple of which is a boundary. Then

$$B_p \subset W_p \subset Z_p \subset C_p .$$

Since ϕ is a chain map, it carries each of these groups into itself. We shall show that the quotient groups C_p/Z_p and Z_p/W_p are free. It will follow then from **Step 1** that

$$\operatorname{tr}(\phi, C_p) = \operatorname{tr}(\phi, C_p/Z_p) + \operatorname{tr}(\phi, Z_p/W_p) + \operatorname{tr}(\phi, W_p) . \qquad (i)$$

[Here we abandon the use of primes to distinguish among the various induced homomorphisms.] We now compute each of these terms.

Step 3: Consider the group C_p/Z_p. The homomorphism $\partial_0 : C_p \to B_{p-1}$ obtained by restricting the range of ∂ is surjective and has kernel equal to Z_p. Therefore it induces an isomorphism of C_p/Z_p with B_{p-1}. Hence C_p/Z_p is free. Also, because ϕ commutes with ∂, it commutes with this isomorphism. Hence

$$\operatorname{tr}(\phi, C_p/Z_p) = \operatorname{tr}(\phi, B_{p-1}) . \qquad (ii)$$

Step 4: Consider the group Z_p/W_p. Consider the projection mappings

$$Z_p \to Z_p/B_p = H_p \to H_p/T_p .$$

This composition is surjective and has kernel W_p. Therefore it induces an isomorphism

$$Z_p/W_p \to H_p/T_p .$$

We conclude that Z_p/W_p is free. Because the projections commute with ϕ, so does this isomorphism. Thus

$$\operatorname{tr}(\phi, Z_p/W_p) = \operatorname{tr}(\phi_*, H_p/T_p) . \qquad (iii)$$

Step 5: Consider the group W_p. We show that

$$\operatorname{tr}(\phi, W_p) = \operatorname{tr}(\phi, B_p) . \qquad (iv)$$

Recall that B_p is a subgroup of W_p. Applying the basic theorem of free abelian groups, we can choose a basis $\alpha_1, \ldots, \alpha_n$ of W_p such that, for some integers $m_1, \ldots, m_k \geq 1$, the elements $m_1\alpha_1, \ldots, m_k\alpha_k$ form a basis for B_p.

Now W_p/B_p is a torsion group. Therefore $k = n$. We compute the trace of ϕ on W_p and B_p. Let

$$\phi(\alpha_j) = \sum_i a_{ij}\alpha_i$$

and

$$\phi(m_j\alpha_j) = \sum_i b_{ij}(m_i\alpha_i)\,,$$

where the summations extend from 1 to n. Then $\mathrm{tr}(\phi, W_p) = \sum_i a_{ii}$ and $\mathrm{tr}(\phi, B_p) = \sum_i b_{ii}$ by definition. Multiplying the first equation by m_j, we conclude that $m_j a_{ij} = b_{ij} m_i$ for all i and j. In particular, $a_{ii} = b_{ii}$ for all i. Thus $\mathrm{tr}(\phi, W_p) = \mathrm{tr}(\phi, B_p)$.

Step 6: To complete the proof, we substitute formulas (ii), (iii), and (iv) into (i) to obtain the equation

$$\mathrm{tr}(\phi, C_p) = \mathrm{tr}(\phi, B_{p-1}) + \mathrm{tr}(\phi_*, H_p/T_p) + \mathrm{tr}(\phi, B_p)\,.$$

If we multiply this equation by $(-1)^p$ and sum over p, then the first and last terms on the right-hand side cancel out in pairs. The desired formula is the result. □

Definition 2.10.2 The *Euler number* of a finite complex K is defined, classically, by the equation

$$\chi(K) = \sum_p (-1)^p \mathrm{rank}(C_p(K))\,.$$

In words, the Euler number is the alternating sum of the number of simplices of K in each dimension.

We next show that the Euler number of K is a topological invariant of $|K|$.

Theorem 2.10.3 *Let K be a finite complex. Let $\beta_p = \mathrm{rank}(H_p(K)/T_p(K))$. This is the betti number of K in dimension p. Then*

$$\chi(K) = \sum_p (-1)^p \beta_p\,.$$

Proof: If $\phi : C_p(K) \to C_p(K)$ is the identity map, then the matrix of ϕ with respect to any basis is the identity matrix. We conclude that $\mathrm{tr}(\phi, C_p) = \mathrm{rank}(C_p)$. Similarly, because ϕ_* is the identity map, $\mathrm{tr}(\phi_*, H_p/T_p) = \mathrm{rank}(H_p/T_p) = \beta_p$. Our formula now follows from Theorem 2.10.1, the Hopf trace theorem. □

This last theorem has a number of consequences. For instance, the fact that $\chi(K) = 2$ if $|K|$ is the boundary of a convex, open region in $\mathbb{R}^3$ (since $|K|$ is homeomorphic to $\mathbb{S}^2$) can be used to show that there are only five such complexes that are regular polyhedra. These give the five classical Platonic solids.

Definition 2.10.4 Let K be a finite complex. Let $h : |K| \to |K|$ be a continuous map. The number

$$\Lambda(h) = \sum_p (-1)^p \mathrm{tr}(h_*, H_p(K)/T_p(K))$$

is called the *Lefschetz number* of h.

Note that, by Theorem 2.7.5, $\Lambda(h)$ depends only on the homotopy class of h. Also, it only depends on the topological space $|K|$, not on the particular complex K. If L is another complex with $|L| = |K|$, then the homomorphism

$$j_* : H_p(L) \to H_p(K)$$

induced by the identity is an isomorphism. The fact that $(h_L)_* = j_*^{-1} \circ (h_K)_* \circ j_*$ implies that $(h_L)_*$ and $(h_K)_*$ have the same trace.

Theorem 2.10.5 (Lefschetz fixed point theorem) *Let K be a finite complex and $h : |K| \to |K|$ a continuous map. If $\Lambda(h) \neq 0$ then h has a fixed point.*

Proof: Assume that h has no fixed point. Then we will prove that $\Lambda(h) = 0$.

Step 1: We show that we may assume that

$$h(\overline{\mathrm{St}}(v, K)) \cap \overline{\mathrm{St}}(v, K) = \emptyset \quad \text{for all vertices } v \text{ of } K. \qquad (*)$$

To begin, let $\epsilon = \min |x - h(x)|$. By the uniform continuity of h, we may select $\delta > 0$ so that when $|x - y| < \delta$ then $|h(x) - h(y)| \leq \epsilon/3$. Let $\lambda = \min\{\delta, \epsilon/2\}$. Then, for any set A of diameter less than λ, both A and $h(A)$ have diameter less than $\epsilon/2$, so they are necessarily disjoint. Replace K by a subdivision of K in which the closed stars have diameter less than λ. Then the condition $(*)$ holds. As noted earlier, this does not change the value of $\Lambda(h)$.

Step 2: Assume that K satisfies $(*)$. Let us choose a subdivision K' of K such that h has a simplicial approximation $f : K' \to K$. We show that if s and σ are simplices of K' and K respectively, such that $s \subset \sigma$, then $f(s) \neq \sigma$.

Suppose instead that $f(s) = \sigma$. Let w be a vertex of s, and let $f(w) = v$, a vertex of σ. The fact that $s \subset \sigma$ implies that $w \in \overline{\mathrm{St}}(v, K)$ so that

$$h(w) \in h(\overline{\mathrm{St}}(v, K)) \,. \qquad (**)$$

On the other hand, we have by the definition of simplicial approximation,

$$h(\mathrm{St}(w, K')) \subset \overline{\mathrm{St}}(f(w), K) \,.$$

This implies that

$$h(w) \in \overline{\mathrm{St}}(v, K) \,. \qquad (***)$$

The combination of $(**)$ and $(***)$ contradicts $(*)$.

Step 3: Now we compute $\Lambda(h)$ by applying Theorem 2.10.1, the Hopf trace theorem.

Let $f : K' \to K$ be a simplicial approximation to h. Let $\lambda : \mathcal{C}(K) \to \mathcal{C}(K')$ be the subdivision operator. Then h_* is induced by the chain map $\phi = f_\# \circ \lambda$ by definition.

We compute the trace of ϕ. Let A be the matrix of ϕ relative to the usual basis for $C_p(K)$, which consists of oriented p-simplices of K. Let σ be a typical basis element. The chain $\lambda(\sigma)$ is a linear combination of oriented simplices s of K' such that $s \subset \sigma$. For any such simplex s, it follows from **Step 2** that $f(s) \neq \sigma$. We conclude that the chain $\phi(\sigma) = f_\#(\lambda(\sigma))$ is a linear combination of oriented p-simplices of K different from σ. The matrix A thus has an entry 0 in the row and column corresponding to σ. It follows that all the diagonal entries of A vanish, so that $\operatorname{tr} A = \operatorname{tr} \phi = 0$.

The Hopf trace theorem tells us now that

$$\Lambda(h) = \sum_p (-1)^p \operatorname{tr}(\phi, C_p(K)) .$$

Since each of the terms in the summation vanishes, we conclude that $\Lambda(h) = 0$. □

In order to apply Lefschetz's theorem, we need the following lemma:

Lemma 2.10.6 *Let K be a finite complex. Let $h : |K| \to |K|$ be a continuous map. If $|K|$ is connected, then $h_* : H_0(K) \to H_0(K)$ is the identity.*

Proof: Let $f : K' \to K'$ be a simplicial approximation to h. If v is a vertex of K, then the subdivision operator λ carries v to a 0-chain carried by the subdivision of v, which is just v itself. Thus $\lambda(v)$ is a multiple of v. Because λ preserves augmentation, $\lambda(v) = v$.

Thus $f_\#\lambda(v) = f_\#(v)$, which is a vertex of K. Because $|K|$ is connected, $f_\#(v)$ is homologous to v. Therefore $h_* = f_* \circ \lambda_*$ equals the identity on $H_0(K)$. □

Theorem 2.10.7 *Let K be a finite complex. Let $h : |K| \to |K|$ be a continuous map. If $|K|$ is acyclic, then h has a fixed point.*

Proof: The group $H_0(K)$ is infinite cyclic, and h_* is the identity on $H_0(K)$. Thus $\operatorname{tr}(h_*, H_0(K)) = 1$. Since all the higher dimensional homology groups vanish, we see that $\Lambda(h) = 1$, So h has a fixed point. □

Theorem 2.10.8 *The antipodal map of $\mathbb{S}^n$ has degree $(-1)^{n+1}$.*

Proof: Let $h : \mathbb{S}^n \to \mathbb{S}^n$ be a map of degree d. We compute $\Lambda(h)$. Now h_* is the identity on 0-dimensional homology. On n-dimensional homology, its matrix is a one-by-one matrix with a single entry $d = \text{degree } f$. Therefore

$$\Lambda(h) = 1 + (-1)^n d\,. \qquad (\dagger)$$

The antipodal map a has no fixed point, so Theorem 2.10.5, the Lefschetz fixed point theorem, tells us that $\Lambda(a) = 0$. Applying (†), we obtain

$$0 = \Lambda(a) = 1 + (-1)^n \deg a\,.$$

It follows that the degree of a is $(-1)^{n+1}$. □

The next definition is used in Exercise 6.

Definition 2.10.9 Suppose that B is a finite collection of polygons in $\mathbb{R}^3$, each two of which intersect in at most a common edge or a common vertex. Each of these polygons is called a *face* of B. Its edges are called *edges* of B, and its vertices are called *vertices* of B.

We say that B is *combinatorially regular* if all the faces of B have the same number (say k) of edges, if each edge of B belongs to exactly two faces of B, and if each vertex of B belongs to the same number (say ℓ) of faces of B.

The triangles making up the boundary of a tetrahedron and the squares making up the boundary of a cube are examples of combinatorially regular collections of polygons.

EXERCISES

1. Show that every map $f : \mathbb{P}^2 \to \mathbb{P}^2$ has a fixed point.

2. It is not true that each continuous map of $\mathbb{S}^0$ to $\mathbb{S}^0$ (the space with two points) has a fixed point. It is also not true that every continuous map of $\mathbb{S}^1$ to $\mathbb{S}^1$ (the circle) has a fixed point. Verify these assertions. What can you say about continuous maps of $\mathbb{S}^2$ to $\mathbb{S}^2$?

3. Let K be a finite complex. Show that if $h : |K| \to |K|$ is homotopic to a constant map, then h has a fixed point.

4. **(a)** Show that there is a map of the Klein bottle S to itself that carries the curve C shown in Figure 2.10.1 homeomorphically onto the curve D, and a map g that carries C into the shaded region E.

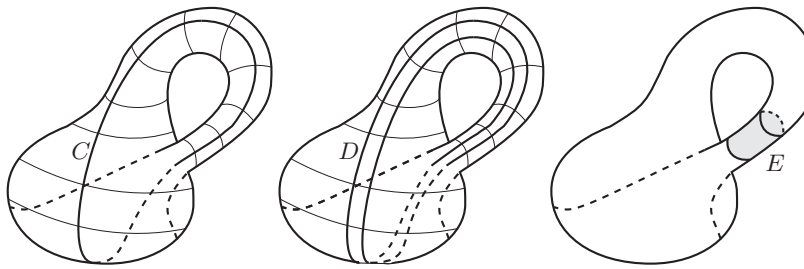


Figure 2.10.1

(b) Let f and g be as in part **(a)**. Show that if $f \simeq f'$ and $g \simeq g'$, then f' and g' have fixed points.

(c) Find a map $h : S \to S$ that has no fixed points.

5. If M is a compact, smooth surface in $\mathbb{R}^n$, and if $v(x)$ is a tangent vector field to M, then it is a standard result of differential geometry that for some $\epsilon > 0$ there is a continuous map

$$F : M \times (-\epsilon, \epsilon) \to M$$

having the property that, for each x_0, the curve

$$t \mapsto F(x_0, t)$$

passes through the point x_0 when $t = 0$ with velocity vector $v(x_0)$. Furthermore, if $v(x) \neq 0$ for all x, then there is a $\delta > 0$ such that $F(x_0, t) \neq x_0$ for all $0 < |t| < \delta$.

(a) Using these facts, show that if M has a non-zero tangent vector field, then $\chi(M) = 0$.

(b) Determine which compact surfaces have non-zero tangent vector fields.

6. Suppose that B is a finite collection of polygons in $\mathbb{R}^3$, each two of which intersect in at most a common edge or a common vertex, as in Definition 2.10.9.

(a) Let $|B|$ be the union of the elements of B. Show that the Euler number of B is given by

$$\chi(B) = (\#\text{ faces}) - (\#\text{ edges}) + (\#\text{ vertices})\,.$$

[**Hint:** Subdivide B into a simplicial complex by starring from an interior point of each face.]

(b) Show that there are only five combinatorially regular collections B such that $|B|$ is homeomorphic to $\mathbb{S}^2$. [**Hint:** Geometric considerations imply that $k \geq 3$ and $\ell \geq 3$.]

(c) There are many combinatorially regular collections B such that $|B|$ is homeomorphic to the torus. But there are only a limited number of possibilities for k (the number of edges) and ℓ (the number of faces). Say what these possibilities are.

7. Let $B \subset \mathbb{R}^N$ be the open unit ball. Give an example of a map $f : B \to B$ which is continuous and which has no fixed point. Now give an example of the same which is one-to-one and onto.

8. Let $\mathbb{B}^N \subset \mathbb{R}^N$ be the closed unit ball. Give an example of a map $f : \mathbb{B}^N \to \mathbb{B}^N$ which is discontinuous and has no fixed point.

9. Let

$$S = \{(\cos t, \sin t, t) : 0 \leq t \leq 4\pi\}.$$

Define

$$E = \{x \in \mathbb{R}^3 : \text{dist}(x, S) \leq 1/10\}.$$

Show that a continuous map $f : E \to E$ will have a fixed point.

3

Relative Homology and the Eilenberg–Steenrod Axioms

3.1 Introduction

Until now we have concentrated primarily on studying the "absolute" simplicial homology groups, though we have defined the relative groups and have proved their topological invariance. Now we shall study the relative groups in more detail. These groups have many uses. For one thing, they arise naturally in many of the applications of topology. A second point is that they are involved in an essential way in expressing those fundamental properties of homology that are called the Eilenberg–Steenrod axioms.

One of the ways that relative homology groups can be useful is for giving information about the absolute homology groups. There are relationships between the relative groups $H_p(K, K_0)$ and the absolute groups $H_p(K)$ and $H_p(K_0)$. For example, the vanishing of $H_p(K, K_0)$ and $H_{p+1}(K, K_0)$ implies that $H_p(K)$ and $H_p(K_0)$ are isomorphic. This fact is not obvious. Formulating this relationship in a precise and general manner is a subtle problem.

In the early days of algebraic topology, the theorems described above were often awkward and wordy. There was a need for the right language to express the ideas. A remarkable algebraic idea due to Eilenberg clarified the matter notably. Really just a new and convenient notation, it is called an "exact sequence" (of groups, or rings, or some other algebraic structure). The usefulness of this concept, in algebra as well as topology, is difficult to overestimate. Obscure algebraic arguments often become quite transparent once they are formulated in terms of exact sequences. Other arguments that were difficult even for professional algebraists become so straightforward that they can be safely left to the reader.

If some terminology in this chapter regarding groups seems unfamiliar, the reader should refer to Section 1.5 for clarification.

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3.2 The Exact Homology Sequence

The relationship between the absolute and relative homology groups is expressed by what is called the "exact homology sequence of a pair." In the present section, we shall study exact sequences of groups (usually abelian groups), and we shall define the exact homology sequence of a pair.

Definition 3.2.1 Consider a sequence (finite or infinite) of groups and homomorphisms

$$\cdots \longrightarrow A_1 \xrightarrow{\phi_1} A_2 \xrightarrow{\phi_2} A_3 \longrightarrow \cdots .$$

This sequence is said to be *exact at* A_2 if

$$\text{image}\,\phi_1 = \text{kernel}\,\phi_2 .$$

If it is exact at each group then it is said simply to be an *exact sequence*. Of course exactness is not defined at the first group or the last group (if these exist).

The items in the next lemma should all be quite familiar to you. Proofs are left as exercises. Because we are considering only abelian groups, we let 0 denote the trivial (one-element) group.

Lemma 3.2.2 *The following facts hold for exact sequences:*

(1) $A_1 \xrightarrow{\phi} A_2 \to 0$ *is exact if and only if* ϕ *is surjective.*

(2) $0 \to A_1 \xrightarrow{\phi} A_2$ *is exact if and only if* ϕ *is injective.*

(3) *Suppose that the sequence*

$$0 \longrightarrow A_1 \xrightarrow{\phi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 0$$

is exact. Such a sequence is called a short exact sequence. *Then* $A_2/\phi(A_1)$ *is isomorphic to* A_3. *The isomorphism is induced by* ψ. *Conversely if* $\psi : A \to B$ *is surjective with kernel* K, *then the sequence*

$$0 \longrightarrow K \xrightarrow{i} A \xrightarrow{\psi} B \longrightarrow 0$$

is exact, where i *denotes the inclusion map.*

(4) *Suppose that the sequence*

$$A_1 \xrightarrow{\alpha} A_2 \xrightarrow{\phi} A_3 \xrightarrow{\beta} A_4$$

is exact. Then the following are equivalent:

(i) α *is surjective;*

(ii) *β is injective;*

(iii) *ϕ is the zero homomorphism.*

(5) *Suppose that the sequence*

$$A_1 \xrightarrow{\alpha} A_2 \longrightarrow A_3 \longrightarrow A_4 \xrightarrow{\beta} A_5$$

is exact. Then so is the induced sequence

$$0 \longrightarrow \operatorname{cok}\alpha \longrightarrow A_3 \longrightarrow \ker\beta \longrightarrow 0\,.$$

Definition 3.2.3 Consider two sequences of groups and homomorphisms having the same index set:

$$\cdots \longrightarrow A_1 \longrightarrow A_2 \longrightarrow \cdots$$

$$\cdots \longrightarrow B_1 \longrightarrow B_2 \longrightarrow \cdots\,.$$

A *homomorphism* of the first sequence into the second sequence is a family of homomorphisms $\alpha_i : A_i \to B_i$ such that each square of maps

$$\begin{array}{ccc} A_i & \longrightarrow & A_{i+1} \\ {\scriptstyle\alpha_i}\downarrow & & \downarrow{\scriptstyle\alpha_{i+1}} \\ B_i & \longrightarrow & B_{i+1} \end{array}$$

commutes. It is an *isomorphism* of sequences if each α_i is an isomorphism.

For example, if a chain complex $\mathcal{C}$ is viewed as a sequence of groups C_i and homomorphisms ∂_i, then a homomorphism of one such sequence $\mathcal{C}$ into another $\mathcal{D}$ is precisely what we have called a chain map of $\mathcal{C}$ into $\mathcal{D}$.

Definition 3.2.4 Consider a short exact sequence

$$0 \longrightarrow A_1 \xrightarrow{\phi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 0\,.$$

The sequence is said to *split* if the group $\phi(A_1)$ is a direct summand in A_2.

The property of being split means that A_2 is the internal direct sum of $\phi(A_1)$ and some other subgroup B. The group B is of course not uniquely determined. Thus the sequence becomes

$$0 \longrightarrow A_1 \xrightarrow{\phi} \phi(A_1) \oplus B \xrightarrow{\psi} A_3 \longrightarrow 0\,,$$

where ϕ defines an isomorphism of A_1 with $\phi(A_1)$ and ψ defines an isomorphism of B with A_3. An equivalent formulation is to state that there is an isomorphism θ such that the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & 0 \\ & & \downarrow{=} & & \downarrow{\theta} & & \downarrow{=} & & \\ 0 & \longrightarrow & A_1 & \xrightarrow[i]{} & A_1 \oplus A_3 & \xrightarrow[\pi]{} & A_3 & \longrightarrow & 0\,. \end{array}$$

In the present instance, $\oplus$ denotes the external direct sum. Also, i is inclusion and π is projection. The map θ is defined by writing $A_2 = \phi(A_1) \oplus B$ and letting θ equal ϕ^{-1} on the first summand and equal to ψ on the second summand.

Theorem 3.2.5 *Let*

$$0 \longrightarrow A_1 \xrightarrow{\phi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 0$$

be exact. Then the following are equivalent:

(1) *The sequence splits.*

(2) *There is a map $p : A_2 \to A_1$ such that $p \circ \phi = i_{A_1}$.*

(3) *There is a map $j : A_3 \to A_2$ such that $\psi \circ j = i_{A_3}$. We have*

$$0 \longrightarrow A_1 \underset{p}{\overset{\phi}{\rightleftarrows}} A_2 \underset{j}{\overset{\psi}{\rightleftarrows}} A_3 \longrightarrow 0\,.$$

Proof: We begin by showing that **(1)** implies **(2)** and **(3)**. It is enough to prove **(2)** and **(3)** for the sequence

$$0 \longrightarrow A_1 \xrightarrow{i} A_1 \oplus A_3 \xrightarrow{\pi} A_3 \longrightarrow 0\,.$$

This is straightforward: We define $p : A_1 \oplus A_3 \to A_1$ as projection and we define $j : A_3 \to A_1 \oplus A_3$ as inclusion.

(**2**) $\Rightarrow$ (**1**): We show that $A_2 = \phi(A_1) \oplus (\ker p)$. First, for $x \in A_2$ we can write $x = \phi p(x) + (x - \phi p(x))$. The first term is obviously in $\phi(A_1)$. The second term is in $\ker p$ because

$$p(x - \phi p(x)) = p(x) - p(\phi p(x)) = p(x) - (p\phi)p(x) = 0\,.$$

Next, if $x \in \phi(A_1) \cap \ker p$, then $x = \phi(y)$ for some y. Hence $p(x) = p\phi(y) = y$. Since $x \in \ker p$, the element y vanishes, so $x = \phi(y)$ vanishes also.

(**3**) $\Rightarrow$ (**1**): We show that $A_2 = (\ker \psi) \oplus j(A_3)$. Since $\ker \psi = \operatorname{im} \phi$, this will suffice for our purposes. For $x \in A_2$, we can write $x = (x - j\psi(x)) + j\psi(x)$. The first term is in $\ker \psi$ because $\psi(x) - (\psi j)(x)\psi(x) = \psi(x) - \psi(x) = 0$. The second term is obviously in $j(A_3)$. Finally, if $x \in (\ker \psi) \cap j(A_3)$, then $x = j(z)$ for some z, hence $\psi(x) = \psi j(z) = z$. Since $x \in \ker \psi$, the element z vanishes so $x = j(z)$ vanishes also. □

Corollary 3.2.6 *Let*

$$0 \longrightarrow A_1 \xrightarrow{\phi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 0$$

be an exact sequence. If A_3 is free abelian, then the sequence splits.

Proof: We choose a basis for A_3, and for each basis element e, we define the value of $j : A_3 \to A_2$ on that basis element to be any element of the nonempty set $\psi^{-1}(e)$. □

Now that we have established some basic facts about exact sequences, we can describe the exact homology sequence of a pair.

First we need to define a certain homomorphism

$$\partial_* : H_p(K, K_0) \to H_{p-1}(K_0)$$

that is induced by the boundary operator and is called the *homology boundary homomorphism*. We construct it as follows: Given a cycle z in $C_p(K, K_0)$, it is the coset modulo $C_p(K_0)$ of a chain d of K whose boundary is carried by K_0. The chain ∂d is automatically a cycle of K_0. We define $\partial_*\{z\} = \{\partial d\}$, where $\{\ \}$ means "homology class of." We will show later in the proof of Lemma 3.3.2 that ∂_* is a well-defined homomorphism.

Algebraically, the construction of ∂_* can be described as follows: Consider the following diagram, where $i : K_0 \to K$ and $\pi : (K, \emptyset) \to (K, K_0)$ are inclusions:

$$\begin{array}{ccc} & C_p(K) & \xrightarrow{\pi_\#} C_p(K, K_0) \\ & \big\downarrow \partial & \\ C_{p-1}(K_0) & \xrightarrow{i_\#} C_{p-1}(K) & \end{array}$$

Now $i_\#$ is inclusion and $\pi_\#$ is projection of $C_p(K)$ onto $C_p(K)/C_p(K_0)$. Given a cycle z of $C_p(K, K_0)$, the chain d of $C_p(K)$ representing it is a chain such that $\pi_\#(d) = z$. We consider ∂d, and note that since it is carried by K_0, it equals $i_\#(c)$ for some $p-1$ chain c of K_0. Now c is actually a cycle. Its homology class is defined to be $\partial_*(\{z\})$. Thus ∂_* is defined by a certain "zig-zag" process: Pull back via $\pi_\#$; apply ∂; pull back via $i_\#$. This description of ∂_* will be useful in the next section.

Now we can state our basic result relating the homology of K, K_0, and (K, K_0).

Definition 3.2.7 A *long exact sequence* is an exact sequence whose index set is the set of integers. That is to say, it is a sequence that is infinite in both directions. It may, however, end with an infinite string of trivial groups.

Theorem 3.2.8 (The exact homology sequence of a pair)
Let K be a complex. Let K_0 be a subcomplex. Then there is a long exact sequence

$$\cdots \longrightarrow H_p(K_0) \xrightarrow{i_*} H_p(K) \xrightarrow{\pi_*} H_p(K, K_0) \xrightarrow{\partial_*} H_{p-1}(K_0) \longrightarrow \cdots,$$

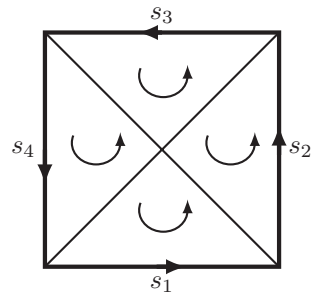


Figure 3.2.1

where $i : K_0 \to K$ *and* $\pi : (K, \emptyset) \to (K, K_0)$ *are inclusions and* ∂_* *is induced by the boundary operator* ∂. *There is a similar exact sequence in reduced homology:*

$$\cdots \longrightarrow \widetilde{H}_p(K_0) \longrightarrow \widetilde{H}_p(K) \longrightarrow H_p(K, K_0) \longrightarrow \widetilde{H}_{p-1}(K_0) \longrightarrow \cdots .$$

The proof of this theorem is basically algebraic in nature. We shall formulate it in a purely algebraic fashion and prove it in the next section. (See Lemma 3.3.2 and the remarks following the proof of that lemma.) For the moment, we shall apply this sequence to several specific examples.

EXAMPLE 3.2.9 Let K be the complex depicted in Figure 3.2.1, whose polytope is a square region. Let K_0 be the subcomplex whose polytope is the boundary of the square. We know from Example 1.10.4 that $H_2(K, K_0)$ is infinite cyclic and is generated by the 2-chain γ that is the sum of the 2-simplices, oriented counterclockwise. Furthermore, $H_1(K_0)$ is infinite cyclic and is generated by the 1-chain $s_1 + s_2 + s_3 + s_4$, which happens to equal ∂_γ. Thus, in this particular case, the boundary homomorphism

$$\partial_* : H_2(K, K_0) \to H_1(K_0)$$

is an isomorphism.

This fact can also be proved by considering the exact homology sequence of the pair (K, K_0). A portion of this sequence is

$$H_2(K) \longrightarrow H_2(K, K_0) \xrightarrow{\partial_*} H_1(K_0) \longrightarrow H_1(K) .$$

The end groups vanish because $|K|$ is contractible. Therefore ∂_* is an isomorphism by Lemma 3.2.2**(1,2)**.

EXAMPLE 3.2.10 Let K be the complex depicted in Figure 3.2.2. The underlying space is an annulus. Let K_0 be the subcomplex of K whose polytope equals the union of the inner and outer edges of the square. In Example 1.10.5 we computed the homology of (K, K_0). We recompute it here using Lemma 3.2.2 and our knowledge of the homology of K and K_0.

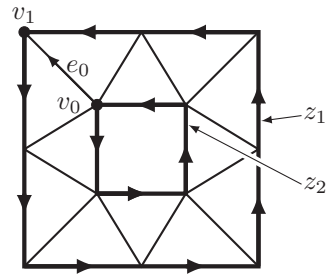


Figure 3.2.2

Consider the following portion of the exact sequence in reduced homology:

$$\begin{array}{ccccccccccccc} 0 & \to & H_2(K,K_0) & \xrightarrow{\partial_*} & H_1(K_0) & \xrightarrow{i_*} & H_1(K) & \xrightarrow{\pi_*} & H_1(K,K_0) & \xrightarrow{\partial_*} & \widetilde{H}_0(K_0) & \to & 0 \\ & & \| & & \| & & \| & & \| & & \| & & \\ 0 & \to & ? & \longrightarrow & \mathbb{Z}\oplus\mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & ? & \longrightarrow & \mathbb{Z} & \to & 0\,. \end{array}$$

The zeros at the ends come from the fact that $|K|$ has the homotopy type of a circle, hence $H_2(K) = \widetilde{H}_0(K) = 0$. Furthermore, $H_1(K)$ is infinite cyclic. It is generated by the cycle z_1 shown in Figure 3.2.2 running around the outer edge of K or else by the cycle z_2 running around the inner edge. These two cycles are homologous. Because $|K_0|$ is topologically the disjoint union of two circles, $H_1(K_0) \cong \mathbb{Z}\oplus\mathbb{Z}$ and has as basis the cosets of z_1 and z_2. At the same time, $\widetilde{H}_0(K_0) \cong \mathbb{Z}$ and is generated by $\{v_1 - v_2\}$.

If $i : K_0 \to K$ is inclusion, then i_* maps both $\{z_1\}$ and $\{z_2\}$ to the same generator of $H_1(K)$. It follows that **(1)** i_* is surjective and **(2)** its kernel is infinite cyclic and is generated by $\{z_1\} - \{z_2\}$.

From the first fact we conclude that π_* is the zero homomorphism hence

$$\partial_* : H_1(K,K_0) \to \widetilde{H}_0(K_0)$$

is an isomorphism. Thus $H_1(K,K_0)$ is infinite cyclic and is generated by the 1-chain e_0 whose boundary is $v_1 - v_0$.

From the second fact we see that since

$$\partial_* : H_2(K,K_0) \to \ker i_*$$

is an isomorphism, the group $H_2(K,K_0)$ is infinite cyclic. Also it is generated by the chain γ that is the sum of the 2-simplices of K, oriented counterclockwise. For we have $|\partial\gamma| = |z_1 - z_2|$ and $z_1 - z_2$ generates $\ker i_*$.

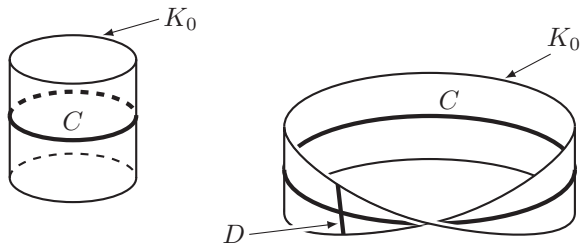


Figure 3.2.3

EXAMPLE 3.2.11 The next two examples we will consider together. Let (K, K_0) denote either the cylinder and its top edge or the Möbius band together with its single edge. In each case $|K_0|$ is a circle. Furthermore, $|K|$ has the homotopy type of a circle. The central circle C shown in Figure 3.2.3 is a deformation retract of $|K|$.

Thus we know the homology of K and K_0. We compute the homology of (K, K_0) from the exact homology sequence, a portion of which is as shown:

$$\begin{array}{ccccccccccc} H_2(K) & \longrightarrow & H_2(K,K_0) & \longrightarrow & H_1(K_0) & \xrightarrow{i_*} & H_1(K) & \longrightarrow & H_1(K,K_0) & \longrightarrow & \widetilde{H}_0(K_0) \\ \| & & \| & & \| & & \| & & \| & & \| \\ 0 & \longrightarrow & ? & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & ? & \longrightarrow & 0\,. \end{array}$$

Everything depends on computing the homomorphism i_*. Since the retraction $r : |K| \to C$ is a homotopy equivalence, it suffices to compute the homomorphism induced by the composite map $r \circ i : |K_0| \to C$, which collapses the edge $|K_0|$ to the central circle C. In the case of the cylinder, this map has degree 1. But in the case of the Möbius band, the induced homomorphism clearly equals multiplication by 2.

Thus $H_2(K, K_0) = H_1(K, K_0) = 0$ in the case of the cylinder. In the case of the Möbius band we have

$$H_2(K, K_0) = 0 \qquad \text{and} \qquad H_1(K, K_0) \cong \mathbb{Z}/2\,.$$

The central circle C represents the non-zero element of $H_1(K, K_0)$ in this case. So also does the chain D, which you may prove as an exercise.

EXERCISES

1. Verify statements **(1)**–**(5)** of Lemma 3.2.2 concerning the properties of exact sequences.

2. **(a)** Suppose that

$$A_1 \xrightarrow{\alpha_1} B_1 \xrightarrow{\beta_1} C_1 \qquad \text{and} \qquad A_2 \xrightarrow{\alpha_2} B_2 \xrightarrow{\beta_2} C_2$$

are exact. Show that

$$A_1 \times A_2 \xrightarrow{\alpha_1 \times \alpha_2} B_1 \times B_2 \xrightarrow{\beta_1 \times \beta_2} C_1 \times C_2$$

is exact.

(b) Generalize part **(a)** to arbitrary direct products.

(c) Generalize part **(a)** to arbitrary direct sums.

3. Show that if K_0 is acyclic, then $H_i(K, K_0) \cong \widetilde{H}_i(K)$.

4. Suppose that the inclusion $|K_0| \to |K|$ is a homotopy equivalence. Show that $H_p(K, K_0) = 0$ for all p.

5. Let (K, K_0) denote the Möbius band and its edge, as in Example 3.2.11. Show that the nonzero element of $H_1(K, K_0)$ is represented by the chain D indicated in Figure 3.2.3.

6. Let S denote the Klein bottle. Let A and C be the usual simple closed curves in S [see Exercise 2 in Section 2.6]. Compute the exact sequences of the pairs (S, A) and (S, C).

7. Show that if

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C \xrightarrow{\gamma} D$$

is exact and $D \neq 0$ then β need not be surjective.

8. Show that if

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C \xrightarrow{\gamma} D$$

is exact and $A \neq 0$ then β need not be injective.

3.3 The Zig-Zag Lemma

In this section we prove the exactness of the homology sequence of a pair. We shall do so by reformulating this result as a theorem about chain complexes and proving it in that form. This first requires a definition.

Definition 3.3.1 Let $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{E}$ be chain complexes. Let 0 denote the trivial chain complex whose groups vanish in every dimension. Let $\phi : \mathcal{C} \to \mathcal{D}$ and $\psi : \mathcal{D} \to \mathcal{E}$ be chain maps. We say that the sequence

$$0 \longrightarrow \mathcal{C} \xrightarrow{\phi} \mathcal{D} \xrightarrow{\psi} \mathcal{E} \longrightarrow 0$$

is *exact*, or that it is a *short exact sequence of chain complexes* if in each dimension p the sequence

$$0 \longrightarrow C_p \xrightarrow{\phi} D_p \xrightarrow{\psi} E_p \longrightarrow 0$$

is an exact sequence of groups.

For example, if K is a complex and K_0 is a subcomplex of K, then the sequence

$$0 \longrightarrow \mathcal{C}(K_0) \longrightarrow \mathcal{C}(K) \longrightarrow \mathcal{C}(K, K_0) \longrightarrow 0$$

is exact just because $C_p(K, K_0) = C_p(K)/C_p(K_0)$ by definition.

Lemma 3.3.2 (The zig-zag lemma) *Suppose that one is given chain complexes $\mathcal{C} = \{C_p, \partial_C\}$, $\mathcal{D} = \{D_p, \partial_D\}$, and $\mathcal{E} = \{E_p, \partial_E\}$. Suppose that also given are chain maps ϕ, ψ such that the sequence*

$$0 \longrightarrow \mathcal{C} \xrightarrow{\phi} \mathcal{D} \xrightarrow{\psi} \mathcal{E} \longrightarrow 0$$

is exact. Then there is a long exact homology sequence

$$\cdots \longrightarrow H_p(\mathcal{C}) \xrightarrow{\phi_*} H_p(\mathcal{D}) \xrightarrow{\psi_*} H_p(\mathcal{E}) \xrightarrow{\partial_*} H_{p-1}(\mathcal{C}) \xrightarrow{\phi_*} H_{p-1}(\mathcal{D}) \longrightarrow \cdots,$$

where ∂_* is induced by the boundary operator on $\mathcal{D}$.

Proof: The proof is of a sort commonly known as "diagram chasing." In the future we shall leave most diagram chasing proofs to the reader.

We shall make use of the following commutative diagram:

$$\begin{array}{ccccccccc}
 & & & & \boxed{d_{p+1}} & & \boxed{e_{p+1}} & & \\
 & & & \overset{\phi}{\longrightarrow} & & \overset{\psi}{\longrightarrow} & & & \\
0 & \longrightarrow & C_{p+1} & & D_{p+1} & & E_{p+1} & \longrightarrow & 0 \\
 & & \partial_C \downarrow & & \partial_D \downarrow & & \partial_E \downarrow & & \\
 & & \boxed{c_p} & & \boxed{d_p} & & \boxed{e_p} & & \\
0 & \longrightarrow & C_p & \overset{\phi}{\longrightarrow} & D_p & \overset{\psi}{\longrightarrow} & E_p & \longrightarrow & 0 \\
 & & \partial_C \downarrow & & \partial_D \downarrow & & \partial_E \downarrow & & \\
 & & \boxed{c_{p-1}} & & & & & & \\
 & & & \overset{\phi}{\longrightarrow} & & \overset{\psi}{\longrightarrow} & & & \\
0 & \longrightarrow & C_{p-1} & & D_{p-1} & & E_{p-1} & \longrightarrow & 0
\end{array}$$

Step 1: First we define ∂_*. Given a cycle e_p in E_p (that is to say, an element of $\ker \partial_E$), choose $d_p \in D_p$ so that $\psi(d_p) = e_p$. [Recall here that ψ is surjective.] The element $\partial_D d_p$ of D_{p-1} lies in $\ker \psi$ since

$$\psi(\partial_D d_p) = \partial_E \psi(d_p) = \partial_E e_p = 0\,.$$

Hence there exists an element c_{p-1} in C_{p-1} such that $\phi(c_{p-1}) = \partial_D d_p$. This because $\ker \psi = \operatorname{im} \phi$. This element is unique because ϕ is injective. Furthermore, c_{p-1} is a cycle. For

$$\phi(\partial_C c_{p-1}) = \partial_D \phi(c_{p-1}) = \partial_D \partial_D d_p = 0\,.$$

Thus $\partial_C c_{p-1} = 0$ because ϕ is injective. Define

$$\partial_*\{e_p\} = \{c_{p-1}\}\,,$$

where $\{\ \}$ means "homology class of."

Step 2: We now show that ∂_* is a well defined homomorphism. For notation, let e_p and e'_p be two elements in the kernel of $\partial_E : E_p \to E_{p-1}$. Choose d_p and d'_p so that $\psi(d_p) = e_p$ and $\psi(d'_p) = e'_p$. Then choose c_{p-1} and c'_{p-1} so that $\phi(c_{p-1}) = \partial_D d_p$ and $\phi(c'_{p-1}) = \partial_D d'_p$.

To show that ∂_* is well defined, we suppose that e_p and e'_p are homologous and show that c_{p-1} and c'_{p-1} are homologous. So suppose that $e_p - e'_p = \partial_E e_{p+1}$. Choose d_{p+1} so that $\psi(d_{p+1}) = e_{p+1}$. Then observe that

$$\begin{aligned}\psi(d_p - d'_p - \partial_D d_{p+1}) &= e_p - e'_p - \partial_E \psi(d_{p+1}) \\ &= e_p - e'_p - \partial_E e_{p+1} \\ &= 0\,.\end{aligned}$$

Therefore we can choose c_p so that $\phi(c_p) = d_p - d'_p - \partial_D d_{p+1}$. Then

$$\phi(\partial_C c_p) = \partial_D \phi(c_p) = \partial_D(d_p - d'_p) - 0 = \phi(c_{p-1} - c'_{p-1})\,.$$

Becuase ϕ is injective, we see that $\partial_C c_p = c_{p-1} - c'_{p-1}$ as desired.

To show that ∂_* is a homomorphism, note that $\psi(d_p + d'_p) = e_p + e'_p$ and $\phi(c_{p-1} + c'_{p-1}) = \partial_D(d_p + d'_p)$. Thus $\partial_*\{e_p + e'_p\} = \{c_{p-1} + c'_{p-1}\}$ by definition. The latter equals $\partial_*\{e_p\} + \partial_*\{e'_p\}$.

Step 3: We prove exactness at $H_p(\mathcal{D})$. Let $\gamma \in H_p(\mathcal{D})$. Because $\psi \circ \phi = 0$, we have $\psi_* \circ \phi_* = 0$. Hence if $\gamma \in \operatorname{im} \phi_*$, then $\psi_*(\gamma) = 0$.

Conversely, let $\gamma = \{d_p\}$ and suppose $\psi_*(\gamma) = 0$. Then $\psi(d_p) = \partial_E e_{p+1}$ for some e_{p+1}. Choose d_{p+1} so that $\psi(d_{p+1}) = e_{p+1}$. Then we have

$$\psi(d_p - \partial_D d_{p+1}) = \psi(d_p) - \partial_E \psi(d_{p+1}) = \psi(d_p) - \partial_E e_{p+1} = 0\,,$$

so $d_p - \partial_D d_{p+1} = \phi(c_p)$ for some c_p. Now, c_p is a cycle, because

$$\phi(\partial_C c_p) = \partial_D \phi(c_p) = \partial_D d_p - 0 = 0$$

and ϕ is injective. Furthermore, we have

$$\phi_*\{c_p\} = \{\phi(c_p)\} = \{d_p - \partial_D d_{p+1}\} = \{d_p\}\,,$$

so $\{d_p\} \in \operatorname{im} \phi_*$, as desired.

Step 4: Next, we prove exactness at $H_p(\mathcal{E})$. For notation, let $\alpha = \{e_p\}$ be an element of $H_p(\mathcal{E})$. Choose d_p so that $\psi(d_p) = e_p$. Then choose c_{p-1} so that $\phi(c_{p-1}) = \partial_D d_p$. Then $\partial_*\alpha = \{c_{p-1}\}$ by definition.

If $\alpha \in \operatorname{im}\psi_*$, then $\alpha = \{\psi(d_p)\}$, where d_p is a cycle of $\mathcal{D}$. Then $\phi(c_{p-1}) = 0$ hence $c_{p-1} = 0$. Therefore $\partial_*\alpha = 0$.

Conversely, suppose that $\partial_*\alpha = 0$. Then $c_{p-1} = \partial_C c_p$ for some c_p. We claim that $d_p - \phi(c_p)$ is a cycle and that $\alpha = \psi_*\{d_p - \phi(c_p)\}$. Hence $\alpha \in \operatorname{im}\psi_*$. By direct computation, we have

$$\partial_D(d_p - \phi(c_p)) = \partial_D d_p - \phi(\partial_C c_p) = \partial_D d_p - \phi(c_{p-1}) = 0$$

and

$$\psi_*\{d_p - \phi(c_p)\} = \{\psi(d_p) - 0\} = \{e_p\} = \alpha\,.$$

Step 5: Finally, we prove exactness at $H_{p-1}(\mathcal{C})$. Let $\beta \in H_{p-1}(\mathcal{C})$. If $\beta \in \operatorname{im}\partial_*$, then $\beta = \{c_{p-1}\}$, where $\phi(c_{p-1}) = \partial_D d_p$ for some d_p by definition. Thus we have

$$\phi_*(\beta) = \{\phi(c_{p-1})\} = \{\partial_D d_p\} = 0\,.$$

Conversely, suppose that $\phi_*(\beta) = 0$. Let $\beta = \{c_{p-1}\}$. Then $\{\phi(c_{p-1})\} = 0$ so that $\phi(c_{p-1}) = \partial_D d_p$ for some d_p. Define $e_p = \psi(d_p)$. Then e_p is a cycle because $\partial_E e_p = \psi(\partial_D d_p) = \psi\phi(c_{p-1}) = 0$. And $\beta = \partial_*\{e_p\}$ by definition. Thus $\beta \in \operatorname{im}\partial_*$. □

Note that nowhere in the proof of this lemma did we assume that the chain complexes involved were free or that they were non-negative. This lemma thus has a much broader range of application than just to chain groups or simplicial complexes. However, when applied to the simplicial chain groups $\mathcal{C}(K_0)$ and $\mathcal{C}(K)$ and $\mathcal{C}(K, K_0)$, we obtain Theorem 3.2.8 for non-reduced homology as an immediate consequence.

To obtain the theorem for reduced homology, we let $\mathcal{E} = \mathcal{C}(K, K_0)$ as before. But we replace $\mathcal{C}$ and $\mathcal{D}$ by the augmented chain complexes $\{\mathcal{C}, \epsilon\}$ and $\{\mathcal{D}, \epsilon\}$, respectively. Note that exactness and commutativity hold in dimension -1 since the inclusion $C_0(K_0) \to C_0(K)$ preserves augmentation:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C_0(K_0) & \longrightarrow & C_0(K) & \longrightarrow & C_0(K, K_0) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \epsilon} & & \downarrow{\scriptstyle \epsilon} & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Z} & \xrightarrow[i_{\mathbb{Z}}]{} & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & 0\,.
\end{array}$$

Application of the zig-zag lemma now gives us Theorem 3.2.8 for reduced homology.

Before considering further applications, let us extract some additional information from the proof of the zig-zag lemma. We have constructed a function assigning to each short exact sequence of chain complexes a long exact

sequence of their homology groups. Now we point out that this assignment is "natural" in the sense that a homomorphism of short exact sequences of chain complexes gives rise to a homomorphism of the corresponding long exact homology sequences.

Theorem 3.3.3 *Suppose one is given the commutative diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{C} & \xrightarrow{\phi} & \mathcal{D} & \xrightarrow{\psi} & \mathcal{E} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & \mathcal{C}' & \xrightarrow{\phi'} & \mathcal{D}' & \xrightarrow{\psi'} & \mathcal{E}' & \longrightarrow & 0\,, \end{array}$$

where the horizontal sequences are exact sequences of chain complexes and α, β, γ are chain maps. Then the following diagram commutes as well:

$$\begin{array}{ccccccccc} \longrightarrow & H_p(\mathcal{C}) & \xrightarrow{\phi_*} & H_p(\mathcal{D}) & \xrightarrow{\psi_*} & H_p(\mathcal{E}) & \xrightarrow{\partial_*} & H_{p-1}(\mathcal{C}) & \longrightarrow \\ & \downarrow{\scriptstyle \alpha_*} & & \downarrow{\scriptstyle \beta_*} & & \downarrow{\scriptstyle \gamma_*} & & \downarrow{\scriptstyle \alpha_*} & \\ \longrightarrow & H_p(\mathcal{C}') & \xrightarrow{\phi'_*} & H_p(\mathcal{D}') & \xrightarrow{\psi'_*} & H_p(\mathcal{E}') & \xrightarrow{\partial'_*} & H_{p-1}(\mathcal{C}') & \longrightarrow\ . \end{array}$$

Proof: Commutativity of the first two squares is immediate, just because commutativity holds already on the chain level. Commutativity of the last square involves examining the definitions of ∂_* and ∂'_*.

Given $\{e_p\} \in H_p(\mathcal{E})$, choose d_p so that $\psi(d_p) = e_p$; also choose c_{p-1} so that $\phi(c_{p-1}) = \partial_D d_p$. Then $\partial_*\{e_p\} = \{c_{p-1}\}$ by definition. Let $e'_p = \gamma(e_p)$. We wish to show that $\partial'_*\{e'_p\} = \alpha_*\{c_{p-1}\}$. Roughly speaking, this follows because each step in the definition of ∂_* commutes: The chain $\beta(d_p)$ is a suitable "pullback" for e'_p since $\psi'\beta(d_p) = \gamma\psi(d_p) = \gamma(e_p) = e'_p$. Then the chain $\alpha(c_{p-1})$ is the pullback of $\partial'_D\beta(d_p)$ since $\phi'\alpha(c_{p-1}) = \beta\phi(c_{p-1}) = \beta(\partial_D d_p) = \partial'_D\beta(d_p)$. Thus $\partial'_*\{e'_p\} = \{\alpha(c_{p-1})\}$ by definition. □

Naturality of the long exact homology sequence is extremely useful. We give one application now. It makes use of the following lemma, whose proof is just diagram chasing. We leave the proof for you.

Lemma 3.3.4 (The Steenrod five lemma) *Suppose one is given a commmutative diagram of abelian groups and homomorphisms:*

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_3} & & \downarrow{\scriptstyle f_4} & & \downarrow{\scriptstyle f_5} \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5 \end{array}$$

where the horizontal sequences are exact. If f_1, f_2, f_4, and f_5 are isomorphisms, then so is f_3.

Lemma 3.3.5 *Let $h : (K, K_0) \to (L, L_0)$ be a simplicial map.*

(a) *The induced homology homomorphisms h_* give a homomorphism of the exact homology sequence of (K, K_0) with that of (L, L_0).*

(b) *If $h_* : H_i(K) \to H_i(L)$ and $h_* : H_i(K_0) \to H_i(L_0)$ are isomorphisms for $i = p$ and $i = p - 1$, then*

$$h_* : H_p(K, K_0) \to H_p(L, L_0)$$

is an isomorphism.

(c) *Both these results hold if absolute homology is replaced throughout by reduced homology.*

Proof: We know that $h_\#$ is a chain map, and the following diagram commutes:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C_p(K_0) & \xrightarrow{i_\#} & C_p(K) & \xrightarrow{\pi_\#} & C_p(K)/C_p(K_0) & \longrightarrow & 0 \\
 & & \downarrow h_\# & & \downarrow h_\# & & \downarrow h_\# & & \\
0 & \longrightarrow & C_p(L_0) & \xrightarrow{i_\#} & C_p(L) & \xrightarrow{\pi_\#} & C_P(L)/C_p(L_0) & \longrightarrow & 0\,.
\end{array}$$

Then **(a)** follows. To deal with the case of reduced homology, we recall that $h_\#$ is augmentation-preserving, so it gives the desired chain map of the augmented chain complexes—provided we define $h_\# : \mathbb{Z} \to \mathbb{Z}$ as the identity on the chain groups in dimension -1. Thus **(a)** holds for reduced homology.

Result **(b)** follows immediately from the Steenrod five lemma. □

The following is an immediate consequence of the last lemma and the results of Chapter 2.

Theorem 3.3.6 *The preceding lemma holds for an arbitrary continuous map $h : (|K|, |K_0|) \to (|L|, |L_0|)$.*

We described the exact homology sequence of a pair in Theorem 3.2.8, and we proved its existence in this section. There is also an exact homology sequence of a triple, as stated in the next theorem. Exercise 1 asks you to prove its existence.

Definition 3.3.7 Let X be a complex. If A is a subcomplex of X and B is a subcomplex of A, then we call (X, A, B) a *simplicial triple* or simply a *triple* if the context makes the meaning clear.

Theorem 3.3.8 (The exact homology sequence of a triple) *Let X be a complex. Let A be a subcomplex of X. Let B be a subcomplex of A. The* exact homology sequence of the triple *is the following:*

$$\cdots \longrightarrow H_i(A, B) \longrightarrow H_i(X, B) \longrightarrow H_i(X, A) \longrightarrow H_{i-1}(A, B) \longrightarrow \cdots .$$

The sequence is natural.

Exercise 2 asks you to prove the next result.[1]

Lemma 3.3.9 (The serpent lemma) *Given a homomorphism of short exact sequences of abelian groups*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow\alpha & & \downarrow\beta & & \downarrow\gamma & & \\ 0 & \longrightarrow & D & \longrightarrow & E & \longrightarrow & F & \longrightarrow & 0\,, \end{array}$$

there is an exact sequence

$$0 \longrightarrow \ker\alpha \longrightarrow \ker\beta \longrightarrow \ker\gamma \longrightarrow \operatorname{cok}\alpha \longrightarrow \operatorname{cok}\beta \longrightarrow \operatorname{cok}\gamma \longrightarrow 0\,.$$

Exercise 7 concerns the Bockstein homomorphism which we define next.

Definition 3.3.10 Given a complex K and a short exact sequence of abelian groups

$$0 \longrightarrow G \to G' \longrightarrow G'' \longrightarrow 0\,,$$

one has a short exact sequence of chain complexes

$$0 \longrightarrow C_p(K;G) \longrightarrow C_p(K;G') \longrightarrow C_p(K;G'') \longrightarrow 0\,.$$

That short exact sequence gives rise to a long exact sequence in homology. The zig-zag homomorphism in this sequence is commonly denoted by

$$\beta_* : H_p(K;G'') \longrightarrow H_{p-1}(K;G)$$

and is called the *Bockstein homomorphism* associated with the given coefficient sequence.

EXERCISES

1. Prove the existence and naturality of the exact homology sequence of a triple.

2. Prove the serpent lemma. [**Hint:** Show how you can apply the zig-zag lemma.]

3. **(a)** Prove the Steenrod five lemma.

(b) Suppose we are given a commutative diagram of abelian groups as in the Steenrod five lemma. Consider the following eight hypotheses:

$$f_i \text{ is univalent for } i = 1, 2, 4, 5\,,$$

$$f_i \text{ is surjective for } i = 1, 2, 4, 5\,,$$

Which of these hypotheses will suffice to prove that f_3 is injective? Which will suffice to prove that f_3 is surjective?

[1] The serpent lemma is sometimes called the snake lemma.

4. Show by example that one can have $H_i(K_0) \cong H_i(L_0)$ and $H_i(K) \cong H_i(L)$ for all i without having $H_i(K, K_0) \cong H_i(L, L_0)$.

5. Let $w * K$ be a cone over K. Show that

$$H_i(w * K, K) \cong \widetilde{H}_{i-1}(K)\,.$$

6. Let K_0 be a subcomplex of K.

(a) If there is a retraction $r : |K| \to |K_0|$, then show that

$$H_p(K) \cong H_p(K, K_0) \oplus H_p(K_0)\,.$$

(b) If the identity map $i : |K| \to |K|$ is homotopic to a map carrying $|K|$ into $|K_0|$, then show that

$$H_p(K_0) \cong H_p(K) \oplus H_{p+1}(K, K_0)\,.$$

(c) If the inclusion $j : |K_0| \to |K|$ is homotopic to a constant, then show that

$$H_p(K, K_0) \cong \widetilde{H}_p(K) \oplus \widetilde{H}_{p-1}(K_0)\,.$$

[**Hint:** Show that the map j extends to a map $f : |w * K_0| \to |K|$. See Corollary 2.8.8. Then apply Exercise 5 above.]

*(d) Show by example that the conclusion of **(c)** does not hold if one assumes only that j_* is the zero homomorphism in reduced homology.

7. (a) Compute the Bockstein homomorphism, β_*, for the coefficient sequences

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0\,,$$

and

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow \mathbb{Z}/4 \longrightarrow \mathbb{Z}/2 \longrightarrow 0\,,$$

when $|K|$ equals $\mathbb{P}^2$.

(b) Repeat **(a)** when $|K|$ is the Klein bottle.

8. Prove Theorem 3.3.6.

3.4 The Mayer–Vietoris Sequence

The exact homology sequence is one useful device for computing homology groups. Another is the Mayer–Vietoris sequence, which we now construct. It is another consequence of the zig-zag lemma.

Theorem 3.4.1 *Let K be a complex. Let K_0 and K_1 be subcomplexes such that $K = K_0 \cup K_1$. Let $A = K_0 \cap K_1$. Then there is an exact sequence*

$$\cdots \longrightarrow H_p(A) \longrightarrow H_p(K_0) \oplus H_p(K_1) \longrightarrow H_p(K) \to H_{p-1}(A) \longrightarrow \cdots .$$

There is a similar exact sequence in reduced homology if A is nonempty.

Definition 3.4.2 The sequence in this last theorem is called the *Mayer–Vietoris sequence* of (K_0, K_1).

Proof: The proof consists of constructing a short exact sequence of chain complexes

$$0 \longrightarrow \mathcal{C}(A) \xrightarrow{\phi} \mathcal{C}(K_0) \oplus \mathcal{C}(K_1) \xrightarrow{\psi} \mathcal{C}(K) \longrightarrow 0 ,$$

and applying the zig-zag lemma.

First we need to define the chain complex in the middle. Its chain group in dimension p is

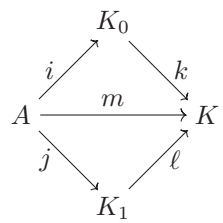
$$C_p(K_0) \oplus C_p(K_1) ,$$

and its boundary operator ∂' is defined by

$$\partial'(d, e) = (\partial_0 d, \partial_1 e) ,$$

where ∂_0 and ∂_1 are the boundary operators in $\mathcal{C}(K_0)$ and $\mathcal{C}(K_1)$ repectively.

Second, we need to define the chain maps ϕ and ψ. We do this as follows. Consider the inclusion mappings in the following commutative diagram:



Define homomorphisms ϕ and ψ by the equations

$$\phi(c) = (i_\#(c), -j_\#(c)) ,$$

$$\psi(d, e) = k_\#(d) + l_\#(e) .$$

It is immediate that ϕ and ψ are chain maps.

Let us check exactness: First note that ϕ is injective since $i_\#$ is just inclusion of chains. Second, we check that ψ is surjective. Given $d \in C_p(K)$, write d as a sum of oriented simplices and let d_0 consist of those terms of d carried by K_0, Then $d - d_0$ is carried by K_1. And $\psi(d_0, d - d_0) = d$.

To check exactness at the middle term, note first that

$$\psi\phi(c) = m_{\#}(c) - m_{\#}(c) = 0\,.$$

Conversely, if $\psi(d, e) = 0$, then $d = -e$ when considered as chains of K. Since d is carried by K_0 and e is carried by K_1, they must both be carried by $K_0 \cap K_1 = A$. Then $(d, e) = (d, -d) = \phi(d)$ as desired.

The homology of the middle chain complex in dimension p equals

$$\begin{aligned} \frac{\ker \partial'}{\operatorname{im} \partial'} &= \frac{\ker \partial_0 \oplus \ker \partial_1}{\operatorname{im} \partial_0 \oplus \operatorname{im} \partial_1} \\ &\cong H_p(K_0) \oplus H_p(K_1)\,. \end{aligned}$$

The Mayer–Vietoris sequence now follows from the zig-zag lemma.

In reduced homology, more work is needed to obtain the Mayer–Vietoris sequence. We replace the chain complexes considered earlier by the corresponding augmented chain complexes. Let ϵ_A, ϵ_0, ϵ_1, and ϵ denote the augmentation maps for $\mathcal{C}(A)$, $\mathcal{C}(K_0)$, $\mathcal{C}(K_1)$, and $\mathcal{C}(K)$ respectively. Consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C_0(A) & \longrightarrow & C_0(K_0) \oplus C_0(K_1) & \longrightarrow & C_0(K) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \epsilon_A} & & \downarrow{\scriptstyle \epsilon_0 \oplus \epsilon_1} & & \downarrow{\scriptstyle \epsilon} & & \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow[\phi]{} & \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow[\psi]{} & \mathbb{Z} & \longrightarrow & 0\,. \end{array}$$

Commutativity and exactness hold at the bottom level when we define $\phi(n) = (n, -n)$ and $\psi(m, n) = m + n$. Each map ϵ_A, $\epsilon_0 \oplus \epsilon_1$, and ϵ is surjective (since A is nonempty). Thus the homology of these respective chain complexes vanishes in dimension -1, and in dimension 0 equals the respective groups $\widetilde{H}_0(A)$, $\widetilde{H}_0(K_0) \oplus \widetilde{H}_0(K_1)$, and $\widetilde{H}_0(K)$. We conclude the proof by applying the zig-zag lemma. □

Lemma 3.4.3 *Let $h : (K, K_0, K_1) \to (L, L_0, L_1)$ be a simplicial map, where $K = K_0 \cup K_1$ and $L = L_0 \cup L_1$. Then h induces a homomorphism of Mayer–Vietoris sequences.*

Proof: One checks immediately that the chain maps $h_{\#}$ induced by h commute with the chain maps ϕ and ψ defined in the preceding proof. Naturality then follows from Theorem 3.3.3. □

The following is an immediate consequence of this last lemma and the results of Chapter 2.

Theorem 3.4.4 *The preceding lemma holds for an arbitrary continuous map $h : (|K|, |K_0|, |K_1|) \to (|L|, |L_0|, |L_1|)$.*

To illustrate how the Mayer–Vietoris sequence is used in practice, we shall now compute the homology of the suspension of a complex. The suspension of a complex appeared earlier in Definition 1.9.8, but for convenience, we repeat it here.

Definition 3.4.5 Let K be a complex. Let $w_0 * K$ and $w_1 * K$ be two cones on K whose polytopes intersect in $|K|$ alone. Then

$$S(K) = (w_0 * K) \cup (w_1 * K)$$

is a complex. It is called a *suspension* of K. Given K, the complex $S(K)$ is uniquely defined up to a simplicial isomorphism.

Theorem 3.4.6 *If K is a complex, then for all p there is an isomorphism*

$$\widetilde{H}_p(S(K)) \to \widetilde{H}_{p-1}(K)\,.$$

Proof: Let $K_0 = w_0 * K$ and $K_1 = w_1 * K$. Then $K_0 \cup K_1 = S(K)$ and $K_0 \cap K_1 = K$. In the reduced Mayer–Vietoris sequence,

$$\widetilde{H}_p(K_0) \oplus \widetilde{H}_p(K_1) \longrightarrow \widetilde{H}_p(S(K)) \longrightarrow \widetilde{H}_{p-1}(K) \longrightarrow \widetilde{H}_{p-1}(K_0) \oplus \widetilde{H}_{p-1}(K_1);$$

both end terms vanish just because K_0 and K_1 are cones. Therefore the middle map is an isomorphism. □

Theorem 3.4.7 *Let K be a complex. Let K_0 and K_1 be subcomplexes of K. Let L_0 and L_1 be subcomplexes of K_0 and K_1 respectively. There exists an exact sequence*

$$\cdots \longrightarrow H_i(K_0 \cap K_1, L_0 \cap L_1) \longrightarrow H_i(K_0, L_0) \oplus H_i(K_1, L_1) \longrightarrow H_i(K_0 \cup K_1, L_0 \cup L_1) \longrightarrow \cdots .$$

Definition 3.4.8 The sequence in this last theorem is called the *relative Mayer–Vietoris sequence.*

EXERCISES

1. Let K be the union of the subcomplexes K_0 and K_1, where $|K_0|$ and $|K_1|$ are connected. What can you say about the homology of K in each of the following cases?
   - **(a)** $K_0 \cap K_1$ is nonempty and acyclic.
   - **(b)** $|K_0| \cap |K_1|$ consists of two points.

(c) $K_0 \cap K_1$ has the homology of $\mathbb{S}^n$, where $n > 0$.

(d) K_0 and K_1 are acyclic.

2. Prove the existence of the relative Mayer–Vietoris sequence as given in Theorem 3.4.7.

3. Show that if d is an integer, and if $n \geq 1$, then there is a map $h : \mathbb{S}^n \to \mathbb{S}^n$ of degree d. [**Hint:** Proceed by induction, using naturality of the Mayer–Vietoris sequence. The case $n = 1$ was considered in Exercise 1 of Section 2.9.]

4. Let $\phi : C_p(K) \to C_{p+1}(S(K))$ be the homomorphism defined in Exercise 1 of Section 1.9. Show that the isomorphism of Theorem 3.4.6 is inverse to ϕ_*.

5. Given a sequence $G_0, G_1, \ldots, G_n$ of finitely generated abelian groups, with G_0 and G_n free and G_0 nontrivial, show that there is a finite complex K of dimension n such that $H_i(K) \cong G_i$ for $i = 0, 1, \ldots, n$. [**Hint:** See Exercise 8 of Section 1.7.]

6. We shall study the homology of $X \times Y$ in Chapter 7. For the present, prove the following—assuming that all the spaces involved are polyhedra.

(a) Show that if $p \in \mathbb{S}^n$, then

$$H_q(X \times \mathbb{S}^n, X \times p) \cong H_{q-n}(X)\,.$$

[**Hint:** Write $\mathbb{S}^n$ as the union of its upper and lower hemispheres, and proceed by induction on n.]

(b) Show that if $p \in Y$, then the homology exact sequence of

$$(X \times Y, X \times p)$$

breaks up into short exact sequences that split.

(c) Prove that

$$H_q(X \times \mathbb{S}^n) \cong H_{q-n}(X) \oplus H_q(X)\,.$$

(d) Compute the homology of $\mathbb{S}^n \times \mathbb{S}^m$.

7. Do Exercise 3 with $\mathbb{S}^n$ replaced by an n-dimensional torus

$$\mathbb{T}^n = \underbrace{\mathbb{S}^1 \times \mathbb{S}^1 \times \cdots \times \mathbb{S}^1}_{n \text{ copies}}\,.$$

3.5 The Eilenberg–Steenrod Axioms

We have defined homology groups for a particular class of spaces—namely, the polyhedra. Historically these were the first homology groups to be defined.

Later various generalized definitions of homology were formulated that applied to more general spaces. These various homology theories had many features in common, and they all gave the same results as simplicial homology theory on the class of polyhedra.

This plethora of homology theories led Eilenberg and Steenrod to axiomatize the notion of a homology theory. They formulated certain crucial properties these theories have in common, and showed that these properties characterize the homology groups completely on the class of polyhedra.

We shall not try to reproduce the axiomatic approach at this point, nor shall we prove that the axioms characterize homology for polyhedra. For this, the reader is referred to Eilenberg and Steenrod's book [ES52]. We shall simply list the seven axioms here, together with an additional axiom that is needed when one deals with non-compact spaces. In the next section, we verify that simplicial homology theory satisfies the axioms.

Definition 3.5.1 Let $\mathcal{A}$ be a class of pairs (X, A) of topological spaces, $X \supset A$, such that

(1) If (X, A) belongs to $\mathcal{A}$, then so do (X, X), $(X, \emptyset)$, (A, A), and $(A, \emptyset)$.

(2) If (X, A) belongs to $\mathcal{A}$, then so does $(X \times I, A \times I)$.

(3) There is a one-point space P such that $(P, \emptyset)$ is in $\mathcal{A}$.

Then we shall call $\mathcal{A}$ an *admissible class of spaces* for a homology theory.

Definition 3.5.2 If $\mathcal{A}$ is admissible, a *homology theory* on $\mathcal{A}$ consists of three functions:

(1) A function H_p defined for each integer p and each pair (X, A) in $\mathcal{A}$ whose value is an abelian group.

(2) A function that, for each integer p and continuous map $h : (X, A) \to (Y, B)$, assigns a homomorphism

$$(h_*)_p : H_p(X, A) \to H_p(Y, B)\,.$$

(3) A function that, for each integer p and pair (X, A) in $\mathcal{A}$, assigns a homomorphism

$$(\partial_*)_p : H_p(X, A) \to H_{p-1}(A)\,.$$

Here A denotes the pair $(A, \emptyset)$.

These functions are to satisfy the following axioms, where all pairs of spaces are in $\mathcal{A}$. As usual, we shall simplify notation and delete the dimensional subscripts on h_* and ∂_*.

Axiom 1: If i is the identity, then i_* is the identity.

Axiom 2: $(k \circ h)_* = k_* \circ h_*$.

Axiom 3: If $f : (X, A) \to (Y, B)$, then the following diagram commutes:

$$\begin{array}{ccc} H_p(X, A) & \xrightarrow{f_*} & H_p(Y, B) \\ \downarrow{\scriptstyle \partial_*} & & \downarrow{\scriptstyle \partial_*} \\ H_{p-1}(A) & \xrightarrow{(f|A)_*} & H_{p-1}(B) \end{array}$$

Axiom 4: [Exactness Axiom] The sequence

$$\cdots \longrightarrow H_p(A) \xrightarrow{i_*} H_p(X) \xrightarrow{\pi_*} H_p(X, A) \xrightarrow{\partial_*} H_{p-1}(A) \longrightarrow \cdots$$

is exact, where $i : A \to X$ and $\pi : X \to (X, A)$ are inclusion maps.

Recall that two maps $h, k : (X, A) \to (Y, B)$ are said to be homotopic (written $h \simeq k$) if there is a map

$$F : (X \times I, A \times I) \to (Y, B)$$

such that $F(x, 0) = h(x)$ and $F(x, 1) = k(x)$ for all $x \in X$.

Axiom 5: [Homotopy Axiom] If h and k are homotopic, then $h_* = k_*$.

Axiom 6: [Excision Axiom] Given (X, A), let U be an open subset of X such that $\overline{U} \subset \operatorname{Int} A$. If $(X \setminus U, A \setminus U)$ is admissible, then inclusion induces an isomorphism

$$H_p(X \setminus U, A \setminus U) \cong H_p(X, A)\,.$$

Axiom 7: [Dimension Axiom] If P is a one-point space, then $H_p(P) = 0$ for $p \neq 0$ and $H_0(P) = \mathbb{Z}$.

Axiom 8: [Compact Support Axiom] If $\alpha \in H_p(X, A)$, then there is an admissible pair (X_0, A_0) with X_0 and A_0 compact, such that α is in the image of the homomorphism $H_p(X_0, A_0) \to H_p(X, A)$ induced by inclusion.

A pair (X_0, A_0) with both X_0 and A_0 compact is called a *compact pair*.

Observe that one can modify Axiom 7 by writing $H_0(P) = G$, where G is a fixed abelian group. What this produces is called "homology with coefficients in G." We shall, for the moment, stick with integer coefficients.

Of all the Eilenberg–Steenrod axioms, the dimension axiom seems to be the most innocuous and the least interesting. But in a certain sense quite the opposite is true. Since Eilenberg and Steenrod's axioms were created, several new mathematical theories have been discovered that resemble homology theory. Cobordism theories in differential topology and K-theory in vector bundle theory are examples of such. Although these new theories were invented for purposes quite different from those for which the homology groups

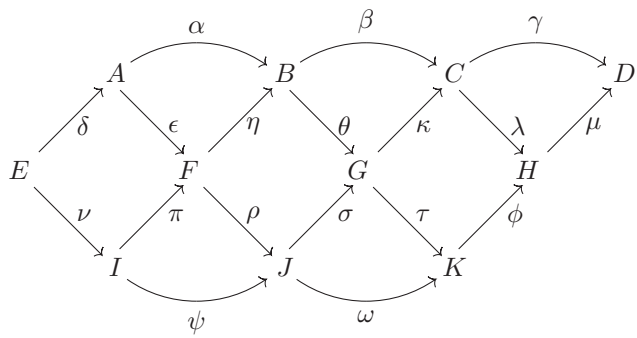
were originally invented, they share many formal properties with homology theory. In particular, they satisfy all the Eilenberg–Steenrod axioms except for the dimension axiom.

Such a theory is nowadays called an *extraordinary homology theory*, or a *generalized homology theory*. It differs from ordinary homology in that a one-point space may have non-zero homology in many dimensions.

This situation illustrates a phenomenon that occurs over and over again in mathematics: A theory formulated for one purpose turns out to have consequences far removed from what its originators envisaged.

The next definition is used in Exercise 1.

Definition 3.5.3 Consider the following commutative diagram of abelian groups and homomorphisms, which is called a *braid*.



This diagram contains the following four sequences, arranged in the form of overlapping sine curves.

$$\left.\begin{aligned} &E \longrightarrow A \longrightarrow B \longrightarrow G \longrightarrow K\,, \\ &E \longrightarrow I \longrightarrow J \longrightarrow G \longrightarrow C \to D\,, \\ &A \longrightarrow F \longrightarrow J \longrightarrow K \longrightarrow H \to D\,, \end{aligned}\right\} \qquad (*)$$
$$I \longrightarrow F \longrightarrow B \longrightarrow C \longrightarrow H\,.$$

If all four sequences are exact, this diagram is called an *exact braid*.

Lemma 3.5.4 (The braid lemma) *In Definition 3.5.3, if the sequences* $(*)$ *are exact and if the composite* $I \to F \to B$ *is zero, then the braid is exact.*

The next definition is used in Exercise 3.

Definition 3.5.5 Let $X = X_1 \cup X_2$. Let $A = X_1 \cap X_2$. We say that (X_1, X_2) is an *excisive couple* for the given homology theory if (X_1, A) and (X, X_2) are admissible and if the inclusion $(X_1, A) \to (X, X_2)$ induces a homology isomorphism.

EXERCISES

1. **(a)** If the braid in Definition 3.5.3 is exact, then there is an isomorphism

$$\Lambda : \frac{\ker \omega}{\operatorname{im} \psi} \to \frac{\ker \beta}{\operatorname{im} \alpha}$$

defined as follows. If $\omega(j) = 0$, then choose f so that $\rho(f) = j$. Then define $\Lambda(j) = \{\eta(f)\}$.

(b) Prove the braid lemma.

2. Suppose that X is a topological space and $X \supset A \supset B$. We call (X, A, B) a *triple* of topological spaces, relying on the context of homology theory to imply the nestedness of the spaces. Using only the axioms for a homology theory, prove exactness of the homology sequence of a triple:

$$\cdots \longrightarrow H_p(A, B) \xrightarrow{\pi} H_p(X, B) \xrightarrow{\eta} H_p(X, A) \xrightarrow{\beta} H_{p-1}(A, B) \longrightarrow \cdots$$

where π and η are induced by inclusion. The map β is the composition

$$H_p(X, A) \xrightarrow{\partial_*} H_{p-1}(A) \xrightarrow{i_*} H_{p-1}(A, B)\,,$$

where ∂_* is given by the axioms and i is inclusion. Assume that the pairs involved are admissible. [**Hint:** Prove that $H_p(A, A) = 0$. Show that $\eta \circ \pi = 0$. Then apply Exercise 1.]

3. Using only the axioms for a homology theory, derive the Mayer–Vietoris sequence in the following manner.

(a) Consider the following diagram of inclusions, and the corresponding homomorphism of long exact homology sequences:

$$\begin{array}{ccccc} A & \xrightarrow{i} & X_1 & \xrightarrow{\alpha} & (X_1, A) \\ {\scriptstyle j}\downarrow & & {\scriptstyle k}\downarrow & & \downarrow{\scriptstyle \gamma} \\ X_2 & \xrightarrow[\ell]{} & X & \xrightarrow[\beta]{} & (X, X_2)\,. \end{array}$$

Given that (X_1, X_2) is excisive, define a sequence

$$\cdots \to H_p(A) \xrightarrow{\phi} H_p(X_1) \oplus H_p(X_2) \xrightarrow{\psi} H_p(X) \xrightarrow{\theta} H_{p-1}(A) \to \cdots$$

by letting

$$\phi(a) = (i_*(a), -j_*(a))\,,$$
$$\psi(x_1, x_2) = k_*(x_1) + \ell_*(x_2)\,,$$

$$\theta(x) = \partial_*(\gamma_*)^{-1}\beta_*(x)\,.$$

Here ∂_* is the boundary homomorphism in the homology exact sequence of (X_1, A). Show that this sequence is exact and is natural with respect to homomorphisms induced by continuous maps. [**Hint:** The proof is by diagram chasing. One begins with a homomorphism of one long exact sequence to another, where every third homomorphism is an isomorphism.]

(b) Show that if X is the polytope of a complex K, and if X_1 and X_2 are polytopes of subcomplexes of K, then (X_1, X_2) is excisive for simplicial theory, and this sequence is the same as the Mayer–Vietoris sequence of Theorem 3.4.1.

(c) Suppose that (X_1, A) and (X, X_2) are admissible. Show that if $\operatorname{Int} X_1$ and $\operatorname{Int} X_2$ cover X and if X_1 is closed in X, then (X_1, X_2) is excisive for any homology theory satisfying the axioms.

3.6 The Axioms for Simplicial Theory

Before showing that simplicial homology theory satisfies the Eilenberg–Steenrod axioms, we must treat several points of the theory with more care than we have shown so far. The axioms involve homology groups that are defined on an admissible class of spaces. Strictly speaking, we have not defined homology groups for *spaces*, but only for simplicial complexes. Given a polyhedron X, there are many different simplicial complexes whose polytopes equal X. Their homology groups are isomorphic to one another in a natural way, but they are nevertheless distinct groups. Similarly, if $h : X \to Y$ is a continuous map where $X = |K|$ and $Y = |L|$, we have defined an induced homomorphism $h_* : H_p(K) \to H_p(L)$. Of course if $X = |M|$ and $Y = |N|$, then we also have an induced homomorphism $H_p(M) \to H_p(N)$, which we also denote by h_*. We have noted this notational ambiguity in an earlier discussion.

The way out of this difficulty is to introduce a new level of abstraction: Given a polyhedron X, we can consider the class of all simplicial complexes whose polytopes equal X, and we can *identify* their homology groups in a natural way. The resulting groups can be called the homology groups of the polyhedron X.

More generally, we can perform this same construction for any space that is homeomorphic to a polyhedron. We give the details in what follows.

Definition 3.6.1 Let A be a subspace of the space X. A *triangulation* of the pair (X, A) is a complex K, a subcomplex K_0 of K, and a homeomorphism

$$h : (|K|, |K_0|) \to (X, A)\,.$$

If such a triangulation exists, then we say that (X, A) is a *triangulable pair*. If A is empty, then consistent with Definition 2.9.7, we say simply that X is a *triangulable space*.

Now let (X, A) be a triangulable pair. We define the simplicial homology $H_p(X, A)$ of this pair as follows. Consider the collection of all triangulations of (X, A). They are of the form

$$h_\alpha : (|K_\alpha|, |C_\alpha|) \to (X, A)\,,$$

where C_α is a subcomplex of K_α.

Now there is some set-theoretic difficulty with the concept of the "set of all triangulations of a pair," as there is with the "set of all sets." We avoid these problems by assuming that each K_α lies in some fixed space $\mathbb{E}^J$. This is justified by noting that if J is large enough, then each K_α has an isomorphic copy lying in $\mathbb{E}^J$. [For instance, we can let J have the cardinality of X itself.] Once we know that all the sets K_α are confined to a fixed $\mathbb{E}^J$, then we can find a large enough index set for K_α.

For fixed p, consider the groups $H_p(K_\alpha, C_\alpha)$. We make sure they are disjoint as sets by forming $H_p(K_\alpha, C_\alpha) \times \{\alpha\}$. Then in the disjoint union

$$\bigcup_\alpha H_p(K_\alpha, C_\alpha) \times \{\alpha\}\,, \qquad (*)$$

we introduce an equivalence relation. We define

$$(x, \alpha) \in H_p(K_\alpha, C_\alpha) \times \{\alpha\}$$

$$(y, \beta) \in H_p(K_\beta, C_\beta) \times \{\beta\}$$

to be equivalent if $(h_\beta^{-1} h_\alpha)_*(x) = y$. We define

$$H_p(X, A)$$

to be the set of equivalence classes in $(*)$.

Now each equivalence class contains *exactly one element* from each group $H_p(K_\alpha, C_\alpha) \times \{\alpha\}$. Thus the map

$$H_p(K_\alpha, C_\alpha) \times \{\alpha\} \to H_p(X, A)$$

that carries each element to its equivalence class is bijective. We make $H_p(X, A)$ into a group by requiring this map to be an isomorphism. This group structure is unambiguous because $(h_\beta^{-1} h_\alpha)_*$ is an isomorphism for each pair of indices α and β.

A continuous map $h : (X, A) \to (Y, B)$ induces a homomorphism in homology as follows: Take any pair of triangulations

$$h_\alpha : (|K_\alpha|, |C_\alpha|) \to (X, A)\,,$$

$$k_\beta : (|L_\beta|, |D_\beta|) \to (Y, B)\,.$$

The map h induces a map $h' : (|K_\alpha|, |C_\alpha|) \to (|L_\beta|, |D_\beta|)$, which in turn gives rise to a homomorphism

$$h'_* : H_p(K_\alpha, C_\alpha) \to H_p(L_\beta, D_\beta)$$

of simplicial homology groups. By passing to equivalence classes, this homomorphism induces a well-defined homomorphism

$$h_* : H_p(X, A) \to H_p(Y, B)\,.$$

In a similar fashion, the boundary homomorphism $\partial_* : H_p(K_\alpha, C_\alpha) \to H_{p-1}(C_\alpha)$ induces a boundary homomorphism

$$\partial_* : H_p(X, A) \to H_{p-1}(A)\,.$$

We now have all the necessary pieces for a homology theory.

First, we note that the class of triangulable pairs forms an admissible class of spaces for a homology theory. If (X, A) is triangulable, then so are (X, X), $(X, \emptyset)$, (A, A), and $(A, \emptyset)$. Of course any one-point space is triangulable. Finally, if (X, A) is triangulable, then so is $(X \times I, A \times I)$ by Lemma 2.7.3.

Theorem 3.6.2 *Simplicial homology theory on the class of triangulable pairs satisfies the Eilenberg–Steenrod axioms.*

Proof: Axioms **1**–**5** and **7** express familiar properties of the homology of simplicial complexes that carry over at once to the homology of triangulable pairs. Only Axioms **6** and **8**, the excision axiom and the compact support axiom, require some thought.

To check the compact support axiom, it suffices to show that it holds for simplicial complexes. Let α be an element of $H_p(K, K_0)$. Let c be a chain of K representing α. Its boundary is carried by K_0. Since c is carried by a finite subcomplex L of K, it can be considered as a cycle of (L, L_0), where $L_0 = K_0 \cap L$. If β is the homology class of c in $H_p(L, L_0)$, then $j_*(\beta) = \alpha$, where j is inclusion of (L, L_0) in (K, K_0). Thus this axiom is verified.

Checking the excision axiom involves a subtlety that may not at first be apparent. The problem is that even though both (X, A) and $(X \setminus U, A \setminus U)$ may be triangulable, the two triangularizations may be entirely unrelated to one another. If this is *not* the case, then the excision axiom follows readily from Theorem 1.10.6, as we now show.

Let $U \subset A \subset X$. Suppose that there is a triangulation

$$h : (|K|, |K_0|) \to (X, A)$$

of the pair (X, A) that induces a triangulation of the subspace $X \setminus U$. This means that $X \setminus U = h(|L|)$ for some subcomplex L of K. Let $L_0 = L \cap K_0$.

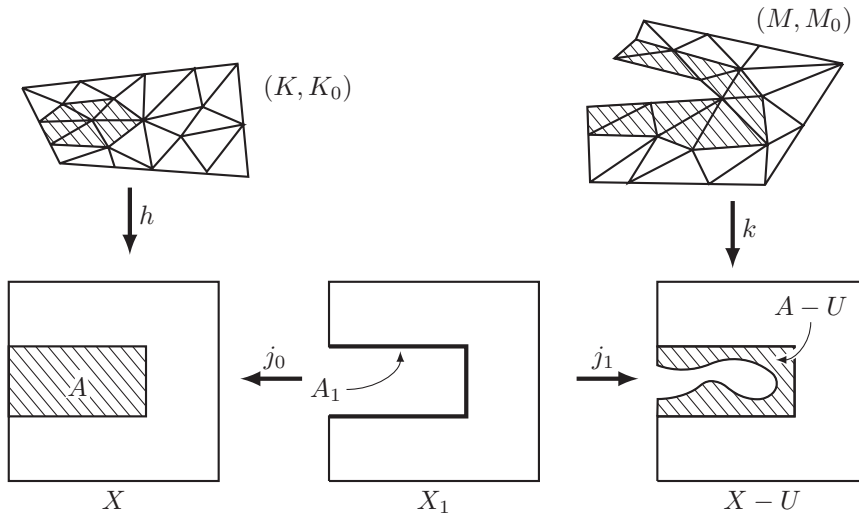


Figure 3.6.1

Then $A \backslash U = h(L_0)$. [It follows that U is open and A is closed.] Theorem 1.10.6 states that inclusion induces an isomorphism

$$H_p(L, L_0) \cong H_p(K, K_0)\,.$$

Thus our result is proved.

Now we shall prove excision in the general case. Let $U \subset \operatorname{Int} A$. Let

$$h : (K, K_0) \to (X, A)$$

$$k : (M, M_0) \to (X \setminus U, A \setminus U)$$

be triangulations of these respective pairs of spaces. Let X_1 denote the closure of $X \setminus A$ and let $A_1 = X_1 \cap A$. We assert that the pair (X_1, A_1) is triangulated by both h and k. Refer to Figure 3.6.1, where the maps j_0 and j_1 both denote inclusions.

To verify this assertion, note that the space $|K| \setminus |K_0|$ is the union of all open simplices $\operatorname{Int} \sigma$ such that $\sigma \in K$ and $\sigma \notin K_0$. Then its closure C is the polytope of the subcomplex of K consisting of all simplices σ of K that are not in K_0, and their faces. The image of the set C under the homeomorphism h equals the closure of $X \setminus A$, which is X_1. Therefore X_1 is triangulated by h. Since both X_1 and A are triangulated by h, so is $A_1 = X_1 \cap A$. Similarly, the closure of $|M| \setminus |M_0|$ is the polytope of a subcomplex of M, and its image under k equals the closure of

$$(X \setminus U) \setminus (A \setminus U) = X \setminus A\,.$$

Thus X_1 is triangulated by k and so is $A_1 = X_1 \cap A$. [Here is where we need the fact that $U \subset \operatorname{Int} A$.]

It follows from the special case already proved that in the diagram of inclusions

$$\begin{array}{ccc} (X_1, A_1) & \xrightarrow{j_0} & (X, A) \\ & {\scriptstyle j_1}\searrow \quad \nearrow{\scriptstyle j} & \\ & (X \setminus U, A \setminus U) & \end{array}$$

both j_0 and j_1 induce isomorphisms. Because the diagram commutes, j_* is also an isomorphism. □

We remark that in the preceding proof we did not need the full strength of the hypothesis of the excision axiom. We did not use the fact that $\overline{U} \subset \operatorname{Int} A$, nor that U is open. Thus for simplicial theory on the class of triangulable pairs, the following stronger version of the excision axiom holds:

Theorem 3.6.3 (Excision in simplicial theory) *Let A be a subspace of X. Let U be a subset of X such that $U \subset \operatorname{Int} A$. If both of the pairs (X, A) and $(X \setminus U, A \setminus U)$ are triangulable, then inclusion induces an isomorphism*

$$H_p(X \setminus U, A \setminus U) \cong H_p(X, A)\,.$$

The compact support axiom states, roughly speaking, that every homology class is compactly supported. It is also true that every homology relation between such classes is compactly supported. More precisely, one has the following useful result, which we shall verify directly for simplicial theory. It may also be derived from the axioms. See the exercises below.

Theorem 3.6.4 *Let $i : (X_0, A_0) \to (X, A)$ be an inclusion of triangulable pairs, where (X_0, A_0) is a compact pair. If $\alpha \in H_p(X_0, A_0)$ and $i_*(\alpha) = 0$, then there are a compact triangulable pair (X_1, A_1) and inclusion maps*

$$(X_0, A_0) \xrightarrow{j} (X_1, A_1) \xrightarrow{k} (X, A)$$

such that $j_*(\alpha) = 0$.

Proof: We may assume that (X, A) is the polytope of a simplicial pair (K, C). Because X_0 is compact, it is contained in the polytope of a finite subcomplex K_0 of K. Then A_0 is contained in the polytope of $C \cap K_0 = C_0$. The theorem thus reduces to the case where

$$i : (K_0, C_0) \to (K, C)$$

is an inclusion of subcomplexes and K_0 is finite.

Let $\alpha \in H_p(K_0, C_0)$ and suppose that $i_*(\alpha) = 0$. Let c_p be a chain of K_0 representing α. Since $i_*(\alpha) = \{i_\#(c_p)\} = 0$, there is a chain d_{p+1} of K such

that $c_p - \partial d_{p+1}$ is carried by C. Choose a finite subcomplex of C carrying $c_p - \partial d_{p+1}$. Let C_1 be the union of this subcomplex and C_0. Then choose a finite subcomplex of K carrying d_{p+1}. Let K_1 be the union of this subcomplex and K_0 and C_1. Then the homomorphism induced by the inclusion

$$(K_0, C_0) \to (K_1, C_1)$$

carries α to 0. □

Theorem 3.6.5 *Given a sequence G_0, G_1, ... of finitely generated abelian groups with G_0 free and non-trivial, there is a complex K such that $H_i(K) \cong G_i$ for each i.*

The proof of this last theorem is left to the exercises.

EXERCISES

1. Prove Theorem 3.6.5. [**Hint:** See Exercise 5 of Section 3.4.]

2. Let $\mathcal{A}$ be either the class of triangulable pairs, or the class of all topological pairs. Prove Theorem 3.6.4, with the word "triangulable" replaced throughout by the word "admissible"—directly from the axioms.

[**Hint:** Show that in the triangulable case one can assume that the triangulation of (X, A) triangulates X_0 as well. Examine the diagram

$$\begin{array}{ccccccc} & & & & H_q(X_0, A_0) & & \\ & & & & \downarrow \ell_* & \searrow & \\ \longrightarrow & H_q(A, A_0) & \longrightarrow & H_q(X, A_0) & \longrightarrow & H_q(X, A) & \longrightarrow \end{array}$$

to find a compact A_1 such that $\ell_*(\alpha)$ is in the image of

$$H_q(A_1, A_0) \longrightarrow H_q(X, A_0)\,.$$

Then the inclusion $(X_0, A_0) \to (X, A_1)$ induces a homomorphism carrying α to 0. Examine the diagram

$$\begin{array}{ccccc} & & H_q(X_0, A_0) & & \\ & & \downarrow m_* & \searrow & \\ H_{q+1}(X, X_0 \cup A_1) & \longrightarrow & H_q(X_0 \cup A_1, A_1) & \longrightarrow & H_q(X, A_1) \end{array}$$

to find a compact X_1 such that $m_*(\alpha)$ is in the image of

$$\partial_* : H_{q+1}(X_1, X_0 \cup A_1) \to H_q(X_0 \cup A_1, A_1)\,.$$

Now you should complete the proof.]

3. Give three distinct examples of triangulable pairs.

3.7 *Categories and Functors

We have seen many references to "induced homomorphisms" and their "functorial properties" and to the naturality of the way we assign one mathematical object to another. This leads us to suspect that there is some common idea underlying all this language. Indeed there is, and we study it in this last section of Chapter 3. It consists mostly of new terminology. The few proofs are elementary and are mostly left to the reader.

Definition 3.7.1 A *category* **C** consists of three things:

(1) A class of *objects* X.

(2) For each ordered pair (X, Y) of objects, a set $\hom(X, Y)$ of *morphisms* f.

(3) A function, called *composition of morphisms*,

$$\hom(X, Y) \times \hom(Y, Z) \to \hom(X, Z)\,,$$

which is defined for every triple (X, Y, Z) of objects.

The image of the pair (f, g) under the composition operation is denoted by $g \circ f$. The following two properties must be satisfied.

Axiom 1 (Associativity): If $f \in \hom(W, X)$ and $g \in \hom(X, Y)$ and $h \in \hom(Y, Z)$, then $h \circ (g \circ f) = (h \circ g) \circ f$.

Axiom 2 (Existence of identities): If X is an object, then there is an element $1_X \in \hom(X, X)$, the *identity morphism*, such that

$$1_X \circ f = f \qquad \text{and} \qquad g \circ 1_X = g$$

for every $f \in \hom(W, X)$ and every $g \in \hom(X, Y)$, where W and Y are arbitrary objects.

One standard example of a category consists of topological spaces and continuous maps, with the usual composition of maps. This example illustrates why we speak of the objects in a category as forming a *class* rather than a

set, for one cannot speak of the "set of all topological spaces" or "the set of all sets" without becoming involved in logical paradoxes. [For example, is the set of all sets a member of itself?] A *class* is something larger than a set, to which we do not apply the usual set-theoretic operations [such as taking the set of all subsets].

Let us note the following: The identity morphism 1_X is unique. For suppose instead that

$$1_X \circ f = f \qquad \text{and} \qquad g = g \circ 1'_X$$

for every $f \in \hom(W, X)$ and $g \in \hom(X, Y)$. Then setting $f = 1'_X$ and $g = 1_X$ we find that

$$1_X \circ 1'_X = 1'_X \qquad \text{and} \qquad 1_X = 1_X \circ 1'_X \,,$$

hence $1'_X = 1_X$.

Definition 3.7.2 Let $f \in \hom(X, Y)$ and $g, g' \in \hom(Y, X)$. If $g \circ f = 1_X$, then we call g a *left inverse* for f. If $f \circ g' = 1_Y$, then we call g; a *right inverse* for f.

Note the following: If f has a left inverse g and a right inverse g' then $g = g'$. For we have

$$(g \circ f) \circ g' = 1_x \circ g' = g' \,,$$
$$g \circ (f \circ g') = g \circ 1_Y = g \,,$$

hence $g = g'$. In this case, the map $g = g'$ is called an *inverse* to f. It is unique.

If f has an inverse, then f is called an *equivalence* in the category in question. A category in which every morphism is an equivalence is called a *groupoid*.

In general, we write $f : X \to Y$ to mean $f \in \hom(X, Y)$. We call X the *domain object* of f, and we call Y the *range object* of f.

Definition 3.7.3 A (covariant) *functor* G from a category $\mathbf{C}$ to a category $\mathbf{D}$ is a function assigning to each object X of $\mathbf{C}$ an object $G(X)$ of $\mathbf{D}$ and to each morphism $f : X \to Y$ of $\mathbf{C}$ a morphism $G(f) : G(X) \to G(Y)$ of $\mathbf{D}$. The following two conditions must be satisfied:

$$G(1_X) = 1_{G(X)} \quad \text{for all } X \,,$$
$$G(g \circ f) = G(g) \circ G(f) \,.$$

That is, a functor must preserve composition and identities. It is immediate that if f is an equivalence in $\mathbf{C}$, then $G(f)$ is an equivalence in $\mathbf{D}$.

EXAMPLE 3.7.4 We list here a number of different categories. In all these examples, composition is either the usual composition of functions or is induced by said composition. Equivalences in some of these categories are given special names. In such cases, the name is listed in parentheses.

(a) Sets and maps (bijective correpondences).

(b) Topological spaces and continuous maps (homeomorphisms).

(c) Topological spaces and homotopy classes of maps (homotopy equivalences).

(d) Simplicial complexes and simplicial maps (simplicial homeomorphisms).

(e) Simplicial complexes and continuous maps of their polytopes.

(f) Simplicial complexes and homotopy classes of continuous maps.

(g) Groups and homomorphisms (isomorphisms).

(h) Chain complexes and chain maps.

(i) Chain complexes and chain-homotopy classes (chain equivalences).

(j) Short exact sequences of abelian groups and homomorphisms of such.

(k) Short exact sequences of chain complexes and homomorphisms of such.

(l) Long exact sequences of abelian groups and homomorphisms of such.

(m) Pairs (X, Y) of topological spaces and pairs (f, g) of continuous maps.

EXAMPLE 3.7.5 We list now several examples of functors.

(a) The correspondence assigning to a pair (X, Y) of spaces the space $X \times Y$, and to a pair (f, g) of continuous maps the map $f \times g$. This is a functor from pairs of spaces to spaces.

(b) The correspondence assigning to a space X its underlying set, and to a continuous map the underlying set map. This is a functor from spaces to sets. It is called the *forgetful functor*—it "forgets" the topological structure involved.

(c) The correspondence $K \to \mathcal{C}(K)$ and $f \to f_{\#}$ is a functor from the category of simplicial complexes and simplicial maps to the category of chain complexes and chain maps.

(d) Given a homology theory, the correspondence $X \to H_p(X)$ and $[h] \to h_*$ is a functor from the category of admissible spaces and homotopy classes of maps to the category of abelian groups and homomorphisms. [Here $[h]$ denotes the homotopy class of h.] This is precisely the substance of the first two Eilenberg–Steenrod axioms and the homotopy axioms.

(e) The zig-zag lemma assigns to each short exact sequence of chain complexes a long exact sequence of their homology groups. The "naturality" property expressed in Theorem 3.3.3 is just the statement that this assignment is a functor.

Definition 3.7.6 Let G and H be two functors from category $\mathbf{C}$ to category $\mathbf{D}$. A *natural transformation* T from G to H is a rule assigning to each object X of $\mathbf{C}$ a morphism

$$T_X : G(X) \to H(X)$$

of $\mathbf{D}$ such that the following diagram commutes for all morphisms $f : X \to Y$ of the category $\mathbf{C}$:

$$\begin{array}{ccc} G(X) & \xrightarrow{T_X} & H(X) \\ \Big\downarrow{\scriptstyle G(f)} & & \Big\downarrow{\scriptstyle H(f)} \\ G(Y) & \xrightarrow{T_Y} & H(Y) \end{array}$$

If, for each X, the morphism T_X is an equivalence in the category $\mathbf{D}$, then T is called a *natural equivalence* of functors.

EXAMPLE 3.7.7 Given a homology theory, let p be fixed, and consider the following two functors defined on admissible pairs:

$$\begin{aligned} G(X, A) &= H_p(X, A); & G(f) &= f_* , \\ H(X, A) &= H_{p-1}(A); & H(f) &= (f|A)_* . \end{aligned}$$

The commutativity of the diagram

$$\begin{array}{ccc} H_p(X, A) & \xrightarrow{\partial_*} & H_{p-1}(A) \\ \Big\downarrow{\scriptstyle f_*} & & \Big\downarrow{\scriptstyle (f|A)_*} \\ H_p(Y, B) & \xrightarrow{\partial_*} & H_{p-1}(B) \end{array}$$

tells us that ∂_* is a natural transformation of the functor G to the functor H. This is precisely the third of the Eilenberg–Steenrod axioms.

EXAMPLE 3.7.8 Consider the category of pairs of spaces and pairs of maps. Let G and H be the functors

$$\begin{aligned} G(X, Y) &= X \times Y; & G(f, g) &= f \times g , \\ H(X, Y) &= Y \times X; & H(f, g) &= g \times f . \end{aligned}$$

Given (X, Y), let $T_{(X,Y)}$ be the homeomorphism of $X \times Y$ with $Y \times X$ that switches coordinates. Then T is a natural equivalence of G with H.

We have until now been dealing with what we call *covariant* functors. There is also a notion of *contravariant functor* which differs only inasmuch as "all the arrows are reversed." Formally, it is defined as follows:

Definition 3.7.9 A *contravariant functor* G from a category $\mathbf{C}$ to a category $\mathbf{D}$ is a rule that assigns to each object X in $\mathbf{C}$ an object $G(X)$ in $\mathbf{D}$ and to each morphism $f : X \to Y$ of $\mathbf{C}$ a morphism

$$G(f) : G(Y) \to G(X)$$

of $\mathbf{D}$ such that $G(1_X) = 1_{G(X)}$. Also

$$G(g \circ f) = G(f) \circ G(g)\,.$$

A *natural transformation* between contravariant functors is defined in the obvious way.

In our discussions here, we have not yet studied any contravariant functors; but we shall do so in the future. We close this section with an example taken from linear algebra.

EXAMPLE 3.7.10 If V is a vector space over $\mathbb{R}$, consider the space $\mathcal{L}(V, \mathbb{R})$ of linear functionals on V (i.e., linear transformations of V into $\mathbb{R}$). It is often called the *dual space* to V. The space $\mathcal{L}(V, \mathbb{R})$ has the structure of a vector space, in the obvious fashion. Now if $f : V \to W$ is a linear transformation from V to another vector space W, then there is a linear transformation

$$f^{\text{tr}} : \mathcal{L}(W, \mathbb{R}) \to \mathcal{L}(V, \mathbb{R})$$

which is often called the *transpose* (or *adjoint*) of f. It is defined as follows: If $\alpha : W \to \mathbb{R}$ is a linear functional on W, then $f^{\text{tr}}(\alpha) : V \to \mathbb{R}$ is the linear functional on V which is the composition

$$V \xrightarrow{f} W \xrightarrow{\alpha} \mathbb{R}\,.$$

The assignment

$$V \to \mathcal{L}(V, \mathbb{R}) \qquad \text{and} \qquad f \to f^{\text{tr}}$$

is a contravariant functor from the category of vector spaces and linear transformations to itself.

EXERCISES

1. Check that each class of objects and morphisms listed in Example 3.7.4 is indeed a category, and that the equivalences are as stated [Only **(c)** and **(i)** require any particular attention.]

2. Functors have appeared already in your study of mathematics, although that terminology may not have been used. Here are some examples. Recall their definitions now.

(a) Consider the category of complexes and equivalence classes of simplicial maps, the equivalence relation being generated by the relation of contiguity. Define a functor to chain complexes and chain homotopy classes of chain maps.

(b) Give a functor from abstract complexes to geometric complexes, called the "geometric realization functor." What are the maps involved?

(c) Give functors from families of abelian groups, indexed with the fixed index set J, to abelian groups, called "direct product" and "direct sum."

(d) In algebra, there is a functor $G \to G/[G,G]$ called the "abelianization functor." Either recall it or look it up.

(e) In topology, there is a functor from completely regular spaces to compact Hausdorff spaces called the Stone-Čech compactification. Either recall it or look it up. Do not forget to deal with maps as well as spaces.

3. Let G and H be the functors assigning to a complex K its oriented chain complex and its ordered chained complex, respectively. To each simplicial map they assign the induced chain map.

(a) There is a natural transformation either of G to H or of H to G. Which?

(b) Show that if you consider G and H as taking values in the category of chain complexes and chain homotopy classes of chain maps, then both natural transformations exist and are natural equivalences.

4. Consider the category of pairs (X,Y) of triangulable spaces such that $X \times Y$ is triangulable, and pairs of continuous maps. Define

$$G(X,Y) = H_p(X \times Y) \qquad \text{and} \qquad G(f,g) = (f \times g)_*\,,$$
$$H(X,Y) = H_p(X) \times H_p(Y) \qquad \text{and} \qquad H(f,g) = f_* \times g_*\,.$$

Define a natural transformation of G to H. Show that it is not a natural equivalence.

5. We have not yet studied a category where the morphisms are other than maps in the usual sense or equivalence classes of maps. Here is an example, for those who are familiar with the fundamental group of a topological space.

Let X be a fixed space. Let $\mathbf{C}$ be the category whose objects are the

points of X. And let $\hom(x_1, x_0)$ consist of path homotopy classes $[\alpha]$ of paths α from x_0 to x_1. The composition operation

$$\hom(x_2, x_1) \times \hom(x_1, x_0) \to \hom(x_2, x_0)$$

is induced by the usual composition of paths $(\beta, \alpha) \to \alpha * \beta$.

(a) Check that **C** is a category in which every morphism is an equivalence, i.e., **C** is a groupoid.

(b) Check that the assignment $x_0 \to \pi_1(X, x_0)$ and $[\alpha] \to \widehat{\alpha}$ is a contravariant functor from **C** to the category of groups and homomorphisms. [Here $\widehat{\alpha}([f]) = [\overline{\alpha} * f * \alpha]$, where $\overline{\alpha}$ is the reverse path to α. [**Note:** The reason we must let $[\alpha]$ represent an element of $\hom(x_1, x_0)$ rather than $\hom(x_0, x_1)$ arises from the awkward fact that when we compose paths we put the first path on the left, while when we compose maps we put the first function on the right.]

6. Give an example of a contravariant functor that arises in the context of abelian groups.

4

Singular Homology Theory

4.1 Introduction

Now we are in some sense going to begin all over again. We construct a new homology theory. As compared with simplicial theory, it is much more "natural." For one thing, the homology groups are defined for arbitrary topological spaces, not only for triangulable ones. For another, the homomorphism induced by a continuous map is defined directly and its functorial properties are proved easily; no difficult results such as the simplicial approximation theorem are needed. The topological invariance of the singular homology groups follows at once.

Unfortunately, the singular homology groups are not immediately computable. One must develop a good deal of singular theory before one can compute the homology of even such a simple space as the sphere $\mathbb{S}^n$. Eventually, when we develop the theory of CW complexes, we will see how singular homology can be computed fairly readily. Alternatively, since we shall show that simplicial and singular homology groups are isomorphic for triangulable spaces, we can always go back to simplicial theory if we want to compute something.

In this chapter, we construct the singular homology groups and prove that they satisfy the Eilenberg–Steenrod axioms on the class of all topological spaces. [The homotopy axiom and the excision axiom, it turns out, require more effort than the others.] Then we construct a specific isomorphism between the singular and simplicial theories that will be useful later.

Finally, we give a number of applications. They include the Jordan curve theorem, theorems about manifolds, and the computation of the homology of real and complex projective spaces.

4.2 The Singular Homology Groups

In this section we define the singular homology groups and derive their elementary properties. First, we introduce some notation.

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Let $\mathbb{R}^\infty$ denote the vector space $\mathbb{E}^J$, where $J = \mathbb{Z}_+$, the set of positive integers. An element of $\mathbb{R}^\infty$ is an infinite sequence of real numbers having only finitely many non-zero components. Let Δ_p denote the p-simplex in $\mathbb{R}^\infty$ having as vertices the origin and the first p standard basis elements, that is,

$$\begin{aligned}
\epsilon_0 &= (0, 0, \dots, 0, 0, \dots), \\
\epsilon_1 &= (1, 0, \dots, 0, 0, \dots), \\
&\vdots \quad \vdots \qquad \vdots \\
\epsilon_p &= (\underbrace{0, 0, \dots, 1,}_{p \text{ entries}} 0, \dots)
\end{aligned}$$

We call Δ_p the *standard p-simplex*. Note that, under this definition, Δ_{p-1} is a face of Δ_p.

If X is a topological space, we define a *singular p-simplex* of X to be a continuous map

$$T : \Delta_p \to X .$$

The word "singular" is used to emphasize that T need not be an imbedding. The map T could be a constant map for instance.

The free abelian group generated by the singular p-simplices of X is denoted by $S_p(X)$ and is called the *singular chain group* of X in dimension p. By our usual convention for free abelian groups [see Section 1.5], we shall denote an element of $S_p(X)$ by a formal linear combination, with integer coefficients, of singular p-simplices.

It is convenient to consider a special type of singular simplex. Given points $a_0, a_1, \dots, a_p$ in some euclidean space $\mathbb{E}^J$, which need not be independent, there is a unique affine map ℓ of Δ_p into $\mathbb{E}^J$ that maps ϵ_i to a_i for $i = 0, 1, \dots, p$. It is defined by the equation

$$\ell(x_1, x_2, \dots, x_p, 0 \dots) = a_0 + \sum_{i=1}^{n} x_i(a_i - a_0) .$$

We call this map the *linear singular simplex* determined by $a_0, a_1, \dots, a_p$, and we denote it by $\ell(a_0, a_1, \dots, a_p)$.

For example, the map $\ell(\epsilon_0, \epsilon_1, \dots, \epsilon_p)$ is simply the inclusion map of Δ_p into $\mathbb{R}^\infty$.

Similarly, if as usual we use the notation $\widehat{v_i}$ to mean the symbol v_i is to be deleted, then the map

$$\ell(\epsilon_0, \epsilon_1, \dots, \widehat{\epsilon_i}, \dots, \epsilon_p)$$

is a map of Δ_{p-1} into $\mathbb{R}^\infty$ that carries Δ_{p-1} by a linear homeomorphism onto the face $(\epsilon_0, \epsilon_1, \dots, \epsilon_{i-1}, \epsilon_{i+1}, \dots, \epsilon_p)$ of Δ_p. We often consider it as a map of Δ_{p-1} into Δ_p rather than into $\mathbb{R}^\infty$. Then, if $T : \Delta_p \to X$, we can form the composition

$$T \circ \ell(\epsilon_0, \epsilon_1, \dots, \widehat{\epsilon_i}, \dots, \epsilon_p) .$$

This is a singular $(p-1)$-simplex of X, which we think of as the "ith face" of the singular p-simplex T.

We now define a homomorphism $\partial : S_p(X) \to S_{p-1}(X)$. If $T : \Delta_p \to X$ is a singular p-simplex, then the set

$$\partial T = \sum_{i=0}^{p} (-1)^p \, T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon}_i, \ldots, \epsilon_p) \, .$$

Thus ∂T equals a formal sum of singular simplices of dimension $p-1$, which are the "faces" of T. We shall verify presently that $\partial^2 = 0$.

If $f : X \to Y$ is a continuous map, then we define the homomorphism $f_\# : S_p(X) \to S_p(Y)$ by the equation $f_\#(T) = f \circ T$, that is, $f_\#(T)$ is defined by the composition

$$\Delta_p \xrightarrow{T} X \xrightarrow{f} Y \, .$$

Theorem 4.2.1 *The homomorphism $f_\#$ commutes with ∂. Furthermore, we have $\partial^2 = 0$.*

Proof: The first statement follows by direct computation:

$$\begin{aligned} \partial f_\#(T) &= \sum_{i=0}^{p} (-1)^i \, (f \circ T) \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon}_i, \ldots, \epsilon_p) \, , \\ f_\#(\partial T) &= \sum_{i=0}^{p} (-1)^i \, f \circ (T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon}_i, \ldots, \epsilon_p)) \, . \end{aligned}$$

To prove the second statement, we first compute ∂ for linear singular simplices. We have

$$\begin{aligned} \partial \ell(a_0, a_1, \ldots, a_p) &= \sum_{i=0}^{p} (-1)^i \, \ell(a_0, a_1, \ldots, a_p) \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon}_i, \ldots, \epsilon_p) \\ &= \sum_{i=0}^{p} (-1)^i \, \ell(a_0, a_1, \ldots, \widehat{a}_i, \ldots, a_p) \, , \end{aligned}$$

where the second equality comes from the fact that a composition of linear maps is linear [see the remark after Lemma 1.3.17]. That $\partial\partial(\ell(a_0, a_1, \ldots, a_p)) = 0$ now follows from the proof that $\partial^2 = 0$ in simplicial theory [Lemma 1.6.9]. Simply take that proof and insert ℓ at appropriate places.

The general result now follows from the facts that

$$\partial\partial(T) = \partial\partial(T_\#(\ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)))$$

and that ∂ commutes with $T_\#$. □

Definition 4.2.2 The family of groups $S_p(X)$ and homomorphisms $\partial : S_p(X) \to S_{p-1}(X)$ is called the *singular chain complex* of X, and is denoted $\mathcal{S}(X)$. The homology groups of this chain complex are called the *singular homology groups* of X, and are denoted $H_p(X)$.

If X is triangulable, this notation overlaps with that introduced in Section 3.6 for the simplicial homology of a triangulable space. We shall prove later in Section 4.7 that the simplicial and singular homology theories are naturally isomorphic, so there is in fact no real ambiguity involved.

The chain complex $\mathcal{S}(X)$ is augmented by the homomorphism $\epsilon : \mathcal{S}_0(X) \to \mathbb{Z}$ defined by setting $\epsilon(T) = 1$ for each 0-simplex $T : \Delta_0 \to X$. It is immediate that, if T is a singular 1-simplex, then $\epsilon(\partial T) = 0$. The homology groups of $\{\mathcal{S}(X), \epsilon\}$ are called the *reduced singular homology groups* of X, and are denoted $\widetilde{H}_p(X)$. If $f : X \to Y$ is a continuous map, then $f_\# : \mathcal{S}_p(X) \to \mathcal{S}_p(Y)$ is an augmentation-preserving chain map, since $f_\#(T)$ is a singular 0-simplex if T is. Thus $f_\#$ induces a homomorphism f_* in both ordinary and reduced singular homology.

If the reduced homology of X vanishes in all dimensions, we say that X is *acyclic* [in singular homology].

Theorem 4.2.3 *If $i : X \to X$ is the identity, then $i_* : H_p(X) \to H_p(X)$ is the identity. If $f : X \to Y$ and $g : Y \to Z$ are continuous maps, then $(g \circ f)_* = g_* \circ f_*$. The same facts hold in reduced homology.*

Proof: These equations hold on the chain level. We have $i_\#(T) = i \circ T = T$, and $(g \circ f)_\#(T) = (g \circ f) \circ T = g \circ (f \circ T) = g_\#(f_\#(T))$. □

The next corollary follows immediately.

Corollary 4.2.4 (Topological invariance of singular homology groups) *If $h : X \to Y$ is a homeomorphism, then h_* is an isomorphism.*

The reader will note how quickly we have proved that the singular homology groups are topological invariants. This is in sharp contrast to the simplicial theory for which it took us most of Chapter 2 to do the same thing.

Following the pattern of simplicial theory, we next compute the zero-dimensional homology groups.

Theorem 4.2.5 *Let X be a topological space. Then $H_0(X)$ is free abelian. If $\{X_\alpha\}$ is the collection of path components of X, and if T_α is a singular 0-simplex with image in X_α, for each α, then the homology classes of the chains T_α form a basis for $H_0(X)$.*

The group $\widetilde{H}_0(X)$ is free abelian; it vanishes if X is path connected. If X is not path connected, let α_0 be a fixed index; then the homology classes of the chains $T_\alpha - T_{\alpha_0}$, for $\alpha \neq \alpha_0$, form a basis for $\widetilde{H}_0(X)$.

Proof: If $T : \Delta_0 \to X$ is any singular 0-simplex of X, then the point $x = T(\Delta_0)$ is in some path component of X, say X_α. There must be a path $f : [0,1] \to X$ from the point x to the point $x_\alpha = T_\alpha(\Delta_0)$. Then f is a singular 1-simplex and $\partial f = T_\alpha - T$. We conclude that an arbitrary singular 0-chain of X is homologous to a chain of the form $\sum n_\alpha T_\alpha$.

We show that no such 0-chain bounds. Suppose that $\sum n_\alpha T_\alpha = \partial d$ for some d. Now each singular 1-simplex in the expression for d has a path connected image, so its image lies in precisely one of the path components X_α. Thus we can write $d = \sum d_\alpha$, where d_α consists of those terms of d carried by X_α. Then ∂d_α lies in X_α as well. It follows that $n_\alpha T_\alpha = \partial d_\alpha$ for each α. Applying ϵ to both sides of this last equation, we conclude that $n_\alpha = 0$, and thus that $d = 0$.

The computation of $\widetilde{H}_0(X)$ proceeds as in the proof of Theorem 1.8.3. □

Still following the pattern of simplicial theory, we compute the homology of a cone-like space. Actually, it is more convenient here to deal with the analogous notion of a star-convex set. This notion was defined for sets in $\mathbb{R}^N$ in Definition 1.2.5. Here we extend it, in the obvious way, to sets in $\mathbb{E}^J$.

Definition 4.2.6 A set X in $\mathbb{E}^J$ is said to be *star convex* relative to the point w of X if, for each $x \in X$ different from w, the line segment from x to w lies in X.

Definition 4.2.7 Suppose $X \subset \mathbb{E}^J$ is star convex relative to w. We define a *bracket operation* on singular chains of X. Let $T : \Delta_p \to X$ be a singular p-simplex of X. Define a singular $(p+1)$-simplex $[T, w] : \Delta_{p+1} \to X$ by letting it carry the line segment from x to ϵ_{p+1}, for x in Δ_p, linearly onto the line segment from $T(x)$ to w. See Figure 4.2.1. We extend the definition to arbitrary p-chains as follows: If $c = \sum n_i T_i$ is a singular p-chain on X, let

$$[c, w] = \sum_i n_i [T_i, w] .$$

This operation is similar to the one we introduced in Section 1.9 for cones, except here we have put the vertex w last instead of first.

Note that, when restricted to the face Δ_p of Δ_{p+1}, the map $[T, w]$ equals the map T. Note also that, if T is the linear singular simplex $\ell(a_0, a_1, \ldots, a_p)$, then $[T, w]$ equals the linear singular simplex $\ell(a_0, a_1, \ldots, a_p, w)$.

We must show that the map $[T, w]$ is continuous. First, we note that the map $\pi : \Delta_p \times I \to \Delta_{p+1}$ defined by the equation $\pi(x, t) = (1-t)x + t\epsilon_{p+1}$ is a quotient map that collapses $\Delta_p \times 1$ to the vertex ϵ_{p+1} and is otherwise one-to-one. The continuous map

$$f : \Delta_p \times I \to X$$

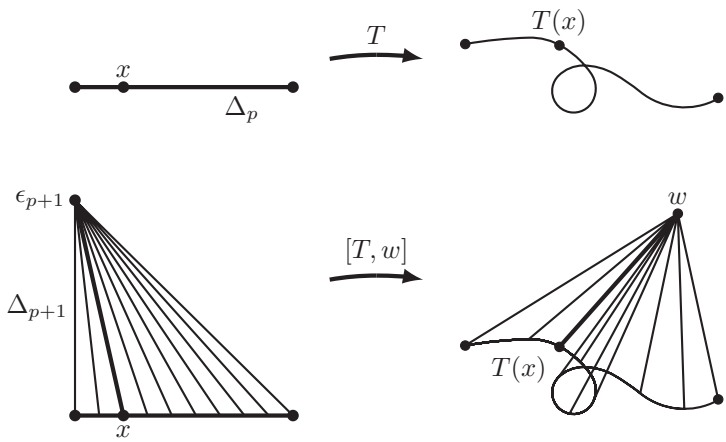


Figure 4.2.1

defined by $f(x,t) = (1-t)T(x)+tw$ is constant on Δ_p; because π is a quotient map, it induces a continuous map of Δ_{p+1} into X. Since π maps the line segment $x \times I$ linearly onto the line segment from x to ϵ_{p+1}, and f maps $x \times I$ linearly onto the line segment from $T(x)$ to w, this induced map equals the singular $(p+1)$-simplex $[T, w]$ defined previously. Thus $[T, w]$ is continuous.

We now compute how the bracket operation and the boundary operator interact.

Lemma 4.2.8 *Let X be star convex with respect to w, and let c be a singular p-chain of X. Then*

$$\partial[c, w] = \begin{cases} [\partial c, w] + (-1)^{p+1}c & \text{if} \quad p > 0, \\ \epsilon(c)T_w - c & \text{if} \quad p = 0, \end{cases}$$

where T_w is the singular 0-simplex mapping Δ_0 to w.

Proof: If T is a singular 0-simplex, then $[T, w]$ maps the simplex Δ_1 linearly onto the line segment from $T(\Delta_0)$ to w. Then we have $\partial[T, w] = T_w - T$, and the second formula follows.

Suppose $p > 0$. It suffices to check the formula when c is a singular p-simplex T. The formula's plausibility when $p = 1$ is illustrated in Figure 4.2.2.

Using the definition of ∂, we compute

$$\partial[T, w] = \sum_{i=0}^{p+1} (-1)^i [T, w] \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i}, \ldots, \epsilon_{p+1}) \,. \qquad (*)$$

For convenience, we will use the notation ℓ_i to denote the linear singular simplex on the right-hand side of $(*)$, that is, we will write

$$\ell_i = \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i}, \ldots, \epsilon_{p+1})\,.$$

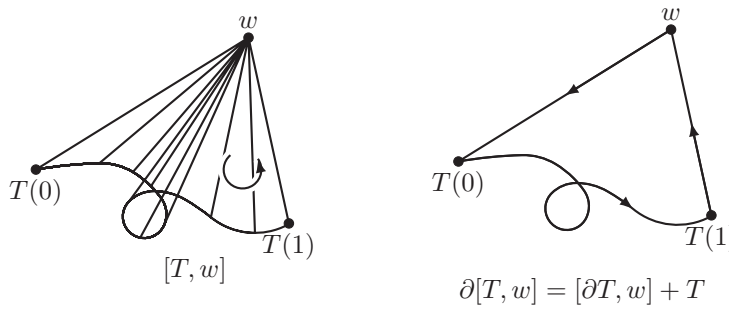


Figure 4.2.2

Observe that ℓ_{p+1} equals the inclusion map of Δ_p into Δ_{p+1}. Since the restriction of $[T, w]$ to Δ_p equals the map T, the last term of $(*)$ equals $(-1)^{p+1}T$.

To complete the proof, we consider the singular simplex $[T, w] \circ \ell_i$ for $i < p+1$. Now the map ℓ_i carries Δ_p by a linear homeomorphism onto the ith face of Δ_{p+1}; it carries $\epsilon_0, \epsilon_1, \ldots, \epsilon_p$ to $\epsilon_0, \epsilon_1, \ldots, \epsilon_{i-1}, \epsilon_{i+1}, \ldots, \epsilon_{p+1}$. Therefore the restriction of ℓ_i to $\Delta_{p-1} = \epsilon_0\epsilon_1 \ldots \epsilon_{p-1}$ carries this simplex by a linear map onto the simplex spanned by $\epsilon_0, \epsilon_1, \ldots, \epsilon_{i-1}, \epsilon_{i+1}, \ldots, \epsilon_p$. Thus we have

$$\ell_i|\Delta_{p-1} = \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i}, \ldots, \epsilon_p)\,. \qquad (**)$$

Now we can compute $[T, w] \circ \ell_i : \Delta_p \to X$. Let x be a general point of Δ_{p-1}. Since $\ell_i : \Delta_p \to \Delta_{p+1}$ is a linear map, it carries the line segment from x to ϵ_p linearly onto the line segment from $\ell_i(x)$ to ϵ_{p+1}. Since $\ell_i(x) \in \Delta_p$, the map $[T, w] : \Delta_{p+1} \to X$ by definition carries this line segment linearly onto the line segment from $T(\ell_i(x))$ to w. Therefore, by definition, we have

$$[T, w] = [T \circ (\ell_i|\Delta_{p-1}), w]\,.$$

Substituting this formula into $(*)$ and using $(**)$, we obtain the equation

$$\begin{aligned} \partial[T, w] &= \sum_{i=0}^{p} (-1)^i [T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i}, \ldots, \epsilon_p), w] + (-1)^{p+1} T \\ &= [\partial T, w] + (-1)^{p+1} T\,. \end{aligned}$$

□

Theorem 4.2.9 *Let X be a subspace of $\mathbb{E}^J$ that is star convex relative to w. Then X is acyclic in singular homology.*

Proof: To show that $\widetilde{H}_0(X) = 0$, let c be a singular 0-chain on X such that $\epsilon(c) = 0$. Then, by the preceding lemma, we have

$$\partial[c, w] = \epsilon(c)T_w - c = -c\,,$$

so c bounds a 1-chain.

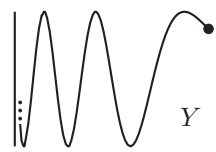
To show that $H_p(X) = 0$ for $p > 0$, let z be a singular p-cycle on X. By the preceding lemma, we have

$$\partial[z, w] = [\partial z, w] + (-1)^{p+1}z = (-1)^{p+1}z\,,$$

so z bounds a $(p+1)$-chain. □

Corollary 4.2.10 *Any simplex is acyclic in singular homology.*

Definition 4.2.11 Figure 4.2.3 shows the *topologist's sine curve*. It is the subspace of $\mathbb{R}^2$ consisting of all points $(x, \sin 1/x)$ for $0 < x \leq 1$, and all points $(0, y)$ for $-1 \leq y \leq 1$. It will be used in Exercise 1(b).



The Topologist's Sine Curve

Figure 4.2.3

EXERCISES

1. **(a)** Suppose $\{X_\alpha\}$ is the set of path components of X. Show that $H_p(X) \cong \bigoplus_\alpha H_p(X_\alpha)$.

(b) Compute the singular homology of the topologist's sine curve, Y.

2. Let X be a compact subspace of $\mathbb{R}^N$, and let $f : X \to Y$.

(a) Let w be the point $(0, 0, \ldots, 0, 1)$ in $\mathbb{R}^N \times \mathbb{R}$. Let C be the union of all line segments joining points of X to w. Show that C is a quotient space of $X \times I$.

(b) Show that if f is homotopic to a constant map, then f_* is the zero homomorphism in reduced homology. [**Hint:** Show that f extends to C.]

(c) Show that if X is contractible, then X is acyclic.

3. Calculate H_1 of the unit disc in $\mathbb{R}^2$.

4. Show that if $U \subset \mathbb{R}^2$ is homeomorphic to the unit disc, then all its homology groups H_p for $p > 0$ are 0.

4.3 The Axioms for Singular Theory

We now introduce the relative singular homology groups. Then we show, in this section and the next, that they satisfy the Eilenberg–Steenrod axioms on the class of all topological pairs.

If X is a topological space and A is a subspace of X, then there is a natural inclusion $S_p(A) \to S_p(X)$. The group of *relative singular chains* is defined by

$$S_p(X, A) = S_p(X)/S_p(A)\,.$$

The boundary operator $S_p(X) \to S_{p-1}(X)$ restricts to the boundary operator of $S_p(A)$; hence it induces a boundary operator

$$\partial : S_p(X, A) \to S_{p-1}(X, A)$$

on relative chains. The family of groups $S_p(X, A)$ and homomorphisms ∂ is called the *singular chain complex* of the pair (X, A) and is denoted $\mathcal{S}(X, A)$. The homology groups of this chain complex are called the *singular homology groups* of the pair (X, A) and are denoted $H_p(X, A)$.

Note that the chain complex $\mathcal{S}(X, A)$ is free. The group $S_p(X, A)$ has as its basis all cosets of the form $T + S_p(A)$, where T is a singular p-simplex whose image set does not lie in A.

If $f : (X, A) \to (Y, B)$ is a continuous map, then the homomorphism $f_\# : S_p(X) \to S_p(Y)$ carries chains of A into chains of B, so it induces a homomorphism [also denoted $f_\#$]

$$f_\# : S_p(X, A) \to S_p(Y, B)\,.$$

This map commutes with ∂, so it in turn induces a homomorphism

$$f_* : H_p(X, A) \to H_p(Y, B)\,.$$

The following theorem is immediate:

Theorem 4.3.1 *If $i : (X, A) \to (X, A)$ is the identity map, then $i_* : H_p(X) \to H_p(X)$ is the identity. If $h : (X, A) \to (Y, B)$ and $k : (Y, B) \to (Z, C)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$.*

Theorem 4.3.2 *There is a homomorphism $\partial_* : H_p(X, A) \to H_{p-1}(A)$, defined for $A \subset X$ and all p, such that the sequence*

$$\cdots \longrightarrow H_p(A) \xrightarrow{i_*} H_p(X) \xrightarrow{\pi_*} H_p(X, A) \xrightarrow{\partial_*} H_{p-1}(A) \longrightarrow \cdots$$

is exact, where i and π are inclusions. The same holds if reduced homology is used for X and A, provided $A \neq \emptyset$.

A continuous map $f : (X, A) \to (Y, B)$ induces a homomorphism of the corresponding exact sequences in singular homology, either ordinary or reduced.

Proof: For the existence of ∂_* and the exact sequence, we apply the "zig-zag lemma" [Lemma 3.3.2] to the short exact sequence of chain complexes

$$0 \longrightarrow S_p(A) \xrightarrow{i_\#} S_p(X) \xrightarrow{\pi_\#} S_p(X, A) \longrightarrow 0\,,$$

where $i : A \to X$ and $\pi : (X, \emptyset) \to (X, A)$ are inclusions. The naturality of ∂_* follows from Theorem 3.3.3, once we note that $f_\#$ commutes with $i_\#$ and $\pi_\#$. The results for reduced homology follow by the same methods used for simplicial theory in the proof of Lemma 3.3.5. □

Theorem 4.3.3 *If P is a one-point space, then $H_p(P) = 0$ for $p \neq 0$ and $H_0(P) = \mathbb{Z}$.*

Proof: This result follows from Theorem 4.2.9, once one notes that a one-point space in $\mathbb{R}^N$ is star convex relative to the one and only point in the space.

An explicit proof is the following: There is exactly one singular simplex $T : \Delta_p \to P$ in each non-negative dimension, so $S_p(P)$ is infinite cyclic for $p \geq 0$. Since

$$T_p \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i}, \ldots, \epsilon_p) = T_{p-1}\,,$$

we have

$$\partial T_p = \sum_{i=0}^{p} (-1)^i T_{p-1} = \begin{cases} 0 & \text{if } p \text{ is odd,} \\ T_{p-1} & \text{if } p \text{ is even.} \end{cases}$$

The chain complex $\mathcal{S}(P)$ is thus of the form

$$\begin{array}{ccccccccc} \xrightarrow{\partial} & S_{2k}(P) & \xrightarrow{\partial} & S_{2k-1}(P) & \xrightarrow{\partial} & S_{2k-2}(P) & \xrightarrow{\partial} \\ & \| & & \| & & \| & \\ \xrightarrow{0} & \mathbb{Z} & \xrightarrow{\cong} & \mathbb{Z} & \xrightarrow{0} & \mathbb{Z} & \xrightarrow{\cong} \end{array}$$

ending with

$$\begin{array}{ccccccc} \xrightarrow{\partial} & S_1(P) & \xrightarrow{\partial} & S_0(P) & \longrightarrow & 0 \\ & \| & & \| & & \\ \xrightarrow{\cong} & \mathbb{Z} & \xrightarrow{0} & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

In dimension $2k-1$, every chain is a boundary; while in dimension $2k>0$, no chain is a cycle. In dimension 0, every chain is a cycle and no chain is a boundary, so the homology is infinite cyclic. □

Theorem 4.3.4 *Given $\alpha \in H_p(X,A)$, there is a compact pair $(X_0,A_0)\subset (X,A)$, such that α is in the image of the homomorphism induced by inclusion*

$$i_*: H_p(X_0,A_0)\to H(X,A)\,.$$

Proof: If $T:\Delta_p\to X$ is a singular simplex, its *minimal carrier* is defined to be the image set $T(\Delta_p)$. The minimal carrier of a singular p-chain $\sum n_iT_i$ [where each $n_i\neq 0$] is the union of the minimal carriers of the T_i. Since the sum is finite, the minimal carrier of a p-chain is compact.

Let c_p be a chain is $S_p(X)$ whose coset modulo $S_p(A)$ represents α. Then ∂c_p is carried by A. Let X_0 be the minimal carrier of c_p and let A_0 be the minimal carrier of ∂c_p. Then c_p can be taken to represent a homology class β in $H_p(X_0,A_0)$ and $i_*(\beta)=\alpha$. □

The preceding theorem shows that singular theory satisfies the compact support axiom. There is an addendum to this theorem, which we prove here. [It can also be derived directly from the axioms; see the exercises of Section 3.6.]

Theorem 4.3.5 *Let (X_0,A_0) be a compact pair, and let $i:(X_0,A_0)\to(X,A)$ be inclusion. If $\alpha\in H_p(X_0,A_0)$ and $i_*(\alpha)=0$, then there are a compact pair (X_1,A_1) and inclusions*

$$(X_0,A_0)\xrightarrow{j}(X_1,A_1)\xrightarrow{k}(X,A)$$

such that $j_*(\alpha)=0$.

Proof: Let c_p be a singular p-chain of X_0 representing α. Then ∂c_p is carried by A_0. By hypothesis, there is a chain d_{p+1} of X such that $c_p-\partial d_{p+1}$ is carried by A. Let X_1 be the union of X_0 and the minimal carrier of d_{p+1}, and let A_1 be the union of A_0 and the minimal carrier of $c_p-\partial d_{p+1}$. □

Thus far everything has been relatively easy. We have verified all the axioms but the homotopy and excision axioms. These two require more work. We verify the homotopy axiom now, while we defer the excision axiom to the next section.

In outline, the proof is similar to the proof of the homotopy axiom for simplicial theory [Theorem 2.7.5]. We consider the inclusion maps $i,j:X\to X\times I$ defined by

$$i(x)=(x,0)\quad\text{and}\quad j(x)=(x,1)\,,$$

and we construct a chain homotopy D between $i_\#$ and $j_\#$. If F is a homotopy between f and g, then $F_\#\circ D$ is a chain homotopy between $f_\#$ and $g_\#$, and the

theorem follows. To construct D in simplicial theory, we used acyclic carriers, but here we need something more general. It is a special case of what we shall later call the "method of acyclic models."

It is plausible that a chain homotopy D between $i_\#$ and $j_\#$ should exist. For example, if T is the singular 1-simplex pictured in Figure 4.3.1, and if $\partial T = T_1 - T_0$, we can take DT_1 and DT_0 to be the "vertical" singular 1-simplices pictured. Then we can take DT to be a 2-chain that covers the shaded region in the figure, having as boundary $j_\#(T) - DT_1 - i_\#(T) + DT_0$.

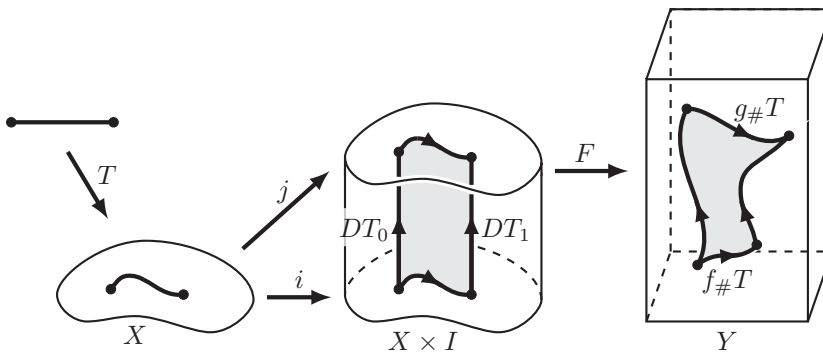


Figure 4.3.1

We state the existence of D as a separate lemma:

Lemma 4.3.6 *For each space X and each non-negative integer p, there exists a homomorphism*

$$D_X : S_p(X) \to S_{p+1}(X \times I)$$

having the following properties:

(a) *If $T : \Delta_p \to X$ is a singular simplex, then*

$$\partial D_X T + D_X \partial T = j_\#(T) - i_\#(T)\,.$$

Here the map $i : X \to X \times I$ carries x to $(x, 0)$ and the map $j : X \to X \times I$ carries x to $(x, 1)$.

(b) *The homomorphism D_X is natural, that is, if $f : X \to Y$ is a continuous map, then the following diagram commutes:*

$$\begin{array}{ccc} S_p(X) & \xrightarrow{D_X} & S_{p+1}(X \times I) \\ \downarrow f_\# & & \downarrow (f \times i_I)_\# \\ S_p(Y) & \xrightarrow{D_Y} & S_{p+1}(Y \times I)\,. \end{array}$$

Proof: We proceed by induction on p. Suppose $p = 0$. Given $T : \Delta_0 \to X$, let x_0 denote the point $T(\Delta_0)$, and define $D_X T : \Delta_1 \to X \times I$ by the equation

$$D_X T(t, 0, 0, \dots) = (x_0, t) \quad \text{for} \quad 0 \le t \le 1.$$

[Recall that $\Delta_1 = \epsilon_0 \epsilon_1$ consists of the points $(t, 0, 0, \dots)$ of $\mathbb{R}^\infty$ with $0 \le t \le 1$.] Properties **(a)** and **(b)** follow easily.

Now suppose that $p > 0$ and that D_X has been defined in dimensions less than p for all spaces X so that properties **(a)** and **(b)** hold. We define $D_X T$ in dimension p *by first defining it in the case where* $X = \Delta_p$ *and* $T = i_p$*, the identity map from* Δ_p *to itself.* This is the heart of the technique: We deal first with a very special space, called the "model space," and a very special singular simplex on that space.

Let $\widehat{\imath}, \widehat{\jmath} : \Delta_p \to \Delta_p \times I$ map x to $(x, 0)$ and $(x, 1)$, respectively. Let c_p be the singular chain on $\Delta_p \times I$ defined by the equation

$$c_p = \widehat{\jmath}_\#(i_p) - \widehat{\imath}_\#(i_p) - D_{\Delta_p}(\partial i_p)\,.$$

The induction hypothesis guarantees that the far-right term of the last equation is well-defined. To see this, note that i_p is a singular simplex on Δ_p, so i_p belongs to $S_p(\Delta_p)$ and ∂i_p belongs to $S_{p-1}(\Delta_p)$.

We use the induction hypothesis again to show that c_p is a cycle by computing

$$\begin{aligned} \partial c_p &= \partial \widehat{\jmath}_\#(i_p) - \partial \widehat{\imath}_\#(i_p) - \partial D_{\Delta_p}(\partial i_p) \\ &= \partial \widehat{\jmath}_\#(i_p) - \partial \widehat{\imath}_\#(i_p) - [\widehat{\jmath}(\partial i_p) - \widehat{\imath}(\partial i_p) - D_{\Delta_p} \partial \partial i_p] = 0\,. \end{aligned}$$

Since $\Delta_p \times I$ is convex, it is acyclic by Theorem 4.2.9. Therefore we can choose an element of $S_{p+1}(\Delta_p \times I)$ whose boundary equals c_p. We denote this element by $D_{\Delta_p}(i_p)$. Then **(a)** holds because

$$\partial D_{\Delta_p}(i_p) = c_p = \widehat{\jmath}_\#(i_p) - \widehat{\imath}_\#(i_p) - D_{\Delta_p}(\partial i_p)\,.$$

Having dealt with the special case of the singular simplex i_p on the space Δ_p, we now consider an arbitrary space X and an arbitrary singular p-simplex $T : \Delta_p \to X$. In this case, we define

$$D_X T = (T \times i_I)_\#(D_{\Delta_p} i_p)\,.$$

See Figure 4.3.2. Intuitively, $D_{\Delta_p} i_p$ is a singular $(p+1)$-chain filling up the entire prism $\Delta_p \times I$, and its boundary is, roughly speaking, the boundary of the prism. The map $(T \times i_I)_\#$ carries this chain over to a singular chain on $X \times I$.

To check that **(a)** holds, we compute

$$\begin{aligned} \partial D_X T &= (T \times i_I)_\# \partial D_{\Delta_p} i_p \\ &= (T \times i_I)_\#(\widehat{\jmath}_\#(i_p) - \widehat{\imath}_\#(i_p) - D_{\Delta_p} \partial i_p) \\ &= j_\#(T) - i_\#(T) - (T \times i_I)_\# D_{\Delta_p}(\partial i_p)\,. \qquad (*) \end{aligned}$$

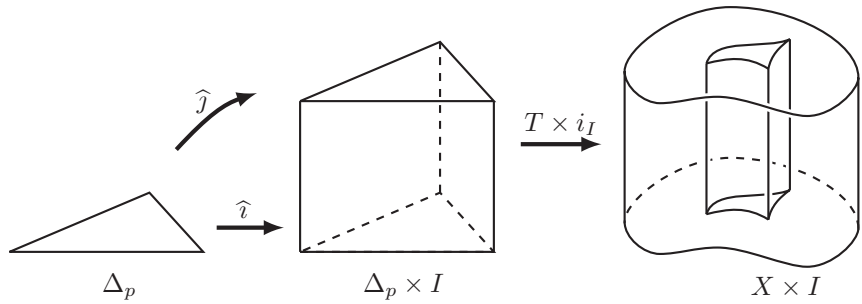


Figure 4.3.2

Now we use naturality. By the induction hypothesis, we have commutativity in the following diagram:

$$\begin{array}{ccc} S_{p-1}(\Delta_p) & \xrightarrow{D_{\Delta_p}} & S_p(\Delta_p \times I) \\ \big\downarrow T_\# & & \big\downarrow (T \times i_I)_\# \\ S_{p-1}(X) & \xrightarrow{D_X} & S_p(X \times I)\,. \end{array}$$

Therefore we have $(T \times i_I)_\# D_{\Delta_p}(\partial i_p) = D_X T_\#(\partial i_p)$, so $(*)$ equals

$$j_\#(T) - i_\#(T) - D_X T_\#(\partial i_p)\,,$$

as desired.

Checking that **(b)** holds is easier. If $f : X \to Y$, then we first note that $(f \circ T) \times i_I = (f \times i_I) \circ (T \times i_I)$. Using the definitions of D_Y and D_X, we have

$$\begin{aligned} D_Y(f_\#(T)) &= D_Y(f \circ T) = ((f \circ T) \times i_I)_\# D_{\Delta_p}(i_p) \\ &= ((f \times i_I) \circ (T \times i_I))_\# D_{\Delta_p}(i_p) = (f \times i_I)_\# D_X T\,. \quad \square \end{aligned}$$

In the proof just given, we gave a direct construction for D_X in the case $p = 0$. Only in the inductive step did we use the "model space" Δ_p. Here is a proof for the case $p = 0$ that looks more like the inductive step. This is the proof we shall later generalize.

Let $p = 0$. We define $D_X T$ first in the case where $X = \Delta_0$ and $T = i_0$, the identity map from Δ_0 to itself. We want the formula

$$\partial D_{\Delta_0}(i_0) = \widehat{\jmath}_\#(i_0) - \widehat{\imath}_\#(i_0)$$

to hold. We note that the right-hand side of this formula is a 0-chain carried by $\Delta_0 \times I$ and when the augmentation map ϵ is applied to it the value is zero. Since the space $\Delta_0 \times I$ is convex, it is acyclic. Therefore, we can choose

$D_{\Delta_0}(i_0)$ to be an element of $S_1(\Delta_0 \times I)$ whose boundary is $\widehat{j}_\#(i_0) - \widehat{i}_\#(i_0)$. Then, for general X and T, we define

$$D_X T = (T \times i_I)_\# D_{\Delta_0}(i_0)\,.$$

Properties **(a)** and **(b)** are proved just as they were in the inductive step.

Theorem 4.3.7 *If $f, g : (X, A) \to (Y, B)$ are homotopic, then $f_* = g_*$. The same fact holds in reduced homology if $A = B = \emptyset$.*

Proof: Let $F : (X \times I, A \times I) \to (Y, B)$ be the homotopy between f and g. Let i, $j : (X, A) \to (X \times I, A \times I)$ be given by $i(x) = (x, 0)$ and $j(x) = (x, 1)$. Let $D_X : S_p(X) \to S_{p+1}(X \times I)$ be the chain homotopy of the preceding lemma. Naturality of D_X with respect to inclusion $A \to X$ shows that the restriction of D_X to $S_p(A)$ equals D_A. Thus D_X carries $S_p(A)$ into $S_{p+1}(A \times I)$, and thus induces a chain homotopy

$$D_{X,A} : S_p(X, A) \to S_{p+1}(X \times I, A \times I)$$

on the relative level. Formula **(a)** of Lemma 4.3.6 holds because $D_{X,A}$ is induced by D_X. Define D to be the composite of $D_{X,A}$ and the homomorphism

$$F_\# : S_{p+1}(X \times I, A \times I) \to S_{p+1}(Y, B)\,.$$

Then we compute

$$\begin{aligned} \partial D &= \partial F_\# D_{X,A} = F_\# \partial D_{X,A} \\ &= F_\#(j_\# - i_\# - D_{X,A}\partial) \\ &= (F \circ j)_\# - (F \circ i)_\# - F_\# D_{X,A}\partial \\ &= f_\# - g_\# - D\partial\,. \end{aligned}$$

□

Definition 4.3.8 Let $f : (X, A) \to (Y, B)$. If there is a map $g : (Y, B) \to (X, A)$ such that $f \circ g$ and $g \circ f$ are homotopic to the appropriate identities *as maps of pairs*, then we call f a *homotopy equivalence*, and we call g a *homotopy inverse* for f.

Theorem 4.3.9 *Let $f : (X, A) \to (Y, B)$.*

> **(a)** *If f is a homotopy equivalence, then f_* is an isomorphism in relative homology.*
>
> **(b)** *More generally, if $f : X \to Y$ and $f|A : A \to B$ are homotopy equivalences, then f_* is an isomorphism in relative homology.*

Proof: Part **(a)** is immediate. To prove **(b)**, we examine the long exact homology sequences of (X, A) and of (Y, B), and the homomorphism f_* carrying one exact sequence to the other. The hypotheses of the theorem tell us that

$$f_* : H_p(X) \to H_p(Y) \quad \text{and} \quad (f|A)_* : H_p(A) \to H_p(B)$$

are isomorphisms. The theorem then follows from the Steenrod five-lemma, Lemma 3.3.4. □

If $f : (X, A) \to (Y, B)$ is a homotopy equivalence, then so are $f : X \to Y$ and $f|A : A \to B$. The converse does not hold, as the following example shows.

EXAMPLE 4.3.10 Consider the inclusion map $j : (\mathbb{B}^n, \mathbb{S}^{n-1}) \to (\mathbb{R}^n, \mathbb{R}^n \setminus \mathbf{0})$. Since $\mathbb{B}^n$ is a deformation retract of $\mathbb{R}^n$ and $\mathbb{S}^{n-1}$ is a deformation retract of $\mathbb{R}^n \setminus \mathbf{0}$, the map j_* is an isomorphism in relative homology. Suppose there existed a map $g : (\mathbb{R}^n, \mathbb{R}^n \setminus \mathbf{0}) \to (\mathbb{B}^n, \mathbb{S}^{n-1})$ that served as a homotopy inverse for j. Then, since $\mathbf{0}$ is a limit point of $\mathbb{R}^n \setminus \mathbf{0}$, the map g necessarily carries $\mathbf{0}$ into $\mathbb{S}^{n-1}$. Hence

$$g \circ j : (\mathbb{B}^n, \mathbb{S}^{n-1}) \to (\mathbb{B}^n, \mathbb{S}^{n-1})$$

carries all of $\mathbb{B}^n$ into $\mathbb{S}^n - 1$, so it induces the trivial homomorphism in homology. On the other hand, this map is by hypothesis homotopic to the identity, so it induces the identity homomorphism of $H_n(\mathbb{B}^n, \mathbb{S}^{n-1})$. We will shortly prove that this last group is non-trivial, so we have a contradiction.

EXERCISES

1. Construct the exact sequence of a triple in singular homology. [See Exercise 1 of Section 3.3 and Exercise 2 of Section 3.5.]
2. Show that if $f : X \to Y$ is homotopic to a constant, then f_* is the zero homomorphism in reduced homology. Conclude that if X is contractible, then X is acyclic in singular homology.
3. Suppose the inclusion $j : A \to X$ is a homotopy equivalence. Show that $H_p(X, A) = 0$.
4. Give an example in which $H_p(X, A) \ncong H_p(Y, B)$, although X has the homotopy type of Y and A has the homotopy type of B. Compare with Theorem 4.3.9.
5. In the proof of Lemma 4.3.6, one has some freedom in choosing the chain $D_{\Delta_p} i_p$. Show that the formula

$$D_{\Delta_p} i_p = \sum_{i=0}^{p} (-1)^p \ell((\epsilon_0, 0), (\epsilon_1, 0), \ldots, (\epsilon_i, 0), (\epsilon_i, 1), (\epsilon_{i+1}, 1), \ldots, (\epsilon_p, 1))$$

   gives a $(p+1)$-chain on $\Delta_p \times I$ that satisfies the requirements of the lemma. Draw a picture when $p = 1$ and when $p = 2$.
6. Show that if $f : (X, A) \to (Y, B)$ is a homotopy equivalence, then both $f : X \to Y$ and $f|A : A \to B$ are homotopy equivalences.

7. Consider the category **C** whose objects are chain complexes. A *morphism of* **C** *of degree* d mapping $\mathcal{C}$ to $\mathcal{C}'$ is a family of homomorphisms $\phi_p : C_p \to C'_{p+d}$. Consider the following functors from the topological category to **C**:

$$G(X) = \mathcal{S}(X); \qquad G(f) = f_{\#};$$
$$H(X) = \mathcal{S}(X \times I); \qquad H(f) = (f \times i_I)_{\#}.$$

Show that the chain homotopy of Lemma 4.3.6 is a natural transformation of G to H.

8. Verify properties **(a)** and **(b)** of Lemma 4.3.6 are satisfied when $p = 0$.

9. Show that if X consists of n points with the discrete topology, then $H_p(X) = 0$ for $p \neq 0$ and $H_0(X) \cong \mathbb{Z}^n$.

10. Give an example of a space X with $H_1(X) = \mathbb{Z}$.

11. Give an example of a space Y with $H_1(Y) = \mathbb{Z} \times \mathbb{Z}$.

4.4 Excision in Singular Homology

In this section we verify the excision axiom for singular homology. The techniques involved in the proof will be useful later in other situations.

One of the facts we proved about simplicial complexes was that we could chop up a finite complex barycentrically into simplices that were as small as desired. We need a similar result for singular chains. To make this precise, we shall use the following definition:

Definition 4.4.1 Suppose one is given a space X and a collection $\mathcal{A}$ of subsets of X whose interiors cover X. A singular simplex of X is said to be $\mathcal{A}$-*small* if its image set lies in an element of $\mathcal{A}$.

Given a singular chain on X, we shall show how to "chop it up" so that all its simplices are $\mathcal{A}$-small. Not surprisingly, what we do is to introduce something like barycentric subdivision in order to accomplish this. We give the details now.

Definition 4.4.2 Let X be a topological space. We define a homomorphism $\mathrm{sd}_X : S_p(X) \to S_p(X)$ by induction. If $T : \Delta_0 \to X$ is a singular 0-simplex, we define $\mathrm{sd}_X T = T$. Now suppose sd_X is defined in dimensions less than p. If $i_p : \Delta_p \to \Delta_p$ is the identity map, let $\widehat{\Delta}_p$ denote the barycenter of Δ_p; then, using the bracket operation of Section 4.2, we define

$$\mathrm{sd}_{\Delta_p} i_p = (-1)^p [\mathrm{sd}_{\Delta_p}(\partial i_p), \widehat{\Delta}_p] .$$

The definition makes sense since Δ_p is star convex relative to $\widehat{\Delta}_p$.

If $T : \Delta_p \to X$ is an arbitrary singular p simplex on X, we define

$$\mathrm{sd}_X T = T_\#(\mathrm{sd}_{\Delta_p} i_p)\,.$$

See Figure 4.4.1. We call sd_X the *barycentric subdivision operator in singular theory.*

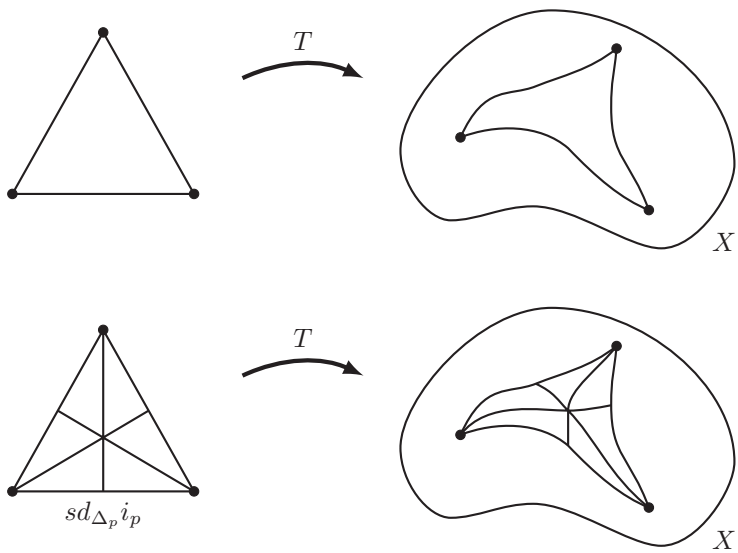


Figure 4.4.1

Lemma 4.4.3 *The homomorphism* sd_X *is an augmentation-preserving chain map, and it is natural in the sense that for any continuous map* $f : X \to Y$ *we have*

$$f_\# \circ \mathrm{sd}_X = \mathrm{sd}_Y \circ f_\#\,.$$

Proof: The map sd_X preserves augmentation because it is the identity in dimension 0. Naturality holds in dimension 0 for the same reason. Naturality holds in positive dimensions by direct computation:

$$\begin{aligned} f_\#(\mathrm{sd}_X T) &= f_\# T_\#(\mathrm{sd}_{\Delta_p} i_p) \\ &= \mathrm{sd}_Y(f \circ T) = \mathrm{sd}_Y(f_\#(T))\,. \end{aligned}$$

Henceforth we shall normally omit the subscript on the operator sd_X, relying on the context to make the meaning clear.

To check that sd is a chain map, we proceed by induction on p. The fact that $\mathrm{sd} \circ \partial = \partial \circ \mathrm{sd}$ holds in dimension 0 is trivial. Assuming the result true in dimensions less than p, we apply Lemma 4.2.8 to compute

$$\begin{aligned} \partial \,\mathrm{sd}\, i_p &= (-1)^p \partial\, [\mathrm{sd}\, \partial i_p, \widehat{\Delta}_p] \\ &= \begin{cases} (-1)^p [\partial \,\mathrm{sd}\, \partial\, i_p, \widehat{\Delta}_p] + \mathrm{sd}\, \partial i_p & \text{if } p > 1, \\ -\epsilon(\mathrm{sd}\, \partial\, i_1) T_0 + \mathrm{sd}\, \partial\, i_1 & \text{if } p = 1, \end{cases} \end{aligned}$$

where T_0 is the 0-simplex whose image point is $\widehat{\Delta}_1$. Now if $p > 1$, we have $\partial \,\mathrm{sd}\, \partial\, i_p = \mathrm{sd}\, \partial\, \partial\, i_p = 0$ by the induction hypothesis. If $p = 1$, we have $\epsilon(\partial \,\mathrm{sd}\, i_1) = \epsilon(\partial\, i_1) = 0$, because sd preserves augmentation. Hence in either case, we have $\partial \,\mathrm{sd}\, i_p = \mathrm{sd}\, \partial\, i_p$. In general, we compute

$$\begin{aligned} \partial \,\mathrm{sd}\, T &= \partial\, T_\#(\mathrm{sd}\, i_p) && \text{by definition,} \\ &= T_\#(\partial \,\mathrm{sd}\, i_p) && \text{because } T_\# \text{ is a chain map,} \\ &= T_\#(\mathrm{sd}\, \partial\, i_p) && \text{by the formula just proved,} \\ &= \mathrm{sd}\, T_\#(\partial\, i_p) && \text{because sd is natural,} \\ &= \mathrm{sd}\, \partial\, (T_\#(i_p)) && \text{because } T_\# \text{ is a chain map,} \\ &= \mathrm{sd}\, \partial\, T\,. \end{aligned}$$

□

Lemma 4.4.4 *Let $T : \Delta_p \to \sigma$ be a linear homeomorphism of Δ_p with the p-simplex σ. Then each term of $\mathrm{sd}\, T$ is a linear homeomorphism of Δ_p with a simplex in the first barycentric subdivision of σ.*

Proof: The lemma is trivial for $p = 0$. Now suppose the lemma is true in dimensions less than p. First we consider the identity linear homeomorphism $i_p : \Delta_p \to \Delta_p$. We have

$$\mathrm{sd}\, i_p = [\mathrm{sd}\, \partial i_p, \widehat{\Delta}_p]\,.$$

Each term in ∂i_p is a linear homeomorphism of Δ_{p-1} with a simplex in Bd Δ_p. By the induction hypotheses, $\mathrm{sd}\, (\partial i_p) = \sum \pm\, T_i$, where T_i is a linear homeomorphism of Δ_{p-1} with a simplex $\widehat{s}_1 \ldots \widehat{s}_p$ in the first barycentric subdivision of Bd Δ_p. Then $[T_i, \widehat{\Delta}_p]$ is by definition a linear homeomorphism of Δ_p with the simplex $\widehat{\Delta}_p \widehat{s}_1 \ldots \widehat{s}_p$, which belongs to the first barycentric subdivision of Δ_p.

Now consider a general linear homeomorphism $T : \Delta_p \to \sigma$. Note that T defines a linear isomorphism of the first barycentric subdivision of Δ_p with the first barycentric subdivision of σ, because it carries the barycenter of Δ_p to the barycenter of σ. By definition, we have $\mathrm{sd}\, T = T_\#(\mathrm{sd}\, i_p)$. Since the composition of linear homeomorphisms is a linear homeomorphism, each term in $\mathrm{sd}\, T$ is a linear homeomorphism of Δ_p with a simplex in the first barycentric subdivision of σ. □

Theorem 4.4.5 *Let $\mathcal{A}$ be a collection of subsets of X whose interiors cover X. Given $T : \Delta_p \to X$, there is an m such that each term of* $\mathrm{sd}^m T$ *is $\mathcal{A}$-small.*

Proof: It follows from the preceding lemma that if L is a linear homeomorphism of Δ_p with the p-simplex σ, then each term of $\mathrm{sd}^m L$ is a linear homeomorphism of Δ_p with a simplex in the mth barycentric subdivision of σ.

Let us cover Δ_p by the open sets $T^{-1}(\mathrm{Int}\ A)$, for $A \in \mathcal{A}$. Let λ be a Lebesgue number for this open cover [see Lemma 2.4.1]. Choose m so that each simplex in the mth barycentric subdivision of Δ_p has diameter less than λ. By the preceding remark, each term of $\mathrm{sd}^m i_p$ is a linear singular simplex on Δ_p whose image set has diameter less than λ. Then each term of $\mathrm{sd}^m T = T_\#(\mathrm{sd}^m i_p)$ is a singular simplex on X whose image set lies in an element of $\mathcal{A}$. □

Having shown how to chop up singular chains so they are $\mathcal{A}$-small, we now show that these $\mathcal{A}$-small singular chains suffice to generate the homology of X. First, we need a lemma.

Lemma 4.4.6 *Let m be given. For each space X, there is a homomorphism $D_X : S_p(X) \to S_{p+1}(X)$ such that for each singular p-simplex T of X,*

$$\partial D_X T + D_X \partial T = \mathrm{sd}^m T - T\,. \qquad (*)$$

Furthermore, D_X is natural, that is, if $f : X \to Y$ then $f_\# \circ D_X = D_Y \circ f_\#$.

Proof: If $T : \Delta_0 \to X$ is a singular 0-simplex, define $D_X T = 0$. Formula $(*)$ and naturality follow trivially. Now for $p > 0$ we assume that D_X is defined, satisfying $(*)$ and naturality, for all spaces in all dimensions less than p. We proceed by a method similar to that used in the proof of Lemma 4.3.6, which will be formalized in the next section as the "method of acyclic models."

We first define $D_X T$ in the special case $X = \Delta_p$ and $T = i_p$, the identity map from Δ_p to itself. Consider the singular p-chain, c_p, on Δ_p given by

$$c_p = \mathrm{sd}^m i_p - i_p - D_{\Delta_p}(\partial\, i_p)\,.$$

By the usual computation and using the hypothesis that $(*)$ holds in dimensions less than p, we see that c_p is a cycle. Since Δ_p is acyclic in singular homology, we can choose $D_{\Delta_p} i_p$ to be an element of S_{p+1} whose boundary equals c_p. Then $(*)$ holds for $X = \Delta_p$ and $T = i_p$.

Given an arbitrary singular p-simplex $T : \Delta_p \to X$, we define

$$D_X T = T_\#(D_{\Delta_p}(i_p))\,.$$

Formula $(*)$ is seen to hold for D_X by the usual direct calculation:

$$\begin{aligned}
\partial\, D_X\, T &= \partial\, T_\#(D_{\Delta_p}(i_p)) = T_\#(\partial\, D_{\Delta_p}(i_p)) \\
&= T_\#(\mathrm{sd}^m\, i_p - i_p - D_{\Delta_p}(\partial\, i_p)) \\
&= \mathrm{sd}^m\, T_\#(i_p) - T_\#(i_p) - D_X T_\#(\partial\, i_p) \\
&= \mathrm{sd}^m\, T - T - D_X\, \partial\, T\,,
\end{aligned}$$

where the next to last equality uses naturality of D_X for the $(p-1)$-chain ∂i_p. Naturality of D_X in dimension p follows directly from the definition. □

Note that the naturality of sd^m and D_X show that if A if a subspace of X, then sd^m and D_X carry $S_p(A)$ into $S_p(A)$ and $S_{p+1}(A)$, respectively. Thus they induce a chain map and a chain homotopy, respectively, on the relative chain complex $\mathcal{S}(X, A)$ as well.

Definition 4.4.7 Let X be a space and let $\mathcal{A}$ be a covering of X. Let $S_p^{\mathcal{A}}(X)$ denote the subgroup of $S_p(X)$ generated by $\mathcal{A}$-small singular simplices. Let $\mathcal{S}^{\mathcal{A}}(X)$ denote the chain complex whose chain groups are the groups $S_p^{\mathcal{A}}(X)$. It is a subchain complex of $\mathcal{S}(X)$, for if the image set of T lies in the element A of $\mathcal{A}$, then so does the image set of each term of ∂T.

Note that each singular 0-simplex is automatically $\mathcal{A}$-small. Hence $S_p^{\mathcal{A}}(X) = S_0(X)$, and ϵ defines an augmentation for $\mathcal{S}^{\mathcal{A}}(X)$. It follows from the preceding remark that both sd^m and D_X carry $\mathcal{S}^{\mathcal{A}}(X)$ into itself, for if the image set of T lies in $A \in \mathcal{A}$, so does the image set of each term in sd^m and D_X.

Theorem 4.4.8 *Let X be a space and let $\mathcal{A}$ be a collection of subsets of X whose interiors cover X. Then the inclusion map $\mathcal{S}^{\mathcal{A}}(X) \to \mathcal{S}(X)$ induces an isomorphism in homology, both ordinary and reduced.*

Proof: The obvious thing to do is to define a chain map $\lambda : \mathcal{S}^{\mathcal{A}}(X) \to \mathcal{S}(X)$ that is a chain-homotopy inverse for the inclusion map. This is not as easy as one might hope. For any particular singular chain, there is an m such that the map sd^m will work, but as the singular chain changes, one may need to take m larger and larger. We avoid this difficulty by using a different method.

Consider the short exact sequence of chain complexes

$$0 \longrightarrow S_p^{\mathcal{A}}(X) \longrightarrow S_p(X) \longrightarrow S_p(X)/S_p^{\mathcal{A}}(X) \longrightarrow 0\,.$$

It gives rise to a long exact sequence in homology (either ordinary or reduced). To prove our theorem, it will suffice to show that the homology of the chain complex $\{S_p(X)/S_p^{\mathcal{A}}(X), \partial\}$ vanishes in every dimension. This we can do. We need only prove the following:

Suppose that c_p is an element of $S_p(X)$ whose boundary belongs to $S_{p-1}^{\mathcal{A}}(X)$. Then there is an element d_{p+1} of $S_{p+1}(X)$ such that $c_p + \partial d_{p+1}$ belongs to $S_p^{\mathcal{A}}(X)$.

Note that c_p is a *finite* formal linear combination of singular p-simplices. In view of Theorem 4.4.5, we can choose m so that each singular simplex appearing in the expression for $\mathrm{sd}^m c_p$ is $\mathcal{A}$-small. Once m is chosen, let D_X be the chain homotopy of Lemma 4.4.6. We show that $c_p + \partial D_X c_p$ belongs to $S_p^{\mathcal{A}}(X)$. Then we are finished.

We know that

$$\partial D_X c_p + D_X \partial c_p = \mathrm{sd}^m c_p - c_p ,$$

or

$$c_p + \partial D_X c_p = \mathrm{sd}^m c_p - D_X \partial c_p .$$

The chain $\mathrm{sd}^m c_p$ is in $S_p^{\mathcal{A}}(X)$, by the choice of m. And since by hypothesis ∂c_p belongs to $S_{p-1}^{\mathcal{A}}(X)$, the chain $D_X \partial c_p$ belongs to $S_p^{\mathcal{A}}(X)$, as noted earlier. □

In fact, there does exist a chain map $\lambda : \mathcal{S}^{\mathcal{A}}(X) \to \mathcal{S}(X)$ that is a chain-homotopy inverse for the inclusion map. A specific formula for λ, involving the maps sd^m, is given in [Vic73], page 207. We shall derive the existence of λ shortly from a more general result. [See the exercises of Section 5.7.]

Corollary 4.4.9 *Let X and $\mathcal{A}$ be as in the preceding theorem. If $B \subset X$, let $S_p^{\mathcal{A}}(B)$ be generated by those singular simplices $T : \Delta_p \to B$ whose image sets lie in elements of $\mathcal{A}$. Let $S_p^{\mathcal{A}}(X, B)$ denote $S_p^{\mathcal{A}}(X)/S_p^{\mathcal{A}}(B)$. Then the inclusion $S_p^{\mathcal{A}}(X, B) \to S_p(X, B)$ induces a homology isomorphism.*

Proof: The inclusions $S_p^{\mathcal{A}}(B) \to S_p(B)$ and $S_p^{\mathcal{A}}(X) \to S_p(X)$ give rise to a homomorphism of the long exact homology sequence derived from

$$0 \longrightarrow \mathcal{S}(B) \longrightarrow \mathcal{S}(X) \longrightarrow \mathcal{S}(X, B) \longrightarrow 0$$

with the one derived from

$$0 \longrightarrow \mathcal{S}^{\mathcal{A}}(B) \longrightarrow \mathcal{S}^{\mathcal{A}}(X) \longrightarrow \mathcal{S}^{\mathcal{A}}(X, B) \longrightarrow 0 .$$

Since inclusion induces an isomorphism of the absolute homology groups of these respective sequences, the Steenrod five-lemma [Lemma 3.3.4] implies that it induces an isomorphism of the relative groups as well. □

Theorem 4.4.10 (Excision for singular theory) *Let A be a subset of X. If U is a subset of X such that $\overline{U} \subset \mathrm{Int}\ A$, then the inclusion*

$$j : (X \setminus U, A \setminus U) \to (X, A)$$

induces an isomorphism in singular homology.

Proof: For the purpose of applying the preceding corollary, we let $\mathcal{A}$ be the collection of sets $\{X \setminus U, A\}$. Now, $X \setminus U$ contains $X \setminus \overline{U}$, and since $\overline{U} \subset \mathrm{Int}\ A$, the interiors of the sets in $\mathcal{A}$ cover X, as required by Corollary 4.4.9. Consider the homomorphisms

$$\frac{S_p(X \setminus U)}{S_p(A \setminus U)} \longrightarrow \frac{S_p^{\mathcal{A}}(X)}{S_p^{\mathcal{A}}(A)} \longrightarrow \frac{S_p(X)}{S_p(A)}$$

induced by inclusion. The second of these homomorphisms induces a homology isomorphism, by the preceding corollary. We show that the first is an isomorphism already on the chain level, and the proof is then complete.

Note first that the map

$$\phi : S_p(X \setminus U) \to S_p^{\mathcal{A}}(X)/S_p^{\mathcal{A}}(A)$$

induced by inclusion is surjective. For if c_p is a chain of $S_p^{\mathcal{A}}(X)$, then each term of c_p has image set lying in either $X \setminus U$ or in A. When we form the coset $c_p + S_p^{\mathcal{A}}(A)$ we can discard those terms lying in A. Thus ϕ is surjective. The kernel of ϕ is

$$S_p(X \setminus U) \cap S_p^{\mathcal{A}}(A) = S_p((X \setminus U) \cap A) = S_p(A \setminus U)\,,$$

as desired. □

Note that this theorem is slightly stronger than the excision axiom proper [see Section 3.5]. In singular theory we do not need to assume U is an *open* subset of X.

As an application, we compute the singular homology of the ball and the sphere.

Theorem 4.4.11 *Let $n \geq 0$. The group $H_i(\mathbb{B}^n, \mathbb{S}^{n-1})$ is infinite cyclic for $i = n$ and vanishes otherwise. The group $\widetilde{H}_i(\mathbb{S}^n)$ is infinite cyclic for $i = n$ and vanishes otherwise. The homomorphism of $\widetilde{H}_n(\mathbb{S}^n)$ with itself induced by the reflection map*

$$\rho_1(x_1, x_2, \ldots, x_{n+1}) = (-x_1, x_2, \ldots, x_{n+1})$$

equals multiplication by -1.

Remark 4.4.12 Recalling Definition 2.9.1, we see that, for $n \geq 1$, the theorem tells us that the reflection map has degree -1.

Proof of the theorem: First we consider the case $n = 0$. Recall that $\mathbb{B}^0$ is the one-point space. Also, we must have $\mathbb{S}^{-1} = \emptyset$, since points of the unit sphere have norm 1. Theorem 4.3.3 then tells us that $H_p(\mathbb{B}^0, \emptyset)$ is infinite cyclic for $p = 0$ and vanishes otherwise.

It is easy to generalize the proof of Theorem 4.3.3 to a space consisting of two points, such as $\mathbb{S}^0 = \{-1, +1\}$. One shows that $H_p(\mathbb{S}^0) = 0$ for $p > 0$ and that $H_0(\mathbb{S}^0) = \mathbb{Z} \oplus \mathbb{Z}$ with generators T_{-1} and T_{+1}. Consequently, $\widetilde{H}_0(\mathbb{S}^0)$ is infinite cyclic with generator $T_{+1} - T_{-1}$. The map ρ_1 on $\mathbb{S}^0$ exchanges the two points -1 and $+1$, so the induced homomorphism of $\widetilde{H}_0(\mathbb{S}^0)$ equals multiplication by -1.

Now we use induction. Suppose the theorem is true in dimension $n - 1$, where $n \geq 1$. It follows from the long exact homology sequence of $(\mathbb{B}^n, \mathbb{S}^{n-1})$ that

$$H_i(\mathbb{B}^n, \mathbb{S}^{n-1}) \cong \widetilde{H}_{i-1}(\mathbb{S}^{n-1})\,,$$

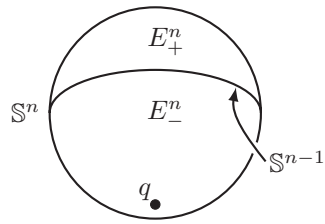


Figure 4.4.2

since $\mathbb{B}^n$ is contractible. Thus $H_i(\mathbb{B}^n, \mathbb{S}^{n-1})$ is infinite cyclic for $i = n$ and vanishes otherwise.

To compute $\widetilde{H}_i(\mathbb{S}^n)$, consider the following homomorphisms,

$$\begin{array}{ccc} H_i(E^n_+, \mathbb{S}^{n-1}) & \xrightarrow{\partial_*} & \widetilde{H}_{i-1}(\mathbb{S}^{n-1}) \\ \downarrow{\scriptstyle k_*} & & \\ H_i(\mathbb{S}^n \setminus q, E^n_- \setminus q) & & \\ \downarrow{\scriptstyle j_*} & & \\ \widetilde{H}_i(\mathbb{S}^n) \xrightarrow{i_*} H_i(\mathbb{S}^n, E^n_-) & & \end{array}$$

where i, j, k are inclusions and $q = (0, 0, \ldots, 0, -1)$ is the "south pole" of $\mathbb{S}^n$. The plan is to show all the homomorphisms i_*, j_*, k_*, and ∂_* to be isomorphisms, and consequently to conclude that $\widetilde{H}_i(\mathbb{S}^n) \cong \widetilde{H}_{i-1}(\mathbb{S}^{n-1})$.

Recall that E^n_+ and E^n_- are the upper and lower hemispheres of $\mathbb{S}^n$, respectively, and $\mathbb{S}^{n-1} = E^n_+ \cap E^n_-$ is the "equator" of $\mathbb{S}^n$. See Figure 4.4.2. Now both E^n_+ and E^n_- are n-balls; in fact, the projection of $\mathbb{R}^{n+1}$ onto $\mathbb{R}^n \times 0$ carries E^n_+ and E^n_- homeomorphically onto $\mathbb{B}^n$. [It also carries $E^n_- \setminus q$ onto $\mathbb{B}^n \setminus \mathbf{0}$.] In particular, E^n_- is contractible, so the long exact homology sequence of $(\mathbb{S}^n, E^n_-)$ shows that i_* is an isomorphism. The excision property shows that j_* is an isomorphism, since $q \in \operatorname{Int} E^n_-$. The map k_* is an isomorphism because $\mathbb{S}^{n-1}$ is a deformation retract of $E^n_- \setminus q$, and E^n_+ is a deformation retract of $\mathbb{S}^n \setminus q$. [Since the pair $(E^n_- \setminus q, \mathbb{S}^{n-1})$ is homeomorphic to the pair $(\mathbb{B}^n \setminus \mathbf{0}, \mathbb{S}^{n-1})$, there is a deformation retraction F_t of $E^n_- \setminus q$ onto $\mathbb{S}^{n-1}$. It extends to a deformation retraction of $\mathbb{S}^n \setminus q$ onto E^n_+, by letting F_t equal the identity on E^n_+, for each t.] The fact that ∂_* is an isomorphism follows from the contractibility of E^n_+.

It follows from the induction hypothesis that $\widetilde{H}_i(\mathbb{S}^n)$ is infinite cyclic for $i = n$ and vanishes otherwise.

Now the reflection map ρ_1 induces a homomorphism of the preceding diagram with itself, since it maps each of q, E^n_-, E^n_+ and $\mathbb{S}^n$ into itself. By the

induction hypothesis, the homomorphism induced by ρ_1 equals multiplication by -1 on $\widetilde{H}_{n-1}(\mathbb{S}^{n-1})$; therefore, it equals the same on $\widetilde{H}_n(\mathbb{S}^n)$. □

The reader should note that, in the proof of the preceding theorem, the only homology groups that were computed directly were those for spaces consisting of one or two points.

The next corollary follows immediately from Theorem 4.4.11.

Corollary 4.4.13 *If $a : \mathbb{S}^n \to \mathbb{S}^n$ is the antipodal map $a(x) = -x$, then a_* equals multiplication by $(-1)^{n+1}$.*

EXERCISE

1. Express the naturality provisions in Lemmas 4.4.3 and 4.4.6 by stating that sd and D_X are natural transformations of certain functors.
2. Let X be the annulus $\{(x, y) : 1 < x^2 + y^2 < 4\}$ in $\mathbb{R}^2$. Let $\mathcal{A}$ be the collection of open discs in the plane. Give an example of a singular simplex of X that is $\mathcal{A}$-small.
3. Repeat Exercise 2 with X being the unit sphere in $\mathbb{R}^3$ and $\mathcal{A}$ being the collection of open cubes with sides parallel to the axes.

4.5 *Acyclic Models

In this section, we assume familiarity with Section 3.7, Categories and Functors. The results of this section will be used when we prove the Eilenberg–Zilber theorem in Section 7.9.

In the two preceding sections, we constructed certain natural chain homotopies D_X. The method was to define D_X first for a particular singular simplex i_p on a particular space Δ_p. To do this, we needed the acyclicity of another space, either $\Delta_p \times I$ [in Section 4.3] or of Δ_p itself [in Section 4.4]. This part of the construction involved an arbitrary choice; everything thereafter was forced by naturality. The resemblance to earlier constructions involving acyclic carriers was strong, but nevertheless, there were differences.

Now we formalize this method for later use. In this section, we state a theorem, sufficiently strong for our purposes, that we shall call the acyclic model theorem. Its formulation is sufficiently abstract to bother some readers. To help you keep your feet on the ground, we will briefly review the meaning of various terms from category theory that we will be using.

Throughout this section, let $\mathbf{C}$ denote an arbitrary category with objects $X, Y, \ldots$ and morphisms $f, g, \ldots$, and let $\mathbf{A}$ denote the category of augmented chain complexes and chain maps of such. For most of the applications we have in mind, $\mathbf{C}$ will be either the topological category [whose objects are topological spaces and whose morphisms are continuous maps] or the category of pairs of spaces and pairs of continuous maps. So you may think only of those categories if you wish.

We will be dealing with functors [see Definition 3.7.3] G and G' from $\mathbf{C}$ to $\mathbf{A}$. Recall that if X is an element of $\mathbf{C}$, then the functors assign to X objects $G(X)$ and $G'(X)$ in $\mathbf{A}$. Thus both $G(X)$ and $G'(X)$ are augmented chain complexes. For each $p \geq 0$, $G_p(X)$ and $G'_p(X)$ are chain groups. If Y is another element of $\mathbf{C}$, then we have two more chain complexes $G(Y)$ and $G'(Y)$, and for each $p \geq 0$, the corresponding chain groups $G_p(Y)$ and $G'_p(Y)$. The two elements X and Y of $\mathbf{C}$ also bring into play the set of morphisms $\hom(X, Y)$ in $\mathbf{C}$. The functors G and G' act on morphisms of $\mathbf{C}$, so for $f \in \hom(X, Y)$, the functors assign to f morphisms $G(f)$ and $G'(f)$ in $\mathbf{A}$. In this case, the morphisms $G(f) : G(X) \to G(Y)$ and $G'(f) : G'(X) \to G'(Y)$ are chain maps of augmented chain complexes [see Definition 1.14.2].

Since we are considering two functors, we can also consider a natural transformation [see Definition 3.7.6] T from G to G'. Such a natural transformation assigns to each $X \in \mathbf{C}$ a morphism [in $\mathbf{A}$ that means a chain map of augmented chain complexes] $T_X : G(X) \to G'(X)$. The requirement of naturality is that the following diagram be commutative when X and Y are two elements of $\mathbf{C}$ and $f \in \hom(X, Y)$:

$$\begin{array}{ccc} G_p(X) & \xrightarrow{T_X} & G'_p(X) \\ {\scriptstyle G(f)}\downarrow & & \downarrow{\scriptstyle G'(f)} \\ G_p(Y) & \xrightarrow{T_Y} & G'_p(Y) \end{array}$$

Because $\mathbf{A}$ is a category of augmented chain complexes, it also makes sense to consider a pair of natural transformations S and T of G to G' and a related natural transformation, called a *natural chain homotopy*, between S and T. A natural chain homotopy D between S and T assigns to X in $\mathbf{C}$ the morphism $D_X : G_p(X) \to G'_{p+1}(X)$ such that

$$\partial' D_X + D_X \partial = T_X - S_X \, .$$

The requirement of naturality is that the following diagram be commutative when X and Y are two elements of $\mathbf{C}$ and $f \in \hom(X, Y)$:

$$\begin{array}{ccc} G_p(X) & \xrightarrow{D_X} & G'_{p+1}(X) \\ {\scriptstyle G(f)}\downarrow & & \downarrow{\scriptstyle G'(f)} \\ G_p(Y) & \xrightarrow{D_Y} & G'_{p+1}(Y) \end{array}$$

Definition 4.5.1 Let G and G' be functors from $\mathbf{C}$ to $\mathbf{A}$, and let X be an object of $\mathbf{C}$. Let $G_p(X)$ denote the p-dimensional group of the augmented chain complex $G(X)$. Let $\mathcal{M}$ be a collection of objects of $\mathbf{C}$, called *models* or *model objects*.

(1) We say that the functor G' is *acyclic* relative to the collection $\mathcal{M}$ if $G'(X)$ is acyclic for each $X \in \mathcal{M}$.

(2) We say that the functor G is *free* relative to the collection $\mathcal{M}$ if for each $p \geq 0$ there are

(a) An index set J_p.

(b) An indexed family $\{M_\alpha\}_{\alpha \in J_p}$ of objects of $\mathcal{M}$.

(c) An indexed family $\{i_\alpha\}_{\alpha \in J_p}$, where $i_\alpha \in G_p(M_\alpha)$ for each α.

The following condition is to hold: Given X, the elements

$$G(f)(i_\alpha) \in G_p(X)$$

are to be *distinct* and form a *basis* for $G_p(X)$ as f ranges over all elements of $\hom(M_\alpha, X)$ and α ranges over J_p.

Since $G_p(X)$ has a basis if the functor G is free, that implies that $G_p(X)$ is a free group [as one would hope when the functor is said to be "free"]. Also, we shall be constructing functions by first doing the construction on the basis elements. If these basis elements were not distinct from each other, it might lead to two constructions producing unequal results for what turned out to be the same basis element.

EXAMPLE 4.5.2 Consider the singular chain complex functor, from the topological category to $\mathbf{A}$. Let $\mathcal{M}$ be the collection $\{\Delta_p : p = 0, 1, \dots\}$. This functor is acyclic relative to $\mathcal{M}$. We show it is free relative to $\mathcal{M}$: For each p, let the index set J_p have only one element, let the corresponding family consist of Δ_p alone, and let the corresponding element of $S_p(\Delta_p)$ be the identity singular simplex i_p. It is immediate that, as T ranges over all continuous maps $\Delta_p \to X$, the elements $T_\#(i_p) = T$ form a basis for $S_p(X)$.

EXAMPLE 4.5.3 Consider the following functor G, defined on the category of topological pairs:

$$(X, Y) \to \mathcal{S}(X \times Y) \quad \text{and} \quad (f, g) \to (f \times g)_\# .$$

Let $\mathcal{M} = \{(\Delta_p, \Delta_q) : p, q = 0, 1, \dots\}$. Then G is acyclic relative to $\mathcal{M}$, since $\Delta_p \times \Delta_q$ is contractible. We show it is free relative to $\mathcal{M}$: For each index p, let J_p consist of a single element, let the corresponding family consist of (Δ_p, Δ_p) alone, and let the corresponding element of $S_p(\Delta_p \times \Delta_p)$ be the diagonal map $d_p(x) = (x, x)$. As f and g range over all maps from Δ_p into X and Y, respectively, $(f \times g)_\#(d_p)$ ranges over all maps $\Delta_p \to X \times Y$, that is, over a basis for $S_p(X \times Y)$.

EXAMPLE 4.5.4 Let G be the functor

$$X \to \mathcal{S}(X \times I) \quad \text{and} \quad f \to (f \times i_I)_{\#}\,.$$

Let $\mathcal{M}$ be the collection $\{\Delta_p : p = 0, 1, \dots\}$. Then G is acyclic relative to $\mathcal{M}$. It is also true that G is free relative to $\mathcal{M}$, but the proof is not as simple as in the previous two examples. Here we let J_p be the set of all continuous functions $\alpha : \Delta_p \to I$. Let the family $\{M_\alpha\}_{\alpha \in J}$ be defined by setting $M_\alpha = \Delta_p$ for each α. For each α, let $i_\alpha \in S_p(M_\alpha \times I)$ be the singular simplex

$$i_\alpha : \Delta_p \to \Delta_p \times I$$

defined by

$$i_\alpha(x) = (x, \alpha(x))\,.$$

As f ranges over all maps of Δ_p into X and α ranges over the set J_p, the element $(f \times i_I)_{\#}(i_\alpha)$ ranges over a basis for $S_p(X \times I)$.

Note that if G is free relative to a collection $\mathcal{M}$, then it is automatically free relative to any larger collection. On the other hand, if G is acyclic relative to $\mathcal{M}$, then it is automatically acyclic relative to any smaller collection. Thus, if we wish G to be both free and acyclic relative to $\mathcal{M}$, then we must choose $\mathcal{M}$ to be just the right size, neither too large nor too small.

Theorem 4.5.5 (Acyclic model theorem) *Let G and G' be functors from the category $\mathbf{C}$ to the category $\mathbf{A}$ of augmented chain complexes and chain maps of augmented chain complexes. Let $\mathcal{M}$ be a collection of objects of $\mathbf{C}$.*

If G is free relative to $\mathcal{M}$ and G' is acyclic relative to $\mathcal{M}$, then the following hold:

- **(a)** *There is a natural transformation T of G to G'.*
- **(b)** *Given two natural transformations S and T of G to G', then there is a natural chain homotopy D between S and T.*

The proof is sketched in Exercises 1 and 2. If Exercise 1 causes too much difficulty, you might switch your attention to Exercise 2. Then you might take some inspiration from the proof of Lemma 4.3.6 and the remarks after that proof regarding the case $p = 0$. Once Exercise 2 is understood, return to Exercise 1.

The method of acyclic models was first formalized by Eilenberg and MacLane in [EM53]. In the immediately following paper in the same issue of the *American Journal of Mathematics*, it was applied by Eilenberg and Zilber in [EZ53].

Corollary 4.5.6 *Let $\mathbf{C}$ be a category, and let G and G' be functors from $\mathbf{C}$ to $\mathbf{A}$. If G and G' are free and acyclic relative to the collection $\mathcal{M}$ of objects of $\mathbf{C}$, then there is a natural transformation $T : G \to G'$; any such transformation is a natural equivalence.*

Proof: We apply the preceding theorem four times. Because G is free and G' is acyclic, T exists. Because G' is free and G is acyclic, there is a natural transformation S of G' to G. Now $S \circ T$ and the identity transformation are two natural transformations of G to G. Because G is free and acyclic, there is a natural chain homotopy of $S \circ T$ to the identity. Similarly, because G' is free and acyclic, there is a natural chain homotopy of $T \circ S$ to the identity. □

EXERCISES

1. Prove part **(a)** of the acyclic model theorem by following these steps:

(a) Fix $M_\alpha \in \mathcal{M}$ and $c \in G_p(M_\alpha)$, where $p \geq 0$. Show that there is a chain $d' \in G_p(M_\alpha)$ such that if you let $d' = T_{M_\alpha}(c)$, then

$$\begin{aligned} \epsilon' \circ T_{M_\alpha}(c) &= \epsilon(c), && \text{for } p = 0, \\ \partial' \circ T_{M_\alpha}(c) &= T_{M_\alpha} \circ \partial(c), && \text{for } p > 0. \end{aligned}$$

Here ϵ' denotes the augmentation in G' and ϵ denotes the augmentation in G. Likewise, ∂' denotes the boundary operator in G' and ∂ denotes the boundary operator in G. [**Hint:** Use induction on p.]

(b) In part **(a)**, you constructed a transformation $T : G \to G'$ that operates on the model spaces $M_\alpha \in \mathcal{M}$, but only on the model spaces. Because G is free, the formula

$$T_X G(f)(i_\alpha) = G'(f) T_{M_\alpha}(i_\alpha), \text{ for all } \alpha \in J_p, \text{ all } f \in \hom(M_\alpha, X),$$

extends the transformation $T : G \to G'$ to all $X \in \mathbf{C}$. Show that this transformation $T : G \to G'$ is natural.

(c) Show that the transformation $T : G \to G'$ defined in part **(b)** satisfies

$$\begin{aligned} \epsilon' \circ T_X &= \epsilon, && \text{for } p = 0, \\ \partial' \circ T_X &= T_X \circ \partial, && \text{for } p > 0. \end{aligned}$$

2. Prove part **(b)** of the acyclic model theorem by following these steps:

(a) Fix $M_\alpha \in \mathcal{M}$ and $c \in G_p(M_\alpha)$, where $p \geq 0$. Show that there is a chain $d' \in G_{p+1}(M_\alpha)$ such that if you let $d' = D_{M_\alpha}(c)$, then

$$\begin{aligned} \partial' \circ D_{M_\alpha}(c) &= (T_{M_\alpha} - S_{M_\alpha})(c), && \text{for } p = 0, \\ \partial' \circ D_{M_\alpha}(c) + D_{M_\alpha}(\partial c) &= (T_{M_\alpha} - S_{M_\alpha})(c), && \text{for } p > 0. \end{aligned}$$

Here ∂' denotes the boundary operator in G' and ∂ denotes the boundary operator in G. [**Hint:** Use induction on p.]

(b) In part **(a)**, you constructed a transformation $D : G \to G'$ that operates as a chain homotopy on the model spaces $M_\alpha \in \mathcal{M}$, but only on the model spaces. Because G is free, the formula

$$D_X G(f)(i_\alpha) = G'(f) D_{M_\alpha}(i_\alpha), \text{ for all } \alpha \in J_p, \quad \text{all } f \in \hom(M_\alpha, X),$$

extends the transformation $D : G \to G'$ to all $X \in \mathbf{C}$. Show that this transformation $D : G \to G'$ is natural.

(d) Show that the transformation $D : G \to G'$ defined in part **(b)** satisfies

$$\begin{aligned} \partial' \circ D_X &= T_X - S_{M_\alpha}, \quad \text{for } p = 0, \\ \partial' \circ D_X + D_X \circ \partial &= T_X - S_X, \quad \text{for } p > 0, \end{aligned}$$

and thus is a natural chain homotopy between S and T.

3. Consider the following functors from the topological category to **A**:

$$\begin{aligned} &G : X \to \mathcal{S}(X) \quad \text{and} \quad f \to f_\# \,, \\ &G' : X \to \mathcal{S}(X \times I) \quad \text{and} \quad f \to (f \times i_I)_\# \,. \end{aligned}$$

(a) Let $i, j : X \to X \times I$ be given by $i(x) = (x, 0)$ and $j(x) = (x, 1)$. Show that the maps $T_X, T'_X : S_p(X) \to S_p(X \times I)$ given by $T_X(T) = i_\#(T)$ and $T'_X(T) = j_\#(T)$ define natural transformations T and T'. Derive Lemma 4.3.6 as a consequence of the acyclic model theorem.

(b) Derive Lemma 4.4.6 from the acyclic model theorem by showing that sd^m and $(i_X)_\#$ are natural transformations of the functor G to itself.

4. Let K and L be simplicial complexes. Suppose Φ is an acyclic carrier from K to L.

Consider the category **C** whose objects are subcomplexes of K and whose morphisms are inclusion maps j of subcomplexes of K. If K_0 is a subcomplex of K, let $\Phi(K_0)$ be the union of the subcomplexes $\Phi(\sigma)$ of L, as σ ranges over all simplices of K_0.

(a) If K_0 and K_1 are subcomplexes of K and $j : K_0 \to K_1$ is inclusion, consider the functors from **C** to **A** given by

$$\begin{aligned} &G : K_0 \to \mathcal{C}(K_0) \quad \text{and} \quad j \to j_\# \,, \\ &G' : K_0 \to \mathcal{C}(\Phi(K_0)) \quad \text{and} \quad j \to \ell_\# \,, \end{aligned}$$

where $\ell : \Phi(K_0) \to \Phi(K_1)$ is the inclusion map. Show that G and G' are indeed functors.

(b) Derive Theorem 1.14.10, the acyclic carrier theorem (geometric version), from the acyclic model theorem. [**Hint:** Let $\mathcal{M}$ consist of those subcomplexes of K whose polytopes are simplices of K.]

5. Give an example of an acyclic functor.

6. Give an example of a free functor.

4.6 Mayer–Vietoris Sequences

If X is the union of two subspaces X_1 and X_2, under suitable hypotheses there is an exact sequence relating the homology of X with that of X_1 and X_2. It is called the Mayer–Vietoris sequence of the pair X_1, X_2. We constructed such a sequence in simplicial theory under the assumption that X_1 and X_2 were polytopes of subcomplexes of a complex. In singular theory, we need an analogous condition:

Definition 4.6.1 Let $X = X_1 \cup X_2$. Let $\mathcal{A} = \{X_1, X_2\}$, and write

$$\mathcal{S}(X_1) + \mathcal{S}(X_2) = \mathcal{S}^{\mathcal{A}}(X)\,.$$

The pth chain group of $\mathcal{S}(X_1) + \mathcal{S}(X_2)$ is $S_p(X_1) + S_p(X_2)$, which is not a direct sum unless X_1 and X_2 are disjoint. We say $\{X_1, X_2\}$ is an *excisive couple* if the inclusion

$$\mathcal{S}(X_1) + \mathcal{S}(X_2) \longrightarrow \mathcal{S}(X)$$

induces an isomorphism in homology.

For singular homology, this definition is equivalent to the one given in Exercise 3 of Section 3.5. The proof of this equivalence is left to Exercise 2.

In view of Theorem 4.4.8, one situation in which $\{X_1, X_2\}$ is excisive occurs when the sets Int X_1 and Int X_2 cover X.

Theorem 4.6.2 *Let $X = X_1 \cup X_2$. Suppose $\{X_1, X_2\}$ is an excisive couple. Let $A = X_1 \cap X_2$. Then there is an exact sequence*

$$\ldots \longrightarrow H_p(A) \xrightarrow{\phi_*} H_p(X_1) \oplus H_p(X_2) \xrightarrow{\psi_*} H_{p-1}(A) \longrightarrow \ldots$$

called the Mayer–Vietoris sequence *of $\{X_1, X_2\}$. The homomorphisms are defined by*

$$\begin{aligned} \phi_*(a) &= (i_*(a), -j_*(a)), \\ \psi_*(x_1, x_2) &= k_*(x_1) + \ell_*(x_2), \end{aligned}$$

where the maps

$$\begin{array}{ccc} A & \xrightarrow{\;i\;} & X_1 \\ {\scriptstyle j}\big\downarrow & & \big\downarrow{\scriptstyle k} \\ X_2 & \xrightarrow{\;\ell\;} & X \end{array}$$

are inclusions. A similar sequence exists in reduced homology if A is nonempty. Both sequences are natural with respect to homomorphisms induced by continuous maps.

Proof: We define a short exact sequence of chain complexes

$$0 \longrightarrow S_p(A) \xrightarrow{\phi} S_p(X_1) \oplus S_p(X_2) \xrightarrow{\psi} S_p(X_1) + S_p(X_2) \longrightarrow 0 \qquad (*)$$

by the equations

$$\begin{aligned} \phi(c) &= (i_\#(c), -j_\#(c)), \\ \psi(c_1, c_2) &= k_\#(c_1) + \ell_\#(c_2). \end{aligned}$$

The map ϕ is injective, while ψ is surjective and its kernel consists of all chains of the form $(c, -c)$, where $c \in S_p(X_1)$ and $-c \in S_p(X_2)$. Exactness follows. From the zig-zag lemma [Lemma 3.3.2], we obtain a long exact sequence in homology. Since the hypotheses of the theorem guarantee that

$$H_p(\mathcal{S}(X_1) + \mathcal{S}(X_2)) \cong H_p(X)\,, \qquad (**)$$

the proof is complete. Exactness of the Mayer–Vietoris sequence in reduced homology when $A \neq \emptyset$ follows by a similar argument.

Now suppose that $f : (X, X_1, X_2) \to (Y, Y_1, Y_2)$ is a continuous map; where $X = X_1 \cup X_2$ and $Y = Y_1 \cup Y_2$ and both $\{X_1, X_2\}$ and $\{Y_1, Y_2\}$ are excisive couples. Since $f_\#$ commutes with inclusions, it commutes with the chain maps ϕ and ψ. Thus $f_\#$ gives a homomorphism of short exact sequences of chain complexes, so that f_* is a homomorphism of the corresponding homology sequences. Finally, we note that the isomorphism $(**)$ commutes with f_*, since it is induced by inclusion. The naturality of the Mayer–Vietoris sequence follows. □

The Mayer–Vietoris sequence has many applications. We give one here, as an illustration.

Recall that if K is a complex and if $w_0 * K$ and $w_1 * K$ are cones on K whose polytopes intersect in $|K|$ alone, then their union is a complex denoted $S(K)$ and called a *suspension* of K. [See Section 3.4.] It is easy to show that the map

$$\pi : |K| \times [-1, 1] \to |S(K)|$$

defined by

$$\pi(x, t) = \begin{cases} (1-t)x + tw_0 & \text{if } t \geq 0, \\ (1+t)x + tw_1 & \text{if } t \leq 0, \end{cases}$$

is a quotient map that collapses $|K| \times 1$ to the point w_0, $|K| \times (-1)$ to w_1, and is otherwise one-to-one. The proof is similar to the corresponding result for cones [see Corollary 2.8.8]. This fact motivates the definition of suspension for an arbitrary topological space.

Definition 4.6.3 Let X be a space. We define the *suspension* of X to be the quotient space of $X \times [-1,1]$ obtained by identifying the subset $X \times 1$ to a point v and identifying the subset $X \times (-1)$ to a point w. The suspension of X is denoted $S(X)$. The points v and w are called the *suspension points* of $S(X)$.

Just as in the case of simplicial theory, one can compute the homology of a suspension by using a Mayer–Vietoris sequence. One has the following theorem:

Theorem 4.6.4 *Let X be a space. For all p, there is an isomorphism*

$$\widetilde{H}_p(S(X)) \longrightarrow \widetilde{H}_{p-1}(X)\,.$$

Proof: Let $\pi : X \times [-1,1] \to S(X)$ be the quotient map. Let $v = \pi(X \times 1)$ and $w = \pi(X \times (-1))$ be the suspension points. Let $X_1 = S(X) \setminus w$ and $X_2 = S(X) \setminus v$. Since both X_1 and X_2 are open in $S(X)$, the pair $\{X_1, X_2\}$ is excisive. We will show that X_1 and X_2 are acyclic. The Mayer–Vietoris sequence then implies that there is an isomorphism

$$\widetilde{H}_p(S(X)) \longrightarrow \widetilde{H}_{p-1}(X_1 \cap X_2)\,.$$

We will also show that there is a homotopy equivalence of $X_1 \cap X_2$ with X, so there is an isomorphism

$$\widetilde{H}_p(X_1 \cap X_2) \longrightarrow \widetilde{H}_{p-1}(X)\,.$$

The theorem follows.

Now $X \times (-1,1]$ is open in $X \times [-1,1]$ and is saturated with respect to π. Therefore, the restricted map

$$\pi' : X \times (-1,1] \to X_1$$

is a quotient map. The map

$$F : X \times (-1,1] \times I \to X \times (-1,1]$$

given by

$$F(x,s,t) = (x,(1-t)s+t)$$

is a deformation retraction of $X \times (-1,1]$ onto $X \times 1$. Since the map $\pi' \circ F$ in the following diagram

$$\begin{array}{ccc} X \times (-1,1] \times I & \xrightarrow{F} & X \times (-1,1] \\ \pi' \times i_I \big\downarrow & & \big\downarrow \pi' \\ X_1 \times I & \xrightarrow{G} & X_1 \end{array}$$

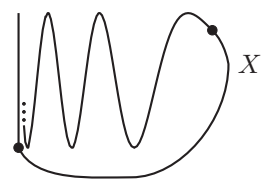
is constant on $X \times 1 \times I$, it induces a continuous map G that is a deformation retraction of X_1 to the point v. [Here we use the fact that $\pi' \times i_I$ is a quotient map, which follows from Theorem 2.8.2.] Thus X_1 is acyclic. A similar proof shows that X_2 is acyclic.

Finally, we note that the restricted map

$$\pi'' : X \times (-1, 1) \to X_1 \cap X_2$$

is a one-to-one quotient map, hence a homeomorphism. Since $X \times 0$ is a deformation retract of $X \times (-1, 1)$, there is a homotopy equivalence of X with $X_1 \cap X_2$. □

Definition 4.6.5 Figure 4.6.1 shows the *closed topologist's sine curve*. It is the union of the topologist's sine curve [see Definition 4.2.11] and an arc that intersects the topologist's sine curve only in the points $(0, -1)$ and $(1, \sin 1)$. It will be used in Exercise 1.



The Closed Topologist's Sine Curve
Figure 4.6.1

EXERCISES

1. Compute the singular homology of the closed topologist's since curve, X, using a suitable Mayer–Vietoris sequence.

2. Show that $\{X_1, X_2\}$ is an excisive couple in singular theory if and only if the inclusion $(X_1, X_1 \cap X_2) \longrightarrow (X, X_2)$ induces an isomorphism in singular homology. [**Hint:** Construct an exact sequence

$$0 \longrightarrow \mathcal{S}(X_2) \longrightarrow \mathcal{S}(X_1) + \mathcal{S}(X_2) \longrightarrow \mathcal{S}(X_1)/\mathcal{S}(X_1 \cap X_2) \longrightarrow 0\,.$$

Compare its derived homology sequence with the homology sequence of $\mathcal{S}(X, X_2)$.]

3. Show that $\mathbb{S}^n$ is homeomorphic to $S(\mathbb{S}^{n-1})$ for $n \geq 1$. Use this fact to compute the singular homology of $\mathbb{S}^n$.

4. State and prove a relative version of the Mayer–Vietoris sequence in singular homology, where the middle group is $H_p(X_1, B_1) \oplus H_p(X_2, B_2)$, assuming $\{X_1, X_2\}$ and $\{B_1, B_2\}$ are excisive. [**Hint:** Include the sequence

$$0 \longrightarrow \mathcal{S}(B_1 \cap B_2) \longrightarrow \mathcal{S}(B_1) \oplus \mathcal{S}(B_2) \longrightarrow \mathcal{S}(B_1) + \mathcal{S}(B_2) \longrightarrow 0$$

as part of the sequence $(*)$ of the proof of Theorem 4.6.2 and use the serpent lemma from Exercise 2 of Section 3.3.]

5. Let $X = X_1 \cup X_2$ and $A = X_1 \cap X_2$. Suppose that X_1 and X_2 are closed in X, and A is a deformation retract of an open set U of X_2. Show that X_1 is a deformation retract of $U \cup X_1$. Show that $\{X_1, X_2\}$ is an excisive couple.

6. Let X_1 be the closed upper half-circle, X_2 be the closed lower half-circle, and X be the full unit circle in the plane. What does the Mayer-Vietoris sequence say in these circumstances?

7. Let X_1 be the closed upper half-sphere, X_2 be the closed lower half-spere, and X be the full unit sphere in 3-dimensional space. What does the Mayer-Vietoris sequence say in these circumstances?

4.7 The Isomorphism between Simplicial and Singular Homology

In this section, we show that if K is a simplicial complex, the simplicial homology groups of K are isomorphic with the singular homology groups of $|K|$. In fact, we show that the isomorphism commutes with induced homomorphisms and with the boundary homomorphism ∂_*, so it is an isomorphism between the two homology theories.

The proof involves a notion we introduced in Section 1.14, that of the ordered chain complex $\{C'_p(K), \partial'\}$ of a simplicial complex K. We proved there that the homology groups of this chain complex, called the ordered homology groups of K, are isomorphic to the usual [oriented] homology groups of K. In this section, we show that the ordered homology groups of K are in turn isomorphic with the singular homology groups of $|K|$, thus showing that the simplicial and singular homology agree for polyhedra.

Recall that an ordered p-simplex of K is a $(p+1)$-tuple $(v_0, v_1, \ldots, v_p)$ of vertices of K [not necessarily distinct] that span a simplex of K, and the group $C'_p(K)$ is the free abelian group generated by the ordered p-simplices. Also recall that

$$\partial'(v_0, v_1, \ldots, v_p) = \sum (-1)^i (v_0, v_1, \ldots, \widehat{v}_i, \ldots, v_p)$$

and that $\epsilon'(v) = 1$ for each vertex v. We now construct a chain map carrying $C'_p(K)$ to $S_p(|K|)$ and prove that it induces an isomorphism in homology.

Definition 4.7.1 Define $\theta : C'_p(K) \to S_p(|K|)$ by the equation

$$\theta((v_0, v_1, \ldots, v_p)) = \ell(v_0, v_1, \ldots, v_p)\,.$$

Then θ assigns, to the ordered simplex $(v_0, v_1, \ldots, v_p)$ of K, the linear singular simplex mapping Δ_p into $|K|$ and carrying ϵ_i to v_i for $i = 0, 1, \ldots, p$.

It is immediate from the definitions that θ is a chain map and that it preserves augmentation. If K_0 is a subcomplex of K, then θ commutes with inclusion, so it maps $C'_p(K_0)$ into $S_p(|K_0|)$, and thus induces a chain map

$$\theta : C'_p(K, K_0) \to S_p(|K|, |K_0|)\,.$$

To show that θ_* is an isomorphism, the following lemma will be useful.

Lemma 4.7.2 *Let $\psi : \mathcal{C} \to \mathcal{C}'$ be a chain map of augmented chain complexes. Then ψ_* is an isomorphism in reduced homology if and only if it is an isomorphism in ordinary homology.*

Proof: We recall the proof that $H_0(\mathcal{C}) \cong \widetilde{H}_0(\mathcal{C}) \oplus \mathbb{Z}$. [See the exercises of Section 1.8.] Begin with the exact sequences

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \ker\epsilon & \longrightarrow & C_0 & \xrightarrow{\epsilon} & \mathbb{Z} & \longrightarrow & 0 \\
 & & \psi\downarrow & & \psi\downarrow & & \downarrow= & & \\
0 & \longrightarrow & \ker\epsilon' & \longrightarrow & C'_0 & \xrightarrow{\epsilon'} & \mathbb{Z} & \longrightarrow & 0\,.
\end{array}$$

Choose $j : \mathbb{Z} \to C_0$ so that $j \circ \epsilon$ is the identity. Define $j' : \mathbb{Z} \to C'_0$ by setting $j' = \psi \circ j$. Then j and j' split the two sequences, so

$$C_0 = \ker\epsilon \oplus \operatorname{im} j \quad \text{and} \quad C'_0 = \ker\epsilon' \oplus \operatorname{im} j'\,.$$

Note that ψ defines an isomorphism of $\operatorname{im} j$ with $\operatorname{im} j'$. Now we have

$$H_0(\mathcal{C}) \cong \frac{\ker\epsilon}{\partial C_1} \oplus \operatorname{im} j \quad \text{and} \quad H_0(\mathcal{C}') \cong \frac{\ker\epsilon'}{\partial' C'_1} \oplus \operatorname{im} j'\,.$$

It follows that $\psi_* : H_0(\mathcal{C}) \to H_0(\mathcal{C}')$ is an isomorphism if and only if ψ induces an isomorphism of $\ker\epsilon/\partial C_1$ with $\ker\epsilon'/\partial' C'_1$. □

Lemma 4.7.3 *Let K_0 be a subcomplex of K. For all p, the chain map θ induces isomorphisms*

(1) $\theta_* : \widetilde{H}_p(\mathcal{C}'(K)) \to \widetilde{H}_p(|K|)$,
(2) $\theta_* : H_p(\mathcal{C}'(K)) \to H_p(|K|)$,
(3) $\theta_* : H_p(\mathcal{C}'(K, K_0)) \to H_p(|K|, |K_0|)$.

Proof: We assume that $K_0 \neq K$, since otherwise **(3)** is trivial.

Step 1. We prove the lemma first when K is finite, by induction on the number n of simplices in K. If $n = 1$, then K consists of a single vertex v. In each dimension p, there is exactly one ordered p-simplex $(v, v, \ldots, v)$ of K and exactly one singular p-simplex $T : \Delta_p \to v$ of $|K|$. Furthermore, we have $\theta(v, v, \ldots, v) = \ell(v, v, \ldots, v) = T$. Then $\theta : \mathcal{C}'(K) \to \mathcal{S}(|K|)$ is an isomorphism already on the chain level. Hence **(1)**, **(2)**, and **(3)** hold.

Now suppose the lemma holds for any complex having fewer than n simplices. Let K have n simplices. We note that it suffices to prove **(1)** for K, for then **(2)** follows by Lemma 4.7.2. Also, since $\theta_* : H_p(\mathcal{C}'(K_0)) \to H_p(|K_0|)$ is an isomorphism by the induction hypothesis, **(3)** follows from the long exact homology sequence and the Steenrod five-lemma [Lemma 3.3.4].

To prove **(1)**, let σ be a simplex of K of maximal dimension. Then σ is not a face of any other simplex, so that the collection of all simplices of K different from σ is a subcomplex K_1 of K having $n-1$ simplices. Let Σ denote the complex consisting of σ and its faces. Let Bd Σ be the collection of proper faces of σ. Consider the following commutative diagram, where i and j are inclusions:

$$\begin{array}{ccc}
\widetilde{H}_p(\mathcal{C}'(K)) & \xrightarrow{\theta_*} & \widetilde{H}_p(|K|) \\
\big\downarrow i_* & & \big\downarrow i_* \\
H_p(\mathcal{C}'(K, \Sigma)) & \xrightarrow{\theta_*} & H_p(|K|, \sigma) \\
\big\uparrow j_* & & \big\uparrow j_* \\
H_p(\mathcal{C}'(K_1, \text{Bd } \Sigma)) & \xrightarrow{\theta_*} & H_p(|K_1|, \text{Bd } \sigma)\,.
\end{array}$$

[If $\dim \sigma = 0$, then Bd Σ and Bd σ are empty.] It follows from the induction hypothesis that the map θ_* on the bottom line of the diagram is an isomorphism. We will show that the vertical maps are isomorphisms. It then follows that the map θ_* on the top line is an isomorphism and that will complete the proof.

First we consider the homomorphisms i_*. Because σ is acyclic in singular homology, the long exact homology sequence of $(|K|, \sigma)$ shows that the right-hand map i_* is an isomorphism. The same argument applies to the map i_* in ordered simplicial homology, since Σ is acyclic in ordered homology, by Lemma 1.14.14.

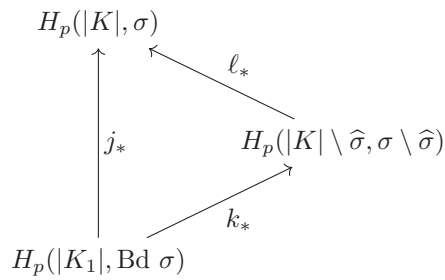
The map j_* is an isomorphism in ordered simplicial homology, because $j_\#$ is an isomorphism already on the chain level; the inclusion map

$$\mathcal{C}'(K_1) \longrightarrow \mathcal{C}'(K) \longrightarrow \mathcal{C}'(K)/\mathcal{C}'(\Sigma)$$

is surjective and carries to zero precisely those chains carried by $K_1 \cap \Sigma =$ Bd Σ. [In fact, j is just an excision map.]

It is tempting to assert that j is also an excision map for singular homology, but this is not true. The domain of j is formed by "excising away" from $|K|$ and σ the set $U = \text{Int } \sigma$. Since $\overline{U} = \sigma$, Theorem 4.4.10 does not apply; we are excising away something too large. Nonetheless, we can excise something smaller—namely, the barycenter $\widehat{\sigma}$ of σ. This we now do.

Consider the diagram



where k and ℓ are inclusions. The map ℓ is an acceptable excision map, since $\widehat{\sigma}$ is closed and is contained in Int σ. [Note that Int σ is open in $|K|$, because $|K_1|$ is closed. Thus the interior of σ in $|K|$, in the sense of point-set topology, equals the "open simplex" Int σ.] Therefore, ℓ_* is an isomorphism. Why is k_* an isomorphism? If $\dim \sigma = 0$, then $\sigma \setminus \widehat{\sigma}$ and Bd σ are empty, and $|K| \setminus \sigma = |K_1|$. So the map k is the identity in this case. If $\dim \sigma > 0$, then we use the fact that Bd σ is a deformation retract of $\sigma \setminus \widehat{\sigma}$. This deformation retraction extends to a deformation retraction F_t of $|K| \setminus \widehat{\sigma}$ onto $|K_1|$ defined by letting F_t equal the identity on $|K_1|$. See Figure 4.7.1. We conclude from Theorem 4.3.9 that k_* is an isomorphism. Then j_* is an isomorphism in singular homology, as desired.

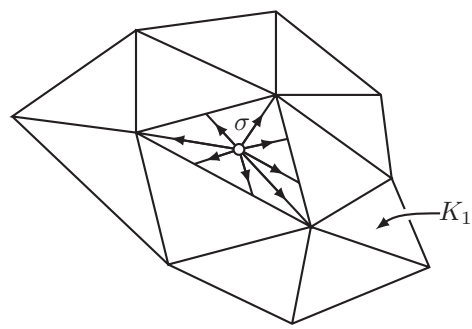


Figure 4.7.1

Step 2. Having proved the lemma when K is finite, we prove it in general. As before, it suffices to prove **(1)** for all K. For then **(2)** follows from the preceding lemma and **(3)** follows from **(2)** and the Steenrod five-lemma.

First, we show that θ_* is surjective. Given $\{z\} \in \widetilde{H}_p(|K|)$, there is a compact subset A of $|K|$ such that the chain z is carried by A. Let L be a finite subcomplex of K such that A is contained in $|L|$. Consider the commutative diagram

$$\begin{array}{ccc} \widetilde{H}_p(\mathcal{C}'(L)) & \xrightarrow[\cong]{\theta_*} & \widetilde{H}_p(|L|) \\ {\scriptstyle i_*}\downarrow & & \downarrow{\scriptstyle j_*} \\ \widetilde{H}_p(\mathcal{C}'(K)) & \xrightarrow{\theta_*} & \widetilde{H}_p(|K|) \end{array}$$

where the vertical maps are induced by inclusion. Then $\{z\}$ lies in the image of j_*. The map θ_* on the top line is an isomorphism by Step 1. Hence $\{z\}$ lies in the image of the map θ_* on the bottom line.

We show that the kernel of θ_* is trivial. Suppose $\{z\} \in \widetilde{H}_p(\mathcal{C}'(K))$ and $\theta_*(\{z\}) = 0$. Then $\theta(z) = \partial d$ for some singular $(p+1)$-chain d of $|K|$. The chain d is carried by a compact subset of $|K|$. Choose a finite subcomplex L of K such that z is carried by L and d is carried by $|L|$. Consider the same commutative diagram as before, but with the new choice of L. Let α denote the homology class of z in $\widetilde{H}_p(\mathcal{C}'(L))$. Because $\theta(z) = \partial d$, where d is carried by $|L|$, we see that $\theta_* : H_p(\mathcal{C}'(K)) \to H_p(|L|)$ carries α to zero. Because this map is an isomorphism, $\alpha = 0$. Hence $\{z\} = i_*(\alpha) = 0$ as well. □

The argument given in Step 2 of the last proof is of the general kind known as a "direct limit argument." It is a standard way of carrying over results that hold for the homology of finite complexes to complexes in general.

We now combine the results of Section 1.14 on ordered homology with the lemma just proved.

Definition 4.7.4 Let K be a complex. We define

$$\eta : \mathcal{C}(K) \to \mathcal{S}(|K|)$$

as follows: Choose a partial ordering of the vertices of K that induces a linear ordering on the vertices of each simplex of K. Orient the simplices of K by using this ordering and define

$$\eta([v_0, v_1, \ldots, v_p]) = \ell(v_0, v_1, \ldots, v_p)\,,$$

where $v_0 < v_1 < \cdots < v_p$ in the given ordering. It is immediate that η is a chain map, that it preserves augmentation, and that it commutes with inclusions. Thus it induces a chain map on the relative level as well.

In fact, η is the composite

$$\mathcal{C}(K) \xrightarrow{\phi} \mathcal{C}'(K) \xrightarrow{\theta} \mathcal{S}(|K|)\,,$$

where θ is the chain map of the preceding lemma and ϕ is the chain equivalence of Theorem 1.14.16. Since both ϕ and θ induce homology isomorphisms, so does η. This result holds for relative homology and reduced homology as well.

The chain map η certainly depends on the chosen ordering of the vertices of K, but the homomorphism η_* does not. We see this because $\eta_* = \theta_* \circ \phi_*$, where θ obviously does not depend on the ordering and ϕ_* does not depend on the ordering by Theorem 1.14.16.

Finally, we note that since the chain map η commutes with inclusions, the naturality of the zig-zag lemma [Lemma 3.3.2] implies that η_* commutes with ∂_*.

We summarize this fact as follows:

Theorem 4.7.5 *The map η_* is a well-defined isomorphism of simplicial with singular homology that commutes with the boundary homomorphism ∂_*.*

Furthermore, one has the following naturality result:

Theorem 4.7.6 *The isomorphism η_* commutes with homomorphisms induced by simplicial maps.*

Proof: Let $f : (K, K_0) \to (L, L_0)$ be a simplicial map. We have already shown that f_* commutes with ϕ_*. [See Theorem 1.14.18.] We now show it commutes with θ_*. In fact, $f_\#$ commutes with θ on the chain level.

Recall that, in ordered homology, the chain map induced by f is defined by the equation

$$f'_\#((w_0, w_1, \ldots, w_p)) = (f(w_0), f(w_1), \ldots, f(w_p)) \,,$$

where $(w_0, w_1, \ldots, w_p)$ is an ordered p-simplex of K. We compute directly that

$$f_\# \theta((w_0, w_1, \ldots, w_p)) = f \circ \ell(w_0, w_1, \ldots, w_p) \,.$$

This map equals the linear map of Δ_p into $|K|$ that carries ϵ_i to w_i for each i, followed by the map f, which carries the simplex spanned by $w_0, w_1, \ldots, w_p$ onto a simplex of L. Since the composite of linear maps is linear, this map equals

$$\begin{aligned} \ell(f(w_0), f(w_1), \ldots, f(w_p)) &= \theta((f(w_0), f(w_1), \ldots, f(w_p))) \\ &= \theta f'_\#((w_0, w_1, \ldots, w_p)) \,. \end{aligned}$$

□

It is *possible* to define the homomorphism of simplicial homology induced by a continuous map h as the composite $\eta_*^{-1} \circ h_* \circ \eta_*$, where h_* denotes the induced homomorphism in singular homology. This would give us the same homomorphism of simplicial homology as we defined in Chapter 2 by use of simplicial approximations. We prove this result as follows:

Theorem 4.7.7 *The isomorphism η_* commutes with homomorphisms induced by continuous maps.*

Proof: Let

$$h : (|K|, |K_0|) \to (|L|, |L_0|)$$

be a continuous map. Let

$$f : (K', K_0') \to (L, L_0) \quad \text{and} \quad g : (K', K_0') \to (K, K_0)$$

be simplicial approximations to h and to the identity, respectively. In simplicial homology, we have

$$h_* = f_* \circ (g_*)^{-1},$$

by definition. Now, in singular homology, the map g_* equals the identity isomorphism, because g is homotopic to the identity map [Theorem 2.7.7]. Similarly, $f_* = h_*$ in singular homology, because f is homotopic to h. Thus in singular homology, we also have the equation $h_* = f_* \circ (g_*)^{-1}$.

Our theorem now follows by applying the preceding theorem to the maps f and g. □

Although we have proved that the chain map η induces an isomorphism in homology, we have not found a chain-homotopy inverse λ for η. The fact that such a λ exists is a consequence of results that we will prove later. [See the exercises of Section 5.7.] A specific formula for λ can be derived using the theory of "regular neighborhoods." [See [ES52].]

EXERCISE

1. Show that if K_1 and K_2 are subcomplexes of a complex K, then $\{|K_1|, |K_2\}$ is an excisive couple in singular homology.
2. Show that the result of Exercise 1 fails if K_1 and K_2 are not subcomplexes.

4.8 *Application: Local Homology Groups and Manifolds

In this section, we define the local homology groups of a space X at a point x of X, and we use these groups to prove several non-trivial facts about manifolds. *Throughout this section, let X denote a Hausdorff space.*

This section will be assumed as background for Chapter 8. It is also used in treating one of the examples in Section 4.12.

Definition 4.8.1 If X is a space and if $x \in X$, then the *local homology groups* of X at x are the singular homology groups

$$H_p(X, X \setminus x)\,.$$

The reason for the term "local" comes from the following lemma.

Lemma 4.8.2 *Let $A \subset X$. If A contains a neighborhood of the point x, then*

$$H_p(X, X \setminus x) \cong H_p(A, A \setminus x)\,.$$

Therefore, if $x \in X$ and $y \in Y$ have neighborhoods U and V, respectively, such that (U, x) and (V, y) are homeomorphic, then the local homology groups of X at x and of Y at y are isomorphic.

Proof: Let U denote the set $X \setminus A$. Because A contains a neighborhood of x,

$$\overline{U} \subset X \setminus x = \text{Int } (X \setminus x)\,.$$

It follows from the excision property that

$$H_p(X, X \setminus x) \cong H_p(X \setminus U, (X \setminus x) \setminus U) = H_p(A, A \setminus x)\,. \qquad \square$$

Let us compute some local homology groups.

EXAMPLE 4.8.3 If $x \in \mathbb{R}^m$, we show that $H_i(\mathbb{R}^m, \mathbb{R}^m \setminus x)$ is infinite cyclic for $i = m$ and vanishes otherwise.

Let B denote a ball of positive radius centered at x. By the preceding lemma,

$$H_i(\mathbb{R}^m, \mathbb{R}^m \setminus x) \cong H_i(B, B \setminus x) \cong H_i(\mathbb{B}^m, \mathbb{B}^m \setminus \mathbf{0})\,.$$

Now $\mathbb{S}^{m-1}$ is a deformation retract of $\mathbb{B}^m \setminus \mathbf{0}$. See Figure 4.8.1. The formula for the deformation is given in the proof of Theorem 2.7.11. Therefore, we have

$$H_i(\mathbb{B}^m, \mathbb{B}^m \setminus \mathbf{0}) \cong H_i(\mathbb{B}^m, \mathbb{S}^{m-1})\,.$$

Theorem 4.4.11 tells us that the group $H_i(\mathbb{B}^m, \mathbb{S}^{m-1})$ is infinite cyclic for $i = m$ and vanishes otherwise.

EXAMPLE 4.8.4 Let $\mathbb{H}^m$ denote the *euclidean half-space*

$$\mathbb{H}^m = \{(x_1, x_2, \ldots, x_m) : x_m \geq 0\}\,.$$

We have Bd $\mathbb{H}^m = \mathbb{R}^{m-1} \times 0$. If $x \in$ Bd $\mathbb{H}^m$, then the group $H_i(\mathbb{H}^m, \mathbb{H}^m \setminus x)$ vanishes for all i. If $x \in \mathbb{H}^m \setminus$ Bd $\mathbb{H}^m$, then $H_i(\mathbb{H}^m, \mathbb{H}^m \setminus x)$ is infinite cyclic for $i = m$ and vanishes otherwise. We prove these facts as follows.

If $x \in \mathbb{H}^m \setminus$ Bd $\mathbb{H}^m$, then this result follows from the previous example and Lemma 4.8.2, once we note that x has a neighborhood that is an open subset of $\mathbb{R}^m$.

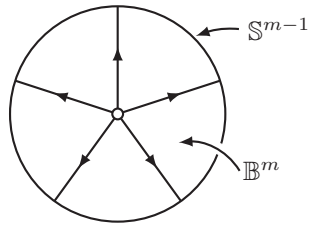


Figure 4.8.1

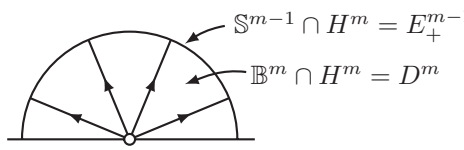


Figure 4.8.2

Suppose $x \in \operatorname{Bd} \mathbb{H}^m$. Without loss of generality, we may assume that $x = \mathbf{0}$. Let $\mathbb{B}^m$ be the unit ball in $\mathbb{R}^m$ centered at $\mathbf{0}$. The set $\mathbb{B}^M \cap \mathbb{H}^m$ contains a neighborhood of $\mathbf{0}$ in $\mathbb{H}^m$. Letting D^m denote the half-ball $\mathbb{B}^m \cap \mathbb{H}^m$, we have

$$H_i(\mathbb{H}^m, \mathbb{H}^m \setminus \mathbf{0}) \cong H_i(D^m, D^m \setminus \mathbf{0}) .$$

See Figure 4.8.2. Now there is a deformation retraction of $\mathbb{B}^m \setminus \mathbf{0}$ onto $\mathbb{S}^{m-1}$. That deformation retraction restricts to a deformation retraction of the punctured half-ball $D^m \setminus \mathbf{0}$ onto the set

$$\mathbb{S}^{m-1} \cap \mathbb{H}^m = E_+^{m-1} .$$

Now D^m is acyclic, being a convex set in $\mathbb{R}^m$, and E_+^{m-1} is acyclic, since it is homeomorphic to $\mathbb{B}^{m-1}$. The long exact exact homology sequence shows that their relative homology vanishes.

Definition 4.8.5 A nonempty Hausdorff space X is called an *m-manifold* if each point of X has a neighborhood homeomorphic with an open subset of euclidean space $\mathbb{R}^m$. It is called an *m-manifold with boundary* if each point of X has a neighborhood homeomorphic with an open subset of the euclidean half-space $\mathbb{H}^m$.

Note that an m-manifold is automatically an m-manifold with boundary. For if x has a neighborhood homeomorphic with an open set in $\mathbb{R}^m$, it has a neighborhood U homeomorphic with an open ball in $\mathbb{R}^m$. Then U is homeomorphic with the open unit ball in $\mathbb{R}^m$ centered at $(0, 0, \ldots, 0, 1)$, which is an open set of $\mathbb{H}^m$.

One often includes in the definition of a manifold the requirement that X have a countable basis, or at least that X be metrizable. We shall not make either assumption.

Manifolds and manifolds with boundary are among the most familiar and important geometric objects. They are the main objects of study in differential geometry and differential topology.

If X is a manifold with boundary and if $h : U \to V$ is a homeomorphism of an open set U in X onto an open set V in $\mathbb{H}^m$, then h is called a *coordinate patch* on X. A point x of X may be mapped by h either into the upper half space $\mathbb{R}^{m-1} \times \mathbb{R}_+$ of $\mathbb{H}^m$ or onto the "edge" $\mathbb{R}^{m-1} \times 0$. The local homology groups distinguish between these two possibilities. For it follows from Example 4.8.4 that if x is mapped into the open upper half space of $\mathbb{H}^m$, then $H_m(X, X \setminus x)$ is infinite cyclic, while if x is mapped into Bd $\mathbb{H}^m$, then $H_m(X, X \setminus x)$ vanishes. This fact leads to the following definition:

Definition 4.8.6 Let X be an m-manifold with boundary. If the point x of X maps to a point of Bd $\mathbb{H}^m$ under one coordinate patch about x, it maps to a point of Bd $\mathbb{H}^m$ under every coordinate patch. Such a point is called a *boundary point* of X. The set of all such points is called the *boundary* of X, and is denoted Bd X. The space $X \setminus$ Bd X is called the *interior* of X and is denoted Int X.

Note that there is nothing in the definition requiring that X have any boundary points. If it does not then Bd X is empty and X is an m-manifold. While Bd X may be empty, the set Int X cannot be. For if $h : U \to V$ is a coordinate patch about a point x of X, then V is open in $\mathbb{H}^m$ and hence contains at least one point of the open upper half space. The corresponding point y of X lies in Int X by definition.

We remark that the space $\mathbb{H}^m$ is itself an m-manifold with boundary, and its boundary is precisely the set $\mathbb{R}^{m-1} \times 0$, which we have already denoted by Bd $\mathbb{H}^m$. Similarly, $\mathbb{R}^m$ is itself an m-manifold.

Definition 4.8.7 Let X be an m-manifold with boundary. It follows from Example 4.8.4 that m is uniquely determined by X, for it is the unique integer such that the group $H_m(X, X \setminus x)$ is non-trivial for at least one x in X. The number m is called [of course] the *dimension* of the manifold with boundary X.

EXAMPLE 4.8.8 The unit ball $\mathbb{B}^n$ in $\mathbb{R}^n$ is an n-manifold with boundary, and Bd $\mathbb{B}^n = \mathbb{S}^{n-1}$. We prove these facts as follows.

If $p \in \mathbb{B}^n \setminus \mathbb{S}^{n-1}$, then the set of all x with $\|x\| < 1$ is an open set of $\mathbb{R}^n$ that contains p. Thus there is a coordinate patch about p.

Now suppose $p \in \mathbb{S}^{n-1}$. We will find a coordinate patch about p. Some coordinate of $p = (p_1, p_2, \ldots, p_n)$ is non-zero. Suppose for convenience that $p_n < 0$. Let U be the open set in $\mathbb{B}^n$ consisting of all points x of $\mathbb{B}^n$ with $x_n < 0$. Define $h : U \to \mathbb{H}^n$ by the equation

$$h(x) = (x_1, x_2, \ldots, x_{n-1}, x_n + f(x)) ,$$

where $f(x) = [1 - x_1^2 - x_2^2 - \cdots - x_{n-1}^2]^{1/2}$. Then you can check that h is a homeomorphism of U onto the open set V of $\mathbb{H}^m$ consisting of all points y with $\|y\| < 1$ and $y_n \geq 0$. This homeomorphism carries p to a point of Bd $\mathbb{H}^n$. [See Figure 4.8.3.]

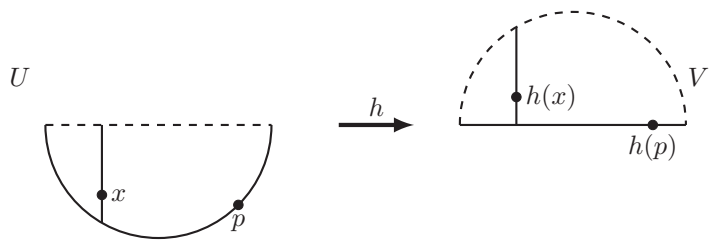


Figure 4.8.3

EXAMPLE 4.8.9 Let σ be an n-simplex. Then σ is an n-manifold with boundary, because there is a homeomorphism of σ with $\mathbb{B}^n$. This homeomorphism carries the union Y of the proper faces of σ onto $\mathbb{S}^{n-1}$. Thus the set Y, which we have been denoting by Bd σ, is the boundary of σ when it is considered as an n-manifold with boundary. Also, the set $\sigma \setminus Y$, which we have been denoting by Int σ, is the interior of σ as a manifold with boundary.

There is a certain overlapping of terminology here, which we should clarify. In general topology, if A is a subset of a space X, then the *interior* of A, denoted Int A, is the union of all open sets of X contained in A. The *boundary* of A, denoted Bd A, is the set $\overline{A} \cap \overline{X \setminus A}$. In the special case where A is an open set of X, it turns out that Bd $A = \overline{A} \setminus A$. We have used this terminology from general topology earlier in this book. For instance, the notion Bd U appeared in Lemma 1.2.7, and the notion Int A was used in formulating the excision axiom.

The concepts of boundary and interior for a manifold with boundary are entirely different. It is unfortunate that the same terminology is commonly used in two different ways. Some authors use ∂X to denote the boundary of a manifold with boundary. But that leads to difficulty when one wishes to distinguish the boundary of the space σ from the simplicial chain $\partial\sigma$. As is usual in mathematics, we will simply endure some ambiguity, relying on the context to make the meaning clear.

It happens that for the subset $\mathbb{B}^n$ of the topological space $\mathbb{R}^n$, its boundary in the sense of general topology is the same as its boundary in the sense of manifolds with boundary. The same remark applies to the subspace $\mathbb{H}^n$ of $\mathbb{R}^n$. But it is safest to think of these cases being the exception rather than the rule.

Now we prove a result about triangulations of manifolds.

Lemma 4.8.10 *Let s be a simplex of the complex K. If x and y are points of* Int s*, then the local homology groups of $|K|$ at x and y are isomorphic.*

Proof: It suffices to prove the theorem when $x = \widehat{s}$ the barycenter of s. Let sd K be the first barycentric subdivision of K. Let K' be a subdivision of K defined exactly as sd K was, except that y is used instead of $\widehat{s}$ when "starring" the subdivision of Bd s from an interior point of s. There is a linear isomorphism of sd K with K' that carries $\widehat{s}$ to y, and carries each remaining vertex of sd K to itself. Then the pair $(|K|, |K| \setminus \widehat{s})$ is homeomorphic with the pair $(|K|, |K| \setminus y)$. □

Recall that a homeomorphism $h : |K| \to M$ is called a *triangulation* of M. Answering the general question of whether or not an arbitrary manifold can be triangulated took about a century of work by various mathematicians. The initial conjecture dates back to an 1899 paper by Poincaré.[1] But it was the 1924 paper [Kne25] by Kneser that gave the problem the name the "Hauptvermutung." The succinct answer to the triangulation question is that all two- and three-dimensional topological manifolds can be triangulated, while for dimensions four and higher there are topological manifolds that cannot be. A more detailed, but still short, discussion can be found in [Man14].

Theorem 4.8.11 *Let M be an m-manifold with boundary. Suppose K is a complex and $h : |K| \to M$ is a homeomorphism. Then $h^{-1}(\text{Bd } M)$ is the polytope of a subcomplex of K.*

Proof: If an open simplex Int s of K intersects the set $h^{-1}(\text{Bd } M)$, it lies in this set by the preceding lemma. Since this set is closed, it must contain s. □

Now we give a final application of local homology groups. We show that the dimension of a finite-dimensional simplicial complex K is a topological invariant of $|K|$. [A different proof, using the notion of "covering dimension," was outlined in the exercises of Sections 2.4 and 2.7.]

First we need two lemmas.

Lemma 4.8.12 *Let L be a complex, and let $v * L$ be a cone over L. Then $|L|$ is a deformation retract of $|v * L| \setminus v$.*

Proof: Consider the quotient map

$$\pi : |L| \times I \to |v * L|$$

defined by $\pi(x, t) = (1 - t)x + tv$. [See Corollary 2.8.8.] Since $|L| \times [0, 1)$ is open in $|L| \times I$ and is saturated with respect to π, the restriction of π,

$$\pi' : |L| \times [0, 1) \to |v * L| \setminus v\,,$$

[1] Poincaré's writings on topology appear in translation in [Poi10]. Pages 105–109 are relevant to triangulation. It is perhaps not surprising that the argument there does not meet modern standards of rigor.

is a quotient map; being one-to-one, it is a homeomorphism. Since $|L| \times 0$ is a deformation retract of $|L| \times [0,1)$, the space L is a deformation retract of $|v * L| \setminus v$. □

Recall the if v is a vertex of K, then St v is the union of the interiors of all simplices that have v as vertex and Lk $v = \overline{\text{St}}\ v \setminus \text{St}\ v$.

Lemma 4.8.13 *Let v be a vertex of the simplicial complex K. Then*

$$H_i(|K|, |K| \setminus v) \cong H_i(\overline{\text{St}}\ v, \text{Lk}\ v)\,.$$

Proof: The set $\overline{\text{St}}\ v$ contains a neighborhood of v. Therefore, it follows from Lemma 4.8.2 that

$$H_i(|K|, |K| \setminus v) \cong H_i(\overline{\text{St}}\ v, \overline{\text{St}}\ v \setminus v)\,.$$

Let L denote the subcomplex of K whose polytope is Lk v. Then $\overline{\text{St}}\ v$ is the polytope of the cone $v * L$, by Lemma 1.9.4. See Figure 4.8.4. The preceding lemma tells us that Lk v is a deformation retract of $\overline{\text{St}}\ v \setminus v$, so our proof is complete. □

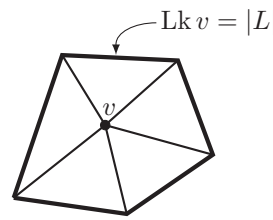


Figure 4.8.4

Theorem 4.8.14 *Let K be a complex of dimension n, and let $X = |K|$. For $p > n$ the local homology groups $H_p(X, X \setminus x)$ vanish, while for $p = n$ one of the groups $H_n(X, X \setminus x)$ is non-trivial for some choice of x.*

Proof: Let σ be an n-simplex of K. Then σ is a face of no other simplex of K, so the set Int σ is in fact an open set of $|K|$. [Its complement is the union of all simplices of K different from σ.] If x is the barycenter $\widehat{\sigma}$ of σ, it follows from Lemma 4.8.2 that

$$\begin{aligned} H_n(X, X \setminus x) &\cong H_n(\sigma, \sigma - \widehat{\sigma}) \\ &\cong H_n(\mathbb{B}^n, \mathbb{B}^n \setminus \mathbf{0})\,. \end{aligned}$$

By Example 4.8.3, this group is infinite cyclic.

Now let x be an arbitrary point of X. We wish to show that $H_p(X, X \setminus x) = 0$ for $p > n$. In view of Lemma 4.8.10, it suffices to consider the case where x is

the barycenter of a simplex of K. Then x is a vertex of the complex $L = \text{sd } K$, and Lemma 4.8.13 applies. We have

$$H_n(X, X \setminus x) \cong H_p(\overline{\text{St}}(x, L), \text{Lk}(x, L)) . \qquad (*)$$

The left-hand side of $(*)$ is a singular homology group, but the right-hand side can be considered in either singular or simplicial homology. Since L is a complex of dimension n, $\overline{\text{St}}(x, L)$ is the polytope of a complex of dimension at most n. Therefore, this group vanishes in simplicial homology for $p > n$, so it vanishes in singular homology as well. □

The next three definitions are used in the exercises.

Definition 4.8.15 A space X is *homogeneous* if given x and y, there is a homeomorphism of X with itself carrying x to y.

Definition 4.8.16 An n-dimensional *homology sphere* [respectively, *homology ball*] is a space or complex having the same homology groups as $\mathbb{S}^n$ [respectively, $\mathbb{B}^n$].

Definition 4.8.17 A *solid torus* is a space homeomorphic to $\mathbb{S}^1 \times \mathbb{B}^2$. Thus a solid torus is a 3-manifold with boundary, and the boundary is homeomorphic to the torus.

EXERCISES

1. Check the details of Example 4.8.8

2. Show that if M and N are manifolds with boundary of dimensions m and n, respectively, then $M \times N$ is a manifold with boundary of dimension $m + n$, and

$$\text{Bd } (M \times N) = (M \times (\text{Bd } N)) \cup ((\text{Bd } M) \times N) .$$

3. Show that a connected manifold is homogeneous. [**Hint:** Define $x \sim y$ if there is a homeomorphism of X with itself carrying x to y. Show that $\sim$ is an equivalence relation, and that the equivalence classes are open.]

4. Let M be an m-manifold with boundary. Suppose $h : |K| \to M$ is a triangulation of M.

(a) Show that if v is a vertex of K, then Lk v is an $(m-1)$-dimensional homology sphere or ball, according to whether $h(v) \in \text{Int } M$ or $h(v) \in \text{Bd } M$.

(b) Show that every simplex of K either has dimension m or is the face of a simplex of dimension m.

(c) Show that an $(m-1)$-simplex s of K is a face of precisely one m-simplex of K if $s \subset h^{-1}(\text{Bd } M)$, and it is a face of precisely two m-simplices of K otherwise.

5. Use the fact that $\mathbb{S}^3$ is homeomorphic to Bd $(\mathbb{B}^2 \times \mathbb{B}^2)$ to write $\mathbb{S}^3$ as the union of two solid tori T_1, T_2 that intersect in their common boundary. Compute the Mayer–Vietoris sequence of T_1, T_2.

6. Let C be the Cantor ternary set [see Definition 2.3.13] in the real line. What is the dimension of C?

7. Give an example of a set in $\mathbb{R}^3$ of dimension 1. How about a set of dimension 2?

4.9 *Application: The Jordan Curve Theorem

Using the basic properties of singular homology, we now prove several classical theorems of topology, including the generalized Jordan curve theorem and the Brouwer theorem on invariance of domain.

Definition 4.9.1 If A is a subspace of X, we say that A *separates* X if the space $X \setminus A$ is not connected.

We are going to be concerned with the case where X is $\mathbb{R}^n$ or $\mathbb{S}^n$, and A is closed in X. Since $X \setminus A$ is then locally path connected, its components and path components are identical. In particular, the group $\widetilde{H}_0(X \setminus A)$ vanishes if and only if $X \setminus A$ is connected. In general, the rank of $\widetilde{H}_0(X \setminus A)$ is one less than the number of components of $X \setminus A$. We will use this fact freely in what follows.

Definition 4.9.2 A space homeomorphic to the unit k-ball $\mathbb{B}^k$ is called a *k-cell.*

Theorem 4.9.3 *Let B be a k-cell in $\mathbb{S}^n$. Then $\mathbb{S}^n \setminus B$ is acyclic. In particular, B does not separate in $\mathbb{S}^n$*

Proof: Let n be fixed. We proceed by induction on k. First consider the case $k = 0$. Then B is a single point. The space $\mathbb{S}^n \setminus B$ is a single point if $n = 0$, while if $n > 0$, it is homeomorphic to $\mathbb{R}^n$. In either case, $\mathbb{S}^n \setminus B$ is acyclic.

We now suppose the theorem holds for a $(k-1)$-cell in $\mathbb{S}^n$. Let B be a k-cell in $\mathbb{S}^n$. Let $h : I^k \to B$ be a homeomorphism.

Step 1. Let B_0 and B_1 be the two k-cells

$$B_0 = h(I^{k-1} \times [0, \tfrac{1}{2}]) \quad \text{and} \quad B_1 = h(I^{k-1} \times [\tfrac{1}{2}, 1])$$

in $\mathbb{S}^n$. Let C be the $(k-1)$-cell $h(I^{k-1} \times (\frac{1}{2}))$ in $\mathbb{S}^n$. See Figure 4.9.1. We show that if α is a non-zero element of $\widetilde{H}_i(\mathbb{S}^n \setminus B)$, then its image is non-zero under at least one of the homomorphisms induced by the inclusion mappings

$$i : (\mathbb{S}^n \setminus B) \to (\mathbb{S}^n \setminus B_0) \quad \text{and} \quad j : (\mathbb{S}^n \setminus B) \to (\mathbb{S}^n \setminus B_1)\,.$$

To prove this fact, let $X = \mathbb{S}^n \setminus C$. By the induction hypothesis, X is acyclic. We write X as the union of the two subspaces

$$X_0 = \mathbb{S}^n \setminus B_0 \quad \text{and} \quad X_1 = \mathbb{S}^n \setminus B_1\,.$$

Since X_0 and X_1 are open in $\mathbb{S}^n$, they are open in X. Set

$$A = X_0 \cap X_1 = (\mathbb{S}^n \setminus B_0) \cap (\mathbb{S}^n \setminus B_1) = \mathbb{S}^n \setminus (B_0 \cup B_1) = \mathbb{S}^n \setminus B\,.$$

So we have an exact Mayer–Vietoris sequence

$$\widetilde{H}_{i+1}(X) \longrightarrow \widetilde{H}_i(A) \longrightarrow \widetilde{H}_i(X_0) \oplus \widetilde{H}_i(X_1) \longrightarrow \widetilde{H}_i(X)\,. \qquad (*)$$

Since X is acyclic, the map in the middle of $(*)$ is an isomorphism. By Theorem 4.6.2, it carries $\alpha \in \widetilde{H}_i(A)$ to $(i_*(\alpha), -j_*(\alpha))$. Therefore, at least one of the elements $i_*(\alpha)$ and $j_*(\alpha)$ is non-trivial.

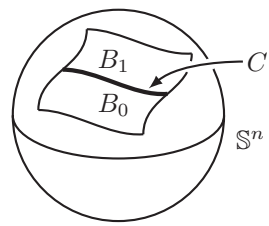


Figure 4.9.1

Step 2. We suppose there exists a non-zero element α in $\widetilde{H}_i(\mathbb{S}^n \setminus B)$ and derive a contradiction.

Let B_0 and B_1 be as in Step 1. Then the image of α in either $\widetilde{H}_i(\mathbb{S}^n \setminus B_0)$ or $\widetilde{H}_i(\mathbb{S}^n \setminus B_1)$ is non-zero. In the former case, we set $a_1 = 0$ and $b_1 = \frac{1}{2}$. Otherwise, we set $a_1 = \frac{1}{2}$ and $b_1 = 1$. Then, setting $D_1 = h(I^{k-1} \times [a_1, b_1])$, we know that the image of α in $\widetilde{H}_i(\mathbb{S}^n \setminus D_1)$ is non-zero.

Now D_1 is the k-cell that is the image under h of $I^{k-1} \times [a_1, b_1]$ with $b_1 - a_1 = 2^{-1}$. We repeat the process by writing D_1 as the union of the two k-cells, again denoted B_0 and B_1, given by

$$B_0 = h(I^{k-1} \times [a_1, b_1 - 2^{-2}]) \quad \text{and} \quad B_1 = h(I^{k-1} \times [a_1 + 2^{-2}, b_2])\,.$$

Arguing as in Step 1, we see that the image of α in either $\widetilde{H}_i(\mathbb{S}^n \setminus B_0)$ or $\widetilde{H}_i(\mathbb{S}^n \setminus B_1)$ is non-zero. In the former case, we set $a_2 = a_1$ and $b_2 = b_1 - 2^{-2}$ and otherwise we set $a_2 = a_1 + 2^{-2}$ and $b_2 = b_1$. Then, setting $D_2 = h(I^{k-1} \times [a_2, b_2])$, we know that the image of α in $\widetilde{H}_i(\mathbb{S}^n \setminus D_2)$ is non-zero.

Continuing similarly, we obtain a sequence of closed intervals $[a_1, b_1] \supset [a_2, b_2] \supset \cdots$, with $b_m - a_m = 2^{-m}$, and k-cells $D_m = h(I^{k-1} \times [a_m, b_m)]$ such that the image of α in $\widetilde{H}_i(\mathbb{S}^n \setminus D_m)$ is non-zero.

Let e be the unique point in $\cap_m [a_m, b_m]$, and set $E = h(I^{k-1} \times \{e\})$. The set E is a $(k-1)$-cell in $\mathbb{S}^n$ that equals the intersection of the nested sequence of k-cells $D_1 \supset D_2 \supset \cdots$. By the induction hypothesis, the group $\widetilde{H}_i(\mathbb{S}^n \setminus E)$ vanishes. In particular, the image of α in this group vanishes. By Theorem 4.3.5, there is a compact subset A of $\mathbb{S}^n \setminus E$ such that the image of α in $\widetilde{H}_i(A)$ vanishes. Since $\mathbb{S}^n \setminus E$ is the union of the open sets

$$\mathbb{S}^n \setminus D_1 \subset \mathbb{S}^n \setminus D_2 \subset \cdots\,,$$

the set A lies in one of them, say in $\mathbb{S}^n \setminus D_m$. But this means that the image of α in $\widetilde{H}_i(\mathbb{S}^n \setminus D_m)$ vanishes, contrary to the construction. □

Theorem 4.9.4 *Let $n > k \geq 0$. Let $h : \mathbb{S}^k \to \mathbb{S}^n$ be an imbedding. Then*

$$\widetilde{H}_i(\mathbb{S}^n \setminus h(\mathbb{S}^k)) = \begin{cases} \mathbb{Z} & \text{if } i = n-k-1, \\ 0 & \text{otherwise.} \end{cases}$$

Proof: Let n be fixed. We prove the theorem by induction on k. First consider the case $k = 0$. Then $h(\mathbb{S}^0)$ consists of two points p and q. Since $\mathbb{S}^n \setminus \{p, q\}$ is homeomorphic to $\mathbb{R}^n \setminus \mathbf{0}$, and $\mathbb{R}^n \setminus \mathbf{0}$ has the homotopy type of $\mathbb{S}^{n-1}$, we see that $\widetilde{H}_i(\mathbb{S}^n \setminus \{p, q\})$ is infinite cyclic for $i = n-1$ and vanishes otherwise.

Now suppose the theorem is true in dimension $k-1$. Let $h : \mathbb{S}^k \to \mathbb{S}^n$ be an imbedding. We construct a certain Mayer–Vietoris sequence. Let X_1 and X_2 be the following open subsets of $\mathbb{S}^n$:

$$X_1 = \mathbb{S}^n \setminus h(E^k_+) \quad \text{and} \quad X_2 = \mathbb{S}^n \setminus h(E^k_-)\,.$$

See Figure 4.9.2. Then let

$$\begin{aligned} X &= X_1 \cup X_2 = \mathbb{S}^n \setminus h(\mathbb{S}^{k-1})\,, \\ A &= X_1 \cap X_2 = \mathbb{S}^n \setminus h(\mathbb{S}^k)\,. \end{aligned}$$

Since X_1 and X_2 are open in X, we have a Mayer–Vietoris sequence

$$\widetilde{H}_{i+1}(X_1) \oplus \widetilde{H}_{i+1}(X_2) \longrightarrow \widetilde{H}_{i+1}(X) \longrightarrow \widetilde{H}_i(A) \longrightarrow \widetilde{H}_i(X_1) \oplus \widetilde{H}_i(X_2)\,.$$

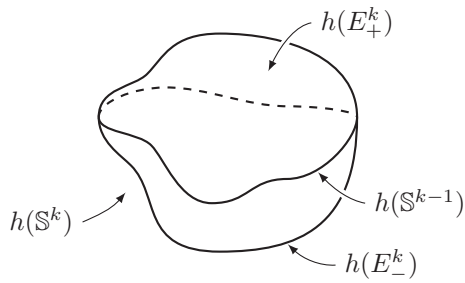


Figure 4.9.2

Both X_1 and X_2 are acyclic by the preceding theorem; therefore, the middle map is an isomorphism. That is,

$$\widetilde{H}_{i+1}(\mathbb{S}^n \setminus h(\mathbb{S}^{k-1})) \cong \widetilde{H}_i(\mathbb{S}^n \setminus h(\mathbb{S}^k))$$

for all i. By the induction hypothesis, the group on the left is infinite cyclic for $i+1 = n-(k-1)-1$ and vanishes otherwise. Hence the group on the right is infinite cyclic for $i = n-k-1$ and vanishes otherwise. □

Theorem 4.9.5 (The generalized Jordan curve theorem) *Let $n > 0$. Let C be a subset of $\mathbb{S}^n$ homeomorphic to the $n-1$ sphere. Then $\mathbb{S}^n \setminus C$ has precisely two components, of which C is the common topological boundary.*

Proof: Applying the preceding theorem to the case $k = n-1$, we see that $\widetilde{H}_0(\mathbb{S}^n \setminus C) \cong \mathbb{Z}$. Thus $\mathbb{S}^n \setminus C$ has precisely two path components [which are the same as its components, as noted just after Definition 4.9.1]. Let W_1 and W_2 be these path components. Because $\mathbb{S}^n$ is locally path connected, the W_i are open in $\mathbb{S}^n$. Thus the topological boundary of W_i is the set $\overline{W}_i \setminus W_i$. We need to show that

$$\overline{W}_1 \setminus W_1 = C = \overline{W}_2 \setminus W_2 \,.$$

It suffices to prove the first of these equations.

Since W_2 is open, no point of W_2 is a limit point of W_1. Therefore we have $\overline{W}_1 \subset \mathbb{S}^n \setminus W_2 = W_1 \cap C$, so $\overline{W}_1 \setminus W_1 \subset C$. We still need to show containment in the other direction.

Given $x \in C$ and a neighborhood U of x, it will suffice to show that U intersects the closed set $\overline{W}_1 \setminus W_1$. Since C is homeomorphic to $\mathbb{S}^{n-1}$, we can write C as the union of two $(n-1)$-cells, C_1 and C_2, such that C_1 is small enough to lie in U. Figure 4.9.3 illustrates the case $n = 2$.

Now C_2 does not separate $\mathbb{S}^n$, so we can choose a path α in $\mathbb{S}^n \setminus C_2$ joining a point p of W_1 to a point q of W_2. Now α must contain a point of $\overline{W}_1 \setminus W_1$, since otherwise α would lie in the union of the disjoint sets W_1 and $\mathbb{S}^n \setminus W_1$

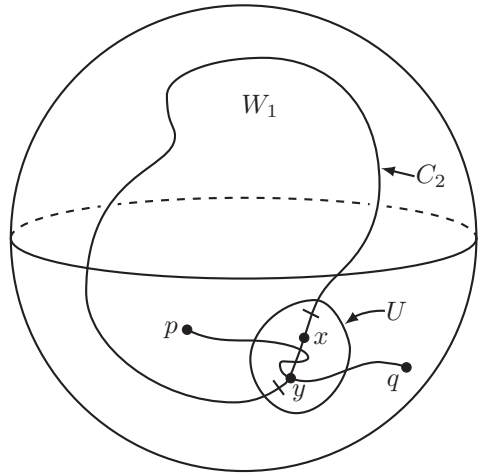
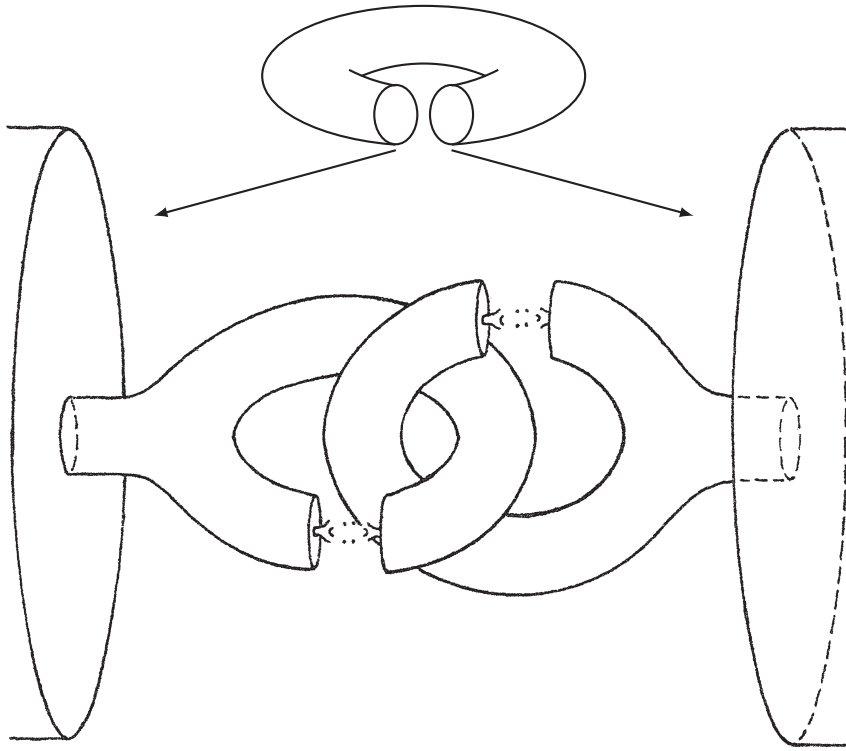


Figure 4.9.3

and contain a point of each of them, contrary to the fact that the image set of α is connected. Let y be a point of $\overline{W}_1 \setminus W_1$ lying on the path α. Then y lies in C. Since y does not lie in C_2, it must lie in C_1 and hence in U. Thus U intersects $\overline{W}_1 \setminus W_1$ in the point y, as desired. □

The equator in $\mathbb{S}^n$ is a homeomorphic copy of $\mathbb{S}^{n-1}$. The two connected components of its compliment are homeomorphic to open balls and their closures are n-cells. So under the hypotheses of the preceding theorem, one might conjecture that if W_1 and W_2 are the components of $\mathbb{S}^n \setminus C$, where C is an $n-1$ sphere, then the sets $\overline{W}_1$ and $\overline{W}_2$ would be n-cells. This is true for $n = 2$, and is known as *Schoenflies Theorem* [a proof may be found in [New51]], but for $n > 2$ additional hypotheses are needed. The result for $n \geq 3$ is known as the *Generalized Schoenflies Theorem*. It was proved independently by M. Brown and B. Mazur. The relevant papers are [Maz59], [Maz61], [Bro60], and [Mor60].

The difficulty in dimensions 3 and higher was discovered by J. W. Alexander in 1924 [in [Ale24]]. He constructed a surface that later came to be known as the *Alexander horned sphere*. The construction starts by removing a section from a torus as shown at the top of Figure 4.9.4. The two holes are patched with discs, so that the surface is homeomorphic to a sphere. Alexander then took two smaller partial tori and attached one to each disc by putting a tube from a small hole in the disc to a hole in the small partial torus that is being added. This process was then repeated infinitely often. The lower part of Figure 4.9.4 showing the smaller partial tori is from Alexander's original paper. [The labels in Alexander's figure have been masked to avoid clutter.] The



Alexander Horned Sphere
Figure 4.9.4

surface resulting at the end of Alexander's construction is homeomorphic to a 2-sphere. On the other hand, as it sits in space, the horned sphere behaves very differently from the familiar round sphere. If you run a loop through one of the apparent holes in one of the partial tori, unlike with an ordinary sphere, that loop cannot be contracted to a point without at some time intersecting the horned sphere.

Classically, the Jordan curve theorem[2] is usually stated for spheres imbedded in $\mathbb{R}^n$ rather than $\mathbb{S}^n$. We prove that version of the theorem now.

Corollary 4.9.6 *Let $n \geq 2$. Let C be a subset of $\mathbb{R}^n$ homeomorphic to $\mathbb{S}^{n-1}$. Then $\mathbb{R}^n \setminus C$ has precisely two components, of which C is the common topological boundary.*

[2]For many decades it has been said that Jordan did not successfully prove the theorem that bears his name. Jordan's original presentation (in French, of course) is in [Jor14], pages 587–594. The reader can study it, and form his or her own opinion regarding the validity of the argument.

Proof: Step 1. First, we show for $n \geq 2$ that if U is a connected open set in $\mathbb{S}^n$, then no point of U separates U.

Let p be a point of U. We will suppose that $U \setminus p$ is not connected, and derive a contradiction. Let $B(p, \epsilon)$ be the open ball centered at p with radius $\epsilon > 0$ chosen small enough that $B(p, \epsilon) \subset U$. Since $B(p, \epsilon) \setminus p$ is homeomorphic to $\mathbb{R}^n \setminus \mathbf{0}$, we have that $B(p, \epsilon) \setminus p$ is connected. Hence $B(p, \epsilon) \setminus p$ lies entirely in one component D of $U \setminus p$. Let E be the union of the other components of $U \setminus p$. There are other components of $U \setminus p$, so E is not empty. Now p is not a limit point of E, since $B(p, \epsilon)$ is a neighborhood of p disjoint from E. Hence the two sets $D \cup \{p\}$ and E are nonempty disjoint open sets whose union is U, contrary to the hypothesis that U is connected.

Step 2. We prove the theorem. Without loss of generality, we can replace $\mathbb{R}^n$ in the statement of the theorem by $\mathbb{S}^n \setminus p$, where p is the north pole of $\mathbb{S}^n$. We then have C a homeomorphic image of $\mathbb{S}^{n-1}$ with $C \subset \mathbb{S}^n \setminus p$. Now, by the preceding theorem, $\mathbb{S}^n \setminus C$ has two components W_1 and W_2. Renaming the components if necessary, we suppose that $p \in W_1$. By the result of Step 1, $W_1 \setminus p$ is connected. Then $W_1 \setminus p$ and W_2 are the components of $(\mathbb{S}^n \setminus p) \setminus C$ and C equals the boundary of W_2 and W_1. □

Theorem 4.9.7 (Invariance of domain) *Let U be an open set in $\mathbb{R}^n$ and let $f : U \to \mathbb{R}^n$ be continuous and one-to-one. Then $f(U)$ is open in $\mathbb{R}^n$ and $f : U \to f(U)$ is a homeomorphism.*

Proof: Without loss of generality, we can replace $\mathbb{R}^n$ by $\mathbb{S}^n$.

Step 1. Given a point y of $f(U)$, we show that $f(U)$ contains a neighborhood of y. This will prove that $f(U)$ is open in $\mathbb{S}^n$.

Let x be the point of U such that $f(x) = y$. Choose an open ball $B(x, \epsilon)$ centered at x with radius $\epsilon > 0$ chosen small enough that the closure of the ball, $\overline{B}(x, \epsilon)$, lies in U. Set $S(x, \epsilon) = \overline{B}(x, \epsilon) \setminus B(x, \epsilon)$. See Figure 4.9.5. Now the set $f(S(x, \epsilon))$ is homeomorphic to $\mathbb{S}^{n-1}$. By Theorem 4.9.5, $f(S(x, \epsilon))$ separates $\mathbb{S}^n$ into two components W_1 and W_2, each open in $\mathbb{S}^n$. The set $f(B(x, \epsilon))$ is connected and disjoint from $f(S(x, \epsilon))$. Therefore, it lies in either W_1 or W_2. Renaming if necessary, we may assume it lies in W_1. Now it must equal all of W_1, for otherwise the set

$$\mathbb{S}^n \setminus [f(B(x, \epsilon)) \cup f(S(x, \epsilon))] = \mathbb{S}^n \setminus f(\overline{B}(x, \epsilon))$$

would consist of all of W_2 along with points of W_1 as well, contradicting Theorem 4.9.3 which tells us that the cell $f(\overline{B}(x, \epsilon))$ does not separate $\mathbb{S}^n$. Therefore, $f(B(x, \epsilon)) = W_1$. We conclude that $f(U)$ contains the neighborhood W_1 of y, as desired.

Step 2. In view of Step 1, f carries any set open in U to a set that is open in $\mathbb{S}^n$ and hence open in $f(U)$. Thus f is a homeomorphism of U with $f(U)$. □

The theorem on invariance of domain is much easier to prove if one assumes that f is continuously differentiable with non-singular Jacobian. In this

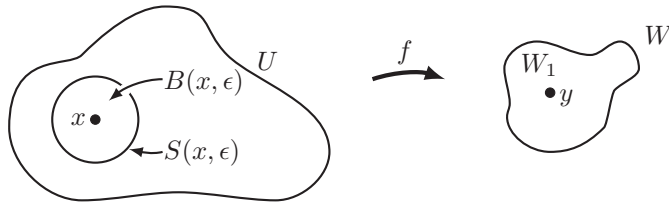


Figure 4.9.5

case, the theorem follows from the inverse function theorem of analysis. The true profundity of invariance of domain is that it has nothing to do with differentiability or Jacobians, but depends only on continuity and injectivity of the map f.

Similarly, the Jordan curve theorem is much easier to prove in the case that the imbedding $f : \mathbb{S}^{n-1} \to \mathbb{R}^n$ is a simplicial map of some complex whose space is homeomorphic to $\mathbb{S}^{n-1}$, or is a differentiable map with Jacobian of maximal rank. The true difficulties appear only when one assumes no more than continuity and injectivity.

EXERCISES

1. Let M be an m-manifold with boundary, as defined in Section 4.8. Use invariance of domain to prove the following.

(a) Show that if a point x of M maps to a point of $\mathbb{R}^{m-1} \times 0$ under one coordinate patch about x, then it does so under every such coordinate patch.

(b) Let U be open in $\mathbb{R}^m$ and let V be open in $\mathbb{R}^n$. Show that if U and V are homeomorphic, then $m = n$.

(c) Show that the number m is uniquely determined by M.

2. **(a)** Let Y be the topologist's sine curve [see Definition 4.2.11]. Show that if $h : Y \to \mathbb{S}^n$ is an imbedding, then $\mathbb{S}^n \setminus h(Y)$ is acyclic.

(b) Let X be the closed topologist's sine curve [see Definition 4.6.5]. Show that if $h : X \to \mathbb{S}^n$ is an imbedding, then $\widetilde{H}_i(\mathbb{S}^n \setminus h(X))$ is infinite cyclic if $i = n - 2$, and vanishes otherwise.

(c) With X as in **(b)**, show that if $h : X \to \mathbb{S}^2$ is an imbedding, then $\mathbb{S}^2 \setminus h(X)$ has precisely two components, of which $h(X)$ is the common topological boundary.

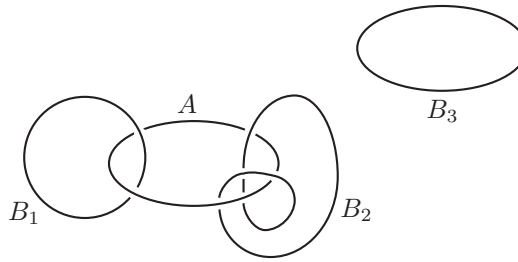


Figure 4.9.6

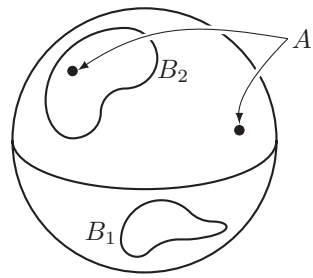


Figure 4.9.7

3. **(a)** Consider $\mathbb{S}^3$ as $\mathbb{R}^3$ with a point at infinity adjoined. Let A, B_1, B_2, and B_3 be the simple closed curves in $\mathbb{S}^3$ pictured in Figure 4.9.6. The map

$$H_1(B_i) \longrightarrow H_1(\mathbb{S}^3 \setminus A)$$

induced by inclusion is a homomorphism of infinite cyclic groups, so it equals multiplication by d_i, where d_i is well-defined up to sign. The integer d_i measures how many times B_i *links* A. What is it in each case? Similarly, determine the integer corresponding to the homomorphism

$$H_1(A) \longrightarrow H_1(\mathbb{S}^3 \setminus B_i)$$

induced by inclusion. Can you formulate a conjecture?

(b) Let A consist of two points of $\mathbb{S}^2$, and let B_1 and B_2 be two simple closed curves in $\mathbb{S}^2$, as pictured in Figure 4.9.7. What are the homomorphisms

$$\widetilde{H}_0(A) \longrightarrow \widetilde{H}_0(\mathbb{S}^2 \setminus B_i) \quad \text{and} \quad H_1(B_i) \longrightarrow H_1(\mathbb{S}^2 \setminus A)$$

induced by inclusion? Formulate a conjecture.

(c) Formulate a conjecture concerning disjoint imbeddings of $\mathbb{S}^p$ and $\mathbb{S}^q$ in $\mathbb{S}^{p+q+1}$.

We shall return to this conjecture later on, after we prove the Alexander duality theorem. [See the exercises of Section 8.12.]

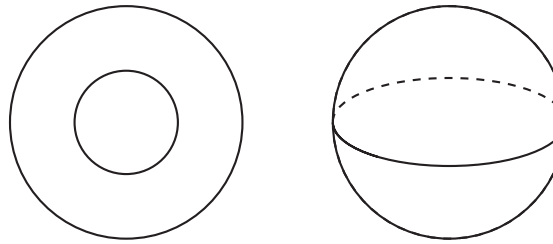


Figure 4.10.1: Different types of holes.

4. A version of the Jordan curve theorem in dimension 3 is true if one assumes some smoothness of the boundary surface. This is a famous result of Mazur and Brown. Discuss the possibilities here.

4.10 The Fundamental Group

In this section we will give a brief introduction to homotopy theory, and look particularly at the fundamental group of a space. One of the motivations for homotopy theory is identifying and classifying the "holes" in a topological space. The annulus has a hole in its center, and so does the unit sphere. See Figure 4.10.1. But one has a sense that these two holes are different in nature. They are, perhaps, holes of different dimensions. Homotopy theory helps us to sort out these ideas.

The basis for homotopy theory is to consider the set of all loops in a given topological space X. Here, a *loop* means a continuous function $\gamma : [0, 1] \to X$ such that $\gamma(0) = \gamma(1)$. See Figure 4.10.2

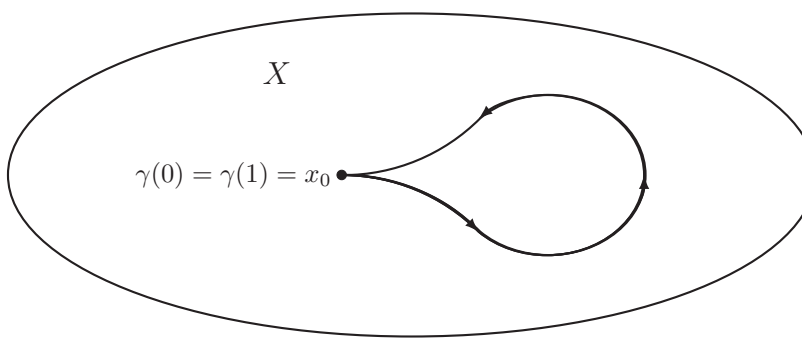


Figure 4.10.2: A loop in X.

Definition 4.10.1 Let X be a space. Fix a point $x_0 \in X$. This fixed point will serve as what is called the *base point*. Let $\gamma_1 : [0,1] \to X$ and $\gamma_2 : [0,1] \to X$ be two loops with

$$\gamma_1(0) = \gamma_1(1) = \gamma_2(0) = \gamma_2(1) = x_0 \,.$$

We say that γ_1 and γ_2 are *base-point homotopic*, written $\gamma_1 \simeq \gamma_2$, when there is a continuous function, called the *homotopy function*,

$$\Gamma : [0,1] \times [0,1] \to X$$

with these properties:

(1) $\Gamma(t,0) = \gamma_1(t)$, $0 \le t \le 1$;
(2) $\Gamma(t,1) = \gamma_2(t)$, $0 \le t \le 1$;
(3) $\Gamma(0,u) = x_0$, $0 \le u \le 1$;
(4) $\Gamma(1,u) = x_0$, $0 \le u \le 1$.

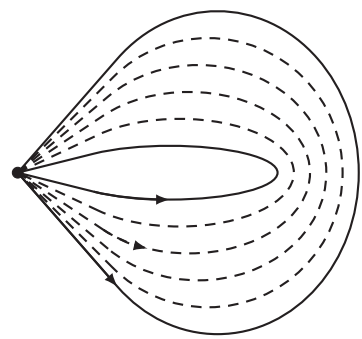


Figure 4.10.3: A pair of homotopic loops.

Figure 4.10.3 illustrates a pair of homotopic loops, and some of the intermediate loops the deformation passes through. Figure 4.10.4 shows a pair of loops in an annular region that cannot be homotopic. [Of course, "looking like they cannot be homotopic" is not the same as proving that they are not homotopic.]

Conditions **(1)** and **(2)** of the definition tell us that γ_1 and γ_2 are homotopic functions, while conditions **(3)** and **(4)** demand that the base point x_0 be maintained throughout the homotopy. Thus Definition 4.10.1 requires more of the homotopy function Γ than is required by the earlier Definition 2.7.1. Nonetheless, we shall use the same notation $\gamma_1 \simeq \gamma_2$, and rely on the context to make clear which definition applies. Further, we shall usually say simply that γ_1 is homotopic to γ_2 without mentioning that a base point is involved.

The base point itself often only plays a tacit role in our discussions and calculations. There is also a homotopy theory that is *not* founded on a base point. But the theory that we present here is the most common.

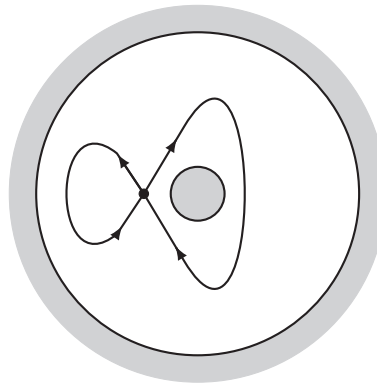


Figure 4.10.4: Two loops in an annular region that cannot be homotopic.

EXAMPLE 4.10.2 Let X be a subset of euclidean space, and let γ_1 and γ_2 be loops in X. Suppose the line segment from $\gamma_1(t)$ to $\gamma_2(t)$ is contained in X, for each $0 \leq t \leq 1$, then $\gamma_1 \simeq \gamma_2$. You have probably already realized that the homotopy function is

$$\Gamma(t,u) = (1-u)\,\gamma_1(t) + u\,\gamma_2(t)\,.$$

Lemma 4.10.3 *The relation $\simeq$ of Definition 4.10.1 defines an equivalence relation on the set of all loops with base point x_0.*

Proof: Reflexivity and symmetry are easy, so we concentrate on transitivity.

Suppose loop γ_1 is homotopic to γ_2 with homotopy map Γ_{12} and also that γ_2 is homotopic to γ_3 with homotopy map Γ_{23}. Now define a new homotopy map Γ_{13} by

$$\Gamma_{13}(t,u) = \begin{cases} \Gamma_{12}(t,2u) & \text{if} \quad 0 \leq u \leq 1/2\,, \\ \Gamma_{23}(t,2u-1) & \text{if} \quad 1/2 \leq u \leq 1\,. \end{cases}$$

When $u = 1/2$, we have $\Gamma_{12}(t,2u) = \gamma_2(t) = \Gamma_{23}(t,2u-1)$, so Γ_{13} is well-defined and continuous. Checking the values of $\Gamma_{13}(t,u)$ when $u = 0$ and $u = 1$, we see that Γ_{13} is a homotopy from γ_1 to γ_3. That confirms the transitivity of the relation. □

Definition 4.10.4 We call the equivalence classes of loops under the relation $\simeq$ the *homotopy classes* for the topological space X. The homotopy class of

γ will be written $[\gamma]$. The collection of homotopy classes is the *first homotopy space*.

Suppose that $[\gamma]$ and $[\eta]$ are two homotopy classes. Then we define the *product* of $[\gamma]$ and $[\eta]$, written $[\gamma] \cdot [\eta]$, to be the equivalence class of the loop $\gamma \cdot \eta$ [also called the *product*] given by

$$\gamma \cdot \eta(t) = \begin{cases} \gamma(2t) & \text{if} \quad 0 \leq t \leq 1/2, \\ \eta(2t-1) & \text{if} \quad 1/2 < t \leq 1. \end{cases}$$

See Figure 4.10.5. We have $[\gamma] \cdot [\eta] = [\gamma \cdot \eta]$. Below we will show that the the first homotopy space with this operation forms a group; we call that group the *first homotopy group* of X, or the *fundamental group* of X. We denote the fundamental group of X with base point x_0 by $\pi_1(X, x_0)$ or simply $\pi_1(X)$. We consider $\pi_1(X)$ to be a multiplicative group, so the identity is denoted by 1.

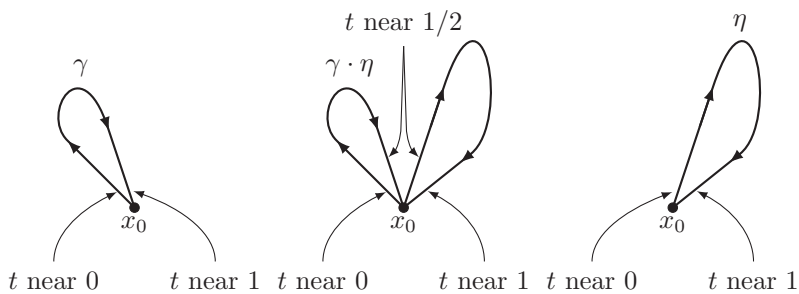


Figure 4.10.5: The product of γ and η.

Of course we need to check that the multiplication defined above is well-defined, and that we really have a group. That is the content of the next lemma.

Lemma 4.10.5 *Let γ_0 be the constant loop $\gamma_0(t) = x_0$ for $0 \leq t \leq 1$.*

(1) *The operation $[\gamma] \cdot [\eta]$ is well-defined.*

(2) *The operation of Definition 4.10.4 is associative.*

(3) *If γ is a loop, then $[\gamma] \cdot [\gamma_0] = [\gamma_0] \cdot [\gamma] = [\gamma]$*

(4) *If γ is a loop and η is defined by $\eta(t) = \gamma(1-t)$ for $0 \leq t \leq 1$, then $[\gamma] \cdot [\eta] = [\eta] \cdot [\gamma] = [\gamma_0]$.*

Proof: Part **(1)** is easily proved as follows: If γ' is another representative from the equivalence class $[\gamma]$ (with homotopy Γ) and η' is another representative from the equivalence class $[\eta]$ (with homotopy Φ) then

$$\Pi(t,u) = \begin{cases} \Gamma(2t,u) & \text{if} \quad 0 \le t \le 1/2, 0 \le u \le 1\,, \\ \Phi(2t-1,u) & \text{if} \quad 1/2 \le t \le 1,\ 0 \le u \le 1/2\,. \end{cases}$$

is a homotopy from $\gamma \cdot \eta$ to $\gamma' \cdot \eta'$. See Figure 4.10.6.

Parts **(2)**, **(3)**, and **(4)** are also easy, and we leave them as exercises. □

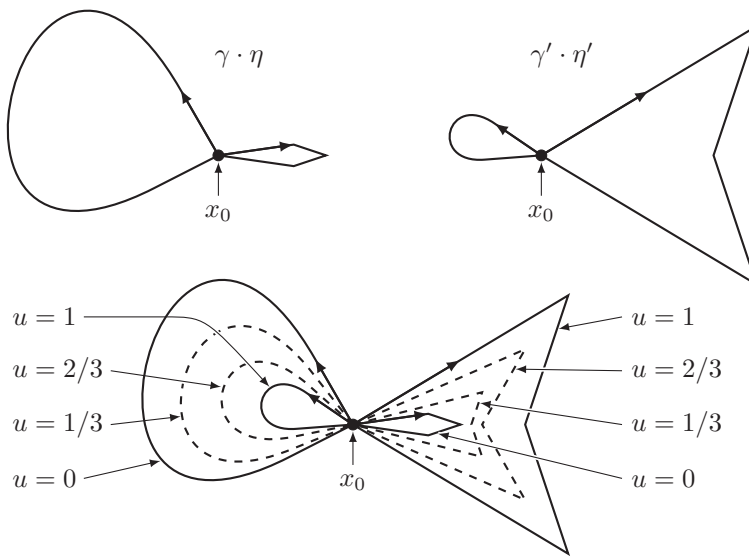


Figure 4.10.6: Homotopy from $\gamma \cdot \eta$ to $\gamma' \cdot \eta'$.

The lemma shows that we do in fact have a well-defined group operation on the set of homotopy classes. There are higher homotopy groups $\pi_k(X)$, but we shall not discuss them here. The important fact about π_1 is that it is *functorial*. This notion is encapsulated in the following theorem.

Theorem 4.10.6 *Let X and Y be topological spaces. Let x_0 be the homotopy base point in X and y_0 be the homotopy base point in Y. Assume that $f : X \to Y$ is a continuous map such that $f(x_0) = y_0$. Then the induced function*

on homotopy given by

$$\begin{array}{rcl} f_* : \pi_1(X) & \to & \pi_1(Y) \\ [\gamma] & \longmapsto & [f \circ \gamma] \end{array}$$

is in fact a group homomorphism. That is to say,

$$f_*([\gamma] \cdot [\mu]) = f_*([\gamma]) \cdot f_*([\mu]) \,.$$

Proof: The mapping is well defined, because if Γ is a homotopy between γ_1 and γ_2, then $f \circ \Gamma$ is a homotopy between $f \circ \gamma_1$ and $f \circ \gamma_2$.

For the canonical homomorphism property, note that

$$f([\gamma] \cdot [\eta]) = \left[\begin{cases} f(\gamma(2t)) & \text{if} \quad 0 \leq t \leq 1/2 \\ f(\eta(2t-1)) & \text{if} \quad 1/2 < t \leq 1 \end{cases} \right] = [f \circ \gamma] \cdot [f \circ \eta] \,. \qquad \square$$

In fact it is not difficult to see that if f is injective, then so is f_* and if f is surjective, then so is f_*. So we have the following corollary.

Corollary 4.10.7 *The fundamental group is a topological invariant of a space.*

Since the fundamental group is a topological invariant, if two spaces are homeomorphic, then their fundamental groups are isomorphic. Contrapositively, this says that if two spaces have different fundamental groups, then they cannot be homeomorphic.

You might recall the elaborate build-up required before we defined the simplicial homology of a space, and how we labored to show that the simplicial homology is a topological invariant. Comparatively speaking, studying homotopy has been quite easy: We defined the fundamental group and proved its topological invariance in just a few pages. Homotopy theory becomes more challenging when we seek to compute a fundamental group.

Proposition 4.10.8 *The fundamental group of the circle is isomorphic to the integers, that is,* $\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$.

Proof: It will be convenient to identify $\mathbb{S}^1$ with

$$\{z \in \mathbb{C} : |z| = 1\} \,.$$

We shall use 1 as our base point. We denote the unit interval $[0, 1]$ by I, so a loop then is a function $\sigma : I \to \mathbb{S}^1$ with $\sigma(0) = \sigma(1) = 1$ and $|\sigma(t)| = 1$ for $t \in I$. It will be convenient to use the function $\phi : \mathbb{R} \to \mathbb{S}^1$ given by

$$\phi(x) = e^{2\pi i x} \,.$$

For each $n \in \mathbb{Z}$, define the loop ρ_n by setting

$$\rho_n(t) = \phi(nt) = e^{2\pi i n t} .$$

We see that if n is non-zero, then ρ_n wraps around the circle n times, in the counterclockwise direction if $n > 0$ and in the clockwise direction if $n < 0$. Of course, ρ_0 stays fixed at the base point.

We claim that every loop σ is homotopic to one of the loops ρ_n. The idea we would like to pursue is creating a function $\sigma' : I \to \mathbb{R}$ such that

$$\sigma = \phi \circ \sigma' .$$

Since $\sigma(1) = 1$, we must have $\sigma'(1) = n$ for some $n \in \mathbb{Z}$. We see then that

$$\Gamma(t, u) = \phi[(1 - u)\sigma'(t) + unt]$$

is a homotopy function from σ to ρ_n.

The function σ' is called a *lifting* of σ. It makes the following diagram commutative, and the diagram itself is suggestive of the name:

$$\begin{array}{ccc} & & \mathbb{R} \\ & \overset{\sigma'}{\nearrow} & \big\downarrow \phi \\ I & \xrightarrow{\sigma} & \mathbb{S}^1 \end{array}$$

We have seen how useful a lifting σ' would be, but we still need to construct one. The equation we are after is

$$\sigma(t) = \exp[2\pi i \sigma'(t)] ,$$

so if we proceed naively, $\sigma'(t) = (2\pi i)^{-1} \log[\sigma(t)]$ seems to be what we want. Of course, the logarithm causes trouble, since we expect most loops to go all the way around the circle. We are going to need a trick to be able to restrict ourselves to a single-valued branch.

The "trick" is to observe that ϕ maps the interval $(-\frac{1}{2}, \frac{1}{2})$ homeomorphically onto $\mathbb{S}^1 \setminus \{-1\}$. Thus we have a well-defined function $\phi^{-1} : \mathbb{S}^1 \setminus \{-1\} \to (-\frac{1}{2}, \frac{1}{2})$. Next we observe that, since ϕ is a continuous function on the compact space I, ϕ is uniformly continuous. Thus we can choose $\epsilon > 0$ such that if $|t - t'| < \epsilon$ holds, then $|\sigma(t) - \sigma(t')| < 1$. That in turn tells us that $\sigma(t)/\sigma(t') \neq -1$, so $\phi^{-1}[\sigma(t)/\sigma(t')]$ is defined.

Now choose K so that $1/K < \epsilon$ holds, and set $p_i = i/K$, for $i = 0, 1, \ldots, K$. We define σ' by setting

$$\sigma'(t) = \sum_{i=0}^{K-1} \phi^{-1} \left(\frac{\sigma(tp_{i+1})}{\sigma(tp_i)} \right) .$$

We can make this definition because $|tp_{i+1} - tp_i| \leq |p_{i+1} - p_i| < \epsilon$ holds. Using the fact that the exponential function ϕ converts sums into products, we see that

$$\phi(\sigma'(t)) = \prod_{i=0}^{K-1} \left(\frac{\sigma(tp_{i+1})}{\sigma(tp_i)} \right) = \frac{\sigma(tp_K)}{\sigma(tp_0)} = \sigma(t)\,.$$

We can also lift a homotopy function $F : I \times I \to \mathbb{S}^1$ with only a small change in the argument: We need to choose K so that $1/K < \epsilon/2$, so that when we define

$$F'(t,u) = \sum_{i=0}^{K-1} \phi^{-1} \left(\frac{F(tp_{i+1}, up_{i+1})}{F(tp_i, up_i)} \right)\,,$$

we will have

$$\begin{aligned} |(tp_{i+1}, up_{i+1}) - (tp_i, up_i)| &\leq |(0, u(p_{i+1} - p_i))| + |(t(p_{i+1} - p_i), 0)| \\ &\leq 2|p_{i+1} - p_i| < \epsilon\,. \end{aligned}$$

If m and n are integers and F is a homotopy function between ρ_m and ρ_n, then the lift F' of F must satisfy $F'(1,0) = m$ and $F'(1,1) = n$. Since $F'(1,u)$ is an integer-valued continuous function, we conclude that $m = n$. Thus the homotopy classes $[\rho_n]$ are distinct.

We leave it as an exercise to complete the proof by verifying the equation $[\rho_m] \cdot [\rho_n] = [\rho_{m+n}]$. □

In contrast to the last theorem, sometimes it is extremely easy to compute a fundamental group.

EXAMPLE 4.10.9 Consider a space X that is homeomorphic to a convex subset of a euclidean space; we shall show that $\pi_1(X)$ is trivial. Since the fundamental group is a topological invariant we may assume the X *is* a convex subset of euclidean space. Pick a base point x_0. Consider any loop γ_1, with base point x_0, and let γ_0 be the constant loop at x_0. The homotopy map of Example 4.10.2 applies to show that $\gamma_0 \simeq \gamma_1$. We know that the homotopy class of γ_0 is the identity in $\pi_1(X)$, so $\pi_1(X) = 1$. [Perhaps we should write $\{1\}$ rather than 1 for the trivial multiplicative group, but our notation should cause no confusion.]

When a topological space X satisfies $\pi_1(X) = 1$, then we say that X is *simply connected*.

EXAMPLE 4.10.10 For $n \geq 2$, we have $\pi_1(\mathbb{S}^n) = 1$. We consider the case $n = 2$ here. To see this, let $\gamma : [0,1] \to \mathbb{S}^2$ be any loop.

We would like to assert that the image of γ lies in a subset U of the sphere that is homeomorphic to a disc in the plane. Unfortunately, that may not be true. Your intuitive idea of a loop may not include some of the really surprising

things that can happen. In 1890, Guiseppe Peano constructed a space-filling curve.[3] So it is possible that the image of γ as given might be *all* of $\mathbb{S}^2$.

The good news is that, no matter how displeasing the behavior of γ as first given, we are free to modify it via a homotopy to ensure that it does lie in a set homeomorphic to a disc in the plane. This can be done. We leave the details for the exercises.

Relation to Homology

Recall from Section 4.2 that a singular p-simplex in a space X is a continuous map

$$T : \Delta_p \to X\,,$$

where Δ_p is the standard p-simplex in $\mathbb{R}^\infty$. The vertices of Δ_p are

$$\epsilon_0 = (0, 0, \ldots, 0, \ldots) \quad \text{and} \quad \epsilon_1, \epsilon_2, \ldots, \epsilon_p\,,$$

where ϵ_i has 1 as its ith coordinate and all other coordinates are 0.

In this subsection, we will be dealing with singular p-simplices of dimension not exceeding 2. For this purpose, the generality above is unnecessary and inconvenient. Accordingly, we let $\Delta_2 \subset \mathbb{R}^2$ be a 2-simplex with vertices $(0,0)$, $(1,0)$, and $(1,1)$, we let $\Delta_1 = [0,1] \subset \mathbb{R}$, and we let $\Delta_0 = 0$. As before, a singular p-simplex is a continuous map $T_p : \Delta_p \to X$, but now Δ_p is a triangle, interval, or singleton. We define the boundary operator ∂ on a singular p-simplex T as follows:

$$\partial T = \begin{cases} 0, & \text{if } T \text{ is a 0-simplex,} \\ T(1) - T(0), & \text{if } T \text{ is a 1-simplex,} \\ S_0 - S_1 + S_2, & \text{if } T \text{ is a 2-simplex,} \end{cases}$$

where

$$\begin{aligned} S_0(t) &= T(1-t, t), \\ S_1(t) &= T(0, t), \\ S_2(t) &= T(t, 0). \end{aligned}$$

You should check that the above definition is in accord with the definition of the boundary operator given in Section 4.2.

We note here that a 1-simplex that has the constant value x is a cycle and it equals the boundary of the 2-simplex that has the constant value x .

The notion of a homotopy can easily be applied to singular 1-simplices, indeed it can be applied to any path in X.

[3] The original paper is [Pea90]. Hilbert provided some helpful figures in [Hil91]. See [Mun00], Section 44, for a description in English.

Definition 4.10.11 Suppose that $a < b \in \mathbb{R}$. Let $p_1, p_2 : [a,b] \to X$ be paths in X with $p_1(a) = p_2(a)$ and $p_1(b) = p_2(b)$. We say that p_1 and p_2 are *homotopic with fixed endpoints*, written

$$p_1 \simeq p_2 \quad \text{rel } \{a,b\}\,,$$

if there is a homotopy function

$$F : [a,b] \times [0,1] \to X$$

with these properties:

(1) $F(t,0) = p_1(t)$, $a \le t \le b$;

(2) $F(t,1) = p_2(t)$, $a \le t \le b$;

(3) $F(a,u) = p_1(a) = p_2(a)$, $0 \le u \le 1$;

(4) $F(b,u) = p_1(b) = p_2(b)$, $0 \le u \le 1$.

Lemma 4.10.12 *Let $x, y \in X$.*

(1) *If p is a singular 1-simplex in X with $\partial p = y - x$ and p^{-1} is given by $p^{-1}(t) = p(1-t)$, then $p + p^{-1}$ bounds.*

(2) *Suppose that p_1 and p_2 are singular 1-simplices in X with $\partial p_i = y - x$. If p_1 and p_2 are homotopic with fixed endpoints, then p_1 and p_2 are homologous.*

Proof: (1) Define the singular 2-simplex T by setting $T(t,u) = p(1-t)$ for $(t,u) \in \Delta_2$. We compute $\partial T = S_0 - S_1 + S_2$, where

$$S_0(t) = T(1-t,t) = p(t)\,, \quad S_1(t) = T(0,t) \equiv y\,, \quad S_2(t) = T(t,0) = p(1-t)\,.$$

Since a constant 1-simplex y is the boundary of the constant 2-simplex, we see that $p + p^{-1}$ bounds.

(2) Suppose that F is a homotopy from p_1 to p_2 as in Definition 4.10.11. To show that p_1 is homologous to p_2, we shall define two singular 2-simplices, T and T', such that $\partial(T + T') = p_2 - p_1 + S$, where S is a boundary.

Setting

$$T(t,u) = F(t,u)\,,$$

for $(t,u) \in \Delta_2$, we compute $\partial T = S_0 - S_1 + S_2$, where

$$S_0(t) = F(1-t,t)\,, \quad S_1(t) = F(0,t) \equiv x\,, \quad S_2(t) = F(t,0) = p_1(t)\,.$$

Setting

$$T'(t,u) = F(t+u, 1-t)$$

for $(t,u) \in \Delta_2$, we compute $\partial T' = S'_0 - S'_1 + S'_2$, where

$$\begin{aligned} S'_0(t) &= T'(t, 1-t) \equiv y\,, \\ S'_1(t) &= T'(0,t) = F(t,1) = p_2(t)\,, \\ S'_2(t) &= T'(t,0) = F(t,1-t)\,. \end{aligned}$$

We have $\partial(T + T') = p_1 - p_2 + y - x + p + p^{-1}$, where $p(t) = F(1-t,t)$ for $t \in [0,1]$. □

Definition 4.10.13

(1) A loop is a map $\gamma : [0,1] \to X$, so it can be considered a singular 1-simplex in X. Treating γ as a 1-simplex, we see that $\partial\gamma = x_0 - x_0 = 0$. So γ is a singular 1-cycle, and as such it determines a homology class. We will denote the homology class of γ by $\{\gamma\}$.

(2) The map

$$\chi : \pi_1(X, x_0) \to H_1(X)$$

is defined by $\chi([\gamma]) = \{\gamma\}$.

The lemma tells us that χ is a well-defined mapping, so after giving one more definition, we will be prepared to state our key result relating homology and homotopy.

Definition 4.10.14 Let G be a multiplicative group. For $a, b \in G$, the element $a^{-1}b^{-1}ab$ is called the *commutator* of a and b. The subgroup generated by the commutators of G is called the *commutator subgroup* of G.

Theorem 4.10.15 *Let X be a topological space and $x_0 \in X$ a base point for homotopy.*

(1) *The map $\chi : \pi_1(X, x_0) \to H_1(X)$ is a homomorphism of groups.*

(2) *If X is path-connected, then χ is surjective.*

(3) *If X is path-connected, then the kernel of the homomorphism is the commutator subgroup of $\pi_1(X, x_0)$.*

(4) *If X is path-connected, then χ is an isomorphism if and only if the fundamental group of X is commutative.*

Of course, it is immediate from the theorem that if C is the kernel of the homomorphism [i.e., the commutator subgroup], then $\pi_1/C \cong H_1$. It is in this sense that the first homology group is the abelianization of the first homotopy group.

We will find the following lemma helpful. It allows us to break a singular 1-simplex into smaller pieces.

Lemma 4.10.16 *Suppose that* $0 = a_0 < a_1 < a_2 < \cdots < a_r = 1$. *If* $T : [0,1] \to X$ *is a singular* 1*-simplex, then* T *is homologous to the singular* 1*-chain*

$$\sum_{i=0}^{r-1} T_i \,,$$

where $T_i(t) = T(\,(1-t)a_i + ta_{i+1}\,)$.

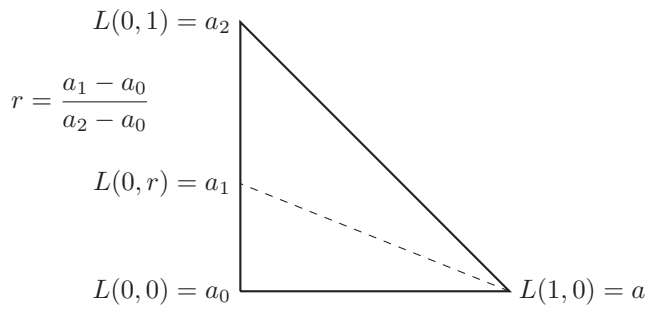


Figure 4.10.7

Proof: It suffices to consider $r = 2$. It is straightforward to construct a function $L(x,y) = \alpha\, x + \beta\, y + \gamma$ having the values indicated in Figure 4.10.7.

Consider the singular 2-simplex $W = T \circ L$. We have $\partial W = S_0 - S_1 + S_2$, where

$$\begin{aligned} S_0(t) &= T \circ L(1-t,t) = T_1(t)\,, \\ S_1(t) &= T \circ L(0,t) = T(t)\,, \\ S_2(t) &= T \circ L(t,0) = T_0(t)\,. \end{aligned}$$

showing that T is homologous to $T_1 + T_2$. □

Proof of Theorem 4.10.15: (1) Show that χ is a homomorphism. We need to prove that

$$\chi([\gamma \cdot \gamma']) = \chi([\gamma]) + \chi([\gamma'])\,.$$

Lemma 4.10.16 tells us that if we consider $\gamma \cdot \gamma'$ as a singular 1-simplex, then it is homologous to singular 1-chain $\gamma + \gamma'$. Thus we have $\{\gamma \cdot \gamma'\} = \{\gamma\} + \{\gamma'\}$.

(2) Assuming that X is path connected, show that χ is surjective.

For each $x \in X$, let $\zeta_x : [0,1] \to X$ be a path with $\zeta_x(0) = x_0$ and $\zeta_x(1) = x$. Also, let $\zeta_x^{-1} : [0,1] \to X$ be the path that runs in the opposite direction, i.e., $\zeta_x^{-1}(t) = \zeta_x(1-t)$.

Let $z = \sum n_i \sigma_i$ be a singular 1-cycle, where each σ_i is a singular 1-simplex. For each i, we form a loop γ_i by setting

$$\gamma_i(t) = \begin{cases} \zeta_{\sigma_i(0)}(3t) & \text{if } 0 \le t \le \frac{1}{3}\,, \\ \sigma_i((3t-1)\epsilon_1) & \text{if } \frac{1}{3} \le t \le \frac{2}{3}\,, \\ \zeta^{-1}_{\sigma_i(1)}(3t-2) & \text{if } \frac{2}{3} \le t \le 1\,. \end{cases}$$

We claim that $\chi\left(\prod_i \gamma_i^{n_i}\right)$ is homologous to z.

By **(1)** , we have

$$\chi\left(\prod_i \gamma_i^{n_i}\right) = \sum n_i \{\gamma_i\}\,.$$

By Lemma 4.10.16, we know that γ_i is homologous to

$$= \zeta_{\sigma_i(0)} + \sigma_i + \zeta^{-1}_{\sigma_i(1)}\,,$$

and $\zeta^{-1}_{\sigma_i(1)}$ is homologous to $-\zeta_{\sigma_i(1)}$, so we have

$$\chi\left(\prod_i \gamma_i^{n_i}\right) = \sum n_i(\zeta_{\sigma_i(0)} - \zeta_{\sigma_i(1)}) + \sum n_i \sigma_i\,.$$

Thus we need to show that

$$0 = \sum n_i(\zeta_{\sigma_i(0)} - \zeta_{\sigma_i(1)})\,. \qquad (*)$$

Because $\sum n_i \sigma_i$ is a cycle, we have

$$0 = \sum n_i(\sigma_i(1) - \sigma_i(0))\,. \qquad (**)$$

Now we compare equations $(*)$ and $(**)$. Consider any point x that occurs among the various points $\sigma_i(1)$ and $\sigma_i(0)$. Equation $(**)$ tells us that the sum of the coefficients of that point add to zero, but the very same sum of coefficients applies to the terms $\zeta_{\sigma_i(1)}$ and $\zeta_{\sigma_i(0)}$ in equation $(*)$. Thus the sum in $(*)$ does also equal zero.

(3) Assuming that X is path connected, show that the kernel of χ is a commutator subgroup. The commutator subgroup, C, is generated by elements of the form $\gamma^{-1} \cdot \eta^{-1} \cdot \gamma \cdot \eta$. If the multiplication is commutative, then the commutator subgroup is obviously trivial, but the fact we need and shall use is that the quotient group π_1/C is commutative.

Because $H_1(X)$ is commutative, χ carries any generator of C to 0, so $C \subset \ker \chi$.

To show the reverse containment, suppose that $\chi(\gamma)$ bounds. That is,

$$\gamma = \partial \left(\sum n_i \sigma_i \right),$$

where the σ_i are singular 2-simplices. Write each $\partial\sigma_i$ as

$$\partial\sigma_i = \alpha_{i0} - \alpha_{i1} + \alpha_{i2},$$

where

$$\alpha_{i0}(t) = \sigma_i(1-t,t), \quad \alpha_{i1}(t) = \sigma_i(0,t), \quad \alpha_{i2}(t) = \sigma_i(t,0).$$

We have

$$\gamma = \sum n_i(\alpha_{i0} - \alpha_{i1} + \alpha_{i2}). \tag{$\dagger$}$$

In the equation ($\dagger$), γ is a singular 1-cycle and the right-hand side is a singular 1-chain, and they are *equal*—not merely homologous. That tells us that, after collecting terms in ($\dagger$), γ occurs with coefficient 1 and all other paths have coefficient 0.

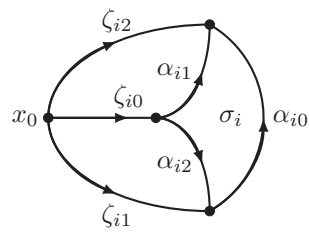


Figure 4.10.8: The simplex σ_i, its edges $\alpha_{i0}, \alpha_{i1}, \alpha_{i2}$, and the paths $\zeta_{i0}, \zeta_{i1}, \zeta_{i2}$ connecting the base point x_0 to the vertices of σ_i.

We need to label the vertices of each 2-simplex σ_i. Set

$$\begin{aligned} x_{i0} &= \sigma_i(0,0) = \alpha_{i1}(0) = \alpha_{i2}(0), \\ x_{i1} &= \sigma_i(1,0) = \alpha_{i0}(0) = \alpha_{i2}(1), \\ x_{i2} &= \sigma_i(0,1) = \alpha_{i0}(1) = \alpha_{i1}(1). \end{aligned}$$

We will use the paths ζ_x from **(2)** above that go from x_0 to x, but to simplify the notation, we shall write ζ_{ij} instead of $\zeta_{x_{ij}}$, and ζ_{ij}^{-1} instead of $\zeta_{x_{ij}}^{-1}$. See Figure 4.10.8.

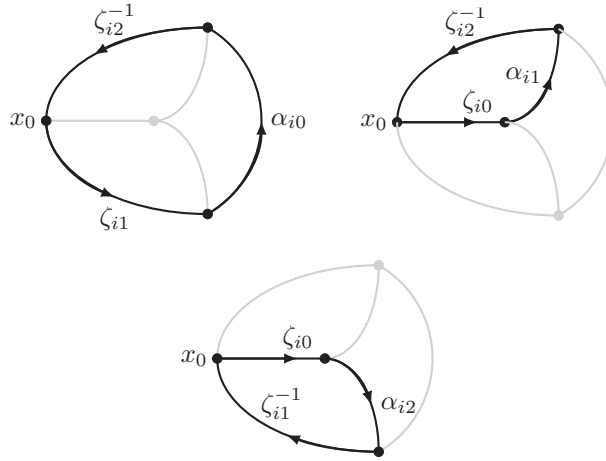


Figure 4.10.9: The loops $\beta_{i0}, \beta_{i1}, \beta_{i2}$.

Consider the loops β_{ji} given by

$$\beta_{j0} = \zeta_{j1} \cdot \alpha_{j0} \cdot \zeta_{j2}^{-1}\,, \quad \beta_{j1} = \zeta_{j0} \cdot \alpha_{j1} \cdot \zeta_{j2}^{-1}\,, \quad \beta_{j2} = \zeta_{j0} \cdot \alpha_{j2} \cdot \zeta_{j1}^{-1}\,,$$

where the $\cdot$ operation is to follow one path by the other from left to right [again we call this operation a *product*]. To obtain a map $[0,1]$ to X, the parametrizations need to be combined appropriately. This amounts to reversing the process in Lemma 4.10.16 where a path was broken into pieces. See Figure 4.10.9.

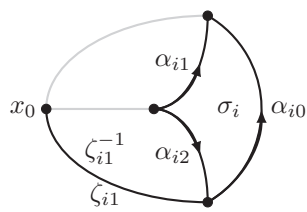


Figure 4.10.10: The loop β_i.

Next we combine those loops to obtain [see Figure 4.10.10]

$$\begin{aligned}\beta_i = \beta_{i0} \cdot \beta_{i1}^{-1} \cdot \beta_{i2} &= \zeta_{i1} \cdot \alpha_{i0} \cdot \zeta_{i2}^{-1} \cdot (\zeta_{i0} \cdot \alpha_{i1} \cdot \zeta_{i2}^{-1})^{-1} \cdot \zeta_{i0} \cdot \alpha_{i2} \cdot \zeta_{i1}^{-1} \\ &= \zeta_{i1} \cdot \alpha_{i0} \cdot \zeta_{i2}^{-1} \cdot \zeta_{i2} \cdot \alpha_{i1}^{-1} \cdot \zeta_{i0}^{-1} \cdot \zeta_{i0} \cdot \alpha_{i2} \cdot \zeta_{i1}^{-1} \\ &\simeq \zeta_{i1} \cdot \alpha_{i0} \cdot \alpha_{i1}^{-1} \cdot \alpha_{i2} \cdot \zeta_{i1}^{-1}\,.\end{aligned}$$

We observe that the loop $\zeta_{i1} \cdot \alpha_{i0} \cdot \alpha_{i1}^{-1} \cdot \alpha_{i2} \cdot \zeta_{i1}^{-1}$ starts at the base point, goes to $x_{i1} = \alpha_{i0}(0)$, then around the boundary of σ_i to $\alpha_{i2}(1) = \alpha_{i0}(0)$, and

finally back to the base point. Now, if we contract Δ_2 down to $(1,0)$, then the effect is to contract σ_i down to $\sigma_i(1,0) = \alpha_{i0}(0)$. We conclude that

$$\beta_i \simeq \zeta_{i1} \cdot \alpha_{i0} \cdot \alpha_{i1}^{-1} \cdot \alpha_{i2} \cdot \zeta_{i1}^{-1} \simeq \zeta_{i1} \cdot \zeta_{i1}^{-1} \simeq x_0 \,.$$

That is, in the fundamental group $[\beta_i] = 1$ holds. Consequently, we have

$$\prod [\beta_i]^{n_i} = 1 \,. \tag{‡}$$

If we could collect terms in (‡) exactly as we did in (†), then we could conclude that $\gamma \simeq 1$. But because the fundamental group is not necessarily commutative, we cannot collect terms freely. On the other hand, in the quotient group $\pi_1(X)/C$ which is commutative, we are able to collect terms at will. Denoting the coset of $[\beta]$ in $\pi(X)/C$ by $\lceil\beta\rceil$, we see from (‡) that

$$\prod \lceil\beta_j\rceil^{n_j} = 1 \,,$$

and from (†) we see that

$$\prod \lceil\beta_j\rceil^{n_j} = \lceil\gamma\rceil \,.$$

Since $\lceil\gamma\rceil = 1$ in $\pi_1(X)/C$, we conclude $\gamma \in C$.

Finally, conclusion **(4)** follows from **(3)**. □

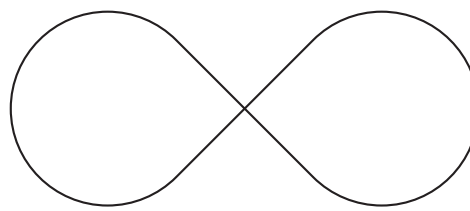


Figure 4.10.11: The figure 8 space.

Proposition 4.10.17 Let X be the figure 8, illustrated in Figure 4.10.11. The fundamental group $\pi_1(X)$ is not abelian.

Proof: We identify $\mathbb{S}^1$ with

$$\{z \in \mathbb{C} : |z| = 1\} \,,$$

as in the proof of Proposition 4.10.8, so that we can then identify X with

$$(\mathbb{S}^1 \times 1) \cup (1 \times \mathbb{S}^1) \,.$$

We let the base point be $x_0 = (1,1)$, the unique point in $(\mathbb{S}^1 \times 1) \cap (1 \times \mathbb{S}^1)$.

Our mental picture of the figure 8 now lies in the cartesian product of two planes: There is a circle in each plane, and they intersect only at $(1,1)$.

Let $\gamma, \eta : I \to X$ be given by

$$\gamma(t) = (e^{2\pi i t}, 1) \quad \text{and} \quad \eta(t) = (1, e^{2\pi i t}) .$$

So we see $\gamma^{-1}, \eta^{-1} : I \to X$ are given by

$$\gamma^{-1}(t) = (e^{2\pi i (1-t)}, 1) \quad \text{and} \quad \eta^{-1}(t) = (1, e^{2\pi i (1-t)}) .$$

For convenience, let us denote the loop $\gamma \cdot \eta \cdot \gamma^{-1} \cdot \eta^{-1}$ by ζ. We see that ζ starts at x_0, goes once around the first circle in the counterclockwise direction, once around the second circle in the counterclockwise direction, once around the first circle in the clockwise direction, and finally once around the second circle in the clockwise direction, ending at x_0.

Arguing by contradiction, we suppose that $\pi_1(X, x_0)$ is commutative, which would imply that $[\gamma \cdot \eta \cdot \gamma^{-1} \cdot \eta^{-1}] = 1$. In $\pi_1(X, x_0)$, the identity 1 is the homotopy class of the constant loop that maps each point of I to $x_0 = (1,1)$. Thus there exists a homotopy function $F : I \times I \to X$ with

$$F(t,u) = \begin{cases} (1,1) & \text{if } u = 0,\ t = 0, \text{ or } t = 1, \\ (\gamma \cdot \eta \cdot \gamma^{-1} \cdot \eta^{-1})(t) & \text{if } u = 1. \end{cases}$$

Motivated by the proof of Proposition 4.10.8, we define $\phi : (\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R}) \to X$ by

$$\phi(w,z) = (e^{2\pi i w}, e^{2\pi i z}) .$$

Arguing as in the proof of Proposition 4.10.8, we lift ζ to $\zeta' : I \to (\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R})$ and we lift F to $F' : I \times I \to (\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R})$. So we have $\zeta = \phi \circ \zeta'$ and $F = \phi \circ F'$. Since ζ has been explicitly defined, we know precisely what its lifting to $(\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R})$ is, namely, the loop that maps the interval $I = [0,1]$ once counterclockwise around the boundary of $I \times I$.

Now we also define $\psi : (\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R}) \to \text{Bd}\ (I \times I)$ by the mapping

$$(x,y) \xrightarrow{\psi} \text{Bd}\ (I \times I) \cap \{(1-t)(x,y) + t(\tfrac{1}{2}, \tfrac{1}{2}) : 0 \le t \le 1\} .$$

[See Figure 4.10.12.] Clearly, ψ is the identity on Bd $(I \times I)$. So $\psi \circ \zeta' : I \to$ Bd $(I \times I)$ is the loop in Bd $(I \times I)$ that goes once around counterclockwise. Of course, Bd $(I \times I)$ is homeomorphic to $\mathbb{S}^1$, so we know from Theorem 4.10.8 that the loop $\psi \circ \zeta'$ is homotopically nontrivial. But $\psi \circ F'$ is a homotopy between $\psi \circ \zeta'$ and the constant loop always equal to $(0,0)$. □

Exercises

1. Prove that the relation $\simeq$ of Definition 4.10.1 is reflexive and symmetric.

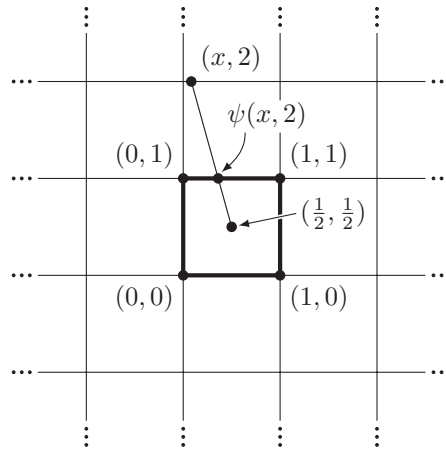


Figure 4.10.12: The map ψ.

2. Prove parts **(2)**, **(3)**, and **(4)** of Lemma 4.10.5.

3. Referring to the proof of Proposition 4.10.8, verify the equation $[\rho_m] \cdot [\rho_n] = [\rho_{m+n}]$.

4. **(a)** Prove that any loop in $\mathbb{R}^n$, $n \geq 2$, is homotopic to a loop contained in a finite union of lines.

(b) Prove that any loop in $\mathbb{S}^n$, $n \geq 2$, is homotopic to a loop contained in a finite union of great circles. This result justifies the assertions made in Example 4.10.9.

5. Let $X = U \cup V$ where U and V are simply connected open sets. Show that if $U \cap V$ is path connected, then X is simply connected.

4.11 More on Quotient Spaces

We reviewed some aspects of the theory of quotient spaces in Section 2.8. We now discuss the topic further. In particular, we consider separation axioms for quotient spaces.

We have already noted that the separation axioms do not behave well for quotient spaces. In general, it is difficult to ensure that a quotient space satisfies any stronger separation axiom than the T_1-axiom. Even the Hausdorff axiom is often hard to verify. We give here three situations in which we can be sure that the quotient space is Hausdorff, indeed, normal.

Theorem 4.11.1 *Let $p : X \to Y$ be a quotient map. If p is a closed map, and if X is normal, then Y is normal.*

Proof: If x is a point of X, then x is closed in X, so the one-point set $p(x)$ is closed in Y [because p is a closed map]. Thus Y is a T_1-space.

Let A and B be disjoint closed sets in Y. Then $p^{-1}(A)$ and $p^{-1}(B)$ are disjoint closed sets in X. Choose disjoint open sets U and V in X about $p^{-1}(A)$ and $p^{-1}(B)$, respectively. The sets $p(U)$ and $p(V)$ need not be disjoint nor open in Y. See Figure 4.11.1. Nonetheless, the sets $C = X \setminus U$ and $D = X \setminus V$ are closed in X, so $p(C)$ and $p(D)$ are closed in Y. Then we claim the sets $Y \setminus p(C)$ and $Y \setminus p(D)$ are disjoint open sets containing A and B, respectively.

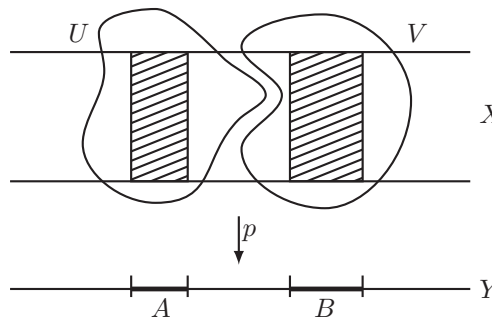


Figure 4.11.1

To show that these sets are disjoint, recall that U and V are disjoint, so De Morgan's law gives us

$$C \cup D = (X \setminus U) \cup (X \setminus V) = X \setminus (U \cap V) = X\,.$$

Thus we have $p(C) \cup p(D) = Y$. Applying De Morgan's law again, we obtain

$$\emptyset = Y \setminus (p(C) \cup p(D)) = (Y \setminus p(C)) \cap (Y \setminus p(D))$$

as desired.

To show that these sets contain A and B, respectively, note first that because C is disjoint from $p^{-1}(A)$, the set $p(C)$ is disjoint from A. Thus $p(C) \subset Y \setminus A$. Taking complements, we have $Y \setminus p(C) \supset A$. Similarly, $Y \setminus p(D)$ contains B. □

If X^* is a partition of X into closed sets, and if the quotient map $p : X \to X^*$ is closed, then X^* is called, classically, an *upper semicontinuous decomposition* of X. Specifically, this means that for each closed set A of X, the set $p^{-1}p(A)$, which is called the *saturation* of A, is also closed in X In such a case, normality of X implies normality of X^*.

Definition 4.11.2 Let X and Y be disjoint topological spaces, let A be a closed subset of X, and let $f : A \to Y$ be a continuous map. We define a certain quotient space as follows: Topologize $X \cup Y$ as the topological sum. Form a quotient space by identifying the set

$$\{y\} \cup f^{-1}(y) \qquad (*)$$

to a point, for each $y \in Y$. That is, partition $X \cup Y$ into the sets in $(*)$ along with the one-point sets $\{x\}$, for $x \in X \setminus A$. We denote this quotient space by $X \cup_f Y$, and call it the *adjunction space determined by* f. See Figure 4.11.2.

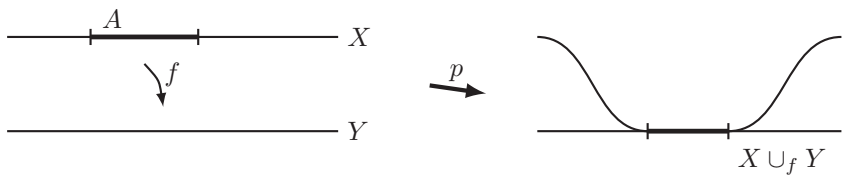


Figure 4.11.2

Let $p : X \cup Y \to X \cup_f Y$ be the quotient map. We show first that p defines a homeomorphism of Y with a closed subspace of $X \cup_f Y$. Obviously, $p|Y$ is continuous and injective. Furthermore, if C is closed in Y, then $f^{-1}(C)$ is closed in X, because $f : A \to Y$ is continuous and A is closed in X. It follows that

$$p^{-1}p(C) = C \cup f^{-1}(C)$$

is closed in $X \cup Y$. Then $p(C)$ is closed in $X \cup_f Y$ by definition of the quotient topology. Thus $p|Y$ carries Y homeomorphically onto the closed subspace $p(Y)$ of $X \cup_f Y$. We normally abuse notation and identify Y with $p(Y)$.

Now if X and Y are T_1-spaces, then $X \cup_f Y$ is also a T_1-space, since each of the equivalence classes is closed in $X \cup Y$. The preceding theorem does not apply to give us further separation properties, for p is not, in general, a closed map. Nonetheless, we can still prove the following.

Theorem 4.11.3 *If X and Y are normal, then the adjunction space $X \cup_f Y$ is normal.*

Proof: Let A be closed in X and let $f : A \to Y$ be continuous. Let B and C be disjoint closed sets in $X \cup_f Y$. Let

$$B_X = p^{-1}(B) \cap X, \qquad C_X = p^{-1}(C) \cap X,$$
$$B_Y = p^{-1}(B) \cap Y, \qquad C_Y = p^{-1}(C) \cap Y.$$

See Figure 4.11.3. Applying Urysohn's lemma, we can choose a continuous function $g : Y \to [0,1]$ that maps B_Y to 0 and C_Y to 1. Then the function $g \circ f : A \to [0,1]$ equals 0 on $A \cap B_X$ and 1 on $A \cap C_X$. We define a function

$$h : A \cup B_X \cup C_X \to [0,1]$$

by letting it equal $g \circ f$ on A, and 0 on B_X, and 1 on C_X. Since A, B_X, and C_X are closed in X, the map h is continuous. By the Tietze extension theorem, we can now extend h to a continuous function k defined on all of X.

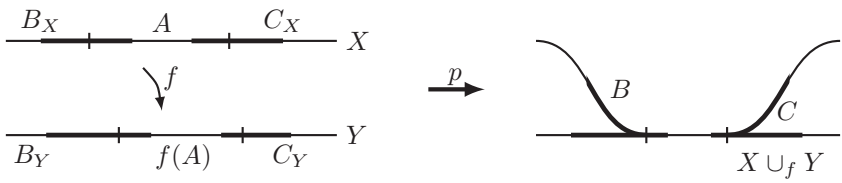


Figure 4.11.3

The function $\ell : X \cup Y \to [0,1]$ that agrees with k on X and agrees with g on Y is continuous. Note that ℓ is constant on each equivalence class, because $k(f^{-1}(y)) = g(y)$ if $y \in f(A)$. Therefore ℓ induces a continuous map of $X \cup_f Y$ into $[0,1]$ that equals 0 on B and equals 1 on C. □

We have already defined what it means for the topology of a given space X to be coherent with a collection of subspaces of X. We wish to deal with the separation axioms in this context.

First, we extend the notion of coherent topology slightly. We suppose that we have a collection of topological spaces $\{X_\alpha\}$ whose union X has no topology. We seek to find conditions under which there *exists* a topology on X, of which the X_α are subspaces, such that this topology is coherent with the subspaces X_α.

Lemma 4.11.4 *Let X be a set that is the union of the topological spaces $\{X_\alpha\}$.*

> **(a)** *If there is a topological space X_T having X as its underlying set, and each X_α is a subspace of X_T, then X has a topology, of which the X_α are subspaces, that is coherent with the X_α. This topology is in general finer than the topology of X_T.*
>
> **(b)** *If, for each pair α, β of indices, the set $X_\alpha \cap X_\beta$ is closed in both X_α and X_β, and inherits the same subspace topology from each of them, then X has a topology coherent with the subspaces X_α. Each X_α is a closed set in this topology.*

Proof: **(a)** By hypothesis, we are given the topological space X_T with underlying set X. Let us define a [possibly different] topological space X_C whose underlying set is X by declaring a set A to be closed in X_C if and only if its intersection with each X_α is a closed subset of X_α. The collection of such sets contains arbitrary intersections and finite unions of its elements, so it does define a topology on the set X.

If A is closed in X_T, then because X_α is a subspace of X_T, the set $A \cap X_\alpha$ is closed in X_α for each α. It follows that A is closed in X_C. Thus the topology of X_C is finer than that of X_T.

We next show that each X_α is a subspace of X_C. It then follows from its definition that X_C is coherent with the subspaces X_α. For this purpose, we show that the collection of closed sets of X_α equals the collection of sets of the form $A \cap X_\alpha$, where A is closed in X_C. By the definition of X_C, if A is closed in X_C, then $A \cap X_\alpha$ is closed in X_α. Conversely, suppose that B is closed in X_α. Because X_α is a subspace of X_T, we have $B = A \cap X_\alpha$ for some set A closed in X_T. Because the topology of X_C is finer than that of X_T, the set A is also closed in X_C. Thus $B = A \cap X_\alpha$ for some A closed in X_C, as desired.

(b) As in part **(a)**, we define the topology X_C on the set X by declaring A to be closed in X_C if and only if $A \cap X_\alpha$ is closed in X_α for each α.

We show that each X_α is a subspace of X_C. It then follows from its definition that X_C is coherent with the subspaces X_α. As before, we show that the collection of closed sets of X_α equals the collection of sets of the form $A \cap X_\alpha$, where A is closed in X_C. First, if A is closed in X_C, then $A \cap X_\alpha$ is closed in X_α. Conversely, suppose B is closed in X_α. Let β be any index. Because $X_\alpha \cap X_\beta$ is a closed set in X_α, the set

$$B \cap X_\beta = X \cap (X_\alpha \cap X_\beta)$$

is a closed set of X_α. Because $X_\alpha \cap X_\beta$ is a subspace of X_α, it is also a closed set in $X_\alpha \cap X_\beta$. Because $X_\alpha \cap X_\beta$ is a closed subspace of X_β, it is a closed set in X_β. Since β was arbitrary, B is by definition a closed set of X_C. Thus $B = B \cap X_\alpha$, where B is closed in X_C, as desired.

Incidentally, we have shown that every closed subset of X_α is also closed in X_C. In particular, X_α is itself closed in X_C. □

EXAMPLE 4.11.5 Let K be a complex in $\mathbb{E}^J$. Since $\mathbb{E}^J$ is a space, $|K|$ has a topology inherited from $\mathbb{E}^J$. Each simplex σ of K is a subspace in this topology. By **(a)** of the preceding lemma, $|K|$ has a topology coherent with the subspaces σ, which is in general finer than the topology $|K|$ inherits from $\mathbb{E}^J$. This is a fact we proved directly in Section 1.3.

Suppose that X is a space whose topology is coherent with its subspaces X_α. In general, even if each of the spaces X_α has nice separation properties, X need not have these properties. Nonetheless, in the special case of a countable union of closed subspaces, one can prove the following:

Theorem 4.11.6 *Let X be a space that is the countable union of certain closed subspaces X_i. If the topology of X is coherent with the spaces X_i, and if each X_i is normal, then X is normal.*

Proof: If p is a point of X, then $\{p\} \cap X_i$ is closed in X_i for each i, so $\{p\}$ is closed in X. Thus X is a T_1-space.

Let A and B be disjoint closed sets in X. Define $Y_0 = A \cup B$, and for $n > 0$, define

$$Y_n = X_1 \cup X_2 \cup \cdots \cup X_n \cup A \cup B\,.$$

We define a function $f_0 : Y_0 \to [0,1]$ by letting f_0 equal 0 on A and equal 1 on B. Since A and B are disjoint and closed, this function f_0 is continuous. Proceeding inductively, we will extend the continuous function $f_n : Y_n \to [0,1]$ to a continuous function on Y_{n+1} that equals 0 on A and equals 1 on B. The space X_{n+1} is normal and $Y_n \cap X_{n+1}$ is closed in X_{n+1}, so we use the Tietze extension theorem to extend the function $f_n|(Y_n \cap X_{n+1})$ to a continuous function $g : X_{n+1} \to [0,1]$. Because Y_n and X_{n+1} are closed subsets of Y_{n+1}, the functions f_n and g combine to define a continuous function $f_{n+1} : Y_{n+1} \to [0,1]$ that extends f_n. The functions $f_0, f_1, \ldots$ combine to define a function $f : X \to [0,1]$ that equals 0 on A and equals 1 on B. Because X has the topology coherent with the subspaces X_n, the map f is continuous. □

The next definition is used in Exercise 3.

Definition 4.11.7 A space is said to be *compactly generated* if its topology is coherent with the collection of its compact subspaces.

EXERCISES

1. Let X be a set which is the union of the topological spaces $\{X_\alpha\}$. Suppose that each set $X_\alpha \cap X_\beta$ inherits the same topology from each of X_α and X_β.

(a) Show that if $X_\alpha \cap X_\beta$ is open in both X_α and X_β, for each pair α, β, then there is a topology on X of which the X_α are subspaces.

(b) Show that in general there is no topology on X of which each X_α is a subspace. [**Hint:** Let A and B be disjoint dense subsets of $\mathbb{R}$, such that the complement of their union is also dense in $\mathbb{R}$. Let A, B, $X_1 = \mathbb{R} \setminus A$, and $X_2 = \mathbb{R} \setminus B$ be topologized as subspaces of $\mathbb{R}$. Let $X_3 = A \cup B$ be topologized as the topological sum of A and B. Let $X = X_1 \cup X_2 \cup X_3$. Compute $\overline{A}$.]

2. Recall that $\mathbb{R}^\infty$ denotes the space $\mathbb{E}^J$, with J equal to the set of positive integers. Each space $\mathbb{R}^n \times \mathbf{0}$ is a subspace of $\mathbb{R}^\infty$. Show that the function $f : \mathbb{R}^\infty \to \mathbb{R}$ given by

$$f(x) = \sum_{i=1}^{\infty} i x_i$$

is not continuous in the usual metric topology of $\mathbb{R}^\infty$, but is continuous in the topology coherent with the subspaces $\mathbb{R}^n \times \mathbf{0}$.

3. Let X be a space. Let X_C denote the set X in the topology coherent with the collection of compact subspaces of X.

(a) Show that a subset B of X is compact in the topology it inherits from X if and only if it is compact in the topology it inherits from X_C. [**Hint:** Show also that these two distinct subspace topologies on B are in fact the same topology.]

(b) Show that X_C is compactly generated. Conclude that $(X_C)_C = X_C$.

(c) Show that inclusion $X_C \longrightarrow X$ induces an isomorphism in singular homology.

(d) In general, let $X \times_C Y$ denote the compactly generated topology $(X \times Y)_C$ derived from $X \times Y$. Show that if K and L are complexes, then the topology of $|K| \times_C |L|$ is coherent with the subspaces $\sigma \times \tau$, for $\sigma \in K$ and $\tau \in L$. Compare with Exercise 6 of Section 2.8. [**Hint:** If $D \subset X \times Y$ is compact, then $D \subset A \times B \subset X \times Y$, where A and B are compact.]

4. Show that Theorem 4.11.6 does not hold for uncountable collections. [**Hint:** Let X be an uncountable well-ordered set with a smallest element 0 and a largest element ω_1, such that $[0, \alpha] = \{x : 0 \leq x \leq \alpha\}$ is countable for each $\alpha < \omega_1$. Then $[0, \omega_1) \times [0, \omega_1]$ is not normal. See [Mun00], Example 2 on page 203.]

5. Find an additional hypothesis that will make Theorem 4.11.6 true for uncountable unions.

4.12 CW Complexes

We have stated that one of the advantages of simplicial homology theory is its effective computability. On the other hand, the amount of labor involved in a straightforward calculation in all but the simplest cases is too large to carry out by hand. Even when we calculated the homology of such simple spaces as the torus and the Klein bottle, in Section 1.7, we did not proceed

straightforwardly. Instead, we used geometric arguments—of a rather ad hoc nature—to reduce the computations to simpler ones.

We now refine these ad hoc techniques into a systematic method for computing homology groups. This method will apply not only to simplicial homology, but to singular homology as well. We introduce in this section a notion of complex more general than that of the simplicial complex. It was invented by J. H. C. Whitehead and introduced in [Whi49]. It is called a "CW complex." In the next section, we show how to assign to each CW complex a certain chain complex, called its "cellular chain complex," which can be used to compute the homology of the underlying space. This chain complex is much simpler and easier to deal with than the singular or simplicial chain complexes. In a final section, we apply these methods to compute, among other things, the homology of real and complex projective spaces.

Definition 4.12.1 Recall that a space is called a *cell* of dimension m if it is homeomorphic with $\mathbb{B}^m$. It is called an *open cell* of dimension m if it is homeomorphic with $\text{Int } \mathbb{B}^n$. In each case the integer m is uniquely determined by the space in question.

Definition 4.12.2 A *CW complex* is a space X and a collection of disjoint open cells e_α whose union is X such that:

(1) X is Hausdorff.

(2) For each open m-cell of the collection, there exists a continuous map $f_\alpha : \mathbb{B}^m \to X$ that maps $\text{Int } \mathbb{B}^m$ homeomorphically onto e_α and carries $\text{Bd } \mathbb{B}^m$ into a finite union of open cells, each of dimension less than m.

(3) A set A is closed in X if $A \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ for each α.

The finiteness part of condition **(2)** was called "closure-finiteness" by J. H. C. Whitehead. Condition **(3)** expresses the fact that X has what he called the "weak topology" relative to the collection $\{e_\alpha\}$. These terms are the origin of the letters "C" and "W" in the phrase "CW complex."

We typically denote $\overline{e}_\alpha \setminus e_\alpha$ by $\dot{e}_\alpha$. We remark that conditions **(1)** and **(2)** imply that f carries $\mathbb{B}^m$ onto $\overline{e}_\alpha$ and $\text{Bd } \mathbb{B}^m$ onto $\dot{e}_\alpha$. To see these facts, note that because f_α is continuous, it carries $\mathbb{B}^m$, which is the closure of $\text{Int } \mathbb{B}^m$, *into* the closure of $f_\alpha(\text{Int } \mathbb{B}^m)$, which is $\overline{e}_\alpha$. Because $f_\alpha(\mathbb{B}^m)$ is compact, it is closed [since X is Hausdorff]. Because it contains e_α it must contain $\overline{e}_\alpha$. Thus we have $f_\alpha(\mathbb{B}^m) = \overline{e}_\alpha$. Finally, because $f_\alpha(\text{Bd } \mathbb{B}^m)$ is disjoint from e_α it must equal $\dot{e}_\alpha$.

We remark also that the converse of **(3)** holds trivially: If A is closed in X, then $A \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ for each α.

The map f_α is called the "characteristic map" for the open cell e_α. Note that the maps f_α are not uniquely specified in the definition of a CW complex. Only the space X and the collection $\{e_\alpha\}$ are specified. We customarily abuse

notation and use the symbol X to refer to both the CW complex and to the underlying space.

A *finite CW complex* X is a CW complex for which the collection of open cells is finite. If X has only finitely many open cells, then the finiteness part of **(2)** is automatic, and condition **(3)** is implied by the other conditions: If the set $A \cap e_\alpha$ is closed in $\overline{e}_\alpha$, then it is closed in X; thus since A is a finite union of such sets, A is also closed in X.

A finite CW complex is of course compact. Conversely, any compact subset A of a CW complex X can intersect only finitely many open cells of X. We leave the proof to the exercises. One needs the finiteness part of condition **(2)** in the proof.

The following lemma is an immediate consequence of our general results about coherent topologies [see Section 2.8].

Lemma 4.12.3 *Let X be a CW complex with the open cells e_α. A function $f : X \to Y$ is continuous if and only if $f|\overline{e}_\alpha$ is continuous for each α. A function $F : X \times I \to Y$ is continuous if and only if $F|(\overline{e}_\alpha \times I)$ is continuous for each α.*

EXAMPLE 4.12.4 Consider the torus as a quotient space of a rectangle, as usual. See Figure 4.12.1. We can express T as a CW complex having a single open 2-cell [the image under π of the interior of the rectangle], two open 1-cells [the images of the open edges], and one 0-cell [the image of the vertices]. Conditions **(1)**, **(2)**, and **(3)** hold at once.

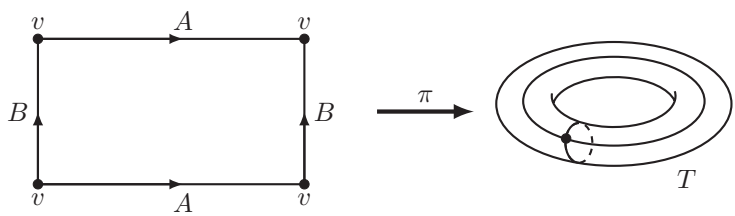


Figure 4.12.1

Similarly, one can express the Klein bottle as a CW complex having the same number of cells in each dimension as the torus. The projective plane can be expressed as a CW complex having one open cell in each dimension 0, 1, and 2.

More generally, the discussion in the exercises of Section 1.7 shows how one can express the n-fold connected sum $T \# T \# \cdots \# T$ as a CW complex having one open cell in dimension 2, one cell in dimension 0, and $2n$ open cells in dimension 1. Likewise, the n-fold connected sum $\mathbb{P}^2 \# \mathbb{P}^2 \# \cdots \# \mathbb{P}^2$, can be expressed as a CW complex having one open cell in dimension 2, one

cell in dimension 0, and n open cells in dimension 1. Finally, the k-fold dunce cap can be expressed as a CW complex with one open cell in each dimension 0, 1, and 2.

EXAMPLE 4.12.5 The quotient space formed from $\mathbb{B}^n$ by collapsing Bd $\mathbb{B}^n$ to a point is homeomorphic to $\mathbb{S}^n$. [We leave the proof to you.] Therefore, $\mathbb{S}^n$ can be expressed as a CW complex having one open n-cell, one 0-cell, and *no other cells at all*. See Figure 4.12.2.

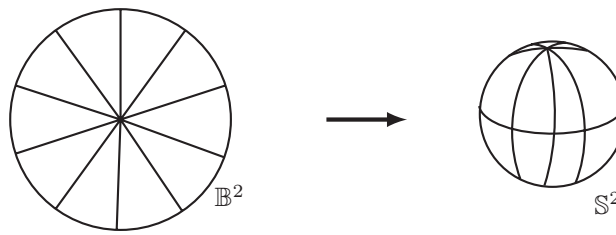


Figure 4.12.2

EXAMPLE 4.12.6 Condition **(2)** does not require that $f_\alpha(\text{Bd } \mathbb{B}^m) = \dot{e}_\alpha$ *equal* the union of a collection of open cells of lower dimension. For example, the space X pictured in Figure 4.12.3 is a CW complex having one open cell in each dimension 0, 1, 2. If e_2 is the open 2-cell, then $\dot{e}_2$ lies in, but does not equal, the union of open cells of lower dimensions.

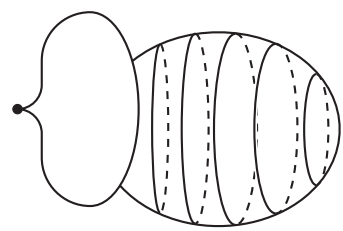


Figure 4.12.3

EXAMPLE 4.12.7 Let K and L be simplicial complexes and suppose that K is locally finite. The space $X = |K| \times |L|$ can be expressed as a CW complex by taking the sets $(\text{Int } \sigma) \times (\text{Int } \tau)$ as its cells, for $\sigma \in K$ and $\tau \in L$. In this case, the characteristic maps

$$f_\alpha : \mathbb{B}^m \to \sigma \times \tau$$

can be taken to be *homeomorphisms*. Furthermore, in this case, $\mathrm{Bd}(\sigma \times \tau)$ *equals* a union of open cells of lower dimension. Condition **(3)** is a consequence of Exercise 6 of Section 2.8. It depends on the local finiteness of K.

The preceding example showed us that $|K| \times |L|$ is a special kind of CW complex for which we now give a formal definition.

Definition 4.12.8 A CW complex X for which the maps f_α can be taken to be homeomorphisms, and for which each set $\dot{e}_\alpha$ equals the union of finitely many open sells of X is called a *regular cell complex*. A regular cell complex X can always be triangulated so each closed cell of X is the polytope of a subcomplex. The proof is similar to the one we gave for $|K| \times I$ in Lemma 2.7.3.

EXAMPLE 4.12.9 Many of the constructions used in topology for forming new spaces from old ones, when applied to CW complexes, give rise to CW complexes. For instance, the product $X \times Y$ of two CW complexes is a CW complex, provided Y is locally compact. Often adjunction spaces can be made into CW complexes. If A is a subcomplex of X, a *cellular map* $f : A \to Y$ is a map carrying each p-cell of A into the union of the open cells of Y of dimension at most p. In this case, one can show that the adjunction space $X \cup_f Y$ is a CW complex.

Definition 4.12.10 Let X be a CW complex. Let Y be a subspace of X that equals a union of open cells of X. Suppose that for each open cell e_α of X contained in Y, its closure is also contained in Y. Then we shall show that Y is a closed set in X and that Y is a CW complex in its own right. It is called a *subcomplex* of X. In particular, the subspace X^p of X that is the union of the open cells of X of dimension at most p satisfies these conditions. It is thus a subcomplex of X, that is called the *p-skeleton* of X.

We now show that Y is indeed a CW complex. First, Y is clearly Hausdorff. If e_α is an open m-cell of X contained in Y, then its characteristic map $f_\alpha : \mathbb{B}^m \to X$ carries $\mathbb{B}^m$ onto $\overline{e}_\alpha$, which is contained in Y by hypothesis. The open cells of X that intersect $f_\alpha(\mathrm{Bd}\ \mathbb{B}^m) = \dot{e}_\alpha$ must lie in Y; thus f_α carries $\mathrm{Bd}\ \mathbb{B}^m$ into the union of finitely many open cells of Y. It only remains to show that Y has the topology specified by **(3)**.

Let $B \subset Y$. Suppose $B \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ for each e_α contained in Y. If e_β is a cell of X not contained in Y, then e_β is disjoint from Y. Hence $Y \cap \overline{e}_\beta \subset \dot{e}_\beta$, so $Y \cap \overline{e}_\beta$ lies in the union of finitely many open cells of Y, say $e_1, e_2, \ldots, e_k$. Then

$$B \cap \overline{e}_\beta = [(B \cap \overline{e}_1) \cup (B \cap \overline{e}_2) \cup \cdots \cup (B \cap \overline{e}_k)] \cap \overline{e}_\beta\,.$$

By hypothesis, $B \cap \overline{e}_i$ is closed in $\overline{e}_i$ and hence in X. Therefore, $B \cap \overline{e}_\beta$ is closed in X, and in particular is closed in $\overline{e}_\beta$. Since β was arbitrary, it follows that B is closed in X, and in particular, is closed in Y, as desired.

It follows that Y has the topology specified by condition **(3)**. It also follows that if B is closed in Y, then B is closed in X. In particular, Y itself is closed in X.

The finiteness part of condition **(2)** is crucial for what we have just proved, as the following example shows.

EXAMPLE 4.12.11 Let X be a 2-simplex σ in the plane, with its usual topology. Break X up into a single open 2-cell Int σ, and infinitely many open 1-cells and 0-cells, as indicated in Figure 4.12.4. Then X satisfies all the conditions for a CW complex except the finiteness part of **(2)**. [Condition **(3)** is trivial, since the closure of Int σ equals X.] Let Y be the union of the 1-cells and 0-cells of X, topologized by declaring C to be closed in Y if $C \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ for each 1-cell and 0-cell e_α. Then Y is a CW complex, but it is not a subspace of X. For the subspace Bd σ of X is compact, and Y is not.

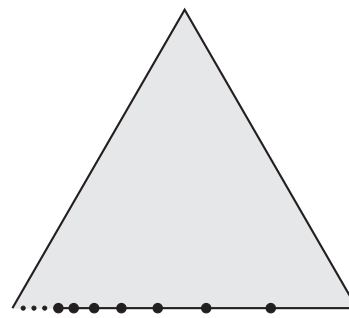


Figure 4.12.4

Definition 4.12.12 If a CW complex X can be triangulated by a complex K in such a way that each skeleton X^p of X is triangulated by a subcomplex of K of dimension at most p, then we say that X is a *triangulable CW complex*.[4]

Each of the CW complexes mentioned in the earlier examples is a triangulable CW complex. We now give an example of one that is not. The proof uses results about local homology groups from Section 4.8.

EXAMPLE 4.12.13 Let A be a subspace of $\mathbb{R}^3$ that is the union of a square and a triangle, such that one of the edges of the triangle coincides with the diagonal D of the square. See Figure 4.12.5. Then A is the space of a complex consisting of three triangles with an edge in common. Now draw a wiggly 1-cell C in the square, intersecting the diagonal in an infinite, totally disconnected set. [An $x \sin(1/x)$ curve will do.] Take the 3-ball $\mathbb{B}^3$ and attach it to A by

[4]The dimensional condition is in fact redundant. See Exercise 2.

a map $f : \text{Bd } \mathbb{B}^3 \to C$ that maps each great circle arc in $\mathbb{S}^2$ from the south pole to the north pole homeomorphically onto the 1-cell C. The resulting adjunction space X is easily seen to be a CW complex; one takes the open simplices of A as the open cells of dimensions 0, 1, and 2, and Int $\mathbb{B}^3$ as the open 3-cell e_3. We show that the space X cannot be triangulated; hence, in particular, it is not a triangulable CW complex.

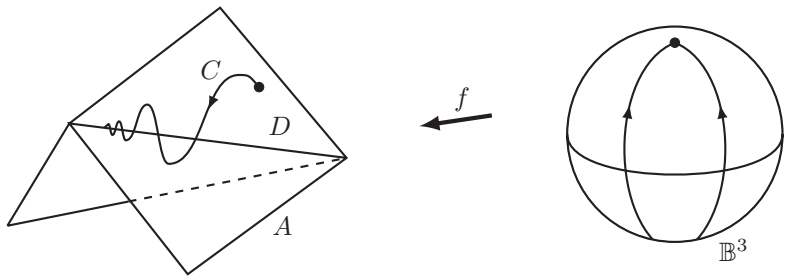


Figure 4.12.5

Suppose $h : |K| \to X$ is a triangulation. First we write X as the disjoint union

$$X = (A \setminus C) \cup C \cup e_3 \,.$$

Now if $x \in e_3$, then $H_3(X, X \setminus x)$ is infinite cyclic, because x has a neighborhood homeomorphic to an open 3-ball. On the other hand, notice that if $x \in A \setminus C$, then $H_3(X, X \setminus x)$ vanishes, because x has a neighborhood lying entirely in the 2-dimensional complex A. [See Theorem 4.8.14.] It follows from Lemma 4.8.10 that if σ is a simplex of K, then $h(\text{Int } \sigma)$ cannot intersect both $A \setminus C$ and e_3. It follows that $h(\sigma)$ lies either in A or in $\overline{e}_3$. Thus A and $\overline{e}_3$ are triangulated by h, and thus so is $A \cap \overline{e}_3 = C$.

On the other hand, we next give similar reasoning that shows that D is triangulated by h. Consider the local homology groups $H_i(A, A \setminus x)$ of A. At each point x interior to D, $H_2(A, A \setminus x) \cong \mathbb{Z} \oplus \mathbb{Z}$. [For A can be expressed as a cone with vertex x and base homeomorphic to a curve with a shape similar to the letter θ, a curve that we denote by θ. So by the long exact homology sequence, $H_2(A, A \setminus x) \cong H_1(\theta) \cong \mathbb{Z} \oplus \mathbb{Z}$.] At each point x of A not in D, either $H_2(A, A \setminus x) \cong \mathbb{Z}$ [if x is interior to one of the triangles] or $H_2(A, A \setminus x) = 0$. We conclude that if $h(\text{Int } \sigma)$ intersects the interior of D, then it lies in D. Thus D is triangulated by h.

It follows that $C \cap D$ is triangulated by h. This is impossible, since $C \cap D$ has infinitely many components and K is a finite complex.

It is often helpful to view a CW complex as a space built up from a collection of closed balls by forming appropriate quotient spaces or to construct

a CW complex in this way, as we did in the preceding example. The following two theorems show how this is done.

The *dimension* of a CW complex X is the largest dimension of a cell of X, if such a maximum exists, otherwise the dimension is said to be infinite.

Theorem 4.12.14

> **(a)** *Suppose X is a CW complex of dimension p. Then X is homeomorphic to an adjunction space formed from X^{p-1} and a topological sum $\sum B_\alpha$ of closed p-balls, by means of a continuous map $g : \sum \mathrm{Bd}\ B_\alpha \to X^{p-1}$. It follows that X is normal.*
>
> **(b)** *Conversely, let Y be a CW complex of dimension at most $p-1$, let $\sum B_\alpha$ be a topological sum of closed p-balls, and let $g : \sum \mathrm{Bd}\ B_\alpha \to Y$ be a continuous map. Then the adjunction space X formed from Y and $\sum B_\alpha$ by means of g is a CW complex, and Y is its $p-1$ skeleton.*

Proof: **(a)** For each cell e_α of X of dimension p one is given the characteristic map $f_\alpha : \mathbb{B}^p \to \overline{e}_\alpha$. Let $B_\alpha = \mathbb{B}^p \times \{\alpha\}$, and let $\sum B_\alpha$ be the topological sum of these disjoint p-balls. Form the topological sum

$$E = X^{p-1} \cup \left(\sum B_\alpha\right),$$

and define $\pi : E \to X$ by letting π equal inclusion on X^{p-1} and the composite

$$B_\alpha = \mathbb{B}^p \times \{\alpha\} \longrightarrow \mathbb{B}^p \xrightarrow{f_\alpha} X$$

on B_α. To prove **(a)**, it will suffice to show that π is a quotient map.

Obviously, π is continuous and surjective. Suppose C is a subset of X and $\pi^{-1}(C)$ is closed in E. Then

(1) $\pi^{-1}(C) \cap X^{p-1} = C \cap X^{p-1}$ is closed in X^{p-1}.

(2) $\pi^{-1}(C) \cap B_\alpha$ is closed in B_α for each α.

The first condition implies that $C \cap \overline{e}_\beta$ is closed in $\overline{e}_\beta$ whenever $\dim e_\beta < p$. Since B_α is compact and π is continuous, the second condition implies that $\pi(\pi^{-1}(C) \cap B_\alpha) = C \cap \overline{e}_\alpha$ is compact. Since X is Hausdorff, $C \cap \overline{e}_\alpha$ is closed in X and hence closed in $\overline{e}_\alpha$, whenever $\dim e_\alpha = p$. Thus C is closed in X, so π is a quotient map, as desired.

Normality of X follows from Theorem 4.11.3; one proceeds by induction on p.

(b) Let $f : Y \cup (\sum B_\alpha) \to X$ be the hypothesized quotient map. Now Y is normal, by part **(a)**, and $\sum B_\alpha$ is normal. It follows from Theorem 4.11.3 that X is normal [and consequently Hausdorff]. As usual, we consider Y to be a subspace of the adjunction space X; then f equals inclusion of Y, and f equals g on $\sum \mathrm{Bd}\ B_\alpha$. We define the open cells of X to be the cells $\{e_\beta\}$ of Y [having

dimension less than p], and the cells $e_\alpha = f(\text{Int } B_\alpha)$ [having dimension p]. Since Int B_α is open in the topological sum $Y \cup (\sum B_\alpha)$, and it is saturated relative to f, the restriction of f to Int B_α is a quotient map. Being one-to-one, it is a homeomorphism. Thus f maps Int B_α homeomorphically onto e_α, so e_α is an open p-cell, as desired.

We check condition **(2)** for a CW complex. We have already noted that the map $f_\alpha = f|B_\alpha$ maps Int B_α homeomorphically onto the set e_α. By construction, f_α maps Bd B_α into Y, which is the union of cells of dimension less than p. Because Bd B_α is compact, the set $f(\text{Bd } B_\alpha)$ is a compact subset of Y. Because Y is a CW complex, it intersects only finitely many open cells of Y. Thus condition **(2)** is satisfied.

Condition **(3)** follows readily. Suppose C is a subset of X and $C \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ for each open cell e_α. We show that $f^{-1}(C)$ is closed in $Y \cup (\sum B_\alpha)$; from this it follows that C is closed in X.

First note that $f^{-1}(C) \cap Y = C \cap Y$. Because $C \cap \overline{e}_\beta$ is closed in $\overline{e}_\beta$ for each cell of dimension less than p, $C \cap Y$ is closed in Y. Similarly, we have

$$f^{-1}(C) \cap B_\alpha = f^{-1}(C \cap \overline{e}_\alpha) \cap B_\alpha \,.$$

We use here the fact that $f(B_\alpha) = \overline{e}_\alpha$. Now $C \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ by hypothesis, and hence closed in X. We apply continuity of f to see that $f^{-1}(C \cap \overline{e}_\alpha) \cap B_\alpha$ is closed in B_α. Thus $f^{-1}(C)$ is closed, as desired. □

Theorem 4.12.15

> **(a)** *Let X be a CW complex. Then X^p is a closed subspace of X^{p+1} for each p and X is the coherent union of the spaces $X^0 \subset X^1 \subset \cdots$. It follows that X is normal.*
>
> **(b)** *Conversely, suppose that X_p is a CW complex for each p, and X_p equals the p-skeleton of X_{p+1} for each p. If X is the coherent union of the spaces X_p, then X is a CW complex having X_p as its p-skeleton.*

Proof: **(a)** Suppose that $C \cap X^p$ is closed in X^p for each p. Then $C \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ for each cell e_α of dimension at most p. Since p was arbitrary, we conclude that C is closed in X. Thus X has the topology coherent with the subspaces X^p. Normality follows from Theorem 4.11.6.

(b) If $p < q$, then $X_p \cap X_q = X_p$ is a closed subspace of both X_p and X_q. Therefore by **(b)** of Lemma 4.11.4, there is a topology on X coherent with the subspaces X_p, and each X_p is closed in X. By the preceding theorem, each space X_p is normal. Therefore, by Theorem 4.11.6, X is normal [consequently, Hausdorff]. Condition **(2)** for a CW complex is trivial. We next check condition **(3)**. Suppose $C \cap \overline{e}_\alpha$ is closed in $\overline{e}_\alpha$ for each cell e_α. Then $C \cap X_p$ is closed in X_p, because X_p is a CW complex. Then C is closed in X because the topology of X is coherent with the spaces X_p. □

Sometimes one begins with a space X and seeks to give it the structure of a CW complex. This we did in Examples 4.12.4, 4.12.5, and 4.12.6. On the other hand, sometimes one seeks to construct *new* spaces that are CW complexes by the process of pasting balls together. This is what we did in Example 4.12.13. The general construction is described in the two theorems we just proved. This construction significantly enlarges the class of spaces about which algebraic topology has something interesting to say.

EXERCISES

1. Let X be a CW complex, and let A be a compact subset of X.

- **(a)** Show that A intersects only finitely many open cells of X. Where do you use "closure-finiteness" in the proof?
- **(b)** Show that A lies in a finite subcomplex of X.

2. Let X be a CW complex. Suppose $h : |K| \to X$ is a triangulation of X that induces a triangulation of each skeleton X^p.

- **(a)** Show that h induces a triangulation of each set $\overline{e}_\alpha$.
- **(b)** Show that if K_p is the subcomplex of K triangulating X^p, then K_p has dimension at most p. [**Hint:** Use local homology groups.]

3. Let X be a CW complex. Show that the topology of X is compactly generated. [See Exercise 3 of Section 4.11.]

4. Let X and Y be a CW complexes. Then $X \times Y$ is the union of open cells $e_\alpha \times e_\beta$, for e_α a cell of X and e_β a cell of Y.

- **(a)** Show that if Y is locally compact, then $X \times Y$ is a CW complex.
- **(b)** Show that $X \times_C Y$ is a CW complex in general. [See Exercise 3 of Section 4.11.]

5. Verify that a regular cell complex can be triangulated so that each closed cell is the polytope of a subcomplex.

6. Show that the n-dimensional torus is a CW-complex.

7. Show that the solid torus in 3-dimensional space is a CW-complex.

4.13 The Homology of CW Complexes

We now show how to compute the singular homology of a CW complex.

Throughout this section, X will denote a CW complex with open cells e_α and characteristic maps f_α. The symbol H_p will denote singular homology

in general, but if it happens that X is a triangulable CW complex, then H_p can also be taken to denote simplicial homology, since there is a natural isomorphism between singular and simplicial theory.

Definition 4.13.1 If X is a CW complex, let

$$D_p(X) = H_p(X^p, X^{p-1}) .$$

Let $\partial : D_p(X) \to D_{p-1}(X)$ be defined to be the composite

$$H_p(X^p, X^{p-1}) \xrightarrow{\partial_*} H_{p-1}(X^{p-1}) \xrightarrow{j_*} H_{p-1}(X^{p-1}, X^{p-2}) ,$$

where j is inclusion. The fact that $\partial^2 = 0$ follows from the fact that

$$H_{p-1}(X^{p-1}) \xrightarrow{j_*} H_{p-1}(X^{p-1}, X^{p-2}) \xrightarrow{\partial_*} H_{p-2}(X^{p-2})$$

is exact. The chain complex $\mathcal{D}(X) = \{D_p(X), \partial\}$ is called the *cellular chain complex* of X.

EXAMPLE 4.13.2 Consider the case where X is the space of a simplicial complex K, and the open cells of X are the open simplices of K. Let H_p denote ordinary simplicial homology. We compute $H_p(X^p, X^{p-1})$. The simplicial chain group $C_i(K^{(p)}, K^{(p-1)})$ vanishes if $i \neq p$, and it equals the chain group $C_p(K^{(p)}) = C_p(K)$ when $i = p$. Therefore, we have

$$H_p(X^p, X^{p-1}) = H_p(K^{(p)}, K^{(p-1)}) = C_p(K) .$$

Furthermore, the boundary operator in the cellular chain complex is just the ordinary simplicial boundary operator of K. It follows that, in this case at least, the cellular chain complex can be used to compute the homology of X.

Our goal is to prove in general that, if X is a CW complex, the cellular chain complex $\mathcal{D}(X)$ behaves very much like the simplicial chain complex $\mathcal{C}(K)$. In particular, we will show that the group $D_p(X)$ is free abelian with a basis consisting of, roughly speaking, the oriented p-cells. And we will show that the chain complex $\mathcal{D}(X)$ can be used to compute the singular homology of X.

We begin with a sequence of lemmas.

Lemma 4.13.3 *Given an open p-cell e_α of X, any characteristic map for e_α*

$$f_\alpha : (\mathbb{B}^p, \mathbb{S}^{p-1}) \to (\overline{e}_\alpha, \dot{e}_\alpha) ,$$

induces an isomorphism in relative homology.

Proof: If $p = 0$, the result is trivial. Let $p > 0$. The point $\mathbf{0}$ is the center of $\mathbb{B}^p$. Let $\widehat{e}_\alpha$ denote $f_\alpha(\mathbf{0})$. Note that, because f_α is a quotient map, so is its restriction

$$f'_\alpha : (\mathbb{B}^p \setminus \mathbf{0}) \to (\overline{e}_\alpha \setminus \widehat{e}_\alpha)$$

to the saturated open set $\mathbb{B}^p \setminus \mathbf{0}$. Now $\mathbb{S}^{p-1}$ is a deformation retract of $\mathbb{B}^p \setminus \mathbf{0}$, so this deformation retraction induces, via the quotient map

$$f'_\alpha \times i_I : (\mathbb{B}^p \setminus \mathbf{0}) \times I \to (\overline{e}_\alpha \setminus \widehat{e}_\alpha) \times I\,,$$

a deformation retraction of $\overline{e}_\alpha \setminus \widehat{e}_\alpha$ onto $\dot{e}_\alpha$. See Figure 4.13.1.

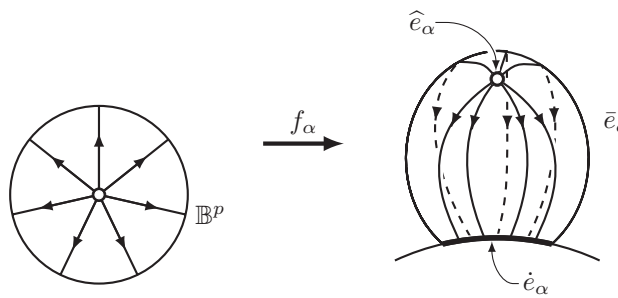


Figure 4.13.1

By Theorem 4.3.9, it follows that the horizontal inclusion maps on the left-hand side of the following diagram induce isomorphisms in homology:

$$\begin{array}{ccccc}
(\mathbb{B}^p, \mathbb{S}^{p-1}) & \longrightarrow & (\mathbb{B}^p, \mathbb{B}^p \setminus \mathbf{0}) & \longleftarrow & (\text{Int } \mathbb{B}^p, \text{Int } \mathbb{B}^p \setminus \mathbf{0}) \\
\downarrow f_\alpha & & \downarrow f_\alpha & & \downarrow f_\alpha \\
(\overline{e}_\alpha, \dot{e}_\alpha) & \longrightarrow & (\overline{e}_\alpha, \overline{e}_\alpha \setminus \widehat{e}_\alpha) & \longleftarrow & (e_\alpha, e_\alpha \setminus \widehat{e}_\alpha
\end{array}$$

Since the horizontal inclusion maps on the right-hand side of the diagram are excision maps, they also induce homology isomorphisms. [On the top line, one excises $\mathbb{S}^{p-1}$, while on the bottom line one excises $\dot{e}_\alpha$.] Now the map f_α : Int $\mathbb{B}^p \to e_\alpha$ is a homeomorphism that carries $\mathbf{0}$ to $\widehat{e}_\alpha$. Therefore the vertical map f_α on the right-hand side of the diagram induces a homology isomorphism. Our result follows. □

Lemma 4.13.4 *Let the map*

$$f : X^{p-1} \cup \sum B_\alpha \to X^p$$

express X^p as the adjunction space obtained from X^{p-1} and a topological sum of p-balls $\sum B_\alpha$ via a map $g : S_\alpha \to X^{p-1}$, where $S_\alpha = \text{Bd } B_\alpha$. Then f induces a homology isomorphism

$$H_i(\textstyle\sum B_\alpha, \sum S_\alpha) \cong H_i(X^p, X^{p-1})\,.$$

Proof: The proof is similar to that of the preceding lemma. The restriction f' of f to the space

$$X^{p-1} \cup \sum (B_\alpha \setminus \mathbf{0}_\alpha)\,,$$

where $\mathbf{0}_\alpha$ is the center of B_α, is a quotient map. Furthermore, there is a deformation retraction of this space onto $X^{p-1} \cup \sum S_\alpha$. This deformation retraction induces, via the quotient map $f' \times i_I$, a deformation retraction of $X^p \setminus \cup \widehat{e}_\alpha$ onto X^{p-1}, where $\widehat{e}_\alpha = f(\mathbf{0}_\alpha)$. One has the following diagram:

$$\begin{array}{ccccc}
\left(\sum B_\alpha, \sum S_\alpha\right) & \longrightarrow & \left(\sum B_\alpha, \sum (B_\alpha \setminus \mathbf{0}_\alpha)\right) & \longleftarrow & \left(\sum \text{Int } B_\alpha, \sum ((\text{Int } B_\alpha) \setminus \mathbf{0}_\alpha)\right) \\
\Big\downarrow f & & \Big\downarrow f & & \Big\downarrow f \\
(X^p, X^{p-1}) & \longrightarrow & (X^p, X^p \setminus \cup \widehat{e}_\alpha) & \longleftarrow & (\cup e_\alpha, \cup (e_\alpha \setminus \widehat{e}_\alpha))
\end{array}$$

The map on the right-hand side is a homeomorphism, being a one-to-one quotient map. The horizontal maps are inclusions, and they induce homology isomorphisms for the same reasons as before. □

Theorem 4.13.5 *The group $H_i(X^p, X^{p-1})$ vanishes for $i \neq p$, and is free abelian for $i = p$. If γ generates $H_p(\mathbb{B}^p, \mathbb{S}^p)$, then the elements $(f_\alpha)_*(\gamma)$ form a basis for $H_p(X^p, X^{p-1})$, as f_α ranges over a set of characteristic maps for the p-cells of X.*

Proof: The preceding lemma tells us that

$$H_i(X^p, X^{p-1}) \cong H_i(\sum B_\alpha, \sum S_\alpha)\,,$$

where $\sum B_\alpha$ is a topological sum of p-balls and $S_\alpha = \text{Bd } B_\alpha$. Because the sets B_α are disjoint open sets in $\sum B_\alpha$, this group is isomorphic to the direct sum $\oplus H_i(B_\alpha, S_\alpha)$. The theorem follows. □

Definition 4.13.6 Given a triple $X \supset A \supset B$ of spaces, one has a short exact sequence of chain complexes

$$0 \longrightarrow \frac{S_p(A)}{S_p(B)} \longrightarrow \frac{S_p(X)}{S_p(B)} \longrightarrow \frac{S_p(X)}{S_p(A)} \longrightarrow 0\,.$$

It gives rise to the following sequence, which is called the *exact homology sequence of a triple*:

$$\cdots \longrightarrow H_p(A, B) \longrightarrow H_p(X, B) \longrightarrow H_p(X, A) \longrightarrow H_{p-1}(A, B) \longrightarrow \cdots\,.$$

The existence of the exact homology sequence of a triple was the content of Theorem 3.3.8. The existence of the exact homology sequence of a triple can also be proved from the axioms for homology theory [see the exercises of

Section 3.5]. As you would expect, a continuous map $f : (X, A, B) \to (Y, C, D)$ induces a homomorphism of the corresponding exact homology sequences.

In the case $(X, A, B) = (X^p, X^{p-1}, X^{p-2})$, the boundary operator ∂_* in the above sequence equals the boundary operator ∂ of the cellular chain complex $\mathcal{D}(X)$. This follows from the fact that ∂_* commutes with the homomorphism j_* induced by inclusion:

$$\begin{array}{ccccc} \longrightarrow & H_p(X^p, X^{p-1}) & \xrightarrow{\partial_*} & H_{p-1}(X^{p-1}, \emptyset) & \longrightarrow \\ & = \downarrow & & \downarrow j_* & \\ \longrightarrow & H_p(X^p, X^{p-1}) & \xrightarrow{\partial_*} & H_{p-1}(X^{p-1}, X^{p-2}) & \longrightarrow \end{array}$$

Using this fact, we now prove that the cellular chain complex of X can be used to compute the homology of X. For later purposes, we are going to prove this theorem in a somewhat more general form. We shall assume that we have a space X that is written as the union of a sequence of subspaces

$$X_0 \subset X_1 \subset X_2 \subset \cdots .$$

Then we form the chain complex in which the p-dimensional chain group is $H_p(X^p, X^{p-1})$ and the boundary operator is the boundary homomorphism in the exact sequence of the triple (X^p, X^{p-1}, X^{p-2}). We shall show that under suitable hypotheses [which are satisfied in the case of a CW complex] this chain complex gives the homology of X.

Definition 4.13.7 If X is a space, a *filtration* of X is a sequence $X_0 \subset X_1 \subset \cdots$ of subspaces of X whose union is X. We say that X is *filtered* by the sequence of subspaces, and X together with its filtration is called a *filtered space*. If X and Y are filtered spaces, a continuous map $f : X \to Y$ such that $f(X_p) \subset Y_p$ for all p is said to be *filtration preserving*.

Theorem 4.13.8 *Let X be filtered by the subspaces $X_0 \subset X_1 \subset \cdots$, and let $X_i = \emptyset$ for $i < 0$. Assume that $H_i(X^p, X^{p-1}) = 0$ for $i \neq p$. Suppose also that given any compact set C in X, there is an n such that $C \subset X_n$. Let $\mathcal{D}(X)$ be the chain complex defined by setting $D_p(X) = H_p(X^p, X^{p-1})$ and letting the boundary operator be the boundary homomorphism ∂_* in the exact sequence of a triple. Then there is an isomorphism*

$$\lambda : H_p(\mathcal{D}(X)) \to H_p(X) .$$

It is natural with respect to homomorphisms induced by filtration-preserving continuous maps.

Proof: As motivation for the proof, let us consider the situation of Example 4.13.2, where X_p is the p-skeleton $K^{(p)}$ of a simplicial complex and H_p denotes simplicial homology. Then $\mathcal{D}(X) = \mathcal{C}(K)$ in this case, and the theorem holds. It is also true in this case that

$$H_p(X) = H_p(K^{(p+1)}, K^{(p-2)}) ,$$

because only chains of dimensions $p+1$, p, and $p-1$ are used in defining $H_p(K)$. We shall show in Steps **1** and **2** below that an analogous result holds in the present situation.

Step 1. We show that the homomorphism

$$i_* : H_p(X_{p+1}) \to H_p(X)$$

induced by inclusion is an isomorphism. For this purpose, we first note that the homomorphisms

$$H_p(X_{p+1}) \longrightarrow H_p(X_{p+2}) \longrightarrow H_p(X_{p+3}) \longrightarrow \cdots$$

induced by inclusion are isomorphisms. This follows by examining the exact sequence

$$H_{p+1}(X_{p+i+1}, X_{p+i}) \longrightarrow H_p(X_{p+i}) \longrightarrow H_p(X_{p+i+1}) \longrightarrow H_p(X_{p+i+1}, X_{p+i})$$

and noting that both end groups vanish if $i \geq 1$, by hypothesis.

We will now show that the fact that i_* is an isomorphism is a consequence of the compact support properties of homology. To show that i_* is surjective, let β be an element of $H_p(X)$. Choose a compact set C in X such that β is in the image of $H_p(C) \to H_p(X)$ under the homomorphism induced by inclusion. Since C is compact, $C \subset X_{p+k}$ for some k. Then β is the image of an element of $H_p(X_{p+k})$ in the diagram

$$H_p(X_{p+1}) \longrightarrow H_p(X_{p+k}) \longrightarrow H_p(X)\,.$$

Because the first of these homomorphisms is an isomorphism, β is the image of an element of $H_p(X_{p+1})$, as desired.

To show that i_* has kernel 0, suppose that $\beta \in H_p(X_{p+1})$ maps to zero in $H_p(X)$. There is a compact set C such that β maps to zero in $H_p(C)$. Again, C lies in X_{p+k} for some k. Then

$$H_p(X_{p+1}) \longrightarrow H_p(X_{p+k})$$

carries β to zero. Because this map is an isomorphism, we have $\beta = 0$.

Step 2. We now show that the homomorphism

$$j_* : H_p(X_{p+1}) \to H_p(X_{p+1}, X_{p-2})$$

induced by inclusion is an isomorphism.

This result will follow once we show that the homomorphisms

$$H_p(X_{p+1}, \emptyset) \longrightarrow H_p(X_{p+1}, X_0) \longrightarrow \cdots \longrightarrow H_p(X_{p+1}, X_{p-2})$$

induced by inclusion are isomorphisms. To prove this fact, consider the exact sequence of the triple (X_{p+1}, X_i, X_{i-1}):

$$H_p(X_i, X_{i-1}) \longrightarrow H_p(X_{p+1}, X_{i-1}) \longrightarrow H_p(X_{p+1}, X_i) \longrightarrow H_{p-1}(X_i, X_{i-1})\,.$$

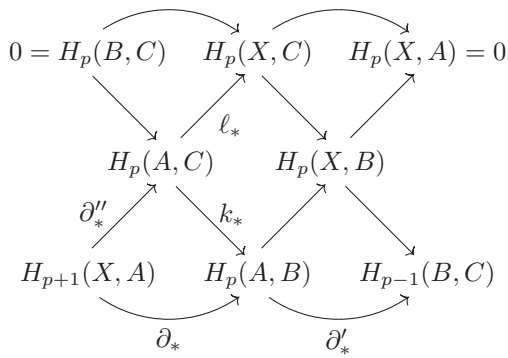
Both end groups vanish for $i \leq p-2$, by hypothesis. Therefore, the middle homomorphism is an isomorphism.

Step 3. We complete the proof of the theorem.

Given a quadruple $X \supset A \supset B \supset C$ of spaces, one has four possible exact sequences of a triple associated with this quadruple. They can be conveniently arranged in the form of four overlapping curves, as in the diagram below. We consider the special case

$$(X, A, B, C) = (X_{p+1}, X_p, X_{p-1}, X_{p-2}) .$$

In this case, the groups in the upper-left and upper-right corners of the diagram, $H_p(B, C)$ and $H_p(X, A)$ vanish by hypothesis.



Now k_* carries $H_p(A, C)$ isomorphically onto $\ker \partial'_*$. Furthermore, the map $\ell_* \circ k_*^{-1}$ carries $\ker \partial'_*$ *onto* $H_p(X, C)$ [because ℓ_* is surjective]. The kernel of $\ell_* \circ k_*^{-1}$ is just $\operatorname{im} \partial_*$ [because the kernel of ℓ_* equals $\operatorname{im} \partial''_*$]. Thus $\ell_* \circ k_*^{-1}$ induces an isomorphism

$$\frac{\ker \partial'_*}{\operatorname{im} \partial_*} \cong H_p(X, C) = H_p(X_{p+1}, X_{p-2}) .$$

[A more general result concerning this diagram was given in Exercise 1 of Section 3.5.]

Combining this result with those of Steps **1** and **2**, we obtain our desired isomorphism

$$H_p(X) \cong H_p(X_{p+1}) \cong H_p(X_{p+1}, X_{p-2}) \cong \frac{\ker \partial'_*}{\operatorname{im} \partial_*} .$$

The latter group equals $H_p(\mathcal{D}(X))$.

Naturality of the isomorphism is easy to check. If $f : X \to Y$ preserves filtrations, the first two of the preceding isomorphisms obviously commute with f_*. Furthermore, f_* carries the diagram of Step **3** for X into the corresponding diagram for Y. It follows that the third isomorphism commutes with f_* as well. □

We now prove an addendum to this theorem in the case where X is triangulable.

Theorem 4.13.9 *Let X be filtered by the subspaces $X_0 \subset X_1 \subset \cdots$. Suppose that X is the space of a simplicial complex K, and each subspace X_p is the space of a subcomplex of K of dimension at most p. Let H_i denote simplicial homology. Assume that $H_i(X_p, X_{p-1}) = 0$ for $i \neq p$. Then $H_p(X_p, X_{p-1})$ equals a subgroup of $C_p(X)$, and the isomorphism λ of the preceding theorem is induced by inclusion.*

Indeed, $H_p(X_p, X_{p-1})$ is the subgroup of $C_p(K)$ consisting of all p-chains of K carried by X_p whose boundaries are carried by X_{p-1}.

Proof: Any compact set in X lies in a finite subcomplex of K, so it lies in X_i for some i. Therefore, the hypotheses of the preceding theorem are satisfied. Since X_p contains no $(p+1)$-simplices, the homology group $H_p(X_p, X_{p-1})$ equals the group of relative p-cycles, which is the kernel of the homomorphism

$$\frac{C_p(X_p)}{C_p(X_{p-1})} \xrightarrow{\partial} \frac{C_{p-1}(X_p)}{C_{p-1}(X_{p-1})}\,.$$

Because X_{p-1} contains no p-simplices, the denominator on the left side is trivial. Thus $H_p(X_p, X_{p-1})$ equals the group of simplicial p-chains of K carried by X_p, whose boundaries are carried by X_{p-1}.

We must check that the isomorphism λ of the preceding theorem is induced by inclusion. Examining the preceding proof, we see that λ is obtained by taking an element of $\ker \partial'_*$ and mapping it into $H_p(X)$ according to the following diagram:

$$\begin{array}{ccccccc} H_p(X_p, X_{p-2}) & \xrightarrow{\ell_*} & H_p(X_{p+1}, X_{p-2}) & \xleftarrow[\cong]{j_*} & H_p(X_{p+1}) & \xrightarrow[\cong]{i_*} & H_p(X) \\ \cong\big\downarrow k_* & & & & & & \\ \ker \partial'_* & & & & & & \end{array}$$

Since each map is induced by inclusion, our theorem follows. □

We now see how strong the analogy is between the homology of simplicial complexes and the homology of CW complexes. Let us introduce some terminology that will make the analogy even stronger.

For each open p-cell e_α of the CW complex X, the group $H_p(\bar{e}_\alpha, \dot{e}_\alpha)$ is infinite cyclic. The two generators of this group will be called the two *orientations* of e_α. An *oriented p-cell* of X is an open p-cell e_α together with an orientation of e_α.

The cellular chain group $D_p(X) = H_p(X_p, X_{p-1})$ is a free abelian group. One obtains a basis for it by orienting each open p-cell e_α of X and passing to the corresponding element of $H_p(X_p, X_{p-1})$. [That is, by taking the image of the orientation under the homomorphism induced by inclusion

$$H_p(\bar{e}_\alpha, \dot{e}_\alpha) \to H_p(X_p, X_{p-1})\,.]$$

The homology of the chain complex $\mathcal{D}(X)$ is isomorphic, by our theorem, with the singular homology of X.

In the special case where X is a triangulable CW complex triangulated by a complex K, and H_p denotes simplicial homology, we interpret these comments as follows: The fact that X_p and X_{p-1} are subcomplexes of K implies that each open p-cell e_α is a union of open simplices of K, so $\overline{e}_\alpha$ is the polytope of a subcomplex of K. The group $H_p(\overline{e}_\alpha, \dot{e}_\alpha)$ equals the group of p-chains carried by $\overline{e}_\alpha$ whose boundaries are carried by $\dot{e}_\alpha$. This group is infinite cyclic, and either generator of this group is called a *fundamental cycle* for $(\overline{e}_\alpha, \dot{e}_\alpha)$.

The cellular chain group $D_p(X)$ equals the group of all simplicial p-chains of X carried by X_p whose boundaries are carried by X_{p-1}. Any such p-chain can be written uniquely as a finite linear combination of fundamental cycles for those pairs $(\overline{e}_\alpha, \dot{e}_\alpha)$ for which $\dim e_\alpha = p$.

Let us interpret these results in some familiar situations.

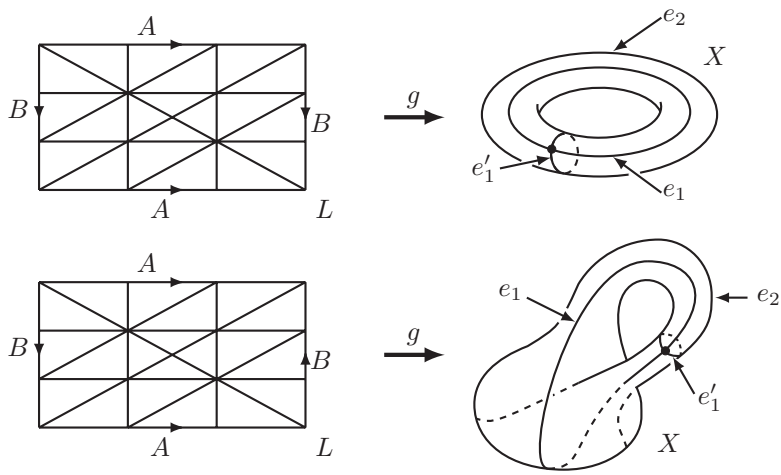


Figure 4.13.2

EXAMPLE 4.13.10 Let X denote either the torus or the Klein bottle, expressed as a quotient space of the rectangle L in the usual way. See Figure 4.13.2.

Then X is a triangulable CW complex, having one open 2-cell e_2, two open 1-cells e_1 and e_1' [that are the images of A and B, respectively], and one 0-cell e_0. Then

$$D_2(X) \cong \mathbb{Z}, \quad D_1(X) \cong \mathbb{Z} \oplus \mathbb{Z}, \quad D_0(X) \cong \mathbb{Z}.$$

Let us find specific generators for these chain groups. The 2-chain d of L that is the sum of all the 2-simplices of L, oriented counterclockwise, is by

inspection a cycle of $(L, \text{Bd } L)$. Because d is a multiple of no other cycle, it is a fundamental cycle for $(L, \text{Bd } L)$. By Lemma 4.13.3, $\lambda = g_\#(d)$ is a fundamental cycle for $(\overline{e}_2, \dot{e}_2)$.

Let c_1 be the sum of the 1-simplices along the top of L oriented as indicated in Figure 4.13.3. Let c_2, c_3, and c_4 denote chains along the other edges of L, as indicated. Now $w_1 = g_\#(c_1)$ is a fundamental cycle for $(\overline{e}_1, \dot{e}_1)$. So is $g_\#(c_3)$, of course. Similarly, $z_1 = g_\#(c_2)$ is a fundamental cycle for $(\overline{e}'_1, \dot{e}'_1)$, as is $g_\#(c_4)$.

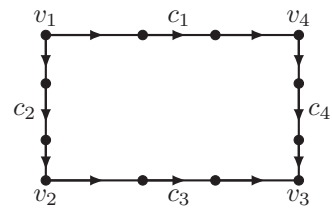


Figure 4.13.3

In terms of these basis elements, it is easy to compute the boundary operator in the cellular chain complex $\mathcal{D}(X)$. We first compute ∂ in the complex L as follows:

$$\partial c_1 = v_4 - v_1, \qquad \partial c_2 = v_2 - v_1,$$
$$\partial d = -c_1 + c_2 + c_3 - c_4.$$

Applying $g_\#$, we see that $\partial w_1 = g_\#(\partial c_1) = 0$ and $\partial z_1 = g_\#(\partial c_2) = 0$ for both the torus and the Klein bottle. In the case of the torus, $\partial\gamma = g_\#(\partial d) = 0$ because $g_\#(c_1) = g_\#(c_3)$ and $g_\#(c_2) = g_\#(c_4)$. In the case of the Klein bottle,

$$\partial\gamma = g_\#(\partial d) = 2g_\#(c_2) = 2z_1$$

because $g_\#(c_1) = g_\#(c_3)$ and $g_\#(c_2) = -g_\#(c_4)$.

Thus the homology of the cellular chain complex $\mathcal{D}(X)$ when X is the torus is

$$H_2(X) \cong \mathbb{Z}, \quad H_1(X) \cong \mathbb{Z} \oplus \mathbb{Z}, \quad H_0(X) \cong \mathbb{Z},$$

while in the case of the Klein bottle, it is

$$H_2(X) = 0, \quad H_1(X) \cong \mathbb{Z} \oplus \mathbb{Z}/2, \quad H_0(X) \cong \mathbb{Z}.$$

Of course, we have carried out these same computations before. But now their justification comes from our general theorems about CW complexes, rather than from the ad hoc arguments we used back in Section 1.7. It is in this sense that our results about CW complexes make systematic the ad hoc computational methods we studied there.

EXAMPLE 4.13.11 Let $\mathbb{S}^n$ be an n-sphere. Assume $n > 1$ for convenience. We can make $\mathbb{S}^n$ into a CW complex having one open cell in dimension n and one cell in dimension 0. It follows that the cellular chain complex of $\mathbb{S}^n$ is infinite cyclic in dimensions n and 0, and vanishes otherwise. Therefore $H_n(\mathbb{S}^n) \cong \mathbb{Z}$ and $H_0(\mathbb{S}^n) \cong \mathbb{Z}$, while $H_i(\mathbb{S}^n) = 0$ for $i \notin \{0, n\}$. A similar computation applies when $n = 1$.

Exercise 4 will ask you to prove the following theorem. But it will also offer some guidance.

Theorem 4.13.12 *Let $\mathcal{C} = \{C_p, \partial\}$ be a non-negative free chain complex such that $H_0(\mathcal{C})$ is free and non-trivial. Then there is a CW complex X whose cellular chain complex is isomorphic to $\mathcal{C}$.*

EXERCISES

1. Recompute the homology of the n-fold connected sums

$$T \# T \# \cdots \# T \quad \text{and} \quad \mathbb{P}^2 \# \mathbb{P}^2 \# \cdots \# \mathbb{P}^2$$

by expressing them as triangulable CW complexes and finding the corresponding cellular chain complexes.

2. Let A be a closed subset of X. Suppose that A is a deformation retract of an open set in X. If X/A is the space obtained by collapsing A to a point, show that

$$H_p(X, A) \cong \widetilde{H}_p(X/A)\,.$$

[**Hint:** Examine the proof of Lemma 4.13.3.]

3. Let X be a CW complex. Let A be a subcomplex. Show that inclusion induces a monomorphism $D_p(A) \to D_p(X)$. The quotient $D_p(X)/D_p(A)$ is denoted $D_p(X, A)$.

(a) Show that if X is a triangulable CW complex, $\mathcal{D}(X, A)$ can be used to compute the simplicial homology of (X, A). [**Hint:** Use the long exact sequence to show that inclusion $D_p(X, A) \to C_p(X, A)$ induces a homology isomorphism.]

(b) Show that in general $\mathcal{D}(X, A)$ can be used to compute the singular homology of (X, A). [**Hint:** Establish that $\mathcal{D}(X, A)$ is isomorphic to the chain complex with pth chain group $H_p(X^p \cup A, X^{p-1} \cup A)$ and with its boundary operator coming from the exact sequence of a triple. Let $X_p = X^p \cup A$ for all integers p. Repeat the proof of Theorem 4.13.8, replacing $H_i(X_p)$ by $H_i(X_p, A)$ and $H_i(X)$ by $H_i(X, A)$.]

4. * Follow the suggested steps to prove Theorem 4.13.12.

Steps:

(a) Show that if $n \geq 1$, given an n-simplex σ and a homomorphism

$$\phi : H_n(\sigma, \text{Bd } \sigma) \to H_n(\mathbb{S}^n, x_0)\,,$$

ϕ is the homomorphism induced by some continuous map. [See Exercise 3 of Section 3.4.]

(b) Show that if X is a CW complex consisting of a collection of n-spheres with a point p in common [$n > 0$], and if $\alpha \in H_n(X, p)$, then there is a map $f : (\mathbb{S}^n, x_0) \to (X, p)$ whose induced homomorphism carries a generator of $H_n(\mathbb{S}^n, x_0)$ to α.

(c) Show that for $p > 0$ one can write $C_p = U_p \oplus Z_p$, where Z_p is the group of p-cycles, and for $p = 0$ one can write $C_0 = \partial U_1 \oplus A$, where A is non-trivial.

(d) Complete the proof.

5. Let $G_0, G_1, \ldots$ be a sequence of abelian groups with G_0 free and non-trivial. Assuming Exercise 4, show that there is a CW complex X such that $H_i(X) \cong G_i$ for all i.

4.14 *Application: Projective Spaces and Lens Spaces

We now apply the theory of CW complexes to compute the homology of certain spaces that are of particular importance in topology and geometry—the projective spaces. We also study those classical 3-manifolds called the lens spaces. The results of this section will be used when we compute the cohomology rings of these spaces in Sections 8.8 and 8.9.

Definition 4.14.1 Let us introduce an equivalence relation on the n-sphere $\mathbb{S}^n$ by defining $x \sim -x$ for each $x \in \mathbb{S}^n$. The resulting quotient space is called *real projective n-space* and is denoted $\mathbb{P}^n$. We will usually drop the designation "real," when the context does not require it.

We show that the quotient map $p : \mathbb{S}^n \to \mathbb{P}^n$ is a closed map. If A is closed in $\mathbb{S}^n$, then the saturation $p^{-1}(p(A))$ of A equals the set $A \cup a(A)$, where $a : \mathbb{S}^n \to \mathbb{S}^n$ is the antipodal map. Because a is a homeomorphism, $A \cup a(A)$ is closed in $\mathbb{S}^n$, so [by the definition of a quotient space] the set $p(A)$ is closed in $\mathbb{P}^n$. Therefore, $\mathbb{P}^n$ is Hausdorff [in fact normal].

If we consider $\mathbb{R}^n$ to be the set of all real sequences $(x_1, x_2, \ldots)$ such that $x_i = 0$ for $i > n$, then $\mathbb{R}^n \subset \mathbb{R}^{n+1}$. As a result, $\mathbb{S}^{n-1} \subset \mathbb{S}^n$. In fact, $\mathbb{S}^{n-1}$ is

the intersection of $\mathbb{S}^n$ with the plane $x_{n+1} = 0$. Now the equivalence relation $x \sim -x$ is the same in $\mathbb{S}^{n-1}$ as it is in $\mathbb{S}^n$. Therefore, $\mathbb{P}^{n-1} \subset \mathbb{P}^n$. In fact, $\mathbb{P}^{n-1}$ is a closed *subspace* of $\mathbb{P}^n$. For if C is a subset of $\mathbb{P}^{n-1}$, then $p^{-1}(C)$ is closed in $\mathbb{S}^{n-1}$ if and only if it is closed in $\mathbb{S}^n$. Hence C is closed in $\mathbb{P}^{n-1}$ if and only if it is closed in $\mathbb{P}^n$.

Theorem 4.14.2 *The space $\mathbb{P}^n$ is a CW complex having one cell in each dimension $0 \leq j \leq n$. Its j-skeleton in $\mathbb{P}^j$.*

Proof: The space $\mathbb{P}^0$ is obtained from the 2-point space $\mathbb{S}^0$ by identifying those two points. Thus $\mathbb{P}^0$ consists of a single point.

We proceed by induction. Suppose we restrict the map $p : \mathbb{S}^n \to \mathbb{P}^n$ to the closed upper hemisphere E^n_+ of $\mathbb{S}^n$. Because $\mathbb{S}^n$ is compact and $\mathbb{P}^n$ is Hausdorff, the map $p'|E^n_+$ is a quotient map, and it maps E^n_+ *onto* $\mathbb{P}^n$ because each equivalence class $\{x, -x\}$ contains at least one point of E^n_+. Its restriction to the open upper hemisphere Int E^n_+ is also a quotient map. Being one-to-one, it is a homeomorphism of Int E^n_+ with $\mathbb{P}^n \setminus \mathbb{P}^{n-1}$. Thus $\mathbb{P}^n \setminus \mathbb{P}^{n-1}$ is an open n-cell. Call it e_n.

Now the map p' carries Bd $E^n_+ = \mathbb{S}^{n-1}$ onto $\mathbb{P}^{n-1}$, which by the induction hypothesis is the union of finitely many open cells of dimension less than n. Thus when we identify E^n_+ with $\mathbb{B}^n$, the map p' becomes a characteristic map for e_n, and the result follows. □

Note that $\mathbb{P}^1$, having one open 1-cell and one 0-cell, is homeomorphic to $\mathbb{S}^1$.

Definition 4.14.3 Consider the increasing sequence $\mathbb{P}^0 \subset \mathbb{P}^1 \subset \dots$ of projective spaces. Their coherent union is denoted by $\mathbb{P}^\infty$, and is called *infinite-dimensional real projective space*. Again, the designation "real" will often be dropped.

Now we perform an analogous construction with the complex numbers replacing the real numbers. Let $\mathbb{C}^n$ be the space of all complex sequences $z = (z_1, z_2, \dots)$ such that $z_i = 0$ for $i > n$. Then $\mathbb{C}^n \subset \mathbb{C}^{n+1}$ for all n. There is an obvious homeomorphism $\rho : \mathbb{C}^{n+1} \to \mathbb{R}^{2n+2}$, which we call the "realification operator," defined by

$$\rho(z_1, z_2, \dots) = (\operatorname{Re} z_1, \operatorname{Im} z_1, \operatorname{Re} z_2, \operatorname{Im} z_2, \dots),$$

where $\operatorname{Re} z_i$ and $\operatorname{Im} z_i$ are the real and imaginary parts of z_i, respectively. Let us define

$$|z| = \|\rho(z)\| = \left[\sum((\operatorname{Re} z_i)^2 + (\operatorname{Im} z_i)^2)\right]^{1/2} = \left[\sum z_i \, \overline{z}_i\right]^{1/2},$$

where $\overline{z}_i$ is the complex conjugate of z_i The subspace of $\mathbb{C}^{n+1}$ consisting of all points z with $|z| = 1$ is called the *complex n-sphere*. It corresponds to the sphere $\mathbb{S}^{2n+1}$ under the operator ρ, so we use the symbol $\mathbb{S}^{2n+1}$ to denote it, whether we consider it in $\mathbb{C}^{n+1}$ or in $\mathbb{R}^{2n+2}$.

Definition 4.14.4 Let us introduce an equivalence relation in the complex n-sphere $\mathbb{S}^{2n+1} \subset \mathbb{C}^{n+1}$ by defining

$$(z_1, z_2, \dots, z_{n+1}, 0, \dots) \sim (\lambda z_1, \lambda z_2, \dots, \lambda z_{n+1}, 0, \dots)$$

for each complex number λ with $|\lambda| = 1$. The resulting quotient space is called *complex projective n-space* and is denoted $\mathbb{C}P^n$.

We show that the quotient map $p : \mathbb{S}^{2n+1} \to \mathbb{C}P^n$ is a closed map. If A is a closed set in $\mathbb{S}^{2n+1}$, then $p^{-1}p(A)$ is the image of $\mathbb{S}^1 \times A$ under the scalar multiplication map $(\lambda, z) \to \lambda z$. Since this map is continuous and $\mathbb{S}^1 \times A$ is compact, its image is compact and therefore closed in $\mathbb{S}^{2n+1}$. Then $p(A)$ is closed in $\mathbb{C}P^n$, by definition.

It follows that $\mathbb{C}P^n$ is Hausdorff [in fact normal].

As in the real case, $\mathbb{C}^n \subset \mathbb{C}^{n+1}$ for all n, so that $\mathbb{S}^{2n-1} \subset \mathbb{S}^{2n+1}$. Then, passing to quotient spaces, we have $\mathbb{C}P^{n-1} \subset \mathbb{C}P^n$. In fact, $\mathbb{C}P^{n-1}$ is a closed subspace of $\mathbb{C}P^n$, by the same argument as before.

Said differently, the elements of $\mathbb{C}P^n$ are equivalence classes of sequences $(z_1, z_2, \dots)$ of complex numbers such that $z_i = 0$ for $i > n+1$. If one sequence in an equivalence class satisfies the equation $z_{n+1} = 0$, so does every member of the equivalence class, and the class in question belongs to $\mathbb{C}P^{n-1}$, by definition.

Theorem 4.14.5 *The space $\mathbb{C}P^n$ is a CW complex of dimension $2n$. It has one open cell in each even dimension $2j$ for $0 \le 2j \le 2n$, and $\mathbb{C}P^j$ is its $2j$-skeleton.*

Proof: The space $\mathbb{C}P^0$ is a single point. In general, we show that $\mathbb{C}P^n \setminus \mathbb{C}P^{n-1}$ is an open $2n$-cell, which we denote e_{2n}. Consider the subset $\mathbb{S}^{2n+1}$ consisting of all points $z = (z_1, z_2, \dots, z_{n+1}, 0, \dots)$ *with z_{n+1} real.* Under the realification operator ρ, this corresponds to the set of all points of $\mathbb{R}^{2n+2}$ of the form

$$(x_1, y_1, x_2, y_2, \dots, x_n, y_n, x_{n+1}, 0, 0, \dots)$$

having euclidean norm 1. *This is simply the unit sphere $\mathbb{S}^{2n}$ in $\mathbb{R}^{2n+1}$,* and it is the equator of $\mathbb{S}^{2n+1}$. If we further restrict the set by requiring z_{n+1} to be real *and* non-negative, then $x_{n+1} \ge 0$ and we obtain the upper hemisphere E^{2n}_+ of this equatorial sphere. The restriction p' of p to E^{2n}_+ is, of course, a quotient map, because the domain is compact and the range is Hausdorff. The boundary of E^{2n}_+ is the sphere $\mathbb{S}^{2n-1}$ obtained by setting $z_{n+1} = 0$. The map p' carries $\mathbb{S}^{2n-1}$ *onto* $\mathbb{C}P^{n-1}$. We shall show that p' maps Int E^{2n}_+ bijectively onto $e_{2n} = \mathbb{C}P^n \setminus \mathbb{C}P^{n-1}$. Then, since it is a one-to-one quotient map, it is a homeomorphism. It follows that e_{2n} is an open $2n$-cell, and that p' is a characteristic map for e_{2n}.

First we show that the map p' : Int $E^{2n}_+ \to e_{2n}$ is surjective. Given a point of $e_{2n} = \mathbb{C}P^n \setminus \mathbb{C}P^{n-1}$, it equals $p(z)$ for some point $z =$

$(z_1, z_2, \ldots, z_{n+1}, 0, \ldots)$ of $\mathbb{S}^{2n+1} \setminus \mathbb{S}^{2n-1}$. Then $|z| = 1$ and $z_{n+1} \neq 0$. Using the polar representation of complex numbers, we write $z_{n+1} = r\, e^{i\theta}$, where $r > 0$. Set $\lambda = e^{-i\theta}$. Then we have

$$\lambda z = (\lambda z_1, \lambda z_2, \ldots, \lambda z_n, r, 0 \ldots) \in E^{2n}_+ .$$

Now $|\lambda| = 1$, so $p(\lambda z) = p(z)$. Thus we have $p(z) \in p(\text{Int } E^{2n}_+)$.

Next we show that the map $p' : \text{Int } E^{2n}_+ \to e_{2n}$ is injective. Suppose that $p(z) = p(w)$, where z and w are in Int E^{2n}_+. Then $w = \lambda z$ with $|\lambda| = 1$. In particular, $w_{n+1} = \lambda z_{n+1}$. Since w_{n+1} and z_{n+1} are real and positive, so is λ. Since $|\lambda| = 1$, we conclude that $\lambda = 1$. Thus $w = z$, as desired. □

We have noted that $\mathbb{C}P^0$ is a single point. The space $\mathbb{C}P^1$ is obtained by attaching a 2-cell to $\mathbb{C}P^0$; therefore, $\mathbb{C}P^1$ is homeomorphic to $\mathbb{S}^2$.

Definition 4.14.6 Consider the increasing sequence $\mathbb{C}P^0 \subset \mathbb{C}P^1 \subset \ldots$ of projective spaces. Their coherent union is denoted by $\mathbb{C}P^\infty$, and is called *infinite-dimensional complex projective space* . By Theorem 4.12.15, it is a CW complex with one cell in each non-negative even dimension, and $\mathbb{C}P^n$ is its $2n$-skeleton.

The homology of complex projective space is exceedingly easy to compute:

Theorem 4.14.7 *The group $H_i(\mathbb{C}P^n)$ is infinite cyclic if i is even and $0 \leq i \leq 2n$, and it vanishes otherwise. The group $H_i(\mathbb{C}P^\infty)$ is infinite cyclic if i is even and $i \geq 0$, and it vanishes otherwise.*

Proof: The cellular chain group $D_i(\mathbb{C}P^n)$ is infinite cyclic if i is even and $0 \leq i \leq 2n$, and otherwise it vanishes. Therefore, every chain of this chain complex is a cycle, and no chain bounds. A similar computation applies to $\mathbb{C}P^\infty$. □

The computation of the homology of real projective space is much more difficult. The cellular chain group $D_k(\mathbb{P}^n)$ is infinite cyclic for $0 \leq k \leq n$, so we cannot avoid a detailed examination of the boundary operator in the cellular complex. Since the open k-cell e_k of $\mathbb{P}^n$ equals $\mathbb{P}^k \setminus \mathbb{P}^{k-1}$, and $\dot{e}_k = \mathbb{P}^{k-1}$, we have $D_k(\mathbb{P}^n) = H_k(\mathbb{P}^k, \mathbb{P}^{k-1})$. Thus we must compute the boundary operator

$$\partial_* : H_{k+1}(\mathbb{P}^{k+1}, \mathbb{P}^k) \to H_k(\mathbb{P}^k, \mathbb{P}^{k-1}) .$$

First we prove a lemma.

Lemma 4.14.8 *Let $p : \mathbb{S}^n \to \mathbb{P}^n$ be the quotient map, where $n \geq 1$. Let $j : \mathbb{P}^n \to (\mathbb{P}^n, \mathbb{P}^{n-1})$ be inclusion. The composite homomorphism*

$$H_n(\mathbb{S}^n) \xrightarrow{p_*} H_n(\mathbb{P}^n) \xrightarrow{j_*} H_n(\mathbb{P}^n, \mathbb{P}^{n-1})$$

is zero if n is even, and multiplication by 2 if n is odd.

Chain-level proof: We assume that $\mathbb{S}^n$ is triangulated so that the antipodal map $a : \mathbb{S}^n \to \mathbb{S}^n$ is simplicial, and that $\mathbb{P}^n$ is triangulated so that $p : \mathbb{S}^n \to \mathbb{P}^n$ is simplicial. [Lemma 4.14.11 will show that this is possible. For now, we assume it.]

We use simplicial homology. Let c_n be a fundamental cycle for $(E^n_+, \mathbb{S}^{n-1})$. Then $p_\#(c_n)$ is a fundamental cycle for $(\mathbb{P}^n, \mathbb{P}^{n-1})$, by Lemma 4.13.3.

Consider the following chain of $\mathbb{S}^n$:

$$\gamma_n = c_n + (-1)^{n-1} a_\#(c_n)\,.$$

It is a cycle, for its boundary is

$$\begin{aligned} \partial\gamma_n &= \partial c_n + (-1)^{n-1} a_\#(\partial c_n) \\ &= \partial c_n + (-1)^{2n-1} \partial c_n = 0\,. \end{aligned}$$

This equation follows from the fact that $a : \mathbb{S}^{n-1} \to \mathbb{S}^{n-1}$ has degree $(-1)^n$, so that $a_\#$, mapping the $(n-1)$-cycle group of $\mathbb{S}^{n-1}$ to itself, equals multiplication by $(-1)^n$. Furthermore, γ_n is a multiple of no other cycle of $\mathbb{S}^n$, for its restriction to E^n_+ is c_n, which is a fundamental cycle for $(E^n_+, \mathbb{S}^{n-1})$. Thus γ_n is a fundamental cycle for $\mathbb{S}^n$.

Finally, we compute

$$p_\#(\gamma_n) = p_\#(c_n + (-1)^{n-1} a_\#(c_n))\,.$$

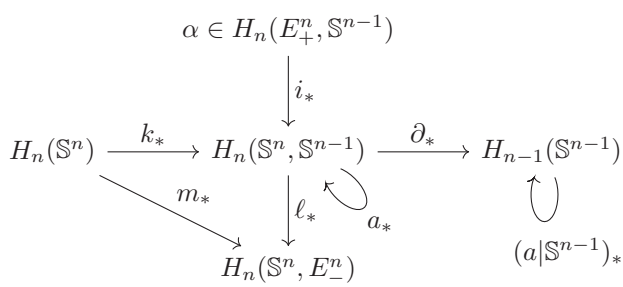
Since $p \circ a = p$, we conclude that

$$p_\#(\gamma_n) = [1 + (-1)^{n-1}] p_\#(c_n)\,.$$

Since γ_n is a fundamental cycle for $\mathbb{S}^n$ and $p_\#(c_n)$ is a fundamental cycle for $(\mathbb{P}^n, \mathbb{P}^{n-1})$, the lemma follows.

Homology-level proof: This proof is similar to the preceding one, except that the computations are carried out on the homology level rather than the chain level. It may seem more complicated, but the ideas are basically the same.

Step 1. Consider the diagram



where i, k, ℓ, and m are inclusions and a is the antipodal map. Let α be a generator of $H_n(E^n_+, \mathbb{S}^{n-1})$. Consider the element

$$\gamma = i_*(\alpha) + (-1)^{n-1} a_* i_*(\alpha)$$

of $H_n(E^n_+, \mathbb{S}^{n-1})$. We will show that there is a generator β of $H_n(\mathbb{S}^n)$ such that $k_*(\beta) = \gamma$.

First, we show that $\gamma = k_*(\beta)$ for some β. Note that

$$\partial_* \gamma = \partial_* i_*(\alpha) + (-1)^{n-1} (a|\mathbb{S}^{n-1})_* (\partial_* i_*(\alpha))$$

because ∂_* is natural. This homology class vanishes, because $(a|\mathbb{S}^{n-1})_*$ equals multiplication by $(-1)^n$. By horizontal exactness, there exists *some* $\beta \in H_n(\mathbb{S}^n)$ such that $k_*(\beta) = \gamma$.

Second, we show that β generates $H_n(\mathbb{S}^n)$. Now m_* is an isomorphism, by the long exact reduced homology sequence. [Recall $n \geq 1$.] Therefore, it suffices to show that $m_*(\beta)$ generates $H_n(\mathbb{S}^n, E^n_-)$. Since $m_*(\beta) = \ell_*(\gamma)$, we shall compute

$$\ell_*(\gamma) = \ell_* i_*(\alpha) + (-1)^{n-1} \ell_* a_* i_*(\alpha) \,.$$

For this purpose, we note first that $\ell_* i_*(\alpha)$ generates $H_n(\mathbb{S}^n, E^n_-)$, because the inclusion map $\ell \circ i$ induces a homology isomorphism. [See the proof of Theorem 4.4.11.] Then we note that $\ell_* a_* i_*(\alpha)$ is trivial. This fact is a consequence of the following commutative diagram, where the unlabeled maps are inclusions:

$$\begin{array}{ccc}
(E^n_+, \mathbb{S}^{n-1}) & \xrightarrow{i} & (\mathbb{S}^n, \mathbb{S}^{n-1}) \\
\downarrow{\scriptstyle a|E^n_+} & & \downarrow{\scriptstyle a} \\
(E^n_-, \mathbb{S}^{n-1}) & \longrightarrow & (\mathbb{S}^n, \mathbb{S}^{n-1}) \\
\downarrow & & \downarrow{\scriptstyle \ell} \\
(E^n_-, E^n_-) & \longrightarrow & (\mathbb{S}^n, E^n_-)
\end{array}$$

It follows that $\ell_*(\gamma)$ generates $H_n(\mathbb{S}^n, E^n_-)$, as desired.

Step 2. We now prove the lemma. Consider the following commutative diagram:

$$\begin{array}{ccccc}
 & & \alpha \in H_n(E^n_+, \mathbb{S}^{n-1}) & & \\
 & & \downarrow{\scriptstyle i_*} & & \\
\beta \in H_n(\mathbb{S}^n) & \xrightarrow{k_*} & H_n(\mathbb{S}^n, \mathbb{S}^{n-1}) & \xrightarrow{a_*} & H_n(\mathbb{S}^n, \mathbb{S}^{n-1}) \\
\downarrow{\scriptstyle p_*} & & \downarrow{\scriptstyle p_*} & \swarrow{\scriptstyle p_*} & \\
H_n(\mathbb{P}^n) & \xrightarrow{j_*} & H_n(\mathbb{P}^n, \mathbb{P}^{n-1}) & &
\end{array}$$

Choose generators α and β as in Step **1**. We wish to compute $j_*p_*(\beta)$. The map

$$p \circ i : (E^n_+, \mathbb{S}^{n-1}) \to (\mathbb{P}^n, \mathbb{P}^{n-1})$$

is a characteristic map for the n-cell of the CW complex $\mathbb{P}^n$, so it induces an isomorphism in homology. Thus $p_*i_*(\alpha)$ is a generator of $H_n(\mathbb{P}^n, \mathbb{P}^{n-1})$. We compute

$$\begin{aligned} j_*p_*(\beta) &= p_*k_*(\beta) = p_*(i_*(\alpha) + (-1)^{n-1}a_*i_*(\alpha)) \\ &= [1 + (-1)^{n-1}]p_*i_*(\alpha)\,. \end{aligned}$$

Here we use the fact that $p \circ a = p$. The lemma follows. □

Theorem 4.14.9 *The homomorphism*

$$\partial_* : H_{n+1}(\mathbb{P}^{n+1}, \mathbb{P}^n) \to H_n(\mathbb{P}^n, \mathbb{P}^{n-1})$$

is zero if n is even, and carries a generator to twice a generator if n is odd.

Proof: The map $p' : (E^{n+1}_+, \mathbb{S}^n) \to (\mathbb{P}^{n+1}, \mathbb{P}^n)$ is a characteristic map for the open $(n+1)$-cell of $\mathbb{P}^{n+1}$; therefore it induces a homology isomorphism. Consider the commutative diagram

$$\begin{array}{ccccc} H_{n+1}(E^{n+1}_+, \mathbb{S}^n) & \xrightarrow[\cong]{\partial_*} & H_n(\mathbb{S}^n) & & \\ \cong\big\downarrow p'_* & & \big\downarrow p_* & & \\ H_{n+1}(\mathbb{P}^{n+1}, \mathbb{P}^n) & \xrightarrow{\partial_*} & H_n(\mathbb{P}^n) & \xrightarrow{j_*} & H_n(\mathbb{P}^n, \mathbb{P}^{n-1}) \end{array}$$

The map ∂_* at the top of the diagram is an isomorphism, by the long exact reduced homology sequence of $(E^{n+1}_+, \mathbb{S}^n)$. By the preceding lemma, $(j \circ p)_*$ is zero if n is even, and multiplication by 2 if n is odd. The theorem follows. □

Theorem 4.14.10 *The homology of projective space is as follows:*

$$\widetilde{H}_i(\mathbb{P}^{2n+1}) \cong \begin{cases} \mathbb{Z}/2 & \text{if } i \text{ is odd and } 0 < i < 2n+1, \\ \mathbb{Z} & \text{if } i = 2n+1, \\ 0 & \text{otherwise.} \end{cases}$$

$$\widetilde{H}_i(\mathbb{P}^{2n}) \cong \begin{cases} \mathbb{Z}/2 & \text{if } i \text{ is odd and } 0 < i < 2n, \\ 0 & \text{otherwise.} \end{cases}$$

$$\widetilde{H}_i(\mathbb{P}^{\infty}) \cong \begin{cases} \mathbb{Z}/2 & \text{if } i \text{ is odd and } 0 < i, \\ 0 & \text{otherwise.} \end{cases}$$

Proof: The cellular chain group $D_i(\mathbb{P}^\infty)$ is infinite cyclic for $i \geq 0$, and the augmented chain complex has the form

$$\cdots \longrightarrow D_{2i}(\mathbb{P}^\infty) \xrightarrow{2} D_{2i-1}(\mathbb{P}^\infty) \xrightarrow{0} \cdots \xrightarrow{0} D_0(\mathbb{P}^\infty) \xrightarrow{1} \mathbb{Z}\,.$$

There are no cycles in even dimensions. In odd dimensions, every element is a cycle, and even multiples of the generator bound. Thus $\tilde{H}_i(\mathbb{P}^\infty)$ is of order 2 if i is positive and odd, and vanishes otherwise.

The computations for $\mathbb{P}^{2n}$ and $\mathbb{P}^{2n+1}$ are similar. □

Now we prove the lemma we used in the earlier chain-level proof.

Lemma 4.14.11 *The spaces $\mathbb{S}^n$ and $\mathbb{P}^n$ can be triangulated so that the antipodal map $a : \mathbb{S}^n \to \mathbb{S}^n$ and the projection map $p : \mathbb{S}^n \to \mathbb{P}^n$ are simplicial.*

Proof: Step 1. We show first that there is a complex L in $\mathbb{R}^{n+1}$ such that each reflection map

$$\rho_i(x_1, x_2, \ldots, x_i, \ldots, x_{n+1}) = (x_1, x_2, \ldots, -x_i, \ldots, x_{n+1})$$

induces a linear isomorphism of L with itself, and we show that there is a triangulation $k : |L| \to \mathbb{S}^n$ that commutes with each map ρ_i.

The result is trivial for $n = 0$. Assume that K is a complex in $\mathbb{R}^n$ and $h : |K| \to \mathbb{S}^{n-1}$ is a triangulation satisfying our hypotheses. Let $w_0 = (0, 0, \ldots, 0, 1)$ and $w_1 = (0, 0, \ldots, 0, -1)$ in $\mathbb{R}^{n+1}$. Let $L = (w_0 * K) \cup (w_1 * K)$. Then ρ_i induces a linear isomorphism of L with itself, for $i = 1, 2, \ldots, n+1$. The triangulation defined by

$$k(y) = (\sqrt{1 - t^2}\, h(x), t)$$

if $y = (1 - t)x + tw_0$, and

$$k(y) = (\sqrt{1 - t^2}\, h(x), -t)$$

if $y = (1 - t)x + tw_1$, commutes with each ρ_i. [This is the same triangulation we used in the proof of Theorem 2.9.8.]

Step 2. Let $k : |L| \to \mathbb{S}^n$ be the triangulation of Step **1**. The antipodal map commutes with k, and induces a linear isomorphism of L with itself. It also induces a linear isomorphism of $\mathrm{sd}^N\ L$ with itself, for any fixed N.

Let us choose N large enough that for any vertex v of $\mathrm{sd}^N\ L$, the closed stars of v and $a(v)$ are disjoint. Then we can use the "vertex-labeling" device of Section 1.4 to construct a complex whose underlying space is homeomorphic to $\mathbb{P}^n$ as follows: Let us label the vertices of $\mathrm{sd}^N\ L$, giving v and $a(v)$ the same label for each vertex v. Let $g : |\mathrm{sd}^N\ L| \to |M|$ be the quotient map obtained from this labeling. Then the map g will identify x and $a(x)$ for each $x \in |L|$, and will identify x with no other point of $|L|$. Because the homeomorphism k commutes with a, it induces a homeomorphism of $|M|$ with $\mathbb{P}^n$ that is our desired triangulation of $\mathbb{P}^n$. □

As a further application of these techniques, we now define a certain class of 3-dimensional manifolds called the "lens spaces" and compute their homology. It is of interest that they form one of the few classes of spaces that have been completely classified up to homeomorphism and up to homotopy type. We discuss this classification later.

Definition 4.14.12 Let n and k be relatively prime positive integers. We construct the lens space $L(n,k)$ as a quotient space of the ball $\mathbb{B}^3$ as follows: Write the general point of $\mathbb{B}^3$ in the form (z,t), where z is complex, t is real, and $|z|^2 + t^2 \leq 1$. Let $\lambda = \exp(2\pi i/n)$. Define $f : \mathbb{S}^2 \to \mathbb{S}^2$ by the equation

$$f(x) = (\lambda^k z, -t)\,.$$

Let us identify each point $x = (z,t)$ of the lower hemisphere E^2_- of $\mathbb{S}^2 = \text{Bd}\,\mathbb{B}^3$ with the point $f(x)$ of the upper hemisphere E^2_+. The resulting quotient space is called the *lens space* $L(n,k)$.

Note that the map $z \longrightarrow \lambda z$ of $\mathbb{C}$ to itself is simply rotation through the angle $2\pi/n$. Thus f equals a rotation of $\mathbb{S}^2$ about the vertical axis through the angle $2\pi k/n$, followed by reflection in the horizontal plane. See Figure 4.14.1.

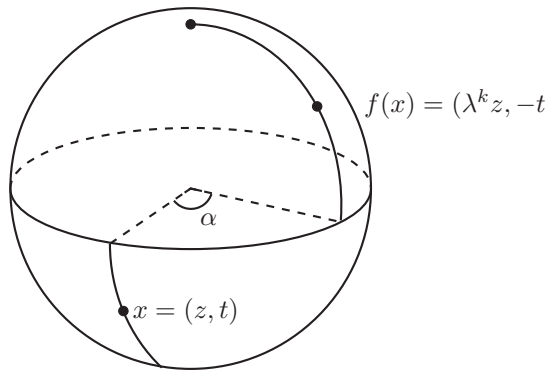


Figure 4.14.1

Let $p : \mathbb{B}^3 \to L(n,k)$ be the quotient map. Each point of Int $\mathbb{B}^3$ is identified only with itself under p, and each point of Int E^2_- is identified with a point of Int E^2_+. Since a point $(z,0)$ of the equator in $\mathbb{S}^2$ belongs to both E^2_- and E^2_+, its equivalence class contains n points—namely, the points

$$(\lambda^k z, 0), (\lambda^{2k}, 0), \ldots, (\lambda^{nk}, 0) = (z,0)\,.$$

Because k and n are relatively prime, these points are distinct and constitute a permutation of the points

$$(\lambda z, 0), (\lambda^2, 0), \ldots, (\lambda^n, 0) = (z,0)\,,$$

which are evenly spaced about the equator $\mathbb{S}^1$.

Theorem 4.14.13 *The space $L(n,k)$ is a CW complex with one cell in each dimension 0, 1, 2, and 3.*

Proof: We first show that the quotient map p is closed, so that $L(n,k)$ is Hausdorff [in fact normal]. Let A be closed in $\mathbb{B}^3$. The saturation $p^{-1}p(A)$ of A is the union of the set A, the following subsets of $\mathbb{S}^2$:

$$f(E^2_- \cap A) \quad \text{and} \quad f^{-1}(E^2_+ \cap A)\,,$$

and the following subsets of $\mathbb{S}^1$:

$$f(A \cap \mathbb{S}^1), f^2(A \cap \mathbb{S}^1), \ldots, f^{n-1}(A \cap \mathbb{S}^1)\,.$$

All these sets are compact, so they are closed in $\mathbb{B}^3$, and so is their union. Since $p^{-1}p(A)$ is closed, so is $p(A)$. Thus p is a closed map.

We give $L(n,k)$ the structure of a CW complex as follows: First, choose the point $a = (1,0)$ on the equator, and let $p(a)$ be the 0-cell e_0 of $L(n,k)$.

Let A denote the smaller arc of $\mathbb{S}^1$ running from a to $b = (\lambda, 0)$. Now $p|A$ is a quotient map, since A is compact and $L(n,k)$ is Hausdorff; $p|A$ identifies a and b, but it is one-to-one on Int A. Thus $p(\text{Int } A)$ is an open 1-cell; we take it to be the open 1-cell e_1 of $L(n,k)$. The map $p|A$ is a characteristic map for e_1. Note that the points of $p^{-1}p(a)$ break up the circle $\mathbb{S}^1$ into n open arcs, each of which is mapped homeomorphically by p onto e_1. See Figure 4.14.2.

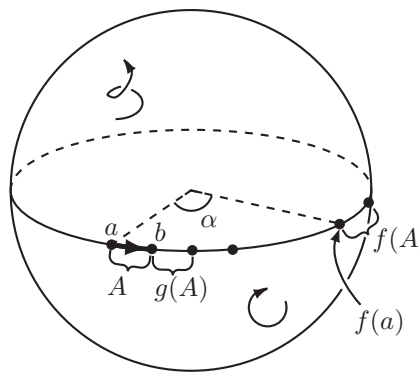


Figure 4.14.2

For similar reasons, the map $p|E^2_-$ is a quotient map. The set $p(\text{Int } E^2_-)$ is an open 2-cell that we take to be the open 2-cell e_2 of $L(n,k)$, and $p|E^2_-$ is a characteristic map for e_2. Finally, the set $p(\text{Int } \mathbb{B}^3)$ is the open 3-cell e_3, and p is its characteristic map.

Note that $p(\mathbb{S}^1)$ equals the 1-skeleton of $L(n,k)$, and $p(\mathbb{S}^2)$ equals the 2-skeleton. □

Now we compute the homology of this lens space.

Theorem 4.14.14 *If $X = L(n,k)$, then the cellular chain complex of X has the form*

$$D_3(X) \xrightarrow{0} D_2(X) \xrightarrow{n} D_1(X) \xrightarrow{0} D_0(X)\,,$$

where each group $D_i(X)$ is infinite cyclic. Therefore, the homology groups of X are

$$H_3(X) \cong \mathbb{Z}, \quad H_2(X) = 0, \quad H_1(X) \cong \mathbb{Z}/n, \quad H_0(X) \cong \mathbb{Z}.$$

Thus the lens spaces $L(n,k)$ and $L(m,\ell)$ cannot be homeomorphic, or even have the same homotopy type, unless $n = m$.

Proof: First, though we leave the proof as an exercise, we observe that $\mathbb{B}^3$ and $L(n,k)$ can be triangulated so that the rotation-reflection map f and the quotient map p are simplicial.

Let A be the arc of $\mathbb{S}^1$ having endpoints $a = (1,0)$ and $b = (\lambda, 0)$, as in the proof of the preceding theorem. Let c_1 be a cycle generating $H_1(A, \mathrm{Bd}\ A)$, let c_2 be a cycle generating $H_2(E^2_-, \mathbb{S}^1)$, and let c_3 be a cycle generating $H_3(\mathbb{B}^3, \mathbb{S}^2)$. Their signs will be chosen shortly. The chains $p_\#(c_1)$, $p_\#(c_2)$, and $p_\#(c_3)$ generate the chain groups $D_1(X)$, $D_2(X)$, and $D_3(X)$, respectively.

Choose the sign of c_1 so the $\partial c_1 = b - a$. The plan for choosing the signs of c_2 and c_3 is as follows: We shall show that the chain

$$z_1 = c_1 + f_\#(c_1) + f^2_\#(c_1) + \cdots + f^{n-1}_\#(c_1) \qquad (*)$$

generates $H_1(\mathbb{S}^1)$. Once that fact has been proved, we note that since ∂c_2 also generates $H_1(\mathbb{S}^1)$, we can choose the sign of c_2 so that $\partial c_2 = z_1$. We shall then show that the chain

$$z_2 = c_2 - f_\#(c_2) \qquad (**)$$

generates $H_2(\mathbb{S}^2)$. Then, given this fact, we can choose the sign of c_3 so that $\partial c_3 = z_2$.

First we consider z_1. Let g be given by $g(z,0) = (\lambda z, 0)$, so on $\mathbb{S}^1$ it equals rotation through angle $2\pi/n$ and agrees with f. Thus we have

$$z_1 = c_1 + g_\#(c_1) + g^2_\#(c_1) + \cdots + g^{n-1}_\#(c_1)\,.$$

Now $\partial c_1 = g_\#(a) - a$. Therefore we have

$$\begin{aligned} \partial z_1 &= [g_\#(a) - a] + [g^2_\#(a) - g_\#(a)] + \cdots + [g^n_\#(a) - g^{n-1}_\#(a)] \\ &= 0\,. \end{aligned}$$

Thus z_1 is a cycle. Because its restriction to A equals c_1, which is a fundamental cycle for $H_1(A, \mathrm{Bd}\ A)$, it is a fundamental cycle for $\mathbb{S}^1$. Thus z_1 generates $H_1(\mathbb{S}^1)$, as we claimed, and we choose the sign of c_2 so that $\partial c_2 = z_1$.

Next, we consider z_2 as in $(**)$. To show that z_2 is a cycle, we compute

$$\partial z_2 = \partial c_2 - f_\#(\partial c_2) = z_1 - f_\#(z_1)\,.$$

Using the formula $(*)$, we see that $f_\#(z_1) = z_1$, so we conclude that $\partial z_2 = 0$. Because the restriction of z_2 to E^2_- is a fundamental cycle for $(E^2_-, \mathbb{S}^1)$, the chain z_2 is a fundamental cycle for $\mathbb{S}^2$, so it generates $H_2(\mathbb{S}^2)$. We now choose the sign of c_3 so that $\partial c_3 = z_2$.

Now we are ready to compute the boundary operators in the cellular chain complex of X. First, we have

$$\partial p_\#(c_1) = p_\#(b) - p_\#(a) = 0\,,$$

so the boundary operator $D_1(X) \to D_0(X)$ is trivial. Second, we have

$$\begin{aligned} \partial p_\#(c_2) &= p_\#(\partial c_2) = p_\#(z_1) = p_\#(c_1 + f_\#(c_1) + f^2_\#(c_1) + \cdots + f^{n-1}_\#(c_1)) \\ &= np_\#(c_1)\,, \end{aligned}$$

because $p(f^j(x)) = p(x)$ for all $x \in \mathbb{S}^1$ and all j. Thus the map $D_2(X) \to D_1(X)$ is multiplication by n. Third, we have

$$\partial p_\#(c_3) = p_\#(\partial c_3) = p_\#(z_2) = p_\#(c_2 - f_\#(c_2)) = 0\,,$$

because $p(f(x)) = p(x)$ for x in E^2_-. Thus $D_3(X) \to D_2(X)$ is trivial. □

Exercise 4 will ask you to prove the following theorem. But it will also offer some guidance.

Theorem 4.14.15 *The lens spaces $L(n,k)$ and $L(n,\ell)$ are homeomorphic if either*

$$k \equiv \pm\ell \pmod{n} \quad \text{or} \quad k\ell \equiv \pm 1 \pmod{n}\,.$$

Remark 4.14.16 The converse of Theorem 4.14.15 is true, but the proof is very difficult.

In a 1935 paper [see [Rei35]], Kurt Reidemeister defined a number associated with a simplicial complex called its *torsion*. The torsion is a *combinatorial invariant*, which means that two complexes that have isomorphic subdivisions necessarily have the same torsion. By computing the torsion for lens spaces, Reidemeister showed that if the simplicial complexes $L(n,k)$ and $L(n,\ell)$ have isomorphic subdivisions, then either $k \equiv \pm\ell \pmod{n}$ or $k\ell \equiv \pm 1 \pmod{n}$.

To complete the proof of the converse, we need to appeal to work of E. E. Moise from the 1950s. Moise wrote a series of papers concerning "affine structures in 3-manifolds." In the fifth of these [see [Moi52]], he proved that two triangulated 3-manifolds that are homeomorphic necessarily have subdivisions that are isomorphic.

The homeomorphism classification of lens spaces is thus known. We shall discuss their homotopy-type classification in Section 8.9.

EXERCISES

1. Show that $\mathbb{P}^n$ is homeomorphic to the quotient space of $\mathbb{B}^3$ obtained by identifying x with $-x$ for each $x \in \mathbb{S}^{n-1}$.

2. **(a)** Show that $\mathbb{P}^n$ is an n-manifold and $\mathbb{C}P^n$ is a $2n$-manifold.

(b) Show more generally that if a finite-dimensional CW complex is homogeneous, then it is a manifold.

3. Let A be a closed region in the plane bounded by a regular n-sided polygon. Let B be the suspension of A. Describe $L(n,k)$ as a quotient space of B. Conclude that $\mathbb{B}^3$ and $L(n,k)$ can be triangulated so that the quotient map is simplicial.

4. Follow the suggested steps to prove Theorem 4.14.15.

Steps:

(a) Prove the case $k \equiv \pm\ell \pmod{n}$ by considering the reflection map $(z,t) \longrightarrow (z,-t)$ in $\mathbb{B}^3$.

(b) For convenience, suppose $1 \leq k < n$. Consider n disjoint 3-simplices

$$a_1b_1c_1d_1,\ a_2b_2c_2d_2,\ \ldots,\ a_nb_nc_nd_n\,,$$

indexed with the elements of $\mathbb{Z}/n$. See Figure 4.14.3. Show that $L(n,k)$ can be obtained from these simplices by first pasting

$$a_ib_id_i \quad \text{to} \quad a_{i+1}b_{i+1}c_{i+1}\,,$$

for each i in $\mathbb{Z}/n$, by a linear homeomorphism that preserves the order of vertices, and then pasting

$$c_id_ib_i \quad \text{to} \quad c_{i+k}d_{i+k}a_{i+k}\,,$$

again by a linear homeomorphism that preserves the order of vertices.

(c) Rewrite the simplices of Step **(b)** in the order

$$a_kb_kc_kd_k,\ a_{2k}b_{2k}c_{2k}d_{2k},\ \ldots,\ a_{nk}b_{nk}c_{nk}d_{nk}\,.$$

See Figure 4.14.4. Carry out the second pasting operation of Step **(b)**, and then carry out the first pasting operation of Step **(b)**. Show that this gives a description of $L(n,\ell)$, where ℓ is the integer between 1 and n such that $k\ell \equiv 1 \pmod{n}$.

(d) Prove the theorem.

5. Show that $L(n,k)$ is a compact 3-manifold.

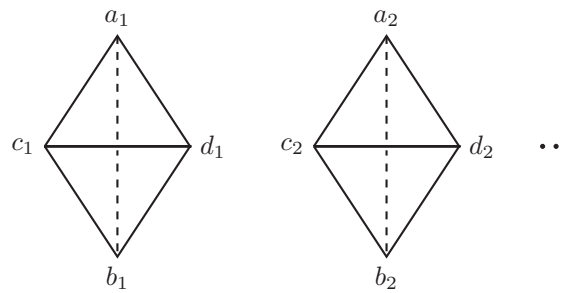


Figure 4.14.3

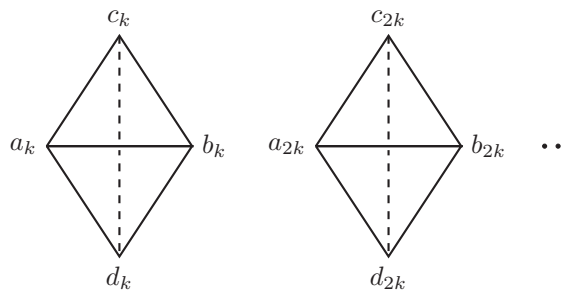


Figure 4.14.4

6. What is the relation between real projective n-space and complex projective n-space?

7. Any object in n-dimensional Euclidean space has an image in real projective n-space. Explain this point. What is the image of the unit ball? What is the image of the unit cube?

5

Cohomology

5.1 Introduction

With each topological space X we have associated a sequence of abelian groups called its *homology groups*. Now we associate with X another sequence of abelian groups, called its *cohomology groups*. These groups were not defined until long after the homology groups. The reason is not hard to understand, for they are geometrically much less natural than the homology groups. Their origins lie in algebra rather than geometry. In a certain algebraic sense [to be made precise later], they are "dual" to the homology groups. In the past, topologists have used such terms as "pseudo-cycle" for representatives of these group elements, implying a certain skepticism as to their legitimacy as objects of study. Eventually, it became clear that these groups are both important in theory and useful in practice. The cohomology groups are more subtle than the homology groups, and much more profound and powerful. We shall learn about them in this chapter.

The duality theorems for manifolds, the connections between topology and differential geometry [de Rham's theorem] and between topology and analysis [cohomology with sheaf coefficients]—all these results are formulated in terms of cohomology. Even such purely topological problems as classifying spaces up to homeomorphism, or maps up to homotopy, are problems about which cohomology has a good deal to say. We will return to some of these problems later.

Throughout this chapter we assume familiarity with the language of categories and functors [Section 3.7].

5.2 The Hom Functor

Associated with any pair of abelian groups A and G is a third abelian group, the group $\mathrm{Hom}\,(A, G)$ of all homomorphisms of A into G. This group will be involved in an essential way in the definition of the cohomology groups. In this section we study some of its properties.

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Definition 5.2.1 If A and G are abelian groups, then the set $\mathrm{Hom}\,(A, G)$ of all homomorphisms of A into G becomes an abelian group if we add two homomorphisms by adding their values in G.

That is, for $\phi, \psi \in \mathrm{Hom}\,(A, G)$, we define $\phi + \psi \in \mathrm{Hom}\,(A, G)$ by setting $(\phi + \psi)(a) = \phi(a) + \psi(a)$, for $a \in A$. The map $\phi + \psi$ is a homomorphism, because $(\phi + \psi)(0) = 0$ and

$$\begin{aligned}(\phi + \psi)(a + b) &= \phi(a + b) + \psi(a + b) \\ &= \phi(a) + \psi(a) + \phi(b) + \psi(b) = (\phi + \psi)(a) + (\phi + \psi)(b)\,.\end{aligned}$$

The identity element of $\mathrm{Hom}\,(A, G)$ is the function mapping A to the identity element of G. The additive inverse of the homomorphism ϕ is the homomorphism that maps each $a \in A$ to $-\phi(a)$. Thus we may refer to Hom as the *group of homomorphisms*, but usually we will simply call it Hom.

EXAMPLE 5.2.2 The group $\mathrm{Hom}\,(\mathbb{Z}, G)$ is isomorphic with G itself. Letting $\mathcal{I}$ denote the isomorphism, we define $\mathcal{I}$ by setting $\mathcal{I}(\phi) = \phi(1)$, for all $\phi \in \mathrm{Hom}\,(\mathbb{Z}, G)$.

In general, if A is a free abelian group of finite rank with basis $e_1, e_2, \ldots, e_n$, then $\mathrm{Hom}\,(A, G)$ is isomorphic with the direct sum $G \oplus G \oplus \cdots \oplus G$ of n copies of G. Again, letting $\mathcal{I}$ denote the isomorphism, we define $\mathcal{I}$ by setting $\mathcal{I}(\phi) = (\phi(e_1), \phi(e_2), \ldots, \phi(e_n))$. This isomorphism depends on the choice of basis, so it is not "natural." Note that if A is free abelian, but with a non-finite basis $\{e_\alpha\}_{\alpha \in J}$, then the mapping $\phi \to (\phi(e_\alpha))_{\alpha \in J}$ carries ϕ to an element of the direct product $\prod_{\alpha \in J} G_\alpha$, not to an element of the direct sum $\bigoplus_{\alpha \in J} G_\alpha$. [See Section 1.5 for definitions.]

We will state these facts formally as a theorem later on.

Definition 5.2.3 Let A, B, and G be abelian groups. The homomorphism $f : A \to B$ gives rise to a *dual homomorphism*

$$\mathrm{Hom}\,(A, G) \xleftarrow{\widetilde{f}} \mathrm{Hom}\,(B, G)$$

going in the reverse direction. The map $\widetilde{f}$ assigns to the homomorphism $\phi : B \to G$ the composite

$$A \xrightarrow{f} B \xrightarrow{\phi} G\,.$$

That is, $\widetilde{f}(\phi) = \phi \circ f$. The composition of homomorphisms yields a homomorphism, so $\widetilde{f}(\phi)$ is an element of $\mathrm{Hom}\,(A, G)$, as claimed.

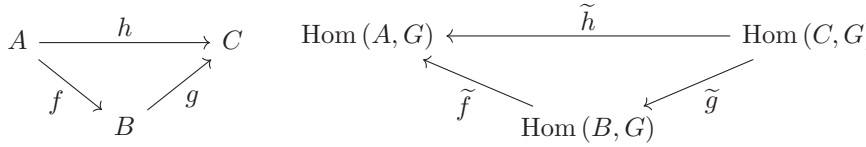
The mapping $f \to \widetilde{f}$ is also a homomorphism, since $\widetilde{f}(0) = 0$ and

$$\begin{aligned}[\widetilde{f}(\phi + \psi)](a) &= (\phi + \psi)(f(a)) = \phi(f(a)) + \psi(f(a)) \\ &= [\widetilde{f}(\phi)](a) + [\widetilde{f}(\psi)](a)\,.\end{aligned}$$

Note that, for fixed G, the assignment

$$A \to \mathrm{Hom}\,(A,G) \quad \text{and} \quad f \to \widetilde{f}$$

defines a *contravariant functor* from the category of abelian groups and homomorphisms to itself. For if $i_A : A \to A$ is the identity homomorphism, then $\widetilde{i}_A(\phi) = \phi \circ i_A = \phi$, so $\widetilde{i}_A$ is the identity map of $\mathrm{Hom}\,(A,G)$, Furthermore, if the left-hand diagram below commutes, then so does the right-hand diagram:



For $\widetilde{h}(\phi) = \phi \circ h = \phi \circ (g \circ f)$, while $\widetilde{f}(\widetilde{g}(\phi)) = \widetilde{f}(\phi \circ g) = (\phi \circ g) \circ f$, by definition.

We list some consequences of this fact:

Theorem 5.2.4 *Let B and C be abelian groups, and let $g : B \to C$ be a homomorphism. Let $\widetilde{g}$ be the dual homomorphism.*

(a) *If g is the zero homomorphism, then so is $\widetilde{g}$.*

(b) *If g is an isomorphism, then so is $\widetilde{g}$.*

(c) *If g is surjective, then $\widetilde{g}$ is injective. That is, exactness of*

$$B \xrightarrow{g} C \longrightarrow 0$$

implies exactness of

$$\mathrm{Hom}\,(B,G) \xleftarrow{\widetilde{g}} \mathrm{Hom}\,(C,G) \longleftarrow 0\,.$$

Proof: Conclusions **(a)** and **(b)** are immediate. To prove **(c)**, assume that g is surjective. Let $\psi \in \mathrm{Hom}\,(C,G)$ and suppose that $\widetilde{g}(\psi) = 0 = \psi \circ g$. Then $\psi(g(b)) = 0$ for every $b \in B$. As b ranges over B, the element $g(b)$ ranges over all elements of C. Thus $\psi(c) = 0$ for every $c \in C$. □

More generally, we have the following result concerning the dual of an exact sequence.

Theorem 5.2.5 *If the sequence*

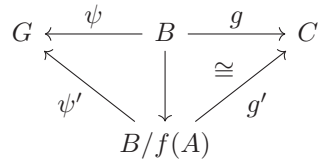
$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact, then the dual sequence

$$\mathrm{Hom}\,(A,G) \xleftarrow{\widetilde{f}} \mathrm{Hom}\,(B,G) \xleftarrow{\widetilde{g}} \mathrm{Hom}\,(C,G) \longleftarrow 0\,.$$

is exact. Furthermore, if f is injective and the first sequence splits [see Definition 3.2.4], then $\widetilde{f}$ is surjective and the second sequence splits.

Proof: Injectivity of $\widetilde{g}$ follows from the preceding theorem. We check exactness at $\mathrm{Hom}\,(B, G)$. Because $h = g \circ f$ is the zero homomorphism, so is $\widetilde{h} = \widetilde{f} \circ \widetilde{g}$. On the other hand, supposing $\widetilde{f}(\psi) = 0$, we shall show that $\psi = \widetilde{g}(\phi)$ for some $\phi \in \mathrm{Hom}\,(C, G)$. Since $\widetilde{f}(\psi) = \psi \circ f$ is the zero homomorphism, ψ vanishes on the group $f(A)$. Thus ψ induces a homomorphism $\psi' : B/f(A) \to G$. Exactness of the original sequence implies that g induces an *isomorphism* $g' : B/f(A) \to C$, as in the following diagram:



The map $\phi = \psi' \circ (g')^{-1}$ is a homomorphism of C into G, and as desired we have

$$\widetilde{g}(\phi) = \phi \circ g = \psi' \circ (g')^{-1} \circ g = \psi\,.$$

Suppose now that f maps A injectively onto a direct summand in B. Let $\pi : B \to A$ be a homomorphism such that $\pi \circ f = i_A$. Then $\widetilde{f} \circ \widetilde{\pi}$ is the identity of $\mathrm{Hom}\,(A, G)$, so $\widetilde{f}$ is surjective and $\pi : \mathrm{Hom}\,(A, G) \to \mathrm{Hom}\,(B, G)$ splits the dual sequence. □

We remark that, in general, exactness of a short exact sequence *does not* imply exactness of the dual sequence. For instance, if $f : \mathbb{Z} \to \mathbb{Z}$ equals multiplication by 2, then the sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{f} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0$$

is exact. But $\widetilde{f}$ is not surjective. Indeed, if $\phi \in \mathrm{Hom}\,(\mathbb{Z}, \mathbb{Z})$, then $\widetilde{f}(\phi) = \phi \circ f$ is a homomorphism that maps $\mathbb{Z}$ into the set of even integers. Thus the image of $\widetilde{f}$ is not all of $\mathrm{Hom}\,(\mathbb{Z}, \mathbb{Z})$.

We have considered Hom as a functor of the first variable alone. But it can also be considered as a functor of both variables. In this case, it has a mixed variance. It is contravariant in the first variable and covariant in the second variable. We formalize this statement as follows:

Definition 5.2.6 Given homomorphisms $\alpha : A \to A'$ and $\beta : G' \to G$, we define the *induced homomorphism of groups of homomorphisms*

$$\mathrm{Hom}\,(\alpha, \beta) : \mathrm{Hom}\,(A', G') \to \mathrm{Hom}\,(A, G)$$

by letting it map the homomorphism $\phi' : A' \to G'$ to the homomorphism

$$\beta \circ \phi' \circ \alpha : A \to G\,.$$

You can check that $\mathrm{Hom}\,(\alpha,\beta)$ is indeed a homomorphism. Functoriality follows easily: The map $\mathrm{Hom}\,(i_A, i_G)$ is the identity. Also, if $\alpha' : A' \to A''$ and $\beta' : G'' \to G'$, then

$$\mathrm{Hom}\,(\alpha' \circ \alpha, \beta \circ \beta') = \mathrm{Hom}\,(\alpha,\beta) \circ \mathrm{Hom}\,(\alpha',\beta')\,,$$

since both sides carry $\phi'' : A'' \to G''$ to $\beta \circ \beta' \circ \phi'' \circ \alpha' \circ \alpha$.

In this notation, what we have called the dual homomorphism $\widetilde{\alpha}$, obtained when we consider Hom as a functor of the first variable alone, is the map $\mathrm{Hom}\,(\alpha, i_G)$.

One can also consider Hom as a functor of the second variable alone. This case we leave to the exercises.

Now we prove some properties of the Hom functor.

Theorem 5.2.7

(a) *One has the following isomorphisms*

$$\begin{aligned} \mathrm{Hom}\,(\textstyle\bigoplus_{\alpha\in J} A_\alpha, G) &\cong \prod_{\alpha\in J} \mathrm{Hom}\,(A_\alpha, G)\,, \\ \mathrm{Hom}\,\left(A, \textstyle\prod_{\alpha\in J} G_\alpha\right) &\cong \prod_{\alpha\in J} \mathrm{Hom}\,(A, G_\alpha)\,. \end{aligned}$$

(b) *There is a natural isomorphism λ of* $\mathrm{Hom}\,(\mathbb{Z}, G)$ *with G.*

(c) *Let m be a positive integer, and let $f_m : \mathbb{Z} \to \mathbb{Z}$ be multiplication by m. Then $\widetilde{f}_m$ equals multiplication by m in* $\mathrm{Hom}\,(\mathbb{Z}, G)$.

(d) *Let m be a positive integer, and let $g_m : G \to G$ be multiplication by m. Then* $\mathrm{Hom}\,(\mathbb{Z}/m, G) \cong \ker(g_m)$.

Proof: Property **(a)** follows immediately from standard facts of algebra concerning homomorphisms of products.

(b) Define $\lambda : \mathrm{Hom}\,(\mathbb{Z}, G) \to G$ by assigning to $\phi \in \mathrm{Hom}\,(\mathbb{Z}, G)$ the value of ϕ at 1. It is easy to see that λ is a homomorphism. Since ϕ is entirely determined by its value at 1, and since this value can be chosen arbitrarily, λ is an isomorphism of $\mathrm{Hom}\,(\mathbb{Z}, G)$ with G.

(c) We compute

$$\widetilde{f}_m(\phi)(x) = \phi(f_m(x)) = \phi(mx) = m\phi(x)\,,$$

so $\widetilde{f}_m(\phi) = m\phi$. Thus $\widetilde{f}_m$ equals multiplication by m in $\mathrm{Hom}\,(\mathbb{Z}, G)$.

(d) Let f_m be as in **(c)**. Begin with the exact sequence

$$\mathbb{Z} \xrightarrow{f_m} \mathbb{Z} \longrightarrow \mathbb{Z}/m \longrightarrow 0\,.$$

By Theorem 5.2.5, the sequence

$$\mathrm{Hom}\,(\mathbb{Z}, G) \xleftarrow{\widetilde{f}_m} \mathrm{Hom}\,(\mathbb{Z}, G) \longleftarrow \mathrm{Hom}\,(\mathbb{Z}/m, G) \longleftarrow 0\,.$$

is exact. Thus $\mathrm{Hom}\,(\mathbb{Z}/m, G)$ is isomorphic to the kernel of $\widetilde{f}_m : \mathrm{Hom}\,(\mathbb{Z}, G) \to \mathrm{Hom}\,(\mathbb{Z}, G)$. By **(c)**, $\widetilde{f}_m$ is multiplication by m in $\mathrm{Hom}\,(\mathbb{Z}, G)$. Using the isomorphism λ from **(b)**, we see that the kernel of $\widetilde{f}_m$ is isomorphic to the kernel of g_m. □

We remark that the isomorphisms given in **(a)** of Theorem 5.2.4 are natural. Specifically, suppose one is given homomorphisms $\phi_\alpha : A_\alpha \to B_\alpha$ and $\psi : H \to G$, then it follows immediately from the definition of the isomorphism that the diagram

$$\begin{array}{ccc} \mathrm{Hom}\,(\bigoplus A_\alpha, G) & \xrightarrow{\cong} & \prod \mathrm{Hom}\,(A_\alpha, G) \\ \Big\uparrow{\scriptstyle \mathrm{Hom}\,(\bigoplus \phi_\alpha, \psi)} & & \Big\uparrow{\scriptstyle \prod \mathrm{Hom}\,(\phi_\alpha, \psi)} \\ \mathrm{Hom}\,(\bigoplus B_\alpha, H) & \xrightarrow{\cong} & \prod \mathrm{Hom}\,(B_\alpha, H) \end{array}$$

commutes. A similar comment applies to the other isomorphism in **(a)**.

The preceding theorem enables us to compute $\mathrm{Hom}\,(A, G)$ whenever A is finitely generated, for $\mathrm{Hom}\,(A, G)$ equals a direct sum of terms of the form $\mathrm{Hom}\,(\mathbb{Z}, G)$ and $\mathrm{Hom}\,(\mathbb{Z}/m, G)$, which we compute by applying the rules

$$\mathrm{Hom}\,(\mathbb{Z}, G) \cong G, \quad \mathrm{Hom}\,(\mathbb{Z}/m) \cong \ker(g_m) \quad [g_m \text{ is multiplication by } m \text{ in } G].$$

When G is also finitely generated, these groups can be written as direct sums of cyclic groups. One needs the following lemma, the proof of which is left to the reader.

Lemma 5.2.8 *Let m and n be positive integers. Suppose that $d = \gcd(m, n)$ and $h_m : \mathbb{Z}/n \to \mathbb{Z}/n$ is multiplication by m; then there is an exact sequence*

$$0 \longrightarrow \mathbb{Z}/d \longrightarrow \mathbb{Z}/n \xrightarrow{h_m} \mathbb{Z}/n \longrightarrow \mathbb{Z}/d \longrightarrow 0\,.$$

Definition 5.2.9 The abelian group G is said to be *divisible* if, for each $x \in G$ and each positive integer n, there exists $y \in G$ such that $ny = x$.

For example, the rational numbers and the real numbers form divisible groups under addition. The next definition gives another example of a divisible group.

Definition 5.2.10 *Let $\mathbb{S}^1$ denote the additive group $\mathbb{R}/\mathbb{Z}$. It is isomorphic to the multiplicative group of complex numbers of unit modulus and is often called the* circle group.

The notion of a divisible group will come up a number of times.

Theorem 5.2.11 *Let G be divisible. If*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

is exact, then so is

$$0 \longleftarrow \mathrm{Hom}\,(A, G) \longleftarrow \mathrm{Hom}\,(B, G) \longleftarrow \mathrm{Hom}\,(C, G) \longleftarrow 0\,.$$

The proof of the theorem is left to Exercise 4.

EXERCISES

1. Show that if T is the torsion subgroup of G, then

$$\mathrm{Hom}\,(G, \mathbb{Z}) \cong \mathrm{Hom}\,(G/T, \mathbb{Z})\,.$$

2. Let G be fixed. Consider the following functor from the category of abelian groups to itself:

$$A \longrightarrow \mathrm{Hom}\,(G, A) \quad \text{and} \quad f \longrightarrow \mathrm{Hom}\,(i_G, f)\,.$$

(a) Show that this functor preserves the exactness of

$$0 \longrightarrow A \longrightarrow B \longrightarrow C\,.$$

(b) Show that this functor preserves split short exact sequences.

3. **(a)** Let m and n be positive integers, and set $d = \gcd(m, n)$. Define $h_m : \mathbb{Z}/n \to \mathbb{Z}/n$ to be multiplication by m. Show that the kernel of h_m is generated by $\{n/d\}$.

(b) Show that a quotient of a cyclic group is cyclic.

(c) Prove Lemma 5.2.8.

4. Follow the suggested steps to prove Theorem 5.2.11.

(a) Prove it suffices to show that if $A \subset B$ and $\phi : A \to G$ is a homomorphism, then ϕ extends to a homomorphism $\psi : B \to G$.

(b) Prove this fact when B is generated by the elements of A and a single additional element b.

(c) Let $\mathcal{B}$ be a collection of subgroups of B that is simply ordered by inclusion. Let $\{\psi_H : H \in \mathcal{B}\}$ be a collection of homomorphisms, where ψ_H maps H into G for each H, such that any two agree on the common part of their domains. Show that the union of the elements of $\mathcal{B}$ is a subgroup of B, and these homomorphisms extend to a homomorphism of this union into G.

(d) Use a Zorn's lemma argument to complete the proof. [See [Mun00], page 70, if you are not familiar with Zorn's lemma.]

5. Let R be a commutative ring with unity element 1. Let A and B be R-modules. [Look ahead into Section 5.9 if you have forgotten the definitions.]

(a) Let $\mathrm{Hom}_R(A, B)$ denote the set of all R-module homomorphisms of A into B. Show that it has the structure of an R-module in a natural way. Show that if f and g are R-module homomorphisms, then so is $\mathrm{Hom}\,(f, g)$.

(b) State and prove the analogue of Theorem 5.2.5 for R-modules. Do the same for Theorem 5.2.7.

(c) Consider the special case where R is a field F. Then A and B are vector spaces over F, and so is $\mathrm{Hom}_F(A, B)$. Show that in this case every exact sequence splits, so the functor Hom_F preserves exact sequences.

6. Show that the product of divisible groups is divisible. Show that the quotient of divisible groups is divisible.

5.3 Simplicial Cohomology Groups

In this section we define the cohomology groups of a simplicial complex, and we compute some elementary examples.

Definition 5.3.1 Let K be a simplicial complex, and let G be an abelian group. The group of *p-dimensional cochains* of K, with coefficients in G, is the group

$$C^p(K; G) = \mathrm{Hom}\,(C_p(K), G)\,.$$

The *coboundary operator* δ is defined to be the dual of the boundary operator $\partial : C_{p+1}(K) \to C_p(K)$. Thus we have

$$C^{p+1}(K; G) \xleftarrow{\delta} C^p(K; G)\,,$$

so that δ *raises* dimension by one. We define $Z^p(K; G)$ to be the kernel of this homomorphism, $B^{p+1}(K; G)$ to be its image, and [noting that $\delta^2 = 0$, because $\partial^2 = 0$],

$$H^p(K; G) = Z^p(K; G)/B^p(K; G)\,.$$

These groups are called the group of *cocycles* the group of *coboundaries* and the *cohomology group* respectively, of K with coefficients in G. We omit G from the notation when G equals the group of integers.

If c^p is a p-dimensional cochain, and c_p is a p-dimensional chain, we commonly use the dual pairing notation $\langle c^p, c_p \rangle$ to denote the value of c^p on c_p, rather than the more familiar functional notation $c^p(c_p)$. In this notation, the definition of the coboundary operator becomes

$$\langle \delta c^p, d_{p+1} \rangle = \langle c^p, \partial d_{p+1} \rangle \,.$$

The definition of cohomology is, as promised, highly algebraic in nature. Is it at all possible to picture the groups involved geometrically? The answer is a qualified "yes," as we now observe.

Recall that the group $C_p(K)$ of p-chains is free abelian. It has a standard basis obtained by orienting the p-simplices of K arbitrarily and using the corresponding elementary chains as a basis. Let $\{\sigma_\alpha\}_{\alpha \in J}$ be this collection of oriented simplices. Then the elements of $C_p(K)$ are represented as finite linear combinations $\sum n_\alpha \sigma_\alpha$ of elementary chains σ_α. Now an element c^p of $\mathrm{Hom}\,(C_p(K), G)$ is determined by its value g_α on each basis element σ_α, and these values may be assigned arbitrarily. There is *no* requirement that c^p vanish on all but finitely many σ_α.

Suppose we let σ_α^* denote the elementary cochain, with $\mathbb{Z}$ coefficients, whose value is 1 on the basis element σ_α and 0 on all other basis elements. Then, for $g \in G$, we let $g\,\sigma_\alpha^*$ denote the cochain whose value is g on σ_α and 0 on all other basis elements. Using this notation, we often represent C^p by the possibly infinite formal sum

$$c^p = \sum g_\alpha \, \sigma_\alpha^* \,.$$

Why is this representation of c^p reasonable? We justify it as follows.

Suppose we let C_α denote the infinite cyclic subgroup of $C_p(K)$ generated by σ_α. Then $C_p(K) = \bigoplus_\alpha C_\alpha$, and as noted earlier in part **(a)** of Theorem 5.2.7,

$$C^p(K; G) = \mathrm{Hom}\,(\textstyle\bigoplus_\alpha C_\alpha, G) \cong \prod_\alpha \mathrm{Hom}\,(C_\alpha, G) \,. \qquad (*)$$

The latter group is a direct product of copies of G. Under the isomorphism $(*)$, the cochain c^p corresponds to the element $(g_\alpha \, \sigma_\alpha^*)_{\alpha \in J}$ of the direct product. Instead of using tuple notation to represent this element of the direct product, we shall use the formal sum notation.

This notation is especially convenient when it comes to computing the coboundary operator δ. We claim that, if $c^p = \sum g_\alpha \, \sigma_\alpha^*$, then

$$\delta c^p = \sum g_\alpha (\delta \sigma_\alpha^*) \,, \qquad (**)$$

just as if we had an honest sum—rather than a formal one. To verify this equation, let us orient each $(p+1)$-simplex τ and show that the right-hand side of $(**)$ makes sense and that both sides of $(**)$ agree when evaluated on τ. Suppose that

$$\partial \tau = \sum_{i=0}^{p+1} \epsilon_i \sigma_{\alpha_i} \,,$$

where $\epsilon_i = \pm 1$ for each i. Then we have

$$\langle \delta c^p, \tau \rangle = \langle c^p, \partial \tau \rangle = \sum_{i=0}^{p+1} \epsilon_i \langle c_p, \sigma_{\alpha_i} \rangle = \sum_{i=0}^{p+1} \epsilon_i g_{\alpha_i} .$$

Also, we have

$$\begin{aligned} \langle g_\alpha (\delta \sigma_\alpha^*), \tau \rangle &= g_\alpha \langle \delta \sigma_\alpha^*, \tau \rangle = g_\alpha \langle \sigma_\alpha^*, \partial \tau \rangle \\ &= \begin{cases} \epsilon_i g_{\alpha_i} & \text{if } \alpha = \alpha_i \text{ for } i = 0, 1, \ldots p+1, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Thus δc^p and $\sum g_\alpha (\delta \sigma_\alpha^*)$ have the same value on τ, so $(**)$ holds.

By $(**)$, in order to compute δc^p, it suffices to compute $\delta \sigma^*$ for each oriented p-simplex σ. That we can compute by using the formula

$$\delta \sigma^* = \sum \epsilon_j \tau_j^* ,$$

where the summation extends over all $(p+1)$-simplices τ_j having σ as a face, and $\epsilon_i = \pm 1$ is the sign with which σ appears in the expression for $\partial \tau_j$. One verifies this formula by simply evaluating both sides on the general $(p+1)$-simplex τ.

Now let us apply these facts to some examples. We make only a few computations here, reserving the general problem of computing cohomology groups until a later section.

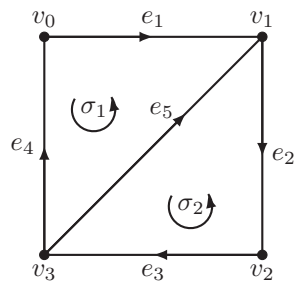


Figure 5.3.1

EXAMPLE 5.3.2 Consider the complex K pictured in Figure 5.3.1. Let us compute the coboundaries of a few cochains. Let $\{v_i\}$ denote the set of vertices, let $\{e_i\}$ denote the edges, oriented as indicated, and let $\{\sigma_i\}$ denote the 2-simplices, oriented as indicated. We compute δe_5^*. It has value 1 on σ_1 and value -1 on σ_2, because e_5 appears in $\partial \sigma_1$ and $\partial \sigma_2$ with signs $+1$ and -1, respectively. Thus

$$\delta e_5^* = \sigma_1^* - \sigma_2^* . \qquad (\dagger)$$

A similar remark shows that

$$\delta v_1^* = e_1^* + e_5^* - e_2^* .$$

Are there any cocycles in this complex? Yes, both σ_1^* and σ_2^* are cocycles, for the trivial reason that K has no 3-simplices. Each of them is a coboundary, since

$$\delta e_1^* = -\sigma_1^* \qquad \text{and} \qquad \delta e_3^* = -\sigma_2^* .$$

The 1-dimensional cochain

$$c^1 = e_1^* + e_5^* - e_3^*$$

is also a cocycle, as you can check. It happens to be a coboundary, since it equals $\delta(v_1^* + v_2^*)$. Similarly, you can check that the 0-cochain

$$c^0 = v_0^* + v_1^* + v_2^* + v_3^*$$

is a cocycle. It cannot be a coboundary because there are no cochains of dimension -1.

Definition 5.3.3 Two cochains are said to be *cohomologous* if their difference is a coboundary. Equation (†) above shows that σ_1^* and σ_2^* are cohomologous.

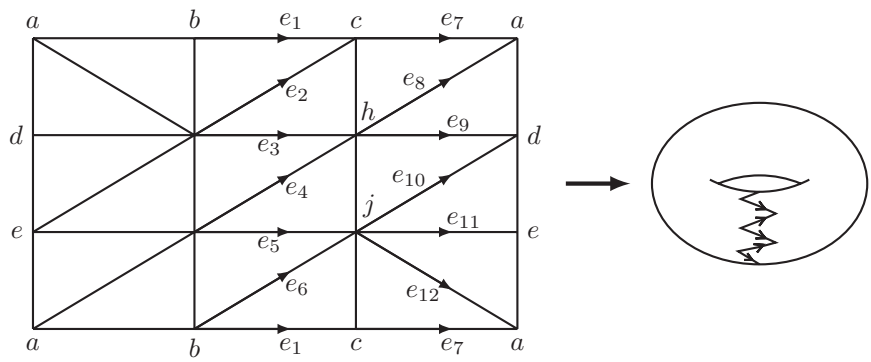


Figure 5.3.2

EXAMPLE 5.3.4 Consider the torus, in a slightly simplified version of our earlier triangulation. See Figure 5.3.2. Consider the 1-cochains

$$z^1 = e_1^* + e_2^* + e_3^* + e_4^* + e_5^* + e_6^*$$

and

$$d^1 = e_7^* + e_8^* + e_9^* + e_{10}^* + e_{11}^* + e_{12}^* .$$

Both are cocycles, as you can check. They are also cohomologous, because

$$\delta(c^* + h^* + j^*) = z^1 - d^1 .$$

This example illustrates the fact that, while we can think of a 1-*cycle* as being a closed curve, the best way to think of a 1-*cocycle* is as a picket fence!

Later we shall compute the cohomology of the torus T and show that the cocycle z^1 represents one of the generators of $H^1(T)$. [See Section 5.8.]

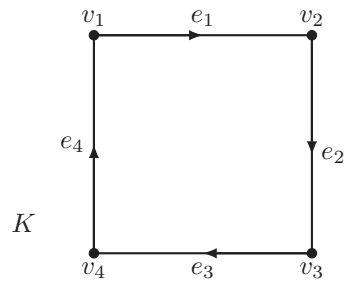


Figure 5.3.3

EXAMPLE 5.3.5 Consider the complex K pictured in Figure 5.3.3. We compute its cohomology groups. The general 0-chain is a sum of the form $c^0 = \sum n_i v_i^*$. Since $\langle \delta c^0, e_i \rangle = \langle c^0, \partial e_i \rangle$, we see that δc^0 has the value $n_2 - n_1$ on e_1, the value $n_3 - n_2$ on e_2, and so on. Thus, if c^0 is a cocycle, then necessarily $n_1 = n_2 = n_3 = n_4$, so c^0 is of the form $n \sum v_i^*$. We conclude that $H^0(K) \cong \mathbb{Z}$, and that $\sum v_i^*$ generates that cohomology group.

Now let c^1 be a 1-cochain. It is a cocycle, trivially. We show that c^1 is cohomologous to some multiple of e_1^*. It suffices to show that e_i^* is cohomologous to e_1^* for $i = 2, 3, 4$, and this can be done directly. For instance, e_3^* is cohomologous to e_1^* because $\delta(v_4^* + v_1^*) = e_3^* - e_1^*$. A similar remark applies to the other e_i.

Furthermore, no multiple of e_1^* is a coboundary: For let z be the cycle $e_1 + e_2 + e_3 + e_4$. Then, for any 0-cochain, we have $\langle \delta c^0, z \rangle = \langle c^0, \partial z \rangle = 0$. But $\langle n e_1^*, z \rangle = n$. Thus $n e_1^*$ is not a coboundary unless $n = 0$.

We conclude that $H^1(K) \cong \mathbb{Z}$, and that e_1^* generates that cohomology group. We can also show that $H^1(K)$ is generated by e_2^* or by e_3^* or by e_4^*.

Note that this same argument applies if K is a general n-gon instead of a quadrilateral.

EXAMPLE 5.3.6 In the preceding example, the homology and cohomology groups of K are equal. Here we consider an example to show that that does not always occur.

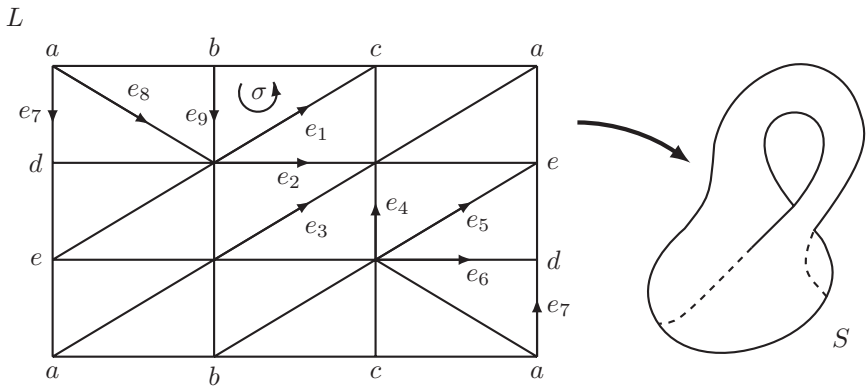


Figure 5.3.4

Let S denote the Klein bottle, represented by the labeled rectangle in Figure 5.3.4. We show that $H^2(S)$ is nontrivial, whereas we know that $H_2(S) = 0$. Orient the 2-simplices of L counterclockwise. Use the induced orientation of the 2-simplices of S and let γ denote their sum. Now γ is not a cycle, because $\partial\gamma = 2z_1$, where $z_1 = [a, d] + [d, e] + [e, a]$.

Let σ denote a single oriented 2-simplex of S, as pictured. Then σ^* is a cocycle of S, because S has no 3-simplices. Furthermore, σ^* is not a coboundary. For if c^1 is an arbitrary 1-cochain, then $\langle \delta c^1, \gamma\rangle = \langle c^1, \partial\gamma\rangle = 2\langle c^1, z_1\rangle$, which is an even integer, while $\langle \sigma^*, \gamma\rangle = 1$. Thus σ^* represents a non-trivial element of $H^2(S)$.

In fact, σ^* represents an element of order 2 in $H^2(S)$. You can check that the coboundary of the 1-cochain $(e_1^* + e_2^* + \cdots + e_9^*)$ pictured in the figure equals $2\sigma^*$.

Now we consider the zero-dimensional cohomology groups and compute them.

Theorem 5.3.7 *Let K be a complex. Then $H^0(K; G)$ equals the group of all 0-cochains c^0 such that $\langle c^0, v\rangle = \langle c^0, w\rangle$ whenever v and w belong to the same component of $|K|$.*

In particular, if $|K|$ is connected, then $H^0(K) \cong \mathbb{Z}$ and that cohomology group is generated by the cochain whose value is 1 on each vertex of K.

Proof: Note that $H^0(K; G)$ equals the group of 0-cocycles, because there are no coboundaries in dimension 0. If v and w belong to the same component of $|K|$, then there is a 1-chain c_1 of K such that $\partial c_1 = v - w$. For any cocycle c^0, we must have

$$0 = \langle \delta c^0, c_1\rangle = \langle c^0, \partial c_1\rangle = \langle c^0, v\rangle - \langle c^0, w\rangle .$$

Conversely, let c^0 be a cochain such that $\langle c^0, v\rangle - \langle c^0, w\rangle = 0$ whenever v and w lie in the same component of $|K|$. Then, for each oriented 1-simplex of K, we have

$$\langle \delta c^0, \sigma\rangle = \langle c^0, \partial\sigma\rangle = 0\,.$$

We conclude that $\delta c^0 = 0$. The theorem follows. □

The preceding theorem shows that in general $H^0(K)$ is isomorphic to a direct *product* of infinite cyclic groups, one for each component of $|K|$. On the other hand, the group $H_0(K)$ is isomorphic to the direct *sum* of this collection of groups. This is another case where the homology and cohomology groups differ.

Definition 5.3.8 Given a complex K, we dualize the standard augmentation

$$C_1(K) \xrightarrow{\partial_1} C_0(K) \xrightarrow{\epsilon} \mathbb{Z}\,,$$

and obtain a homomorphism $\widetilde{\epsilon}$,

$$C^1(K;G) \xleftarrow{\delta_1} C^0(K;G) \xleftarrow{\widetilde{\epsilon}} G\,,$$

called a *coaugmentation*. It is injective, and $\delta_1 \circ \widetilde{\epsilon} = 0$. We define the *reduced cohomology* of K by setting $\widetilde{H}^q(K;G) = H^q(K;G)$ if $q > 0$, and

$$\widetilde{H}^0(K;G) = \ker \delta_1 / \operatorname{im} \widetilde{\epsilon}\,.$$

Theorem 5.3.9 *If $|K|$ is connected, then $\widetilde{H}^0(K;G) = 0$. More generally, for any complex K,*

$$H^0(K;G) \cong \widetilde{H}^0(K;G) \oplus G\,.$$

Proof: If $|K|$ is connected, then $\widetilde{H}^0(K)$ vanishes, so

$$C_1(K) \longrightarrow C_0(K) \longrightarrow \mathbb{Z} \longrightarrow 0$$

is exact. It follows that

$$C^1(K;G) \longleftarrow C^0(K;G) \longleftarrow G \longleftarrow 0$$

is exact, so $\widetilde{H}^0(K;G)$ vanishes. The rest of the theorem we leave as an exercise. □

EXERCISES

1. Consider the complex K of Example 5.3.2. Find a basis for the cocycle group in each dimension. Show that the cohomology of K vanishes in positive dimensions.

2. Check the computations of Examples 5.3.2, 5.3.3, and 5.3.4.

3. **(a)** Suppose one is given homomorphisms

$$C_1 \xrightarrow{\phi} C_0 \xrightarrow{\epsilon} \mathbb{Z}\,,$$

where $\epsilon \circ \phi = 0$ and ϵ is an epimorphism. Consider the dual sequence

$$\mathrm{Hom}\,(C_1, G) \xleftarrow{\widetilde{\phi}} \mathrm{Hom}\,(C_0, G) \xleftarrow{\widetilde{\epsilon}} \mathrm{Hom}\,(\mathbb{Z}, G)\,.$$

Show that $\mathrm{im}\,\widetilde{\epsilon}$ is a direct summand in $\mathrm{Hom}\,(C_0, G)$ and hence in $\ker \widetilde{\phi}$. [**Hint:** See the exercises of Section 1.8.]

(b) Conclude that

$$\ker \widetilde{\phi} \cong \frac{\ker \widetilde{\phi}}{\mathrm{im}\,\widetilde{\epsilon}} \oplus G\,,$$

so that in particular,

$$H^0(K; G) \cong \widetilde{H}^0(K; G) \oplus G\,.$$

4. Let K be the complex whose space is the real line and whose vertices are the integers. Let $\sigma_n = [n, n+1]$. Show that $H^1(K) = 0$ by finding a specific cochain whose coboundary is

$$\sum_{i=-\infty}^{\infty} a_i \sigma_i^*\,.$$

5.* Let K be a finite complex.

(a) Use the theorem on standard bases for chain complexes [Section 1.12] to express $H^p(K)$ in terms of the betti numbers and torsion coefficients of K.

(b) Express $H^p(K; G)$ in terms of the betti numbers and torsion coefficients of K. The answer will involve the groups G, G/mG, and $\ker(G \xrightarrow{m} G)$, the kernel of multiplication by m in G.

(c) Compute $H^p(X; G)$ if X is the torus, the Klein bottle, or projective space.

6. Calculate the cohomology of the 2-sphere. It is more difficult to calculate the cohomology of the higher-dimensional spheres. Why?

5.4 Relative Cohomology

Continuing our discussion of simplicial cohomology, we define relative cohomology groups. We also consider the homomorphism induced by a simplicial

map and the long exact sequence in cohomology. As we shall see, in some respects, relative cohomology is similar to relative homology, while in other respects, it is rather different.

Definition 5.4.1 Let K be a complex, let K_0 be a subcomplex of K, and let G be an abelian group. We define the group of *relative cochains* in dimension p by setting

$$C^p(K, K_0; G) = \mathrm{Hom}\,(C_p(K, K_0); G)\,.$$

The *relative coboundary* operator δ is defined to be the dual of the relative boundary operator. We define $Z^p(K, K_0; G)$ to be the kernel of the homomorphism

$$\delta : C^p(K, K_0; G) \to C^{p+1}(K, K_0; G)\,,$$

$B^{p+1}(K, K_0; G)$ to be its image, and

$$H^p(K, K_0; G) = Z^p(K, K_0; G)/B^p(K, K_0; G)\,.$$

These groups are called the group of *relative cocycles*, the group of *relative coboundaries*, and the *relative cohomology group*, respectively.

The examples in the preceding section have given us some idea of how to picture cochains and cocycles. How shall we picture a *relative* cochain or cocycle? Here the situation is rather different from the situation in homology. We explain the difference as follows:

For chains, we had an exact sequence

$$0 \longrightarrow C_p(K_0) \xrightarrow{i} C_p(K) \xrightarrow{j} C_p(K, K_0) \longrightarrow 0\,,$$

where $C_p(K_0)$ is a subgroup of $C_p(K)$, and $C_p(K, K_0)$ is their quotient. The sequence splits because the relative chain group is free. Therefore the sequence

$$0 \longleftarrow C^p(K_0; G) \xleftarrow{\widetilde{i}} C^p(K; G) \xleftarrow{\widetilde{j}} C^p(K, K_0; G) \longleftarrow 0$$

is exact and splits. This leads to the following surprising fact:

It is natural to consider $C^p(K, K_0; G)$ as a *subgroup* of $C^p(K; G)$, and to consider $C^p(K_0; G)$ as a *quotient group* of $C_p(K; G)$.

Let us examine this matter more closely. A relative cochain is a homomorphism mapping $C_p(K, K_0)$ into G. The group of such homomorphisms corresponds precisely to the group of all homomorphisms of $C_p(K)$ into G that vanish on the subgroup $C_p(K_0)$. This is just a subgroup of the group of *all* homomorphisms of $C_p(K)$ into G. Thus $C^p(K, K_0; G)$ can be naturally considered to be the subgroup of $C^p(K; G)$ consisting of those cochains that vanish on every oriented simplex of K_0. In some sense, $C^p(K, K_0; G)$ is the group of those cochains of K that are "carried by" $K \setminus K_0$. The coboundary operator maps this subgroup of $C^p(K; G)$ into itself: Suppose that c^p vanishes

on every simplex of K_0. If τ is a $(p+1)$-simplex of K_0, then $\partial\tau$ is carried by K_0, so

$$\langle \delta c^p, \tau \rangle = \langle c^p, \partial\tau \rangle = 0\,.$$

Thus the map $\widetilde{j}$ can be interpreted as an inclusion map. To interpret $\widetilde{i}$, we note that it carries the cochain c^p of K to the cochain $c^p \circ i$, which is simply the restriction of c^p to $C_p(K_0)$. We summarize these results as follows:

If we begin with the sequence

$$0 \longrightarrow C_p(K_0) \xrightarrow{i} C_p(K) \xrightarrow{j} C_p(K, K_0) \longrightarrow 0\,,$$

then the dual of the projection map j is an inclusion map $\widetilde{j}$, and the dual of the inclusion map i is a restriction map $\widetilde{i}$.

Let us now consider the homomorphism of cohomology induced by a simplicial map.

Recall that if $f : (K, K_0) \to (L, L_0)$ is a simplicial map, then one has a corresponding chain map

$$f_\# : C_p(K, K_0) \to C_p(L, L_0)\,.$$

The dual of $f_\#$ maps cochains to cochains; we usually denote it by $f^\#$. Because $f_\#$ commutes with ∂, the map $f^\#$ commutes with δ. The dual of the equation $f_\# \circ \partial = \partial \circ f_\#$ is the equation $\delta \circ f^\# = f^\# \circ \delta$. Hence, $f^\#$ carries cocycles to cocycles and coboundaries to coboundaries. It is called a *cochain map*, and it induces a homomorphism of cohomology groups

$$H^p(K, K_0; G) \xleftarrow{f^*} H^p(L, L_0; G)\,.$$

Functoriality holds, even on the cochain level. For if i is the identity map, then $i_\#$ is the identity and so is $i^\#$. Similarly, the equation $(g \circ f)_\# = g_\# \circ f_\#$, when dualized, gives the equation $(g \circ f)^\# = f^\# \circ g^\#$.

Just as in the case of homology, one has a long exact sequence in cohomology involving the relative groups. But again, there are a few differences.

Theorem 5.4.2 *Let K be a complex, and let K_0 be a subcomplex. There exists an exact sequence*

$$\cdots \longleftarrow H^p(K_0; G) \longleftarrow H^p(K; G) \longleftarrow H^p(K, K_0; G) \xleftarrow{\delta^*} H^{p-1}(K_0; G) \longleftarrow \cdots\,.$$

A similar sequence exists in reduced cohomology if K_0 is not empty. A simplicial map $f : (K, K_0) \to (L, L_0)$ induces a homomorphism of long exact cohomology sequences.

Proof: This theorem follows from applying the zig-zag lemma [Lemma 3.3.2] to the diagram

$$\begin{array}{ccccccccc} 0 & \longleftarrow & C^{p+1}(K_0;G) & \xleftarrow{\widetilde{\imath}} & C^{p+1}(K;G) & \xleftarrow{\widetilde{\jmath}} & C^{p+1}(K,K_0;G) & \longleftarrow & 0 \\ & & \uparrow\delta & & \uparrow\delta & & \uparrow\delta & & \\ 0 & \longleftarrow & C^{p}(K_0;G) & \xleftarrow{\widetilde{\imath}} & C^{p}(K;G) & \xleftarrow{\widetilde{\jmath}} & C^{p}(K,K_0;G) & \longleftarrow & 0\,. \end{array}$$

Since i and j commute with ∂, the dual maps $\widetilde{\imath}$ and $\widetilde{\jmath}$ commute with δ. The fact that δ^* *raises* dimension by 1 follows from the proof of the zig-zag lemma. [If you turn the page upside down, the arrows look like those in the proof of the zig-zag lemma.]

The sequence in reduced cohomology is derived similarly, once one adjoins the sequence at the bottom of the diagram.

$$\begin{array}{ccccccccc} 0 & \longleftarrow & C^0(K_0;G) & \longleftarrow & C^0(K;G) & \longleftarrow & C^0(K,K_0;G) & \longleftarrow & 0 \\ & & \uparrow\widetilde{\epsilon} & & \uparrow\widetilde{\epsilon} & & \uparrow & & \\ 0 & \longleftarrow & G & \longleftarrow & G & \longleftarrow & 0 & \longleftarrow & 0 \end{array}$$

Since $\widetilde{\epsilon}$ is injective, no nontrivial cohomology groups appear in dimension -1. □

Now let us compute cohomology groups in a few examples, using this sequence.

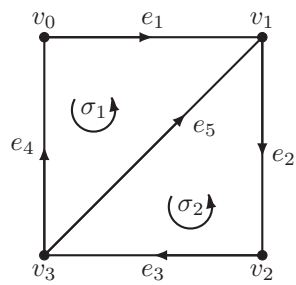


Figure 5.4.1

EXAMPLE 5.4.3 Consider the case of a 2-dimensional square region K modulo its boundary K_0 as pictured in Figure 5.4.1. We treat the group of relative cochains as those cochains of K that are carried by $K \setminus K_0$. Both σ_1^* and σ_2^* are such cochains, and each is a cocycle [trivially]. There is only one cochain e_5^* in dimension 1 carried by $K \setminus K_0$, and its coboundary is

$$\delta e_5^* = \sigma_1^* - \sigma_2^*\,.$$

Thus the group $H^2(K, K_0)$ is infinite cyclic, and is generated by the cohomology class $\{\sigma_1^*\} = \{\sigma_2^*\}$.

The group $H^1(K, K_0)$ vanishes because the only 1-cochain carried by $K \setminus K_0$ is not a cocycle. The group $H^0(K, K_0)$ vanishes because there are no 0-cochains carried by $K \setminus K_0$.

To summarize, we have computed $H^2(K, K_0) \cong \mathbb{Z}$, $H^1(K, K_0) = 0$, and $H^0(K, K_0) = 0$.

The preceding theorem gives us the exact sequence

$$H^2(K_0) \longleftarrow H^2(K) \longleftarrow H^2(K, K_0) \xleftarrow{\delta^*} H^1(K_0) \longleftarrow H^1(K) \longleftarrow H^1(K, K_0)\,.$$

In Example 5.3.5, we found that $H^1(K_0) \cong \mathbb{Z}$. Since there are no 2-cochains carried by K_0, we have $H^2(K_0) = 0$. Let us insert into the exact sequence the information we already have:

$$\begin{array}{ccccccccccc} H^2(K_0) & \longleftarrow & H^2(K) & \longleftarrow & H^2(K, K_0) & \xleftarrow{\delta^*} & H^1(K_0) & \longleftarrow & H^1(K) & \longleftarrow & H^1(K, K_0) \\ \| & & \| & & \| & & \| & & \| & & \| \\ 0 & \longleftarrow & ? & \longleftarrow & \mathbb{Z} & \xleftarrow{\delta^*} & \mathbb{Z} & \longleftarrow & ? & \longleftarrow & 0\,. \end{array}$$

In addition to the unknown groups, we also do not know the action of the homomorphism $\delta^* : H^1(K_0) \to H^2(K, K_0)$.

The group $H^1(K_0)$ is generated by the cocycle e_1^*. To compute $\delta^*\{e_1^*\}$, we first pull e_1^* back to K by considering it as a cochain of K, then take its coboundary δe_1^* in K, and finally consider the result as a cocycle of K modulo K_0. By direct computation, $\delta e_1^* = -\sigma_1^*$ as cochains of K. Since σ_1^* *generates* $H^2(K, K_0)$, as we proved above, it follows that δ^* is an isomorphism.

Now that we know that δ^* is an isomorphism, we can conclude that both of the unknown groups in this exact sequence must vanish.

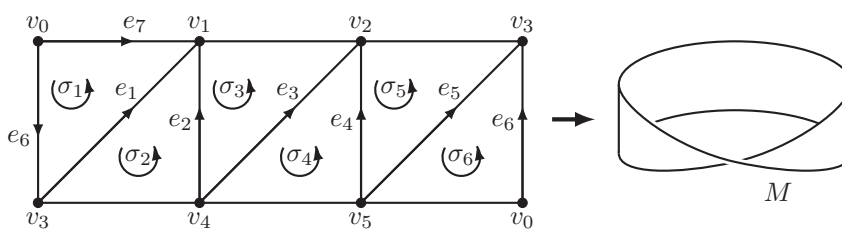


Figure 5.4.2

EXAMPLE 5.4.4 Consider the Möbius band M modulo its edge E, as pictured in Figure 5.4.2. We calculate the cohomology of (M, E) and of M.

Each of the cochains σ_i^* is a cocycle [trivially], so they form a basis for the group $Z^2(M, E)$ of relative 2-cocycles. The six 1-cochains $e_1^*, e_2^*, \ldots, e_6^*$ form a basis for the group $C^1(M, E)$ of relative 1-cochains, because the other 1-simplices of M lie in the edge E. It is convenient to replace these bases for $Z^2(M, E)$ and $C^1(M, E)$ by different bases. Let us take

$$\sigma_1^*, \quad \sigma_1^* - \sigma_2^*, \quad \sigma_2^* - \sigma_3^*, \quad \sigma_3^* - \sigma_4^*, \quad \sigma_4^* - \sigma_5^*, \quad \sigma_5^* - \sigma_6^*$$

as a basis for $Z^2(M, E)$, and

$$e_1^*, \quad e_2^*, \quad e_3^*, \quad e_4^*, \quad e_5^*, \quad e_1^* + e_2^* + \cdots + e_6^*$$

as a basis for $C^1(M, E)$. Then we can calculate $H^2(M, E)$ readily. We see that $\delta e_i^* = \sigma_i^* - \sigma_{i+1}^*$ for $i = 1, 2, \ldots, 5$, and $\delta(e_1^* + e_2^* + \cdots + e_6^*) = 2\sigma_1^*$. Thus we have $H^2(M, E) \cong \mathbb{Z}/2$, and the group is generated by σ_1^*.

The group $H^1(M, E)$ vanishes, because there are no relative 1-cocycles: This follows from the fact that δ carries our chosen basis for $C^1(M, E)$ to a basis for a subgroup of $C^2(M, E)$. [As another proof, you can also show directly that $n_1 e_1^* + n_2 e_2^* + \cdots + n_6 e_6^*$ is a cocycle only if $n_i = 0$ for all i.]

To summarize, we have computed $H^2(M, E) \cong \mathbb{Z}/2$ and $H^1(M, E) = 0$. We have $H^0(M, E) = 0$, because there are no 0-cochains carried by $M \setminus E$.

Theorem 5.4.2 gives us the exact sequence

$$H^2(E) \longleftarrow H^2(M) \longleftarrow H^2(M, E) \xleftarrow{\delta^*} H^1(E) \longleftarrow H^1(M) \longleftarrow H^1(M, E)\,.$$

The proof that $H^1(E)$ is infinite cyclic is easy, since E is a hexagon. [See Example 5.3.5.] Since there are no 2-cochains carried by E, we have $H^2(E) = 0$. Let us insert into the exact sequence the information we already have:

$$\begin{array}{ccccccccccccc}
H^2(E) & \longleftarrow & H^2(M) & \longleftarrow & H^2(M, E) & \xleftarrow{\delta^*} & H^1(E) & \longleftarrow & H^1(M) & \longleftarrow & H^1(M, E) \\
\| & & \| & & \| & & \| & & \| & & \| \\
0 & \longleftarrow & ? & \longleftarrow & \mathbb{Z}/2 & \xleftarrow{\delta^*} & \mathbb{Z} & \longleftarrow & ? & \longleftarrow & 0\,.
\end{array}$$

In addition to the unknown groups, we also do not know the action of the homomorphism $\delta^* : H^1(E) \to H^2(M, E)$.

Let us compute δ^*. The group $H^1(E)$ is generated by e_i^*, where e_i is any oriented 1-simplex of E. We choose e_7 as indicated in Figure 5.4.2. Then, by direct computation, we find $\delta^* e_7 = -\sigma_1^*$, so $\delta^*$is surjective. It follows at once that $H^2(M) = 0$ and $H^1(M) \cong \mathbb{Z}$.

Next we give the definition of a "pseudo manifold," a notion that will occur many times.

Definition 5.4.5 A simplicial pair (K, K_0) is called a *relative pseudo n-manifold* if the following hold:

(1) The closure of $|K| \setminus |K_0|$ equals a union of n-simplices.

(2) Each $(n-1)$-simplex of K that is not in K_0 is a face of exactly two n-simplices of K.

(3) Given two n-simplices σ and σ' of K that are not in K_0, there is a sequence of n-simplices of K, not in K_0,

$$\sigma = \sigma_0,\ \sigma_1,\ \ldots,\ \sigma_k = \sigma'$$

such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$-simplex not in K_0, for each i.

If $K_0 = \emptyset$, then we call K a *pseudo n-manifold*.

In Section 1.6 in which we defined the homology groups, we required that the simplices of a complex be oriented, but the orientation of one simplex need not have any effect on the choice of orientation for any other simplex. In fact, it is useful to know whether or not the top dimensional simplices can be coherently oriented.

Definition 5.4.6 Let (K, K_0) be a relative pseudo n-manifold.

(1) Suppose that $\sigma \neq \sigma'$ are n-simplices of K, not in K_0, and the sequence of n-simplices of K, not in K_0,

$$\sigma = \sigma_0,\ \sigma_1,\ \ldots,\ \sigma_k = \sigma'$$

such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$-simplex not in K_0, for each i, has no repetitions. [Exercise 5(a) asks you to show such a sequence exists.] Once σ is oriented, there are unique orientations of each σ_i such that $\partial\sigma_{i-1} + \partial\sigma_i$ has coefficient 0 on $\sigma_{i-1} \cap \sigma_i$ for each i. The resulting orientation of $\sigma_k = \sigma'$ is said to be *induced* by the given orientation of σ relative to the given sequence.

(2) Let σ be an n-simplex of K that is not in K_0. Fix an orientation for σ. If, for every $\sigma' \neq \sigma$, the orientation of σ' induced by that of σ is independent of the sequence joining them, then (K, K_0) is said to be *orientable*. Otherwise, (K, K_0) is said to be *non-orientable*.

EXERCISES

1. Let M denote the Möbius band as in Example 5.4.4. Find a cocycle generating $H^1(M)$.

2. Consider the cylinder C modulo one edge E.

(a) Compute $H^i(C, E)$.

(b) Use the long exact cohomology sequence of the pair (C, E) to compute $H^i(C)$.

3. Which of the following spaces are pseudo manifolds in their familiar triangulations?

(a) $\mathbb{S}^1$

(b) $\mathbb{S}^2$

(c) The letter θ.

(d) The union of $\mathbb{S}^2$ and a circle that intersects $\mathbb{S}^2$ in one point.

(e) The union of two copies of $\mathbb{S}^2$ with a point in common.

(f) $\mathbb{S}^2$ with the north and south poles identified.

4. Show that the compact 2-manifolds, in their usual triangulations, are pseudo manifolds. [We will see later that *any* connected, triangulated n-manifold is a pseudo n-manifold.]

5. Let (K, K_0) be a relative pseudo n-manifold.

(a) Given n-simplices $\sigma \neq \sigma'$, neither in K_0, show that there exists a sequence

$$\sigma = \sigma_0,\ \sigma_1,\ \ldots,\ \sigma_k = \sigma',$$

with no repetitions, such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$-simplex not in K_0, for each i.

(b) Show that, if K is finite, then

$$H_n(K, K_0) \cong \mathbb{Z}, \quad H^n(K, K_0) \cong \mathbb{Z} \qquad \text{if } (K, K_0) \text{ is orientable,}$$

$$H_n(K, K_0) = 0, \quad H^n(K, K_0) \cong \mathbb{Z}/2 \quad \text{if } (K, K_0) \text{ is non-orientable.}$$

[**Hint:** If γ is the sum of all the n-simplices of K not in K_0, oriented arbitrarily, then for each relative $(n-1)$-cochain c^{n-1}, the number $\langle \delta c^{n-1}, \gamma \rangle$ is even. Therefore, σ^* does not cobound.]

(c) Conclude that, in the finite case at least, orientability is independent of the choice of σ, and in fact depends only on the topological pair $(|K|, |K_0|)$, not on the particular triangulation involved.

(d) Show that, if K is finite, then

$$H_n(K, K_0; \mathbb{Z}/2) \cong \mathbb{Z}/2 \cong H^n(K, K_0; \mathbb{Z}/2).$$

5.5 Cohomology Theory

Now that we have some feeling for what the simplicial cohomology groups look like, let us deal with cohomology theory more generally. We construct

both the simplicial and the singular cohomology *theories*, we show they are naturally isomorphic for triangulable spaces, and we verify the cohomology versions of the Eilenberg–Steenrod axioms.

First, let us work on the level of chain complexes.

The Cohomology of a Chain Complex

Let $\mathcal{C} = \{C_p, \partial\}$ be a chain complex, and let G be an abelian group. We define the *p-dimensional cochain group* of $\mathcal{C}$, with coefficients in G, by

$$C^p(\mathcal{C}; G) = \text{Hom}\,(C_p, G)\,.$$

We define the *coboundary operator* δ to be the dual of the boundary operator. It follows that $\delta^2 = 0$. The family of groups and homomorphisms $\{C^p(\mathcal{C}; G), \delta\}$ is called the *cochain complex* of $\mathcal{C}$ with coefficients in G. As usual, the kernel of the homomorphism

$$\delta : C^p(\mathcal{C}; G) \to C^{p+1}(\mathcal{C}; G)$$

is denoted $Z^p(\mathcal{C}; G)$, its image is denoted $B^{p+1}(\mathcal{C}; G)$, and the *cohomology group* of $\mathcal{C}$ in dimension p, with coefficients in G, is defined by the equation

$$H^p(\mathcal{C}; G) = Z^p(\mathcal{C}; G)/B^p(\mathcal{C}; G)\,.$$

If $\{\mathcal{C}, \epsilon\}$ is an augmented chain complex, then one has a corresponding cochain complex

$$\cdots \longleftarrow C^1(\mathcal{C}; G) \xleftarrow{\delta_1} C^0(\mathcal{C}; G) \xleftarrow{\widetilde{\epsilon}} \text{Hom}\,(\mathbb{Z}, G)\,,$$

where $\widetilde{\epsilon}$ is injective. We define the *reduced cohomology groups* of $\mathcal{C}$ by setting $\widetilde{H}^q(\mathcal{C}; G) = H^q(\mathcal{C}; G)$ if $q > 0$, and

$$\widetilde{H}^0(\mathcal{C}; G) = \ker \delta_1 / \text{im}\ \widetilde{\epsilon}\,.$$

It is easy to see that if $\widetilde{H}_0(\mathcal{C})$ vanishes, then $\widetilde{H}^0(\mathcal{C}; G)$ vanishes as well, for exactness of $C_1 \longrightarrow C_0 \longrightarrow \mathbb{Z} \longrightarrow 0$ implies exactness of the dual sequence. In general [see the exercises of Section 5.3], we have the equation

$$H^0(\mathcal{C}; G) \cong \widetilde{H}^0(\mathcal{C}; G) \oplus G\,.$$

Definition 5.5.1 Suppose that $\mathcal{C} = \{C_p, \partial\}$ and $\mathcal{C}' = \{C'_p, \partial'\}$ are chain complexes. Suppose that $\phi : \mathcal{C} \to \mathcal{C}'$ is a chain map, so that $\partial' \circ \phi = \phi \circ \partial$. Then the dual homomorphism

$$C^p(\mathcal{C}; G) \xleftarrow{\widetilde{\phi}} C^p(\mathcal{C}'; G)$$

commutes with δ. Such a homomorphism is called a *cochain map*. It carries cocycles to cocycles, and coboundaries to coboundaries, so it induces a homomorphism of cohomology groups,

$$H^p(\mathcal{C};G) \xleftarrow{\phi^*} H^p(\mathcal{C}';G)\,.$$

The assignment

$$\mathcal{C} \longrightarrow H^p(\mathcal{C};G) \quad \text{and} \quad \phi \longrightarrow \phi^*$$

satisfies the usual functorial properties. In fact, they hold already on the chain level.

If $\mathcal{C} = \{C_p, \epsilon\}$ and $\mathcal{C}' = \{C'_p, \epsilon'\}$ are augmented chain complexes and if $\phi : \mathcal{C} \to \mathcal{C}'$ is an augmentation preserving chain map, then $\epsilon' \circ \phi = \epsilon$ so that $\widetilde{\phi} \circ \widetilde{\epsilon}' = \widetilde{\epsilon}$ holds. In this case, $\widetilde{\phi}$ induces a homomorphism of reduced as well as ordinary cohomology.

Suppose now that $\phi, \psi : \mathcal{C} \to \mathcal{C}'$ are chain maps, and that D is a chain homotopy between them, so that

$$D\partial + \partial' D = \phi - \psi\,.$$

Then $\widetilde{D} : C^{p+1}(\mathcal{C}';G) \to C^p(\mathcal{C};G)$ is a homomorphism satisfying the equation

$$\delta\widetilde{D} + \widetilde{D}\delta' = \widetilde{\phi} - \widetilde{\psi}\,.$$

It is said to be a *cochain homotopy* between $\widetilde{\phi}$ and $\widetilde{\psi}$. If such a $\widetilde{D}$ exists, it follows at once that the induced cohomology homomorphisms ϕ^* and ψ^* are equal. For, given any cocycle z_p, we have

$$\widetilde{\phi}(z^p) - \widetilde{\psi}(z^p) = \delta\widetilde{D}z^p + 0\,.$$

This observation has the following consequence.

Theorem 5.5.2 *Let $\mathcal{C}$ and $\mathcal{C}'$ be chain complexes, and let $\phi : \mathcal{C} \to \mathcal{C}'$ be a chain equivalence. Then ϕ_* and ϕ^* are isomorphisms of homology and cohomology, respectively. If $\mathcal{C}$ and $\mathcal{C}'$ are augmented and ϕ is augmentation-preserving, then ϕ_* and ϕ^* are isomorphisms of reduced homology and cohomology, respectively.*

Proof: Since ϕ is a chain equivalence, there is a chain map $\psi : \mathcal{C}' \to \mathcal{C}$ such that $\phi \circ \psi$ and $\psi \circ \phi$ are chain homotopic to identity maps. Then $\widetilde{\psi} \circ \widetilde{\phi}$ and $\widetilde{\phi} \circ \widetilde{\psi}$ are cochain homotopic to identity maps, so $\psi^* \circ \phi^*$ and $\phi^* \circ \psi^*$ equal the identity maps of $H^p(\mathcal{C}')$ and $H^p(\mathcal{C})$, respectively.

The same argument holds for reduced cohomology. □

Finally, suppose that

$$0 \longrightarrow \mathcal{C} \xrightarrow{\phi} \mathcal{D} \xrightarrow{\psi} \mathcal{E} \longrightarrow 0$$

is a short exact sequence of chain complexes that *splits in each dimension.* [For example, this occurs when $\mathcal{E}$ is free.] Then the dual sequence

$$0 \longleftarrow C^p(\mathcal{C};G) \xleftarrow{\widetilde{\phi}} C^p(\mathcal{D};G) \xleftarrow{\widetilde{\psi}} C^p(\mathcal{E};G) \longleftarrow 0$$

is exact. Applying the zig-zag lemma [Lemma 3.3.2], we obtain a long exact sequence in cohomology

$$\cdots \longleftarrow H^p(\mathcal{C};G) \xleftarrow{\phi^*} H^p(\mathcal{D};G) \xleftarrow{\psi^*} H^p(\mathcal{E};G) \xleftarrow{\delta^*} H^{p-1}(\mathcal{C};G) \longleftarrow \cdots,$$

where δ^* is induced in the usual manner by the coboundary operator. This sequence is natural in the sense that if f is a homomorphism of short exact sequences of chain complexes, then $\widetilde{f}$ is a homomorphism of the dual sequences, and f^* is a homomorphism of the corresponding long exact cohomology sequences.

The Axioms for Cohomology

Now we state the cohomology versions of the Eilenberg–Steenrod axioms.

Given an admissible class $\mathcal{A}$ of pairs of spaces (X, A) and an abelian group G, a *cohomology theory* on $\mathcal{A}$, with coefficients in G, consists of the following:

(1) A function defined for each integer p and each pair (X, A) in $\mathcal{A}$ whose value is an abelian group $H^p(X, A; G)$.

(2) A function that, for (X, A) and (Y, B) in $\mathcal{A}$, assigns to each continuous map $h : (X, A) \to (Y, B)$ and each integer p a homomorphism

$$H^p(X, A; G) \xleftarrow{h^*} H^p(Y, B; G).$$

(3) A function that assigns to each pair (X, A) in $\mathcal{A}$ and each integer p a homomorphism

$$H^p(X, A; G) \xleftarrow{\delta^*} H^{p-1}(A; G).$$

The following axioms are to be satisfied:

Axiom 1. If i is the identity, then i^* is the identity.

Axiom 2. $(k \circ h)^* = h^* \circ k^*$.

Axiom 3. δ^* is a natural transformation of functors.

Axiom 4. [Exactness Axiom] The following sequence is exact, where i and j are inclusions:

$$\cdots \longleftarrow H^p(A; G) \xleftarrow{i^*} H^p(X; G) \xleftarrow{j^*} H^p(X, A; G) \xleftarrow{\delta^*} H^{p-1}(A; G) \longleftarrow \cdots.$$

Axiom 5. [Homotopy Axiom] If $h \simeq k$, then $h^* = k^*$.

Axiom 6. [Excision Axiom] Given (X, A), let U be an open set in X such that $\overline{U} \subset \text{Int } A$. If $(X \setminus U, A \setminus U)$ is admissible, then the inclusion j induces a cohomology isomorphism

$$H^p(X \setminus U, A \setminus U; G) \xleftarrow{j^*} H^p(X, A; G)\,.$$

Axiom 7. [Dimension Axiom] If P is a one-point space, then $H^p(P; G) = 0$ for $p \neq 0$ and

$$H^0(P; G) \cong G\,.$$

The axiom of compact support has no counterpart in cohomology theory.

Singular Cohomology Theory

Now we consider singular theory and show that it satisfies the axioms.

The *singular cohomology groups* of a topological pair (X, A), with coefficients in the abelian group G, are defined by the equation

$$H^p(X, A; G) = H^p(\mathcal{S}(X, A); G)\,,$$

where $\mathcal{S}(X, A)$ is the singular chain complex of (X, A). As usual, we delete A from the notation if $A = \emptyset$, and we delete G if it equals the group of integers.

The reduced cohomology groups are defined by the equation

$$\widetilde{H}^p(X; G) = \widetilde{H}^p(\mathcal{S}(X); G)\,,$$

relative to the standard augmentation for $\mathcal{S}(X)$.

Given a continuous map $h : (X, A) \to (Y, B)$, there is a chain map

$$h_\# : S_p(X, A) \to S_p(Y, B)\,,$$

defined by $h_\#(T) = h \circ T$. We customarily denote the dual chain map by $h^\#$ [so $h^\#$ is the induced cochain map]. It induces a homomorphism

$$H^p(X, A; G) \xleftarrow{h^*} H^p(Y, B; G)\,.$$

[The same holds in reduced cohomology if A and B are empty, since $h_\#$ is augmentation-preserving.] The functorial properties [Axioms **1** and **2**] hold even on the cochain level. The short exact sequence of chain complexes

$$0 \longrightarrow S_p(A) \longrightarrow S_p(X) \longrightarrow S_p(X, A) \longrightarrow 0$$

splits because $S_p(X, A)$ is free. Therefore the zig-zag lemma gives us an exact sequence

$$\cdots \longleftarrow H^p(A; G) \longleftarrow H^p(X; G) \longleftarrow H^p(X, A; G) \xleftarrow{\delta^*} H^{p-1}(A; G) \longleftarrow \cdots\,.$$

A continuous map $h : (X, A) \to (Y, B)$ induces a homomorphism of long exact cohomology sequences by the naturality property of the zig-zag lemma. [A similar result holds for reduced cohomology if A is non-empty.] Thus axioms **3** and **4** hold.

If $h, k : (X, A) \to (Y, B)$ are homotopic, then $h_\#$ and $k_\#$ are chain homotopic, as we proved in Theorem 4.3.7. Then $h^\#$ and $k^\#$ are cochain homotopic, so that $h^* = k^*$.

To compute the cohomology of a one-point space P, we recall [see Theorem 4.3.3] that the singular chain complex of P has the form

$$\cdots \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0\,.$$

The cochain complex of P then has the form

$$\cdots \xleftarrow{\cong} G \xleftarrow{0} G \xleftarrow{\cong} G \xleftarrow{0} G \longleftarrow 0\,.$$

It follows that $H^i(P; G)$ is isomorphic to G if $i = 0$, and vanishes otherwise.

Finally, we come to the excision property of singular cohomology. Here is a point where the arguments we gave for homology do not go through automatically for cohomology. Suppose that

$$j : (X \setminus U, A \setminus U) \to (X, A)$$

is an excision map. If we had shown that $j_\#$ is a chain equivalence, then there would be no difficulty, for $j^\#$ would be a cochain equivalence, and it would follow that j^* is an isomorphism. But we proved only the weaker result that j_* is an isomorphism. [See Theorem 4.4.10.] So we have some work to do to carry this result over to cohomology.

What we need is the following fact, which will be proved in the next section: *Let $\mathcal{C}$ and $\mathcal{D}$ be free chain complexes, and let $\phi : \mathcal{C} \to \mathcal{D}$ be a chain map that induces a homology isomorphism in all dimensions. Then ϕ induces a cohomology isomorphism in all dimensions, for all coefficient groups G.*

The excision property of singular cohomology is an immediate consequence: Given $U \subset A \subset X$, with $\overline{U} \subset \operatorname{Int} A$, consider the inclusion map

$$j : (X \setminus U, A \setminus U) \to (X, A)\,.$$

Since the chain complexes involved are free, and since $j_\#$ induces an isomorphism in homology, it induces an isomorphism in cohomology as well.

Note that singular cohomology, like singular homology, satisfies an excision property slightly stronger than that stated in the axiom. One needs to have $\overline{U} \subset \operatorname{Int} A$, but one does not need U to be open, in order for excision to hold.

Simplicial Cohomology Theory

We have already dealt with some aspects of simplicial cohomology in Section 5.4. We have defined the cohomology groups of a simplicial pair (K, K_0),

and we have shown how a simplicial map f induces a homomorphism of these groups. Just as with homology, showing that an arbitrary continuous map induces a homomorphism requires a bit of work.

The construction follows the pattern of Sections 2.2 through 2.6. First, we recall that if f and g are two simplicial approximations of the same continuous map, then they are contiguous, so the corresponding chain maps $f_\#$ and $g_\#$ are chain homotopic. It follows that $f^\#$ and $g^\#$ are cochain homotopic, so that the cohomology homomorphisms f^* and g^* are equal. Furthermore, if K' is a subdivision of K, and if $g : (K', K'_0) \to (K, K_0)$ is a simplicial approximation to the identity, then $g_\#$ is a chain equivalence, so that $g^\#$ is a cochain equivalence, and g^* is an isomorphism.

The homomorphism induced by the continuous map

$$h : (|K|, |K_0|) \to (|L|, |L_0|)$$

is defined as follows: Choose a subdivision K' of K so that h has a simplicial approximation $f : (K', K'_0) \to (L, L_0)$. Choose $g : (K', K'_0) \to (K, K_0)$ to be a simplicial approximation to the identity. Finally, define

$$h^* = (g^*)^{-1} \circ f^* .$$

To verify that h^* is independent of the choices involved, and to check its functorial properties, involves arguments very similar to those given in Section 2.6, where we verified the corresponding properties for the homomorphism h_*. In fact, one can use exactly the same diagrams as appear in that section by simply reversing the arrows for all the induced homomorphisms! We leave the details for the reader.

Verifying the Eilenberg–Steenrod axioms is now straightforward. The existence of the long exact cohomology sequence we have already noted. Naturality of the sequence reduces to proving naturality in the case of a simplicial map, which we have already done. The homotopy axiom, the simplicial version of the excision axiom, and the dimension axiom follow just as they did in the case of singular cohomology. Nothing new of interest occurs.

The topological invariance, indeed the homotopy-type invariance, of the simplicial cohomology groups follows at once.

The Isomorphism between Simplicial and Singular Cohomology

Let K be a simplicial complex. In Section 4.7, we defined a chain map

$$\eta : \mathcal{C}(K) \to \mathcal{S}(|K|)$$

that induces an isomorphism in homology. Although it depends on a choice of a partial ordering for the vertices of K, the induced homomorphism η_* does not. Because η commutes with inclusions of subcomplexes, it induces a homomorphism on the relative groups, which is an isomorphism as well. We now show that the same result holds for cohomology.

Theorem 5.5.3 *Let (K, K_0) be a simplicial pair. Then η induces a cohomology isomorphism*

$$H^p(\mathcal{C}(K, K_0); G) \xleftarrow{\eta^*} H^p(\mathcal{S}(|K|, |K_0|); G)$$

that is independent of the choice of partial ordering of the vertices of K. It commutes with δ^* and with homomorphisms induced by continuous maps.

Proof: The chain map η carries the oriented simplex $[v_0, v_1, \ldots, v_p]$ of K to the linear singular simplex $\ell(v_0, v_1, \ldots, v_p)$ of K, provided $v_0 < v_1 < \cdots < v_p$ in the chosen ordering. Because the chain complexes involved are free and η induces a homology isomorphism, it induces a cohomology isomorphism as well. Furthermore, because η commutes with inclusions, it induces a homomorphism of relative cohomology. This homomorphism commutes with the coboundary operator δ^*, by the naturality of the zig-zag lemma. Therefore it is an isomorphism in relative cohomology, by the Steenrod five-lemma [Lemma 3.3.4].

To prove the rest of the theorem, we must examine the definition of η more closely. The map η was defined as the composite

$$C_p(K) \xrightarrow{\phi} C'_p(K) \xrightarrow{\theta} S_p(|K|)\,,$$

where $\phi([v_0, v_1, \ldots, v_p])$ equals the ordered simplex $(v_0, v_1, \ldots, v_p)$ if $v_0 < v_1 < \cdots < v_p$ in the chosen ordering, and

$$\theta((w_0, w_1, \ldots, w_p)) = \ell(w_0, w_1, \ldots, w_p)\,.$$

In Section 1.14, we defined a chain-homotopy inverse to ϕ by the equation

$$\psi((w_0, w_1, \ldots, w_p)) = \begin{cases} [w_0, w_1, \ldots, w_p] & \text{if the } w_i \text{ are distinct,} \\ 0 & \text{otherwise.} \end{cases}$$

The maps ψ and θ do not depend on the ordering, although ϕ does. The fact that η^* is independent of the ordering follows from the equation $\eta^* = (\psi^*)^{-1} \circ \theta^*$.

To show that η^* commutes with induced homomorphisms, we first consider the case of a simplicial map f. We showed in proving Theorem 1.14.18 that $f_\#$ commutes with ψ, and we showed in proving Theorem 4.7.6 that $f_\#$ commutes with θ. It follows that $f^\#$ commutes with the dual of ψ and θ, so that f^* commutes with η^*. To extend this result to arbitrary continuous maps, one follows the pattern of Theorem 4.7.7. □

When a space is not compact, it is often convenient to look at the functions that have compact support.

Definition 5.5.4 Let K be a complex. Let $C^p_c(K; G)$ denote the group of homomorphisms of $C_p(K)$ into G that vanish on all but finitely many oriented simplices of K. These homomorphisms are called *cochains with compact*

support. Provided $\delta : C_c^p(K;G) \to C_c^{p+1}(K;G)$ [see Exercise 4], the cohomology groups $H_c^p(K;G)$ exist and are called *cohomology groups with compact support*.

Similarly, if $h : X \to Y$ is continuous, then the inverse image of a compact subset of Y, while closed, need not be compact. Again it may be convenient to restrict the class of functions.

Definition 5.5.5 A map $h : X \to Y$ is said to be *proper* if for each compact subset D of Y, the set $h^{-1}(D)$ is a compact subset of X. A *proper homotopy* is a homotopy that is itself a proper map.

EXERCISES

1. If X is a path-connected space, show that $H^0(X) \cong \mathbb{Z}$. Find a generating cocycle.

2. State and prove a Mayer–Vietoris theorem in simplicial cohomology. Can you prove it for arbitrary coefficients?

3. **(a)** Suppose $A_1, A_2 \subset X$. Show that if $\{A_1, A_2\}$ is an excisive couple, then the inclusion

$$\mathcal{S}(A_1) + \mathcal{S}(A_2) \longrightarrow \mathcal{S}(A_1 \cup A_2)$$

induces an isomorphism in cohomology as well a homology.

(b) State and prove a Mayer–Vietoris theorem in singular cohomology.

4. **(a)** Show that if the complex K is locally finite, then δ maps C_c^p to C_c^{p+1}.

(b) If K is the complex whose space is $\mathbb{R}$ and whose vertices are the integers, show that $H_c^1(K) \cong \mathbb{Z}$ and $H_c^0(K) = 0$.

(c) Show that if $|K|$ is connected and non-compact, then $H_c^0(K) = 0$.

5. **(a)** Show that the assignment

$$K \longrightarrow H_c^p(K;G) \qquad \text{and} \qquad h \longrightarrow h^*$$

defines a functor from the category of locally finite simplicial complexes and proper continuous maps of their polytopes to abelian groups and homomorphisms. [**Hint:** If h is proper, so is any simplicial approximation to h. If f and g are contiguous and proper, and D is the chain homotopy between them, show that $\widetilde{D}$ carries finite cochains to finite cochains. If $\lambda : \mathcal{C}(K) \to \mathcal{C}(K')$

is the subdivision operator and if $g : K' \to K$ is a simplicial approximation to the identity, show that the duals of λ and $g_\#$ carry finite cochains to finite cochains. Prove the same for the dual of the chain homotopy between $\lambda \circ g_\#$ and the identity.]

(b) Show that if

$$h, k : |K| \to |L|$$

are properly homotopic, then $h^* = k^*$ as homomorphisms of cohomology with compact support. [**Hint:** Show that if D is the chain homotopy constructed in proving Theorem 2.7.5, then $\widetilde{D}$ carries finite cochains to finite cochains.]

(c) Extend the results of **(a)** and **(b)** to relative cohomology. Does there exist a long exact sequence of cohomology with compact support? If so, is it natural with respect to induced homomorphisms?

(d) Is there an excision theorem for cohomology with compact support?

6.* Repeat Exercise 5 for the homology groups based on infinite chains, that were introduced in the Exercises of Section 1.6.

7. Refer to Exercise 5 for terminology. Give an example of a proper, continuous mapping of the open unit ball in 3-dimensional Euclidean space to itself which is *not* a homeomorphism.

8. For subsets of Euclidean space, give a characterization of proper mappings in terms of sequences of points that converge to the boundary.

5.6 The Cohomology of Free Chain Complexes

Until now, we have computed the cohomology groups only for a few simple spaces. We wish to compute them more generally. We shall prove that, for a CW complex X, the cellular chain complex $\mathcal{D}(X)$, which we know can be used to compute the homology of X, can be used to compute the cohomology as well. This we shall prove in Section 5.8. The proof depends on two theorems about free chain complexes that we prove in this section.

The first theorem states that, for free chain complexes $\mathcal{C}$ and $\mathcal{D}$, *any* homomorphism $H_p(\mathcal{C}) \to H_p(\mathcal{D})$ of homology groups is induced by a chain map ϕ. The second theorem states that if a chain map $\phi : \mathcal{C} \to \mathcal{D}$ of free chain complexes induces an isomorphism in homology, then it induces an isomorphism in cohomology as well. An independent proof of the second theorem will be given in Chapter 7, so you can simply assume this theorem for the time being if you wish.

Definition 5.6.1 A short exact sequence of abelian groups,

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0\,,$$

where A and B are free, is called a *free resolution* of C. Any abelian group C has a free resolution: one can take B to be the free abelian group generated by the elements of C, and take A to be the kernel of the natural projection $B \longrightarrow C$. This last construction gives what is known as the *canonical free resolution* of C.

Free resolutions have the following useful property: Suppose one is given the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{\phi} & B & \xrightarrow{\psi} & C & \longrightarrow & 0 \\ & & & & & & \downarrow\gamma & & \\ 0 & \longrightarrow & A' & \xrightarrow{\phi'} & B' & \xrightarrow{\psi'} & C' & \longrightarrow & 0 \end{array}$$

where the horizontal sequences are exact and A and B are free. In this situation, there exist homomorphisms $\alpha : A \to A'$ and $\beta : B \to B'$ that make this diagram commute.

The proof is straightforward. Choose a basis for B. If b is a basis element, let β map b into any element of the set $(\psi')^{-1}(\gamma(\psi(B)))$, which is non-empty because ψ' is surjective. A bit of diagram chasing shows that β carries im ϕ into im ϕ'. Because ϕ' is a monomorphism, β induces a homomorphism $\alpha : A \to A'$.

Now we state and prove the first of our basic theorems.

Theorem 5.6.2 *Let $\mathcal{C}$ and $\mathcal{C}'$ be free chain complexes. If $\gamma : H_p(\mathcal{C}) \to H_p(\mathcal{C}')$ is a homomorphism defined for all p, then there is a chain map $\phi : \mathcal{C} \to \mathcal{C}'$ that induces γ.*

Indeed, if $\beta : Z_p \to Z'_p$ is any homomorphism of groups of cycles inducing γ, then β extends to a chain map.

Proof: Let Z_p denote the p-cycles, and B_p the p-boundaries, in the chain complex $\mathcal{C}$. Similarly, let Z'_p and B'_p be the p-cycles and p-boundaries of $\mathcal{C}'$. Since these groups are free abelian, there exist, for all p, homomorphisms α, β making the following diagram commute:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & B_p & \longrightarrow & Z_p & \longrightarrow & H_p(\mathcal{C}) & \longrightarrow & 0 \\ & & \downarrow\alpha & & \downarrow\beta & & \downarrow\gamma & & \\ 0 & \longrightarrow & B'_p & \longrightarrow & Z'_p & \longrightarrow & H_p(\mathcal{C}') & \longrightarrow & 0 \end{array}$$

We seek to extend β to a chain map ϕ of C_p into C'_p. For that purpose, consider the short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Z_p & \longrightarrow & C_p & \xrightarrow{\partial_0} & B_{p-1} & \longrightarrow & 0 \\ & & \beta\downarrow & & \phi\downarrow & & \downarrow\alpha & & \\ 0 & \longrightarrow & Z'_p & \longrightarrow & C'_p & \xrightarrow{\partial'_0} & B'_{p-1} & \longrightarrow & 0 \end{array}$$

Because B_{p-1} and B'_{p-1} are free, the sequences split. Choose subgroups U_p and U'_p so that

$$C_p = Z_p \oplus U_p \qquad \text{and} \qquad C'_p = Z'_p \oplus U'_p \,.$$

Then $\partial_0 : U_p \to B_{p-1}$ and $\partial'_0 : U'_p \to B'_{p-1}$ are isomorphisms. We define $\phi : C_p \to C'_p$ by letting it equal $\beta : Z_p \to Z'_p$ on the summand Z_p, and the map $U_p \to U'_p$ induced by α on the summand U_p. Then the first square commutes, automatically. The second square commutes for any element of U_p, by definition. And it commutes for any element of Z_p by exactness, for

$$\alpha(\partial_0 z_p) = \alpha(0) = 0 \qquad \text{and} \qquad \partial'_0\phi(z_p) = \partial'_0(\beta(z_p)) = 0 \,.$$

Thus it commutes for any element of C_p.

Now we show that ϕ is a chain map. Consider the diagram

$$\begin{array}{ccccccc} C_p & \xrightarrow{\partial_0} & B_{p-1} & \longrightarrow & Z_{p-1} & \longrightarrow & C_{p-1} \\ \downarrow\phi & & \downarrow\alpha & & \downarrow\beta & & \downarrow\phi \\ C'_p & \xrightarrow{\partial'_0} & B'_{p-1} & \longrightarrow & Z'_{p-1} & \longrightarrow & C'_{p-1} \end{array}$$

where the unlabeled maps are inclusions. The middle square commutes by the definition of α and β. The two end squares commute as we have just proved. Thus ϕ is a chain map.

The fact that ϕ induces the original homology homomorphism γ follows from the definition of β. □

Corollary 5.6.3 *Suppose that $\{\mathcal{C}, \epsilon\}$ and $\{\mathcal{C}', \epsilon'\}$ are free augmented chain complexes. If $\gamma : \widetilde{H}_p(\mathcal{C}) \to \widetilde{H}_p(\mathcal{C}')$ is a homomorphism defined for all p, then γ is induced by an augmentation-preserving chain map $\phi : \mathcal{C} \to \mathcal{C}'$.*

Proof: Consider the augmented chain complexes obtained from $\mathcal{C}$ and $\mathcal{C}'$. They have $\mathbb{Z}$ as their (-1)-dimensional groups and ϵ, ϵ', respectively, as the boundary operators from dimension 0 to dimension -1. We define β to equal the identity in dimension -1 [where the homology vanishes], and to be any homomorphism of the groups of cycles inducing γ in other dimensions. The preceding theorem applies. The resulting chain map ϕ will automatically preserve augmentation. □

Before we state and prove the second of our basic theorems, we consider a special case.

Lemma 5.6.4 *Let*

$$0 \longrightarrow \mathcal{C} \xrightarrow{\phi} \mathcal{D} \longrightarrow \mathcal{E} \longrightarrow 0$$

be an exact sequence of free chain complexes. If ϕ induces homology isomorphisms in all dimensions, then it induces cohomology isomorphisms as well.

Proof: The existence of the long exact sequence in homology and the fact that ϕ_* is an isomorphism imply that $H_p(\mathcal{E}) = 0$ for all p.

Let $B_p \subset Z_p \subset E_p$ denote the boundaries, cycles, and chains of $\mathcal{E}$, respectively, in dimension p. The short exact sequence

$$0 \longrightarrow Z_p \longrightarrow E_p \xrightarrow{\partial} B_{p-1} \longrightarrow 0$$

splits because B_{p-1} is free. Furthermore, $Z_p = B_p$ because the homology of $\mathcal{E}$ vanishes. Therefore, we can write $E_p = B_p \oplus U_p$, where ∂ maps B_p to zero and carries U_p isomorphically onto B_{p-1}. Now

$$\mathrm{Hom}\,(E_p, G) \cong \mathrm{Hom}\,(B_p, G) \oplus \mathrm{Hom}\,(U_p, G)\,.$$

You can check that δ induces a homomorphism that carries $\mathrm{Hom}\,(B_p, G)$ isomorphically onto $\mathrm{Hom}\,(U_{p+1}, G)$ and carries $\mathrm{Hom}\,(U_p, G)$ to zero. Then $\mathrm{Hom}\,(U_p, G)$ represents the cocycle group of $\mathcal{E}$ and it equals the image of δ. Thus we have $H^p(\mathcal{E}; G) = 0$. □

The above special case of our theorem is all we actually used in the preceding section. Later, we shall need the general version. The general case is reduced to the special case by means of the following lemma.

Lemma 5.6.5 *Let $\mathcal{C}$ and $\mathcal{D}$ be free chain complexes. Let $\phi : \mathcal{C} \to \mathcal{D}$ be a chain map. There is a free chain complex $\mathcal{D}'$ along with injective chain maps $i : \mathcal{C} \to \mathcal{D}'$ and $j : \mathcal{D} \to \mathcal{D}'$ such that j induces homology isomorphisms in all dimensions, and the diagram*

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\phi} & \mathcal{D} \\ & \searrow^{i} & \downarrow j \\ & & \mathcal{D}' \end{array}$$

commutes up to chain homotopy. Furthermore, the quotients $\mathcal{D}'/\mathrm{im}\ i$ and $\mathcal{D}'/\mathrm{im}\ j$ are free.

Proof: The definition of $\mathcal{D}'$ is one that we simply "pull out of a hat." Later, we shall explain its geometric motivation. It is sometimes called the "algebraic mapping cylinder" of ϕ.

Define $\mathcal{D}'$ to be the chain complex whose chain group in dimension p is given by

$$\mathcal{D}'_p = C_p \oplus D_p \oplus C_{p-1}\,.$$

Let the boundary operator in $\mathcal{D}'$ be defined by the equations

$$\begin{aligned}\partial'(c_p,0,0) &= (\partial c_p,0,0)\,,\\ \partial'(0,d_p,0) &= (0,\partial d_p,0)\,,\\ \partial'(0,0,c_{p-1}) &= (-c_{p-1},\phi(c_{p-1}),-\partial c_{p-1})\,.\end{aligned}$$

You can check without difficulty that $\partial'\circ\partial'=0$. It is clear from the definition that the natural inclusions $i : C_p \to D'_p$ and $j : D_p \to D'_p$ are chain maps. It is also clear that D'_p and the quotients $D'_p/\mathrm{im}\ i$ and $D'_p/\mathrm{im}\ j$ are free.

We define the chain homotopy $D : C_p \to D'_{p+1}$ by the equation

$$D(c_p) = (0,0,c_p)\,.$$

You can check that it satisfies the equation

$$\partial' D + D\partial = j\circ\phi - i\,.$$

Finally, we show that j induces an isomorphism in homology. For that purpose, it suffices to show that the homology of the chain complex $\mathcal{D}'/\mathcal{D}$ vanishes. The pth chain group of $\mathcal{D}'/\mathcal{D}$ is isomorphic to $C_p \oplus C_{p-1}$, and the induced boundary operator ∂'' is given by the equation

$$\partial''(c_p,c_{p-1}) = (\partial c_p - c_{p-1}, -\partial c_{p-1})\,.$$

If (c_p,c_{p-1}) is a cycle of this chain complex, then it follows in particular that $\partial c_p - c_{p-1}=0$. We compute directly that

$$(c_p,c_{p-1}) = (c_p,\partial c_p) = -\partial''(0,c_p)\,.$$

Thus (c_p,c_{p-1}) is a boundary. □

Theorem 5.6.6 *Let $\mathcal{C}$ and $\mathcal{D}$ be free chain complexes. Let $\phi : \mathcal{C}\to\mathcal{D}$ be a chain map. If ϕ induces homology isomorphisms in all dimensions, then ϕ induces cohomology isomorphisms in all dimensions.*

Proof: Given ϕ, let $i : \mathcal{C}\to\mathcal{D}'$ and $j : \mathcal{D}\to\mathcal{D}'$ be as in the preceding lemma. One has exact sequences of free chain complexes

$$0 \longrightarrow \mathcal{C} \xrightarrow{\ i\ } \mathcal{D}' \longrightarrow \mathcal{D}'/\mathcal{C} \longrightarrow 0\,,$$

$$0 \longrightarrow \mathcal{D} \xrightarrow{\ j\ } \mathcal{D}' \longrightarrow \mathcal{D}'/\mathcal{D} \longrightarrow 0\,.$$

The map j induces a homology isomorphism by the preceding lemma, while i induces a homology isomorphism because both j and ϕ do and i is chain homotopic to $j\circ\phi$. Therefore, by Lemma 5.6.4, i and j induce cohomology isomorphisms i^* and j^*, respectively, in all dimensions. Since i is chain homotopic to $j\circ\phi$, we have $i^* = \phi^*\circ j^*$. Therefore, ϕ^* is a cohomology isomorphism as well. □

Corollary 5.6.7 *Let $\mathcal{C}$ and $\mathcal{D}$ be free chain complexes. If $H_p(\mathcal{C}) \cong H_p(\mathcal{D})$ for all p, then $H^p(\mathcal{C}; G) \cong H^p(\mathcal{D}; G)$ for all p and G.*

Remark 5.6.8 Let us explain the geometric motivation underlying the definition of the chain complex $\mathcal{D}'$.

In homotopy theory, there is a standard construction for, roughly speaking, replacing an arbitrary continuous map $h : X \to Y$ by an imbedding of X in a space that is homotopy equivalent to Y. More precisely, there is a space Y' and imbeddings i and j such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{h} & Y \\ & \searrow^{i} & \downarrow{\scriptstyle j} \\ & & Y' \end{array}$$

commutes up to homotopy, and such that j is a homotopy equivalence. It follows at once that h is a homotopy equivalence if and only if i is. In this way problems concerning the map h are reduced to problems concerning the imbedding i.

This construction is called the *mapping cylinder construction*. We describe it here, and explain how the chain complex $\mathcal{D}'$ is an algebraic analogue.

Given $h : X \to Y$, let us form a quotient space from the disjoint union of $X \times I$ and Y by identifying each point $(x, 0)$ of $X \times 0$ with the point $h(x)$ of Y. The resulting adjunction space Y' is called the *mapping cylinder* of h. We picture it as looking something like a top hat. See Figure 5.6.1

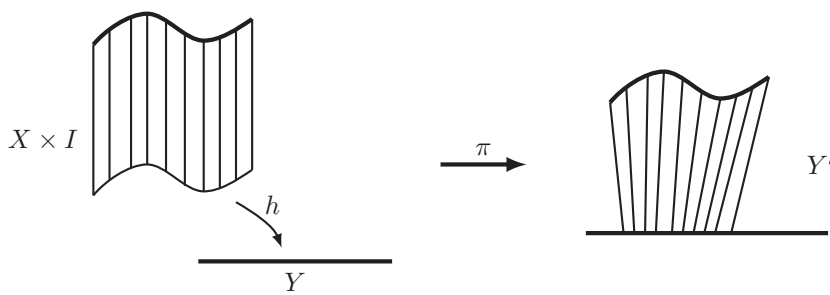


Figure 5.6.1

Let $\pi : (X \times I) \cup Y \to Y'$ be the quotient map. The restriction of π to Y defines an imbedding j of Y in Y', and the map $i(x) = \pi(x, 1)$ defines an imbedding of X in Y'. Clearly $j(Y)$ is a deformation retract of Y': One simply pushes down the top hat onto $j(Y)$. Just as clearly, the map $i : X \to Y'$ is homotopic to the map $j \circ h$: again, one simply pushes $i(X)$ down.

Now we seek to imitate this construction algebraically. So let us suppose for convenience that Y' is triangulated in such a way that $i(X)$ and $j(Y)$ are subcomplexes, and so is each set $\pi(\sigma \times I)$, for $\sigma \in X$. Let us identify X with $i(X)$, and Y with $j(Y)$, for simplicity of notation. Now $\mathcal{C}(X)$ plays the role of the chain complex $\mathcal{C}$, $\mathcal{C}(Y)$ plays the role of $\mathcal{D}$, and $\mathcal{C}(Y')$ plays the role of $\mathcal{D}'$. The map $h_\# : \mathcal{C}(X) \to \mathcal{C}(Y)$ plays the role of the chain map ϕ.

What does the chain complex $\mathcal{D}'$ look like algebraically? Suppose we break Y' up into the cells of a CW complex. The open p-cells will consist of open p-simplices Int σ_p of X, the open p-simplices of Int τ_p of Y, and the open cells of the form $\pi(\text{Int } \sigma_{p-1} \times \text{Int } I)$ that lie "between" X and Y. Then the pth chain group of Y' is essentially just

$$C_p(X) \oplus C_p(Y) \oplus C_{p-1}(X)\,,$$

since the group of "in between" cells is isomorphic to $C_{p-1}(X)$. How does the boundary operator of Y' act on these chains? Clearly, it acts just like ∂_X in X, and ∂_Y in Y. What does it do to a cell of the third kind? In the space $X \times I$, it is easy to see that

$$\partial(\sigma \times I) = \sigma \times 0 - \sigma \times 1 \pm (\partial\sigma) \times I\,.$$

[One has to be careful with signs.] When $X \times 0$ is pasted onto Y via h, this formula becomes

$$\partial'(\text{Int } \sigma \times \text{Int } I) = h_\#(\sigma) - \sigma \pm (\partial\sigma) \times I\,.$$

Finally, when we identify the p-chains of Y' with the group $C_p(X) \oplus C_p(Y) \oplus C_{p-1}(X)$, this formula becomes

$$\partial'(0, 0, \sigma_{p-1}) = (-\sigma_{p-1}, h_\#(\sigma_{p-1}), \pm\partial\sigma_{p-1})\,.$$

[The last sign must in fact be $-$, in order to guarantee that $\partial' \circ \partial' = 0$.] By now it should be quite clear what the connection is between the algebraic and topological mapping cylinders!

Let us give one application of this theorem. It is a formula that relates cohomology with homology. It will be generalized in Chapter 7.

Definition 5.6.9 If $\mathcal{C} = \{C_i, \partial\}$ is a chain complex, then there is a map

$$\text{Hom}\,(C_p, G) \times C_p \to G$$

that carries the pair (c^p, c_p) to the element $\langle c^p, c_p\rangle$ of G. It is bilinear, and it is called the *evaluation map*. It induces a bilinear map

$$H^p(\mathcal{C}; G) \times H_p(\mathcal{C}) \to G\,,$$

that we call the *Kronecker index*. The image of α^p and β_p under the map is also denoted by $\langle \alpha^p, \beta_p\rangle$.

Lemma 5.6.10 *The Kronecker index is well-defined.*

Proof: Let z_p and z^p be a p-cycle and a p-cocycle, respectively. Let d_{p+1} and d^{p-1} be a $(p+1)$-chain and a $(p-1)$-cochain, respectively. Then we have

$$\begin{aligned}&\langle z^p + \delta d^{p-1}, z_p + \partial d^{p+1}\rangle \\ &\quad = \langle z^p, z_p\rangle + \langle \delta d^{p-1}, z_p\rangle + \langle z^p, \partial d^{p+1}\rangle + \langle \delta d^{p-1}, \partial d^{p+1}\rangle \\ &\quad = \langle z^p, z_p\rangle + \langle d^{p-1}, \partial z_p\rangle + \langle \delta z^p, d^{p+1}\rangle + \langle d^{p-1}, \partial^2 d^{p+1}\rangle = \langle z^p, z_p\rangle\,.\end{aligned}$$

□

Definition 5.6.11 It is convenient to define the *Kronecker map*

$$\kappa : H^p(\mathcal{C}; G) \to \mathrm{Hom}\,(H_p(\mathcal{C}), G)$$

to be the map that sends α to the homomorphism $\langle \alpha, \cdot\,\rangle$. Formally, we define

$$(\kappa\alpha^p)(\beta_p) = \langle \alpha^p, \beta_p\rangle\,.$$

The map κ is a homomorphism because the Kronecker index is linear in the first variable. We leave it to you to check that κ is natural. [See Exercise 2.]

The following lemma is elementary. Its proof makes no use of the theorems of this section.

Lemma 5.6.12 *Let $\mathcal{C}$ be a free chain complex. Then there is a natural exact sequence*

$$0 \longleftarrow \mathrm{Hom}\,(H_p(\mathcal{C}), G) \xleftarrow{\kappa} H^p(\mathcal{C}; G) \longleftarrow \ker\kappa \longleftarrow 0\,.$$

It splits, but not naturally.

Proof: We shall construct a homomorphism

$$\lambda^* : \mathrm{Hom}\,(H_p(\mathcal{C}), G) \to H^p(\mathcal{C}; G)$$

such that $\kappa \circ \lambda^*$ is the identity. It follows that κ is surjective and that the sequence splits.

Let B_p, Z_p, and C_p denote the boundaries, cycles, and chains, respectively, in $\mathcal{C}$. We begin with the projection homomorphism

$$\pi : Z_p \to Z_p/B_p = H_p(\mathcal{C})\,.$$

The exact sequence $0 \to Z_p \to C_p \to B_{p-1} \to 0$ shows that Z_p is a direct summand in C_p. Therefore, π extends to a homomorphism $\lambda : C_p \to H_p(\mathcal{C})$. Define a chain complex $\mathcal{E}$ by letting $E_p = H_p(\mathcal{C})$ and letting all boundary operators in $\mathcal{E}$ vanish. Then

$$\begin{aligned}H_p(\mathcal{E}) &= E_p = H_p(\mathcal{C})\,,\\ H^p(\mathcal{E}; G) &= \mathrm{Hom}\,(E_p, G) = \mathrm{Hom}\,(H_p(\mathcal{C}), G)\,.\end{aligned}$$

Because the boundary operators in $\mathcal{E}$ vanish, $\lambda(\partial c_{p+1}) = \{\partial c_{p+1}\} = 0$, so the map $\lambda : \mathcal{C} \to \mathcal{E}$ is a chain map. The induced homomorphism in homology

$$\lambda_* : H_p(\mathcal{C}) \to H_p(\mathcal{E}) = H_p(\mathcal{C})$$

is the identity map [and hence an isomorphism]. For if z_p is a p-cycle, then

$$\lambda_*(\{z_p\}) = \lambda(z_p) = \pi(z_p) = \{z_p\}\,.$$

The induced homomorphism in cohomology

$$H^p(\mathcal{C}; G) \xleftarrow{\lambda^*} H^p(\mathcal{E}; G) = \mathrm{Hom}\,(H_p(\mathcal{C}), G)$$

is not in general an isomorphism.

Now the composite $\kappa \circ \lambda^*$ is the identity map of $\mathrm{Hom}\,(H_p(\mathcal{C}), G)$, for if $\{z_p\} \in H_p(\mathcal{C})$ and $\gamma \in \mathrm{Hom}\,(H_p(\mathcal{C}), G)$, then

$$\begin{aligned}(\kappa\lambda^*(\gamma))(\{z_p\}) &= \langle \lambda^*(\gamma), \{z_p\}\rangle = \langle \widetilde{\lambda}(\gamma), z_p\rangle \\ &= \langle \gamma, \lambda(z_p)\rangle = \gamma(\{z_p\})\,.\end{aligned}$$

□

Theorem 5.6.13 *Let $\mathcal{C}$ be a free chain complex. If $H_p(\mathcal{C})$ is free for all p, then κ is an isomorphism for all p.*

Proof: Let $\lambda : \mathcal{C} \to \mathcal{E}$ be as in the preceding lemma. Since the homology of $\mathcal{C}$ is free, $\mathcal{E}$ is a free chain complex and Theorem 5.6.6 applies. Since the chain map $\lambda : \mathcal{C} \to \mathcal{E}$ induces homology isomorphisms λ_* in all dimensions, λ^* is an isomorphism. The fact that $\kappa \circ \lambda^*$ is the identity implies that κ is an isomorphism as well. □

This last theorem says that if the homology of $\mathcal{C}$ is free in all dimensions, then the cohomology group $H^p(\mathcal{C}; G)$ can be considered in a natural way to be the dual group $\mathrm{Hom}\,(H_p(\mathcal{C}), G)$ of the homology group $H_p(\mathcal{C})$. We will generalize this theorem in Chapter 7. All one in fact needs is for the single group $H_{p-1}(\mathcal{C})$ to be free.

EXERCISES

1. Check that the operator ∂' defined in the proof of Lemma 5.6.5 satisfies $\partial' \circ \partial' = 0$.

2. **(a)** Show that the Kronecker map κ is natural with respect to homomorphisms induced by chain maps. That is, show that if $\phi : \mathcal{C} \to \mathcal{D}$ is a chain map, then the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Hom}\,(H_p(\mathcal{C}),G) & \xleftarrow{\kappa} & H^p(\mathcal{C};G) \\ \uparrow{\scriptstyle \widetilde{\phi}_*} & & \uparrow{\scriptstyle \phi^*} \\ \mathrm{Hom}\,(H_p(\mathcal{D}),G) & \xleftarrow{\kappa} & H^p(\mathcal{D};G) \end{array}$$

(b) Naturality of the Kronecker index itself is awkward to formulate, since it is covariant in one variable and contravariant in the other. Let $\phi : \mathcal{C} \to \mathcal{D}$ be a chain map. Show that if $\alpha \in H^p(\mathcal{D};G)$ and $\beta \in H_p(\mathcal{C})$, then

$$\langle \phi^*(\alpha), \beta \rangle = \langle \alpha, \phi_*(\beta) \rangle \,.$$

3. Let X and Y be spaces such that $H_n(X)$, $H^n(X)$, $H_n(Y)$, and $H^n(Y)$ are infinite cyclic. Let $f : X \to Y$ be a continuous map. Show that if

$$f_* : H_n(X) \to H_n(Y)$$

equals multiplication by d, then, up to sign, so does

$$f^* : H^n(Y) \to H^n(X) \,.$$

[**Hint:** Show that κ is an isomorphism.]

4. Since arbitrary choices are made in the definition of the homomorphism $\lambda : C_p \to H_p(\mathcal{C})$ in the proof of Lemma 5.6.12, one would not expect the splitting homomorphism

$$\lambda^* : \mathrm{Hom}\,(H_p(\mathcal{C}),G) \to H^p(\mathcal{C};G)$$

to be natural. Show that it is in fact not natural as follows:

(a) Find free chain complexes $\mathcal{C}$, $\mathcal{D}$, and a chain map $\phi : \mathcal{C} \to \mathcal{D}$ such that for no choices of λ does the following diagram commute:

$$\begin{array}{ccc} \mathrm{Hom}\,(H_p(\mathcal{C}),G) & \xrightarrow{\lambda^*} & H^p(\mathcal{C};G) \\ \uparrow{\scriptstyle \widetilde{\phi}_*} & & \uparrow{\scriptstyle \phi^*} \\ \mathrm{Hom}\,(H_p(\mathcal{D}),G) & \xrightarrow{\lambda^*} & H^p(\mathcal{D};G) \end{array}$$

***(b)** Find spaces X and Y and a continuous map $f : X \to Y$ such that setting $\mathcal{C} = \mathcal{S}(X)$, $\mathcal{D} = \mathcal{S}(Y)$, and $\phi = f_\#$ realizes the situation of **(a)**.

5. Find a free resolution of $\mathbb{Z}/n$.

5.7 *Chain Equivalences in Free Chain Complexes

Results of this section will be used in Section 7.6 and in Section 7.10 in proving the naturality of certain exact sequences.

We now prove a second version of Theorem 5.6.6. Again we assume we are given a chain map $\phi : \mathcal{C} \to \mathcal{D}$ of free chain complexes that induces homology isomorphisms in all dimensions. We show that if $\mathcal{C}$ and $\mathcal{D}$ satisfy the fairly mild additional condition that both vanish below a certain dimension, then it follows not only that ϕ^* is an isomorphism, but that the chain map ϕ is itself a chain equivalence. The proof involves the algebraic mapping cylinder constructed in the last section.

First we need an elementary lemma, that for later use is stated in slightly greater generality than we presently need.

Lemma 5.7.1 *Let $\mathcal{E}$ and $\mathcal{F}$ be non-negative chain complexes. Suppose E_p is free for $p > 0$ and $H_p(\mathcal{F}) = 0$ for $p > 0$. Then any two chain maps $f, g : \mathcal{E} \to \mathcal{F}$ that agree in dimension 0 are chain homotopic.*

Proof: Define $D : E_0 \to F_1$ to be zero. Then the equation $\partial D + D\partial = g - f$ holds in dimension 0, because $g = f$ in dimension 0. Suppose D is defined in dimension $p-1$, where $p > 0$. Choose a basis for E_p. If e is a basis element, then $g(e) - f(e) - D(\partial e)$ is well-defined, and it is a cycle [by the usual computation]. Define $D(e)$ to be an element of F_{p+1} whose boundary equals this cycle. □

As you might suspect, one can derive this last result from the acyclic carrier theorem. But care must be exercised, so it is not worth the effort.

Theorem 5.7.2 *Let $\mathcal{C}$ and $\mathcal{D}$ be free chain complexes that vanish below a certain dimension. Let $\phi : \mathcal{C} \to \mathcal{D}$ be a chain map. If ϕ induces homology isomorphisms in all dimensions, then ϕ is a chain equivalence.*

Proof: We return to the chain complex $\mathcal{D}'$ defined in the proof of Lemma 5.6.5. The inclusion mapping

$$i : C_p \to D'_p = C_p \oplus D_p \oplus C_{p-1}$$

is chain homotopic to $j \circ \phi$, where $j : D_p \to D'_p$ is inclusion. Since j and ϕ induce homology isomorphisms, so does i. It follows that the chain complex $\mathcal{E} = \mathcal{D}'/\mathcal{C}$ has vanishing homology in all dimensions. Now we have

$$E_p \cong D_p \oplus C_{p-1}\,.$$

The induced boundary operator satisfies the formula

$$\partial(d_p, c_{p-1}) = (\partial d_p + \phi(c_{p-1}), -\partial c_{p-1})\,.$$

We apply the preceding lemma to the chain complex $\mathcal{E}$ and any two chain maps f and g from $\mathcal{E}$ to itself. The chain complex $\mathcal{E}$ vanishes below a certain dimension [since $\mathcal{C}$ and $\mathcal{D}$ do]; we may as well take this dimension to be dimension 1. We know that E_p is free and $H_p(\mathcal{E}) = 0$, for all p. Since any two chain maps agree trivially in dimension 0, they are chain homotopic. In particular there is a chain homotopy between the identity map and the zero chain map. That is, there is a homomorphism $D : E_p \to E_{p+1}$ satisfying the equation

$$\partial D + D\partial = \text{identity}\,.$$

Hidden in this formula are all the chain maps and chain homotopies we need.

Recalling that $E_p \cong D_p \oplus C_{p-1}$, we define homomorphisms θ, ψ, λ, and μ by the equations

$$\begin{aligned} D(d_p, 0) &= (\theta(d_p), \psi(d_p)) &&\in D_{p+1} \oplus C_p\,, \\ D(0, c_{p-1}) &= (\lambda(c_{p-1}), \mu(c_{p-1})) &&\in D_{p+1} \oplus C_p\,. \end{aligned}$$

Then we do some computation. First, we compute

$$\begin{aligned} D\partial(d_p, 0) &= (\theta(\partial d_p), \psi(\partial d_p))\,, \\ \partial D(d_p, 0) &= (\partial\theta(d_p) + \phi\psi(d_p), -\partial\psi(d_p))\,. \end{aligned}$$

Adding these two equations, we obtain the equations

$$\begin{aligned} d_p &= \theta(\partial d_p) + \partial\theta(d_p) + \phi\psi(d_p)\,, \\ 0 &= \psi(\partial d_p) - \partial\psi(d_p)\,. \end{aligned}$$

The second equation says that $\psi : D_p \to C_p$ is a chain map, and the first says that $\theta : D_p \to D_{p+1}$ is a chain homotopy between $\phi \circ \psi$ and the identity. Second, we compute

$$\begin{aligned} D\partial(0, c_{p-1}) &= D(\phi(c_{p-1}), 0) - D(0, \partial c_{p-1}) \\ &= (\theta\phi(c_{p-1}), \psi\phi(c_{p-1})) - (\lambda(\partial c_{p-1}), \mu(\partial c_{p-1}))\,, \\ \partial D(0, c_{p-1}) &= \partial(\lambda(c_{p-1}), \mu(c_{p-1})) \\ &= (\partial\lambda(c_{p-1}) + \phi\mu(c_{p-1}), -\partial\mu(c_{p-1}))\,. \end{aligned}$$

Adding the second coordinates of these equations, we obtain the equation

$$c_{p-1} = \psi\phi(c_{p-1}) - \mu(\partial c_{p-1}) - \partial\mu(c_{p-1})\,,$$

which says that μ is a chain homotopy between $\psi \circ \phi$ and the identity.

Thus our theorem is proved. □

EXERCISES

1. Let X be a space. Let $\mathcal{A}$ be a collection of subsets of X whose interiors cover X. Show that the inclusion
$$i : \mathcal{S}^{\mathcal{A}}(X) \to \mathcal{S}(X)$$
is a chain equivalence.
2. Let $\eta : \mathcal{C}(K) \to \mathcal{S}(|K|)$ be the chain map of Section 4.7 that induces an isomorphism of simplicial and singular homology. Show η is a chain equivalence.
3. Give a rigorous definition of chain equivalence.

5.8 The Cohomology of CW Complexes

Now we can compute the cohomology groups of some familiar spaces, and we can find specific cocycles that generate these groups. The basic theorem we shall use in carrying out these computations is the following:

Theorem 5.8.1 *Let X be a CW complex. Let $\mathcal{D}(X)$ be its cellular chain complex. Then*
$$H^p(\mathcal{D}(X); G) \cong H^p(X; G)$$
for all p and G. If X is a triangulable CW complex, triangulated by a complex K, then the isomorphism is induced by the inclusion $\mathcal{D}(X) \to \mathcal{C}(X)$.

Proof: Both $\mathcal{D}(X)$ and $\mathcal{S}(X)$ are free chain complexes. Since their homology groups are isomorphic, so are their cohomology groups, by Corollary 5.6.7. In the case where X is triangulable, the inclusion map $i : \mathcal{D}(X) \to \mathcal{C}(X)$ induces the homology isomorphism in question. [See Theorem 4.13.9.] Then i induces a cohomology isomorphism as well. □

Corollary 5.8.2 *Suppose $n > 0$. Then the following hold:*
$$\begin{aligned} H^i(\mathbb{S}^n; G) &\cong G \quad \text{for } i = 0 \text{ and } i = n, \\ H^i(\mathbb{B}^n, \mathbb{S}^{n-1}; G) &\cong G \quad \text{for } i = n. \end{aligned}$$
These cohomology groups vanish for other values of i.

Proof: The first statement follows from the fact that the cellular chain complex of $\mathbb{S}^n$ is infinite cyclic in dimensions 0 and n and vanishes otherwise, and all the boundary operators vanish. The second then follows from the long exact sequence in reduced cohomology, using the fact that the reduced cohomology of $\mathbb{B}^n$ vanishes, since it is contractible. □

EXAMPLE 5.8.3 Let X denote either the torus T or the Klein bottle S, expressed as a CW complex having one open cell in dimension 2, two in dimension 1 and one in dimension 0. We computed the cellular chain complex of X, in Example 4.13.10, to be of the form

$$\cdots \longrightarrow 0 \longrightarrow \mathbb{Z} \xrightarrow{\partial_2} \mathbb{Z} \oplus \mathbb{Z} \xrightarrow{\partial_1} \mathbb{Z} \longrightarrow 0\,.$$

Let γ generate $D_2(X)$, and let w_1 and z_1 be a basis for $D_1(X)$. We know ∂_2 and ∂_1 vanish in the case of the torus. Passing to the dual sequence, we compute

$$H^2(T;G) \cong G, \quad H^1(T;G) \cong G \oplus G, \quad H^0(T;G) \cong G.$$

In the case of the Klein bottle, we know that ∂_1 vanishes, and that we can choose w_1 and z_1 so that $\partial_2 \gamma = 2z_1$. The dual sequence is of the form

$$\cdots \longleftarrow 0 \longleftarrow G \xleftarrow{\delta_2} G \oplus G \xleftarrow{\delta_1} G \longleftarrow 0\,.$$

Here $\mathrm{Hom}\,(D_1(S), G) \cong G \oplus G$, where the first summand represents those homomorphisms $\phi : D_1(S) \to G$ that vanish on z_1, and the second represents those homomorphisms $\psi : D_1(S) \to G$ that vanish of w_1. Because ∂_1 is trivial, so is its dual δ_1. We compute δ_2 as follows:

$$\begin{aligned} \langle \delta_2 \phi, \gamma \rangle &= \langle \phi, \partial_2 \gamma \rangle = 2\langle \phi, z_1 \rangle = 0, \\ \langle \delta_2 \psi, \gamma \rangle &= \langle \psi, \partial_2 \gamma \rangle = 2\langle \psi, z_1 \rangle. \end{aligned}$$

Thus δ_2 carries the first summand to zero, and equals multiplication by 2 on the second summand. We conclude that

$$H^2(S;G) \cong G/2G, \quad H^1(S;G) \cong G \oplus \ker(G \xrightarrow{2} G), \quad H^0(S;G) \cong G$$

[here $G \xrightarrow{2} G$ denotes the map given by multiplication by 2]. In particular, we have

$$H^2(S) \cong \mathbb{Z}/2\,, \quad H^1(S) \cong \mathbb{Z}\,, \quad H^0(S) \cong \mathbb{Z}\,,$$

and

$$H^2(S;\mathbb{Z}/2) \cong \mathbb{Z}/2\,, \quad H^1(S;\mathbb{Z}/2) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2\,, \quad H^0(S) \cong \mathbb{Z}/2\,.$$

The computations given in the preceding example are typical. Once one has the cellular chain complex $\mathcal{D}(X)$, computing the cohomology groups is not hard.

Nonetheless, there is an associated problem that is more difficult—namely, the problem of finding specific simplicial cocycles that generate these groups. In the next section, when we study cup products, we shall need to have such cocycles at hand. How can one find them?

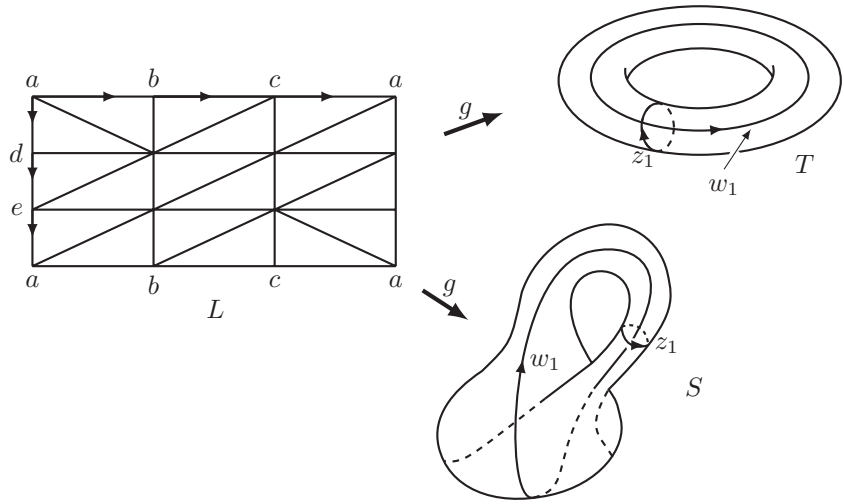


Figure 5.8.1

In the case of *homology*, finding simplicial *cycles* of X that generate the homology is not difficult. Let us represent T and S as quotient spaces of the rectangle L, as pictured in Figure 5.8.1. The chain d of L that is the sum of all the 2-simplices of L, oriented counterclockwise, is a fundamental cycle for $(L, \text{Bd } L)$, so its image $\gamma = g_{\#}(d)$ generates the cellular chain group $D_2(X)$. Similarly, the chains

$$w_1 = [a,b] + [b,c] + [c,a]\,,$$
$$z_1 = [a,d] + [d,e] + [e,a]\,,$$

are a basis for the cellular chain group $D_1(X)$. As we know, these chains are cycles representing certain elements of the homology of $\mathcal{D}(X)$. Now *because* the isomorphism $H_i(\mathcal{D}(X)) \cong H_i(X)$ is induced by the inclusion $i : \mathcal{D}(X) \to \mathcal{C}(X)$, *these same chains*, considered now as chains in $\mathcal{C}(X)$, represent elements of the simplicial homology of X.

Cohomology is not so easy. What happens in this case? Of course, we can find generators for the cohomology of the cellular chain complex $\mathcal{D}(X)$. The homomorphism $\lambda : D_2(X) \to \mathbb{Z}$ that maps the fundamental cycle γ to 1 generates $\text{Hom}\,(D_2(X), \mathbb{Z}) \cong \mathbb{Z}$. And the homomorphisms $\phi, \psi : D_1(X) \to \mathbb{Z}$ defined by

$$\langle \phi, w_1 \rangle = 1 \quad \text{and} \quad \langle \phi, z_1 \rangle = 0,$$
$$\langle \psi, w_1 \rangle = 0 \quad \text{and} \quad \langle \psi, z_1 \rangle = 1,$$

are a basis for the group $\mathrm{Hom}\,(D_1(X), \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}$. For the torus, δ vanishes, so λ generates $H^2(\mathcal{D}(T))$ and ϕ and ψ represent a basis for $H^1(\mathcal{D}(T))$. For the Klein bottle, we have $\delta\phi = 0$ and $\delta\psi = 2\lambda$, so λ represents the non-zero element of $H^2(\mathcal{D}(S)) \cong \mathbb{Z}/2$ and ϕ represents a generator of $H^1(\mathcal{D}(S)) \cong \mathbb{Z}$.

Now, unlike the situation in homology, the homomorphisms λ, ϕ, and ψ *cannot* be "considered" as cocycles of the simplicial complex X. For the inclusion map $i : D_p(X) \to C_p(X)$ induces a homomorphism in the *opposite* direction

$$\mathrm{Hom}\,(D_p(X), \mathbb{Z}) \xleftarrow{\widetilde{i}} \mathrm{Hom}\,(C_p(X), \mathbb{Z})\,.$$

This homomorphism is a *restriction* map. To find the cocycles of the simplicial complex X that generate the simplicial cohomomology of X, we must *pull back* λ, ϕ, and ψ to cocycles

$$z^2 : C_2(X) \to \mathbb{Z}\,,$$
$$w^1, w^2; C_1(X) \to \mathbb{Z}\,,$$

whose *restrictions* to the subgroups $D_2(X)$ and $D_1(X)$ equal λ, ϕ, and ψ, respectively.

There is no general procedure for finding such cocycles. But in the present case, since we know that a cocycle is supposed to look like a picket fence, we can find the desired cocycles without too much difficulty, as we now show.

EXAMPLE 5.8.4 *Generators for the cohomology of the torus.* We represent T as a quotient space of the rectangle, as in the preceding example. Let w_1 and z_1 be as in that example. The cochains w^1 and z^1 pictured in Figure 5.8.2 are cocycles of T, by *direct computation*. Furthermore, when evaluated on the cycles w_1 and z_1 that generate $D_1(X)$, we have

$$\langle w^1, w_1\rangle = 1 \quad \text{and} \quad \langle w^1, z_1\rangle = 0,$$
$$\langle z^1, w_1\rangle = 0 \quad \text{and} \quad \langle z^1, z_1\rangle = 1\,.$$

Thus w^1 is a pull-back of ϕ, and z^1 is a pull-back of ψ. Therefore they represent a basis for $H^1(T)$.

Similarly, if σ is any 2-simplex of T, oriented counterclockwise, then σ^* is a cocyle of T. Because $\langle \sigma^*, \gamma\rangle = 1$, the restriction of σ^* to the subgroups $D_2(X)$ of $C_2(X)$ equals λ, so σ^* is a pull-back of λ. Thus σ^* represents a generator of $H^2(T)$. [More generally, if all 2-simplices of T are oriented counterclockwise, then the cochain $\sum n_i\sigma_i^*$ of T is a pull-back of λ if and only if $\sum n_i = 1$.]

EXAMPLE 5.8.5 *Generators for the cohomology of the Klein bottle, with integer coefficients.* We follow the pattern of the preceding example.

By switching the labels d and e on the right-hand side of the rectangle in Figure 5.8.2 that illustrates the cochain w^1, we obtain Figure 5.8.3 representing the Klein bottle and illustrating the cochain w^1. Then w^1 still represents

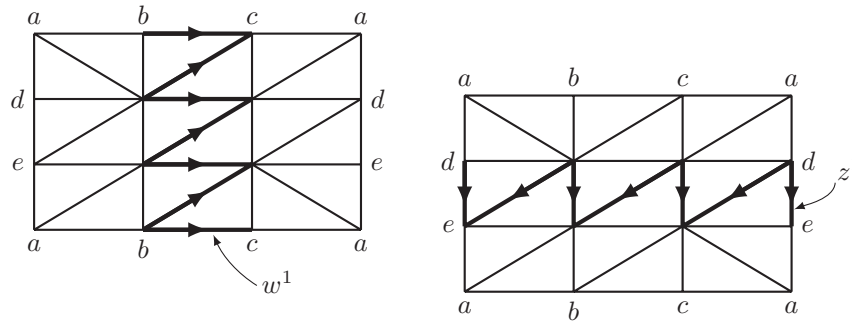


Figure 5.8.2

a cocycle. It generates $H^1(S)$. Also, the cochain σ^* represents the non-zero element of $H^2(S)$. [More generally, if all 2-simplices of T are oriented counterclockwise, then the cochain $\sum n_i \sigma_i^*$ represents the non-zero element of $H^2(S)$ if and only if $\sum n_i$ is odd.]

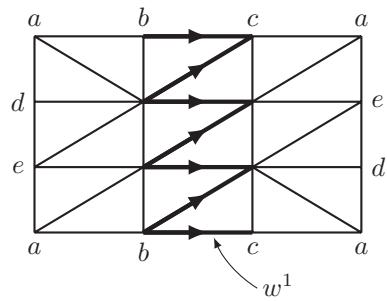


Figure 5.8.3

EXAMPLE 5.8.6 *Generators for the cohomology of the Klein bottle, with* $\mathbb{Z}/2$ *coefficients.* Recall from Example 5.8.3 that the cohomology groups are given by

$$H^2(S;\mathbb{Z}/2) \cong \mathbb{Z}/2\,, \quad H^1(S;\mathbb{Z}/2) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2\,, \quad H^0(S) \cong \mathbb{Z}/2\,.$$

The pattern of the preceding argument applies to show that the cochains w^1 and z^1 of Figure 5.8.4 generate the 1-dimensional cohomology. Notice that the arrows that were shown in Figures 5.8.2 and 5.8.3 have been erased, because in the group $\mathbb{Z}/2$ we have $1 = -1$. The cochain σ^* [or more generally $\sigma_1^* + \sigma_2^* + \cdots + \sigma_k^*$, where k is odd] generates $H^2(S;\mathbb{Z}/2)$.

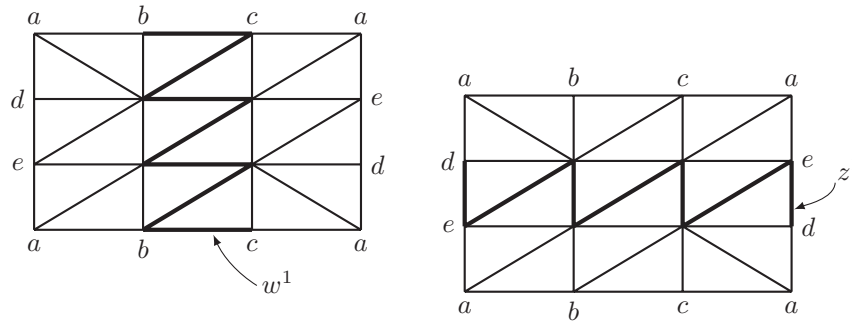


Figure 5.8.4

EXAMPLE 5.8.7 *The cohomology of* $\mathbb{P}^2$ *with* $\mathbb{Z}/2$ *coefficients.* One has

$$H^i(\mathbb{P}^2; \mathbb{Z}/2) \cong \mathbb{Z}/2 \quad \text{for } i = 0, 1, 2.$$

If σ is a 2-simplex, then the cochain σ^* generates the 2-dimensional group. Also, the cocycle pictured in Figure 5.8.5 generates $H^1(\mathbb{P}^2; \mathbb{Z}/2)$, for its value is 1 on the cycle

$$[a, b] + [b, c] + [c, d] + [d, e] + [e, f] + [f, a]$$

that generates $D_1(X)$.

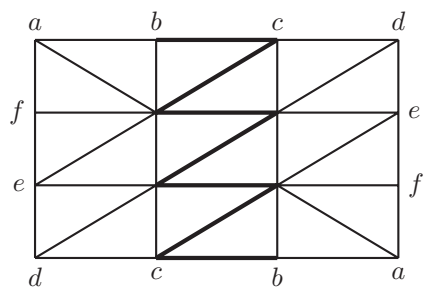


Figure 5.8.5

EXERCISES

1. Compute the cohomology of $\mathbb{P}^n$ and $\mathbb{P}^\infty$ with integer coefficients, with $\mathbb{Z}/2$ coefficients, and with rational coefficients.
2. Compute the cohomology of $\mathbb{C}P^n$ and $\mathbb{C}P^\infty$.
3. Compute $H^i(T \# T)$. Find simplicial cocyles that generate the cohomology, if $T \# T$ is triangulated as in the exercises of Section 1.7.
4. Compute the cohomology for $\mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$, with $\mathbb{Z}/2$ coefficients. Find simplicial cocycles that generate the cohomology.
5. Compute the cohomology of the Klein bottle S with $\mathbb{Z}/6$ coefficients. Find representative cocycles as in Examples 5.8.5 and 5.8.6.
6. Compute the cohomology of the 5-fold dunce cap X with $\mathbb{Z}$ and $\mathbb{Z}/5$ coefficients. [See Exercise 6 of Section 1.7.] Triangulate X and find cocycles that generate the cohomology.
7. Compute the cohomology of the lens space $L(n, k)$ with $\mathbb{Z}$ and $\mathbb{Z}/n$ coefficients.
8. Triangulate $\mathbb{S}^2$ and the torus T. Let $f : T \to \mathbb{S}^2$ be a simplicial map. Show that if $f_* : H_2(T) \to H_2(\mathbb{S}^2)$ equals multiplication by d, then so does $f^* : H^2(\mathbb{S}^2) \to H^2(T)$, by comparing the values of $f_\#$ and $f^\#$ on generators. Compare with Exercise 3 of Section 5.6.
9. Compute the cohomology of the torus with $\mathbb{Z}/6$ coefficients.

5.9 Cup Products

The results of Section 5.6 tell us that, if the homology groups fail to distinguish between two spaces, then the cohomology groups will fail as well. One might be tempted to ask, "Why bother with cohomology? What possible use can it be?"

There are several answers to this question. One answer is that cohomology appears naturally when one studies the problem of classifying, up to homotopy, maps of one space into another. Another answer is that cohomology is involved when one integrates differential forms on manifolds. Yet another answer is the one given in this section. We show that the cohomology groups have an additional algebraic structure—that of a *ring*—and that this ring will distinguish between spaces when the groups themselves will not.

We shall define the ring structure of cohomology by giving a specific cochain formula for the multiplication operation. Historically, this is how the ring structure was first obtained. The formula was discovered about 1936 by

Alexander, Čech, and Whitney. At the time, it seemed very mysterious. Its geometric meaning was not at all clear. Furthermore, it was very puzzling why there was a multiplication operation in cohomology, but not in homology.

In the case of a compact orientable *manifold*, it had been known for some time that its homology had a ring structure. The multiplication in this ring was called the *intersection product*, and it has a clear geometric meaning. But all attempts to generalize this multiplication to more general spaces failed. It was not until 1942, when Lefschetz gave a new definition of the multiplication operation in cohomology, that it became clear why, for general spaces, there exists a cohomology ring, but not a homology ring. It also became clear, about the same time, what the relationship was between the homology and cohomology rings when both were defined: The Poincaré duality theorem showed that the two rings were isomorphic. We shall return to these matters later in Sections 7.11 and 8.9.

Review of Rings, Modules, and Fields

We begin by reviewing some basic facts from algebra concerning rings and modules.

A *ring* R is an abelian group, written additively, with a multiplication operation, denoted $\cdot$, satisfying two axioms:

(1) Associativity $\alpha \cdot (\beta \cdot \gamma) = (\alpha \cdot \beta) \cdot \gamma$.

(2) Distributivity $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$,

$(\alpha + \beta) \cdot \gamma = \alpha \cdot \gamma + \beta \cdot \gamma$.

[Absent parentheses, multiplications are performed before additions.]

If $\alpha \cdot \beta = \beta \cdot \alpha$ for all $\alpha, \beta \in R$, then R is said to be *commutative*. If there is an element 1 in R such that $\alpha \cdot 1 = 1 \cdot \alpha = \alpha$ for all $\alpha \in R$, then 1 is called a *unity element* in R. If R has a unity element, then this element is easily seen to be unique. Furthermore, one has $(-1) \cdot \alpha = -\alpha$ and $0 \cdot \alpha = 0$ for all $\alpha \in R$.

If R is a commutative ring with unity, and if R satisfies the additional condition that for every $\alpha \neq 0$, there is a β such that $\alpha \cdot \beta = 1$, then R is called a *field*.

EXAMPLE 5.9.1 Examples of rings include, among many others, the following:

(i) The integers $\mathbb{Z}$.

(ii) The set $\mathbb{Z}/n$ of integers modulo n.

(iii) The set of $n \times n$ matrices with integer coefficients.

(iv) The set of polynomials with integer coefficients.

Examples of fields include:

(v) $\mathbb{Z}/p$, where p is prime.

(vi) The rationals $\mathbb{Q}$.

(vii) The real numbers $\mathbb{R}$.

(viii) The complex numbers $\mathbb{C}$.

Now suppose A is an additive group and R is a commutative ring with unity. We say that A has the structure of a *module* over R, or A is an *R-module*, if there is a binary operation $R \times A \to A$ [written as scalar multiplication] such that for $\alpha, \beta \in R$ and $a, b \in A$ we have

(1) $\alpha(a+b) = \alpha a + \alpha b$.

(2) $(\alpha+\beta)a = \alpha a + \beta a$.

(3) $\alpha(\beta a) = (\alpha \cdot \beta)a$.

(4) $1a = a$.

If A and B are R-modules, then a *module homomorphism* is a homomorphism $\phi : A \to B$ such that $\phi(\alpha a) = \alpha\phi(a)$ holds, for $\alpha \in R$ and $a \in A$. The kernel and cokernel of such a homomorphism have natural R-module structures. In the special case where R is a field F, we call A a *vector space* over F, and we call the homomorphism ϕ a *linear transformation*.

We shall not have much occasion to deal with modules in this book. Our primary concern will be with rings and vector spaces.

EXAMPLE 5.9.2 Given R, it can always be considered as an R-module over itself. More generally, the cartesian product R^n becomes an R-module if we define

$$\alpha(\beta_1, \beta_2, \ldots, \beta_n) = (\alpha\beta_1, \alpha\beta_2, \ldots, \alpha\beta_n)\,.$$

EXAMPLE 5.9.3 If G is an abelian group, then G has a natural structure as a $\mathbb{Z}$-module by defining ng to be the n-fold sum $\underbrace{g+g+\cdots+g}_{n \text{ copies}}$, as usual.

Cup Products

Throughout this section and the next, R will denote a commutative ring with unity element 1.

Definition 5.9.4 Let X be a topological space. Let $S^p(X;R) = \mathrm{Hom}\,(S_p(X), R)$ denote the group of singular p-chains of X, with coefficients in R. We define a map

$$S^p(X;R) \times S^q(X;R) \xrightarrow{\smile} S^{p+q}(X;R)$$

by the following equation: If $T : \Delta_{p+q} \to X$ is a singular $(p+q)$-simplex, then

$$\langle c^p \smile c^q, T\rangle = \langle c^p, T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle c^q, T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_{p+q})\rangle\,. \quad (*)$$

The cochain $c^p \smile c^q$ is called the *cup product* of the cochains c^p and c^q.

Recall that $\ell(w_0, w_1, \ldots, w_p)$ is the linear singular simplex mapping ϵ_i to w_i for $i = 0, 1, \ldots, p$. The map $T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)$ is simply the restriction of T to what we might call the "front p-face" Δ_p of Δ_{p+q}. It is a singular p-simplex on X. Similarly, $T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_{p+q})$ is the restriction of T to the "back q-face" of Δ_{p+q}. It is a singular q-simplex on X. The multiplication indicated on the right-hand side of $(*)$ is multiplication in the ring R.

What this mysterious formula means remains to be seen.

Theorem 5.9.5 *The cup product of cochains is bilinear and associative. The cochain z^0 whose value is 1 on each singular 0-simplex acts as a unity element. Furthermore, the following coboundary formula holds:*

$$\delta(c^p \smile c^q) = (\delta c^p) \smile c^q + (-1)^p c^p \smile (\delta c^q) \qquad (**)\,.$$

Proof: Bilinearity is immediate, since two cochains are added by adding their values, and multiplication in R is distributive. Associativity is also easily verified: Suppose T is a singular $(p+q+r)$-simplex. The value of $(c^p \smile c^q) \smile c^r$ on T and $c^p \smile (c^q \smile c^r)$ on T both equal the product of

$$\begin{aligned}
&\langle c^p, T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle\,,\\
&\langle c^q, T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_{p+q})\rangle\,, \quad \text{and}\\
&\langle c^r, T \circ \ell(\epsilon_{p+q}, \epsilon_{p+q+1}, \ldots, \epsilon_{p+q+r})\rangle\,.
\end{aligned}$$

The fact that $z^0 \smile c^p = c^p \smile z^0 = c^p$ follows directly from the definition.

To check the coboundary formula, we compute the value of both sides of $(**)$ on $T : \Delta_r \to X$, where we let $r = p + q + 1$ for convenience. The cochain $(\delta c^p) \smile c^q$ evaluated on T equals

$$\sum_{i=0}^{p+1} (-1)^i \langle c^p, T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i}, \ldots, \epsilon_{p+1})\rangle \cdot \langle c^q, T \circ \ell(\epsilon_{p+1}, \epsilon_{p+2}, \ldots, \epsilon_r)\rangle\,, \quad (\dagger)$$

while $(-1)^p\, c^p \smile (\delta c^q)$ evaluated on T equals

$$(-1)^p \sum_{i=0}^{q+1} (-1)^i \langle c^p, T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle c^q, T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \widehat{\epsilon}_{p+i}, \ldots, \epsilon_r)\rangle\,. \quad (\ddagger)$$

The last term in $(\dagger)$ equals

$$(-1)^{p+1} \langle c^p, T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle c^q, T \circ \ell(\epsilon_{p+1}, \epsilon_{p+2}, \ldots, \epsilon_r)\rangle\,,$$

and the first term in (‡) equals

$$(-1)^p \langle c^p, T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p) \rangle \cdot \langle c^q, T \circ \ell(\epsilon_{p+1}, \epsilon_{p+2} \ldots, \epsilon_r) \rangle \,.$$

When we add together (†) and (‡), the last term of (†) cancels the first term of (‡), and what remains is precisely the formula for $\langle c^p \smile c^q, \partial T \rangle$. □

Theorem 5.9.6 *The cochain cup product induces a cohomology operation*

$$H^p(X; R) \times H^q(X; R) \xrightarrow{\smile} H^{p+q}(X; R)$$

that is bilinear and associative. The cohomology class $\{z^0\}$ *acts as a unity element.*

Proof: If z^p and z^q are cocycles, then their cup product is a cocycle as well, since

$$\delta(z^p \smile z^q) = \delta z^p \smile z^q + (-1)^p z^p \smile \delta z^q = 0 \,.$$

The cohomology class of this product depends only on the cohomology classes of z^p and z^q, since

$$\begin{aligned}
&(z^p + \delta d^{p-1}) \smile (z^q + \delta d^{q-1}) \\
&\quad = z^p \smile z^q + z^p \smile \delta d^{q-1} + \delta d^{p-1} \smile z^q + \delta d^{p-1} \smile \delta d^{q-1} \\
&\quad = z^p \smile z^q + (-1)^p \, \delta(z^p \smile d^{q-1}) + \delta(d^{p-1} \smile z^q) + \delta(d^{p-1} \smile \delta d^{q-1}) \,.
\end{aligned}$$

Finally, $\{z^0\}$ is a unity element, since z^0 is already a unity element at the cochain level. □

Theorem 5.9.7 *If* $h : X \to Y$ *is a continuous map, then* h^* *preserves cup products.*

Proof: In fact, the cochain map $h^\#$ preserves cup products of cochains. We see this as follows: If T is a singular $(p+q)$-simplex, then we have

$$\begin{aligned}
\langle h^\#(c^p \smile c^q), T \rangle &= \langle c^p \smile c^q, h_\#(T) \rangle \\
&= \langle c^p, h \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p) \rangle \cdot \langle c^q, h \circ T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_{p+q}) \rangle \\
&= \langle h^\#(c^p), T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p) \rangle \cdot \langle h^\#(c^q), T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_{p+q}) \rangle \\
&= \langle h^\#(c^p) \smile h^\#(c^q), T \rangle \,.
\end{aligned}$$

□

Definition 5.9.8 Let $H^*(X; R)$ denote the external direct sum $\bigoplus H^i(X; R)$. The cup product operation makes this group into a ring with a unity element. It is called the *cohomology ring* of X with coefficients in R.

In Exercise 4, you will be asked to show that the cup product, as a function of each variable separately, is an R-module homomorphism. This means that $H^*(X; R)$ is what is sometimes called a *ring with operators* R. In the special case where R is a field F, it is called an *algebra* over F.

If $h : X \to Y$ is a continuous map, then h^* is a ring homomorphism. Therefore a homotopy equivalence induces a ring isomorphism. It follows that the cohomology ring is a topological invariant, in fact, a homotopy-type invariant.

Commutativity

We have not yet discussed whether or not the cohomology ring is commutative. In fact it is not, in general. Instead, it has a property commonly called *anticommutativity*. Specifically, if $\alpha^p \in H^p(X;R)$ and $\beta^q \in H^q(X;R)$, then

$$\alpha^p \smile \beta^q = (-1)^{pq}\, \beta^q \smile \alpha^p$$

holds. We shall not prove this formula now, for the proof will become much easier when we give an alternative definition of the cup product later on.

In the next section, we shall compute the cohomology ring in several specific cases. But first, let us introduce several generalized versions of the cup product.

Cup Products with General Coefficients

Let G be an arbitrary abelian group. Then we note that the cup product formula makes sense if we interpret it as a function

$$S^p(X) \times S^q(X;G) \xrightarrow{\smile} S^{p+q}(X;G)\,.$$

In this case, c^p is a cochain with integral values, c^q is a cochain with values in G, and the multiplication on the right-hand side of the formula $(*)$ is the usual product operation sending (n, g) to ng. Bilinearity is immediate. Associativity holds when it makes sense—that is, when it involves a map

$$S^p(X) \times S^q(X) \times S^r(X;G) \xrightarrow{\smile} S^{p+q+r}(X;G)\,.$$

The cochain z^0 whose value is 1 on each 0-simplex T acts as a left unity element. The proof of the coboundary formula is unchanged, as is the proof that the homomorphism h^* induced by a continuous map preserves cup products.

Therefore we have a well-defined cup product operation

$$H^p(X) \times H^q(X;G) \xrightarrow{\smile} H^{p+q}(X;G)\,.$$

The most general cup product operation usually considered begins with a bilinear map $\alpha : G \times G' \to G''$, called a "coefficient pairing." Using this map to replace the multiplication operation in $(*)$, one has a well-defined cup product

$$H^p(X;G) \times H^q(X;G') \xrightarrow{\smile} H^{p+q}(X;G'')\,.$$

We shall not need this degree of generality.

Relative Cup Products

Sometimes one wishes to define a cup product operation on relative cohomology groups. One can use the same cup product formula $(*)$. The relative cup products we shall need are the following:

$$H^p(X, A; R) \times H^q(X; R) \longrightarrow H^{p+q}(X, A; R)\,,$$
$$H^p(X, A; R) \times H^q(X, A; R) \longrightarrow H^{p+q}(X, A; R)\,.$$

The second of these is in fact simply a restriction of the first. The existence of these cup products is easy to demonstrate. For if $c^p : S_p(X) \to R$ vanishes on all singular p-simplices carried by A, then $c^p \smile c^q$ automatically vanishes on all $(p+q)$-simplices carried by A. The coboundary formula holds just as before, so one has an induced operation on the cohomology level. Bilinearity and associativity are immediate, as is the fact that the homomorphism induced by a continuous map preserves cup products. The class $\{z^0\} \in H^0(X; R)$ acts as a right unity element for the first of these operations.

The most general relative cup product operation is a bilinear map

$$H^p(X, A; R) \times H^q(X, B; R) \longrightarrow H^{p+q}(X, A \cup B; R)\,.$$

It is defined whenever $\{A, B\}$ *is an excisive couple.* See the exercises.

EXERCISES

1. Let A be a path component of X, let B be the union of the remaining path components of X, and assume $B \neq \emptyset$. Let c^0 be the cochain whose value is 1 on each $T : \Delta_0 \to A$ and 0 on each $T : \Delta_0 \to B$. Show that c^0 is a cocycle, and describe $\{c^0\} \smile \beta^p$ for a general cohomology class β^p.

2. Let $A \subset X$, and let $i : A \to X$ be inclusion. Let $\eta \in H^q(X; R)$, and let $\eta|A$ denote $i^*(\eta) \in H^q(A; R)$. Show that the following diagram commutes:

$$\begin{array}{ccccccc}
H^{p+1}(X, A; R) & \longleftarrow & H^p(A; R) & \longleftarrow & H^p(X; R) & \longleftarrow & H^p(X, A; R) \\
\downarrow{\scriptstyle \smile \eta} & & \downarrow{\scriptstyle \smile (\eta|A)} & & \downarrow{\scriptstyle \smile \eta} & & \downarrow{\scriptstyle \smile \eta} \\
H^{p+q+1}(X, A; R) & \longleftarrow & H^{p+q}(A; R) & \longleftarrow & H^{p+q}(X; R) & \longleftarrow & H^{p+q}(X, A; R)
\end{array}$$

What happens if you replace $\smile \eta$ with $\eta \smile$ throughout?

3. Show that, if $\{A, B\}$ is an excisive couple, then the cup product formula induces a bilinear map

$$H^p(X, A; R) \times H^q(X, B; R) \longrightarrow H^{p+q}(X, A \cup B; R)\,.$$

Interpret associativity in this case.

4. **(a)** If G is an abelian group, show that the group $\mathrm{Hom}\,(G, R)$ can be given the structure of an R-module by defining

$$\langle \alpha\, \phi, g\rangle = \alpha \cdot \langle \phi, g\rangle\,,$$

for $\phi \in \mathrm{Hom}\,(G, R)$, $\alpha \in R$, and $g \in G$. Show that if $f : G \to G'$ is a homomorphism, then $\widetilde{f}$ is an R-module homomorphism.

(b) Give $S^p(X; R) = \mathrm{Hom}\,(S_p(X), R)$ the structure of an R-module as in **(a)**. Show that δ is an R-module homomorphism, so that $H^p(X; R)$ has the structure of an R-module.

(c) Show that if $h : X \to Y$ is a continuous map, then h^* is an R-module homomorphism.

(d) Show that the cup product, as a function of each variable separately, is an R-module homomorphism. [So $H^*(X; R)$ is a ring with operators R, and in case R is a field F, it is an algebra over F.]

5. Refer to Definition 5.9.8. Give three examples of algebras that arise naturally in mathematics.

5.10 Cohomology Rings of Surfaces

There is no general method for computing the singular cohomology ring of X, even if X is a CW complex. The reason is not hard to find: It turns out that the cellular complex of X does not *determine* the cohomology ring of X. That is, two CW complexes can have isomorphic cellular chain complexes without having isomorphic cohomology rings!

Therefore, to compute cup products, we turn to simplicial cohomology. In this section, we define a simplicial cup product formula that corresponds to the previous formula for singular cochains under the standard isomorphism of simplicial theory with singular theory. Then we use this formula to compute a number of examples.

Definition 5.10.1 Given a complex K, choose a partial ordering of the vertices of K that linearly orders the vertices of each simplex of K. Define

$$C^p(K; R) \times C^q(K; R) \xrightarrow{\smile} C^{p+q}(K; R)$$

by the formula

$$\langle c^p \smile c^q, [v_0, v_1, \ldots, v_{p+q}]\rangle = \langle c^p, [v_0, v_1, \ldots, v_p]\rangle \cdot \langle c^q, [v_p, v_{p+1}, \ldots, v_{p+q}]\rangle$$

if $v_0 < v_1 < \cdots < v_{p+q}$ in the given ordering.

The similarity of this last formula to the corresponding formula in singular theory is very strong.

Theorem 5.10.2 *Given an ordering of the vertices of the complex K, the corresponding simplicial cup product is bilinear and associative. The cochain z^0 whose value is* 1 *on each vertex of K acts as a unity element. The coboundary formula*

$$\delta(c^p \smile c^q) = (\delta c^p) \smile c^q + (-1)^p c^p \smile (\delta c^q)$$

of Theorem 5.9.5 holds also for the simplicial cup product.

If $\eta : C_p(K) \to S_p(|K|)$ is the chain map of Section 4.7, determined by the given ordering, then its dual $\widetilde{\eta}$ carries the singular cup product to the simplicial cup product.

Proof: The proofs are straightforward. Only the coboundary formula requires comment. One can prove it by using the same computations used in proving Theorem 5.9.5; only small changes of notation are needed.

As an alternative argument, one can use the fact that η carries basis elements to basis elements, η is injective, and its image is a direct summand in $S_p(|K|)$. Hence its dual $\widetilde{\eta}$ is surjective. Given simplicial cochains c^p and c^q, they can thus be pulled back to singular cochains of $|K|$, say $\widetilde{c}^p$ and $\widetilde{c}^q$. We know that the coboundary formula holds in singular theory for $\widetilde{c}^p \smile \widetilde{c}^q$. Since $\widetilde{\eta}$ preserves both cup products and coboundaries, the same coboundary formula must hold in simplicial theory for $c^p \smile c^q$. □

Theorem 5.10.3 *The simplicial cup product induces an operation*

$$H^p(K; R) \times H^q(K; R) \xrightarrow{\smile} H^{p+q}(K; R)$$

that is bilinear and associative. It is independent of the ordering of the vertices of K. The cohomology class $\{z^0\}$ acts as a unity element. If $h : |K| \to |L|$ is a continuous map, then h^* preserves cup products.

Proof: The existence of $\smile$ as an operation on cohomology follows from the coboundary formula as before. [See the proof of Theorem 5.9.6.] The chain map η induces an isomorphism η^* of singular and simplicial cohomology that preserves cup products. Since η^* is independent of the chosen ordering in K, so is the cup product in simplicial cohomology.

Because h^* preserves cup products in singular cohomology, and η^* commutes with h^*, the homomorphism h^* in simplicial theory necessarily preserves cup products. □

Note that in general, if $f : K \to L$ is a simplicial map, the cochain map $f^{\#}$ need not preserve cup products on the chain level. That is because the simplicial cup product is defined using a particular ordering of the vertices, and the simplicial map f need not preserve the ordering of vertices in K and L. This would be a serious problem if one wanted to work entirely within oriented simplicial theory. It would be difficult to prove naturality of the cohomology cup product.

We remark that the more general cup products

$$H^p(K) \times H^q(K; G) \xrightarrow{\smile} H^{p+q}(K; G)$$

and

$$H^p(K, A; R) \times H^q(K, B; R) \xrightarrow{\smile} H^{p+q}(K, A \cup B; R)$$

exist in simplicial theory just as they do in singular theory. The relative cup product is in fact easier to define in simplicial theory than in singular theory, for if c^p vanishes on $C_p(A)$ and c^q vanishes on $C_q(B)$, then $c^p \smile c^q$ automatically vanishes on C_{p+q}, because a simplex of $A \cup B$ must lie in either A or B. [Of course, $\{A, B\}$ is excisive in this case, so the singular cup product is defined as well. See the exercises of Section 4.7.]

We need some terminology, and then we shall compute some examples.

Definition 5.10.4 Since the cohomology ring has a unity element in dimension 0, multiplication by this element is never trivial. It can happen though that every product of positive-dimensional cohomology classes vanishes. In that case, we say that the cohomology ring is the *trivial ring*.

Here are some cohomology rings that are not trivial.

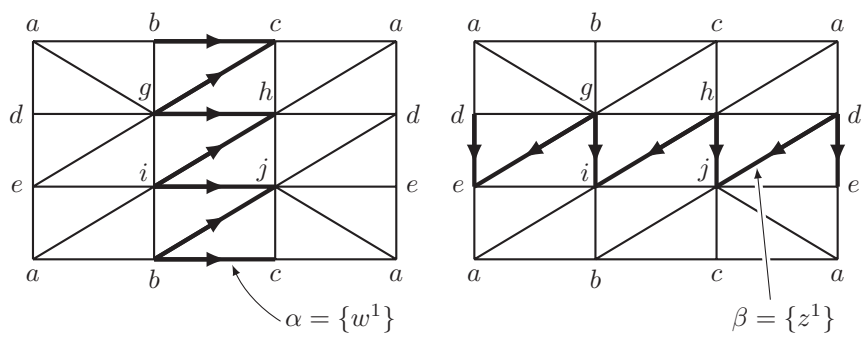


Figure 5.10.1

EXAMPLE 5.10.5 Consider the torus T. Let w^1 and z^1 denote the cocyles pictured in Figure 5.10.1. We know that $\alpha = \{w^1\}$ and $\beta = \{z^1\}$ generate

$H^1(T)$. If we orient each 2-simplex counterclockwise, then $\Lambda = \{\sigma^*\}$ generates $H^2(T) \cong \mathbb{Z}$, where σ is any oriented 2-simplex of T. In general, if $\sigma_1, \sigma_2, \ldots, \sigma_k$ are oriented 2-simplices of T, then the cochain $\sum n_i \sigma_i^*$ is cohomologous to $(\sum n_i)\sigma^*$.

Order the vertices of T alphabetically. Using this ordering, we compute the value of $w^1 \smile z^1$ on each oriented 2-simplex σ. Note that $\langle w^1 \smile z^1, \sigma\rangle = 0$ unless a face of σ is in the carrier of w^1 and a face of σ is also in the carrier of z^1. Thus the only possible non-zero values occur when σ is one of the simplices $[g,h,i]$ or $[h,i,j]$. We compute

$$\langle w^1 \smile z^1, [g,h,i]\rangle = \langle w^1, [g,h]\rangle \cdot \langle z^1, [h,i]\rangle = 1 \cdot 1 = 1\,,$$
$$\langle w^1 \smile z^1, [h,i,j]\rangle = \langle w^1, [h,i]\rangle \cdot \langle z^1, [i,j]\rangle = (-1) \cdot 0 = 0\,.$$

Thus we have $w^1 \smile z^1 = [g,h,i]^*$. Now the orientation of $[g,h,i]$ is clockwise. Therefore, in terms of our standard generators, $\alpha \smile \beta = -\Lambda$.

A similar computation shows that

$$\langle z^1 \smile w^1, [g,h,i]\rangle = \langle z^1, [g,h]\rangle \cdot \langle w^1, [h,i]\rangle = 0 \cdot (-1) = 0\,,$$
$$\langle z^1 \smile w^1, [h,i,j]\rangle = \langle z^1, [h,i]\rangle \cdot \langle w^1, [i,j]\rangle = 1 \cdot 1 = 1\,,$$

so that $z^1 \smile w^1 = [h,i,j]^*$. Thus we have $\beta \smile \alpha = \Lambda$, as anticommutivity would have led us to expect.

A similar direct computation shows that $\alpha \smile \alpha = 0$ and $\beta \smile \beta = 0$.

Alternatively, we note that w^1 is cohomologous to the cochain y^1 pictured in Figure 5.10.2. Since no 2-simplex has one face in the carrier of w^1 and another face in the carrier of y^1, necessarily $w^1 \smile y^1 = 0$. Hence $\alpha \smile \alpha = 0$. A similar argument shows that $\beta \smile \beta = 0$.

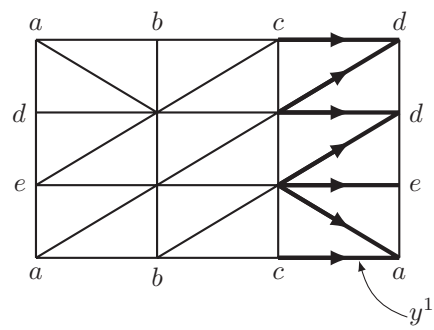


Figure 5.10.2

Another alternative computation proceeds by noting that anticommutivity implies that $\alpha \smile \alpha = -(\alpha \smile \alpha)$. Since $H^2(T)$ has no elements of order 2, $\alpha \smile \alpha$ must vanish. A similar remark applies to $\beta \smile \beta$.

We can specify the ring structure of $H^*(T)$ by writing its multiplication table in terms of generators for $H^*(T)$. Omitting the unity element, we have the following table:

$\smile$	α	β	Λ
α	0	$-\Lambda$	0
β	Λ	0	0
Λ	0	0	0

The last row and column in the table vanish for dimensional reasons.

EXAMPLE 5.10.6 Consider now the Klein bottle S. Let us compute the cohomology ring with $\mathbb{Z}/2$ coefficients. We know that $H^1(S;\mathbb{Z}/2)$ is generated by the cocyles w^1 and z^1 pictured in Figure 5.10.3. Furthermore, $H^2(S;\mathbb{Z}/2)$ is generated by σ^*, for any 2-simplex σ. [We omit orientations, since $1 = -1$ in $\mathbb{Z}/2$.] We let $\alpha = \{w^1\}$, $\beta = \{z^1\}$, and $\Lambda = \{\sigma^*\}$. Some of the computations we carried out in Example 5.10.5 apply, provided we reduce coefficients modulo 2. In particular, we have

$$w^1 \smile z^1 = [g,h,i]^* \quad \text{and} \quad w^1 \smile y^1 = 0\,,$$

where y^1 is the cochain pictured in Figure 5.10.4. [This figure is like Figure 5.10.2, without the arrows, because the coefficients are in $\mathbb{Z}/2$.] We conclude that $\alpha \smile \beta = \Lambda$ and $\alpha \smile \alpha = 0$.

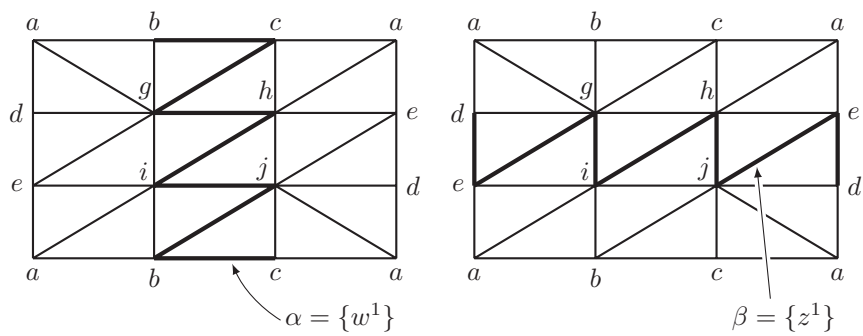


Figure 5.10.3

Computing $z^1 \smile z^1$ must be done directly, since we cannot pull it off itself as we did with w^1. [Why?] The cochain $z^1 \smile z^1$ has value 1 on $[d,e,g]$, 1 on $[e,g,i]$, and 1 on $[d,e,j]$. It vanishes on all other 2-simplices. Thus it is cohomologous to $3\sigma = \sigma$. We conclude that $\beta \smile \beta = \Lambda$.

Omitting the unity element, we have the following multiplication table for $H^*(S;\mathbb{Z}/2)$:

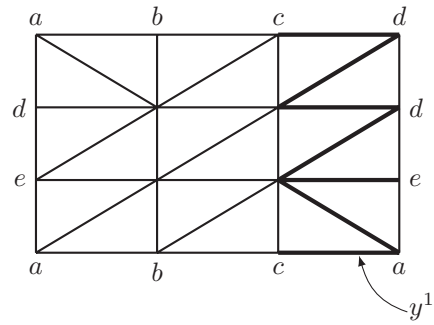


Figure 5.10.4

$\smile$	α	β	Λ
α	0	Λ	0
β	Λ	Λ	0
Λ	0	0	0

EXAMPLE 5.10.7 Consider the connected sum $\mathbb{P}^2 \# \mathbb{P}^2$. We compute its cohomology ring with $\mathbb{Z}/2$ coefficients. Let us express $\mathbb{P}^2 \# \mathbb{P}^2$ as a CW complex X having one cell in dimension 0, one cell in dimension 2, and two cells in dimension 1. See Figure 5.10.5. Fundamental cycles for the 1-cells are

$$w_1 = [a,b] + [b,c] + [c,a] \quad \text{and} \quad z_1 = [a,d] + [d,e] + [e,a]\,.$$

A fundamental cycle γ for the 2-cell is the sum of all the 2-simplices, oriented counterclockwise. By direct computation, $\partial w_1 = \partial x_1 = 0$, and $\partial\gamma = -2w_1 - 2z_1$. When we pass to the dual cochain complex $\mathrm{Hom}\,(\mathcal{D}(X), \mathbb{Z}/2)$, all coboundary operators vanish [since we are using $\mathbb{Z}/2$ coefficients]. Thus, in singular cohomology, we have

$$H^1(\mathbb{P}^2 \# \mathbb{P}^2; \mathbb{Z}/2) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2\,,$$
$$H^2(\mathbb{P}^2 \# \mathbb{P}^2; \mathbb{Z}/2) \cong \mathbb{Z}/2\,.$$

Passing now to simplicial cohomology, we see that the cocycles w^1 and z^1 pictured in the figure, restricted to $D_1(X)$, serve as a basis for $\mathrm{Hom}\,(D_1(X), \mathbb{Z}/2)$, since

$$\langle w^1, w_1\rangle = 1 \quad \text{and} \quad \langle w^1, z_1\rangle = 0\,,$$
$$\langle z^1, w_1\rangle = 0 \quad \text{and} \quad \langle z^1, z_1\rangle = 1\,.$$

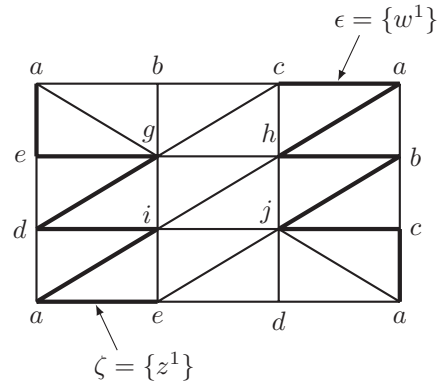


Figure 5.10.5

The classes $\epsilon = \{w^1\}$ and $\zeta = \{z^1\}$ thus generate $H^1(\mathbb{P}^2 \# \mathbb{P}^2; \mathbb{Z}/2)$. The class $\Lambda = \{\sigma^*\}$ generates $H^2(\mathbb{P}^2 \# \mathbb{P}^2; \mathbb{Z}/2)$ [where σ is any 2-simplex]. Direct computation shows that

$$\begin{aligned} w^1 \smile z^1 &= 0\,, \\ w^1 \smile w^1 &= [a, c, j]^*\,, \\ z^1 \smile z^1 &= [a, e, g]^*\,. \end{aligned}$$

Thus $H^*(\mathbb{P}^2 \# \mathbb{P}^2; \mathbb{Z}/2)$ has the multiplication table

$\smile$	ϵ	ζ	Λ
ϵ	Λ	0	0
ζ	0	Λ	0
Λ	0	0	0

But we know that $\mathbb{P}^2 \# \mathbb{P}^2$ *is homeomorphic to the Klein bottle* S. [See Figure 1.7.9.] Thus their cohomology rings are isomorphic, even though this multiplication table is quite unlike the one we computed in Example 5.10.6. We leave it to you to construct an isomorphism between these two rings.

This example illustrates the following, perhaps unfortunate, but important fact: *In general, examining the multiplication tables of two rings will not allow you to determine immediately whether or not the two rings are isomorphic.*

EXAMPLE 5.10.8 Consider the space X pictured in Figure 5.10.6. It is the union of two two topological circles and a topological 2-sphere, all three with a point in common. It is called the *wedge product* $\mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^2$. The space X

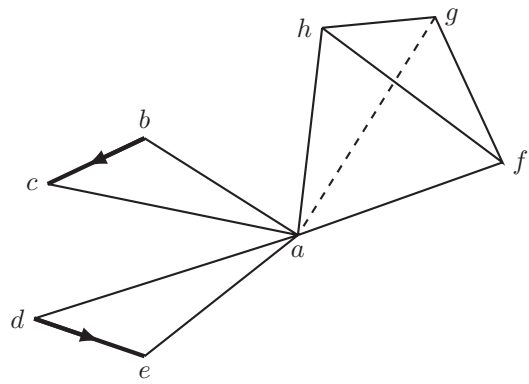


Figure 5.10.6

can be represented as a CW complex with one cell in dimension 0, one cell in dimension 2, and two cells in dimension 1. When we write fundamental cycles

$$\begin{aligned} w_1 &= [a,b] + [b,c] + [c,a]\,, \\ z_1 &= [a,d] + [d,e] + [e,a]\,, \\ z_2 &= \partial[a,f,g,h] \end{aligned}$$

for these cells, we see that the boundary operators in the cellular chain complex all vanish. *Therefore, the cellular chain complex $\mathcal{D}(X)$ of X is isomorphic to the cellular chain complex $\mathcal{D}(T)$ of the torus T.*

It follows that the homology groups and cohomology groups of X are isomorphic to those of T. On the other hand, their cohomology *rings* are not isomorphic. For it is easy to see that the cohomology ring of X is trivial, as we now show.

Consider the cocycles

$$w^1 = [b,c]^* \qquad \text{and} \qquad z^1 = [d,e]^*\,.$$

Since the cycles w_1 and z_1 are a basis for the chain group $D_1(X)$, the cocycles w^1 and z^1 give the dual basis for the cochain group $\mathrm{Hom}\,(D_1(X), \mathbb{Z})$. All the cup products $w^1 \smile z^1$, $w^1 \smile w^1$, and $z^1 \smile z^1$ vanish, because no 2-simplex has a face in the carriers of either w^1 or z^1.

The preceding examples should convince you that, as a rule, cup products are not easy to compute. The difficulty is that one must go down to the simplicial level and find specific representative cocycles, in order to use the cup product formula.

As a consequence, any theorem that tells us something about cup products in general is likely to be a useful theorem. We shall prove two such theorems in later chapters. One will tell us something about the cohomology ring of a product space $X \times Y$. The other will give us information about the cohomology ring of a manifold, which will in particular enable us to compute the cohomology rings of projective spaces.

EXERCISES

Throughout these exercises, let T denote the torus, and let S denote the Klein bottle.

1. Let $f : \mathbb{S}^2 \to T$ be continuous. Show that $f^* : H^2(T) \to H^2(\mathbb{S}^2)$ is trivial, and conclude that $f_* : H_2(\mathbb{S}^2) \to H_2(T)$ is trivial. What can you say about a continuous map $g : T \to \mathbb{S}^2$?

2. If $f : X \to Y$, show that

$$f^* : H^2(Y; \mathbb{Z}/2) \to H^2(X; \mathbb{Z}/2)$$

is trivial in the following cases:

(a) $X = \mathbb{S}^2$ and $Y = S$.

(b) $X = S$ and $Y = T$.

(c) $X = T$ and $Y = S$.

3. Give multiplication tables for the following cohomology rings:

(a) $H^*(\underbrace{T \# T \# \cdots \# T}_{n \text{ copies}})$.

(b) $H^*(\mathbb{P}^2; \mathbb{Z}/2)$.

(c) $H^*(\underbrace{\mathbb{P}^2 \# \mathbb{P}^2 \# \cdots \# \mathbb{P}^2}_{n \text{ copies}} ; \mathbb{Z}/2)$.

4. Define an isomorphism between the ring of Example 5.10.5 and the ring of Example 5.10.6.

5. Consider the cohomology rings of the Klein bottle S and the space $\mathbb{P}^2 \vee \mathbb{S}^1$.

(a) Show that these rings are isomorphic with integer coefficients.

(b) Show that these rings are not isomorphic with $\mathbb{Z}/2$ coefficients.

[**Note:** It does not suffice to show that they have different multiplication tables.]

6. Compute the cohomology ring of the 3-fold dunce cap with $\mathbb{Z}/3$ coefficients.

7. **(a)** Let (M, E) denote the Möbius band and its edge. Compute the cup product operations

$$H^*(M, E; \mathbb{Z}/2) \times H^*(M; \mathbb{Z}/2) \longrightarrow H^*(M, E; \mathbb{Z}/2)\,.$$
$$H^*(M, E; \mathbb{Z}/2) \times H^*(M, E; \mathbb{Z}/2) \longrightarrow H^*(M, E; \mathbb{Z}/2)\,.$$

(b) Repeat **(a)** when M is the cylinder $\mathbb{S}^1 \times I$ and $E = \mathbb{S}^1 \times \text{Bd } I$.



Figure 5.10.7

8. Let A be the union of two once-linked circles in $\mathbb{S}^3$, and let B be the union of two unlinked circles in $\mathbb{S}^3$. See Figure 5.10.7. Show that the cohomology groups of $\mathbb{S}^3 \setminus A$ and $\mathbb{S}^3 \setminus B$ are isomorphic, but the cohomology rings are not.

9. What can you say about homology in Exercise 8?

6

Homology with Coefficients

6.1 Introduction

After having studied cohomology with arbitrary coefficients, we now return to a subject that was briefly introduced for the simplicial theory in Chapter 1—homology with arbitrary coefficients.

To start with, we introduce an algebraic functor called the tensor product and study its properties. This functor plays a role in homology theory that is similar to that played by the Hom functor for cohomology. After that introduction, we study homology with arbitrary coefficients.

6.2 Tensor Products

If A and B are abelian groups, then of course their Cartesian product $A \times B$ is an abelian group, and one often considers homomorphisms of the group $A \times B$ into a third abelian group C. Sometimes it is important to consider functions from $A \times B$ to C that are *bilinear*, that is, functions that are homomorphisms when considered as function *of each variable separately*. In this section, we define an abelian group called the *tensor product* of A and B, denoted $A \otimes B$, that has the property that *bilinear* functions from $A \times B$ to C can naturally be considered as *homomorphisms* from $A \otimes B$ to C, and conversely. Thus the study of bilinear functions is reduced to the familiar study of homomorphisms.

Definition 6.2.1 Let A and B be abelian groups. Let $F(A, B)$ be the free abelian group generated by the set $A \times B$. Let $R(A, B)$ be the subgroup generated by all elements of the form

$$(a + a', b) - (a, b) - (a', b)\,,$$
$$(a, b + b') - (a, b) - (a, b')\,,$$

for $a, a' \in A$ and $b, b' \in B$. The subgroup $R(A, B)$ is called the *relations subgroup*. We define

$$A \otimes B = F(A, B)/R(A, B)\,,$$

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and call it the *tensor product* of A and B. The coset of the pair (a, b) is denoted by $a \otimes b$.

Any function f from the set $A \times B$ to the abelian group C determines a unique homomorphism of $F(A, B)$ into C, because the elements of $A \times B$ are the basis elements for $F(A, B)$. This function f is bilinear if and only if it maps the subgroup $R(A, B)$ to zero. Thus every homomorphism of $A \otimes B$ into C gives rise to a bilinear function from $A \times B$ into C, and conversely.

Note that any element of $F(A, B)$ is a finite linear combination of pairs (a, b), so any element of $A \otimes B$ is a finite linear combination of elements of the form $a \otimes b$. Note that this does not tell us that $a \otimes b$ is a typical *element* of $A \otimes B$. Rather, it tells us that $a \otimes b$ is a typical *generator* of $A \otimes B$.

Now we observe a few properties of the tensor product. By the definition, we have

$$(a + a') \otimes b = a \otimes b + a' \otimes b\,,$$
$$a \otimes (b + b') = a \otimes b + a \otimes b'\,,$$

for $a, a' \in A$ and $b, b' \in B$. Since

$$a \otimes b = (0 + a) \otimes b = 0 \otimes b + a \otimes b$$

holds, we conclude that $0 \otimes b = 0$, for all $b \in B$. Similarly, we have $a \otimes 0 = 0$, for all $a \in A$. It follows that

$$(-a) \otimes b = -(a \otimes b) = a \otimes (-b)\,,$$

since by adding $a \otimes b$ to each expression we obtain zero. An immediate consequence is that the relation

$$(na) \otimes b = n(a \otimes b) = a \otimes (nb)$$

holds, for any integer n.

Definition 6.2.2 Let $f : A \to A'$ and $g : B \to B'$ be homomorphisms. There is a unique homomorphism

$$f \otimes g : A \otimes B \to A' \otimes B'$$

defined by the property that $(f \otimes g)(a \otimes b) = f(a) \otimes g(b)$, for all $a \in A$ and $b \in B$. This homomorphism is called the *tensor product* of f and g.

This fact is an immediate consequence of the fact that the function from $A \times B$ into $A' \otimes B'$ carrying (a, b) to $f(a) \otimes g(b)$ is bilinear, as you can check.

We also leave it to you to check the following:

Lemma 6.2.3 *The function mapping (A, B) to $A \otimes B$ [objects to objects] and mapping (f, g) to $f \otimes g$ [morphisms to morphisms] is a covariant functor from the category of pairs of abelian groups and homomorphisms to the category of abelian groups and homomorphisms.*

Later we will show how to compute tensor products. For the present, we merely note the following:

Theorem 6.2.4 *There is an isomorphism*

$$\mathbb{Z} \otimes G \cong G$$

that maps $n \otimes g$ to ng. It is natural with respect to homomorphisms of G.

Proof: The function mapping $\mathbb{Z} \times G$ to G that sends (n, g) to ng is bilinear, so it induces a homomorphism $\phi : \mathbb{Z} \otimes G \to G$ sending $n \otimes g$ to ng.

Let $\psi : G \to \mathbb{Z} \otimes G$ be defined by the equation $\psi(g) = 1 \otimes g$. Then ψ is a homomorphism. For $g \in G$, we have

$$\phi\psi(g) = \phi(1 \otimes g) = g\,,$$

while on a typical generator $n \otimes g$ of $\mathbb{Z} \otimes G$, we have

$$\psi\phi(n \otimes g) = \psi(ng) = 1 \otimes (ng) = n \otimes g\,.$$

Thus ψ is an inverse for ϕ.

Naturality is a consequence of the commutativity of the following diagram:

$$\begin{array}{ccc} \mathbb{Z} \otimes G & \xrightarrow{\cong} & G \\ {\scriptstyle i_{\mathbb{Z}} \otimes f}\downarrow & & \downarrow{\scriptstyle f} \\ \mathbb{Z} \otimes H & \xrightarrow{\cong} & H \end{array}$$

□

Before we derive some general properties of tensor products, let us first note a common fallacy. Suppose A' is a subgroup of A, and B' is a subgroup of B. Then it is tempting to assume that $A' \otimes B'$ can be considered a subgroup of $A \otimes B$, *but this is not correct in general.* While the inclusion mappings $i : A' \to A$ and $j : B' \to B$ do give rise to a homomorphism

$$i \otimes j : A' \otimes B' \to A \otimes B\,,$$

this homomorphism is not necessarily injective. As an example, let $A = \mathbb{Q}$, the additive group of rational numbers, and let $B = \mathbb{Z}/2$. Then $A \otimes B = \mathbb{Q} \otimes \mathbb{Z}/2$ is in fact trivial, because in $\mathbb{Q} \otimes \mathbb{Z}/2$ we have

$$a \otimes b = (a/2) \otimes 2b = (a/2) \otimes 0 = 0\,.$$

For the subgroups, we let $A' = \mathbb{Z}$ and $B' = B = \mathbb{Z}/2$. Then $A' \otimes B' = \mathbb{Z} \otimes \mathbb{Z}/2$ which is non-trivial, so it cannot be a subgroup of the trivial group $A \otimes B = \mathbb{Q} \otimes \mathbb{Z}/2$.

Although tensor products of injective maps are not in general injective, tensor products of *surjective* maps are always surjective. That is the substance of the following lemma.

Lemma 6.2.5 *Suppose the homomorphisms $\phi : B \to C$ and $\phi' : B' \to C'$ are surjective. Then*

$$\phi \otimes \phi' : B \otimes B' \to C \otimes C'$$

is surjective, and its kernel is the subgroup of $B \otimes B'$ generated by all elements of the form $b \otimes b'$ for which $b \in \ker \phi$ or $b' \in \ker \phi'$.

Proof: Let G denote the subgroup of $B \otimes B'$ generated by all elements of the form $b \otimes b'$ for which $b \in \ker \phi$ or $b' \in \ker \phi'$. Clearly, $\phi \otimes \phi'$ maps G to zero, so it induces a homomorphism

$$\Phi : (B \otimes B')/G \to C \otimes C' .$$

We show that Φ is an isomorphism by defining an inverse Ψ for Φ.

We begin by defining a function

$$\psi : C \times C' \to (B \otimes B')/G$$

by the rule $\psi(c, c') = b \otimes b' + G$, where b is chosen so that $\phi(b) = c$ and b' is chosen so that $\phi'(b') = c'$. To show that ψ is well-defined, suppose $\phi(b_0) = c$ and $\phi(b'_0) = c'$. Then we have

$$b \otimes b' - b_0 \otimes b'_0 = ((b - b_0) \otimes b') + (b_0 \otimes (b' - b'_0)) . \qquad (*)$$

The element in $(*)$ lies in G because $b - b_0 \in \ker \phi$ and $b' - b'_0 \in \ker \phi'$. Thus ψ is well-defined. It follows from its definition that ψ is bilinear, so it induces a homomorphism

$$\Psi : C \otimes C' \to (B \otimes B')/G .$$

It is straightforward to check that $\Phi \circ \Psi$ and $\Psi \circ \Phi$ are identity maps. □

Just as we did with the Hom functor, we consider what "tensoring" does to an exact sequence.

Theorem 6.2.6 *Suppose the sequence*

$$A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

is exact. Then the sequence

$$A \otimes G \xrightarrow{\phi \otimes i_G} B \otimes G \xrightarrow{\psi \otimes i_G} C \otimes G \longrightarrow 0$$

is exact. If ϕ is injective and the first sequence splits, then $\phi \otimes i_G$ is injective and the second sequence splits.

Proof: The preceding lemma implies that $\psi \otimes i_G$ is surjective, and that its kernel is the subgroup D of $B \otimes G$ generated by all elements of the form $b \otimes g$ for $b \in \ker \psi$. The image of $\phi \otimes i_G$ is the subgroup E generated by all elements

of the form $\phi(a) \otimes g$. Since the image of ϕ equals the kernel of ψ, we have $D = E$.

Suppose that ϕ is injective and the sequence splits. Let $p : B \to A$ be a homomorphism such that $p \circ \phi = i_A$. Then we have

$$(p \otimes i_G) \circ (\phi \otimes i_G) = i_A \otimes i_G = i_{A \otimes G}\,,$$

so $\phi \otimes i_G$ is injective and $p \otimes i_G$ splits the tensored sequence. □

Corollary 6.2.7 *There is a natural isomorphism*

$$\mathbb{Z}/m \otimes G \cong G/mG\,.$$

Proof: Let ϕ_m be multiplication by m [both in $\mathbb{Z}$ and in G, as the context dictates]. We take the sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\phi_m} \mathbb{Z} \longrightarrow \mathbb{Z}/m \longrightarrow 0$$

and tensor it with G to obtain the exact sequence

$$\mathbb{Z} \otimes G \xrightarrow{\phi_m \otimes i_G} \mathbb{Z} \otimes G \longrightarrow \mathbb{Z}/m \otimes G \longrightarrow 0\,.$$

Applying Theorem 6.2.4, we have the exact sequence

$$G \xrightarrow{\phi_m} G \longrightarrow \mathbb{Z}/m \otimes G \longrightarrow 0\,,$$

and the corollary follows. □

Now we prove some additional properties of tensor products.

Theorem 6.2.8 *One has the following natural isomorphisms*

(a) $A \otimes B \cong B \otimes A$.
(b) $(\bigoplus A_\alpha) \otimes B = \bigoplus (A_\alpha \otimes B)$,
(c) $A \otimes (\bigoplus B_\alpha) = \bigoplus (A \otimes B_\alpha)$,
(d) $A \otimes (B \otimes C) \cong (A \otimes B) \otimes C$.

Proof: We construct the isomorphisms, and leave naturality for you to check.

(a) The map $A \times B \to B \times A$ sending (a, b) to (b, a) induces an isomorphism of $F(A, B)$ with $F(B, A)$ that carries $R(A, B)$ to $R(B, A)$.

(b) We apply Lemma 1.5.1. By hypothesis, there are homomorphisms

$$j_\beta : A_\beta \to \bigoplus A_\alpha \qquad \text{and} \qquad \pi_\beta : \bigoplus A_\alpha \to A_\beta$$

such that $\pi_\beta \circ j_\alpha$ is trivial if $\alpha \neq \beta$ and equals the identity if $\alpha = \beta$. Let

$$f_\beta = j_\beta \otimes i_B : A_\beta \otimes B \to (\bigoplus A_\alpha) \otimes B\,,$$
$$g_\beta = \pi_\beta \otimes i_B : (\bigoplus A_\alpha) \otimes B \to A_\beta \otimes B\,.$$

Then $g_\beta \circ f_\alpha$ is trivial if $\alpha \neq \beta$ and equals the identity if $\alpha = \beta$. Now $(\bigoplus A_\alpha) \otimes B$ is generated by elements of the form $a \otimes b$, where $a \in \bigoplus A_\alpha$ and $b \in B$. Since a is, in turn, equal to a finite sum of elements of the form $j_\alpha(a_\alpha)$, we see that $(\bigoplus A_\alpha) \otimes B$ is generated by the groups $f_\alpha(A_\alpha \otimes B)$. The existence of the isomorphism of **(b)** follows from Lemma 1.5.1.

(c) The existence of the isomorphism in **(c)** follows from **(b)** and commutativity.

(d) To define a homomorphism mapping the tensor product $A \otimes (B \otimes C)$ into an abelian group G, we must define a bilinear function f on the *set* $A \times (B \otimes C)$. In order that f be linear in the second variable, it must for fixed $a \in A$ come from a bilinear map of the set $a \times B \times C$ into G. We conclude that a map f of the set $A \times B \times C$ into G defines a homomorphism from $A \otimes (B \otimes C)$ into G precisely if it is linear in each of the three variables separately. Such a function is called a *multilinear function*. A similar argument shows that homomorphisms of $(A \otimes B) \otimes C$ into G are obtained in exactly the same way.

Now consider the functions

$$f(a, b, c) = a \otimes (b \otimes c)\,,$$
$$g(a, b, c) = (a \otimes b) \otimes c\,.$$

These are multilinear functions from $A \times B \times C$ to $A \otimes (B \otimes C)$ and $(A \otimes B) \otimes C$, respectively. They induce homomorphisms

$$(A\otimes)B \otimes C \xrightarrow{F} A \otimes (B \otimes C) \xrightarrow{G} (A \otimes B) \otimes C\,,$$

respectively. Both $F \circ G$ and $G \circ F$ act as identity maps on generators of these groups, so they are identity maps. □

Corollary 6.2.9 *If*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

is exact and G is torsion-free, then

$$0 \longrightarrow A \otimes G \longrightarrow B \otimes G \longrightarrow C \otimes G \longrightarrow 0$$

is exact.

Proof: Step 1. We show first that the theorem holds if G is free. The sequence

$$0 \longrightarrow A \otimes \mathbb{Z} \longrightarrow B \otimes \mathbb{Z} \longrightarrow C \otimes \mathbb{Z} \longrightarrow 0 \qquad (**)$$

is exact, because $D \otimes \mathbb{Z}$ is naturally isomorphic to D for all D. Therefore

$$0 \longrightarrow A \otimes G \longrightarrow B \otimes G \longrightarrow C \otimes G \longrightarrow 0$$

is exact; for by the preceding theorem this sequence is isomorphic to a direct sum of sequences of the type $(**)$ and direct sums of exact sequences are exact.

Step 2. We prove the following fact:
Let $a_1, a_2, \ldots, a_k \in A$ and $b_1, b_2, \ldots, b_k \in B$. Suppose the element $\sum a_i \otimes b_i$ of $A \otimes B$ vanishes. Then there are finitely generated subgroups A_0 and B_0 of A and B containing $\{a_1, a_2, \ldots, a_k\}$ and $\{b_1, b_2, \ldots, b_k\}$, respectively, such that the sum $\sum a_i \otimes b_i$ vanishes when considered as an element of $A_0 \otimes B_0$.

Recall that $A \otimes B$ equals the quotient of $F(A, B)$ by the relations subgroup $R(A, B)$. The equation $\sum a_i \otimes b_i = 0$ means that the element $\sum(a_i, b_i)$ of $F(A, B)$ lies in $R(A, B)$. That is, it can be written as a finite linear combination of elements of $R(A, B)$, and thus as a finite linear combination of generators of $R(A, B)$. Recall also that each generator of $R(A, B)$ has the form

$$(a + a', b) - (a, b) - (a', b) \qquad \text{or} \qquad (a, b + b') - (a, b) - (a, b')\,. \qquad (\dagger)$$

Collecting together all the first components of terms in $(\dagger)$ occurring in the finite expression for $\sum(a_i, b_i)$, we obtain a finite set $A' \subset A$. Similarly, we collect together all the second components of terms in $(\dagger)$ occurring in the finite expression for $\sum(a_i, b_i)$ to form a finite set $B' \subset B$.

Let A_0 denote the subgroup of A generated by $A' \cup \{a_1, a_2, \ldots, a_k\}$, and let B_0 denote the subgroup of B generated by $B' \cup \{b_1, b_2, \ldots, b_k\}$. Then when we consider $\sum(a_i, b_i)$ as an element of $F(A_0, B_0)$, we see that it lies in $R(A_0, B_0)$. Thus $\sum a_i \otimes b_i$ vanishes when considered as an element of $A_0 \otimes B_0$.

Step 3. We complete the proof. Suppose

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

is exact and G is torsion-free. By Theorem 6.2.6, what we need to show is that $\phi \otimes i_G$ is injective. The typical element of $A \otimes G$ is a finite sum $\sum a_i \otimes g_i$. Suppose it lies in the kernel of $\phi \otimes i_G$. Then $\sum \phi(a_i) \otimes g_i$ vanishes in $B \otimes G$. By **Step 2**, we can choose finitely generated subgroups B_0 of B and G_0 of G, such that the sum $\sum \phi(a_i) \otimes g_i$ vanishes when considered as an element of $B_0 \otimes G_0$. Applying the map $B_0 \otimes G_0 \to B \otimes G_0$ induced by inclusion, we see that $\sum \phi(a_i) \otimes g_i$ vanishes when considered as an element of $B \otimes G_0$.

Because G_0 is a subgroup of G, G_0 is torsion-free. Therefore since G_0 is finitely generated, *it is free*. **Step 1** then tells us that the sequence

$$0 \longrightarrow A \otimes G_0 \xrightarrow{\phi \otimes i_{G_0}} B \otimes G_0 \xrightarrow{\psi \otimes i_{G_0}} C \otimes G_0 \longrightarrow 0$$

is exact. We conclude that $\sum a_i \otimes g_i$ must vanish when it is considered as an element of $A \otimes G_0$. Applying the map $A \otimes G_0 \to A \otimes G$ induced by inclusion, we see that it also vanishes when considered as an element of $A \otimes G$. □

The theorems we have proved enable us to compute the group $A \otimes B$ when A is finitely generated. For we know that $\otimes$ commutes with direct sums, and we have the rules

$$\mathbb{Z} \otimes G \cong G \qquad \text{and} \qquad \mathbb{Z}/m \otimes G \cong G/mG\,,$$

from Theorem 6.2.4 and Corollary 6.2.7, respectively. In particular, tensor products of free abelian groups are free abelian. For later use, we state this fact formally as follows:

Theorem 6.2.10 *If A is free abelian with basis $\{a_i\}$ and B is free abelian with basis $\{b_j\}$, then $A \otimes B$ is free abelian with basis $\{a_i \otimes b_j\}$.*

Proof: Let $\langle a_i \rangle$ and $\langle b_j \rangle$ denote the infinite cyclic subgroups of A and B generated by a_i and b_j, respectively. Then we have

$$A = \bigoplus \langle a_i \rangle \qquad \text{and} \qquad B = \bigoplus \langle b_j \rangle\,.$$

It follows that $A \otimes B \cong \bigoplus (\langle a_i \rangle \otimes \langle b_j \rangle)$. Now $\mathbb{Z} \otimes \mathbb{Z}$ is infinite cyclic and is generated by $1 \otimes 1$; likewise $\langle a_i \rangle \otimes \langle b_j \rangle$ is infinite cyclic and is generated by $a_i \otimes b_j$. The theorem follows. □

Tensor Products of Modules

Let R be a commutative ring with unity element. If A and B are modules over R, then as was mentioned in the exercises of Section 5.2, the group $\mathrm{Hom}_R(A, B)$ of all module homomorphisms of A into B has the structure of an R-module. An analogous situation obtains for the tensor product functor. We shall in fact be interested only in the special case where R is a field, but we may as well consider the general case for the time being.

Let A and B be modules over the ring R. As before, let $F(A, B)$ be the free abelian group generated by the set $A \times B$. But now we let the *relations subgroup* $R(A, B)$ be generated by elements of the form

$$\begin{aligned} &(a + a', b) - (a, b) - (a', b)\,,\\ &(a, b + b') - (a, b) - (a, b')\,,\\ &(\alpha a, b) - (a, \alpha b)\,, \quad \text{for } \alpha \in R\,. \end{aligned}$$

Then $F(A, B)/R(A, B)$ has the structure of a module over R when, for $\alpha \in R$, we define the map of $F(A, B)$ to itself by $\alpha(a, b) = (\alpha a, b)$. This map carries $R(A, B)$ into itself. To see this, observe that when we apply $\alpha \in R$ to the first of the listed generators for $R(A, B)$, we have

$$\begin{aligned} &(\alpha(a + a'), b) - (\alpha a, b) - (\alpha a', b)\\ &\quad = (\alpha a + \alpha a', b) - (\alpha a, b) - (\alpha a', b)\,. \end{aligned}$$

A similar remark applies to the second of the listed generators. For the third, we have

$$\begin{aligned}&\beta(\alpha a, b) - \beta(a, \alpha b)\\ &\quad = (\alpha(\beta a), b) - (\beta a, \alpha b)\,,\end{aligned}$$

which is in $R(A, B)$ by definition.

Thus we have an induced operation on the quotient F/R. The module properties are easily verified. We shall denote the resulting module by $A \otimes_R B$, and call it the *tensor product of A and B over the ring R*. In addition to the usual relations in $A \otimes B$, in $A \otimes_R B$ one has the relation

$$\alpha(a \otimes b) = (\alpha a) \otimes b = a \otimes (\alpha b)\,.$$

Now let us consider a set map $f : A \times B \to C$ that is a module homomorphism in each variable separately. Since it maps $R(A, B)$ to zero, it induces a homomorphism

$$g : A \otimes_R B \to C\,,$$

which is actually a module homomorphism, as you can check.

Note that $A \otimes_R B$ is smaller than $A \otimes B$, because it is isomorphic to the quotient group of $A \otimes B$ by the subgroup generated by all terms of the form

$$(\alpha a) \otimes b - a \otimes (\alpha b)\,.$$

If $f : A \to A'$ and $g : B \to B'$ are module homomorphisms, then there is a module homomorphism

$$f \otimes g : A \otimes_R B \to A' \otimes_R B'$$

mapping $a \otimes b$ to $f(a) \otimes g(b)$ for all a, b, because the map sending (a, b) to $f(a) \otimes g(b)$ is a module homomorphism in each variable separately.

The theorems of this section generalize to the tensor product of modules. Proofs are left as exercises.

EXERCISES

1. Let G be an abelian group with torsion subgroup T. Let H be a divisible group [see Definition 5.2.9]. Show that

$$G \otimes H \cong (G/T) \otimes H\,.$$

[**Note:** The subgroup T is not necessarily a direct summand in G!]

2. Show that the additive group $\mathbb{Q}$ is torsion-free but not free. [**Hint:** Compute $\mathbb{Q} \otimes \mathbb{Z}/2$.]

3. Show that if A and B are $\mathbb{Z}$-modules, then

$$A \otimes_{\mathbb{Z}} B = A \otimes B\,.$$

4. Let A be an R-module. Show that there is an R-module isomorphism

$$R \otimes_R A \cong A\,.$$

5. Prove the analogues of Theorem 6.2.6 and Theorem 6.2.8 for $\otimes_R$.

6. Let A, B, and C be vector spaces over a field F.

(a) Show that $\otimes_F$ preserves exact sequences of vector spaces. [**Hint:** Every such sequence splits.]

(b) Show that if A and B have vector space bases $\{a_i\}$ and $\{b_j\}$, respectively, then $\{a_i \otimes b_j\}$ is a vector space basis for $A \otimes_F B$.

7. If A and B are vector spaces over $\mathbb{Q}$, show that

$$A \otimes_{\mathbb{Q}} B = A \otimes B\,.$$

8. The tensor product is a universal object for bilinear maps. Explain what this statement means.

6.3 Homology with Arbitrary Coefficients

We now use the tensor product functor to give a general definition of homology groups with arbitrary coefficients. The treatment will follow the pattern of Section 5.5, where we dealt with cohomology theory with arbitrary coefficients.

First, we work on the level of chain complexes. Then we specialize to singular theory, and finally to simplicial theory, at which point we show that the definition of homology with arbitrary coefficients we use here is equivalent to the one given in Section 1.11.

Homology of a chain complex

Let G be an abelian group. Let $\mathcal{C} = \{C_p, \partial\}$ be a chain complex. The chain complex $\mathcal{C} \otimes G$ consists of the *chain groups with coefficients* $C_p \otimes G$ and the boundary operators $\partial \otimes i_G$. We denote the pth homology group of the chain complex $\mathcal{C} \otimes G$ by $H_p(\mathcal{C}; G)$, and we call it the pth *homology group of* $\mathcal{C}$ *with coefficients in* G.

If $\{\mathcal{C}, \epsilon\}$ is an augmented chain complex, then one has the corresponding chain complex obtained from $\mathcal{C} \otimes G$ by adjoining the group $\mathbb{Z} \otimes G \cong G$ in dimension -1, and using $\epsilon \otimes i_G$ as the boundary operator from dimension 0 to dimension -1. Its homology groups are denoted $\widetilde{H}_p(\mathcal{C}; G)$ and are called the

reduced homology groups of $\mathcal{C}$ with coefficients in G. If $\widetilde{H}_0(\mathcal{C})$ vanishes, then so does $\widetilde{H}_0(\mathcal{C};G)$, since exactness of the sequence $C_1 \longrightarrow C_0 \longrightarrow \mathbb{Z} \longrightarrow 0$ implies exactness of

$$C_1 \otimes G \longrightarrow C_0 \otimes G \longrightarrow \mathbb{Z} \otimes G \longrightarrow 0\,.$$

In general, we have the equation

$$H_0(\mathcal{C};G) \cong \widetilde{H}_0(\mathcal{C};G) \oplus G\,.$$

Note that if G is the group of integers, then $\mathcal{C} \otimes G$ is naturally isomorphic with $\mathcal{C}$. Thus the usual homology of $\mathcal{C}$ can be thought of as "homology with coefficients in $\mathbb{Z}$."

If $\phi : \mathcal{C} \to \mathcal{D}$ is a chain map, then so is $\phi \otimes i_G : \mathcal{C} \otimes G \to \mathcal{D} \otimes G$. For convenience, the induced homology homomorphism is denoted by

$$\phi_* : H_p(\mathcal{C};G) \to H_p(\mathcal{D};G)\,,$$

rather than by $(\phi \otimes i_G)_*$. If $\mathcal{C}$ and $\mathcal{D}$ are augmented and ϕ is augmentation-preserving, then $\phi \otimes i_G$ induces a homomorphism of reduced as well as ordinary homology.

If $\phi, \psi : \mathcal{C} \to \mathcal{C}'$ are chain maps, and if D is a chain homotopy between them, then $D \otimes i_G$ is a chain homotopy between $\phi \otimes i_G$ and $\psi \otimes i_G$. If follows that if ϕ and ψ are chain homotopic, then ϕ_* and ψ_* are equal as homomorphisms of homology with arbitrary coefficients. It also follows that if ϕ is a chain equivalence, then so is $\phi \otimes i_G$.

Finally, suppose one has a short exact sequence

$$0 \longrightarrow \mathcal{C} \longrightarrow \mathcal{D} \longrightarrow \mathcal{E} \longrightarrow 0$$

of chain complexes *that splits in each dimension.* Then the tensored sequence

$$0 \longrightarrow \mathcal{C}_p \otimes G \longrightarrow \mathcal{D}_p \otimes G \longrightarrow \mathcal{E}_p \otimes G \longrightarrow 0$$

is exact. Applying the zig-zag lemma [Lemma 3.3.2], one obtains the long exact sequence

$$\cdots \longrightarrow H_p(\mathcal{C};G) \longrightarrow H_p(\mathcal{D};G) \longrightarrow H_p(\mathcal{E};G) \xrightarrow{\partial_*} H_{p-1}(\mathcal{C};G) \longrightarrow \cdots\,,$$

where ∂_* is induced by $\partial \otimes i_G$. This sequence is natural with respect to homomorphisms induced by chain maps.

Singular Homology

We define the *singular homology groups of a topological pair (X, A) with coefficients in G*, by the equation

$$H_p(X, A; G) = H_p(\mathcal{S}(X, A); G)\,.$$

As usual, we delete A from the notation if A is empty. We define reduced homology groups by the equation

$$\widetilde{H}_p(X;G) = H_p(\{\mathcal{S}(X), \epsilon\}; G)\,,$$

where ϵ is the usual augmentation for $\mathcal{S}(X)$.

Since $S_p(X, A)$ is free abelian, the group $S_p(X, A) \otimes G$ is a direct sum of copies of G. Indeed, if $\{T_\alpha\}$ is the family consisting of those singular p-simplices of X whose image sets do not lie in A, then the cosets modulo $S_p(A)$ of the singular simplices T_α form a basis for $S_p(X, A)$. Therefore each element of $S_p(X, A) \otimes G$ can be represented uniquely by a finite sum $\sum T_\alpha \otimes g_\alpha$. This last sum is our usual way of representing a singular p-chain with coefficients in G. The *minimal carrier* of such a chain is the union of the sets $T_\alpha(\Delta_p)$, where the union is taken over those α for which $g_\alpha \neq 0$. It is of course a compact set.

A continuous map $h : (X, A) \to (Y, B)$ gives rise to a chain map

$$h_\# \otimes i_G : \mathcal{S}(X, A) \otimes G \to \mathcal{S}(Y, B) \otimes G\,.$$

We sometimes denote this map simply by $h_\#$, and the induced homology homomorphism by h_*. Functoriality is immediate; in fact, it holds on the chain level.

The short exact sequence of chain complexes

$$0 \longrightarrow S_p(A) \longrightarrow S_p(X) \longrightarrow S_p(X, A) \longrightarrow 0$$

splits because $S_p(X, A)$ is free. Therefore, one has a long exact homology sequence with coefficients in G. This long exact sequence is natural with respect to homomorphisms induced by continuous maps.

If the maps $h, k : (X, A) \to (Y, B)$ are homotopic, then by the proof of Theorem 4.3.7 there is a chain homotopy between $h_\#$ and $k_\#$. Then $h_\# \otimes i_G$ and $k_\# \otimes i_G$ are chain homotopic, so h_* and k_* are equal as maps of homology with coefficients in G.

Direct consideration of the singular chain complex of a one-point space P leads to the result that $H_i(P; G) = 0$ for $i \neq 0$ and $H_0(P; G) \cong G$.

The compact support property of singular homology carries over at once to the same property for singular homology with arbitrary coefficients.

The only property of singular homology that requires extra care is the excision property.

Let (X, A) be a topological pair, and let U be a subset of X such that $\overline{U} \subset \text{Int } A$. We know that the inclusion

$$j : (X \setminus U, A \setminus U) \to (X, A)$$

induces an isomorphism in ordinary homology. We wish to show it induces an isomorphism in homology with arbitrary coefficients as well.

One way of doing this is to use Theorem 5.7.2, which implies that $j_\#$ is a chain equivalence. [Note that the chain complexes involved are free and vanish below a certain dimension.] Alternatively, one can prove the following, which is an analogue of Theorem 5.6.6.

Theorem 6.3.1 *If $\mathcal{C}$ and $\mathcal{D}$ are free chain complexes, and if the chain map $\phi : \mathcal{C} \to \mathcal{D}$ induces homology isomorphisms in all dimensions, then so does the chain map*

$$\phi \otimes i_G : \mathcal{C} \otimes G \to \mathcal{D} \otimes G\,.$$

Proof: **Step 1.** We first consider the case where we have a short exact sequence

$$0 \longrightarrow \mathcal{C} \longrightarrow \mathcal{D} \longrightarrow \mathcal{E} \longrightarrow 0$$

of free chain complexes. We know that $H_p(\mathcal{E}) = 0$ for all p, and we wish to prove that $H_p(\mathcal{E}; G) = 0$ for all p. As in the proof of Lemma 5.6.4, we can write $E_p = B_p \oplus U_p$, where ∂ maps B_p to zero and carries U_p isomorphically onto B_{p-1}. Then we have

$$E_p \otimes G \cong (B_p \otimes G) \oplus (U_p \otimes G)\,,$$

and $\partial \otimes i_G$ maps $B_p \otimes G$ to zero and carries $U_p \otimes G$ isomorphically onto $B_{p-1} \otimes G$. It follows that $H_p(\mathcal{E}; G) = 0$ for all p.

Step 2. The general case now follows from Lemma 5.6.5. Given $\phi : \mathcal{C} \to \mathcal{D}$, there are a chain complex $\mathcal{D}'$ and injective chain maps $i : \mathcal{C} \to \mathcal{D}'$ and $j : \mathcal{D} \to \mathcal{D}'$ such that j induces homology isomorphisms in all dimensions and $j \circ \phi$ is chain homotopic to i. Furthermore, $\mathcal{D}'$, $\mathcal{D}'/\text{im } i$, and $\mathcal{D}'/\text{im } j$ are free. If ϕ_* is an isomorphism in ordinary homology, so is $i_* = j_* \circ \phi_*$. Then by **Step 1**, both i and j induce isomorphisms of homology with arbitrary coefficients. Hence ϕ does the same. □

Simplicial Homology

Let (K, K_0) be a simplicial pair, and let G be an abelian group. We define the simplicial homology of (K, K_0) with coefficients in G by the equation

$$H_p(K, K_0; G) = H_p(\mathcal{C}(K, K_0); G)\,.$$

We define the reduced groups by the equation

$$\widetilde{H}_p(K; G) = H_p(\{\mathcal{C}(K), \epsilon\}; G)\,,$$

where ϵ is the usual augmentation for $\mathcal{C}(K)$.

Now the group $C_p(K, K_0) \otimes G$ is the direct sum of copies of G, one copy for each p-simplex of K not in K_0. Indeed, if we orient the p-simplices σ_α of K not in K_0 arbitrarily, then each element of $C_p(K, K_0) \otimes G$ can be represented uniquely by a finite sum $\sum \sigma_\alpha \otimes g_\alpha$. Its boundary is then represented by $\sum(\partial\sigma_\alpha) \otimes g_\alpha$.

The connection between the current definition of homology with arbitrary coefficients and the definition we gave in Section 1.11 can now be made clear. In that section, we represented a simplicial p-chain c_p with coefficients in G by a finite formal sum $c_p = \sum g_\alpha \sigma_\alpha$, and its boundary was defined by the formula

$$\partial c_p = \sum g_\alpha (\partial \sigma_\alpha)\,.$$

It follows that the chain complex $\mathcal{C}(K;G)$ defined in Section 1.11 is isomorphic to the chain complex $\mathcal{C}(K) \otimes G$. Henceforth, we shall use the latter chain complex in dealing with simplicial homology with arbitrary coefficients.

The existence of induced homomorphisms and the long exact homology sequence, and the verification of the Eilenberg–Steenrod axioms, is so straightforward that we omit the details. The argument follows the pattern given in Section 5.5 for simplicial cohomology.

The Isomorphism between Simplicial and Singular Theory

We have the chain map of Section 4.7,

$$\eta : C_p(K, K_0) \to S_p(|K|, |K_0|)\,,$$

which induces an isomorphism in ordinary homology. It follows from Theorem 6.3.1 that it induces an isomorphism in homology with arbitrary coefficients as well. The fact that it is independent of the ordering of vertices used in defining η, and the fact that it commutes with the boundary homomorphism ∂_* and with homomorphisms induced by continuous maps, follow as in the proof of Theorem 5.5.3.

The Homology of CW Complexes

If X is a CW complex with cellular chain complex $\mathcal{D}(X)$, we know that $H_p(\mathcal{D}(X)) \cong H_p(X)$ for all p. It follows from Theorem 5.6.2 that there is a chain map inducing this isomorphism. It then follows from Theorem 6.3.1 that

$$H_p(\mathcal{D}(X) \otimes G) \cong H_p(X;G)\,.$$

Thus the cellular chain complex of X can be used to compute homology with arbitrary coefficients. If X is a triangulable CW complex, then the chain map inducing this isomorphism is the inclusion map

$$j : \mathcal{D}(X) \to \mathcal{C}(X)\,.$$

The proof of the following will be left to the exercises:

Theorem 6.3.2 *Let $\mathcal{C}$ be a free chain complex. Then there is a natural exact sequence*

$$0 \longrightarrow H_p(\mathcal{C}) \otimes G \xrightarrow{\phi} H_p(\mathcal{C} \otimes G) \longrightarrow \operatorname{cok}\, \phi \longrightarrow 0\,,$$

where ϕ is induced by inclusion. The sequence splits, but not naturally. If $H_i(\mathcal{C})$ is free for all i, then ϕ is an isomorphism.

This last result is the analogue for homology of Lemma 5.6.12 and Theorem 5.6.13. It will be generalized in the next chapter.

EXERCISES

1. If $\{\mathcal{C}, \epsilon\}$ is an augmented chain complex, show that

$$H_0(\mathcal{C}; G) \cong \widetilde{H}_0(\mathcal{C}; G) \oplus G\,.$$

[**Hint:** The sequence $0 \to \ker \epsilon \to C_0 \to \mathbb{Z} \to 0$ splits.]

2. Use the cellular chain complexes to compute the homology, with general coefficients G, of the following spaces:

(a) $T \# T$.
(b) $\mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$.
(c) $\mathbb{P}^N$.
(d) the k-fold dunce cap.

3. Prove theorem 6.3.2.

4. Let R be a commutative ring with unity. Let $\mathcal{C}$ be a chain complex. Show that $C_p \otimes R$ can be given the structure of an R-module by defining

$$\alpha(c_p \otimes \beta) = c_p \otimes (\alpha\beta)$$

for $\alpha, \beta \in R$. Show that $H_p(\mathcal{C}; R)$ has the structure of an R-module, and that chain maps induce R-module homomorphisms. Show that ∂_* is an R-module homomorphism.

7

Homological Algebra

7.1 Introduction

We have already seen, in Chapter 5, that two spaces with isomorphic homology groups have isomorphic cohomology groups as well. This fact leads one to suspect that the cohomology groups of a space are in some way determined by the homology groups. In this chapter this suspicion is confirmed. We show precisely *how* the cohomology groups *with arbitrary coefficients* are determined by the homology groups with *integer coefficients*. The theorem involved is called the Universal Coefficient Theorem for Cohomology. Its statement involves not only the Hom functor introduced in Section 5.2, but also a new functor called "Ext" that we shall introduce in the next section.

Similarly, we know that if two spaces have isomorphic integral homology groups, then the same is true for homology with arbitrary coefficients. Just as with the cohomology groups, it turns out that the homology groups with arbitrary coefficients are determined by the homology groups with integer coefficients. The theorem involved is called the Universal Coefficient Theorem for Homology. It involves not only the tensor product functor introduced in Section 6.2, but also a new functor called the "torsion product" that we shall introduce in Section 7.4.

The functors mentioned above form part of a general subject called homological algebra. Although its origins are topological in nature, it has come to have an independent existence nowadays within the field of algebra, having applications to many problems that are purely algebraic. Our interest in it is confined to its connections with topology.

These functors also play a role when one comes to study the homology of a product space $X \times Y$. It turns out that the homology of $X \times Y$ is determined by the homology of X and of Y. The relationship is expressed in the form of an exact sequence called the Künneth sequence. It involves both the tensor and torsion functors.

There is a similar sequence for cohomology that holds if the homology is finitely generated. It has applications to cup products in general, and to computing the cohomology ring of $X \times Y$ in particular.

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7.2 The Ext Functor

Associated with the functor $\mathrm{Hom}\,(A, B)$ is another functor of two variables called $\mathrm{Ext}\,(A, B)$. Like the Hom functor, it is contravariant in the first variable and covariant in the second variable. This means that, given homomorphisms $\gamma : A \to A'$ and $\delta : B' \to B$, there is a homomorphism

$$\mathrm{Ext}\,(\gamma, \delta) : \mathrm{Ext}\,(A', B') \to \mathrm{Ext}\,(A, B)\,,$$

and the usual functorial properties hold.

Defining the functor Ext requires some preliminary work, but its crucial property is easy to remember. We express it in the form of a theorem.

Theorem 7.2.1 *There is a function that assigns to each free resolution [see Definition 5.6.1]*

$$0 \longrightarrow R \xrightarrow{\phi} F \xrightarrow{\psi} A \longrightarrow 0$$

of the abelian group A and each abelian group B an exact sequence

$$0 \longleftarrow \mathrm{Ext}\,(A, B) \xleftarrow{\pi} \mathrm{Hom}\,(R, B) \xleftarrow{\widetilde{\phi}} \mathrm{Hom}\,(F, B) \xleftarrow{\widetilde{\psi}} \mathrm{Hom}\,(A, B) \longleftarrow 0\,.$$

This function is natural, in the sense that a homomorphism

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \longrightarrow & F & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & R' & \longrightarrow & F' & \longrightarrow & A' & \longrightarrow & 0 \end{array}$$

of free resolutions and a homomorphism $\delta : B' \to B$ of abelian groups gives rise to a homomorphism of exact sequences:

$$\begin{array}{ccccccccc} 0 & \longleftarrow & \mathrm{Ext}\,(A, B) & \longleftarrow & \mathrm{Hom}\,(R, B) & \longleftarrow & \mathrm{Hom}\,(F, B) & \longleftarrow & \mathrm{Hom}\,(A, B) \longleftarrow 0 \\ & & \mathrm{Ext}\,(\gamma,\delta)\uparrow & & \mathrm{Hom}\,(\alpha,\delta)\uparrow & & \mathrm{Hom}\,(\beta,\delta)\uparrow & & \mathrm{Hom}\,(\gamma,\delta)\uparrow \\ 0 & \longleftarrow & \mathrm{Ext}\,(A', B') & \longleftarrow & \mathrm{Hom}\,(R', B') & \longleftarrow & \mathrm{Hom}\,(F', B') & \longleftarrow & \mathrm{Hom}\,(A', B') \longleftarrow 0\,. \end{array}$$

We shall prove this theorem shortly. It will then be used to derive the other properties of the Ext functor and to compute it.

Before we define the Ext functor, we need a lemma.

Lemma 7.2.2 *Suppose one is given a homomorphism*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \xrightarrow{\phi} & F & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & R' & \xrightarrow[\phi']{} & F' & \longrightarrow & A' & \longrightarrow & 0 \end{array}$$

of free resolutions of A and A', respectively, and a homomorphism $\delta : B' \to B$. Then there is a unique homomorphism $\epsilon : \operatorname{cok} \widetilde{\phi}' \to \operatorname{cok} \widetilde{\phi}$ making the following diagram of exact sequences commute:

$$\begin{array}{ccccccccccc} 0 & \longleftarrow & \operatorname{cok} \widetilde{\phi} & \longleftarrow & \operatorname{Hom}(R,B) & \xleftarrow{\widetilde{\phi}} & \operatorname{Hom}(F,B) & \longleftarrow & \operatorname{Hom}(A,B) & \longleftarrow & 0 \\ & & \uparrow{\scriptstyle \epsilon} & & \uparrow{\scriptstyle \operatorname{Hom}(\alpha,\delta)} & & \uparrow{\scriptstyle \operatorname{Hom}(\beta,\delta)} & & \uparrow{\scriptstyle \operatorname{Hom}(\gamma,\delta)} & & \\ 0 & \longleftarrow & \operatorname{cok} \widetilde{\phi}' & \longleftarrow & \operatorname{Hom}(R',B') & \xleftarrow[\widetilde{\phi}']{} & \operatorname{Hom}(F',B') & \longleftarrow & \operatorname{Hom}(A',B') & \longleftarrow & 0\,. \end{array}$$

The homomorphism ϵ is independent of the choice of α and β.

Proof: Functoriality of Hom shows that the two right squares of the preceding diagram commute. Therefore, $\operatorname{Hom}(\alpha, \delta)$ induces a homomorphism ϵ of cokernels.

What remains to be shown is that ϵ is independent of the choice of α and β. Suppose $\{\alpha', \beta', \gamma\}$ is another homomorphism of the two given free resolutions. Consider the free resolution of A as a chain complex $\mathcal{A}$, indexed so that A is the 0-dimensional group. Do the same for A', thus obtaining a chain complex $\mathcal{A}'$. Then $\{\alpha, \beta, \gamma\}$ and $\{\alpha', \beta', \gamma\}$ are chain maps of $\mathcal{A}$ to $\mathcal{A}'$. By exactness, the homology groups of $\mathcal{A}$ and $\mathcal{A}'$ vanish. The cohomology groups need not vanish. Indeed, the group $\operatorname{cok} \widetilde{\phi}$ is just the 2-dimensional cohomology group $H^2(\mathcal{A}; B)$, and similarly $\operatorname{cok} \widetilde{\phi}' = H^2(\mathcal{A}'; B)$. The map ϵ is just the cohomology homomorphism induced by the chain map $\{\alpha, \beta, \gamma\}$.

We see that the hypotheses of Lemma 5.7.1 are satisfied by the chain complexes $\mathcal{A}$ and $\mathcal{A}'$. Therefore there is a chain homotopy D between the chain map $\{\alpha, \beta, \gamma\}$ and the chain map $\{\alpha', \beta', \gamma\}$. Then $\operatorname{Hom}(D, \delta)$ is a cochain homotopy between the corresponding cochain maps. It follows that they induce the same homomorphism ϵ of 2-dimensional cohomology groups. □

Definition 7.2.3 The homomorphism ϵ constructed in the preceding lemma is said to be *induced* by γ and δ and the free resolutions involved.

We show that a version of functoriality holds. First, we show that *composites* behave correctly. Let α, β, γ, δ, and ϵ be as in Lemma 7.2.2, and suppose that

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R' & \longrightarrow & F' & \longrightarrow & A' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \beta'} & & \downarrow{\scriptstyle \gamma'} & & \\ 0 & \longrightarrow & R'' & \longrightarrow & F'' & \longrightarrow & A'' & \longrightarrow & 0 \end{array}$$

is another homomorphism of free resolutions, with $\delta' : B'' \to B'$ another homomorphism of abelian groups. Let ϵ' be the homomorphism induced by γ' and δ'. The fact that

$$\mathrm{Hom}\,(\gamma, \delta) \circ \mathrm{Hom}\,(\gamma', \delta') = \mathrm{Hom}\,(\gamma' \circ \gamma, \delta \circ \delta')$$

and the similar equations for α, α' and β, β', imply that $\epsilon \circ \epsilon'$ is the homomorphism induced by $\gamma' \circ \gamma$ and $\delta \circ \delta'$.

Second, we show that if $\gamma = i_A$ and $\delta = i_B$ are identity maps, then the homomorphism ϵ that they induce is an *isomorphism*. Certainly, this is true in the situation where the two free resolutions are the same, since α and β can be chosen to be identity maps, so that ϵ is the identity. But it is also true in the situation

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \xrightarrow{\phi} & F & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle i_A} & & \\ 0 & \longrightarrow & R' & \xrightarrow{\phi'} & F' & \longrightarrow & A & \longrightarrow & 0\,, \end{array}$$

as we now prove. Let $\epsilon : \mathrm{cok}\ \widetilde{\phi}' \to \mathrm{cok}\ \widetilde{\phi}$ be the map induced by (i_A, i_B), relative to these free resolutions. Choose α' and β' so that the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R' & \xrightarrow{\phi'} & F' & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \beta'} & & \downarrow{\scriptstyle i_A} & & \\ 0 & \longrightarrow & R & \xrightarrow{\phi} & F & \longrightarrow & A & \longrightarrow & 0\,. \end{array}$$

[Here we use the fact that F is free.] Let $\epsilon' : \mathrm{cok}\ \widetilde{\phi} \to \mathrm{cok}\ \widetilde{\phi}'$ be the map induced by (i_A, i_B), relative to these free resolutions. By the previous remarks, both composites $\epsilon \circ \epsilon'$ and $\epsilon' \circ \epsilon$ equal identity maps. Thus ϵ is an isomorphism.

It follows from these comments that, given A and B, if one chooses *any* free resolution of A, the group $\mathrm{cok}\ \widetilde{\phi}$ will be independent, up to isomorphism, of the choice. This fact leads us to the following definition of a "canonical version" of $\mathrm{cok}\ \widetilde{\phi}$, that we shall call $\mathrm{Ext}\,(A, B)$.

Definition 7.2.4 Suppose A and B are abelian groups. Let $F(A)$ denote the free abelian group generated by the elements of A, and let $R(A)$ be the kernel of the natural projection $F(A) \to A$. The sequence

$$0 \longrightarrow R(A) \xrightarrow{\phi} F(A) \longrightarrow A \longrightarrow 0$$

is called the *canonical free resolution* of A. [The canonical free resolution was also defined earlier in Definition 5.6.1.] The group

$$\operatorname{cok} \widetilde{\phi} = \operatorname{Hom}(R(A), B)/\widetilde{\phi}(\operatorname{Hom}(F(A), B))$$

is denoted $\operatorname{Ext}(A, B)$. This group is sometimes called the *group of extensions of B by A* [for an explanation of this terminology see Chapter 3 of [ML63], but we will simply call it Ext].

If $\gamma : A \to A'$ and $\delta : B' \to B$ are homomorphisms, then we extend γ to a homomorphism of the canonical free resolution of A with that of A', and define

$$\operatorname{Ext}(\gamma, \delta) : \operatorname{Ext}(A', B') \to \operatorname{Ext}(A, B)$$

to be the homomorphism induced by γ and δ relative to these free resolutions.

The previous remarks show that Ext is a functor of two variables, contravariant in the first variable and covariant in the second variable.

Now we prove the basic theorem.

Proof of Theorem 7.2.1: The exact sequence

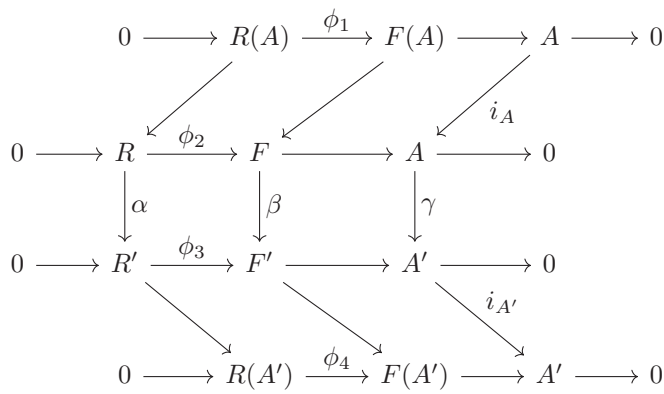
$$0 \longrightarrow R \xrightarrow{\phi} F \longrightarrow A \longrightarrow 0$$

gives rise to the exact sequence

$$0 \longleftarrow \operatorname{cok} \widetilde{\phi} \longleftarrow \operatorname{Hom}(R, B) \longleftarrow \operatorname{Hom}(F, B) \longleftarrow \operatorname{Hom}(A, B) \longleftarrow 0\,.$$

In view of the preceding remarks, there is an isomorphism of $\operatorname{cok} \widetilde{\phi}$ with $\operatorname{Ext}(A, B)$ that is induced by (i_A, i_B). We use this isomorphism to replace $\operatorname{cok} \widetilde{\phi}$ by $\operatorname{Ext}(A, B)$ in this exact sequence.

It remains to check naturality. Let α, β, γ define a homomorphism of free resolutions of A and A', respectively, as in the statement of the theorem. Let $\delta : B' \to B$. Consider the following diagram:



There are homomorphisms

$$\operatorname{cok} \widetilde{\phi}_4 \xrightarrow{\epsilon_3} \operatorname{cok} \widetilde{\phi}_3 \xrightarrow{\epsilon_2} \operatorname{cok} \widetilde{\phi}_2 \xrightarrow{\epsilon_1} \operatorname{cok} \widetilde{\phi}_1$$

induced by $(i_{A'}, i_{B'})$, (γ, δ), and (i_A, i_B), respectively. Both ϵ_3 and ϵ_1 are isomorphisms. By functoriality, the composite $\epsilon_1 \circ \epsilon_2 \circ \epsilon_3$ is the unique homomorphism induced by

$$(\, i_{A'} \circ \gamma \circ i_A,\ i_B \circ \delta \circ i_{B'} \,) = (\gamma, \delta)\,,$$

relative to the canonical free resolutions, so it must equal $\operatorname{Ext}(\gamma, \delta)$. Therefore the diagram

$$\begin{array}{ccccc}
\operatorname{Ext}(A,B) & \xleftarrow[\cong]{\epsilon_1} & \operatorname{cok} \widetilde{\phi}_2 & \longleftarrow & \operatorname{Hom}(R,B) \\
{\scriptstyle \operatorname{Ext}(\gamma,\delta)}\uparrow & & \uparrow{\scriptstyle \epsilon_2} & & \uparrow{\scriptstyle \operatorname{Hom}(\alpha,\delta)} \\
\operatorname{Ext}(A',B') & \xrightarrow[\cong]{\epsilon_3} & \operatorname{cok} \widetilde{\phi}_3 & \longleftarrow & \operatorname{Hom}(R',B')
\end{array}$$

commutes, and the proof is complete. □

We now prove some further properties of the Ext functor.

Theorem 7.2.5 **(a)** *There are natural isomorphisms*

$$\begin{aligned}
\operatorname{Ext}\left(\textstyle\bigoplus_{\alpha\in J} A_\alpha, B\right) &\cong \textstyle\prod_{\alpha\in J} \operatorname{Ext}(A_\alpha, B)\,, \\
\operatorname{Ext}\left(A, \textstyle\prod_{\alpha\in J} B_\alpha\right) &\cong \textstyle\prod_{\alpha\in J} \operatorname{Ext}(A, B_\alpha)\,.
\end{aligned}$$

(b) $\operatorname{Ext}(A, B) = 0$ *if* A *is free.*

(c) *Given* B*, there is an exact sequence*

$$0 \longleftarrow \operatorname{Ext}(\mathbb{Z}/m, B) \longleftarrow B \xleftarrow{\phi_m} B \longleftarrow \operatorname{Hom}(\mathbb{Z}/m, B) \longleftarrow 0\,,$$

where ϕ_m *is multiplication by* m*.*

Proof: In the proof, we shall use the fact that direct sums and direct products of exact sequences are exact, and the fact that a direct sum [but not a direct product] of free abelian groups is free abelian.

(a) Let

$$0 \longrightarrow R_\alpha \longrightarrow F_\alpha \longrightarrow A_\alpha \longrightarrow 0$$

be a free resolution of A_α. Then

$$0 \longrightarrow \textstyle\bigoplus_{\alpha\in J} R_\alpha \longrightarrow \bigoplus_{\alpha\in J} F_\alpha \longrightarrow \bigoplus_{\alpha\in J} A_\alpha \longrightarrow 0$$

is a free resolution of $\bigoplus_{\alpha\in J} A_\alpha$. Both of the sequences

$$0 \longleftarrow \mathrm{Ext}\,(A_\alpha, B) \longleftarrow \mathrm{Hom}\,(R_\alpha, B) \longleftarrow \mathrm{Hom}\,(F_\alpha, B) \longleftarrow \mathrm{Hom}\,(A_\alpha, B) \longleftarrow 0\,,$$

$$0 \leftarrow \mathrm{Ext}\,(\oplus A_\alpha, B) \leftarrow \mathrm{Hom}\,(\oplus R_\alpha, B) \leftarrow \mathrm{Hom}\,(\oplus F_\alpha, B) \leftarrow \mathrm{Hom}\,(\oplus A_\alpha, B) \leftarrow 0\,,$$

are exact by Theorem 7.2.1. The direct product of sequences of the first type gives a sequence that is naturally isomorphic with the second sequence in its three right-hand terms. Therefore there is an isomorphism between the left-hand terms as well. This isomorphism $\mathrm{Ext}\,(\oplus A_\alpha, B) \to \prod \mathrm{Ext}\,(A_\alpha, B)$ is in fact natural, by a standard argument.

Similarly, if

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$$

is a free resolution of A, then the sequences

$$0 \longleftarrow \mathrm{Ext}\,(A, B_\alpha) \longleftarrow \mathrm{Hom}\,(R, B_\alpha) \longleftarrow \mathrm{Hom}\,(F, B_\alpha) \longleftarrow \mathrm{Hom}\,(A, B_\alpha) \longleftarrow 0\,,$$

$$0 \leftarrow \mathrm{Ext}\,(A, \prod B_\alpha) \leftarrow \mathrm{Hom}\,(R, \prod B_\alpha) \leftarrow \mathrm{Hom}\,(F, \prod B_\alpha) \leftarrow \mathrm{Hom}\,(A, \prod B_\alpha) \leftarrow 0\,,$$

are exact. Since the direct product of sequences of the first type agrees with the second in the three right-hand terms [the isomorphisms in question being natural], these sequences agree in the left-hand terms as well. Again, the isomorphism is natural.

(b) To check **(b)**, we recall that a free resolution of A splits if A is free. Then the dual sequence is exact, so $\mathrm{Ext}\,(A, B)$ vanishes.

(c) To prove **(c)**, one begins with the free resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\phi_m} \mathbb{Z} \longrightarrow \mathbb{Z}/m \longrightarrow 0\,.$$

Applying Theorem 7.2.1, we obtain the exact sequence

$$0 \leftarrow \mathrm{Ext}\,(\mathbb{Z}/m, B) \leftarrow \mathrm{Hom}\,(\mathbb{Z}, B) \xleftarrow{\phi_m} \mathrm{Hom}\,(\mathbb{Z}, B) \leftarrow \mathrm{Hom}\,(\mathbb{Z}/m, B) \leftarrow 0\,,$$

from which **(c)** follows. □

This last theorem enables us to compute $\mathrm{Ext}\,(A, B)$ when A is finitely generated. Such computations utilize the fact that Ext commutes with finite direct sums, and the rules

$$\mathrm{Ext}\,(\mathbb{Z}, G) = 0 \quad \text{and} \quad \mathrm{Ext}\,(\mathbb{Z}/m, G) \cong G/mG\,.$$

If we apply the Hom functor to a short exact sequence with a group G [either on the left or the right], the resulting sequence may fail to be exact. In some sense, the Ext functor measures the extent to which exactness fails. One has the following theorem:

Theorem 7.2.6 *There are functors assigning to each short exact sequence of abelian groups*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0\,,$$

and each abelian group G, two exact sequences:

$$0 \leftarrow \mathrm{Ext}(A,G) \leftarrow \mathrm{Ext}(B,G) \leftarrow \mathrm{Ext}(C,G) \leftarrow$$
$$\mathrm{Hom}(A,G) \leftarrow \mathrm{Hom}(B,G) \leftarrow \mathrm{Hom}(C,G) \leftarrow 0\,,$$

$$0 \rightarrow \mathrm{Hom}(G,A) \rightarrow \mathrm{Hom}(G,B) \rightarrow \mathrm{Hom}(G,C) \rightarrow$$
$$\rightarrow \mathrm{Ext}(G,A) \rightarrow \mathrm{Ext}(G,B) \rightarrow \mathrm{Ext}(G,C) \rightarrow 0\,.$$

Exercise 4 should guide you through a proof of this theorem. The next definition is used in that proof:

Definition 7.2.7 Let G be an abelian group. An exact sequence

$$0 \longrightarrow G \longrightarrow H \longrightarrow K \longrightarrow 0\,,$$

where H and K are divisible [see Definition 5.2.9], is called an *injective resolution* of G.

EXERCISES

1. Show that if B is divisible [see Definition 5.2.9] , then $\mathrm{Ext}\,(A,B) = 0$.

2. Compute $\mathrm{Hom}\,(A,B)$ and $\mathrm{Ext}\,(A,B)$ if

$$A = \mathbb{Z} \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/6\,, \quad B = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}/9 \oplus \mathbb{Z}/12\,.$$

3. (a) If A is finitely generated, compute $\mathrm{Hom}\,(A,\mathbb{R}/\mathbb{Z})$ and $\mathrm{Ext}\,(A,\mathbb{R}/\mathbb{Z})$ in terms of the betti number and the torsion coefficients of A. [Recall from Definition 5.2.10 that the additive group $\mathbb{R}/\mathbb{Z}$ is called the circle group, because it is isomorphic to the multiplicative group of complex numbers of unit modulus.]

(b) Again assume A is finitely generated. Compute $\mathrm{Hom}\,(A,\mathbb{Q})$ and $\mathrm{Ext}\,(A,\mathbb{Q})$ in terms of the betti number and the torsion coefficients of A.

4. Follow the suggested steps to prove Theorem 7.2.6.

Steps:

(a) To obtain the second sequence, let

$$0 \longrightarrow R \longrightarrow F \longrightarrow G \longrightarrow 0$$

be a free resolution of G, and apply the serpent lemma [Lemma 3.3.9] to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Hom}\,(F,A) & \longrightarrow & \mathrm{Hom}\,(F,B) & \longrightarrow & \mathrm{Hom}\,(F,C) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{Hom}\,(R,A) & \longrightarrow & \mathrm{Hom}\,(R,B) & \longrightarrow & \mathrm{Hom}\,(R,C) & \longrightarrow & 0\,. \end{array}$$

(b) Given an abelian group G, show that there is an injective resolution of G. [**Hint:** It suffices to find H, since a quotient of a divisible group is divisible. Write $G = F/R$, where F is free, and construct a monomorphism into a direct sum of copies of $\mathbb{Q}$.]

(c) Show that there is a functor assigning to each injective resolution

$$0 \longrightarrow G \longrightarrow H \longrightarrow K \longrightarrow 0$$

of G, and each abelian group A, an exact sequence

$$0 \to \mathrm{Hom}\,(A,G) \to \mathrm{Hom}\,(A,H) \to \mathrm{Hom}\,(A,K) \to \mathrm{Ext}\,(A,G) \to 0\,.$$

(d) Use the serpent lemma, Lemma 3.3.9, to derive the first sequence of the theorem.

5. Refer to Exercise 3. Show that $\mathbb{S}^1$ really does model a circle. Show that this group is commutative. Show that it is not finitely generated.

7.3 The Universal Coefficient Theorem for Cohomology

We now show how the homology groups of a space determine the cohomology groups. The answer is expressed in the form of a short exact sequence involving the groups H^p, H_p, and H_{p-1}, and the functors Hom and Ext.

We already know that if $\mathcal{C}$ is a free chain complex, then there is a natural exact sequence

$$0 \longleftarrow \mathrm{Hom}\,(H_p(\mathcal{C}),G) \xleftarrow{\kappa} H^p(\mathcal{C};G) \longleftarrow \ker\kappa \longleftarrow 0\,.$$

[See Lemma 5.6.12.] We now identify the group $\ker \kappa$, and show that it depends only on the groups $H_{p-1}(\mathcal{C})$ and G.

Theorem 7.3.1 (The universal coefficient theorem for cohomology)

Let $\mathcal{C}$ be a free chain complex, and let G be an abelian group. There is an exact sequence

$$0 \longleftarrow \mathrm{Hom}\,(H_p(\mathcal{C}), G) \xleftarrow{\kappa} H^p(\mathcal{C}; G) \longleftarrow \mathrm{Ext}\,(H_{p-1}(\mathcal{C}), G) \longleftarrow 0\,,$$

that is natural with respect to homomorphisms induced by chain maps. It splits, but not naturally.

Proof: Step 1. Let C_p, Z_p, and B_p denote the groups of p-chains, p-cycles, and p-boundaries of $\mathcal{C}$, respectively. Consider the exact sequence

$$0 \longrightarrow Z_p \xrightarrow{i} C_p \xrightarrow{\partial_0} B_{p-1} \longrightarrow 0\,.$$

It splits because B_{p-1} is free. Define a chain complex $\mathcal{Z}$ by letting its p-dimensional group be Z_p and letting its boundary operator be the restriction of ∂. Then all the boundary operators in $\mathcal{Z}$ vanish. Similarly, define a chain complex $\mathcal{D}$ by letting $D_p = B_{p-1}$ and by letting its boundary operator be the restriction of ∂, so all boundary operators in $\mathcal{D}$ vanish.

We then have a [split] short exact sequence of chain complexes

$$0 \longrightarrow Z_p \xrightarrow{i} C_p \xrightarrow{\partial_0} D_p \longrightarrow 0\,.$$

The map i is a chain map by definition, and the map ∂_0 is a chain map trivially [since $\partial \circ \partial = 0$].

Applying the Hom functor, we obtain a short exact sequence of cochain complexes and hence a long exact cohomology sequence, of which we consider five terms:

$$H^{p+1}(\mathcal{D}; G) \xleftarrow{\beta} H^p(\mathcal{Z}; G) \xleftarrow{i^*} H^p(\mathcal{C}; G) \xleftarrow{\partial_0^*} H^p(\mathcal{D}; G) \xleftarrow{\beta} H^{p-1}(\mathcal{Z}; G)\,. \qquad (*)$$

To avoid confusion, we use β instead of δ^* to denote the zig-zag homomorphism [see Lemma 3.3.2].

Step 2. We now identify the terms of this sequence and the homomorphism β. Since the chain complex $\mathcal{D}$ has trivial boundary operators, so does the corresponding cochain complex. Therefore, the group $H^{p+1}(\mathcal{D}; G)$ equals the cochain group $\mathrm{Hom}\,(D_{p+1}, G) = \mathrm{Hom}\,(B_p, G)$. For similar reasons, we have $H^p(\mathcal{Z}; G) = \mathrm{Hom}\,(Z_p, G)$.

Thus β is a map

$$\mathrm{Hom}\,(B_p, G) \xleftarrow{\beta} \mathrm{Hom}\,(Z_p, G)\,.$$

We show that it is simply the dual $\widetilde{j}_p$ of the inclusion map $j_p : B_p \to Z_p$.

One obtains the zig-zag homomorphism β by tracing through the diagram

$$\begin{array}{ccc} & & \operatorname{Hom}(C_{p+1}, G) \xleftarrow{\widetilde{\partial}_0} \operatorname{Hom}(B_p, G) \\ & & \uparrow \delta \\ \operatorname{Hom}(Z_p, G) & \xleftarrow{\widetilde{\imath}} & \operatorname{Hom}(C_p, G) \end{array}$$

from left to right, as follows: Let f be an element of $\operatorname{Hom}(Z_p, G)$. Note that $\delta f = 0$, because all coboundary operations in $\mathcal{Z}$ vanish. Pull f back via $\widetilde{\imath}$ to an element g of $\operatorname{Hom}(C_p, G)$. [That is, extend f to a homomorphism $g : C_p \to G$.] Form δg, and pull it back via $\widetilde{\partial}_0$ to an element of $\operatorname{Hom}(B_p, G)$. We show that $\widetilde{\jmath}_p(f)$ is such a pull-back of δg. That is, we show that $\widetilde{\partial}_0 \widetilde{\jmath}_p(f) = \delta g$. Then our result is proved.

We compute

$$\langle \delta g, c_{p+1} \rangle = \langle g, \partial c_{p+1} \rangle = \langle f, \partial c_{p+1} \rangle ,$$

the last equation holding because ∂c_{p+1} is in Z_p. Similarly, we have

$$\langle \widetilde{\partial}_0 \widetilde{\jmath}(f), c_{p+1} \rangle = \langle f, j_p(\partial_0 c_{p+1}) \rangle = \langle f, \partial c_{p+1} \rangle .$$

Step 3. We now prove the existence of the exact sequence of the theorem. The five term exact sequence $(*)$ gives rise to the short exact sequence

$$0 \longleftarrow \ker \widetilde{\jmath}_p \longleftarrow H^p(\mathcal{C}; G) \longleftarrow \operatorname{cok} \widetilde{\jmath}_{p-1} \longleftarrow 0 . \qquad (**)$$

We need to identify the kernel and the cokernel of the map

$$\operatorname{Hom}(B_p, G) \xleftarrow{\widetilde{\jmath}_p} \operatorname{Hom}(Z_p, G) .$$

Now the sequence

$$0 \longrightarrow B_p \xrightarrow{j_p} Z_p \xrightarrow{\pi} H_p(\mathcal{C}) \longrightarrow 0$$

is a free resolution of $H_p(\mathcal{C})$. Therefore, the sequence

$$0 \leftarrow \operatorname{Ext}(H_p(\mathcal{C}), G) \leftarrow \operatorname{Hom}(B_p, G) \xleftarrow{\widetilde{\jmath}_p} \operatorname{Hom}(Z_p, G) \xleftarrow{\widetilde{\pi}} \operatorname{Hom}(H_p(\mathcal{C}), G) \leftarrow 0 \qquad (***)$$

is exact. The kernel and cokernel of $\widetilde{\jmath}_p$ are now apparent. The existence of the exact sequence of our theorem is thus established.

Step 4. We check naturality of the sequence. Let $\phi : \mathcal{C} \to \mathcal{C}'$ be a chain map. Let C'_p, Z'_p, and B'_p denote the groups of p-chains, p-cycles, and p-boundaries of $\mathcal{C}'$, respectively. The chain maps

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Z_p & \xrightarrow{i} & C_p & \xrightarrow{\partial_0} & B_{p-1} & \longrightarrow & 0 \\ & & \downarrow \phi & & \downarrow \phi & & \downarrow \phi & & \\ 0 & \longrightarrow & Z'_p & \xrightarrow{i'} & C'_p & \xrightarrow{\partial'_0} & B'_{p-1} & \longrightarrow & 0 \end{array}$$

define a homomorphism of short exact sequences, for ϕ commutes with i because i is inclusion, and ϕ commutes with ∂_0 because ϕ is a chain map. It follows that ϕ induces a homomorphism to the five-term exact sequence $(*)$ from the corresponding sequence for $\mathcal{C}'$, and hence induces a homomorphism to the short exact sequence $(**)$ from the corresponding sequence for $\mathcal{C}'$. It only remains to comment that the exact sequence $(***)$ is natural with respect to homomorphisms induced by chain maps, so the isomorphisms of $\ker \widetilde{j}_p$ and $\operatorname{cok} \widetilde{j}_{p-1}$ with the appropriate Hom and Ext groups are natural as well.

Step 5. To complete the proof, we show that the homomorphism

$$\operatorname{Hom}(H_p(\mathcal{C}), G) \longleftarrow H^p(\mathcal{C}; G)$$

of our exact sequence equals the Kronecker map κ. Then the splitting of the sequence is a consequence of Lemma 5.6.12.

The homomorphism in our exact sequence equals the composite $(\widetilde{\pi})^{-1} \circ i^*$ of the following diagram:

$$\begin{array}{ccc} \operatorname{Hom}(Z_p, G) \supset \ker \widetilde{j}_p & \xleftarrow{i^*} & H^p(\mathcal{C}; G) \\ \widetilde{\pi} \uparrow \cong & \swarrow \kappa & \\ \operatorname{Hom}(H_p(\mathcal{C}), G) & & \end{array}$$

where $\widetilde{j}_p$ is the dual of the inclusion map $j_p : B_p \to Z_p$, i^* is induced by the incusion $i : Z_p \to C_p$, and $\widetilde{\pi}$ is the dual of the projection $\pi : Z_p \to H_p(\mathcal{C})$. We need to show that the diagram commutes.

Let $\{z^p\}$ be an element of $H^p(\mathcal{C}; G)$. We show that

$$\widetilde{\pi}\kappa(\{z^p\}) = i^*(\{z^p\}) = \widetilde{\imath}(z^p)\,,$$

and the proof is complete. Let $z_p \in Z_p$, and compute

$$\begin{aligned} [\widetilde{\pi}\kappa(\{z^p\})](z_p) &= [\kappa(\{z^p\}](\pi(z_p)) = \langle \{z^p\}, \{z_p\} \rangle \\ &= \langle z^p, z_p \rangle = \langle z^p, i(z_p) \rangle = \langle \widetilde{\imath}(z^p), z_p \rangle\,. \end{aligned}$$ □

Corollary 7.3.2 *Let (X, A) be a topological pair. There is an exact sequence*

$$0 \longleftarrow \operatorname{Hom}(H_p(X, A), G) \longleftarrow H^p(X, A; G) \longleftarrow \operatorname{Ext}(H_{p-1}(X, A), G) \longleftarrow 0\,,$$

which is natural with respect to homomorphisms induced by continuous maps. It splits, but not naturally.

Corollary 7.3.3 *Let $\mathcal{C}$ and $\mathcal{D}$ be free chain complexes. Let $\phi : \mathcal{C} \to \mathcal{D}$ be a chain map. If $\phi_* : H_i(\mathcal{C}) \to H_i(\mathcal{D})$ is an isomorphism for $i = p$ and $i = p-1$, then*

$$H^p(\mathcal{C}; G) \xleftarrow{\phi^*} H^p(\mathcal{D}; G)$$

is an isomorphism.

Proof: Apply naturality of the universal coefficient sequence and the Steenrod five lemma [Lemma 3.3.4]. □

This last corollary provides an alternative proof of Theorem 5.6.6. In fact, it proves something more than is stated in that theorem, since we do not need to assume that ϕ_* is an isomorphism in *all* dimensions.

The Universal Coefficient Theorem with Field Coefficients

There is a second version of the universal coefficient theorem that is often useful. It concerns the case where the coefficient group is a field F. It relates cohomology with homology if *both* are taken with coefficients in F.

We have already remarked that if $\mathcal{C}$ is a chain complex and F is a field, then both

$$\text{Hom}\,(C_p, F) \quad \text{and} \quad C_p \otimes F$$

have, in a natural way, the structure of vector spaces over F. Given $\alpha \in F$, and $c^p \in \text{Hom}\,(C_p, F)$, and $c_p \in C_p$, one defines

$$\langle \alpha c^p, c_p \rangle = \alpha \cdot \langle c^p, c_p \rangle\,,$$
$$\alpha(c_p \otimes \beta) = c_p \otimes (\alpha\beta)\,.$$

[See the exercises of Section 5.9 and Section 6.3.] Both δ and ∂ are vector space homomorphisms [i.e., linear transformations], so that

$$H^p(\mathcal{C}; F) \quad \text{and} \quad H_p(\mathcal{C}; F)$$

also have the structure of vector spaces over F. We show that they are in fact dual vector spaces.

We recall that if A and B are vector spaces over F, then $\text{Hom}_F(A, B)$ is also a vector space over F. It is the set of linear transformations of A into B. If $f : A \to A'$ and $g : B' \to B$ are linear transformations, then $\text{Hom}\,(f, g)$ is also a linear transformation. [See the exercises of Section 5.2.]

We wish to reprove the universal coefficient theorem for cohomology in this context. One difficulty is that we have several possible contenders for the title of the space of cochains—namely,

$$\text{Hom}\,(C_p, F)\,, \quad \text{Hom}\,(C_p \otimes F, F)\,, \quad \text{Hom}_F(C_p \otimes F, F)\,.$$

By definition, the first of these gives the cohomology $H^p(\mathcal{C}; F)$ of $\mathcal{C}$ with coefficients in F. We shall show that the third does, as well.

Lemma 7.3.4 *Let $\mathcal{C}$ be a chain complex. Let*

$$\omega : \text{Hom}\,(C_p, F) \to \text{Hom}_F(C_p \otimes F, F)$$

be defined by the equation

$$\langle \omega(f), c_p \otimes \alpha \rangle = \langle f, c_p \rangle \cdot \alpha\,, \qquad (\dagger)$$

where $f \in \text{Hom}\,(C_p, F)$, and $c_p \in C_p$, and $\alpha \in F$. Then ω is a vector space isomorphism that commutes with δ.

Remark 7.3.5 Strictly speaking, the formula (†) defines $\omega(f)$ as a function on the cartesian product $C_p \times F$. One then observes that the function is bilinear, thus it does define a function on $C_p \otimes F$. Often this step is not mentioned.

Proof: To check that $\omega(f)$ is a linear transformation, we compute

$$\begin{aligned}\langle \omega(f), \alpha(c_p \otimes \beta)\rangle &= \langle \omega(f), c_p \otimes \alpha\beta\rangle = \langle f, c_p\rangle \cdot (\alpha\beta) \\ &= \alpha \cdot (\langle f, c_p\rangle \cdot \beta) = \alpha \cdot \langle \omega(f), c_p \otimes \beta\rangle\,.\end{aligned}$$

We show that the map ω is injective. Suppose $\omega(f)$ is the zero linear transformation. Then in particular

$$\langle \omega(f), c_p \otimes 1\rangle = 0 = \langle f, c_p\rangle \cdot 1$$

for all $c_p \in C_p$. This implies that f is the zero homomorphism.

We show that the map ω is surjective. Let $\phi : C_p \otimes F \to F$ be a linear transformation. Let us define $f : C_p \to F$ by the equation $f(c_p) = \phi(c_p \otimes 1)$. It follows that f is a homomorphism of abelian groups, because $f(0) = 0$ and

$$\begin{aligned}f(c_p + d_p) &= \phi((c_p + d_p) \otimes 1) \\ &= \phi(c_p \otimes 1 + d_p \otimes 1) = f(c_p) + f(d_p)\,.\end{aligned}$$

Furthermore, $\omega(f) = \phi$ holds, since

$$\begin{aligned}\langle \omega(f), c_p \otimes \alpha\rangle &= \langle f, c_p\rangle \cdot \alpha = \alpha \cdot \phi(c_p \otimes 1)\,, \\ \phi(c_p \otimes \alpha) &= \phi(\alpha \cdot (c_p \otimes 1)) = \alpha \cdot \phi(c_p \otimes 1)\,.\end{aligned}$$

This last equation holds *because* ϕ is a linear transformation, not just a homomorphism of abelian groups.

We show the map ω commutes with δ. We compute

$$\begin{aligned}\langle \delta\omega(f), c_{p+1} \otimes \alpha\rangle &= \langle \omega(f), (\partial \otimes i_F)(c_{p+1} \otimes \alpha)\rangle \\ &= \langle f, \partial c_{p+1}\rangle \cdot \alpha = \langle \delta f, c_{p+1}\rangle \cdot \alpha \\ &= \langle \omega(\delta f), c_{p+1} \otimes \alpha\rangle\,.\end{aligned}$$

□

Theorem 7.3.6 (Universal coefficient theorem with field coefficients) *Let $\mathcal{C}$ be a free chain complex, and let F be a field. Then there is a natural vector space isomorphism*

$$\mathrm{Hom}_F(H_p(\mathcal{C}; F), F) \longleftarrow H^p(\mathcal{C}; F)\,.$$

Proof: We imitate the proof of the universal coefficient theorem. First, we note that if

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

is a short exact sequence of vector spaces over F and linear transformations, then for any vector space V over F, the dual sequence

$$0 \longleftarrow \mathrm{Hom}_F(A, V) \longleftarrow \mathrm{Hom}_F(B, V) \longleftarrow \mathrm{Hom}_F(C, V) \longleftarrow 0$$

is exact. The proof is easy, for since any vector space has a basis, the first sequence splits.

Let $\mathcal{E}$ denote the chain complex $\mathcal{C} \otimes F$. Then $E_p = C_p \otimes F$ is a vector space over F. Let B_p and Z_p denote boundaries and cycles, respectively, in the chain complex $\mathcal{E}$. They are also vector spaces over F. Consider the short exact sequence of vector spaces

$$0 \to Z_p \longrightarrow E_p \longrightarrow B_{p-1} \longrightarrow 0\,.$$

It gives rise to the dual sequence

$$0 \longleftarrow \mathrm{Hom}_F(Z_p, F) \longleftarrow \mathrm{Hom}_F(E_p, F) \longleftarrow \mathrm{Hom}_F(B_{p-1}, F) \longleftarrow 0\,.$$

We apply the zig-zag lemma [Lemma 3.3.2], as before. The cochain complex in the middle is isomorphic to the cochain complex Hom (C_p, F), by the preceding lemma. Therefore, by the same argument as used before, we obtain an exact sequence

$$0 \longleftarrow \ker \widetilde{j}_p \longleftarrow H^p(\mathcal{C}; F) \longleftarrow \mathrm{cok}\ \widetilde{j}_{p-1} \longleftarrow 0\,,$$

where $j_p : B_p \to Z_p$ is inclusion.

Now the proof takes a different tack. Consider the exact sequence

$$0 \longrightarrow B_p \xrightarrow{j_p} Z_p \longrightarrow H_p(\mathcal{E}) \longrightarrow 0\,.$$

Because it is a sequence of vector spaces and linear transformations, the dual sequence is exact:

$$0 \longleftarrow \mathrm{Hom}_F(B_p, F) \xleftarrow{\widetilde{j}_p} \mathrm{Hom}_F(Z_p, F) \longleftarrow \mathrm{Hom}_F(H_p(\mathcal{E}), F) \longleftarrow 0\,.$$

Therefore $\mathrm{cok}\ \widetilde{j}_p = 0$ and

$$\ker \widetilde{j}_p \cong \mathrm{Hom}_F(H_p(\mathcal{C}; F), F)\,,$$

proving the theorem. □

Corollary 7.3.7 *If (X, A) is a topological pair, then there is a natural vector space isomorphism*

$$\mathrm{Hom}_F(H_p(X, A; F), F) \longleftarrow H^p(X, A; F)\,.$$

Theorem 7.3.6 shows that if F is a field, then the vector space $H^p(X, A; F)$ can be identified in a natural way with the *dual* vector space of the vector space $H_p(X, A; F)$. In the case where the dimension of $H_p(X, A; F)$ is finite,

this means that $H^p(X, A; F)$ and $H_p(X, A; F)$ are isomorphic as vector spaces [though not naturally].

In differential geometry, it is common to deal with compact manifolds and to use the field of reals as coefficients. Because the cohomology and homology vector spaces are dual in this case, differential geometers sometimes treat homology and cohomology as if they were the same object. Unfortunately, this practice can lead to confusion.

The next definition is used in Exercise 4.

Definition 7.3.8 A homomorphism $\alpha : G \to G'$ gives rise to homomorphisms

$$\alpha_* : H_i(\mathcal{C}; G) \to H_i(\mathcal{C}; G'),$$
$$\alpha^* : H^i(\mathcal{C}; G) \to H^i(\mathcal{C}; G').$$

They are called *coefficient homomorphisms*.

EXERCISES

1. Let $\mathcal{C}$ be a free chain complex. Show that if $H_{p-1}(\mathcal{C})$ is free or if G is divisible [see Definition 5.2.9], then the Kronecker map κ is an isomorphism.

2. Use the universal coefficient theorem to compute the cohomology with general coefficients G of $T \# T$, $\mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$, $\mathbb{P}^N$, and the k-fold dunce cap.

3. Assume $H_i(X)$ is finitely generated for all i.

(a) Compute $H^p(X; G)$ in terms of the betti numbers and torsion coefficients of X. Compare with Exercise 5 of Section 5.3.

(b) Repeat **(a)** when G is the circle group $\mathbb{S}^1 = \mathbb{R}/\mathbb{Z}$.

(c) Repeat **(a)** when $G = \mathbb{Q}$.

4. Show that the universal coefficient sequence for cohomology is natural with respect to coefficient homomorphisms.

5. Let (K, K_0) be a relative pseudo n-manifold. [See Definition 5.4.5.] Let K be finite. Show that the torsion subgroup of $H_{n-1}(K, K_0)$ vanishes if (K, K_0) is orientable, and has order 2 otherwise.

6. Explain why the splitting in Theorem 7.3.1 is not natural.

7. Explain why the splitting in Corollary 7.3.2 is not natural.

7.4 Torsion Products

Associated with the functor Hom and derived from it is another functor, called Ext. Both were involved in the statement of the universal coefficient theorem for cohomology. Similarly, associated with the tensor product functor and derived from it is a second functor, which we call the torsion product. Both of *these* functors will be involved in the statement of the universal coefficient theorem for homology. Construction of the torsion product is so similar to construction of the Ext functor that we abbreviate some of the details.

The torsion product is a functor that assigns to an ordered pair A, B of abelian groups an abelian group $A * B$, and to an ordered pair of homomorphisms $\gamma : A \to A'$ and $\delta : B \to B'$ a homomorphism

$$\gamma * \delta : A * B \to A' * B' .$$

Like the tensor product, it is covariant in both variables.

The crucial property of the torsion product is expressed in the following theorem, the proof of which we shall give later:

Theorem 7.4.1 *There is a function that assigns to each free resolution*

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$$

of the abelian group A, and to each abelian group B, an exact sequence

$$0 \longrightarrow A * B \longrightarrow R \otimes B \longrightarrow F \otimes B \longrightarrow A \otimes B \longrightarrow 0 .$$

This function is natural, in the sense that a homomorphism of a free resolution of A to a free resolution of A', and a homomorphism of B to B', induce a homomorphism of the corresponding exact sequences.

First we prove a lemma.

Lemma 7.4.2 *Given homomorphism of free resolutions*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \xrightarrow{\phi} & F & \xrightarrow{\psi} & A & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & R' & \xrightarrow{\phi'} & F' & \xrightarrow{\psi'} & A' & \longrightarrow & 0 \end{array}$$

and a homomorphism $\delta : B \to B'$, there exists a unique homomorphism

$$\epsilon : \ker(\phi \otimes i_B) \to \ker(\phi' \otimes i_{B'}$$

making the following diagram commute:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \ker(\phi \otimes i_B) & \longrightarrow & R \otimes B & \xrightarrow{\phi \otimes i_B} & F \otimes B & \xrightarrow{\psi \otimes i_B} & A \otimes B \longrightarrow 0 \\ & & \downarrow \epsilon & & \downarrow \alpha \otimes \delta & & \downarrow \beta \otimes \delta & & \downarrow \gamma \otimes \delta \\ 0 & \longrightarrow & \ker(\phi' \otimes i_{B'}) & \longrightarrow & R' \otimes B' & \xrightarrow{\phi' \otimes i_{B'}} & F' \otimes B' & \xrightarrow{\psi' \otimes i_{B'}} & A' \otimes B' \longrightarrow 0 \end{array}$$

The homomorphism ϵ is independent of the choice of α and β.

Proof: Functoriality of $\otimes$ shows that the two right-hand squares of the preceding diagram commute. Therefore, $\alpha \otimes \delta$ induces a homomorphism ϵ of kernels.

The proof that ϵ is independent of the choice of α and β proceeds as in the proof of Lemma 7.2.2. Let $\{\alpha', \beta', \gamma\}$ be another choice. We treat $\{\alpha, \beta, \gamma\}$ and $\{\alpha', \beta', \gamma\}$ as chain maps of one chain complex $\mathcal{A}$ to another $\mathcal{A}'$. There is a chain homotopy between the corresponding chain maps of $\mathcal{A} \otimes B$ to $\mathcal{A}' \otimes B'$. Thus they induce the same homomorphism

$$\epsilon : H_2(\mathcal{A}; B) \longrightarrow H_2(\mathcal{A}'; B')\,,$$

and we recognize $H_2(\mathcal{A}; B) = \ker(\phi \otimes i_B)$ while $H_2(\mathcal{A}'; B') = \ker(\phi' \otimes i_{B'})$. □

The homomorphism ϵ is said to be *induced* by γ and δ, relative to the free resolutions involved.

Just as in Section 7.2, we see that ϵ depends functorially on γ and δ. That is, the composite of the homomorphisms induced by (γ, δ) and by (γ', δ') is the homomorphism induced by $(\gamma \circ \gamma', \delta \circ \delta')$. Also the homomorphism induced by (i_A, i_B) is an isomorphism.

Definition 7.4.3 Given A, let

$$0 \longrightarrow R(A) \xrightarrow{\phi} F(A) \longrightarrow A \longrightarrow 0$$

be the canonical free resolution of A. The group $\ker(\phi \otimes i_B)$ is denoted $A * B$, and is called the *torsion product* of A and B. If $\gamma : A \to A'$ and $\delta : B \to B'$ are homomorphisms, then we extend γ to a homomorphism of canonical free resolutions, and define

$$\gamma * \delta : A * B \to A' * B'$$

to be the homomorphism induced by γ and δ relative to these free resolutions.

The preceding remarks show that the torsion product is a functor of two variables, covariant in both.

Proof of Theorem 7.4.1: In fact the proof is now straightforward. Given any free resolution

$$0 \longrightarrow R \xrightarrow{\phi} F \longrightarrow A \longrightarrow 0\,,$$

the preceding remarks show that (i_A, i_B) induces an isomorphism of $\ker(\phi \otimes i_B)$ with $A * B$. Thus the exact sequence of the theorem exists. Naturality is proved as in the proof of Theorem 7.2.1. □

One property possessed by the torsion product that is not possessed by Ext is commutativity. To prove it, we need the following lemma.

Lemma 7.4.4 *There is a function assigning to each short exact sequence of abelian groups*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

and each abelian group D, an exact sequence

$$0 \longrightarrow D * A \longrightarrow D * B \longrightarrow D * C \longrightarrow D \otimes A \longrightarrow D \otimes B \longrightarrow D \otimes C \longrightarrow 0\,.$$

This function is natural with respect to homomorphisms of short exact sequences and abelian groups.

Proof: This result is analogous to Theorem 7.2.6. Let

$$0 \longrightarrow R \xrightarrow{\phi} F \longrightarrow D \longrightarrow 0$$

be a free resolution of D. Because R and F are free, we have horizontal exactness in the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R \otimes A & \longrightarrow & R \otimes B & \longrightarrow & R \otimes C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \phi \otimes i_A} & & \downarrow{\scriptstyle \phi \otimes i_B} & & \downarrow{\scriptstyle \phi \otimes i_C} & & \\ 0 & \longrightarrow & F \otimes A & \longrightarrow & F \otimes B & \longrightarrow & F \otimes C & \longrightarrow & 0 \end{array}$$

[See Corollary 6.2.9.] Treating the diagram as a short exact sequence of chain complexes, we apply the zig-zag lemma [Lemma 3.3.2] to obtain an exact sequence in homology that has the form

$$0 \to \ker(\phi \otimes i_A) \to \ker(\phi \otimes i_B) \to \ker(\phi \otimes i_C) \to \operatorname{cok}(\phi \otimes i_A) \to \operatorname{cok}(\phi \otimes i_B) \to \operatorname{cok}(\phi \otimes i_C) \to 0\,.$$

[This result is an instance of the serpent lemma, Lemma 3.3.9.] Theorem 7.4.1 enables us to identify these terms as follows:

$$\ker(\phi \otimes i_A) \cong D * A \quad \text{and} \quad \operatorname{cok}(\phi \otimes i_A) \cong D \otimes A\,,$$

and similar results hold for B and C.

The naturality of the zig-zag lemma [Lemma 3.3.2] and the naturality of the sequence in Theorem 7.4.1 together give us naturality of this exact sequence. □

Theorem 7.4.5 **(a)** *There is a natural isomorphism*

$$A * B \cong B * A\,.$$

(b) *There are natural isomorphisms*

$$(\bigoplus A_\alpha) * B \cong \bigoplus (A_\alpha * B)\,,$$
$$A * (\bigoplus B_\alpha) \cong \bigoplus (A * B_\alpha)\,.$$

(c) $A * B = 0$ *if* A *or* B *is torsion-free.*

(d) *Given* B*, there is an exact sequence*

$$0 \longrightarrow (\mathbb{Z}/m) * B \longrightarrow B \xrightarrow{\phi_m} B \longrightarrow (\mathbb{Z}/m) \otimes B \longrightarrow 0\,,$$

where ϕ_m *is multiplication by* m*.*

Proof: Note first that if B is torsion-free, then when one tensors a free resolution of A with B, exactness is preserved, by Corollary 6.2.9. It follows that $A * B = 0$.

(a) By applying the preceding lemma to the free resolution

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0\,,$$

one obtains a six-term exact sequence. The first terms

$$0 \longrightarrow B * R \longrightarrow B * F$$

vanish because R and F are torsion-free. What remains is the exact sequence

$$0 \longrightarrow B * A \longrightarrow B \otimes R \longrightarrow B \otimes F \longrightarrow B \otimes A \longrightarrow 0\,.$$

In its last three terms, this sequence is naturally isomorphic to the sequence of Theorem 7.4.1,

$$0 \longrightarrow A * B \longrightarrow R \otimes B \longrightarrow F \otimes B \longrightarrow A \otimes B \longrightarrow 0\,.$$

Therefore, the first terms are isomorphic as well. Naturality is straightforward.

(b) Let

$$0 \longrightarrow R_\alpha \longrightarrow F_\alpha \longrightarrow A_\alpha \longrightarrow 0 \qquad (*)$$

be a free resolution of A_α. Then

$$0 \longrightarrow \bigoplus R_\alpha \longrightarrow \bigoplus F_\alpha \longrightarrow \bigoplus A_\alpha \longrightarrow 0 \qquad (**)$$

is a free resolution of $\bigoplus A_\alpha$. We tensor the sequence in $(*)$ with B and sum to obtain

$$0 \longrightarrow \bigoplus (A_\alpha * B) \longrightarrow \bigoplus (R_\alpha \otimes B) \longrightarrow \bigoplus (A_\alpha \times B) \longrightarrow 0\,. \qquad (\dagger)$$

Likewise, we tensor the sequence in $(**)$ with B to obtain

$$0 \longrightarrow (\bigoplus A_\alpha) * B \longrightarrow (\bigoplus R_\alpha) \otimes B \longrightarrow (\bigoplus F_\alpha) \otimes B \longrightarrow (\bigoplus A_\alpha) \otimes B \longrightarrow 0\,. \quad (\ddagger)$$

Since the last three terms of the sequences in (†) and (‡) are naturally isomorphic, the first terms are isomorphic as well. Naturality is straightforward. This proves the first isomorphism of **(b)**, and the second follows from commutativity.

(c) Part **(c)** follows from the remark made at the beginning of this proof and commutativity.

(d) We begin with the free resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\phi_m} \mathbb{Z} \longrightarrow \mathbb{Z}/m \longrightarrow 0\,.$$

Tensoring with B, we obtain

$$0 \longrightarrow (\mathbb{Z}/m) * B \longrightarrow \mathbb{Z} \otimes B \xrightarrow{\phi_m \otimes i_B} \mathbb{Z} \otimes B \longrightarrow (\mathbb{Z}/m) \otimes B \longrightarrow 0\,,$$

from which the sequence in **(d)** follows. □

Remark 7.4.6 We have studied the Ext functor and the torsion product functor only for abelian groups. They are derived from the functors Hom and $\otimes$, respectively. If, instead of abelian groups, one deals with modules over a ring R, then one has the functors Hom_R and $\otimes_R$ [mentioned in Section 6.2]. It is natural to ask whether one can also introduce the Ext functor and the torsion product functor for modules. The answers to these questions are a large part of the subject matter of homological algebra. [See [ML63].]

Much as in the case of abelian groups, one begins with the notion of a free R-module. Then, given an R-module A, one lets F_0 denote the free module generated by the elements of A. There is a natural epimorphism

$$\phi : F_0 \to A\,.$$

The difficulty that immediately arises is that in the case of a general ring R, a submodule of a free module is *not necessarily free.* Thus there need be no short exact sequence

$$0 \longrightarrow S \longrightarrow F \longrightarrow A \longrightarrow 0\,,$$

where S and F are free. Nevertheless, one *can* let F_1 be the free R-module generated by the elements of $\ker \phi$ and thereby obtain an exact sequence

$$F_1 \longrightarrow F_0 \longrightarrow A \longrightarrow 0\,,$$

where F_1 and F_0 are free. Continuing in this way, one obtains an exact sequence

$$\cdots \xrightarrow{\phi_{k+1}} F_k \xrightarrow{\phi_k} F_{k-1} \xrightarrow{\phi_{k-1}} \cdots \xrightarrow{\phi_1} F_0 \longrightarrow A \longrightarrow 0\,,$$

where each F_i is free. Such a sequence is called a *free resolution* of the R-module A. Applying the functor Hom_R, one obtains the sequence

$$\cdots \xleftarrow{\widetilde{\phi}_2} \mathrm{Hom}_R(F_1, G) \xleftarrow{\widetilde{\phi}_1} \mathrm{Hom}_R(F_0, G) \longleftarrow \mathrm{Hom}_R(A, G) \longleftarrow 0\,.$$

Exactness holds for the two right-hand terms. For $n \geq 1$, we define

$$\mathrm{Ext}^n_R(A, G) = \ker \widetilde{\phi}_{n+1} / \mathrm{im}\ \widetilde{\phi}_n\,.$$

Thus we obtain a sequence of Ext groups. If $R = \mathbb{Z}$, then Ext^n_R vanishes for $n > 1$ and equals Ext for $n = 1$.

A similar construction, using $\otimes_R$, gives one a sequence of groups $\mathrm{Tor}^R_n(A, G)$. Many of the theorems we have proved for Ext and the torsion product generalize to statements about these new groups. There are numerous applications to both topology and algebra.

The theorems of this section enable us to compute tensor and torsion products when the groups are finitely generated. Below we summarize the rules for these products, as well as those for Hom and Ext [as usual ϕ_m denotes multiplication by m]:

$\mathbb{Z} \otimes G \cong G$	$\mathbb{Z} * G = 0$
$\mathrm{Hom}\,(\mathbb{Z}, G) \cong G$	$\mathrm{Ext}\,(\mathbb{Z}, G) = 0$
$\mathbb{Z}/m \otimes G \cong G/mG$	$\mathbb{Z}/m * G \cong \ker(G \xrightarrow{\phi_m} G)$
$\mathrm{Hom}\,(\mathbb{Z}/m, G) \cong \ker(G \xrightarrow{\phi_m} G)$	$\mathrm{Ext}\,(\mathbb{Z}/m, G) \cong G/mG$

The above rules imply the following special cases, where $d = \gcd(m, n)$:

$\mathbb{Z}/m \otimes \mathbb{Z} \cong \mathbb{Z}/m$	$\mathbb{Z}/m * \mathbb{Z} = 0$
$\mathrm{Hom}\,(\mathbb{Z}/m, \mathbb{Z}) = 0$	$\mathrm{Ext}\,(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m$
$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/m * \mathbb{Z}/n \cong \mathrm{Hom}\,(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathrm{Ext}\,(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/d$	

EXERCISES

1. Compute $A \otimes B$ and $A * B$ if

$$A = \mathbb{Z} \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/6\,, \qquad B = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}/9 \oplus \mathbb{Z}/12\,.$$

2. Let A be finitely generated.

(a) If $\mathbb{S}^1$ is the circle group, compute $A \otimes \mathbb{S}^1$ and $A * \mathbb{S}^1$ in terms of the betti numbers and torsion coefficients of A.

(b) Repeat **(a)** with $\mathbb{S}^1$ replaced by $\mathbb{Q}$.

3. Suppose that $\phi : A \to B$ is a monomorphism. Show that

$$A * C \xrightarrow{\phi * i_C} B * C$$

is also a monomorphism. Conclude that if $A' \subset A$ and $B' \subset B$, then $A' * B'$ can be naturally considered as a subgroup of $A * B$.

4. Let T_A and T_B be the torsion subgroups of A and B, respectively. Use Lemma 7.4.4 to show that inclusion induces an isomorphism

$$A * T_B \longrightarrow A * B\,.$$

Conclude that $A * B \cong T_A * T_B$.

5. Show that, in general, $A * B$ is a torsion group. [**Hint:** Consider the sequence

$$0 \longrightarrow A * T_B \longrightarrow R \otimes T_B \longrightarrow F \otimes T_B \longrightarrow A \otimes T_B \longrightarrow 0\,.]$$

6. Prove naturality in Theorem 7.4.5(a) and (b).

7.5 The Universal Coefficient Theorem for Homology

As with the cohomology group $H^p(X;G)$, there is a theorem expressing the homology group $H_p(X;G)$ in terms of the homology groups $H_p(X)$ and $H_{p-1}(X)$. The statements of the two theorems are similar, except that here the tensor product and the torsion product replace the Hom and Ext functors, and the arrows are reversed.

Theorem 7.5.1 (The universal coefficient theorem for homology) *Let $\mathcal{C}$ be a free chain complex, and let G be an abelian group. There is an exact sequence*

$$0 \longrightarrow H_p(\mathcal{C}) \otimes G \longrightarrow H_p(\mathcal{C};G) \longrightarrow H_{p-1}(\mathcal{C}) * G \longrightarrow 0\,,$$

that is natural with respect to homomorphisms induced by chain maps. It splits, but not naturally.

One can give a direct proof of this theorem that is very similar to the proof of the universal coefficient theorem for cohomology. Instead, we shall derive the theorem from the more general Künneth theorem that we shall prove in Section 7.8.

Corollary 7.5.2 *Let (X, A) be a topological pair. Then there is an exact sequence*

$$0 \longrightarrow H_p(X, A) \otimes G \longrightarrow H_p(X, A; G) \longrightarrow H_{p-1}(X, A) * G \longrightarrow 0\,,$$

that is natural with respect to homomorphisms induced by chain maps. It splits, but not naturally.

Corollary 7.5.3 *Let $\mathcal{C}$ and $\mathcal{D}$ be free chain complexes, and let $\phi : \mathcal{C} \to \mathcal{D}$ be a chain map. If $\phi_* : H_i(\mathcal{C}) \to H_i(\mathcal{D})$ is an isomorphism for $i = p$ and $i = p-1$, then*

$$\phi_* : H_p(\mathcal{C}; G) \to H_p(\mathcal{D}; G)$$

is an isomorphism for arbitrary G.

Proof: This result follows from naturality of the universal coefficient sequence and the Steenrod five lemma [Lemma 3.3.4]. □

This last corollary provides an alternative proof of Theorem 6.3.1.

EXERCISES

1. Let $\mathcal{C}$ be a free chain complex. Show that if $H_{p-1}(\mathcal{C})$ or G is torsion-free, then $H_p(\mathcal{C}) \otimes G \cong H_p(\mathcal{C}; G)$.

2. Use the universal coefficient theorem to compute the homology, with general coefficients G, of the following spaces:

- **(a)** $T \# T$.
- **(b)** $\mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$.
- **(b)** $\mathbb{P}^N$.
- **(c)** the k-fold dunce cap.

Compare with Exercise 2 of Section 6.3.

3. Assume that $H_i(X)$ is finitely generated for all i.

- **(a)** Compute $H_p(X; G)$ in terms of the betti numbers and torsion coefficients of X.
- **(b)** Repeat **(a)** when G is the circle group $\mathbb{S}^1$.
- **(c)** Repeat **(a)** when $G = \mathbb{Q}$.

4. Let $\mathcal{C}$ be a free chain complex. Show that if $H_i(\mathcal{C})$ is finitely generated for all i, then

$$H_p(\mathcal{C}; \mathbb{Z}/n) \cong H^p(\mathcal{C}; \mathbb{Z}/n)\,.$$

[The isomorphism is not natural.]

5. Show that if K is a finite complex, then the homology group $H_n(K)$ is uniquely determined by the groups $H_i(K; \mathbb{Z}/p^k)$, as p^k ranges over all prime powers and i ranges from 0 to n.

6. Show that if K is a finite complex, then the homology group $H_p(K)$ is uniquely determined by the groups $H_p(K; \mathbb{S}^1)$ and $H_{p+1}(K; \mathbb{S}^1)$.

7. Show that if T is the torus and S is the Klein bottle, and if $f : T \to S$, then

$$f_* : H_2(T; \mathbb{Z}/2) \to H_2(S; \mathbb{Z}/2)$$

is trivial.

8. Show by example that the splitting of the sequence in Corollary 7.5.2 is not natural.

***9.** Prove Theorem 7.5.1. [**Hint:** Let $\mathcal{Z}$, $\mathcal{C}$, and $\mathcal{D}$ be as in the proof of Theorem 7.3.1. One has the exact sequence

$$\cdots \longrightarrow H_{p+1}(\mathcal{D}; G) \xrightarrow{\beta} H_p(\mathcal{Z}; G) \longrightarrow H_p(\mathcal{C}; G) \longrightarrow H_p(\mathcal{D}; G) \longrightarrow \cdots\,.$$

Show that β is induced by the inclusion $j_p : B_p \to Z_p$.]

10. Show that the splitting in Theorem 7.5.1 is not natural.

7.6 *Other Universal Coefficient Theorems

We have stated two universal coefficient theorems that express the homology and cohomology groups with arbitrary coefficients in terms of homology with integer coefficients. One might ask whether for this purpose cohomology and homology are symmetric in some sense—that is, whether these same groups can be expressed in terms of *cohomology* with integer coefficients. In fact, there are two short exact sequences that do this. They do not hold in complete generality, but apply only if the homology groups $H_i(\mathcal{C})$ are finitely generated. We consider these theorems now. We will be using the results of Section 5.7.

The exact sequences in question are derived by reindexing the groups of the cochain complex $\{\mathrm{Hom}\,(C_i, \mathbb{Z}), \delta\}$ so as to make it into a chain complex, and then applying the universal coefficient theorems we have already considered. The difficulty with this approach is that the group $\mathrm{Hom}\,(C_i, \mathbb{Z})$ is not in general free, so some restrictive hypotheses are needed.

Theorem 7.6.1 *Let $\mathcal{C}$ be a free chain complex, and let G be an abelian group. If $\mathcal{C}$ is finitely generated in each dimension, then there are exact sequences*

$$0 \longleftarrow \mathrm{Hom}\,(H^p(\mathcal{C}), G) \longleftarrow H_p(\mathcal{C}; G) \longleftarrow \mathrm{Ext}\,(H^{p+1}(\mathcal{C}), G) \longleftarrow 0\,,$$
$$0 \longrightarrow H^p(\mathcal{C}) \otimes G \longrightarrow H^p(\mathcal{C}; G) \longrightarrow H^{p+1}(\mathcal{C}) * G \longrightarrow 0\,,$$

that are natural with respect to homomorphisms induced by chain maps. They split, but not naturally.

Proof: Step 1. Since C_p is free and finitely generated, the cochain complex $\{\mathrm{Hom}\,(C_p, \mathbb{Z}), \delta\}$ is free. We define a chain complex $\mathcal{E}$ by letting

$$E_p = \mathrm{Hom}\,(C_{-p}, \mathbb{Z})$$

and taking as the boundary operator ∂_E in $\mathcal{E}$ the usual coboundary operator δ of $\mathcal{C}$. One then has the universal coefficient sequences

$$0 \longleftarrow \mathrm{Hom}\,(H_{-p}(\mathcal{E}), G) \longleftarrow H^{-p}(\mathcal{E}; G) \longleftarrow \mathrm{Ext}\,(H_{-p-1}(\mathcal{E}), G) \longleftarrow 0\,,$$
$$0 \longrightarrow H_{-p}(\mathcal{E}) \otimes G \longrightarrow H_{-p}(\mathcal{E}; G) \longrightarrow H_{-p-1}(\mathcal{E}) * G \longrightarrow 0\,,$$

where for convenience we have replaced p by $-p$ throughout. It is immediate from the definition that $H_{-p}(\mathcal{E}) = H^p(\mathcal{C})$. We shall show that there are natural isomorphisms

$$H_{-p}(\mathcal{E}; G) \cong H^p(\mathcal{C}; G)\,, \qquad (*)$$

$$H^{-p}(\mathcal{E}; G) \cong H_p(\mathcal{C}; G)\,, \qquad (**)$$

from which the theorem follows.

Step 2. First we show that, if A is free of finite rank, then there is a natural isomorphism

$$\Phi : \mathrm{Hom}\,(A, \mathbb{Z}) \otimes G \to \mathrm{Hom}\,(A, G)\,.$$

Initially, we define Φ on pairs (ϕ, g) in the cartesian product of $\mathrm{Hom}\,(A, \mathbb{Z})$ and G by the equation

$$[\Phi(\phi, g)](a) = \phi(a) \cdot g\,.$$

After observing that $\Phi(\phi, g)$ is indeed an element of $\mathrm{Hom}\,(A, G)$, we then note that Φ is bilinear. Thus Φ can be considered a homomorphism of $\mathrm{Hom}\,(A, \mathbb{Z}) \otimes G$ into $\mathrm{Hom}\,(A, G)$.

Now, let $a_1, a_2, \ldots, a_k$ be a basis for A, and let $\widetilde{a}_1, \widetilde{a}_2, \ldots, \widetilde{a}_k$ be the dual basis for $\mathrm{Hom}\,(A, \mathbb{Z})$ defined by

$$\widetilde{a}_i(a_j) = \delta_{ij}\,,$$

where δ_{ij} is the Kronecker delta function. The group $\mathrm{Hom}\,(A, \mathbb{Z}) \otimes G$ is a direct sum of k copies of G, the ith copy consisting of all elements of the form

$\widetilde{a}_i \otimes g$, for $g \in G$. The group $\mathrm{Hom}\,(A, G)$ is also a direct sum of k copies of G. In this case, the ith copy consists of all homomorphisms that vanish on a_j for $j \neq i$. These two copies of G are isomorphic under Φ, since $\Phi(\widetilde{a}_i \otimes g)$ is the homomorphism of A into G that maps a_i to g and maps a_j to 0 for $j \neq i$.

Step 3. To prove $(*)$, we recall that $E_{-p} = \mathrm{Hom}\,(C_p, \mathbb{Z})$ and apply **Step 2** to obtain an isomorphism

$$E_{-p} \otimes G \cong \mathrm{Hom}\,(C_p, G)\,.$$

The naturality of the isomorphism means that $\partial_E \otimes i_G$ corresponds to δ. Thus we have $H_{-p}(\mathcal{E}; G) \cong H^p(\mathcal{C}; G)$, as desired.

Step 4. We show that, if A is free of finite rank, then there is a natural isomorphism

$$e : A \to \mathrm{Hom}\,(\mathrm{Hom}\,(A, \mathbb{Z}), \mathbb{Z})\,,$$

defined by the equation

$$[e(a)](\phi) = \phi(a)\,,$$

where $a \in A$ and $\phi \in \mathrm{Hom}\,(A, \mathbb{Z})$.

First, we leave it to you to check that $e(a)$ is a homomorphism of $\mathrm{Hom}\,(A, \mathbb{Z})$ into $\mathbb{Z}$, and then to check that e itself is a homomorphism. To prove that e is an isomorphism, choose a basis $a_1, a_2, \ldots, a_k$ for A, and let $\widetilde{a}_1, \widetilde{a}_2, \ldots, \widetilde{a}_k$ be the dual basis for $\mathrm{Hom}\,(A, \mathbb{Z})$, as above in **Step 2**. Let the corresponding dual basis for $\mathrm{Hom}\,(\mathrm{Hom}\,(A, \mathbb{Z}), \mathbb{Z})$ be $\widetilde{\widetilde{a}}_1, \widetilde{\widetilde{a}}_2, \ldots, \widetilde{\widetilde{a}}_k$. All the groups involved are free abelian of rank k. The homomorphism e carries a_i to $\widetilde{\widetilde{a}}_i$, since for all j, we have

$$[e(a_i)](\widetilde{a}_j) = \widetilde{a}_j(a_i) = \delta_{ij} = \widetilde{\widetilde{a}}_i(\widetilde{a}_j)\,.$$

Step 5. To prove $(**)$, we apply **Step 4**, then **Step 2**, to define natural isomorphisms

$$A \otimes G \longrightarrow \mathrm{Hom}\,(\mathrm{Hom}\,(A, \mathbb{Z}), \mathbb{Z}) \otimes G \longrightarrow \mathrm{Hom}\,(\mathrm{Hom}\,(A, \mathbb{Z}), G)\,.$$

Setting $A = C_p$, we have a natural isomorphism

$$C_p \otimes G \longrightarrow \mathrm{Hom}\,(\mathrm{Hom}\,(C_p, \mathbb{Z}), G) = \mathrm{Hom}\,(E_{-p}, G)\,.$$

By naturality, the boundary operator $\partial \otimes i_G$ corresponds to the coboundary operator $\widetilde{\partial}_E$, so one has a natural isomorphism

$$H_p(\mathcal{C}; G) \cong H^{-p}(\mathcal{E}; G)\,,$$

as desired. □

The preceding theorem applies to the homology of a finite simplicial complex or a finite CW complex, but one can obtain the following more general theorem with a bit more effort.

Theorem 7.6.2 *Let $\mathcal{C}$ be a free chain complex, and let G be an abelian group. If $\mathcal{C}$ vanishes below a certain dimension and if the homology of $\mathcal{C}$ is finitely generated in each dimension, then there are exact sequences*

$$0 \longleftarrow \mathrm{Hom}\,(H^p(\mathcal{C}), G) \longleftarrow H_p(\mathcal{C}; G) \longleftarrow \mathrm{Ext}\,(H^{p+1}(\mathcal{C}), G) \longleftarrow 0\,,$$
$$0 \longrightarrow H^p(\mathcal{C}) \otimes G \longrightarrow H^p(\mathcal{C}; G) \longrightarrow H^{p+1}(\mathcal{C}) * G \longrightarrow 0\,,$$

that are natural with respect to homomorphisms induced by chain maps. They split, but not naturally.

We need the following lemma:

Lemma 7.6.3 *If $\mathcal{C}$ is a free chain complex such that $H_i(\mathcal{C})$ is finitely generated for each i, then there is a free chain complex $\mathcal{C}'$ that is finitely generated in each dimension, and that has homology isomorphic to that of $\mathcal{C}$. If $\mathcal{C}$ vanishes below a certain dimension, then so does $\mathcal{C}'$.*

Proof: Let β_p be the betti number of $H_p(\mathcal{C})$, and let $t_1^{(p)}, t_2^{(p)}, \ldots, t_{k_p}^{(p)}$ be its torsion coefficients. Let U_p, V_p, and W_p be free abelian groups, where U_p has rank k_{p-1}, V_p has rank β_p, and W_p has rank k_p. Set

$$C'_p = U_p \oplus V_p \oplus W_p\,.$$

Let $\partial' : C'_{p+1} \to C'_p$ vanish on $V_{p+1} \oplus W_{p+1}$. Choose bases for U_{p+1} and W_p, and define $\partial' : U_{p+1} \to W_p$ to be the homomorphism whose matrix relative to these bases is

$$\begin{bmatrix} t_1^{(p)} & 0 & \cdots & 0 \\ 0 & t_2^{(p)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & t_{k_p}^{(p)} \end{bmatrix}.$$

Then ∂' satisfies the requirements of the lemma. □

Proof of Theorem 7.6.2: Choose a chain complex $\mathcal{C}'$ as in the preceding lemma. In view of Theorem 5.6.2, we can choose a chain map $\phi : \mathcal{C}' \to \mathcal{C}$ inducing the given isomorphism $H_i(\mathcal{C}') \longrightarrow H_i(\mathcal{C})$. It follows from the universal coefficient theorems already stated [or from Theorems 5.6.6 and 6.3.1] that ϕ also induces isomorphisms

$$H_i(\mathcal{C}'; G) \cong H_i(\mathcal{C}; G) \qquad \text{and} \qquad H^i(\mathcal{C}'; G) \cong H^i(\mathcal{C}; G)$$

for all i and G. Now the universal coefficient sequences in question hold for $\mathcal{C}'$. Using the isomorphisms induced by ϕ, we can replace $\mathcal{C}'$ by $\mathcal{C}$ in these exact sequences.

It is a consequence of naturality that the resulting exact sequences for $\mathcal{C}$ are independent of the choices of $\mathcal{C}'$ and ϕ; a fact that we now prove.

Let $\theta : \mathcal{C} \to \mathcal{D}$ be a chain map, where $\mathcal{C}$ and $\mathcal{D}$ are free chain complexes vanishing below a certain dimension, and for which $H_i(\mathcal{C})$ and $H_i(\mathcal{D})$ are finitely generated for each i. We wish to show that θ induces homomorphisms of the universal coefficient sequences of our theorem.

Here we are going to need to establish the fact that the chain maps we are considering have the stronger property of being chain equivalences: Let $\phi : \mathcal{C}' \to \mathcal{C}$ and $\psi : \mathcal{D}' \to \mathcal{D}$ be chain maps that induce homology isomorphisms in all dimensions, as in the preceding lemma. Here $\mathcal{C}'$ and $\mathcal{D}'$ are free, finitely generated in each dimension, and vanishing below a certain dimension. Theorem 5.7.2 tells us that there are chain-homotopy inverses $\lambda : \mathcal{C} \to \mathcal{C}'$ and $\mu : \mathcal{D} \to \mathcal{D}'$ to ϕ and ψ, respectively.

Now ϕ induces an isomorphism of the exact sequences for $\mathcal{C}$ with the exact sequences for $\mathcal{C}'$. Indeed, this is how we *obtained* the exact sequences for $\mathcal{C}$. A similar remark applies to ψ. Consider the diagram

$$\begin{array}{ccc} \mathcal{C}' & \underset{\lambda}{\overset{\phi}{\rightleftarrows}} & \mathcal{C} \\ & & \downarrow \theta \\ \mathcal{D}' & \underset{\mu}{\overset{\psi}{\rightleftarrows}} & \mathcal{D}\,. \end{array}$$

By the naturality property of Theorem 7.6.1, the composite $\mu \circ \theta \circ \phi$ induces homomorphisms of the exact sequences for $\mathcal{C}'$ with those for $\mathcal{D}'$. It follows that $\psi \circ (\mu \circ \theta \circ \phi) \circ \lambda$ induces homomorphisms of the exact sequences for $\mathcal{C}$ with those for $\mathcal{D}$. But this map is chain homotopic to θ.

Applying this result to the special case in which $\mathcal{C} = \mathcal{D}$ and θ is the identity, we see that the exact sequences are independent of the choices of $\mathcal{C}'$ and ϕ. □

Corollary 7.6.4 *If (X, A) is a topological pair such that $H_i(X, A)$ is finitely generated for each i, then there are natural exact sequences*

$$0 \longleftarrow \mathrm{Hom}\,(H^p(X, A), G) \longleftarrow H_p(X, A; G) \longleftarrow \mathrm{Ext}\,(H^{p+1}(X, A), G) \longleftarrow 0\,,$$
$$0 \longrightarrow H^p(X, A) \otimes G \longrightarrow H^p(X, A; G) \longrightarrow H^{p+1}(X, A) * G \longrightarrow 0\,.$$

They split, but not naturally.

The argument used to prove Lemma 7.6.3 might make one suspect that there is a functorial version of the result. Here is that formulation:

Theorem 7.6.5 *Let $\mathbf{C}$ be the category whose objects are free chain complexes $\mathcal{C}$, each of which vanishes below some dimension, and is such that $H_i(\mathcal{C})$ is finitely generated for each i. Let $\mathbf{C}'$ be the subcategory of $\mathbf{C}$ whose objects are those objects of $\mathbf{C}$ whose chain groups are finitely generated in each dimension. The morphisms in both cases are chain-homotopy classes of chain maps.*

If $J : \mathbf{C}' \to \mathbf{C}$ is the inclusion functor, then there is a functor $G : \mathbf{C} \to \mathbf{C}'$ such that $G \circ J$ is the identity functor of $\mathbf{C}'$ and $J \circ G$ is naturally equivalent to the identity functor of $\mathbf{C}$.

EXERCISE

1. Prove Theorem 7.6.5.
2. Is there a functorial version of Theorem 7.6.2? If so, can you prove it?

7.7 Tensor Products of Chain Complexes

In this section, we introduce an algebraic device that will be used for studying the homology of a product space. We define the tensor product of two chain complexes, and show how to make it into a chain complex.

Definition 7.7.1 Let $\mathcal{C} = \{C_p, \partial\}$ and $\mathcal{C}' = \{C'_p, \partial'\}$ be chain complexes. We define their *tensor product* $\mathcal{C} \otimes \mathcal{C}'$ to be the chain complex whose chain group in dimension M is defined by

$$(\mathcal{C} \otimes \mathcal{C}')_m = \bigoplus_{p+q=m} C_p \otimes C'_q \,,$$

and whose boundary operator $\bar{\partial}$ is given by the equation

$$\bar{\partial}(c_p \otimes c'_q) = \partial c_p \otimes c'_q + (-1)^p c_p \otimes \partial' c'_q \,.$$

Formally, we define $\bar{\partial}$ as a function on the set $C_p \times C'_q$; then, since it is bilinear, it induces a homomorphism of the tensor product.

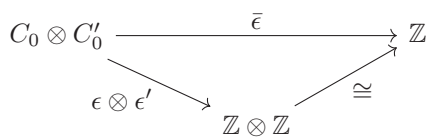
It is straightforward to check that $\bar{\partial}^2 = 0$. It is also straightforward to check that if $\phi : \mathcal{C} \to \mathcal{D}$ and $\phi' : \mathcal{C}' \to \mathcal{D}'$ are chain maps, then so is $\phi \otimes \phi'$.

Note that if $\mathcal{C}$ and $\mathcal{C}'$ are free chain complexes, then so is $\mathcal{C} \otimes \mathcal{C}'$. Indeed, if $\{a^p_\alpha\}$ is a basis for C_p and $\{b^q_\beta\}$ is a basis for C'_q, then the collection

$$\{a^p_\alpha \otimes b^q_\beta : p + q = m\}$$

is a basis for the group $(\mathcal{C} \otimes \mathcal{C}')_m$.

Definition 7.7.2 Let $\{\mathcal{C}, \epsilon\}$ and $\{\mathcal{C}', \epsilon'\}$ be augmented chain complexes. We augment $\mathcal{C} \otimes \mathcal{C}'$ by the homomorphism $\bar{\epsilon}$ that is defined to be $\epsilon \otimes \epsilon'$ followed by the isomorphism mapping $\mathbb{Z} \otimes \mathbb{Z}$ to $\mathbb{Z}$. Thus the following diagram commutes:



You can check that $\bar{\epsilon}$ is surjective, and that $\bar{\epsilon} \circ \bar{\partial} = 0$.

The next theorem describes the geometric motivation underlying our definition of $\mathcal{C} \otimes \mathcal{C}'$.

Theorem 7.7.3 *If K and L are simplicial complexes, and if $|K| \times |L|$ is triangulated so that each cell $\sigma \times \tau$ is the polytope of a subcomplex, then*

$$\mathcal{C}(K) \otimes \mathcal{C}(L) \cong \mathcal{D}(K \times L)\,.$$

Thus $\mathcal{C}(K) \otimes \mathcal{C}(L)$ can be used to compute the homology of $|K| \times |L|$.

Proof: Let $\mathcal{D}(K \times L)$ denote the cellular chain complex of the CW complex $X = |K| \times |L|$, and let M denote the underlying simplicial complex of X. Because X is triangulated, we can use simplicial homology in the definition of the chain group $D_m(K \times L)$. This chain group is a free abelian group, with one basis element of each cell $\sigma^p \times \tau^q$ for which $p+q = m$. This basis element is a fundamental cycle for $\sigma^p \times \tau^q$.

When we orient the simplices of K and L, they form bases for $\mathcal{C}(K)$ and $\mathcal{C}(L)$, respectively. A basis for the m-dimensional chain group of $\mathcal{C}(K) \otimes \mathcal{C}(L)$ consists of all the elements $\sigma^p \otimes \tau^q$ for which $p + q = m$.

It follows that the *groups*

$$(\mathcal{C}(K) \otimes \mathcal{C}(L))_m \qquad \text{and} \qquad D_m(K \times L)$$

are isomorphic, for they are both free abelian and there is a one-to-one correspondence ϕ between their bases. That correspondence ϕ assigns to the basis element $\sigma^p \otimes \tau^q$ of $(\mathcal{C}(K) \otimes \mathcal{C}(L))_m$ a fundamental cycle for the $(p+q)$-cell $\sigma \times \tau$. We need to show that this isomorphism can be chosen so that it preserves boundary operators, and thus is an isomorphism of *chain complexes* as well.

We now proceed by induction. A basis element for $(\mathcal{C}(K) \otimes \mathcal{C}(L))_0$ is of the form $v \otimes w$, where v is a vertex of K and w is a vertex of L. Define $\phi(v \otimes w)$ to be a fundamental 0-cycle for $v \times w$, that is, $\phi(v \otimes w) = v \times w$.

The augmentation $\bar{\epsilon}$ for $\mathcal{C}(K) \otimes \mathcal{C}(L)$ satisfies the equation $\bar{\epsilon}(v \otimes w) = 1$ whenever v is a vertex of K and w is a vertex of L. On the other hand, the chain complex $\mathcal{D}(K \times L)$ is augmented as a subchain complex of $\mathcal{C}(M)$, so $\epsilon(v \times w) = 1$. Thus the map ϕ is augmentation-preserving, since $\epsilon \circ \phi = \bar{\epsilon}$ holds for 0-chains.

For the induction step, we assume that

$$\phi : (\mathcal{C}(K) \otimes \mathcal{C}(L))_i \to D_i(K \times L)$$

is defined in dimensions i less than m, where $m \geq 1$, and that $\phi \circ \bar{\partial} = \partial \circ \phi$ in those dimensions. We suppose further that for $p+q < m$, the chain $\phi(\sigma^p \otimes \tau^q)$ is a fundamental cycle for the cell $\sigma^p \times \tau^q$.

We need to prove that we can indeed define ϕ in dimension m so that ϕ maps a basis element $\sigma^p \otimes \tau^q$, where $p+q=m$, to a fundamental cycle for the $p+q$ cell $\sigma^p \times \tau^q$.

Consider such a basis element $\sigma^p \otimes \tau^q$, where $p+q=m$. Let

$$c_{m-1} = \phi(\bar{\partial}(\sigma^p \otimes \tau^q))\,.$$

Then c_{m-1} is an $(m-1)$-chain of the simplicial complex M underlying X.

First, we note that if $m=1$, then $c_0 \in \ker \epsilon$, for

$$\epsilon(c_0) = \epsilon\phi(\bar{\partial}c_1) = \bar{\epsilon}(\bar{\partial}c_1) = 0$$

holds. Second, we note that if $m > 1$, then c_{m-1} is a cycle. For, by the induction hypothesis, we have

$$\partial c_{m-1} = \partial\phi(\bar{\partial}(\sigma^p \otimes \tau^q)) = \phi\bar{\partial}(\bar{\partial}(\sigma^p \otimes \tau^q)) = 0\,.$$

Now we show that c_{m-1} is carried by Bd $(\sigma \times \tau)$. Let us write $\partial\sigma = \sum s_i$ if $\dim\sigma > 0$, where the s_i are oriented simplices of dimension $p-1$. Similarly, we write $\partial\tau = \sum t_j$ if $\dim\tau > 0$. Then we have the formula

$$c_{m-1} = \sum_i \phi(s_i \otimes \tau) + (-1)^p \sum_j \phi(\sigma \otimes t_j)\,,$$

if $\dim\sigma > 0$ and $\dim\tau > 0$ hold. If $\dim\sigma = 0$, then the first summation is missing from this last formula, while if $\dim\tau = 0$, then the second summation is missing. Since the chain $\phi(s_i \otimes \tau)$ is carried by $s_i \times \tau$, and $\phi(\sigma \otimes t_j)$ is carried by $\sigma \times t_j$, we see that the cycle c_{m-1} is carried by Bd $(\sigma \times \tau)$.

Now Bd $(\sigma \times \tau)$ is topologically an $m-1$ sphere. We shall show that c_{m-1} is a fundamental cycle for this $m-1$ sphere. That is, it generates the group $\widetilde{H}_{m-1}(\text{Bd }(\sigma \times \tau))$, which is infinite cyclic and equals the group of $m-1$ cycles if $m > 1$, and the group $\ker\epsilon$ if $m = 1$.

We have either $\dim\sigma > 0$ or $\dim\tau > 0$. We assume the former. Then $\partial\sigma = \sum s_i$ holds, and the restriction of c_{m-1} to the cell $s_i \times \tau$ equals $\phi(s_i \otimes \tau)$, which by hypothesis, is a fundamental cycle for this cell. It follows that c_{m-1} is not zero, and that it is not a multiple of another cycle. Thus c_{m-1} is a generator of the infinite cyclic group $\widetilde{H}_{m-1}(\text{Bd }(\sigma \times \tau))$.

Now consider the exact homology sequence of the pair $(\sigma \times \tau, \text{Bd }(\sigma \times \tau))$. Because $\sigma \times \tau$ is acyclic, both $H_m(\sigma \times \tau)$ and $H_{m-1}(\sigma \times \tau)$ vanish. So the relevant part of the sequence reduces to

$$0 \longrightarrow H_m(\sigma \times \tau, \text{Bd }(\sigma \times \tau)) \xrightarrow{\partial_*} \widetilde{H}_{m-1}(\text{Bd }(\sigma \times \tau)) \longrightarrow 0\,.$$

Because homology groups equal the groups of cycles in this instance, the sequence is actually

$$0 \longrightarrow Z_m(\sigma \times \tau, \text{Bd }(\sigma \times \tau)) \longrightarrow \widetilde{Z}_{m-1}(\text{Bd }(\sigma \times \tau)) \longrightarrow 0\,,$$

where $\widetilde{Z}_{m-1}$ denotes $\ker \epsilon$ if $m = 1$. We define $\phi(\sigma^p \otimes \tau^q)$ to be the unique fundamental cycle for the cell $\sigma \times \tau$ such that

$$\partial\phi(\sigma^p \otimes \tau^q) = c_{m-1}\,.$$

Then we have $\partial\phi = \phi\bar{\partial}$, and our result is proved. □

EXERCISES

1. **(a)** Show that $\bar{\partial}^2 = 0$.

(b) Show that if $\mathcal{C}$ and $\mathcal{C}'$ are augmented by ϵ and ϵ', respectively, then $\mathcal{C} \otimes \mathcal{C}'$ is augmented by $\bar{\epsilon}$ as given in Definition 7.7.2.

2. Let $\phi, \psi : \mathcal{C} \to \mathcal{D}$ and $\phi', \psi' : \mathcal{C}' \to \mathcal{D}'$ be chain maps.

(a) Show that $\phi \otimes \phi' : \mathcal{C} \otimes \mathcal{C}' \to \mathcal{D} \otimes \mathcal{D}'$ is a chain map.

(b) Show that if D is a chain homotopy of ϕ to ψ, then $D \otimes \phi'$ is a chain homotopy of $\phi \otimes \phi'$ to $\psi \otimes \phi'$.

(c) If D is a chain homotopy of ϕ' to ψ', find a chain homotopy of $\psi \otimes \phi'$ to $\psi \otimes \psi'$.

(d) Show that if ϕ and ϕ' are chain equivalences, then so is $\phi \otimes \phi'$.

3. Our formula for a boundary operator in $\mathcal{C} \otimes \mathcal{C}'$ may seem rather arbitrary. It is. There are other formulas that would do as well.

Let K and L be complexes. Let $\widetilde{\partial}$ be a boundary operator for $\mathcal{C}(K) \otimes \mathcal{C}(L)$ such that the following hold:

(i) If K_0 and L_0 are subcomplexes of K and L, respectively, then $\widetilde{\partial}$ maps $\mathcal{C}(K_0) \otimes \mathcal{C}(L_0)$ to itself.

(ii) If v is a vertex of K and w is a vertex of L, then

$$\widetilde{\partial} = \pm\partial_K \otimes i \quad \text{on} \quad C_1(K) \otimes C_0(w)\,,$$

and

$$\widetilde{\partial} = \pm i \otimes \partial_L \quad \text{on} \quad C_0(v) \otimes C_1(L)\,.$$

Assuming that $X = |K| \times |L|$ is triangulated as in Theorem 7.7.3, show that the chain complex $\{\mathcal{C}(K) \otimes \mathcal{C}(L), \widetilde{\partial}\}$ is isomorphic to the cellular chain complex $\mathcal{D}(X)$.

4. Show that the formula

$$\widehat{\partial}(c_p \otimes c'_q) = (-1)^q \partial c_p \otimes c'_q + c_p \otimes \partial c'_q$$

defines a boundary operator for $\mathcal{C} \otimes \mathcal{C}'$ that satisfies the requirements of Exercise 3.

7.8 The Künneth Theorem

Now we prove a basic theorem that enables us to compute the homology of the tensor product of two chain complexes. As you might expect, the proof is highly algebraic.

The first thing we show, under suitable hypotheses, is the existence of a natural monomorphism of the tensor product $H_p(\mathcal{C}) \otimes H_q(\mathcal{C}')$ into the group $H_{p+q}(\mathcal{C} \otimes \mathcal{C}')$.

Definition 7.8.1 We define the *tensor homomorphism for homology*

$$\Theta : H_p(\mathcal{C}) \otimes H_q(\mathcal{C}') \to H_{p+q}(\mathcal{C} \otimes \mathcal{C}')$$

as follows: If z_p is a p-cycle of $\mathcal{C}$ and z'_q is a q-cycle of $\mathcal{C}'$, let

$$\Theta\left(\{z_p\} \otimes \{z'_q\}\right) = \{z_p \otimes z'_q\}\,.$$

Using the formula for the boundary operator $\bar{\partial}$, we see that $z_p \otimes z'_q$ is a cycle of $\mathcal{C} \otimes \mathcal{C}'$. To show that Θ is well-defined, one computes

$$\begin{aligned}(z_p + \partial d_{p+1}) \otimes z'_q &= z_p \otimes z'_q + \bar{\partial}(d_{p+1} \otimes z'_q)\,.\\ z_p \otimes (z'_q + \partial' d'_{q+1}) &= z_p \otimes z'_q + (-1)^p \bar{\partial}(z_p \otimes d'_{q+1})\,.\end{aligned}$$

Because Θ is induced by the inclusion map $Z_p \otimes Z'_q \to (\mathcal{C} \otimes \mathcal{C}')_{p+q}$, it is natural with respect to chain maps.

Lemma 7.8.2 *If $\mathcal{C}$ and $\mathcal{C}'$ are chain complexes such that in each dimension the cycles form a direct summand in the chains, then*

$$\Theta : \bigoplus_{p+q=m} H_p(\mathcal{C}) \otimes H_q(\mathcal{C}') \to H_m(\mathcal{C} \otimes \mathcal{C}')$$

is a monomorphism, and its image is a direct summand.

Remark 7.8.3 The hypothesis of the lemma is satisfied when $\mathcal{C}$ and $\mathcal{C}'$ are free chain complexes.

Proof of the Lemma: We shall define a homomorphism λ in the opposite direction to Θ, such that $\lambda \circ \Theta$ equals the identity. That suffices.

Let Z_p denote the group of p-cycles in $\mathcal{C}$, and let Z'_q denote the group of q-cycles in $\mathcal{C}'$. We begin with the natural projections $Z_p \to H_p(\mathcal{C})$ and $Z'_q \to H_q(\mathcal{C}')$. Because the cycles form a direct summand in the chains, these maps can be extended to homomorphisms

$$\phi : C_p \to H_p(\mathcal{C}) \qquad \text{and} \qquad \phi' : C'_q \to H_q(\mathcal{C}')\,.$$

Let $\mathcal{E}$ be the chain complex whose pth group is $H_p(\mathcal{C})$, and whose boundary operators vanish. Similarly, let $\mathcal{E}'$ be the chain complex whose qth group is $H_q(\mathcal{C}')$, and whose boundary operators vanish. Then $\phi : \mathcal{C} \to \mathcal{E}$ and $\phi' : \mathcal{C}' \to \mathcal{E}'$ are chain maps, as you can check. Since

$$\phi \otimes \phi' : \mathcal{C} \otimes \mathcal{C}' \to \mathcal{E} \otimes \mathcal{E}'$$

is a chain map, it induces a homomorphism

$$\lambda : H_m(\mathcal{C} \otimes \mathcal{C}') \to H_m(\mathcal{E} \otimes \mathcal{E}')\,.$$

Since the boundary operator in $\mathcal{E} \otimes \mathcal{E}'$ vanishes, $H_m(\mathcal{E} \otimes \mathcal{E}')$ equals the m-dimensional chain group

$$(\mathcal{E} \otimes \mathcal{E}')_m = \bigoplus\nolimits_{p+q=m} H_p(\mathcal{C}) \otimes H_q(\mathcal{C}')\,.$$

It is trivial to check that $\lambda \circ \Theta$ is the identity, because we have

$$\begin{aligned} \lambda\Theta\left(\{z_p\} \otimes \{z'_q\}\right) &= \lambda\{z_p \otimes z'_q\} \\ &= \phi(z_p) \otimes \phi'(z'_q) = \{z_p\} \otimes \{z'_q\}\,. \end{aligned}$$ □

Now we prove our main theorem, which identifies the group cok Θ.

Theorem 7.8.4 (The Künneth theorem for chain complexes) *If $\mathcal{C}$ is a free chain complex and $\mathcal{C}'$ is a chain complex, then there is an exact sequence*

$$0 \to \bigoplus\nolimits_{p+q=m} H_p(\mathcal{C}) \otimes H_q(\mathcal{C}') \xrightarrow{\Theta} H_m(\mathcal{C} \otimes \mathcal{C}') \hookrightarrow \bigoplus\nolimits_{p+q=m} H_{p-1}(\mathcal{C}) * H_q(\mathcal{C}') \to 0\,,$$

which is natural with respect to homomorphisms induced by chain maps. If the cycles of $\mathcal{C}'$ are a direct summand in the chains, then the sequence splits, but not naturally.

Proof: Let B_p, Z_p, and C_p be the p-boundaries, p-cycles, and p-chains, respectively, of $\mathcal{C}$. Define B'_p, Z'_p, and C'_p similarly for $\mathcal{C}'$.

Step 1. We begin with the exact sequence

$$0 \longrightarrow Z_p \xrightarrow{j} C_p \xrightarrow{\partial_0} B_{p-1} \longrightarrow 0\,,$$

which splits because B_{p-1} is free. Let $\mathcal{Z}$ be the chain complex whose p-dimensional group is Z_p, and whose boundary operators vanish. Let $\mathcal{D}$ be the chain complex whose p-dimensional group D_p equals B_{p-1}, and whose

boundary operators vanish. Then the preceding sequence becomes a short exact sequence of chain complexes.

Now tensor with C'_q and sum over $p+q=m$. Exactness is retained, because the original sequence splits and direct sums of exact sequences are exact. One obtains the exact sequence

$$0 \longrightarrow \bigoplus(Z_p \otimes C'_q) \xrightarrow{j \otimes i'} \bigoplus(C_p \otimes C'_q) \xrightarrow{\partial_0 \otimes i'} \bigoplus(D_p \otimes C'_q) \longrightarrow 0\,,$$

where we use i' instead of $i_{\mathcal{C}'}$ to denote the identity map of $\mathcal{C}'$. Since the maps involved are chain maps, this is a short exact sequence of chain complexes

$$0 \longrightarrow (\mathcal{Z} \otimes \mathcal{C}')_m \longrightarrow (\mathcal{C} \otimes \mathcal{C}')_m \longrightarrow (\mathcal{D} \otimes \mathcal{C}')_m \longrightarrow 0\,.$$

The boundary operator in $\mathcal{C} \otimes \mathcal{C}'$ is denoted as usual by $\overline{\partial}$. The boundary operators in the end groups have the form $\pm i \otimes \partial'$, where i is the identity map in $\mathcal{C}$. For instance, in $\mathcal{Z} \otimes \mathcal{C}'$, one computes

$$\overline{\partial}(z_p \otimes c'_q) = (-1)^p z_p \otimes \partial' c'_q\,,$$

since $\partial z_p = 0$. A similar remark applies to $\mathcal{D} \otimes \mathcal{C}'$.

We obtain from this short exact sequence of chain complexes a long exact homology sequence, of which we consider five terms:

$$H_{m+1}(\mathcal{D} \otimes \mathcal{C}') \xrightarrow{\beta_{m+1}} H_m(\mathcal{Z} \otimes \mathcal{C}') \longrightarrow H_m(\mathcal{C} \otimes \mathcal{C}') \longrightarrow H_m(\mathcal{D} \otimes \mathcal{C}') \xrightarrow{\beta_m} H_{m-1}(\mathcal{Z} \otimes \mathcal{C}')\,,$$

where β_m denotes the zig-zag homomorphism [see Lemma 3.3.2]. This sequence is natural with respect to homomorphisms induced by chain maps. Consider the induced sequence

$$0 \longrightarrow \operatorname{cok} \beta_{m+1} \longrightarrow H_m(\mathcal{C} \otimes \mathcal{C}') \longrightarrow \ker \beta_m \longrightarrow 0\,. \qquad (*)$$

It is also natural. We identify the terms of this short exact sequence.

Before proceeding, let us recall that if B is a subgroup of C, and if A is free, then the map

$$A \otimes B \longrightarrow A \otimes C$$

induced by inclusion is injective. Hence in this case, we can consider $A \otimes B$ as a subgroup of $A \otimes C$. We shall use this fact freely in what follows.

Step 2. We compute the group $H_m(\mathcal{Z} \otimes \mathcal{C}')$.

We show that there is an isomorphism, induced by inclusion and given by

$$\textstyle\bigoplus_{p+q=m} Z_p \otimes H_q(\mathcal{C}') \cong H_m(\mathcal{Z} \otimes \mathcal{C}')\,.$$

The chain group of $\mathcal{Z} \otimes \mathcal{C}'$ in dimension m is

$$(\mathcal{Z} \otimes \mathcal{C}')_m = \bigoplus_{p+q=m} Z_p \otimes C'_q \,.$$

To compute the cycles and boundaries of $\mathcal{Z} \otimes \mathcal{C}'$, we begin with the exact sequence

$$0 \longrightarrow Z'_q \xrightarrow{\ j\ } C'_q \xrightarrow{\ \partial'_0\ } B'_{q-1} \longrightarrow 0 \,.$$

Since Z_p is free, when we tensor with Z_p, we have horizontal and vertical exactness in the following diagram:

$$\begin{array}{ccccccccc} & & & & & & 0 & & \\ & & & & & & \downarrow & & \\ 0 & \longrightarrow & Z_p \otimes Z'_q & \longrightarrow & Z_p \otimes C'_q & \xrightarrow{\ i \otimes \partial'_0\ } & Z_p \otimes B'_{q-1} & \longrightarrow & 0 \,. \\ & & & & & \searrow^{i \otimes \partial'} & \downarrow & & \\ & & & & & & Z_p \otimes C'_{q-1} & & \end{array}$$

It follows that the kernel of $i \otimes \partial'$ is the subgroup $Z_p \otimes Z'_q$ of $C_p \otimes C'_q$, and the image is the subgroup $Z_p \otimes B'_{q-1}$ of $Z_p \otimes C'_{q-1}$. Hence the groups of cycles and boundaries of $\mathcal{Z} \otimes \mathcal{C}'$ in dimension m are the respective groups

$$\bigoplus_{p+q=m} Z_p \otimes Z'_q \qquad \text{and} \qquad \bigoplus_{p+q=m} Z_p \otimes B'_q \,.$$

Now consider the exact sequence

$$0 \longrightarrow B'_q \longrightarrow Z'_q \longrightarrow H_q(\mathcal{C}') \longrightarrow 0 \,.$$

Tensoring and summing, we obtain the exact sequence

$$0 \longrightarrow \bigoplus_{p+q=m} Z_p \otimes B'_q \longrightarrow \bigoplus_{p+q=m} Z_p \otimes Z'_q \longrightarrow \bigoplus_{p+q=m} Z_p \otimes H_q(\mathcal{C}') \longrightarrow 0 \,.$$

Since the first two groups of this sequence are the m-boundaries and m-cycles of $\mathcal{Z} \otimes \mathcal{C}'$, the last group is isomorphic to $H_m(\mathcal{Z} \otimes \mathcal{C}')$, as claimed.

Step 3. An entirely similar argument shows that the m-cycle group of $\mathcal{D} \otimes \mathcal{C}'$ is the group

$$\bigoplus_{p+q=m} D_p \otimes Z'_q \,,$$

and that inclusion induces an isomorphism

$$\bigoplus_{p+q=m} D_p \otimes H_q(\mathcal{C}') \cong H_m(\mathcal{D} \otimes \mathcal{C}') \,.$$

Step 4. We now show that the zig-zag homomorphism β_m is induced by inclusion $j : B_p \to Z_p$. More precisely, we show that the following diagram commutes [recall that $D_{p+1} = B_p$ and that i' denotes the identity map of $\mathcal{C}'$]:

$$\begin{array}{ccc} H_{m+1}(\mathcal{D} \otimes \mathcal{C}') & \xrightarrow{\beta_{m+1}} & H_{m+1}(\mathcal{Z} \otimes \mathcal{C}') \\ \uparrow\cong & & \uparrow\cong \\ \bigoplus_{p+q=m} D_{p+1} \otimes H_q(\mathcal{C}') & \xrightarrow{j \otimes i'_*} & \bigoplus_{p+q=m} Z_p \otimes H_q(\mathcal{C}') \end{array}$$

The map β_{m+1} is defined via the following zig-zag diagram:

$$\begin{array}{ccccc} & & \bigoplus_{p+q=m} C_{p+1} \otimes C'_q & \xrightarrow{\partial_0 \otimes i'} & \bigoplus_{p+q=m} D_{p+1} \otimes C'_q \\ & & \downarrow\overline{\partial} & & \\ \bigoplus_{r+s=m} Z_r \otimes C'_s & \xrightarrow{j \otimes i'} & \bigoplus_{r+s=m} C_r \otimes C'_s & & \end{array}$$

One begins with an element of the group of $(m+1)$-cycles of $\mathcal{D} \otimes \mathcal{C}'$. By the previous step, this group is

$$\bigoplus_{p+q=m} D_{p+1} \otimes Z'_q \,.$$

Since $D_{p+1} = B_p$, a typical generator for this group has the form $b_p \otimes z'_q$, where $b_p \in B_p$, $z'_q \in Z_q$, and $p + q = m$. Pulling back to $\mathcal{C} \otimes \mathcal{C}'$, one obtains an element $c_{p+1} \otimes z'_q$, where $\partial_0 c_{p+1} = b_p$. Applying $\overline{\partial}$, we obtain the element

$$\begin{aligned} \overline{\partial}(c_{p+1} \otimes z'_q) &= \partial c_{p+1} \otimes z'_q \pm c_{p+1} \otimes \partial' z'_q \\ &= b_p \otimes z'_q \,. \end{aligned}$$

Finally, we pull this back to $\mathcal{Z} \otimes \mathcal{C}'$, obtaining $b_p \otimes z'_q$, which is the same element with which we started, thus verifying that the map is induced by inclusion.

Step 5. Now we can identify the kernel and cokernel of β_m.

Begin with the free resolution

$$0 \longrightarrow B_p \longrightarrow Z_p \longrightarrow H_p(\mathcal{C}) \longrightarrow 0 \,.$$

Tensoring with $H_q(\mathcal{C}')$, we obtain the sequence

$$0 \to H_p(\mathcal{C}) * H_q(\mathcal{C}') \to B_p \otimes H_q(\mathcal{C}') \to Z_p \otimes H_q(\mathcal{C}') \to H_p(\mathcal{C}) \otimes H_q(\mathcal{C}') \to 0 \,.$$

The map in the middle is induced by inclusion. Summing over $p + q = m$ and applying **Step 4**, we obtain the sequence

$$0 \to \bigoplus_{p+q=m} H_p(\mathcal{C}) \otimes H_q(\mathcal{C}') \to H_{m+1}(\mathcal{D} \otimes \mathcal{C}') \xrightarrow{\beta_{m+1}}$$
$$\hookrightarrow H_m(\mathcal{Z} \otimes \mathcal{C}') \to \bigoplus_{p+q=m} H_p(\mathcal{C}) * H_q(\mathcal{C}') \to 0\,,$$

The kernel and cokernel of β_{m+1} are now apparent. From this sequence and sequence $(*)$ of **Step 1**, the theorem follows. Because both sequences are natural, the Künneth sequence is also natural.

Finally, we note that the map

$$\text{cok } \beta_{m+1} \to H_m(\mathcal{C} \otimes \mathcal{C}')$$

in sequence $(*)$ is induced by the inclusion $\mathcal{Z} \otimes \mathcal{C}' \to \mathcal{C} \otimes \mathcal{C}'$. Therefore the first map of the Künneth sequence is just the map Θ of Lemma 7.8.2. □

Corollary 7.8.5 *Let $\mathcal{C}$, $\mathcal{C}'$, $\mathcal{D}$, and $\mathcal{D}'$ be chain complexes with $\mathcal{C}$ and $\mathcal{D}$ free. If $\phi : \mathcal{C} \to \mathcal{D}$ and $\phi' : \mathcal{C}' \to \mathcal{D}'$ are chain maps that induce homology isomorphisms in all dimensions, then $\phi \otimes \phi' : \mathcal{C} \otimes \mathcal{C}' \to \mathcal{D} \otimes \mathcal{D}'$ induces homology isomorphisms in all dimensions.*

Proof: We apply naturality of the Künneth sequence and the Steenrod five lemma [Lemma 3.3.4]. □

EXAMPLE 7.8.6 Let K and L be simplicial complexes, with either K or L locally finite. The results of the previous section show that the chain complex $\mathcal{C}(K) \otimes \mathcal{C}(L)$ can be used to compute the homology of $|K| \times |L|$. Since $\mathcal{C}(K)$ and $\mathcal{C}(L)$ are free, the Künneth theorem implies that

$$H_m(|K| \times |L|) \cong \bigoplus_{p+q=m} [H_p(K) \otimes H_q(L) \oplus H_{p-1}(K) * H_q(L)]\ .$$

EXAMPLE 7.8.7 Applying the preceding example to the product $\mathbb{S}^r \times \mathbb{S}^s$, we obtain

$$H_m(\mathbb{S}^r \times \mathbb{S}^s) \cong \bigoplus_{p+q=m} H_p(\mathbb{S}^r) \otimes H_q(\mathbb{S}^s)\,.$$

Hence we have

$$H_k(\mathbb{S}^r \times \mathbb{S}^s) \cong \begin{cases} 0 & \text{if } k \notin \{0, r, s, r+s\}, \\ \mathbb{Z} & \text{if } k \in \{0, r+s\}, \end{cases}$$

$$H_r(\mathbb{S}^r \times \mathbb{S}^s) \cong H_s(\mathbb{S}^r \times \mathbb{S}^s) \cong \begin{cases} \mathbb{Z} \oplus \mathbb{Z} & \text{if } r = s, \\ \mathbb{Z} & \text{if } r \neq s, \end{cases}$$

EXAMPLE 7.8.8 Given $\alpha \in H_p(K)$ and $\beta \in H_q(L)$, it is geometrically clear that there should be a corresponding element in $H_{p+q}(|K| \times |L|)$. The image of $\alpha \otimes \beta$ in $H_{p+q}(|K| \times |L|)$ under the natural monomorphism Θ is called the *homology cross product* of α and β, and is denoted $\alpha \times \beta$. In some sense,

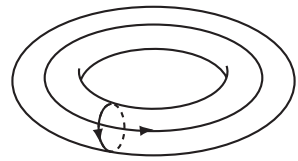


Figure 7.8.1

it behaves like a cartesian product. For example, in the torus $T = \mathbb{S}^1 \times \mathbb{S}^1$ the Künneth theorem tells us that the cross product of the two 1-dimensional homology classes pictured in Figure 7.8.1 is the generator of the 2-dimensional homology. This is quite plausible, because if $z = \sum s_i$ is a fundamental cycle for a triangulation K of $\mathbb{S}^1$, then $z \otimes z = \sum s_i \otimes s_j$ represents the sum of all the 2-cells of $K \times K$, suitably oriented.

It is geometrically less clear why, given a cycle z in K representing a torsion element in $H_p(K)$ of order k, and a cycle z' in L representing a torsion element in $H_q(L)$ of order ℓ, there should be *two* elements in the homology of $|K| \times |L|$ of order $n = \gcd(k, \ell)$. Of course one is their cross product in dimension $p+q$. The other is an element in dimension $p + q + 1$ arising from $H_p(K) * H_q(L)$. Where does this unexpected element come from?

Let us write

$$kz = \partial d \quad \text{and} \quad \ell z' = \partial' d' \,,$$

where d and d' are chains of K and L, respectively. The unexpected element in dimension $p + q + 1$ is exhibited by computing

$$\begin{aligned} \overline{\partial}(d \otimes d') &= (kz) \otimes d' + (-1)^{p+1} d \otimes (\ell z') \\ &= n \left[\frac{k}{n} (z \otimes d') + (-1)^{p+1} \frac{\ell}{n} (d \otimes z') \right] . \end{aligned}$$

The expression in brackets is thus a cycle of dimension $p+q+1$, and n times it is a bounday! This is the element we were seeking.

The Universal Coefficient Theorem for Homology

We now derive the universal coefficient theorem for homology as a corollary of the Künneth theorem.

Proof of Theorem 7.5.1: Let $\mathcal{C}$ be a free chain complex, and let G be an abelian group. We define a chain complex $\mathcal{C}'$ by the equation

$$C'_p = \begin{cases} G & \text{if } p = 0, \\ 0 & \text{if } p \neq 0. \end{cases}$$

All the boundary operators in $\mathcal{C}'$ vanish. The chain complex $\mathcal{C}'$ is not in general free, but the cycles do [trivially] form a direct summand in the chains, in each dimension.

We compute

$$(\mathcal{C} \otimes \mathcal{C}')_m = \bigoplus_{p+q=m} C_p \otimes C'_q = C_m \otimes G\,,$$
$$\overline{\partial}(c_m \otimes g) = \partial c_m \otimes g + 0\,.$$

Thus $\{\mathcal{C} \otimes \mathcal{C}', \overline{\partial}\}$ equals the chain complex $\{\mathcal{C} \otimes G, \partial \otimes i_G\}$. We now apply the Künneth theorem, and use the fact that $H_0(\mathcal{C}') = G$, and that $H_p(\mathcal{C}') = 0$, for $p \neq 0$. The universal coefficient theorem for homology is an immediate consequence. □

The Künneth Theorem with Field Coefficients

Suppose that $\mathcal{E}$ and $\mathcal{E}'$ are chain complexes whose chain groups are vector spaces over a field F, and whose boundary operators are linear transformations. In this situation, we can form the chain complex $\mathcal{E} \otimes_F \mathcal{E}'$, whose chain group in dimension m is

$$\bigoplus_{p+q=m} (E_p \otimes_F E'_q)\,.$$

It is a vector space over F. [See Section 6.2.] The chain map $\overline{\partial}$ is a linear transformation, so its kernel and image are vector spaces.

The Künneth theorem simplifies considerably in this case:

Theorem 7.8.9 (The vector space Künneth theorem) *If the chain complexes $\mathcal{E}$ and $\mathcal{E}'$ are vector spaces over F, and the boundary operators are vector space homomorphisms, then $H_p(\mathcal{E})$ and $H_q(\mathcal{E}')$ are vector spaces over F, and there is a natural isomorphism of vector spaces*

$$\bigoplus_{p+q=m} H_p(\mathcal{E}) \otimes_F H_q(\mathcal{E}') \longrightarrow H_m(\mathcal{E} \otimes_F \mathcal{E}')\,.$$

Proof: If $\otimes$ is replaced by $\otimes_F$, then the proof of the Künneth theorem proceeds through its first four steps unchanged. The first change appears in **Step 5**, where we take the sequence

$$0 \longrightarrow B_p \longrightarrow Z_p \longrightarrow H_p(\mathcal{E}) \longrightarrow 0$$

and tensor it with $H_q(\mathcal{E}')$. Because we are in the context of vector spaces, we obtain the sequence

$$0 \longrightarrow B_p \otimes_F H_q(\mathcal{E}') \xrightarrow{\beta_{m+1}} Z_p \otimes_F H_q(\mathcal{E}') \longrightarrow H_p(\mathcal{E}) \otimes_F H_q(\mathcal{E}') \longrightarrow 0\,.$$

That is, *no torsion product term appears.* As a result, the homomorphism β_{m+1} has vanishing kernel for all m. □

Theorem 7.8.10 (The Künneth theorem with field coefficients) *If $\mathcal{C}$ and $\mathcal{C}'$ are free chain complexes, and if F is a field, then there is a natural isomorphism*

$$\bigoplus_{p+q=m} H_p(\mathcal{C}; F) \otimes_F H_q(\mathcal{C}'; F) \longrightarrow H_m(\mathcal{C} \otimes_F \mathcal{C}'; F)\,.$$

Proof: Applying the preceding theorem to the chain complexes $\mathcal{E} = \mathcal{C} \otimes F$ and $\mathcal{E}' = \mathcal{C}' \otimes F$, which are vector spaces over F, we obtain a natural isomorphism

$$\bigoplus_{p+q=m} H_p(\mathcal{E}) \otimes_F H_q(\mathcal{E}') \longrightarrow H_m(\mathcal{E} \otimes_F \mathcal{E}')\,.$$

By definition, $H_p(\mathcal{E}) = H_p(\mathcal{C}; F)$ and $H_q(\mathcal{E}') = H_p(\mathcal{C}'; F)$. It remains to show that there is a natural vector space isomorphism

$$f : \mathcal{E} \otimes_F \mathcal{E}' \to (\mathcal{C} \otimes \mathcal{C}') \otimes F$$

that is also a chain map. Once that is done our theorem will be proved.

We begin constructing the isomorphism by defining

$$f : C_p \times F \times C'_q \times F \to C_p \otimes C'_q \otimes F$$

by the equation

$$f(x, \alpha, y, \beta) = (x \otimes y) \otimes \alpha\beta\,.$$

This map is multilinear, so it induces a homomophism

$$f' : (C_p \otimes F) \otimes (C'_q \otimes F) \to C_p \otimes C'_q \otimes F\,.$$

In fact, we claim f' is a vector space homomorphism in each variable separately. For instance, we examine what happens when an element of the first space is multiplied by $\gamma \in F$. We have

$$\begin{aligned} f'(\gamma(x \otimes \alpha) \otimes (y \otimes \beta)) &= f'((x \otimes \gamma\alpha) \otimes (y \otimes \beta)) \\ &= x \otimes y \otimes (\gamma\alpha)\beta \\ &= \gamma(x \otimes y \otimes \alpha\beta) = \gamma f'((x \otimes \alpha) \otimes (y \otimes \beta))\,. \end{aligned}$$

Thus f' induces a homomorphism

$$f'' : (C_p \otimes F) \otimes_F (C'_q \otimes F) \to C_p \otimes C'_q \otimes F\,.$$

To show that f'' is an isomorphism, let $\{x_\alpha\}$ be a basis for $\mathcal{C}$ and let $\{y_\beta\}$ be a basis for $\mathcal{C}'$. Then $\{x_\alpha \otimes y_\beta\}$ is a basis for $\mathcal{C} \otimes \mathcal{C}'$. The elements $\{x_\alpha \otimes 1\}$ and $\{y_\beta \otimes 1\}$ form a basis for $\mathcal{C} \otimes F$ and $\mathcal{C}' \otimes F$, respectively. Hence the elements $\{(x_\alpha \otimes 1) \otimes_F (y_\beta \otimes 1)\}$ form a vector space basis for $(\mathcal{C} \otimes F) \otimes_F (\mathcal{C}' \otimes F)$. The images under f'' of the basis elements $\{(x_\alpha \otimes 1) \otimes_F (y_\beta \otimes 1)\}$ are the elements $\{(x_\alpha \otimes y_\beta) \otimes 1\}$, which form a vector space basis for $(\mathcal{C} \otimes \mathcal{C}') \otimes F$. □

EXAMPLE 7.8.11 If K and L are simplicial complexes, with K or L locally finite, then there is a vector space isomorphism

$$\bigoplus_{p+q=m} H_p(K;F) \otimes_F H_q(L;F) \longrightarrow H_m(|K| \times |L|;F)\,.$$

We would like to use the preceding theorem. First recall that Theorem 7.7.3 gives us

$$H_i(\mathcal{C}(K) \otimes \mathcal{C}(L)) \cong H_i(|K| \times |L|)$$

for all i. The same holds for arbitrary coefficients by Theorems 5.6.2 and 6.3.1. Now the preceding theorem applies.

EXERCISES

1. Show that the Künneth theorem fails in reduced homology.

2. Let K and L be simplicial complexes, with K or L locally finite. Then $|K| \times |L|$ is triangulable, as usual. Let K_0 and L_0 be subcomplexes of K and L, respectively. Show there is a chain map that induces homology isomorphisms

$$\mathcal{C}(K, K_0) \otimes \mathcal{C}(L, L_0) \longrightarrow \mathcal{D}(K \times L, (K_0 \times L) \cup (K \times L_0))\,.$$

Derive a Künneth sequence in relative simplicial homology.

3. Let $\mathcal{C}$ be a free chain complex. Show there is a natural exact sequence

$$0 \to \bigoplus_{p+q=m} H_p(\mathcal{C}) \otimes H_q(\mathcal{C}';G) \longrightarrow H_m(\mathcal{C} \otimes \mathcal{C}';G) \longrightarrow \bigoplus_{p+q=m} H_{p-1}(\mathcal{C}) * H_q(\mathcal{C}';G) \to 0\,.$$

Show that this sequence splits provided $\mathcal{C}'$ is free and either G is torsion-free or $H_i(\mathcal{C}')$ is torsion-free for all i.

4. Provide the details of the proof of Corollary 7.8.5.

7.9 The Eilenberg–Zilber Theorem

Now we show that, for an arbitrary pair of topological spaces, the chain complex $\mathcal{S}(X) \otimes \mathcal{S}(Y)$ can be used to compute the singular homology of $X \times Y$. The crucial tool for the proof is the acyclic model theorem.

Lemma 7.9.1 *If $\{\mathcal{C}, \epsilon\}$ and $\{\mathcal{C}', \epsilon'\}$ are acyclic augmented chain complexes, and if $\mathcal{C}$ is free, then $\{\mathcal{C} \otimes \mathcal{C}', \bar{\epsilon}\}$ is acyclic.*

Proof: We shall show that $H_m(\mathcal{C} \otimes \mathcal{C}')$ is infinite cyclic if $m = 0$, and vanishes otherwise. This will prove that $\mathcal{C} \otimes \mathcal{C}'$ is acyclic relative to any augmentation.

Since $\mathcal{C}$ is free, we can apply the Künneth theorem for chain complexes [Theorem 7.8.4] to obtain the exact sequence

$$0 \to \bigoplus_{p+q=m} H_p(\mathcal{C}) \otimes H_q(\mathcal{C}') \longrightarrow H_m(\mathcal{C} \otimes \mathcal{C}') \longrightarrow \bigoplus_{p+q=m} H_{p-1}(\mathcal{C}) * H_q(\mathcal{C}') \to 0\,.$$

For all p and q, we have

$$H_p(\mathcal{C}) * H_q(\mathcal{C}') = 0\,,$$

because each group either vanishes or is infinite cyclic. Therefore we have

$$H_m(\mathcal{C} \otimes \mathcal{C}') \cong \bigoplus_{p+q=m} H_p(\mathcal{C}) \otimes H_q(\mathcal{C}')\,.$$

For each positive m, at least one of the groups $H_p(\mathcal{C})$ or $H_q(\mathcal{C}')$ vanishes, so $H_m(\mathcal{C} \otimes \mathcal{C}') = 0$. For $m = 0$, we have

$$H_m(\mathcal{C} \otimes \mathcal{C}') \cong H_0(\mathcal{C}) \otimes H_0(\mathcal{C}') \cong \mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}\,. \qquad \square$$

Theorem 7.9.2 (The Eilenberg–Zilber theorem) *For every pair X, Y of topological spaces, there are chain maps*

$$\mathcal{S}(X) \otimes \mathcal{S}(Y) \underset{\nu}{\overset{\mu}{\rightleftarrows}} \mathcal{S}(X \times Y)$$

that are chain-homotopy inverse to each other. The maps are natural with respect to chain maps induced by continuous maps.

Proof: Consider the category of pairs of topological spaces and pairs of continuous maps. We define two functors from this category to the category of augmented chain complexes as follows:

$$G(X,Y) = \mathcal{S}(X) \otimes \mathcal{S}(Y)\,, \qquad G'(X,Y) = \mathcal{S}(X \times Y)\,,$$
$$G(f,g) = f_\# \otimes g_\#\,, \qquad G'(f,g) = (f \times g)_\#\,.$$

It is easy to check the functorial properties.

Let $\mathcal{M}$ be the collection of all pairs (Δ_p, Δ_q), where Δ_k denotes the standard k-simplex. We show that G and G' are both acyclic *and* free relative to the collection of models $\mathcal{M}$. Then the theorem follows immediately from the acyclic model theorem [Theorem 4.5.5].

The augmented chain complex $\{\mathcal{S}(\Delta_p \times \Delta_q), \epsilon\}$ is acyclic because $\Delta_p \times \Delta_q$ is a contractible space. The chain complex $\{\mathcal{S}(\Delta_p) \otimes \mathcal{S}(\Delta_q), \overline{\epsilon}\}$ is acyclic by the preceding lemma.

We show that G' is free: Given a non-negative integer m let the indexed family of objects from $\mathcal{M}$ corresponding to this integer consist of the single pair (Δ_m, Δ_m), and let the corresponding element of $S_m(\Delta_m, \Delta_m)$ be the diagonal map $d : \Delta_m \to \Delta_m \times \Delta_m$. We shall show that, as (f, g) ranges over all pairs of continuous maps,

$$(f, g) : (\Delta_m, \Delta_m) \to (X, Y)\,,$$

the elements

$$[G'(f, g)](d) = (f \times g)_{\#}(d) = (f \times g) \circ d$$

are distinct and range over a *basis* for the group $S_m(X \times Y)$. Observe that

$$(f \times g) \circ d : \Delta_m \to X \times Y$$

is just the map T given in coordinates by $T(z) = (f(z), g(z))$. For each pair (f, g) of maps, we have a different singular simplex T, and every singular simplex on $X \times Y$ is of this form. Thus these elements do form a basis for $S_m(\Delta_m, \Delta_m)$.

Now we show that G is free. Given a non-negative integer m, let the indexed family of objects from $\mathcal{M}$ corresponding to this integer be the family

$$\{(\Delta_p, \Delta_q) : p + q = m\}\,.$$

The index set in this case consists of all pairs (p, q) of non-negative integers for which $p + q = m$. Let the element of $(\mathcal{S}(\Delta_p) \otimes \mathcal{S}(\Delta_q))_m$, corresponding to the index (p, q) be the element $i_p \otimes i_q$, where $i_k : \Delta_k \to \Delta_k$ denotes the identity singular simplex on the space Δ_k. We need to show that, as (p, q) ranges over its index set and (f, g) ranges over all maps

$$(f, g) : (\Delta_p, \Delta_q) \to (X, Y)\,,\,,$$

the elements

$$[G(f, g)](i_p \otimes i_q) = (f_{\#} \otimes g_{\#})(i_p \otimes i_q) = f \otimes g$$

are distinct and form a basis for the group $(\mathcal{S}(X) \otimes \mathcal{S}(Y))_m$. We know that $S_p(X)$ is free with the set of all singular simplices $f : \Delta_p \to X$ as a basis, and similarly, $S_q(Y)$ has as basis the set of all singular simplices $g : \Delta_q \to Y$. By Theorem 6.2.10, $S_p(X) \otimes_q (Y)$ has the elements $f \otimes g$ as a basis, and the proof is now complete. □

As a corollaries we have the next two theorems.

Theorem 7.9.3 (The Künneth theorem for topological spaces) *Given topological spaces X and Y, there is an exact sequence*

$$0 \to \bigoplus_{p+q=m} H_p(X) \otimes H_q(Y) \longrightarrow H_m(X \times Y) \hookrightarrow \bigoplus_{p+q=m} H_{p-1}(X) * H_q(Y) \to 0\,.$$

The sequence is natural with respect to homomorphisms induced by continuous maps. It splits but not naturally.

The monomorphism

$$H_p(X) \otimes H_q(Y) \longrightarrow H_m(X \times Y)$$

of the preceding theorem is called the *homology cross product*. It equals the composite

$$H_p(X) \otimes H_q(Y) \xrightarrow{\Theta} H_m(\mathcal{S}(X) \otimes \mathcal{S}(Y)) \xrightarrow{\mu_*} H_m(X \times Y)\,,$$

where Θ is induced by inclusion and μ is the Eilenberg–Zilber chain equivalence. We shall not have much occasion to use the homology cross product, but a similar cross product in cohomology will prove to be quite important, as we shall see in the next two sections.

Theorem 7.9.4 *Let X and Y be topological spaces, and let F be a field. There is a natural isomorphism*

$$\bigoplus_{p+q=m} H_p(X;F) \otimes H_q(Y;F) \to H_m(X \times Y;F)\,.$$

For later applications, it will be convenient to have a specific formula for one of the Eilenberg–Zilber chain equivalences. Here is such a formula:

Theorem 7.9.5 (The Eilenberg–Zilber chain equivalence ν.) *Let $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ be projections. Define*

$$\nu : S_m(X \times Y) \to \bigoplus_{p+q=m} S_p(X) \otimes S_q(Y)$$

by the equation

$$\nu(T) = \sum_{i=0}^{m} [\pi_1 \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_i)] \otimes [\pi_2 \circ T \circ \ell(\epsilon_i, \epsilon_{i+1}, \ldots, \epsilon_m)]\,.$$

Then ν is a natural chain equivalence that preserves augmentation.

Proof: That ν is natural and augmentation-preserving is left to the exercises.

The fact that ν is a chain map follows from a messy, but straightforward, computation that is also left to the exercises. One finds that

$$\begin{aligned}
&\overline{\partial}\nu(T) - \nu(\partial T) \\
&\quad = \sum_{i=1}^{m} (-1)^i [\pi_1 \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i})] \otimes [\pi_2 \circ T \circ \ell(\epsilon_i, \epsilon_{i+1}, \ldots, \epsilon_m)] \\
&\qquad + \sum_{i=1}^{m-1} (-1)^i [\pi_1 \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_i)] \otimes [\pi_2 \circ T \circ \ell(\widehat{\epsilon_i}, \epsilon_{i+1}, \ldots, \epsilon_m)]\,.
\end{aligned}$$

These terms cancel in pairs.

It then follows from the proof of the Eilenberg–Zilber theorem that ν is a chain equivalence. □

The original proof of the Eilenberg–Zilber theorem did not use acyclic models. Instead, it used the chain map ν of the last theorem, and a chain map μ defined by a similar formula. One wrote down formulas to show that both $\nu \circ \mu$ and $\mu \circ \nu$ were chain homotopic to identity maps.

EXERCISES

1. Show that there is a non-natural isomorphism

$$H_m(X \times Y) \cong \bigoplus_{p+q=m} H_p(X; H_q(Y)) .$$

2. **(a)** Let $A \subset X$ and $B \subset Y$. Use the Eilenberg–Zilber theorem to show that there is a natural chain equivalence

$$\mathcal{S}(X, A) \otimes \mathcal{S}(Y, B) \longrightarrow \frac{\mathcal{S}(X \times Y)}{\mathcal{S}(X \times B) + \mathcal{S}(A \times Y)} .$$

(b) State and prove a Künneth theorem in relative singular homology, assuming $\{X \times B, A \times Y\}$ is excisive.

3. Let ν be the chain map defined in Theorem 7.9.5. Show that it is natural and preserves augmentation. Check that ν is a chain map.

4. **(a)** Let X and Y be CW complexes. Show that the tensor product $\mathcal{D}(X) \otimes \mathcal{D}(Y)$ of their cellular chain complexes can be used to compute the homology of $X \times Y$.

*(b) Show that in fact $\mathcal{D}(X) \otimes \mathcal{D}(Y)$ and $\mathcal{S}(X) \otimes \mathcal{S}(Y)$ are chain equivalent.

(c) Compute the homology of $\mathbb{P}^2 \times \mathbb{P}^3$ by computing the groups and boundary operators in the chain complex $\mathcal{D}(\mathbb{P}^2) \otimes \mathcal{D}(\mathbb{P}^3)$.

(d) Check the results of **(c)** by using the Künneth theorem.

5. Compute the homology of the following spaces:

(a) $\mathbb{S}^2 \times \mathbb{P}^5$

(b) $\mathbb{P}^3 \times \mathbb{P}^5$

(c) $\mathbb{S}^3 \times L(n, k)$

(d) $L(n, k) \times L(m, j)$

(e) $\mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{S}^3$

(f) $\mathbb{C}P^2 \times \mathbb{C}P^3$

6. Suppose that M is a compact, connected n-manifold such that $H_n(M) \cong \mathbb{Z}$. There are relations among the betti numbers and torsion coefficients of M. Use the results of the preceding exercise to formulate a conjecture concerning these relations.

7. Verify the functorial properties of the functors at the beginning of the proof of Theorem 7.9.2.

7.10 *The Künneth Theorem for Cohomolgy

One can always compute the cohomology of a product space by first computing its homology from the Künneth theorem, and then applying the universal coefficient theorem for cohomology. But one might hope for a version of the Künneth theorem that applies directly to cohomology. We shall prove such a theorem in this section. It has applications to computing the cohomology ring of a product space, which we shall study in the next section.

First we introduce the cross product operation in cohomology. We give only the definition here. Discussion of its properties will be postponed to the next section.

Throughout this section, let R be a commutative ring with unity element.

Let $\mathcal{C}$ and $\mathcal{C}'$ be chain complexes. Corresponding to the chain complex $\mathcal{C}$, there is a cochain complex $\mathrm{Hom}\,(\mathcal{C}, R)$ with p-dimensional group $\mathrm{Hom}\,(C_p, R)$. Similarly, one has the cochain complex $\mathrm{Hom}\,(\mathcal{C}', R)$. We make the tensor product of these cochain complexes into a cochain complex in the way one might expect: The m-dimensional cochain group of $\mathrm{Hom}\,(\mathcal{C}, R) \otimes \mathrm{Hom}\,(\mathcal{C}', R)$ is the group

$$\bigoplus\nolimits_{p+q=m} \mathrm{Hom}\,(C_p, R) \otimes \mathrm{Hom}\,(\mathcal{C}'_q, R)\,.$$

The coboundary operator is given by the rule

$$\overline{\delta}(\phi^p \otimes \psi^q) = \delta\phi^p \otimes \psi^q + (-1)^p \phi^p \otimes \delta'\psi^q\,.$$

Put simply, this cochain complex is obtained by first applying the Hom functor, then forming a tensor product. One could reverse the order of operations and obtain the cochain complex $\mathrm{Hom}\,((\mathcal{C}\otimes\mathcal{C}'), R)$. Its group in dimension m is

$$\mathrm{Hom}\,((\mathcal{C} \otimes \mathcal{C}')_m, R)\,.$$

Its coboundary operator is the dual of $\overline{\partial}$, so we denote it by $\widetilde{\overline{\partial}}$.

There is a natural homomorphism of the first of these chain complexes to the second, described as follows:

Definition 7.10.1 Let $p + q = m$. The homomorphism

$$\theta : \mathrm{Hom}\,(C_p, R) \otimes \mathrm{Hom}\,(\mathcal{C}'_q, R) \to \mathrm{Hom}\,((\mathcal{C} \otimes \mathcal{C}')_m, R)$$

is defined by the equation

$$\langle \theta(\phi^p \otimes \psi^q), c_r \otimes c'_s \rangle = \langle \phi^p, c_r \rangle \cdot \langle \psi^q, c'_s \rangle \,.$$

Here we make the convention that $\langle \phi^p, c_r \rangle = 0$ if $p \neq r$, and likewise $\langle \psi^q, c'_s \rangle = 0$ if $q \neq s$. Said differently, we agree that any element ϕ^p of $\mathrm{Hom}\,(C_p, R)$ is automatically extended, without change of notation, to a homomorphism of $\bigoplus C_r$ into R, by letting it be zero on all summands other than C_p.

Lemma 7.10.2 *The homomorphism θ is a natural cochain map.*

Proof: Let $\phi \in \mathrm{Hom}\,(C_p, R)$ and $\psi \in \mathrm{Hom}\,(C'_q, R)$, where $p + q = m$. Let $r + s = m + 1$. We compute

$$\begin{aligned} \langle \theta(\overline{\delta}(\phi \otimes \psi)), c_r \otimes c'_s \rangle &= \langle \theta(\delta\phi \otimes \psi + (-1)^p \phi \otimes \delta'\psi), c_r \otimes c'_s \rangle \\ &= \langle \delta\phi, c_r \rangle \cdot \langle \psi, c_s \rangle + (-1)^p \langle \phi, c_r \rangle \cdot \langle \delta'\psi, c'_s \rangle \,, \qquad (*) \end{aligned}$$

and

$$\begin{aligned} \langle \widetilde{\overline{\partial}}\, \theta(\phi \otimes \psi), c_r \otimes c'_s \rangle &= \langle \theta(\phi \otimes \psi), \overline{\partial}(c_r \otimes c'_s) \rangle \\ &= \langle \phi, \partial c_r \rangle \cdot \langle \psi, c_s \rangle + (-1)^r \langle \phi, c_r \rangle \cdot \langle \psi, \partial' c'_s \rangle \,. \qquad (**) \end{aligned}$$

The expressions in the lines $(*)$ and $(**)$ are equal except for the signs $(-1)^p$ and $(-1)^r$ on the final terms. In fact this is not a difficulty, because if $p \neq r$, then $\langle \phi, c_r \rangle = 0$, and both these final terms vanish.

It is apparent from its definition that if $f : C_p \to D_p$ and $g : C'_q \to D'_q$ are homomorphisms, then

$$\theta \circ (\widetilde{f} \otimes \widetilde{g}) = (\widetilde{f \otimes g}\,) \circ \theta$$

holds. Thus θ is natural. □

Now we define a homomorphism in cohomology analogous to the map Θ that appeared in the homology version of the Künneth theorem.

Definition 7.10.3 We define the *tensor homomorphism for homology*

$$\Theta : H^p(\mathcal{C}; R) \otimes H^q(\mathcal{C}'; R) \to H^m(\mathcal{C} \otimes \mathcal{C}'; R) \,,$$

where $m = p + q$, as follows: Let ϕ and ψ be cocycles of dimensions p and q, respectively. Then define

$$\Theta(\{\phi\} \otimes \{\psi\}) = \{\theta(\phi \otimes \psi)\} \,.$$

Note that $\theta(\phi \otimes \psi)$ is a cocycle, because

$$\widetilde{\overline{\partial}}\, \theta(\phi \otimes \psi) = \theta(\overline{\delta}(\phi \otimes \psi)) = \theta(\delta\phi \otimes \psi \pm \phi \otimes \delta'\psi) \,,$$

which vanishes, because ϕ and ψ are cocycles. A similar computation shows that Θ is well-defined.

Definition 7.10.4 Let X and Y be topological spaces. Suppose that $m = p + q$. We define the *cohomology cross product*

$$\times : H^p(X; R) \otimes H^q(Y; R) \to H^m(X \times Y; R)$$

to be the composite

$$H^p(X; R) \otimes H^q(Y; R) \xrightarrow{\Theta} H^m(\mathcal{S}(X) \otimes \mathcal{S}(Y); R) \xrightarrow{\nu^*} H^m(X \times Y; R)\,,$$

where ν is the Eilenberg–Zilber chain equivalence of Theorem 7.9.5. The image of $\alpha^p \otimes \beta^q$ under this homomorphism is denoted $\alpha^p \times \beta^q$.

Let us note that, as was the case with the cup product, the homomorphism Θ can be defined for coefficients other than R. We give one such generalization in the following definition:

Definition 7.10.5 Let X and Y be topological spaces, and let G be an abelian group. Suppose that $m = p + q$. A cohomology cross product

$$\times : H^p(X) \otimes H^q(Y; G) \to H^m(X \times Y; G)$$

is defined as the composite $\nu^* \circ \Theta$ when Θ is induced by the chain map

$$\langle \theta(\phi^p \otimes \psi^q), c_r \otimes c'_s \rangle = \langle \phi^p, c_r \rangle \cdot \langle \psi^q, c'_s \rangle\,,$$

where $\phi^p \in \mathrm{Hom}\,(C_p, \mathbb{Z})$ and $\psi^q \in \mathrm{Hom}\,(C_q, G)$.

One should also note that both $\mathrm{Hom}\,(C_p, R)$ and $\mathrm{Hom}\,(C'_q, R)$ have R-module structures. From the definition of θ, it follows that

$$\theta(\alpha\phi \otimes \psi) = \theta(\phi \otimes \alpha\psi) = \alpha\theta(\phi \otimes \psi)\,.$$

This tells us that θ induces a mapping

$$\theta : \mathrm{Hom}\,(C_p, R) \otimes_R \mathrm{Hom}\,(C'_q, R) \to \mathrm{Hom}\,((\mathcal{C} \otimes \mathcal{C}')_m, R)$$

that is a homomorphism of R-modules. The case of particular interest to us occurs when R is a field F. In this case, θ is a linear transformation of vector spaces.

In both these cases, one has an induced homomorphism Θ on the cohomology level, and thus a cohomology cross product operation as well. In one case, the cross product is a homomorphism

$$\times : H^p(X) \otimes H^q(Y; G) \to H^m(X \times Y; G)\,.$$

In the other case, it is an R-module homomorphism

$$\times : H^p(X; R) \otimes_R H^q(Y; R) \to H^m(X \times Y; R)\,.$$

The most general cross product is defined relative to an arbitrary *coefficient pairing*, which is a homomorphism $A \otimes_R A' \to A''$ of R-modules. We shall not need this degree of generality.

The Künneth Theorem for Cohomology with Integer Coefficients

We will now prove the Künneth theorem for cohomology with integer coefficients. The idea of the proof is to re-index the groups $\mathrm{Hom}\,(C_p, \mathbb{Z})$ so as to make $\mathrm{Hom}\,(\mathcal{C}, \mathbb{Z})$ into a chain complex, and then apply the Künneth theorem that we have already proved. One difficulty is that $\mathrm{Hom}\,(C_p, \mathbb{Z})$ is in general not free. So some restrictive hypotheses are needed.

An additional difficulty is that, by definition, the cohomology of $\mathcal{C} \otimes \mathcal{C}'$ is computed from the cochain complex $\mathrm{Hom}\,((\mathcal{C} \otimes \mathcal{C}')_m, \mathbb{Z})$, and that is not the same as the cochain complex

$$\mathrm{Hom}\,(\mathcal{C}, \mathbb{Z}) \otimes \mathrm{Hom}\,(\mathcal{C}', \mathbb{Z})$$

to which we are going to apply the Künneth theorem. This is where the homomorphism θ comes into play, as we see in the following lemma:

Lemma 7.10.6 *Let $\mathcal{C}$ and $\mathcal{C}'$ be chain complexes that vanish below a certain dimension. If $\mathcal{C}$ is free and finitely generated in each dimension, then*

$$\theta : \bigoplus_{p+q=m} \mathrm{Hom}\,(C_p, \mathbb{Z}) \otimes \mathrm{Hom}\,(C'_q, \mathbb{Z}) \to \mathrm{Hom}\,((\mathcal{C} \otimes \mathcal{C}')_m, \mathbb{Z})$$

is an isomorphism.

Proof: For purposes of this proof, we break θ up into a composite of two maps:

$$\mathrm{Hom}\,(C_p, \mathbb{Z}) \otimes \mathrm{Hom}\,(C'_q, \mathbb{Z}) \xrightarrow{M} \mathrm{Hom}\,(C_p \otimes C'_q, \mathbb{Z})\,,$$

and

$$\bigoplus_{p+q=m} \mathrm{Hom}\,(C_p \otimes C'_q, \mathbb{Z}) \xrightarrow{e} \mathrm{Hom}\,\left(\bigoplus_{p+q=m} C_p \otimes C'_q, \mathbb{Z}\right).$$

The map M is given by the equation

$$\langle M(\phi^p \otimes \psi^q), c_p \otimes c'_q \rangle = \langle \phi^p, c_p \rangle \cdot \langle \psi^q, c'_q \rangle\,.$$

The map e is defined as follows: If $F : C_p \otimes C'_q \to \mathbb{Z}$, then $e(F)$ is the map of $\bigoplus_{r+s=m} C_r \otimes C'_s$ into $\mathbb{Z}$ that equals F on the summand $C_p \otimes C'_q$ and vanishes on all other summands.

Step 1. Because C_p is free and of finite rank, it follows that

$$M : \mathrm{Hom}\,(C_p, \mathbb{Z}) \otimes \mathrm{Hom}\,(C'_q, \mathbb{Z}) \to \mathrm{Hom}\,(C_p \otimes C'_q, \mathbb{Z})$$

is an isomorphism. To see this, let $a_1, a_2, \ldots, a_k$ be a basis for C_p, and let $\widetilde{a}_1, \widetilde{a}_2, \ldots, \widetilde{a}_k$ be the dual basis for $\mathrm{Hom}\,(C_p, \mathbb{Z})$. The group

$$\mathrm{Hom}\,(C_p, \mathbb{Z}) \otimes \mathrm{Hom}\,(C'_q, \mathbb{Z})$$

is the direct sum of k copies of $\mathrm{Hom}\,(C'_q, \mathbb{Z})$, the ith summand being the subgroup

$$\langle \widetilde{a}_i \rangle \otimes \mathrm{Hom}\,(C'_q, \mathbb{Z})\,,$$

where $\langle \widetilde{a}_i \rangle$ is the infinite cyclic group generated by $\widetilde{a}_i$. Also $C_p \otimes C'_q$ is the direct sum of k copies of C'_q, the ith summand being $\langle a_i \rangle \otimes C'_q$. Thus $\mathrm{Hom}\,(C_p \otimes C'_q, \mathbb{Z})$ is the direct sum of k copies of $\mathrm{Hom}\,(C'_q, \mathbb{Z})$, the ith summand consisting of all homomorphisms that vanish on $\langle a_j \rangle \otimes C'_q$ for $j \neq i$. Since M is an isomorphism of $\langle \widetilde{a}_i \rangle \otimes \mathrm{Hom}\,(C'_q, \mathbb{Z})$ with the ith direct summand of $\mathrm{Hom}\,(C_p \otimes C'_q, \mathbb{Z})$, we see that M is an isomorphism.

Step 2. The map e is clearly injective. The image of e consists of those maps of $\bigoplus_{p+q=m} C_p \otimes C'_q$ into $\mathbb{Z}$ that vanish on all but finitely many summands. Since $\mathcal{C}$ and $\mathcal{C}'$ vanish below a certain dimension, $\bigoplus_{p+q=m} C_p \otimes C'_q$ has only finitely many summands. So under our hypotheses, e is surjective, as well as injective. □

Theorem 7.10.7 (The Künneth theorem for cohomology with integer coefficients) *Let $\mathcal{C}$ and $\mathcal{C}'$ be chain complexes that vanish below a certain dimension. If $\mathcal{C}$ is free and finitely generated in each dimension, then there is a natural exact sequence*

$$0 \to \bigoplus_{p+q=m} H^p(\mathcal{C}) \otimes H^q(\mathcal{C}') \xrightarrow{\Theta} H^m(\mathcal{C} \otimes \mathcal{C}') \to \bigoplus_{p+q=m} H^{p+1}(\mathcal{C}) * H^q(\mathcal{C}') \to 0\,.$$

If $\mathcal{C}'$ is free and finitely generated in each dimension, then the sequence splits—but not naturally.

Proof: Let $\mathcal{E}$ and $\mathcal{E}'$ be the chain complexes whose chain groups in dimension $-p$ are defined by the equations

$$E_{-p} = \mathrm{Hom}\,(C_p, \mathbb{Z}) \quad \text{and} \quad E'_{-p} = \mathrm{Hom}\,(C'_p, \mathbb{Z})\,,$$

and whose boundary operators are δ and δ', respectively. Since $\mathcal{E}$ is free, the Künneth theorem applies. There is an exact sequence

$$0 \to \bigoplus_{-p-q=-m} (H_{-p}(\mathcal{E}) \otimes H_{-q}(\mathcal{E}')) \longrightarrow H_{-m}(\mathcal{E} \otimes \mathcal{E}') \to \bigoplus_{-p-q=-m} (H_{-p-1}(\mathcal{E}) * H_{-q}(\mathcal{E}')) \to 0\,.$$

Now we have $H_{-p}(\mathcal{E}) = H^p(\mathcal{C})$ and $H_{-q}(\mathcal{E}') = H^q(\mathcal{C}')$, by definition. Furthermore, by the preceding lemma, there is a natural isomorphism of cochain complexes

$$(\mathcal{E} \otimes \mathcal{E}')_{-m} = (\mathrm{Hom}\,(\mathcal{C}, \mathbb{Z}) \otimes \mathrm{Hom}\,(\mathcal{C}', \mathbb{Z}))_m \xrightarrow{\theta} \mathrm{Hom}\,((\mathcal{C} \otimes \mathcal{C}')_m, \mathbb{Z})\,.$$

It induces a natural isomorphism

$$H_{-m}(\mathcal{E} \otimes \mathcal{E}') \longrightarrow H^m(\mathcal{C} \otimes \mathcal{C}')\,.$$

Combining these facts, we obtain the Künneth sequence for cohomology. Because the maps

$$H_{-p}(\mathcal{E}) \otimes H_{-q}(\mathcal{E}') \longrightarrow H_{-m}(\mathcal{E} \otimes \mathcal{E}') \longrightarrow H^m(\mathcal{C} \otimes \mathcal{C}')$$

are induced by inclusion and by θ, respectively, their composite equals Θ.

If $\mathcal{C}'$ is free and finitely generated in each dimension, then $\mathcal{E}'$ is free and the sequence splits. □

Corollary 7.10.8 *In the preceding theorem, the hypothesis that $\mathcal{C}$ is finitely generated in each dimension can be replaced by the hypothesis that $H_i(\mathcal{C})$ is finitely generated for each i. Similarly, the hypothesis that $\mathcal{C}'$ is finitely generated in each dimension can be replaced by the hypothesis that $H_i(\mathcal{C}')$ is finitely generated for each i.*

Proof: The proof is similar to that of Theorem 7.6.2. If $\mathcal{C}$ is free and vanishes below a certain dimension, and if $H_i(\mathcal{C})$ is finitely generated for each i, then by Lemma 7.6.3 we can choose a free chain complex $\mathcal{D}$ that vanishes below a certain dimension, that is finitely generated in each dimension, and is such that $H_i(\mathcal{D}) \cong H_i(\mathcal{C})$ for each i. By Theorem 5.6.2, there is a chain map $\phi : \mathcal{C} \to \mathcal{D}$ that induces homology isomorphisms in all dimensions. By Theorem 5.7.2, the chain map ϕ is a chain equivalence. Then

$$\phi \otimes i' : \mathcal{C} \otimes \mathcal{C}' \to \mathcal{D} \otimes \mathcal{C}'$$

is also a chain equivalence. [See the exercises of Section 7.7.]

Since the Künneth theorem holds for $\mathcal{D} \otimes \mathcal{C}'$, it holds for $\mathcal{C} \otimes \mathcal{C}'$ as well. Naturality of the sequence follows as in the proof of Theorem 7.6.2.

One proceeds similarly if $H_i(\mathcal{C}')$ is finitely generated for each i. One replaces $\mathcal{C}'$ by a chain-equivalent chain complex $\mathcal{D}'$ that is finitely generated in each dimension. □

Again we have a Künneth theorem for topological spaces.

Theorem 7.10.9 (The Künneth theorem for the cohomology of topological spaces) *Let X and Y be topological spaces. If $H_i(X)$ is finitely generated for each i, then there is a natural exact sequence*

$$0 \to \bigoplus_{p+q=m} H^p(X) \otimes H^q(Y) \xrightarrow{\times} H^m(X \times Y) \longrightarrow \bigoplus_{p+q=m} H^{p+1}(X) * H^q(Y) \to 0\,.$$

If $H_i(Y)$ is finitely generated for each i, then the sequence splits—but not naturally.

The Künneth Theorem for Cohomology with Field Coefficients

Just as with homology, the Künneth theorem simplifies considerably if the cohomology groups have their coefficients in a field.

Theorem 7.10.10 The Künneth theorem for cohomology with coefficients in a field) *Let F be a field. If $\mathcal{C}$ and $\mathcal{C}'$ are chain complexes that vanish below a certain dimension, if $\mathcal{C}$ is free, and if $H_i(\mathcal{C})$ is finitely generated in each dimension, then there is a natural isomorphism*

$$\Theta : \bigoplus_{p+q=m} (H^p(\mathcal{C}; F) \otimes_F H^q(\mathcal{C}'; F)) \to H^m(\mathcal{C} \otimes \mathcal{C}'; F)\,.$$

Proof: By the usual argument, we may assume that $\mathcal{C}$ itself is finitely generated in each dimension. The chain complexes $\mathcal{E}$ and $\mathcal{E}'$ defined by

$$E_{-p} = \mathrm{Hom}\,(C_p, F) \quad \text{and} \quad E'_{-q} = \mathrm{Hom}\,(C'_q, F)$$

are vector spaces over F. By the vector space Künneth theorem, Theorem 7.8.9, there exists a natural vector space isomorphism

$$\bigoplus_{p+q=m} H_{-p}(\mathcal{E}) \otimes_F H_{-q}(\mathcal{E}') \to H_{-m}(\mathcal{E} \otimes_F \mathcal{E}')\,.$$

Since $\mathcal{C}$ and $\mathcal{C}'$ vanish below a certain dimension and $\mathcal{C}$ is free and finitely generated in each dimension, the proof of Lemma 7.10.6 goes through essentially unchanged to show that the map θ defines a vector space isomorphism of $\mathcal{E} \otimes_F \mathcal{E}'$ with $\mathrm{Hom}\,(\mathcal{C} \otimes \mathcal{C}', F)$ that is a cochain map. The theorem follows. □

We also have the following corollary:

Corollary 7.10.11 *Let F be a field. If X and Y are topological spaces, and if $H_i(X)$ is finitely generated for each i, then there is a natural vector space isomorphism*

$$\bigoplus_{p+q=m} H^p(X; F) \otimes_F H^q(Y; F) \xrightarrow{\times} H^m(X \times Y; F)\,.$$

EXERCISES

1. State and prove a Künneth theorem in relative singular cohomology. [See Exercise 2 of Section 7.9.]

2. Compute the cohomology groups of the following spaces. You may use either the Künneth theorem, or the universal coefficient theorem applied to the results of Exercise 5 of Section 7.9.

(a) $\mathbb{S}^2 \times \mathbb{P}^5$
(b) $\mathbb{P}^3 \times \mathbb{P}^5$
(c) $\mathbb{S}^3 \times L(n,k)$
(d) $L(n,k) \times L(m,j)$
(e) $\mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{S}^3$
(f) $\mathbb{C}P^2 \times \mathbb{C}P^3$

3. Suppose that M is a compact, connected n-manifold such that $H_n(M) \cong \mathbb{Z}$. There are relations between the homology and cohomology groups of M. Compare the computations of the preceding exercise with those you made in Exercise 5 of Section 7.9 to formulate a conjecture concerning these relations.

4. Compute the cohomology groups of the following spaces.

(a) $\mathbb{S}^1 \times \mathbb{P}^3$
(b) $\mathbb{P}^2 \times \mathbb{P}^4$
(c) $\mathbb{S}^2 \times L(n,k)$

7.11 *Application: The Cohomology Ring of a Product Space

In the preceding section, we defined the cohomology cross product and used it in obtaining a Künneth theorem in cohomology. In this section, we explore some of its properties, and use it to obtain information about the ring structure of cohomology.

We shall state our results first for the cross product with coefficients in the ring R [Definition 7.10.4], and then explain to what extent they hold for the cross product with $(\mathbb{Z}, G)$ coefficients [Definition 7.10.5].

We let X and Y be topological spaces.

Definition 7.11.1 The cross product is defined as the composite of the homomorphisms Θ and ν^* of cohomology. Because these homomorphisms are induced by the cochain maps θ and $\widetilde{\nu}$, respectively, the cross product is induced by the cochain map $\widetilde{\nu} \circ \theta$. Accordingly, we define

$$c^p \times c^q = \widetilde{\nu}\theta(c^p \otimes c^q)\,,$$

and call it the *cochain cross product*.

The cochain cross product induces the cohomology cross product. We compute it as follows: For $m = p + q$, if $T : \Delta_m \to X \times Y$, then

$$\begin{aligned}\langle c^p \times c^q, T\rangle &= \langle \theta(c^p \otimes c^q), \nu(T)\rangle \\ &= \sum_{i=0}^{m} \langle c^p, \pi_1 \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_i)\rangle \cdot \langle c^q, \pi_2 \circ T \circ \ell(\epsilon_i, \epsilon_{i+1}, \ldots, \epsilon_m)\rangle\,.\end{aligned}$$

The only non-zero term in the summation is the term for which $i = p$. Thus we have the following lemma:

Lemma 7.11.2 *The cochain cross product is given by the formula*

$$\langle c^p \times c^q, T\rangle = \langle c^p, \pi_1 \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle c^q, \pi_2 \circ T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_m)\rangle\,.$$

This last result holds for $(\mathbb{Z}, G)$ coefficients as well. It says that the value of $c^p \times c^q$ on T equals the value of c^p on the front face of the first component of T times the value of c_q on the back face of the second component of T.

We now derive some properties of the cross products.

Theorem 7.11.3 **(a)** *If $\lambda : X \times Y \to Y \times X$ is the map that reverses coordinates, then*

$$\lambda^*(\beta^q \times \alpha^p) = (-1)^{pq} \alpha^p \times \beta^q\,.$$

(b) *In $H^*(X \times Y \times Z; R)$, we have the equation $(\alpha \times \beta) \times \gamma = \alpha \times (\beta \times \gamma)$.*

(c) *Let $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ be the projection maps. Let 1_X and 1_Y be the unity elements of the rings $H^*(X; R)$ and $H^*(Y; R)$, respectively. Then we have*

$$\pi_1^*(\alpha^p) = \alpha^p \times 1_Y \quad \text{and} \quad \pi_2^*(\beta^q) = 1_X \times \beta^q\,.$$

Proof: (a) Consider the chain maps

$$\begin{array}{ccc} (\mathcal{S}(X) \otimes \mathcal{S}(Y))_m & \xleftarrow{\nu} & S_m(X \times Y) \\ \downarrow{\scriptstyle \omega} & & \downarrow{\scriptstyle \lambda_\#} \\ (\mathcal{S}(Y) \otimes \mathcal{S}(X))_m & \xleftarrow{\nu} & S_m(Y \times X) \end{array}$$

where we define ω by the equation

$$\omega(c_p \otimes c_q) = (-1)^{pq}(c_q \otimes c_p)\,.$$

Then ω is a chain map, as you can check. [Without the $(-1)^{pq}$ it would not be a chain map.] We shall show that the diagram commutes up to chain homotopy.

The chain maps $\lambda_\#$ and ν are known to preserve augmentation, and they are natural with respect to homomorphisms induced by continuous maps. One checks readily that ω has the same properties. Consider the functors

$$\begin{aligned} G(X,Y) &= \mathcal{S}(X \times Y)\,, & G'(X,Y) &= \mathcal{S}(Y) \otimes \mathcal{S}(X)\,, \\ G(f,g) &= (f \times g)_\#\,, & G'(f,g) &= g_\# \otimes f_\#\,. \end{aligned}$$

We have already shown that these functors are free and acyclic relative to the collection of model objects $\mathcal{M} = \{\Delta_p, \Delta_q\}$. Therefore, any two natural transformations of G to G' are naturally chain homotopic. The chain maps $\omega \circ \nu$ and $\nu \circ \lambda_\#$ are two such natural transformations, as we noted. Therefore they are chain homotopic.

Consider the dual diagram

$$\begin{array}{ccccc} \bigoplus S^p(X;R) \otimes S^q(Y;R) & \xrightarrow{\theta} & \mathrm{Hom}\,((\mathcal{S}(X) \otimes \mathcal{S}(Y))_m, R) & \xrightarrow{\widetilde{\nu}} & S^m(X \times Y; R) \\ \uparrow \eta & & \uparrow \widetilde{\omega} & & \uparrow \lambda_\# \\ \bigoplus S^q(Y;R) \otimes S^p(X;R) & \xrightarrow{\theta} & \mathrm{Hom}\,((\mathcal{S}(Y) \otimes \mathcal{S}(X))_m, R) & \xrightarrow{\widetilde{\nu}} & S^m(Y \times X; R) \end{array}$$

where we define η by the equation

$$\eta(c^q \otimes c^p) = (-1)^{pq} c^p \otimes c^q\,.$$

The first square of this diagram commutes, as you can check. The second square commutes up to cochain homotopy. Therefore, the induced cohomology diagram actually commutes. Then **(a)** follows.

(b) & (c) While it is possible to prove **(b)** and **(c)** using acyclic model arguments, it is easier to use the cochain product formula of Lemma 7.11.2. In particular, the proof of **(b)** is trivial if one uses this formula.

To verify **(c)**, let z^p be a cocycle of X that represents the element α^p of $H^p(X;R)$. Then $\alpha^p \times 1_Y$ is represented by the cochain $z^p \times z^0$, where z^0 is the cocycle of Y whose value is 1 on each singular 0-simplex of Y. If $T : \Delta_p \to X \times Y$ is a singular p-simplex, we compute

$$\begin{aligned} \langle z^p \times z^0, T\rangle &= \langle z^p, \pi_1 \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle z^0, \pi_2 \circ T \circ \ell(\epsilon_p)\rangle \\ &= \langle z^p, \pi_1 \circ T\rangle \cdot 1 = \langle \pi_1^\#(z^p), T\rangle\,. \end{aligned}$$

Therefore $\alpha^p \times 1_Y = \pi_1^*(\alpha^p)$, as claimed. The proof for $1_X \times \beta^q$ is similar. □

We note that all three of these properties, suitably interpreted, hold for the other version of the cross product. Anticommutativity **(a)** holds if one uses $(\mathbb{Z}, G)$ coefficients on the right. Associativity **(b)** holds if two of the coefficients groups are $\mathbb{Z}$ and one is G. The first statement in **(c)** holds if $i_Y \in H^0(Y)$ and $\alpha^p \in H^p(X;G)$, since then $\alpha^p \times 1_Y \in H^p(X \times Y; G)$ and the statement makes sense. A similar remark applies to $1_X \times \beta^q$.

The similarity of the cochain cross product formula to the formula for the cup product may make you suspect that there is some relation between the two. That relation is described in the next theorem.

Theorem 7.11.4 *If $d : X \to X \times X$ is the diagonal map, given by $d(x) = (x, x)$, then*

$$d^*(\alpha^p \times \beta^q) = \alpha^p \smile \beta^q$$

holds, for both versions of the cohomology cross product.

Proof: Let z^p and z^q be representative cocycles for α^p and β^q, respectively. If $T : \Delta_{p+q} \to X$ is a singular $(p+q)$-simplex, then we compute

$$\begin{aligned}\langle d^{\#}(z^p \times z^q), T\rangle &= \langle z^p \times z^q, d \circ T\rangle \\ &= \langle z^p, \pi_1 \circ (d \circ T) \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle z^q, \pi_2 \circ (d \circ T) \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_m)\rangle \\ &= \langle z^p \smile z^q, T\rangle\,,\end{aligned}$$

where we have used the fact that $\pi_1 \circ d = i_X = \pi_2 \circ d$. □

This last theorem is due to Lefschetz. It shows why cohomology has a ring structure, but homology does not. Both homology and cohomology have cross products, but only in cohomology can one form the composite of the cross product with the homomorphism induced by the diagonal map. In homology one has the diagram

$$H_p(X) \otimes H_q(X) \xrightarrow{\times} H_{p+q}(X \times X) \xleftarrow{d_*} H_{p+q}(X)\,,$$

and we see that the mapping d_* goes in the wrong direction!

Corollary 7.11.5 *Cup products are anticommutative in that*

$$\alpha^p \smile \beta^q = (-1)^{pg}\beta^q \smile \alpha^p$$

holds, for both versions of the cup product.

Proof: Let d be the diagonal map as before, and let $\lambda : X \times X \to X \times X$ be the map that reverses coordinates. Then since $\lambda \circ d = d$, we compute

$$\begin{aligned}\alpha^p \smile \beta^q = d^*(\alpha^p \times \beta^q) &= (\lambda \circ d)^*(\alpha^p \times \beta^q) \\ &= d^*((-1)^{pq}\beta^q \smile \alpha^p) = (-1)^{pg}\beta^q \smile \alpha^p\,.\end{aligned}$$ □

Next we compute cup products in the cohomology of a product space.

Theorem 7.11.6 *In the cohomology ring $H^*(X \times Y; R)$, the following formula holds:*

$$(\alpha \times \beta) \smile (\alpha' \times \beta') = (-1)^{pq}(\alpha \smile \alpha') \times (\beta \smile \beta')\,,$$

where $p = \dim \alpha'$ and $q = \dim \beta$.

Proof: Let $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ be the projection maps.

Step 1. First, we prove the formula when β and α' are the unity elements of their respective cohomology rings.

Let z^p be a cocycle representing α, and let z^q be a cocycle representing β'. Then $\alpha \times \beta'$ is represented by $z^p \times z^q$. We compute

$$\begin{aligned}\langle z^p \times z^q, T\rangle &= \langle z^p, \pi_1 \circ T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle z^q, \pi_2 \circ T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_m)\rangle \\ &= \langle \pi_1^{\#}(z^p), T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \epsilon_p)\rangle \cdot \langle \pi_2^{\#}(z^q), T \circ \ell(\epsilon_p, \epsilon_{p+1}, \ldots, \epsilon_m)\rangle \\ &= \langle \pi_1^{\#}(z^p) \smile \pi_2^{\#}(z^q), T\rangle\,.\end{aligned}$$

Thus we have the desired formula

$$\alpha \times \beta' = \pi_1^*(\alpha) \smile \pi_2^*(\beta') = (\alpha \times 1_Y) \smile (1_X \times \beta')\,.$$

Step 2. Next, we prove the formula when β and β' both equal the unity element of their cohomology ring.

By naturality of cup products, we have

$$\pi_1^*(\alpha) \smile \pi_1^*(\alpha') = \pi_1^*(\alpha \smile \alpha')\,.$$

This last formula tells us that

$$(\alpha \times 1_Y) \smile (\alpha' \times 1_Y) = (\alpha \smile \alpha') \times 1_Y\,.$$

Step 3. Finally, we prove the general case.

We compute

$$\begin{aligned}(\alpha \times \beta) \smile (\alpha' \times \beta') &= (\alpha \times 1_Y) \smile (1_X \times \beta) \smile (\alpha' \times 1_Y) \smile (1_X \times \beta') \\ &= (-1)^{pq}(\alpha \times 1_Y) \smile (\alpha' \times 1_Y) \smile (1_X \times \beta) \smile (1_X \times \beta') \\ &= (-1)^{pq}(\alpha \smile \alpha') \times (\beta \smile \beta')\,.\end{aligned}$$

□

The preceding theorem enables us to compute the cup product of certain elements of the cohomology ring $H^*(X \times Y; R)$ in terms of the cup product operations in $H^*(X)$ and $H^*(Y)$. In the case where the Künneth theorem for cohomology holds, we can state this fact more formally, as we do in Theorem 7.11.8 below.

Definition 7.11.7 Given X and Y, we give the tensor product

$$H^*(X; R) \otimes_R H^*(Y; R)$$

the structure of a ring by defining

$$(\alpha \otimes \beta) \smile (\alpha' \otimes \beta') = (-1)^{pq}(\alpha \smile \alpha') \otimes (\beta \smile \beta')\,,$$

if $\beta \in H^p(Y; R)$ and $\alpha' \in H^q(X; R)$.

You can check that this operation is well-defined, and satisfies the axioms for a ring. In particular, the signs come out right for associativity.

Theorem 7.11.8 *If $H_i(X)$ is finitely generated for all i, then the cohomology cross product defines a monomorphism of rings*

$$\times : H^*(X) \otimes H^*(Y) \to H^*(X \times Y)\,.$$

If F is a field, then

$$\times : H^*(X;F) \otimes_F H^*(Y;F) \to H^*(X \times Y;F)$$

defines an isomorphism of algebras.

EXAMPLE 7.11.9 Consider the cohomology ring of $\mathbb{S}^n \times \mathbb{S}^m$, where $n, m \geq 1$. Let $\alpha^n \in H^n(\mathbb{S}^n)$ and $\beta^m \in H^m(\mathbb{S}^m)$ be generators. Then $H^*(\mathbb{S}^n \times \mathbb{S}^m)$ is free of rank 4, with basis 1×1, $\alpha \times 1$, $1 \times \beta$, and $\alpha \times \beta$. The only non-trivial product of positive dimensional elements is $(\alpha \times 1) \smile (1 \times \beta) = \alpha \times \beta$.

In the special case $n = m = 1$, this space is the torus, and this is the cohomology ring we have already computed in Example 5.10.5. Let us picture these cohomology classes. Represent the torus by the usual diagram, as in Figure 7.11.1. If X denotes the subspace $abca$, then a generator α for its 1-dimensional cohomology is represented by the cocycle $x^1 = [b,c]^*$. Since the projection map π_1 is simplicial, the cocycle $\pi_1^\#(x^1)$ is the cocycle w^1 of T, as pictured. This cocycle represents the cohomology class $\alpha \times 1$. Similarly, the cocycle z^1 pictured in Figure 7.11.2 represents the cohomology class $1 \times \beta$, where β is represented by the cocycle $[d,e]^*$ of the space $Y = adea$. We have $(\alpha \times 1) \smile (1 \times \beta) = \alpha \times \beta$, and this class generates $H^2(T)$.

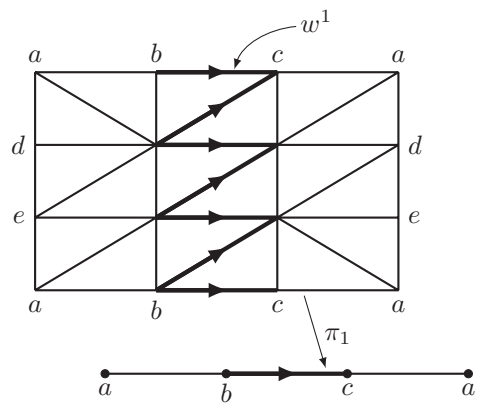


Figure 7.11.1

EXAMPLE 7.11.10 Consider the cohomology ring of $\mathbb{P}^2 \times \mathbb{P}^2$, with $\mathbb{Z}/2$ coefficients. Let $\alpha \in H^1(\mathbb{P}^2;\mathbb{Z}/2)$ be the nonzero element. Then $\alpha^2 = \alpha \smile \alpha$

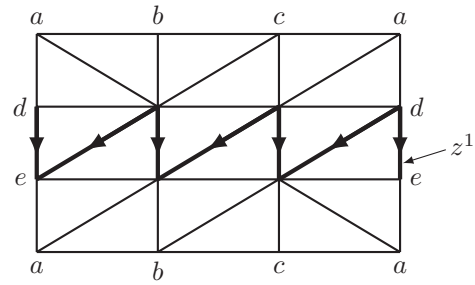


Figure 7.11.2

is nonzero. [See the exercises of Section 5.10.] By Theorem 7.11.8, the vector space $H^*(\mathbb{P}^2 \times \mathbb{P}^2; \mathbb{Z}/2)$ has dimension 9 with the following basis elements:

$$\begin{array}{lllll}
\dim 0 & 1 \times 1\,, & & & \\
\dim 1 & \alpha \times 1, & 1 \times \alpha\,, & & \\
\dim 2 & \alpha^2 \times 1, & \alpha \times \alpha, & 1 \times \alpha^2\,, & \\
\dim 3 & \alpha^2 \times \alpha, & \alpha \times \alpha^2\,, & & \\
\dim 4 & \alpha^2 \times \alpha^2\,. & & &
\end{array}$$

The multiplication table is easily written down. No signs are involved because the coefficient field is $\mathbb{Z}/2$.

The following definition will be used in the Exercise 3:

Definition 7.11.11 If M and N are connected n-manifolds, then the *connected sum*, $M \# N$, is obtained by removing an open n-ball from each of M and N and pasting together what remains along their boundaries. For the manifolds in Exercise 3, the resulting manifold is well-defined up to homeomorphism. In general, one must bring in the issue of "orientation."

EXERCISES

1. (a) Suppose that $A \subset X$ and $B \subset Y$. Show that if $\{X \times B, A \times Y\}$ is excisive, then there is a relative cohomology cross product

$$\times : H^p(X, A) \otimes H^q(Y, B) \to H^{p+q}(X \times Y, (X \times B) \cup (A \times Y))$$

for our usual sets of coefficients.

(b) Prove Theorem 7.11.4 and Corollary 7.11.5 for relative cup products.

2. **(a)** Suppose that the cohomology cross product is defined as in this section, and that cup products are *defined* by the formula

$$\alpha \smile \beta = d^*(\alpha \times \beta)\,,$$

where d is the diagonal map. Use properties of the cross product to show that $\smile$ is bilinear and associative, and that the class $\{z_0\}$ acts as a unity element.

(b) Suppose that cup products are defined as in Section 5.9, and that cohomology cross products are *defined* by the formula

$$\alpha \times \beta = \pi_1^*(\alpha) \smile \pi_2^*(\beta)\,,$$

where $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are the projections. Use properties of cup products to show that the cross product is bilinear and associative, and that

$$\pi_1^*(\alpha) = \alpha \times 1_Y \quad \text{and} \quad \alpha \smile \beta = d^*(\alpha \times \beta)\,.$$

3. Compute the following cohomology rings:

(a) $H^*(\mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{S}^3)$.

(b) $H^*((T \# T) \times \mathbb{S}^3)$, where T is the torus.

(c) $H^*((\mathbb{S}^1 \times \mathbb{S}^5) \# (\mathbb{S}^2 \times \mathbb{S}^4))$.

4. Let M be a compact, connected n-manifold with $H^n(M) \cong \mathbb{Z}$. Use the results of the preceding exercise to formulate a conjecture about the cup product operation

$$H^k(M) \otimes H^{n-k}(M) \to H^n(M)\,.$$

5. Let S denote the Klein bottle and let T denote the torus. Compute the following cohomology rings:

(a) $H^*(T \times S)$.

(b) $H^*(T \times S; \mathbb{Z}/2)$.

6. Verify, in the proof of part **(a)** of Theorem 7.11.3, that ω is a chain map.

8

Duality in Manifolds

8.1 Introduction

As we have mentioned before, manifolds are among the most familiar and important geometric objects in mathematics. It took quite some time for mathematicians to produce a rigorous definition of the manifold concept. It was Hermann Weyl, in his book on Riemann surfaces,[1] who finally nailed down the idea.

Any theorems that we can prove about manifolds are likely to be useful, not only in topology, but also in differential geometry and other branches of mathematics. The duality theorems are examples of such theorems.

We already know one relationship that holds between the homology and cohomology groups of an arbitrary topological space; it is expressed by the universal coefficient theorem for cohomology. For manifolds, there is a second such relationship. It is expressed in the famous Poincaré duality theorem. If we think of the universal coefficient theorem as expressing a sort of algebraic duality between cohomology and homology, then Poincaré duality can be thought of as fundamentally geometric in nature. This duality does not hold for an arbitrary space, but depends specifically on properties possessed by manifolds.

There is a certain amount of interplay between these geometric and algebraic types of duality. This interplay has some interesting applications—we shall apply it to the problem of computing the cohomology ring of a manifold.

There are further duality theorems. They bear the names of Lefschetz, Alexander, and Pontryagin. We shall study them in due course. All are concerned with algebraic-topological properties of manifolds.

Throughout this discussion, we assume familiarity with local homology groups [Section 4.8].

[1] *Die Idee der Riemannschen Fläche*, [Teubner, Leipzig-Berlin, 1913]. An English translation of the third edition is available in [Wey64].

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8.2 The Join of Two Complexes

In this section we introduce the notion of the join of two complexes, and consider some of it properties.

Definition 8.2.1 Let K and L be nonempty complexes in some euclidean space $\mathbb{E}^J$. Let $s = v_0v_1 \dots v_m$ be the general simplex of K and let $t = w_0w_1 \dots w_n$ be the general simplex of L. Suppose that, whenever $s \in K$ and $t \in L$, then the points $v_0, v_1, \dots, v_m, w_0, w_1, \dots, w_n$ are independent. Then we let

$$s * t = v_0v_1 \dots v_m\, w_0w_1 \dots w_n$$

denote the simplex that they span. If the collection consisting of all the simplices $s * t$ and their faces is a simplicial complex, then this complex is called the *join* of K and L and is denoted by $K * L$.

If K consists of a single vertex v, then $v * L$ is the cone over L, as already defined. If K consists of two points, then $K * L$ is the suspension of L and is denoted $S(L)$. Conditions under which $K * L$ exists in general are given in the following lemma.

Lemma 8.2.2 *Let K and L be disjoint nonempty complexes in $\mathbb{E}^J$.*

> **(a)** *If $K * L$ exists, then its polytope equals the union of all line segments joining points of $|K|$ to points of $|L|$. Two such line segments intersect in at most a common endpoint.*
>
> **(b)** *Conversely, if every pair of line segments joining points of $|K|$ to points of $|L|$ intersect in at most a common endpoint, then $K * L$ exists.*

Proof:

Step 1: We will prove the following: Let $\sigma = v_0v_1 \dots v_mw_0w_1 \dots w_n$ be a simplex in $\mathbb{E}^J$. Then σ equals the union of all line segments joining points of $s = v_0v_1 \dots v_m$ to points of $t = w_0w_1 \dots w_n$. Two such line segments intersect in at most a common endpoint. Furthermore, $\operatorname{Int}\sigma$ equals the union of all open line segments joining points of $\operatorname{Int} s$ to points of $\operatorname{Int} t$. This result was given as an exercise in Section 1.2. We now give a proof.

Let us now prove the statement about $\operatorname{Int}\sigma$ first. Suppose that $p = \lambda x + (1 - \lambda)y$, where $0 < \lambda < 1$ and $x \in \operatorname{Int} s$ and $y \in \operatorname{Int} t$. Then $x = \sum \alpha_i v_i$ and $y = \sum \beta_j w_j$ with $\alpha_i > 0$ and $\beta_j > 0$ and $\sum \alpha_i = \sum \beta_j = 1$. Thus

$$p = \sum (\lambda\alpha_i)v_i + \sum (1 - \lambda)\beta_j w_j\,,$$

where the coefficients are positive and their sum is 1. Thus p is in $\operatorname{Int}\sigma$. Conversely, suppose that $p \in \operatorname{Int}\sigma$ so that

$$p = \sum \gamma_i v_i + \sum \delta_j w_j \,,$$

where $\gamma_i > 0$ and $\delta_j > 0$ and $\sum \gamma_i + \sum \delta_j = 1$. Set $\lambda = \sum \gamma_i$. Set $\alpha_i = \gamma_i/\lambda$ and set $\beta_j = \delta_j/(1-\lambda)$, then

$$p = \lambda \sum \alpha_i v_i + (1-\lambda) \sum \beta_j w_j \,.$$

Thus p lies interior to a line segment joining a point of $\operatorname{Int} s$ and a point of $\operatorname{Int} t$.

Next we will prove the statement about σ. Since σ is convex, it contains all line segments joining points of s to points of t. Conversely, suppose that $x \in \sigma$ and x is in neither s nor t. Then x lies interior to a face σ' of σ that has vertices in common with both s and t. By the result of the preceding paragraph, x lies on an open line segment joining a point of $\sigma' \cap s$ with a point of $\sigma' \cap t$.

Finally, we show that any two of these line segments intersect in at most a common endpoint. Suppose that p is a point lying on two such line segments—that is to say, p equals

$$\lambda \sum \alpha_i v_i + (1-\lambda) \sum \beta_j w_j = \ \lambda' \sum \alpha_i' v_i + (1-\lambda') \sum \beta_j' w_j \,,$$

where $0 \le \lambda, \lambda' \le 1$ and $\alpha_i \ge 0$, $\beta_j \ge 0$, $\alpha_i' \ge 0$, $\beta_j' \ge 0$, and $\sum \alpha_i = \sum \beta_j = \sum \alpha_i' = \sum \beta_j' = 1$. Since the sum of the coefficients on either side of this equation is 1, we conclude from the independence of the vertices of σ that

$$\lambda \alpha_i = \lambda' \alpha_i'$$

and

$$(1-\lambda)\beta_j = (1-\lambda')\beta_j' \,.$$

Summing the first of these, we see that $\lambda = \lambda'$. If $\lambda = 0$ or $\lambda = 1$, then p is an endpoint of each line segment. Otherwise, these two equations imply that $\alpha_i = \alpha_i'$ and $\beta_j = \beta_j'$ and thus the two line segments coincide.

Step 2: Now we prove part **(a)** of the lemma. Consider the distinct line segments xy and $x'y'$, where

$$x \in \operatorname{Int} s \text{ and } x' \in \operatorname{Int} s' \ \text{ with } \ s, s' \in K \,,$$

$$y \in \operatorname{Int} t \text{ and } y' \in \operatorname{Int} t' \ \text{ with } \ t, t' \in L \,.$$

Since the interior of the line segment xy lies in $\operatorname{Int}(s * t)$, we have the inclusions

$$xy \subset (\operatorname{Int} s) \cup (\operatorname{Int}(s * t)) \cup (\operatorname{Int} t) \,,$$

$$x'y' \subset (\operatorname{Int} s') \cup (\operatorname{Int}(s' * t')) \cup (\operatorname{Int} t') \,.$$

Because $K * L$ is a complex, these six open simplices are disjoint if $s \neq s'$ and $t \neq t'$, whence xy and $x'y'$ are disjoint. If $s = s'$ and $t \neq t'$, then they intersect only in the set $\operatorname{Int} s$, so the line segments intersect in at most the endpoints x, x'. If $s \neq s'$ and $t = t'$, then it follows from **Step 1** that the line segments intersect in at most a common endpoint.

Step 3: We now prove part **(b)** of the Lemma. First we show that if $v_0 v_1 \dots v_m \in K$ and $w_0 w_1 \dots w_n \in L$, then the points $v_0, v_1, \dots, v_m, w_0, w_1, \dots, w_n$ are linearly independent. Consider the linear singular simplex

$$\ell = \ell(v_0, v_1, \dots, v_m, w_0, w_1, \dots, w_n)\,,$$

mapping Δ_{m+n+1} into $\mathbb{E}^J$. This map carries each line segment joining a point of $\epsilon_0 \epsilon_1 \dots \epsilon_m$ to a point of $\epsilon_{m+1} \dots \epsilon_{m+n+1}$ linearly onto a line segment joining a point of $v_0 v_1 \dots v_m$ to a point of $w_0 w_1 \dots w_n$. Then, by hypothesis, ℓ is injective. As a result, the plane spanned by these points must have dimension $m + n + 1$. Otherwise, the linear map ℓ could not be injective. Hence $v_0, v_1, \dots, v_m, w_0, w_1, \dots, w_n$ are independent.

It follows that $K * L$ exists. For if two simplices of $K * L$ have interiors that intersect, then one of them must be of the form $s * t$. Each point of $\operatorname{Int} s * t$ lies on an open line segment joining a point of $|K|$ to a point of $|L|$. If $\operatorname{Int} s * t$ and $\operatorname{Int} s' * t'$ intersect, then two such open line segments must intersect. If $\operatorname{Int} s * t$ and $\operatorname{Int} s'$ intersect in a point y, then y lies interior to one such line segment and is also the endpoint of another. Similarly if $\operatorname{Int} s * t$ and $\operatorname{Int} t'$ intersect. Each of these situations is contrary to our hypotheses. □

Now suppose that K and L are complete, but they are not situated in euclidean space in such a way that $K * L$ is defined. Then one can find complexes K_0 and L_0 isomorphic to K and L, respectively, such that $K_0 * L_0$ is defined. For instance, one can take K_0 and L_0 to be disjoint subcomplexes of the standard simplex Δ^J for some index J.

Lemma 8.2.3 *Suppose that $K * L$ exists, and suppose that K is locally finite. Then the map*

$$\pi : |K| \times |L| \times I \to |K * L|$$

defined by

$$\pi(x, y, t) = (1 - t)x + ty$$

is a quotient map. For each $x \in |K|$ and $y \in |L|$, it collapses $x \times |L| \times 0$ to a point and $|K| \times y \times 1$ to a point. Otherwise, it is one-to-one.

Proof: The topology of $|K| \times |L| \times I$ is coherent with the subspaces $\sigma \times \tau \times I$ for $\sigma \in K$ and $\tau \in L$. To prove this assertion, we apply the results of Section 2.8. The topology of $|L| \times I$ is coherent with the subspaces $\tau \times I$ for $\tau \in L$. Because $|K|$ is locally compact Hausdorff, the topology of $|K| \times |L| \times I$ is thus coherent with the subspaces $|K| \times \tau \times I$. Because $\tau \times I$ is locally compact Hausdorff

(in fact compact), the topology of $|K| \times \tau \times I$ is in turn coherent with the subspaces $\sigma \times \tau \times I$ for $\sigma \in K$ and $\tau \in L$. Thus A is closed in $|K| \times |L| \times I$ if and only if its intersection with each space $\sigma \times \tau \times I$ is closed in that space.

Now π is continuous as a map of $\sigma \times \tau \times I$ onto $\sigma * \tau$ by definition. [Both are subspaces of euclidean space.] And inclusion of $\sigma * \tau$ in $|K * L|$ is continuous. Thus $\pi | \sigma \times \tau \times I$ is continuous. Because the topology of $|K| \times |L| \times I$ is coherent with these subspaces, we see that π is continuous.

To show that π is a quotient map, suppose that $\pi^{-1}(C)$ is closed in $|K| \times |L| \times I$. Then its intersection with $\sigma \times \tau \times I$ is closed and hence compact. We conclude that $C \cap (\sigma * \tau)$ is compact and therefore closed. Hence C is closed in $K * L$.

The fact that π carries out precisely the indicated identifications follows from Lemma 8.2.2. □

Notation 8.2.4 Let X and Y be topological spaces. We shall write $X \approx Y$ to indicate that X and Y are homeomorphic.

Corollary 8.2.5 *Suppose that $K * L$ and $M * N$ are defined, where K is locally finite. If $|K| \approx |M|$ and $|L| \approx |N|$, then $|K * L| \approx |M * N|$.*

Proof: Since $|K|$ is locally compact, so is $|M|$. Thus M is locally finite. If $h : |K| \to |M|$ and $k : |L| \to |N|$ are homeomorphisms, then $h \times k \times i_I$ induces, via the quotient maps indicated in the preceding lemma, a homeomorphism of $|K * L|$ with $|M * N|$ that equals h on $|K|$ and k on $|L|$. □

Lemma 8.2.6 *Let J, K, and L be complexes. Assume that $J * K$ and $(J * K) * L$ exist. Then $J * K = K * J$ and $(J * K) * L = J * (K * L)$.*

Proof: The symmetry of the definition shows that $J * K = K * J$. Similarly, if $(J * K) * L$ is defined then for $v_0 v_1 \dots v_m \in J$ and $w_0 w_1 \dots w_n \in K$ and $x_0 x_1 \dots x_p \in L$ the points

$$v_0, v_1, \dots, v_m, w_0, w_1, \dots, w_n, x_0, x_1, \dots, x_p$$

are independent and the collection of simplices they span, along with faces of such simplices, forms a complex. But this is just the complex $J * (K * L)$. □

Theorem 8.2.7 *Assume that $K * L$ exists. If $|L| \approx \mathbb{S}^{n-1}$, then for all i,*

$$\widetilde{H}_{i+n}(K * L) \cong \widetilde{H}_i(K)\,.$$

Proof: In the case $n = 1$, $|L|$ consists of two points and the complex $K * L$ is just the suspension of K. The existence of the isomorphism in question is a consequence of the Mayer-Vietoris sequence [see Theorem 3.4.6].

In general, we suppose the theorem true in dimension n and prove inductively that it is true in dimension $n + 1$.

In view of Corollary 8.2.5, it suffices to prove the theorem for any particular complex L whose space is homeomorphic to $\mathbb{S}^n$. Choose a complex J such that $|J| = \mathbb{S}^{n-1}$ and let $L = j * \{w_0, w_2\}$ be a suspension of J. Then it is easy to see that $|L| \approx \mathbb{S}^n$. Now

$$\begin{aligned} \widetilde{H}_i(K) &\cong \widetilde{H}_{i+n}(K * J) \\ &\cong \widetilde{H}_{i+n+1}((K * J) * [w_0, w_1]) \\ &\cong \widetilde{H}_{i+n+1}(K * L)\,. \end{aligned}$$

Here the first equivalence is by the inductive hypothesis, the second is as noted earlier, and the third is by Lemma 8.2.6. □

Now we apply these results to study local properties of arbitrary simplicial complexes.

Definition 8.2.8 Let s be a simplex of the complex K. The *star* of s in K, denoted by $\operatorname{St} s$, is the union of the interiors of all simplices of K having s as a face. The closure of $\operatorname{St} s$ is denoted by $\overline{\operatorname{St}}\, s$. It is the union of all simplices of K having s as a face and is called the *closed star* of s in K. The *link* of s in K, denoted by $\operatorname{Lk} s$, is the union of all simplices of K lying in $\overline{\operatorname{St}}\, s$ that are disjoint from s.

We have already given these definitions when s is a vertex. See Section 1.3. In general, if $s = v_0 v_1 \dots v_n$, then $\operatorname{St} s$ is the open subset of K consisting of those points x for which the barycentric coordinate of x on each of $v_0, v_1, \dots, v_n$ is positive. That is,

$$\operatorname{St} s = \operatorname{St} v_0 \cap \operatorname{St} v_1 \cap \cdots \cap \operatorname{St} v_n\,.$$

Observe that $\operatorname{St} s$ consists of all simplices of K of the form $s * t$. Also the link of s equals the union of all the faces t of such simplices. The sets $\overline{\operatorname{St}}\, s$ and $\operatorname{Lk} s$ are thus polytopes of subcomplexes of K. We often use the notation $\overline{\operatorname{St}}\, s$ and $\operatorname{Lk} s$ to denote these complexes as well as their polytopes.

EXAMPLE 8.2.9 In the two-dimensional complex pictured in Figure 8.2.1, the link of the vertex g consists of the hexagon $a\,b\,c\,d\,e\,f\,a$ and the vertex h. The link of the 1-simplex ag consists of the two vertices b and f. The link of the vertex h is the vertex g.

In the 3-dimensional complex pictured in Figure 8.2.2, the link of the 1-simplex fg is the pentagon $F = a\,b\,c\,d\,e\,a$ and the link of the vertex f is the cone $P * g$. The link of the 1-simplex ab is the 1-simplex $f\,g$. And the link of the vertex a is the union of the 2-simplices $b\,f\,g$ and $e\,f\,g$.

Keep the preceding example in mind as we prove the following lemma:

Lemma 8.2.10 *Let K be a complex. Let s be a simplex of K. Then*

$$\overline{\operatorname{St}}\, s = s * \operatorname{Lk} s\,,$$

$$\overline{\operatorname{St}}\, s - \operatorname{St} s = \operatorname{Bd} s * \operatorname{Lk} s\,.$$

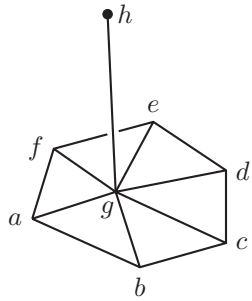


Figure 8.2.1

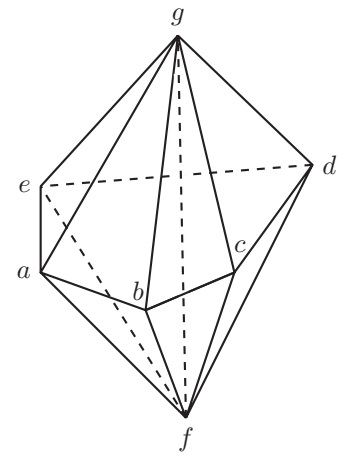


Figure 8.2.2

Note that here we are considering s and $\operatorname{Bd} s$ and $\operatorname{Lk} s$ not only as subspaces of $|K|$ but also as subcomplexes of K. In order that these equations should hold even if $\operatorname{Lk} s$ or $\operatorname{Bd} s$ is empty, we make the convention that $K * \emptyset = \emptyset * K = K$ for all K.

Proof: The first equation is immediate from the definitions. Also $\overline{\operatorname{St}} s$ is the union of all simplices of K of the form $s * t$. And $\operatorname{Lk} s$ is the union of all the faces t of such simplices.

On the other hand, if a simplex of K lies in $\overline{\operatorname{St}} s$ but not in $\operatorname{St} s$, then it must be a face of $s * t$ of the form $s' * t'$, where s' is a *proper* face of s and t' is a face of t. Since t lies in $\operatorname{Lk} s$, so does t'. Hence $s' * t'$ lies in $\operatorname{Bd} s * \operatorname{Lk} s$. Conversely, every simplex of this form lies in $\overline{\operatorname{St}} s$ but not in $\operatorname{St} s$. □

The next definition extends Definition 8.2.1. It will be used in Exercise 2.

Definition 8.2.11 If X and Y are topological spaces then let us define $X * Y$ to be the quotient space of $X \times Y \times I$ obtained by identifying each set $x \times Y \times 0$ to a point and each set $X \times y \times 1$ to a point.

EXERCISES

1. Let s denote the simplex fg in Figure 8.2.2. Describe $\widehat{s} * \operatorname{Lk} s$ and $\widehat{s} * \operatorname{Bd} s$ and $\operatorname{Bd} s * \operatorname{Lk} s$.

2. **(a)** Show that the maps

$$i(x) = x \times Y \times 0 \quad \text{and} \quad j(y) = X \times y \times 1$$

define embeddings of X and Y respectively into $X * Y$.

(b) Show that if $X \approx X'$ and $Y \approx Y'$, then $X * Y \approx X' * Y'$.

(c) Show that $X * Y \approx Y * X$.

3.* Show that there is a split short exact sequence

$$0 \to \widetilde{H}_{p+1}(X * Y) \to \widetilde{H}_p(X \times Y) \to \widetilde{H}_p(X) \oplus \widetilde{H}_p(Y) \to 0\,.$$

4.* Show that $(X * Y) * Z \approx X * (Y * Z)$. [**Hint:** The obvious proof does not work. You can check that the quotient maps

$$(X \times Y \times I_1) \times Z \times I_2 \to (X * Y) \times Z \times I_2 \to (X * Y) * Z$$

and

$$X \times (Y \times Z \times I_2) \times I_1 \to X \times (Y * Z) \times I_1 \to X * (Y * Z)$$

carry out different identifications. Instead, in the space $X \times Y \times Z \times \Delta_2$, introduce the following relations:

$$\begin{aligned} (x, y, z, w) &\sim (x', y, z, w) \quad \text{if} \quad w \in \epsilon_1\epsilon_2\,, \\ (x, y, z, w) &\sim (x, y', z, w) \quad \text{if} \quad w \in \epsilon_0\epsilon_2\,, \\ (x, y, z, w) &\sim (x, y, z', w) \quad \text{if} \quad w \in \epsilon_0\epsilon_1\,. \end{aligned}$$

Let W be the quotient space. Let π be the quotient map. Let $f : I_1 \times I_2 \to \Delta_2$ map $t \times I_2$ linearly onto the line segment from $t \in \epsilon_0\epsilon_1 = [0, 1]$ to ϵ_2. Show that

$$X \times Y \times Z \times I_1 \times I_2 \xrightarrow{\text{id} \times f} X \times Y \times Z \times \Delta_2 \xrightarrow{\pi} W$$

carries out the same identifications as the first of the preceding quotient maps. Conclude that W is homeomorphic to $(X * Y) * Z$. Show that W is also homeomorphic to $X * (Y * Z)$.]

5. Let K be a triangle and L be a triangle. What is the join of K and L?

6. Write a figure 8 as the join of two simplices.

8.3 Homology Manifolds

In this section we shall define homology manifolds and obtain some of their local properties. The class of homology manifolds includes all topological manifolds, so it is a broad and important collection of spaces. The triangulable homology manifolds will be the basic objects involved in the various duality theorems.

For convenience, we shall deal with the case of relative homology manifolds in the present section, even though we shall not actually prove the duality theorems in the relative case until Section 8.10.

We begin now by recording a convention. We shall be dealing with triangulable spaces. If X is such a space with a specific triangulation, then we shall frequently abuse notation and make no notational distinction between the space and the complex triangulating it. For instance, we may refer to a point of [the space] X or to a simplex of [the complex] X. The meaning in each instance will be made clear from context.

Definition 8.3.1 A topological pair (X, A) is called a *relative homology n-manifold* if for each point x of X not in A, the local homology group $H_i(X, X \setminus x)$ vanishes if $i \neq n$ and is infinite cyclic if $i = n$. In the case where A is empty, we refer to X simply as a *homology n-manifold*.

If M is an n-manifold with boundary, then the pair $(\mathrm{M}, \mathrm{Bd}\, M)$ is a relative homology n-manifold (see Section 4.8). More generally, if (X, A) is any pair such that $X \setminus A$ is an n-manifold, then (X, A) is a relative homology n-manifold.

Now if (X, A) is a relative homology n-manifold that is triangulated, then the local homology properties of X are reflected in the homology groups of the links of simplices of X. The connection is presented in the following lemma.

Lemma 8.3.2 *Let s be a k-simplex of the complex K. Let $\widehat{s}$ be its barycenter. Then*

$$H_i(|K|, |K| \setminus \widehat{s}\,) \cong \begin{cases} \widetilde{H}_{i-k-1}(\mathrm{Lk}\, s) & \text{if} \quad \mathrm{Lk}\, s \neq \emptyset\,, \\ H_i(s, \mathrm{Bd}\, s) & \text{if} \quad \mathrm{Lk}\, s = \emptyset\,. \end{cases}$$

Proof: If $\mathrm{Lk}\, s = \emptyset$, then s is a face of no other simplex of K. Hence $\mathrm{Int}\, s$ is open in $|K|$. It follows that

$$H_i(|K|, |K| \setminus \widehat{s}\,) \cong H_i(s, s - \widehat{s}\,) \cong H_i(s, \mathrm{Bd}\, s)\,.$$

The first isomorphism holds by excision. The second holds because $\mathrm{Bd}\, s$ is a deformation retract of $s - \widehat{s}$.

Now suppose that $\mathrm{Lk}\, s \neq \emptyset$. Since the barycenter of s lies in the open set $\mathrm{St}\, s$, we may excise the complement of $\overline{\mathrm{St}}\, s$ to obtain an isomorphism

$$H_i(|K|, |K| \setminus \widehat{s}\,) \cong H_i(\overline{\mathrm{St}}\, s, \overline{\mathrm{St}}\, s - \widehat{s}\,)\,.$$

From basic results about convex sets we know that $|s| = |\widehat{s} * \operatorname{Bd} s|$. From Lemma 8.2.10 it then follows that

$$|\overline{\operatorname{St}}\, s| = |s * \operatorname{Lk} s| = |\widehat{s} * \operatorname{Bd} s * \operatorname{Lk} s|\,.$$

By the previously discussed convention, these equations hold even if $\operatorname{Bd} s$ is empty.

It follows now that $|\overline{\operatorname{St}}\, s|$ is acyclic, being as it is a cone with vertex $\widehat{s}$. Now if we delete the vertex of a cone, what is left can be collapsed onto the base of the cone by a deformation retract. [See Lemma 4.8.12.] In particular, $|\operatorname{Bd} s * \operatorname{Lk}, s|$ is a deformation retract of $\overline{\operatorname{St}}\, s - \widehat{s}$. We conclude that

$$\begin{aligned} H_i(|K|, |K| - \widehat{s}) &\cong H_i(\overline{\operatorname{St}}\, s, \overline{\operatorname{St}}\, s - \widehat{s}) \\ &\cong \widetilde{H}_{i-1}(\overline{\operatorname{St}}\, s - \widehat{s}) \\ &\cong \widetilde{H}_{i-1}(\operatorname{Bd} s * \operatorname{Lk} s) \\ &\cong \widetilde{H}_{i-k-1}(\operatorname{Lk} s)\,. \end{aligned}$$

The last isomorphism is an equality if $\operatorname{Bd} s = \emptyset$, in which case $k = 0$. Otherwise, it holds by Theorem 8.2.7 since $\operatorname{Bd} s$ is topologically a $k-1$ sphere. □

Theorem 8.3.3 *Let (X, A) be a triangulated relative homology n-manifold. Let s be a k-simplex of X that is not in A. If $\operatorname{Lk} s$ is empty, then $k = n$. If $\operatorname{Lk} s$ is non-empty, then it has the homology of an $n-k-1$ sphere, where $n-k-1 \geq 0$. In either case, $k \leq n$.*

Proof: Since s is not in A, its barycenter $\widehat{s}$ lies in $X \setminus A$. Therefore the local homology of X at $\widehat{s}$ is infinite cyclic in dimension n, and vanishes otherwise.

If $\operatorname{Lk} s$ is empty, then by the preceding lemma, $H_n(s, \operatorname{Bd} s) \cong H_n(X, X \setminus \widehat{s})$, which is nontrivial. Hence $\dim s = n$. If $\operatorname{Lk} s$ is nonempty, then its reduced homology is infinite cyclic in dimension $n-k-1$ and vanishes otherwise. This implies in particular that $n-k-1 \geq 0$ or that $k < n$. □

Corollary 8.3.4 *Let (X, A) be a triangulated relative homology n-manifold.*

(a) *The closure of $X \setminus A$ equals a union of n-simplices.*

(b) *Every $n-1$ simplex s of X which is not in A is a face of precisely two n-simplices of X.*

Proof: (a) Let s be a k-simplex of X which is not in A. The preceding theorem shows that $k \leq n$. If $k < n$ then $\operatorname{Lk} s$ is a homology $n-k-1$ sphere, so it contains an $n-k-1$ simplex t. Then s is in fact a face of the n-simplex $s * t$.

(b) If s is an $(n-1)$-simplex of X that is not in A, then $\operatorname{Lk} s$ is a homology 0-sphere. Since $\overline{\operatorname{St}}\, s$ has dimension n and $\overline{\operatorname{St}}\, s = s * \operatorname{Lk} s$, the complex $\operatorname{Lk} s$ must have dimension 0. This means that $\operatorname{Lk} s$ consists of precisely two points. Hence **(b)** holds. □

This theorem shows that a triangulated relative homology n-manifold (X, A) satisfies the first two conditions in the definition of a relative pseudo manifold [see Definition 5.4.5]. If the space $X \setminus A$ is connected, then (X, A) satisfies the third condition as well. A proof of this last fact is outlined in the exercises at the end of the section. The result can also be derived from the duality theorems, as we shall see below.

EXAMPLE 8.3.5 Let X be the disjoint union of a torus T and a two-sphere $\mathbb{S}^2$. Let $S(X)$ be the suspension of X. Let $A = \{w_0, w_1\}$ consist of the suspension points. See Figure 8.3.1. Then $(S(X), A)$ is a relative homology 3-manifold since $S(X) \setminus A$ is homeomorphic to $X \times (-1, 1)$, which is a 3-manifold.

The subspaces $(S(T), A)$ and $(S(\mathbb{S}^2), A)$ are also relative homology 3-manifolds. They are relative pseudo 3-manifolds as well. The space $S(S^2)$ is a homology 3-manifold but the space $S(T)$ is not. For the link of w_0 in $S(T)$ is not a homology 2-sphere but is instead a torus.

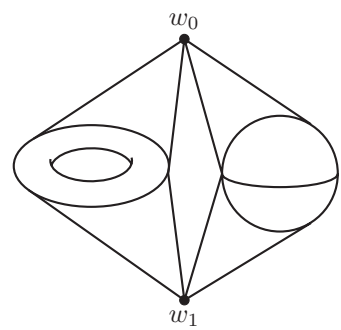


Figure 8.3.1

EXAMPLE 8.3.6 The pictures that we usually draw of manifolds may lead you to conjecture the following: If M is a triangulated topological n-manifold, then the link of each k-simplex of M is a topological $(n - k - 1)$-sphere.

Certainly, this result is true for our usual triangulations of 2-manifolds—for instance, the links of 1-simplices are 0-spheres and the links of vertices are 1-spheres. This conjecture was one of long standing until it was resolved by work of Robert D. Edwards and James W. Cannon. [See [Can79].] The details are complicated, and we omit them.

EXAMPLE 8.3.7 There do exist homology n-manifolds that are not topological manifolds. We outline the construction of one such, but the proof uses tools that we have not yet studied.

The basic fact that one needs is that there is a compact, triangulated 3-manifold M that is a homology 3-sphere, but whose fundamental group $\pi_1(M)$ does not vanish. The construction can be found in [ST80]. The fundamental group of M is the icosahedral group.

Now we form the suspension $S(M) = M * \{w_0, w_1\}$. This space is a homology 4-manifold. Since $S(M) \setminus \{w_0, w_1\}$ is homeomorphic to the 4-manifold $M \times (-1, 1)$, the local homology conditions are satisfied at all points except possibly at the suspension points. Since $\operatorname{Lk} w_0 = \operatorname{Lk} w_1 = M$, and M is a homology 3-sphere, the local homology conditions are also satisfied at w_0 and w_1.

Then one needs a separate argument to show that in a triangulated topological n-manifold, the link of each vertex, while it may not be topologically an $(n-1)$-sphere, must have the homotopy type of an $(n-1)$-sphere. Since $\pi_1(M) \neq 0$, M cannot have the homotopy type of a 3-sphere. Hence $S(M)$ cannot be a topological 4-manifold.

The proofs of the following theorem and corollary will be left to the exercises.

Theorem 8.3.8 *If (X, A) is a triangulated relative homology n-manifold with $X \setminus A$ connected, then (X, A) is a relative pseudo n-manifold.*

Corollary 8.3.9 *If (X, A) be a triangulated relative homology n-manifold, and if s is a complex of X not in A, then $\operatorname{Lk} s$ is a finite complex.*

EXERCISES

Each of the first four exercises depends on the preceding ones.

1. Let (X, A) be a triangulated pair. Suppose that for each k-simplex s of X that is not in A we have: **(i)** $k \leq n$, **(ii)** $H_i(\operatorname{Lk} s) \cong H_i(S^{n-k-1})$ if $k < n$. Show that (X, A) is a relative homology n-manifold.
2. Let (X, A) be a triangulated relative homology n-manifold. If s is a k-simplex of X that is not in A, with $k < n$, then show that $\operatorname{Lk} s$ is a homology $n-k-1$ manifold. [**Hint:** Consider $\operatorname{Lk} s$ as a subcomplex of X. Then if $t \in \operatorname{Lk} s$, show that $\operatorname{Lk}(t, \operatorname{Lk}(s, X)) = \operatorname{Lk}(t * s, X)$.]
3. Prove Theorem 8.3.8. [**Hint:** The result is trivial if $n = 1$. Now suppose it true for dimensions less than n. Define $\sigma \sim \sigma'$ if there is a sequence
$$\sigma = \sigma_0, \sigma_1, \ldots, \sigma_k = \sigma'$$
of n-simplices of X not in A such that $\sigma_i \cap \sigma_{k+1}$ is an $(n-1)$-simplex not in A for each n. Let X_1 be the union of the elements of one equivalence class. Let X_2 be the union of the elements of the

remaining equivalence classes. Let s be a simplex of the complex X such that $s \notin A$ and $s \subset X_1 \cap X_2$. If $k = \dim s$, then $k < n-1$. Show that $\operatorname{Lk} s$ is a pseudo $(n-k-1)$-manifold. Conclude that every two n-simplices of $\overline{\operatorname{St}}\, s$ are equivalent.]

4. Prove Corollary 8.3.9. [**Hint:** If X is a pseudo m-manifold and $H_m(X) \neq 0$, then X must be finite.]

5. Can you give an example of a relative homology n-manifold in which the first entry X is a circle? Why or why not?

6. Can you give an example of a relative homology n-manifold in which the first entry X is a 2-sphere? Why or why not?

8.4 The Dual Block Complex

For simplicity we are going to restrict ourselves in the next few sections to the case of a homology n-manifold X. The techniques that we use will reappear later when we consider the case of a relative homology manifold.

There is associated with the triangulated homology manifold X a certain partition of X into subsets that are [almost] open cells such that X becomes [almost] a regular cell complex. The subsets are not actual topological cells but only homological cells. Since we cannot call them "cells," we shall instead call them "blocks." And we shall call the collection of these blocks the *dual block decomposition of* X. Just as there was with cell complexes, there is a chain complex associated to this block complex that can be used to compute the homology and cohomology of X. We will call it the "dual chain complex." It will be a crucial tool in proving our duality theorems.

Recall from Section 2.3 that we use the notation $\sigma_1 \succ \sigma_2$ to mean σ_2 is a proper face of σ_1.

Definition 8.4.1 Let X be a locally finite simplicial complex, and let $\operatorname{sd} X$ denote its first barycentric subdivision Lemma 2.3.9 tells us that the simplices of $\operatorname{sd} X$ are of the form

$$\widehat{\sigma}_{i_1} \widehat{\sigma}_{i_2} \cdots \widehat{\sigma}_{i_k}\,, \tag{$*$}$$

where

$$\sigma_{i_1} \succ \sigma_{i_2} \succ \cdots \succ \sigma_{i_k}\,. \tag{$**$}$$

The vertices of a simplex of $\operatorname{sd} X$ will be ordered as in $(*)$ so that $(**)$ is satisfied. So we can keep track of things we shall say that $\widehat{\sigma}_{i_1}$ is the *largest vertex* of the simplex, and that $\widehat{\sigma}_{i_k}$ is the *smallest vertex* of the simplex.

EXAMPLE 8.4.2 Let X a triangular region in the plane with vertices a, b, c. Consider X as a 2-dimensional simplicial complex. The vertices of $\operatorname{sd} X$ are

$$\widehat{a} = a, \quad \widehat{b} = b, \quad \widehat{c} = c, \quad \widehat{ab}, \quad \widehat{bc}, \quad \widehat{ca}, \quad \widehat{abc}.$$

The vertex $\widehat{abc}$ is necessarily the largest vertex of any simplex of which it is a vertex. The only simplex with $\widehat{abc}$ as its smallest vertex is $\widehat{abc}$ itself.

The vertex $\widehat{ab}$ is the largest vertex of the 1-simplices $\widehat{ab}\,\widehat{a}$, $\widehat{ab}\,\widehat{b}$, and of the 0-simplex $\widehat{ab}$ itself. The vertex $\widehat{ab}$ is the smallest vertex of the 1-simplex $\widehat{abc}\,\widehat{ab}$ and of the 0-simplex $\widehat{ab}$ itself.

Definition 8.4.3 Given a simplex σ of X, observe that the union of all open simplices of $\operatorname{sd} X$ of which $\widehat{\sigma}$ is the largest vertex is just $\operatorname{Int}\sigma$. On the other hand, we define $D(\sigma)$ to be the union of all open simplices of $\operatorname{sd} X$ of which $\widehat{\sigma}$ is the smallest vertex. This set is called the *block dual to* σ.

The blocks $D(\sigma)$ will play a role similar to that of the open cells of a CW complex. We call the closure $\overline{D}(\sigma)$ of $D(\sigma)$ the *closed block dual to* σ. It equals the union of all simplices of $\operatorname{sd} X$ of which $\widehat{\sigma}$ is the final vertex. It is the polytope of a subcomplex of $\operatorname{sd} X$. We let $\dot{D}(\sigma) = \overline{D}(\sigma) - D(\sigma)$.

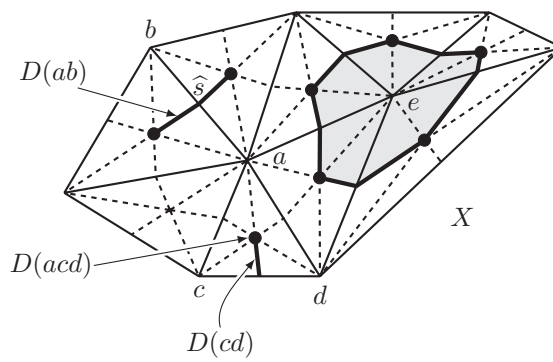


Figure 8.4.1

EXAMPLE 8.4.4 Let X be the 2-dimensional complex pictured in Figure 8.4.1. The complex $\operatorname{sd} X$ is indicated by dotted lines. The block dual to any 2-simplex σ consists of its barycenter $\widehat{\sigma}$ alone. The block $D(ab)$ dual to the 1-simplex $s = ab$ consists of its barycenter $\widehat{s}$ and the two open line segments joining $\widehat{s}$ to the barycenters of the two triangles having ab as a face. The block dual to the vertex e is the shaded region indicated. The corresponding closed block consists of this region plus its boundary. Note the interesting fact that $\dot{D}(e)$ is the union of the lower-dimensional blocks $D(s_i)$ as s_i ranges over all 1-simplices and 2-simplices having e as a face. This situation will hold in general, as we shall soon see.

EXAMPLE 8.4.5 Let X be the complex pictured in Figure 8.4.2. It is the join of the polygon $c\,d\,e\,f\,c$ with the line segment ab. You can picture mentally

what the complex $\operatorname{sd} X$ looks like. The block dual to each 3-simplex of X is its barycenter. The closed block dual to the 2-simplex abc consists of the two line segments joining the barycenter of abc with the barycenters of the two 3-simplices having abc as a face. The closed block $\overline{D}(ab)$ dual to the 1-simplex ab is the closed octagonal region pictured in Figure 8.4.3. The set $\dot{D}(ab)$ is its boundary. It is the union of the blocks dual to the 2-simplices and 3-simplices of X having ab as a face.

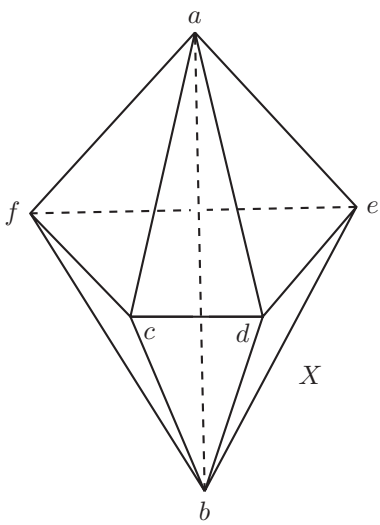


Figure 8.4.2

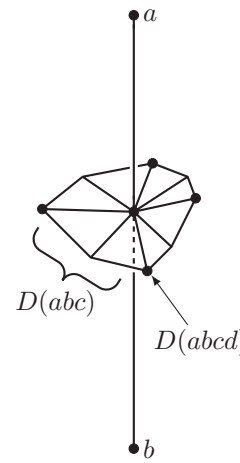


Figure 8.4.3

Keep these two examples in mind as we prove the following theorem.

Theorem 8.4.6 *Let X be a locally finite simplicial complex that consists entirely of n-simplices and their faces. Let σ be a k-simplex of X. Then:*

(a) *The dual blocks are disjoint and their union is $|X|$.*

(b) *$\overline{D}(\sigma)$ is the polytope of a subcomplex of $\operatorname{sd} X$ of dimension $n-k$.*

(c) *$\dot{D}(\sigma)$ is the union of all blocks $D(\tau)$ for which τ has σ as a proper face. These blocks have dimension less than $n-k$.*

(d) *$\overline{D}(\sigma)$ equals the cone $|\dot{D}(\sigma) * \widehat{\sigma}|$.*

(e) *If $H_i(X, X \setminus \widehat{\sigma}) \cong \mathbb{Z}$ for $i = n$ and vanishes otherwise, then $(\overline{D}(\sigma), \dot{D}(\sigma))$ has the homology of an $n-k$ cell modulo its boundary.*

Proof: (a) The open simplices of $\operatorname{sd} X$ are disjoint, and each one lies in precisely one block $D(\sigma)$—namely the one such that $\widehat{\sigma}$ is its final vertex.

(b) Given σ of dimension k, let σ' be an n-simplex of X having σ as a face. There is a sequence of simplices of X

$$\sigma' = \sigma_n \succ \sigma_{n-1} \succ \cdot \succ \sigma_k = \sigma$$

such that each simplex in the sequence has dimension one less than the preceding simplex. Then the simplex $\widehat{\sigma}_n \widehat{\sigma}_{n-1} \cdots \widehat{\sigma}_k$ of $\operatorname{sd} X$ has dimension $n-k$ and lies in $\overline{D}(\sigma)$. Clearly, no simplex of $\operatorname{sd} X$ of larger dimension can have σ_k as its final vertex since X has no simplices of dimension greater than n.

(c)–**(d)** Now $\overline{D}(\sigma)$ is the union of all simplices of $\operatorname{sd} X$ whose final vertex is $\widehat{\sigma}$. They are of the form

$$\widehat{\sigma}_{i_1} \widehat{\sigma}_{i_2} \cdots \widehat{\sigma}_{i_p} \widehat{\sigma}\,.$$

The intersection of such a simplex with $\dot{D}(\sigma)$ consists of the face of this simplex obtained by deleting $\widehat{\sigma}$. Thus $\dot{D}(\sigma)$ consists of all simplices of the form

$$\widehat{\sigma}_{i_1} \widehat{\sigma}_{i_2} \cdots \widehat{\sigma}_{i_p}$$

for which $\sigma_{i_p} \succ \sigma$. Now the interior of such a simplex is contained in $D(\sigma_{i_p})$. Conversely, any open simplex of $\operatorname{sd} X$ lying in $D(\sigma_{i_p})$ is of this form. It follows that $\dot{D}(\sigma)$ is the union of the blocks $D(\sigma_{i_p})$ for $\sigma_{i_p} \succ \sigma$. It also follows that $|\overline{D}(\sigma)| = |\dot{D}(\sigma) * \widehat{\sigma}|$.

(e) If $\dim \sigma = n$, then $D(\sigma)$ consists of the single point $\widehat{\sigma}$, which is of course a 0-cell. Suppose that $\dim \sigma = k < n$. In the complex $\operatorname{sd} X$, the closed star $\overline{\operatorname{St}}(\widehat{\sigma}, \operatorname{sd} X)$ equals the union of all simplices of the form

$$\tau = \widehat{\sigma}_{i_1} \cdots \widehat{\sigma}_{i_p} \widehat{\sigma}_{i_{p+1}} \widehat{\sigma}_{i_q}\,,$$

where $\sigma_{i_{p+1}} = \sigma$ and $\sigma_{i_j} \succ \sigma_{i_{j+1}}$ for all j. The face of τ spanned by the vertices to the left of $\widehat{\sigma}$ in this sequence is the typical simplex of $\dot{D}(\sigma)$. The face spanned by $\widehat{\sigma}$ and the vertices to its right in this sequence is the typical simplex of $\operatorname{sd} \sigma$. We conclude that

$$\begin{aligned} \overline{\operatorname{St}}(\widehat{\sigma}, \operatorname{sd} X) &= \dot{D}(\sigma) * \operatorname{sd} \sigma \\ &= \dot{D}(\sigma) * \widehat{\sigma} * \operatorname{sd}(\operatorname{Bd} \sigma)\,. \end{aligned}$$

Now it is also true that

$$\overline{\operatorname{St}}(\widehat{\sigma}, \operatorname{sd} X) = \widehat{\sigma} * \operatorname{Lk}(\widehat{\sigma}, \operatorname{sd} X)\,.$$

We conclude that

$$\operatorname{Lk}(\widehat{\sigma}, \operatorname{sd} X) = \dot{D}(\sigma) * \operatorname{sd}(\operatorname{Bd} \sigma)\,.$$

Since $\operatorname{sd}(\operatorname{Bd} \sigma)$ is topologically a $(k-1)$-sphere (or is empty if $k = 0$), we see that

$$\widetilde{H}_{i+k}(\operatorname{Lk}(\widehat{\sigma}, \operatorname{sd} X) \cong \widetilde{H}_i(\dot{D}(\sigma))\,.$$

Because the local homology of X at $\widehat{\sigma}$ is infinite cyclic in dimension n and vanishes otherwise, we see that $\mathrm{Lk}(\widehat{\sigma}, \mathrm{sd}\, X)$ is a homology $(n-1)$-sphere by Lemma 8.3.2. Therefore $\dot{D}(\sigma)$ is a homology $n-k-1$ sphere.

Since $\overline{D}(\sigma)$ is acyclic [being a cone], the long exact homology sequence gives us an isomorphism

$$H_i(\overline{D}(\sigma), \dot{D}(\sigma)) \cong \widetilde{H}_{i-1}(\dot{D}(\sigma))\,.$$

We conclude that $(\overline{D}(\sigma), \dot{D}(\sigma))$ has the homology of an $(n-k)$-cell modulo its boundary. □

Definition 8.4.7 Let X be a locally finite complex that is a homology n-manifold.[2] Then the preceding theorem applies to each simplex σ of X. The collection of dual blocks $D(\sigma)$ will be called the *dual block decomposition* of X. The union of the blocks of dimension at most p will be denoted by X_p and called the *dual p-skeleton* of X. The *dual chain complex* $\mathcal{D}(X)$ of X is defined by letting its chain group in dimension p be the group

$$D_p(X) = H_p(X_p, X_{p-1})\,.$$

Its boundary operator is the homomorphism ∂_* in the exact sequence of the triple (X_p, X_{p-1}, X_{p-2}).

Being the polytope of a subcomplex of $\mathrm{sd}\, X$, the dual p-skeleton X_p will sometimes be treated as a complex rather than as a space. In particular, we shall for convenience use simplicial homology in the definition of $D_p(X)$.

For all practical purposes, the dual chain complex will play the same role in computing homology and cohomology as the cellular chain complex did for a CW complex. Repeating the pattern of Section 4.13, we prove the following:

Theorem 8.4.8 *Let X be a locally finite complex that is a homology n-manifold. Let X_p be the dual p-skeleton of X. Let $\mathcal{D}(X)$ be the dual chain complex of X.*

> **(a)** *The group $H_i(X_p, X_{p-1})$ vanishes for $i \neq p$ and is a free abelian group for $i = p$. A basis when $i = p$ is obtained by choosing generators for the groups $H_p(\overline{D}(\sigma), \dot{D}(\sigma))$ as $D(\sigma)$ ranges over all p-blocks of X, and taking their images under the homomorphisms induced by inclusion in $H_p(X_p, X_{p-1})$.*
>
> **(b)** *The dual chain complex $\mathcal{D}(X)$ can be used to compute the homology of X. Indeed, $D_p(X)$ equals the subgroup of $C_p(\mathrm{sd}\, X)$ consisting of those chains carried by X and whose boundaries are also carried by X_{p-1}. The inclusion map $D_p(X) \to C_p(\mathrm{sd}\, X)$ induces a homology isomorphism. Therefore it also induces homology and cohomology isomorphisms with arbitrary coefficients.*

[2] In fact every complex that is a homology n-manifold is locally finite.

Proof: The proof of **(a)** follows the pattern of Lemma 4.13.4, but is easier. Because $\overline{D}(\sigma)$ is a cone with vertex $\widehat{\sigma}$, its base $\dot{D}(\sigma)$ is a deformation retract of $\overline{D}(\sigma) \setminus \widehat{\sigma}$ for each σ. These deformation retractions induce a deformation retraction of $X_p \setminus \cup_p \widehat{\sigma}$ onto X_{p-1}. Here $\cup_p \widehat{\sigma}$ denotes the union of the barycenters of all simplices σ of X of dimension $n - p$.

If $\Sigma_p(\overline{D}(\sigma), \dot{D}(\sigma))$ denotes the topological sum of the p-blocks of X, we have the commutative diagram

$$\begin{array}{ccccc}
\Sigma_p(\overline{D}(\sigma), \dot{D}(\sigma)) & \longrightarrow & \Sigma_p(\overline{D}(\sigma), \overline{D}(\sigma) \setminus \widehat{\sigma}) & \longleftarrow & \Sigma_p(D(\sigma), D(\sigma) \setminus \widehat{\sigma}) \\
\downarrow & & \downarrow & & \downarrow = \\
(X_p, X_{p-1}) & \longrightarrow & (X_p, X_p \setminus \cup_p \widehat{\sigma}) & \longleftarrow & \cup (D(\sigma, D(\sigma) \setminus \widehat{\sigma})
\end{array}$$

where the horizontal maps induce homology isomorphisms in singular homology and the vertical maps are inclusions. Statement **(a)** follows for singular theory and hence for simplicial theory.

Statement **(b)** is an immediate consequence of Theorem 4.13.9. □

We now have the basic tools that we shall use to prove the Poincaré duality theorem.

EXERCISES

1. Consider the usual triangulation of the torus T.

(a) Sketch the blocks of the dual block decomposition of T. Note that they actually make T into a regular cell complex.

(b) Let $\mathcal{C}(T)$ denote the simplicial chain complex of T and let $\mathcal{D}(T)$ denote the dual chain complex. Orient the 2-simplices σ of T counterclockwise. Orient the 1-simplices e arbitrarily. Define an isomorphism

$$\phi : C^2(T) \to D_0(T)$$

by setting

$$\phi(\sigma^*) = D(\sigma) = \widehat{\sigma}$$

for each oriented 2-simplex σ. Define an isomorphism

$$\phi : C^1(T) \to D_1(T)$$

by letting $\phi(e^*)$ equal a fundamental cycle for $(\overline{D}(e), \dot{D}(e))$ for each oriented 1-simplex e. Show that ϕ may be so chosen that

$$\partial \phi(e^*) = \phi(\delta e^*) \, .$$

(c) Define

$$\phi : C^0(T) \to D_2(T)$$

by letting $\phi(v^*)$ equal a fundamental cycle of $(\overline{D}(V), \dot{D}(v))$. Show that if the 2-cell $\overline{D}(v)$ is oriented counterclockwise then

$$\partial\phi(v^*) = \phi(\delta v^*)\,.$$

(d) Conclude that $H^p(T) \cong H_{2-p}(T)$ for all p [which we knew already].

This exercise indicates how Poincaré duality is proved.

2. Repeat Exercise 1 for the Klein bottle, using $\mathbb{Z}/2$ coefficients for both homology and cohomology. Note that the proof is easier, since no signs are involved.

3. What can you say if the torus and the Klein bottle in Exercises 1 and 2 are replaced by the sphere?

4. What can you say if the objects of Exercises 1 and 2 are replaced by a double torus [with two holes rather than one]?

8.5 Poincaré Duality

The Poincaré duality theorem is one of the most profound results in all of topology. In its original form, it stated that for a compact, orientable, triangulated n-manifold X, the betti numbers in dimensions k and $n-k$ were equal and the torsion numbers in dimensions k and $n-k-1$ were equal. Nowadays, it is stated in a different but equivalent formulation. Today we state that the homology and cohomology groups of X in dimensions k and $n-k$ are isomorphic.

There are also versions of Poincaré duality that hold when X is not orientable, or when X is not compact. We shall discuss these too.

We shall give two proofs of Poincaré duality. The first of these, which we give in the present section, is rather straightforward and intuitive. The second, which we will consider in Section 8.7, will provide valuable information on how the duality isomorphism behaves with respect to continuous maps. It also gives information about cup products on manifolds.

Definition 8.5.1 Let X be a compact triangulated homology n-manifold. We say that X is *orientable* if it is possible to orient all the n-simplices σ_i of X so that their sum $\gamma = \sum \sigma_i$ is a cycle. Such a cycle γ will be called an *orientation cycle* for X.

In case X is not compact, but is triangulated by a locally finite complex, we extend this definition by saying that X is *orientable* if it is possible to orient the σ_i so that the [possibly infinite] chain $\sum \sigma_i$ is a cycle.

In either case, if it is not possible to so orient the σ_i, then we say X is *non-orientable*.

If X is not connected, then each component of X is itself a homology n-manifold. In order that X be orientable, it is necessary and sufficient that each component of X be orientable. A sum of orientation cycles for each component of X is an orientation cycle for X. We will show that orientability of X does not depend on the particular triangulation of X. [See Corollary 8.5.5.]

Theorem 8.5.2 (Poincaré duality—first version) *Let X be a compact, triangulated homology n-manifold. If X is orientable, then for all p there is an isomorphism*

$$H^p(X;G) \cong H_{n-p}(X;G)\,,$$

where G is an arbitrary coefficient group. If X is non-orientable, then there is for all p an isomorphism

$$H^p(X;\mathbb{Z}/2) \cong H_{n-p}(X;\mathbb{Z}/2)\,.$$

Proof: We shall use the simplicial chain complex $\mathcal{C}(X)$ of X to compute the cohomology of X. Also we use the dual chain complex $\mathcal{D}(X)$ of X to compute the homology of X. There is a one-to-one correspondence between the p-simplices of X and the dual $n-p$ blocks of X that maps each simplex to its dual block. Hence the free abelian groups $C^p(X)$ and $D_{n-p}(X)$ are isomorphic. This is by an isomorphism ϕ that carries the basis element σ^* for $C^p(X)$ [where σ is an oriented p-simplex] to a generator of $H_{n-p}(\overline{D}(\sigma), \dot{D}(\sigma))$. If X is orientable then we shall show that the sign of $\phi(\sigma^*)$ may be so chosen that the following diagram commutes:

$$\begin{array}{ccc} C^{p-1}(X) & \xrightarrow{\phi} & D_{n-p+1}(X) \\ \delta\downarrow & & \downarrow\partial \\ C^p(X) & \xrightarrow{\phi} & D_{n-p}(X)\,. \end{array}$$

This will prove the existence of the Poincaré duality isomorphism in the case of integer coefficients.

To define ϕ, we proceed as follows. First, orient the n-simplices of X so that their sum γ is a cycle. Orient the other simplices of X arbitrarily.

We begin by defining ϕ in dimension n. The oriented n-simplices σ of X form a basis for $C_n(X)$. Their duals σ^* form a basis for $C^n(X)$. The dual block $\overline{D}(\sigma)$ of the n-simplex σ is the 0-cell $\widehat{\sigma}$. We define $\phi(\sigma^*) = \widehat{\sigma}$. Note that $\widehat{\sigma}$ is a generator of the group $H_0(\widehat{\sigma})$. Thus we have defined an isomorphism

$$\phi : C^n(X) \to D_0(X)\,.$$

We next define ϕ in dimension $n-1$. Let s be an oriented $(n-1)$-simplex of X. We wish to define $\phi(s^*)$ so that

$$\partial\phi(s^*) = \phi(\delta s^*)\,.$$

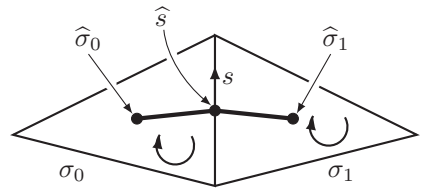


Figure 8.5.1

Now s is a face of exactly two n-simplices σ_0 and σ_1 of X. Since γ is a cycle, they are oriented so that $\partial\sigma_0 + \partial\sigma_1$ has coefficient zero on s. Suppose the indexing to be chosen so that $\partial\sigma_0$ has coefficient -1 on s and $\partial\sigma_1$ has coefficient 1 on s. Then $\delta s^* = \sigma_1^* - \sigma_0^*$ so that

$$\phi(\delta s^*) = \widehat{\sigma}_1 - \widehat{\sigma}_0$$

by definition. The 1-block $\overline{D}(s)$ is the union of the line segments from $\widehat{s}$ to $\widehat{\sigma}_0$ and $\widehat{\sigma}_1$. We now define

$$\phi(s^*) = [\widehat{\sigma}_0, \widehat{s}] + [\widehat{s}, \widehat{\sigma}_1]\,.$$

See Figure 8.5.1. Then $\phi(s^*)$ is a fundamental cycle for the 1-cell $(\overline{D}(s), \dot{D}(s))$ as desired. Also

$$\partial\phi(s^*) = \widehat{\sigma}_1 - \widehat{\sigma}_0 = \phi(\delta s^*)\,.$$

In general, suppose that ϕ is defined in dimension $p+1 < n$. We wish to define it in dimension p in such a way that for each oriented p-simplex s we have

$$\partial\phi(s^*) = \phi(\delta s^*)\,.$$

Given the oriented p-simplex s, we compute

$$\delta s^* = \sum \epsilon_i \sigma_i^*\,,$$

where the sum extends over all $(p+1)$-simplices σ_i of which s is a face. Also $\epsilon_i = \pm 1$ is the coefficient of s in the expression for $\partial\sigma_i$. Thus

$$\phi(\delta s^*) = \sum \epsilon_i \phi(\sigma_i^*)\,.$$

We shall prove that $\phi(\delta s^*)$ is a fundamental cycle for $\dot{D}(s)$.

The chain $\phi(\sigma_i^*)$ is by hypothesis a fundamental cycle for the block $\overline{D}(\sigma_i)$ dual to σ_i. This block lies in $\dot{D}(s)$ since $\sigma_i \succ s$. Thus $\phi(\sigma_i^*)$ is carried by $\dot{D}(s)$ for each i. Then the chain $\phi(\delta s^*)$ is carried by $\dot{D}(s)$ also. In addition, $\phi(\delta s^*)$ is a cycle, since $\partial\phi(\delta s^*) = \phi\delta\delta s^*$ by the induction hypothesis. Now $\dot{D}(s)$ is a homology $n-p-1$ sphere, where $n-p-1 > 0$. The cycle $\phi(\delta s^*)$ is non-trivial, and it is not a multiple of another cycle. This is because its restriction to the block $\overline{D}(\sigma_i)$ equals $\phi(\sigma_i^*)$, which is a fundamental cycle for this block modulo $\dot{D}(\sigma_i)$. Hence $\phi(\delta s^*)$ represents a generator of $H_{n-p-1}(\dot{D}(s))$ as asserted.

Consider the exact sequence

$$0 \to H_{n-p}(\overline{D}(s), \dot{D}(s)) \xrightarrow{\partial_*} H_{n-p-1}(\dot{D}(s)) \to 0\,.$$

For dimensional reasons, homology groups equal the groups of cycles. Define $\phi(s^*)$ to be the fundamental cycle for $(\overline{D}(s), \dot{D}(s))$ whose boundary equals the generator $\phi(\delta s^*)$. Then ϕ is defined for dimension p and $\partial\phi(s^*) = \phi(\delta s^*)$ as desired.

The theorem is thus proved for the case of integer coefficients. To prove it for arbitrary coefficients, we note that the isomorphisms

$$\mathrm{Hom}(C_p(X), G) \cong \mathrm{Hom}(C_p(X), \mathbb{Z}) \otimes G \xrightarrow{\phi \otimes i_G} D_{n-p}(X) \otimes G$$

commute with δ and ∂. [The first of these isomorphisms was proved in Step 2 of Theorem 7.6.1.]

Finally, we consider the non-orientable case. Roughly speaking, the same proof goes through. One defines an isomorphism

$$\mathrm{Hom}(C_p(X), \mathbb{Z}/2) \xrightarrow{\phi} D_{n-p}(X) \otimes \mathbb{Z}/2$$

by induction. If σ is an n-simplex, then we let σ^* denote the cochain whose value is $[1] \in \mathbb{Z}/2$ on σ and 0 on all other simplices. We then define $\phi(\sigma^*) = \partial \otimes [1]$. If s is an $n-1$ simplex that is a face of the n-simplices σ_0 and σ_1, then we define $\phi(s^*) = ([\widehat{\sigma}_0, \widehat{s}] + [\widehat{s}, \widehat{\sigma}_1] \otimes [1]$. Signs do not matter because the coefficient group is $\mathbb{Z}/2$. The rest of the proof goes through without change, with the phrase "fundamental cycle" replaced by "unique non-trivial cycle with $\mathbb{Z}/2$ coefficients." □

Let us note that there was nothing special in this proof about using the simplicial chain complex to compute the cohomology of X and the dual chain complex to compute the homology of X. One could just as well have done it the other way. For since ϕ is an isomorphism, then the dual of ϕ is an isomorphism:

$$\mathrm{Hom}(\mathrm{Hom}(C_p(X), \mathbb{Z}), \mathbb{Z}) \xrightarrow{\widetilde{\phi}} \mathrm{Hom}(D_{n-p}(X), \mathbb{Z})$$

that carries the operator δ in $\mathcal{D}(X)$ to the dual of δ in $\mathcal{C}(X)$. Since there is a natural isomorphism

$$C_p(X) \cong \mathrm{Hom}(\mathrm{Hom}(C_p(X), \mathbb{Z}), \mathbb{Z})$$

that carries ∂ to the dual of δ [see Step 4 of Theorem 7.6.1], the result is an isomorphism

$$C_p(X) \leftarrow \mathrm{Hom}(D_{n-p}(X), \mathbb{Z})$$

that commutes with δ and ∂.

Corollary 8.5.3 *Let X be a compact, triangulated homology n-manifold. If X is connected, then for any two n-simplices σ and σ' of X, there is a sequence*

$$\sigma = \sigma_0, \sigma_1, \ldots, \sigma_k = \sigma'$$

of n-simplices of X such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$ simplex of X for each i.

Proof: Let us define $\sigma \sim \sigma'$ if there is such a sequence connecting them. This relation is clearly an equivalence relation. The sum $\sigma\sigma_i$ of the n-simplices in any one equivalence class is a cycle if its coefficient group is taken to be $\mathbb{Z}/2$. This is because, given an $(n-1)$-simplex s of X, the two n-simplices of which it is a face are equivalent, so either both of them appear in this sum, or neither does. In any case, $\delta(\sum \sigma_i)$ has coefficient $[0] \in \mathbb{Z}/2$ on s.

We conclude that there is only one equivalence class, since otherwise X would have more than one non-trivial n-cycle over $\mathbb{Z}/2$. This would contradict the fact that

$$H_n(X; \mathbb{Z}/2) \cong H^0(X; \mathbb{Z}/2) \cong \mathbb{Z}/2\,. \qquad \square$$

Notice that this corollary shows that a compact connected triangulated homology manifold is a pseudo manifold. [See Corollary 8.3.4 and the comment following.]

Corollary 8.5.4 *Let X be a compact triangulated homology n-manifold. If X is connected, then $H_n(X) \cong \mathbb{Z}$ for X orientable and $H_n(X) = 0$ for X non-orientable.*

Proof: If X is orientable, then $H_n(X) \cong H^0(X) \cong \mathbb{Z}$. Conversely, we show that if $H_n(X) \neq 0$ then X is orientable. Orient the n-simplices of X arbitrarily. Suppose that z is a non-trivial n-cycle of X with integer coefficients. Now if σ_i and σ_{i+1} are n-simplices of X with an $n-1$ face in common, and if z has coefficient m on σ_i, then it must have coefficient $\pm m$ on σ_{i+1}. In view of the preceding corollary, z must have the same coefficient, up to sign, on each n-simplex of X. Suppose this coefficient has absolute value m. Then dividing z by m gives us an orientation cycle for X. $\square$

Corollary 8.5.5 *Let X be a compact, triangulated homology n-manifold. Then X is orientable if and only if for each component X_i of X, we have $H_n(X_i) \cong \mathbb{Z}$.*

It follows that orientability of X does not depend on the triangulation of X.

The following corollary depends on the universal coefficient theorems proved in the last chapter.

Corollary 8.5.6 *Let X be a compact triangulated homology n-manifold. Assume that X is connected.*

(a) *If X is orientable, then $H_{n-1}(X)$ has no torsion. And, for all G,*

$$H_n(X;G) \cong G \cong H^n(X;G)\,.$$

(b) *If X is non-orientable, then the torsion subgroup of $H_{n-1}(X)$ has order 2. Also, for all G,*

$$H_n(X;G) \cong \ker(\phi_2) \quad \text{and} \quad H^n(X;G) \cong G/2G\,,$$

where $\phi_2 : G \to G$ is multiplication by 2. In particular, $H_n(X) = 0$ and $H^n(X) \cong \mathbb{Z}/2$.

Proof: (a) The computation when X is orientable is immediate, since

$$H^n(X;G) \cong H_0(X;G) \quad \text{and} \quad H_n(X;G) \cong H^0(X;G)$$

by Poincaré duality. Each of the 0-dimensional groups is isomorphic to G because X is connected. It follows that $H_{n-1}(X)$ has no torsion. This is because

$$H^n(X) \cong \mathrm{Hom}(H_n(X),\mathbb{Z}) \oplus \mathrm{Ext}(H_{n-1}(X),\mathbb{Z})\,.$$

Since $\mathrm{Ext}(H_{n-1}(X),\mathbb{Z})$ is isomorphic to the torsion subgroup of $H_{n-1}(X)$ [by Theorem 7.2.5], this torsion subgroup must vanish.

(b) Now suppose that X is non-orientable. We shall compute $H_n(X;G)$. Let $\sum \sigma_i$ the sum of all the n-simplices of X, oriented arbitrarily. If $g \in G$ has order 2, then $\sum g\sigma_i$ is clearly a cycle of X since each $(n-1)$-simplex is a face of exactly two n-simplices. Conversely, let z be a non-trivial n-cycle of X with G coefficients. If z has coefficient g on the oriented n-simplex σ of X, then z must have coefficient $\pm g$ on each n-simplex having an $(n-1)$-face in common with σ. Otherwise, z would not be a cycle. Application of Corollary 8.5.3 shows that z has coefficient $\pm g$ on every n-simplex of X. Thus $z = g(\sum \epsilon_i\sigma_i)$ where $\epsilon_i = \pm 1$ and the sum extends over all n-simplices of X. Now the chain $\sum \epsilon_i\sigma_i$ cannot be a cycle since that would imply that X is orientable. Hence $\partial(\sum \epsilon_i\sigma_i)$ has a non-zero coefficient on at least one $(n-1)$-simplex s of X. This coefficient is ± 2 since s is a face of exactly two n-simplices of X. We conclude that ∂z has coefficient $\pm 2g$ on s which means [since z is a cycle] that g has order 2. In particular, $g = -g$ so that $z = g(\sum \epsilon_i\sigma_i) = \sum g\sigma_i$.

Thus the n-cycles of X consist of all chains of the form $\sum g\sigma_i$ where g is 0 or has order 2. Since the group of cycles equals the homology group in dimension n, it follows that

$$H_n(X;G) \cong \ker(\phi_2)\,.$$

We now show that the torsion subgroup $T_{n-1}(X)$ of $H_{n-1}(X)$ is isomorphic to $\mathbb{Z}/2$. Since $H_n(X) = 0$, it follows from the universal coefficient theorem that

$$H_{n-1}(X) * G \cong H_n(X; G) \cong \ker(\phi_2)\,.$$

Hence

$$T_{n-1}(X) * G \cong \ker(\phi_2) \cong \mathbb{Z}/2 * G\,.$$

In particular,

$$T_{n-1}(X) * \mathbb{Z}/m \cong \begin{cases} 0 & \text{if} \quad m \text{ is odd}\,, \\ \mathbb{Z}/2 & \text{if} \quad m \text{ is even}\,. \end{cases}$$

It follows that T_{n-1} has no odd torsion coefficient $2p+1$ [set $m = 2p+1$ in this expression to obtain a contradiction]. Also it has only one even torsion coefficient (set $m = 2$). And this torsion coefficient is not of the form $2k$ for $k > 1$ [set $m = 2k$]. Hence $T_{n-1}(X) \cong \mathbb{Z}/2$.

Finally, it follows from the universal coefficient theorem for cohomology that

$$H^n(X; G) \cong \text{Ext}(H_{n-1}(X), G) \cong G/2G\,. \qquad \square$$

Definition 8.5.7 Let X be a homology n-manifold [not necessarily compact] that is triangulated by a locally finite complex. We say that X is *orientable* if it is possible to orient the n-simplices σ_i of X so that the [possibly infinite] chain $\sum \sigma_i$ is a cycle.

Theorem 8.5.8 (Poincaré duality—the non-compact case) *Let X be a homology n-manifold that is triangulated by a locally finite complex.*

(1) *For all p, there are isomorphisms*

$$H^p_c(X; \mathbb{Z}/2) \cong H_{n-p}(X; \mathbb{Z}/2) \quad \text{and} \quad H^p(X; \mathbb{Z}/2) \cong H^\infty_{n-p}(X; \mathbb{Z}/2)\,,$$

where H^p_c denotes cohomology with compact support [see Definition 5.5.4] and H^∞_{n-p} denotes homology based on infinite chains [see Definition 1.6.14].

(2) *If X is orientable, then for all p there are isomorphisms*

$$H^p_c(X; G) \cong H_{n-p}(X; G) \quad \text{and} \quad H^p(X; G) \cong H^\infty_{n-p}(X; G)\,,$$

where G is an arbitrary coefficient group.

EXERCISES

1. Let M and N be compact, connected, triangulable n-manifolds, and let $f : M \to N$. Show that if M is orientable and N is non-orientable, then

$$f_* : H_n(M; \mathbb{Z}/2) \to H_n(N : \mathbb{Z}/2)$$

is trivial.

2. Let X be a compact, triangulable, orientable manifold with ith betti number $\beta_i(X)$. Recall that Definition 2.10.2 gave the following equation for the *Euler number* of X:

$$\chi(X) = \sum (-1)^i \beta_i(X)$$

(a) Show that if $\dim X$ is odd, then $\chi(X) = 0$.

(b) If $\dim X$ equals $2k$, then express $\chi(X)$ in terms of β_i for $i \leq k$.

(c) Give applications to tangent vector fields. [See Exercise 5 in Section 2.10.]

3. Let X be a homology n-manifold (not necessarily compact) that is triangulated by a locally finite complex.

(a) Prove Theorem 8.5.8(1).

(b) Show that if X is connected, then X is a pseudo n-manifold.

(c) Prove Theorem 8.5.8(2). Conclude that orientability of X depends only on the underlying topological space, not on the particular triangulation.

4. Show that the Klein bottle is not orientable.

5. Look up the term "cross cap" in a topology text or on the Internet. Show that the cross cap is not orientable.

8.6 Cap Products

To present our second proof of Poincaré duality, we must first treat a topic that is purely algebraic in nature. What we shall describe here is a certain operation called the *cap product* of a cohomology class and a homology class. Its definition is formally similar to the definition of cup product that we discussed in Section 5.9. Its usefulness, which lies in its connection with the duality theorems, will become apparent shortly.

Throughout this section, let R be a commutative ring with unit element.

Definition 8.6.1 Let X be a topological space. We define a map

$$S^p(X;R) \otimes S_{p+q}(X;R) \xrightarrow{\frown} S_q(X;R)$$

as follows: If $T : \Delta_{p+q} \to X$ is a singular simplex on X, and if $\alpha \in R$, then

$$c^p \frown (T \otimes \alpha) = T \circ I(\epsilon_0, \ldots, \epsilon_q) \otimes \alpha \cdot \langle c^p, T \circ I(\epsilon_q, \ldots, \epsilon_{p+q}) \rangle .$$

This chain is called the *cap product* of the cochain c^p and the chain $T \otimes \alpha$.

Roughly speaking, $c^p \frown (T \otimes \alpha)$ equals the restriction of T to the front q-face of Δ_{p+q} with coefficient the product of α and the value of c^p on the back p-face of T.

As with the cup product, there are alternative versions of the cap product. Among these are maps

$$S^p(X;G) \otimes S_{p+q}(X;\mathbb{Z}) \xrightarrow{\frown} S_q(X;G)\,,$$

$$S^p(X;F) \otimes S_{p+q}(X;F) \xrightarrow{\frown} S_q(X;F)\,,$$

where G is an abelian group and F is a field. These are all given by the preceding cap product formula. These additional versions of the cap product will also be of use to us in what follows.

Theorem 8.6.2 *The cap product is bilinear and is natural with respect to continuous maps. It satisfies the boundary formula*

$$\partial(d^p \frown c_{p+q}) = (-1)^q(\delta d^p \frown c_{p+q}) + d^p \frown \partial c_{p+q}\,,$$

and is related to the cup product by the formula

$$c^p \frown (d^q \frown e_{p+q+r}) = (c^p \smile d^q) \frown e_{p+q+r}\,.$$

Proof: The bilinear property is easy to check. The naturality property is harder to formulate than to prove. Given a continuous map $f : X \to Y$, consider the following diagram:

$$\begin{array}{ccc} S^p(X) \otimes S_{p+q}(X) & \xrightarrow{\frown} & S_q(X) \\ \uparrow f^\# \quad \downarrow f_\# & & \downarrow f^\# \\ S^p(Y) \otimes S_{p+q}(Y) & \xrightarrow{\frown} & S_q(Y) \end{array}$$

[We omit the coefficients for convenience.] Naturality states that if $c^p \in S^p(Y)$ and $d_{p+q} \in S_{p+q}(X)$, then

$$f_\#(f^\#(c^p) \frown d_{p+q}) = c^p \frown f_\#(d_{p+q})\,.$$

One checks this formula easily by direct computation. The proofs of the coboundary and cup product formulas are straightforward but messy. We leave the details for the reader. □

These formulas in fact hold for all three versions of the cap product. In the case of the coefficient pairing $G \otimes \mathbb{Z} \to G$, one must add the proviso that the cup product formula makes sense. This will occur if two of c^p, d^q, and e_{p+q+r} have coefficients in $\mathbb{Z}$ and the other has coefficients in G.

Theorem 8.6.3 *The cap product induces a homomorphism*

$$H^p(X;R) \otimes H_{p+q}(X;R) \xrightarrow{\frown} H_q(X;R)\,.$$

This is natural, and it satisfies the formula

$$\alpha^p \frown (\beta^q \frown \gamma_{p+q+r}) = (\alpha^p \smile \beta^q) \frown \gamma_{p+q+r}\,.$$

The same result holds for the other versions of the cap product.

Proof of the Theorem: First we note that if d^p is a cocycle and c_{p+q} is a cycle, then $d^p \frown c_{p+q}$ is a cycle, because

$$\partial(d^p \frown c_{p+q}) = (-1)^q \delta d^p \frown c_{p+q} + d^p \frown \partial c_{p+q} = 0\,.$$

Next we show that $\frown$ carries the kernel of the natural projection

$$Z^p(X;R) \otimes Z_{p+q}(X;R) \to H^p(X;R) \otimes H_{p+q}(X;R)$$

into $B_q(X;R)$. This kernel is generated by the elements of the two groups $B^p \otimes Z_{p+q}$ and $Z^p \otimes B_{p+q}$. The boundary formula shows that

$$(\text{cobdry}) \frown (\text{cycle}) = (-1)^{q+1} \delta d^{p-1} \frown z_{p+q} = \partial(d^{p-1} \frown z_{p+q})\,,$$

$$(\text{cocycle}) \frown (\text{bdry}) = z^p \frown \partial c_{p+q+1} = \partial(z^p \frown c_{p+q+1})\,,$$

both of which are boundaries—as desired. Naturality, and the cup product formula, are immediate. □

Relation to the Kronecker Index

Naturality of the cap product on the homology-cohomology level is expressed by a diagram of the following form [where we omit the coefficients for convenience]:

$$\begin{array}{ccc} H^p(X) \otimes H_{p+q}(X) & \longrightarrow & H_q(X) \\ \uparrow f^* \quad \downarrow f_* & & \downarrow f_* \\ H^p(Y) \otimes H_{p+q}(Y) & \longrightarrow & H_q(Y) \end{array}$$

The similarity of this diagram to the corresponding diagram for the Kronecker index makes one suspect that there is some relation between the two. There is indeed, and we discuss it now.

Recall that the Kronecker index was defined in Section 5.6 as the homomorphism

$$\langle\ ,\ \rangle : H^p(X;G) \otimes H_p(X) \to G$$

induced by evaluation of a cochain on a chain.

Theorem 8.6.4 *Let $\epsilon_* : H_0(X;G) \to G$ be the homomorphism induced by the augmentation map ϵ, which is an isomorphism if X is path connected. Then the Kronecker index equals the composite*

$$H^p(X;G) \otimes H_p(X) \xrightarrow{\frown} H_0(X;G) \xrightarrow{\epsilon_*} G\,.$$

Proof: The augmentation map $\epsilon : C_0 \to \mathbb{Z}$ gives rise to a homomorphism $C_p \otimes G \to \mathbb{Z} \otimes G \cong G$ that carries boundaries to zero. Hence it induces a homology homomorphism ϵ_*. We compute directly as follows: If $T : \Delta_p \to X$, then

$$\begin{aligned} \epsilon(c^p \frown T) &= \epsilon[T \circ \ell(\epsilon_0) \otimes \langle c^p, T \circ \ell(\epsilon_0, \ldots, \epsilon_p)\rangle] \\ &= \langle c^p, T\rangle \,. \end{aligned}$$

The theorem follows. □

It was convenient to define the Kronecker map

$$\kappa : H^p(X; G) \to \mathrm{Hom}(H_p(X), G)$$

as the map sending α to the homomorphism $\langle \alpha, \cdot \rangle$. Its naturality was easier to formulate than that of the Kronecker index. There is a similar version of the cap product. Its naturality is expressed by the commutative diagram

$$\begin{array}{ccc} H_{p+q}(X; R) & \xrightarrow{\frown} & \mathrm{Hom}(H^p(X; R), H_q(X; R)) \\ \downarrow f_* & & \downarrow \mathrm{Hom}(f^*, f_*) \\ H_{p+q}(Y; R) & \xrightarrow{\frown} & \mathrm{Hom}(H^p(Y; R), H_q(Y; R)) \,, \end{array}$$

where the horizontal maps carry the homology class γ_{p+q} to the homomorphism $\frown\, \gamma_{p+q}$. This is the formulation of naturality that we often use in practice.

Simplicial Cap Product

Just as with cup products, one needs a formula that holds for simplicial theory if one wants to readily compute. By now it may be clear how to obtain such a formula.

Definition 8.6.5 Given a complex K, choose a partial ordering of the vertices of K that linearly orders the vertices of each simplex. We define a homomorphism

$$C^p(K; R) \otimes C_{p+q}(K; R) \xrightarrow{\frown} C_q(K; R) \,,$$

called the *simplicial cap product*, by the formula

$$c^p \frown ([v_0, v_1, \ldots, v_{p+q}] \otimes \alpha) = [v_0, v_1, \ldots, v_q] \otimes \alpha \cdot \langle c^p, [v_q, v_{q+1}, \ldots, v_{p+q}]\rangle \,.$$

Theorem 8.6.6 *The simplicial cap product satisfies the boundary formula and the cup product formula of Theorem 8.6.2. It induces a homomorphism in simplicial theory given by*

$$H^p(K; R) \otimes H_{p+q}(K; R) \to H_q(K; R)$$

that corresponds to the singular cap product under the standard isomorphism between simplicial and singular theory.

Proof: Let $\eta : C_p(K) \to S_p(|K|)$ be the chain map defined in Section 4.7 that induces an isomorphism of simplicial theory with singular theory. Since η is a monomorphism onto a direct summand, its dual $\widetilde{\eta}$ is surjective.

We first show that η commutes with $\frown$ on the chain level. That is, given $c^p \in C^p(K;R)$, we choose $d^p \in S^p(|K|;R)$ with $\widetilde{\eta}(d^p) = c^p$. And we verify that, for $c_{p+q} \in C_{p+q}(K;R)$, we have

$$\eta(c^p \frown c_{p+q}) = d^p \frown \eta(c_{p+q})\,.$$

To prove this formula, we assume that $v_0 < v_1 < \cdots < v_{p+q}$ and compute

$$\begin{aligned}\eta(c^p \frown [v_0,\ldots,v_{p+q}]\otimes\alpha) &= \ell(v_0,\ldots,v_q)\otimes\alpha\cdot(c^p,[v_q,\ldots,v_{p+q}])\\ &= \ell(v_0,\ldots,v_q)\otimes\alpha\cdot(d^p,\ell(v_q,\ldots,v_{p+q}))\\ &= d^p\frown\eta([v_0,\ldots,v_{p+q}]\otimes\alpha)\,.\end{aligned}$$

This last equation uses the fact that

$$\ell(v_0,\ldots,v_{p+q})\circ\ell(\epsilon_0,\ldots,\epsilon_q) = \ell(v_0,\ldots,v_q)\,.$$

The boundary formula and cup product formula may be proved by direct computation, as in Theorem 8.6.2. Alternatively, they can be derived from the similar formulas for singular theory. For example, given c^p, let $\widetilde{\eta}(d^p) = c^p$ as before. Then if we apply η to both sides of the desired equation

$$\partial(c^p\frown c_{p+q}) = (-1)^q\delta c^p\frown c_{p+q} + c^p\frown\partial c_{p+q}\,, \qquad (*)$$

we obtain

$$\partial(d^p\frown\eta(c_{p+q})) = (-1)^q\delta d^p\frown\eta(c_{p+q}) + d^p\frown\partial\eta(c_{p+q})\,.$$

Since η is a monomorphism, $(*)$ must hold. A similar argument applies to the cup product formula, using the fact that $\widetilde{\eta}$ commutes with the cup product. □

Relative Cap Products

Just as with the cup product, there is a relative version of the cap product. It will be useful in what follows. The most general relative cap product is a bilinear map

$$H^p(X,A)\otimes H_{p+q}(X,A\cup B)\to H_q(X,B)\,,$$

which is defined when $\{A,B\}$ is an excisive couple. [Here, as usual, we shall omit the coefficients for convenience.] We shall in fact be interested only in the cases where either A or B is empty, or $A = B$. That is, we are concerned with the following operations:

$$H^p(X)\otimes H_{p+q}(X,B)\xrightarrow{\frown} H_q(X,B)\,,$$

$$H^p(X,A)\otimes H_{p+q}(X,A)\xrightarrow{\frown} H_q(X)\,,$$

$$H^p(X,A)\otimes H_{p+q}(X,A)\xrightarrow{\frown} H_q(X,A)\,.$$

The existence of these cap products is easy to demonstrate. Let $c^p \in S^p(X)$ and $d_{p+q} \in S_{p+q}(X)$. To obtain the first operation, we note simply that if d_{p+q} is carried by B, then so is $c^p \frown d_{p+q}$ by definition. To check the second, we note that if c^p vanishes on all chains carried by A, and if d_{p+q} is carried by A, then $c^p \frown d_{p+q} = 0$. The third one equals the second one followed by the projection $H_q(X) \to H_q(X, A)$.

The boundary formula and the cup product formula hold as before. Therefore one has a natural bilinear map on the level of relative homology and cohomology. One has similar relative cap products in simplicial theory. One simply repeats the arguments already given for the absolute cap product.

EXERCISES

1. Verify the naturality formula of Theorem 8.6.2 by setting $d_{p+q} = T \otimes \alpha$ and computing directly.

2. Verify the boundary formula of Theorem 8.6.2 by setting $c_{p+q} = T \otimes \alpha$ and adding the expressions for $(-1)^q \delta d^p \frown c_{p+q}$ and $d^p \frown \partial c_{p+q}$. Most of the terms cancel; one has remaining the expression

$$\sum_{i \le q} (-1)^i T \circ \ell(\epsilon_0, \epsilon_1, \ldots, \widehat{\epsilon_i}, \ldots, \epsilon_q) \otimes \alpha \cdot \langle d^p, T \circ \ell(\epsilon_q, \ldots, \epsilon_{p+1}) \rangle ,$$

where the qth term is left over from $(-1)^q \delta d^p \frown c_{p+q}$ and the terms for which $i < q$ are left over from $d^p \frown \partial c_{p+q}$.

3. Verify the cup product formula of Theorem 8.6.2 by setting $c_{p+q+r} = T \otimes \alpha$ and computing directly.

4. Let T denote the torus. Let α and β be the usual generators for $H^1(T)$. Let Γ and Λ generate $H_2(T)$ and $H^2(T)$ respectively. [See Example 5.10.5.]

- **(a)** Compute $\alpha \frown \Gamma$ and $\beta \smile \Gamma$. Draw pictures.
- **(b)** Compute $\Lambda \frown \Gamma$.
- **(c)** Verify that $\alpha \frown (\beta \frown \Gamma) = (\alpha \smile \beta) \frown \Gamma$.

5. Let $\mathbb{P}^2$ be the projective plane. Let $\Gamma_{(2)}$ denote the non-zero element of $H_2(\mathbb{P}^2; \mathbb{Z}/2)$. Compute the values of the homomorphism $\frown \Gamma_{(2)}$ on the groups $H^i(\mathbb{P}^2; \mathbb{Z}/2)$ for $i = 1, 2$.

6. Let (X, A) be a triangulated pair. Let $\alpha \in H_n(X, A)$. Then $\partial_* \alpha$ is an element of $H_{n-1}(A)$. Consider the following diagram:

$$\begin{array}{ccccccc}
H^p(X, A) & \longrightarrow & H^p(X) & \longrightarrow & H^p(A) & \longrightarrow & H^{p+1}(X, A) \\
\downarrow \frown \alpha & & \downarrow \frown \alpha & & \downarrow \frown \partial_* \alpha & & \downarrow \frown \alpha \\
H_{n-p}(X) & \longrightarrow & H_{n-p}(X, A) & \longrightarrow & H_{n-p-1}(A) & \longrightarrow & H_{n-p-1}(X) .
\end{array}$$

Show that each of the squares commutes up to sign. What are the signs involved?

7. Give a definition of the cap product using category theory.

8.7 A Second Proof of Poincaré Duality

Now we give a second proof of the Poincaré duality theorem. This time we construct the duality isomorphism using cap products. There are several interesting consequences of this procedure, including a naturality formula for the duality isomorphism and a proof of its independence of the triangulation involved.

Definition 8.7.1 Let X be a compact, triangulable homology n-manifold. [Note that no specific triangulation of X is assumed.] Let X_i be a connected component of X. If X is orientable, then $H_n(X_i)$ is infinite cyclic. A generator $\Gamma^{(i)}$ of $H_n(X_i)$ is called an *orientation class* for X_i. The image of the classes $\Gamma^{(i)}$ under the isomorphism induced by the inclusions

$$\otimes H_n(X_i) \cong H_n(X)$$

is called an *orientation class* for X and is denoted by Γ. Similarly, if X is not necessarily orientable, and if $\Gamma^{(i)}_{(2)}$ is the unique non-trivial element of $H_n(X_i;\mathbb{Z}/2)$, then the image of these elements in $H_n(X;\mathbb{Z}/2)$ is denoted by $\Gamma_{(2)}$ and is called an *orientation class* for X over $\mathbb{Z}/2$.

If X is given a specific triangulation, then $\Gamma^{(i)}$ is represented by the sum of all the n-simplices of the connected component X_i suitably oriented. Similarly, $\Gamma_{(2)}$ is represented by the sum of the n-simplices of X, each with coefficient $[1] \in \mathbb{Z}/2$.

Theorem 8.7.2 (Poincaré duality—second version) *Let X be a compact triangulable homology n-manifold. If X is orientable, and if $\Gamma \in H_n(X)$ is an orientation class for X, then*

$$H^p(X;G) \xrightarrow{\frown \Gamma} H_{n-p}(X;G)$$

is an isomorphism for arbitrary G. Whether X is orientable or not,

$$H^p(X;\mathbb{Z}/2) \xrightarrow{\frown \Gamma_{(2)}} H_{n-p}(X;\mathbb{Z}/2)$$

is an isomorphism.

Proof: The idea of the proof is the same as before. We choose a triangulation of X, partition X into its dual blocks, and construct an isomorphism

$$\phi : C^p(X) \to D_{n-p}(X)$$

that commutes with δ and ∂. The difference is that we are going to construct ϕ by way of the cap product formula.

Step 1: We first prove the following elementary fact: Let K be a complex. Let $g : \mathrm{sd}\, K \to K$ be a simplicial approximation to the identity. Then, given $\sigma \in K$, there is exactly one simplex t of $\mathrm{sd}\, \sigma$ that g maps onto σ. The others are mapped onto proper faces of σ.

Recall that if $\sigma \in K$ then, since $\widehat{\sigma}$ lies interior to σ, the map g must carry $\widehat{\sigma}$ to one of the vertices of σ.

We now proceed by induction on the dimension of σ. If σ is a vertex v, then $g(v) = v$ and our result holds. If $\sigma = v_0 v_1$, then $g(v_0) = v_0$ and $g(v_1) = v_1$, while $g(\widehat{\sigma})$ equals either v_0 or v_1. In either case, g collapses one of the simplices $v_0\widehat{\sigma}$ or $v_1\widehat{\sigma}$ to a point and maps the other onto $v_0 v_1$.

In general, let s be a k-simplex $v_0 v_1 \dots v_k$. By the induction hypothesis, each $k-1$ face $s_i = v_0 v_1 \dots \widehat{v}_i \dots v_k$ of σ contains exactly one $k-1$ simplex t_i of $\mathrm{sd}\, K$ that is mapped by g onto s_i. Thus every k-simplex of $\mathrm{sd}\, \sigma$ is collapsed by g except possibly for k-simplices of the form $\widehat{\sigma} * t_i$ for $i = 0, \dots, k$. Now g carries $\widehat{\sigma}$ to one of the vertices of σ. Suppose that $g(\widehat{\sigma}) = v_i$. Since

$$g(t_i) = s_i = v_0 v_1 \dots \widehat{v}_i \dots v_k\,,$$

the map g carries $\widehat{\sigma} * t_j$ onto σ. And for $i \neq j$ it collapses each simplex $\widehat{\sigma} * t_i$ onto s_i.

Step 2: Here is another elementary fact: Let K be a complex. Let $\mathrm{sd} : C_p(K) \to C_p(\mathrm{sd}\, K)$ be the barycentric subdivision chain map defined in Section 2.5. If σ is an oriented p-simplex of K, then $\mathrm{sd}\, \sigma$ is the sum of all the p-simplices of $\mathrm{sd}\, K$ lying in σ and suitably oriented.

We proceed now by induction. The formula $\mathrm{sd}(v) = v$ shows the result to be true in dimension 0. In general, let σ have dimension p. Now the chain $\partial\sigma$ is the sum of the $p-1$ faces of σ, suitably oriented. Then $\mathrm{sd}(\partial\sigma)$ is by the induction hypothesis the sum of all the $p-1$ simplices of $\mathrm{sd}\, K$ lying in $\mathrm{Bd}\, \sigma$ and suitably oriented. It follows that

$$\mathrm{sd}\, \sigma = [\widehat{\sigma}, \mathrm{sd}\, (\partial\sigma)]$$

is the sum of all the p-simplices of $\mathrm{sd}\, K$ lying in σ and suitably oriented.

Step 3: Now let X be the manifold of our theorem. Choose a specific triangulation of X. Let γ be the cycle representing Γ relative to this triangulation. In the complex $\mathrm{sd}\, X$, we shall use the standard partial ordering of the vertices

[see Definition 8.4.1]. We orient the simplices of $\operatorname{sd} X$ by using this ordering. Orient the simplices of X arbitrarily.

By **Step 2**, the chain $\operatorname{sd}\gamma$ is the sum of all the oriented n-simplices of $\operatorname{sd} X$ with signs ± 1. Because it is a cycle, $\operatorname{sd}\gamma$ is an orientation cycle for $\operatorname{sd} X$.

Choose $g : \operatorname{sd} X \to X$ to be a simplicial approximation to the identity. Consider the diagram

$$\begin{array}{ccc} C^p(X) & \overset{\psi}{\dashrightarrow} & D_{n-p}(X) \\ \downarrow{\scriptstyle g^\#} & & \downarrow{\scriptstyle j} \\ C^p(\operatorname{sd} X) & \xrightarrow{\frown \operatorname{sd}\gamma} & C_{n-p}(\operatorname{sd} X) \end{array}$$

where j is inclusion of the dual chain complex of X into the simplicial chain complex of $\operatorname{sd} X$. We shall prove that the composite of $g^\#$ and $\frown \operatorname{sd}\gamma$ carries $C^p(X)$ isomorphically onto the subgroup $D_{n-p}(X)$ of $C_{n-p}(\operatorname{sd} X)$. It follows that there is a unique isomorphism ψ making this diagram commute.

First we prove the preliminary result that if σ is an oriented p-simplex of X, then $g^\#(\sigma^*) \frown \operatorname{sd}\gamma$ equals the sum of all $(n-p)$-simplices of $\operatorname{sd} X$ lying in the block $\overline{D}(\sigma)$ with coefficients ± 1.

We prove this fact by direct computation. Given σ, there is by **Step 1** exactly one p-simplex t of $\operatorname{sd} X$ that is mapped onto σ by g. Then

$$g^\#(\sigma^*) = \pm t^* \ ,$$

the sign depending on the chosen orientation. Let us compute the cap product $t^* \frown \operatorname{sd}\gamma$.

Recall the formula for the simplicial cap product. Given an ordering of the vertices of a complex K, and given a cochain c^p of K and a $(p+q)$-simplex τ of K, the chain $C^p \frown \tau$ equals the "front q-face" of γ multiplied by the value of c^p on the "back p-face" of τ.

Now t is a p-simplex of $\operatorname{sd} X$ of the form

$$t = [\widehat{s}_p, \ldots, \widehat{s}_0] \, ,$$

where $s_p = \sigma$ and s_i has dimension i for each i. Furthermore, $\operatorname{sd}\gamma$ is the sum of all the n-simplices of $\operatorname{sd} X$. Each is of the form

$$\tau = [\widehat{\sigma}_n, \ldots, \widehat{\sigma}_0] \, ,$$

where σ_i has dimension i for each i. The chain $t^* \frown \tau$ vanishes unless t equals the back face of τ. That is, unless

$$t = [\widehat{s}_p, \ldots, \widehat{s}_0] = [\widehat{\sigma}_p, \ldots, \widehat{\sigma}_0] \, .$$

In this case, $t^* \frown \tau$ equals $[\widehat{\sigma}_n, \ldots, \widehat{\sigma}_p]$. It follows that $t^* \frown \operatorname{sd}\gamma$ is the sum, with signs ± 1, of all simplices of $\operatorname{sd} X$ of the form

$$[\widehat{\sigma}_n, \ldots, \widehat{\sigma}_p]$$

for which $\sigma_p = s_p = \sigma$. These are precisely the $(n-p)$-simplices of $\overline{D}(\sigma)$.

Now we can prove our result. We show that $g^{\#}(\sigma^*) \frown \mathrm{sd}\,\gamma$ is in fact a *fundamental cycle* for the block $(\overline{D}(\sigma), \dot{D}(\sigma))$.

We know that $g^{\#}(\sigma^*) \frown \mathrm{sd}\,\gamma$ is a non-trivial $(n-p)$-chain carried by $\overline{D}(\sigma)$, and that it is not a multiple of any other chain [since its coefficients are all ± 1]. To show it is a fundamental cycle, it suffices to show that its boundary is carried by $\dot{D}(\sigma)$. For this, we compute

$$\begin{aligned}\partial(g^{\#}(\sigma^*) \frown \mathrm{sd}\,\gamma) &= (-1)^{n-p}\delta g^{\#}(\sigma^*) \frown \mathrm{sd}\,\gamma + 0 \\ &= (-1)^{n-p} g^{\#}(\delta\sigma^*) \frown \mathrm{sd}\,\gamma\,. \qquad (*)\end{aligned}$$

Now $\delta\sigma^*$ is a sum of terms of the form $\pm\sigma^*_{p+1}$, where σ_{p+1} is a $(p+1)$-simplex of X having σ as a face. By the result just proved, the chain $g^{\#}(\sigma^*_{p+1}) \frown \mathrm{sd}\,\gamma$ is carried by the block $\overline{D}(\sigma_{p+1})$ dual to σ_{p+1}. Since $\sigma_{p+1} \succ \sigma$, this block is contained in $\dot{D}(\sigma)$. Thus $g^{\#}(\delta\sigma^*) \frown \mathrm{sd}\,\gamma$ is carried by $\dot{D}(r)$, as desired.

Step 4: We have thus proved that the isomorphism

$$\psi : C^p(X) \to D_{n-p}(X)$$

exists, and that it carries the basis element $\sigma^* A$ to a fundamental cycle for the dual block to σ. Therefore ψ resembles the isomorphism ϕ of Theorem 8.5.2. The only remaining question is whether it commutes with δ and ∂. That it does, but only up to sign. Formula $(*)$ above gives us the equation

$$\partial\psi(c^p) = (-1)^{n-p}\psi(\delta c^p)\,.$$

Now the sign of $(-1)^{n-p}$ does not affect the groups of cocycles and boundaries. It follows that ψ induces an isomorphism of $H^p(X)$ with $H_{n-p}(\mathcal{D}(X))$, just as ϕ did. [In fact, ψ differs from ϕ only by a sign. See Exercise 1 below.]

Now since the inclusion $\mathcal{D}(X) \to \mathcal{C}(\mathrm{sd}\,X)$ induces a homology isomorphism, it follows that the composite map

$$H^p(X) \xrightarrow{\quad g^*\quad} H^p(\mathrm{sd}\,X) \xrightarrow{\frown\{\mathrm{sd}\,\gamma\}} H_{n-p}(\mathrm{sd}\,X)$$

is an isomorphism. Then if we identify the homology of $\mathrm{sd}\,X$ with that of X, by passing to the homology of the triangulable space X, for instance, or to singular homology and cohomology, we see that since g^* is the identity map, this isomorphism can be expressed in the form

$$H^p(X) \xrightarrow{\frown\Gamma} H_{n-p}(X)\,.$$

Step 5: To treat the case of arbitrary coefficients, one checks that the composite

$$C^p(X) \otimes G \cong C^p(X;G) \xrightarrow{g^\#} C^p(\mathrm{sd}\, X;G) \xrightarrow{\frown \mathrm{sd}\,\gamma} D_{n-p}(X) \otimes G$$

equals the isomorphism $C^p(X) \to D_{n-p}(X)$ of **Step 4** tensored with the identity map of G. Hence it is an isomorphism.

Step 6: In the case where X is not necessarily orientable, the preceding proof works if one reduces all the coefficients mod 2. **Steps 1** and **2** proceed unchanged. The proof in **Step 3** shows that $g^\#(\sigma^*) \frown \mathrm{sd}\,\gamma_{(2)}$ is the nontrivial $(n-p)$-cycle (with $\mathbb{Z}/2$ coefficients) of the block $(\overline{D}(\sigma), \dot{D}(\sigma))$. **Step 4** is easier, since no signs are involved. □

Our proof of this theorem used simplicial homology throughout. But the fact that simplicial theory and singular theory are isomorphic, by an isomorphism that preserves cap products, means that this theorem holds for singular theory as well. In particular, the isomorphism given by $\frown \Gamma$ depends only on the homology class Γ, not on any particular triangulation involved.

The preceding theorem gives us information about how the Poincaré duality isomorphism behaves with respect to continuous maps. Specifically, one has the following result, whose proof is an immediate consequence of the naturality of the cap product.

Theorem 8.7.3 *Let X and Y be compact, connected, triangulable, orientable homology n-manifolds. Let Γ_X and Γ_Y be orientation classes for X and Y, respectively. Given $f : X \to Y$, let d be the integer such that $f_*(\Gamma_X) = d \cdot \Gamma_y$. Then the following diagram commutes:*

$$\begin{array}{ccc} H^p(X;G) & \xrightarrow{\frown \Gamma_X} & H_{n-k}(X;G) \\ \downarrow f^* & & \downarrow f_* \\ H^k(Y;G) & \xrightarrow{\frown d\Gamma_Y} & H_{n-k}(Y;G)\,. \end{array}$$

In particular, if $d = \pm 1$, then f^* is injective and f_* is surjective. The same result holds for $G = \mathbb{Z}/2$ without requiring X and Y to be orientable.

□

Definition 8.7.4 The integer d appearing in the statement of this theorem is called the *degree* of f, written $d(f)$, relative to the orientation cycles Γ_X and Γ_Y. Since Γ_X and Γ_Y are uniquely determined up to sign, so is $d(f)$. If $X = Y$ and if $\Gamma_X = \Gamma_Y$, then d is uniquely determined.

The second version of Poincaré duality, in which the isomorphism is obtained by use of cap products, has consequences that go far beyond the naturality diagram just proved. Note that everything in the statement of Theorem 8.7.2 makes sense even if X is not triangulable—provided one knows what

one means by an orientation preserving class for X. This fact suggests that by using cap products in singular theory one might obtain a proof of Poincaré duality for an arbitrary compact topological manifold without assuming triangulability. This is in fact the case. The interested reader may consult [Dol95] [or alternatively, [GH81], [Vic94], or [Spa94]].

EXERCISES

1. Given the orientation cycle γ for X, let ψ be constructed as in the proof of Theorem 8.7.2.

(a) Define ϕ by the rule

$$\phi(c^p) = (-1)^{\alpha(p)}\psi(c^p)\,,$$

where

$$\alpha(4k+n) = 0\,, \quad \alpha(4k+n+2) = 1\,,$$
$$\alpha(4k+n+1) = 0, \quad \alpha(4k+n+3) = 1$$

for all k. Show that ϕ satisfies the condition $\phi\delta = \partial\phi$.

(b) Show that if σ is an n-simplex of X, oriented as it appears in the cycle γ, then $\psi(\sigma^*) = \widehat{\sigma}$. [**Hint:** Note that $\epsilon(\psi(\sigma^*)) = \epsilon(g_\#(\psi(\sigma^*))) = \epsilon(\sigma^* \frown \gamma)$.]

(c) Conclude that the homomorphism ϕ of **(a)** is identical with that in the proof of Theorem 8.5.2.

2. Let X and Y be compact, connected, triangulable homology n-manifolds. Let $f : X \to Y$.

(a) Show that if X and Y are orientable and if f has non-zero degree, then $\beta_i(X) \geq \beta_i(Y)$, where β_i denotes the ith betti number.

(b) If $f_* : H_n(X; \mathbb{Z}/2) \to H_n(Y; \mathbb{Z}/2)$ is non-trivial, then what can you say about $H_i(X; \mathbb{Z}/2)$ and $H_i(Y, \mathbb{Z}/2)$?

3. Let X denote the n-fold connected sum of tori. Let $X_0 = \mathbb{S}^2$. Show that there is a map $f : X_n \to X_m$ of non-zero degree if and only if $n \geq m$.

4. Let X_n be as in Exercise 3. Let Y_n be the n-fold connected sum of projective planes, with $n \geq 1$. Given f, consider the condition

$$f_* \text{ is non-trivial as a map of } H_2 \text{ with } \mathbb{Z}/2 \text{ coefficients.} \qquad (*)$$

(a) Show that there is a map $f : Y_n \to Y_m$ satisfying $(*)$ if and only if $n \geq m$.

(b) Show that there is no map $f : X_n \to Y_m$ satisfying $(*)$.

The case of maps $f : Y_m \to X_n$ will be considered in the exercises of the next section.

8.8 *Application: Cohomology Rings of Manifolds

In this section, we assume familiarity with the projective spaces [Section 4.14]. We also use the universal coefficient theorem for cohomology [Section 7.3].

We now apply the Pioncaré duality theorem to the problem of computing the cohomology ring of a manifold. While we cannot compute the cohomology ring of every compact manifold, the theorem we shall prove is strong enough to enable us to compute the cohomology rings of the real and complex projective spaces. This computation leads, in turn, to proofs of such classical theorems of topology as the Borsuk-Ulam theorem and the so-called "Ham-Sandwich Theorem."

Definition 8.8.1 Let A and B be free abelian groups of the same rank. Let C be an infinite cyclic group. We say that a homomorphism

$$f : A \otimes B \to C$$

is a *dual pairing* if there are bases $a_1, \ldots, a_m$ for A and $b_1, \ldots, b_m$ for B so that

$$f(a_i \otimes b_j) = \delta_{ij}\gamma$$

for all $i, j = 1, \ldots, m$, where γ is a generator of C.

Theorem 8.8.2 *Let X be a compact, connected, triangulable, orientable homology n-manifold. Let $T^k(X)$ denote the torsion subgroup of $H^k(X)$. The cup product operation induces a homomorphism*

$$\frac{H^k(X)}{T^k(X)} \otimes \frac{H^{n-k}(X)}{T^{n-k}(X)} \to H^n(X)$$

that is a dual pairing.

Proof: Let $\alpha \in H^k(X)$ and $\beta \in H^{n-k}(X)$. If α is an element of finite order, then so is $\alpha \smile \beta$. Since $H^p(X)$ is infinite cyclic, we see that $\alpha \smile \beta = 0$. A similar remark applies if β has finite order. Therefore [by Lemma 6.2.5], the cup product induces a homomorphism

$$\frac{H^k(X)}{T^k(X)} \otimes \frac{H^{n-k}(X)}{T^{n-k}(X)} \to H^n(X)\,.$$

Choose a generator Λ of $H^n(X)$. Then choose a generator Γ of $H_n(X)$ by choosing its sign so that the isomorphism

$$H^n(X) \xrightarrow{\frown \Gamma} H_0(X) \xrightarrow{\epsilon_*} \mathbb{Z} \qquad (*)$$

maps Λ to 1. Then $\epsilon_*(\Lambda \frown \Gamma) = 1$.

To prove the theorem, we shall find elements $\alpha_1, \ldots, \alpha_m$ of $H^k(X)$ and $\beta_1, \ldots, \beta_m$ of $H^{n-k}(X)$ such that

$$\alpha_i \smile \beta_j = \delta_{ij}\Lambda$$

for $i, j = 1, \ldots, m$ and such that their cosets modulo torsion, denoted by $\{\alpha_i\}$ and $\{\beta_j\}$ respectively, form bases for $H^k(X)/T^k(X)$ and $H^{n-k}(X)/T^{n-k}(X)$, respectively.

The idea of the proof is this. First, there is a *geometric* duality isomorphism

$$\frown \Gamma : H^{n-k}(X) \to H_k(X)\,,$$

obtained from the cap product with Γ. This isomorphism induces an isomorphism of $H^{n-k}(X)/T^{n-k}(X)$ with $H_k(X)/T_k(X)$.

Second, there is an *algebraic* duality isomorphism κ^* obtained from the Kronecker map κ. It is obtained from the exact sequence

$$0 \longleftarrow \mathrm{Hom}(H_k(X), \mathbb{Z}) \xleftarrow{\kappa} H^k(X) \longleftarrow \mathrm{Ext}(H_{k-1}(X), \mathbb{Z}) \longleftarrow 0\,,$$

as follows: The left-hand group is isomorphic to $\mathrm{Hom}(H_k(X)/T_k(X), \mathbb{Z})$ and is thus free, while the right-hand group is a torsion group. Therefore there exists an induced isomorphism

$$\mathrm{Hom}\left(\frac{H_k(X)}{T_k(X)}, \mathbb{Z}\right) \xleftarrow{\kappa^*} \frac{H^k(X)}{T^k(X)}\,.$$

It is related to the cap product by the formula

$$[\kappa^*\{\alpha\}]\{\beta\} = [\kappa(\alpha)](\beta) = \langle \alpha, \beta \rangle = \epsilon_*(\alpha \frown \beta)\,.$$

Combining these two types of duality will give us our theorem.

Consider now the sequence of isomorphisms

$$\frac{H^{n-k}}{T^{n-k}} \xrightarrow[\cong]{\frown \Gamma} \frac{H_k}{T_k} \cong \mathrm{Hom}\left(\frac{H_k}{T_k}, \mathbb{Z}\right) \xleftarrow[\cong]{\kappa^*} \frac{H^k}{T^k}\,.$$

[Here we delete X from the notation—just for convenience.] Choose a basis $\{\beta_1\}, \{\beta_2\}, \ldots, \{\beta_m\}$ for H^{n-k}/T^{n-k}. Let $\{\gamma_1\}, \{\gamma_2\}, \ldots, \{\gamma_m\}$ be the corresponding basis for H_k/T_k, defined by the equation $\beta_j \frown \Gamma = \gamma_j$ for all j. Let $\gamma_1^*, \gamma_2^*, \ldots, \gamma_m^*$ be the corresponding dual basis for $\mathrm{Hom}(H_k/T_k, \mathbb{Z})$, defined by the equation $\gamma_i^*(\{\gamma_j\}) = \delta_{ij}$. Finally, let $\{\alpha_1\}, \{\alpha_2\}, \ldots, \{\alpha_m\}$ be the

corresponding basis for H^k/T^k, defined by the equation $\kappa^*(\{\alpha_i\}) = \gamma_i^*$ for all i.

To show that $\alpha_i \smile \beta_j = \delta_{ij}\Lambda$, it suffices [applying the isomorphism of $H^n(X)$ with $\mathbb{Z}$ given by $(*)$] to see that

$$\epsilon_*((\alpha_i \smile \beta_j) \frown \Gamma) = \delta_{ij}\epsilon_*(\Lambda \frown \Gamma) = \delta_{ij}\,.$$

We compute as follows:

$$\begin{aligned}\delta_{ij} &= \gamma_i^*(\{\gamma_j\}) \\ &= [\kappa^*\{\alpha_i\}](\{\beta_j\} \frown \Gamma) \\ &= \langle \alpha_i, \beta_j \frown \Gamma\rangle \\ &= \epsilon_*(\alpha_i \frown (\beta_j \frown \Gamma)) \\ &= \epsilon_*((\alpha_i \smile \beta_j) \frown \Gamma)\,. \end{aligned}$$ □

A similar theorem holds if the coefficients form a field.

Theorem 8.8.3 *Let X be a compact, connected, triangulable homology n-manifold. Let F be a field. Assume that F equals $\mathbb{Z}/2$ if X is non-orientable. Let Λ generate $H^n(X;F)$. There are (vector space) bases $\alpha_1,\ldots,\alpha_m$ for $H^k(X;F)$ and $\beta_1,\ldots,\beta_m$ for $H^{n-k}(X;F)$ such that, for $i,j = 1,\ldots,m$,*

$$\alpha_i \smile \beta_j = \delta_{ij}\Lambda\,.$$

Proof: In the orientable case, choose Γ so that the isomorphism

$$H^n(X;F) \xrightarrow{\frown\,\Gamma} H_0(X;F) \xrightarrow{\epsilon_*} F$$

carries Λ to $1 \in F$. In the non-orientable case, $\Gamma_{(2)}$ is unique. Then proceed as before using the following diagram, in which Γ' denotes either Γ or $\Gamma_{(2)}$ depending on the case:

$$H^{n-k}(X;F) \underset{\cong}{\xrightarrow{\frown\,\Gamma'}} H_k(X;F) \cong \mathrm{Hom}\,(H_k(X;F),F) \underset{\cong}{\xleftarrow{\kappa^*}} H^k(X;F)\,.$$ □

EXAMPLE 8.8.4 Let us consider the torus once again. By Theorem 8.8.2, there is a basis α_1,α_2 for $H^1(T)$ and a basis β_1,β_2 for $H^1(T)$ such that $\alpha_i \smile \beta_j = \delta_{ij}\Lambda$ where Λ is a generator of $H^2(T)$. The two "picket fence" cocycles w^1 and z^1 pictured in Figure 8.8.1 give us a basis for $H^1(T)$ and the indicated cocycle σ^* generates $H^2(T)$. Recall that $z^1 \smile w^1 = \sigma^*$. The conclusion of the theorem holds if we let α_1 and α_2 be the cohomology classes of z^1 and w^1 respectively. And we let β_1 and β_2 be the cohomology classes of w^1 and $-z^1$ respectively. Further, we let Λ be the cohomology class of σ^*. These computations were carried out in Example 5.10.5.

The proof of the theorem tells us that this is the way things ought to work.

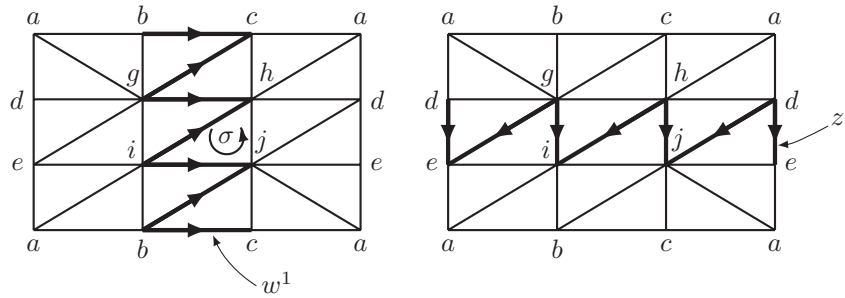


Figure 8.8.1

If one begins with the cocycle w^1, then the geometric [Poincaré] duality map ψ carries it to the cycle c_1 pictured in Figure 8.8.2. [We know that the cycle "dual" to w^1 is carried by the blocks dual to the simplices appearing in the carrier of w^1. To check that ψ carries w^1 to c, rather than to $-c$, requires some care.] The cycle c_1 is homologous to one represented by the cell B along the side edge of the rectangle. Under the algebraic duality κ^*, this cycle B in turn corresponds to a cocycle whose value is 1 on B and 0 on the cycle A going around the torus the other way. The cocycle z^1 is such a cocycle. Thus w^1

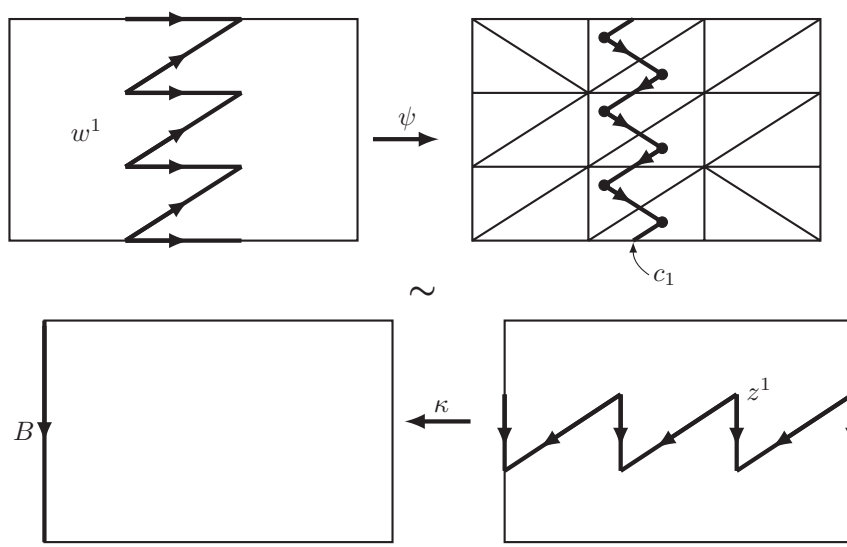


Figure 8.8.2

is carried to z^1 under our combined geometric-algebraic duality isomorphism. This is just as expected, since $\{z^1 \smile w^1\} = \Lambda$ while $z^1 \smile z^1 = w^1 \smile w^1 = 0$.

Now we apply these results to real projective space $\mathbb{P}^n$. We know that $\mathbb{P}^n$ is triangulable [see Lemma 4.14.11], so Poincaré duality applies. The cellular chain complex of $\mathbb{P}^n$ is infinite cyclic in each dimension $k = 0, 1, \ldots, n$. And the boundary operator is either 0 or multiplication by 2. Therefore $H^k(\mathbb{P}^n; \mathbb{Z}/2)$ is a vector space of dimension 1 for $k = 0, 1, \ldots, n$. It remains to calculate cup products in the ring $H^*(\mathbb{P}^n; \mathbb{Z}/2)$.

In the next theorem and its proof, we will use the following notation for repeated cup products:

$$u^n = \underbrace{u \smile u \smile \cdots \smile u}_{n \text{ copies}}$$

Theorem 8.8.5 *If u is the non-zero element of $H^1(\mathbb{P}^n, \mathbb{Z}/2)$, then u^k is the non-zero element of $H^k(\mathbb{P}^n; \mathbb{Z}/2)$ for $k = 2, \ldots, n$. Thus $H^*(\mathbb{P}^n; \mathbb{Z}/2)$ is a truncated polynomial algebra over $\mathbb{Z}/2$ with one generator u in dimension 1, truncated by setting $u^{n+1} = 0$.*

Proof: We proceed by induction, beginning with $n = 2$. The vector space $H^1(\mathbb{P}^2; \mathbb{Z}/2)$ has dimension 1. If u is its non-zero element, then by Theorem 8.8.3 we have that $u^2 = u \smile u$ must be non-zero.

Suppose that the theorem is true for dimension $n - 1$. The inclusion map $j : \mathbb{P}^{n-1} \to \mathbb{P}^n$ induces homology and cohomology isomorphisms [over $\mathbb{Z}/2$] in dimensions less than n. Therefore, by the induction hypothesis, if $u \in H^1(\mathbb{P}^n; \mathbb{Z}/2)$ is non-zero, then so are $u^2, \ldots, u^{n-1}$. [Recall that j^* is a ring homomorphism.] It remains to show that $u^n \neq 0$.

By Theorem 8.8.3, there is a basis α_1 for $H^1(\mathbb{P}^n; \mathbb{Z}/2)$ and a basis β_1 for $H^{n-1}(\mathbb{P}^n; \mathbb{Z}/2)$ such that $\alpha_1 \smile \beta_1$ generates $H^n(\mathbb{P}^n; \mathbb{Z}/2)$. Since u and u^{n-1} are the unique non-zero elements of these groups, respectively, we must have $\alpha_1 = u$ and $\beta_1 = u^{n-1}$. Thus $u^n = u \smile u^{n-1} \neq 0$ as desired. □

Corollary 8.8.6 *We see that $H^*(\mathbb{P}^\infty; \mathbb{Z}/2)$ is a polynomial algebra over $\mathbb{Z}/2$ with a single one-dimensional generator.*

Proof: This corollary follows from the fact that the inclusion $j : \mathbb{P}^n \to \mathbb{P}^\infty$ induces a cohomology isomorphism (over $\mathbb{Z}/2$) in dimensions less than or equal to n, and the fact that j^* preserves cup products. □

We now apply this result to prove several classical theorems of topology.

Definition 8.8.7 A map $f : \mathbb{S}^n \to \mathbb{S}^n$ is said to be *antipode preserving* if $f(-x) = -f(x)$ for all x in $\mathbb{S}^n$.

Theorem 8.8.8 *If $f : \mathbb{S}^n \to \mathbb{S}^m$ is continuous and antipode-preserving, then $n \leq m$.*

Proof: Let $a_n = (1, 0, \ldots, 0) \in \mathbb{S}^n$. We call this the "base point" of $\mathbb{S}^n$. Let F denote the field $\mathbb{Z}/2$.

Step 1: Let $\pi_n : \mathbb{S}^n \to \mathbb{P}^n$ be the usual quotient map. We prove that if $\alpha : I \to \mathbb{S}^n$ is any path from a_n to its antipode $-a_n$, then $\pi_n \circ \alpha$ represents the non-zero element of $H_1(\mathbb{P}^n; F)$. See Figure 8.8.3.

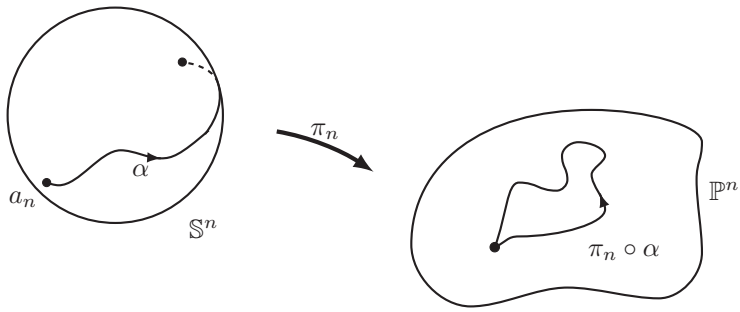


Figure 8.8.3

This conclusion is easy to prove if α is the standard path

$$\beta(t)(\cos \pi t, \sin \pi t, 0, \ldots, 0)\,.$$

For then β is a homeomorphism of $(I, \operatorname{Bd} I)$ with $(E^1_+, \mathbb{S}^0)$, where D^1_+ is the upper half of $\mathbb{S}^1$. And π_1 maps $(E^1_+, \mathbb{S}^0)$ onto $(\mathbb{P}^1, \mathbb{P}^0)$, collapsing $\mathbb{S}^0$ to a point. By our basic results about CW complexes, the image under $(\pi_1)_*$ of a generator of $H_1(E^1_+, \mathbb{S}^0)$ is a fundamental cycle for the 1-cell of $\mathbb{P}^n$. The identity map, considered as a singular simplex $i : \Delta_1 \to I$, generates $H_1(I, \operatorname{Bd} I; F)$. Therefore the singular simplex $\pi_1 \circ \beta \circ i = \pi_1 \circ \beta$ generates $H_1(\mathbb{P}^n; F)$.

To show the same result for a general path α, we proceed as follows. Consider the singular 1-chain $\alpha - \beta$, where β is the standard path as before. It is a singular cycle of $\mathbb{S}^n$ since its boundary vanishes. In the case $n > 1$, $H_1(\mathbb{S}^n) = 0$, so that $\alpha - \beta$ must bound some 2-chain d. Thus

$$\pi_n \circ \alpha - \pi_n \circ \beta = \partial(\pi_n \circ d)\,,$$

so $\pi_n \circ \alpha$ and $\pi_n \circ \beta$ are homologous. Hence they are also homologous as cycles with F coefficients.

If $n = 1$ then we use the fact that the map $\pi_1 : \mathbb{S}^1 \to \mathbb{P}^1$ has degree 2. Since $\alpha - \beta$ is a singular cycle of $\mathbb{S}^1$, the cycle $\pi_1 \circ \alpha - \pi_1 \circ \beta$ represents an even multiple of the generator of $H_1(\mathbb{P}^1)$. In particular, it represents the zero element of $H_1(\mathbb{P}^1; F)$. Then $\pi_1 \circ \alpha$ and $\pi_1 \circ \beta$ are homologous as cycles with F coefficients.

Step 2: Let $f : \mathbb{S}^n \to \mathbb{S}^m$ be an antipode-preserving, continuous map. Choose a rotation of $\mathbb{S}^m$, say $\rho : \mathbb{S}^m \to \mathbb{S}^m$, that carries the point $f(a_n)$ to the base point a_m of $\mathbb{S}^m$. Then the map $g = \rho \circ f$ is continuous and antipode-preserving, and carries a_n to a_m. Via the quotient maps π_n and π_m, the map g induces a continuous map $h : \mathbb{P}^n \to \mathbb{P}^m$:

$$\begin{array}{ccc} \mathbb{S}^n & \xrightarrow{\ g\ } & \mathbb{S}^m \\ \big\downarrow \pi_n & & \big\downarrow \pi_m \\ \mathbb{P}^n & \xrightarrow{\ h\ } & \mathbb{P}^m . \end{array}$$

We shall show that $h_* : H_1(\mathbb{P}^n; F) \to H_1(\mathbb{P}^m; F)$ is non-trivial.

Let α be any path in $\mathbb{S}^n$ from a_n to $-a_n$. Since g preserves antipodes, $g(-a_n) = -a_m$ so $g \circ \alpha$ is a path in $\mathbb{S}^m$ from a_m to $-a_m$. Now $h_\#$ carries the cycle $\pi_n \circ \alpha$ to the cycle $\pi_m \circ g \circ \alpha$. Thus, by **Step 1**, the homomorphism h_* carries the non-zero element of $H_1(\mathbb{P}^n; F)$ to the non-zero element of $H_1(\mathbb{P}^m; F)$.

Step 3: Now we prove the theorem. Using naturality of the isomorphism

$$\mathrm{Hom}_F(H_1(\mathbb{P}^k; F), F) \xleftarrow{\kappa^*} H^1(\mathbb{P}^k; F)$$

for $k = m, n$, it follows from **Step 2** that the homomorphism h^* of cohomology

$$H^1(\mathbb{P}^n; F) \xleftarrow{h^*} H^1(\mathbb{P}^m; F)$$

is non-trivial. Let $u \in H^1(\mathbb{P}^m; F)$ be non-trivial. Then $h^*(u) \in H^1(\mathbb{P}^n; F)$ is non-trivial. Because h^* is a ring homomorphism, $h^*(u^n) = (h^*(u))^n$. By the preceding theorem, the latter element is non-trivial. Therefore u^n must be non-trivial. It follows that $m \geq n$. □

Theorem 8.8.9 (Borsuk–Ulam) *If $h : \mathbb{S}^n \to \mathbb{R}^n$ is a continuous map, then $h(x) = h(-x)$ for at least one $x \in \mathbb{S}^n$.*

Proof: If $h(x) \neq h(-x)$ for all $x \in \mathbb{S}^n$ then the function $f : \mathbb{S}^n \to \mathbb{S}^{n-1}$ given by

$$f(x) = \frac{h(x) - h(-x)}{\|h(x) - h(-x)\|}$$

is continuous and antipode preserving. Such a function does not exist. □

Theorem 8.8.10 (Baby ham sandwich theorem) *Let A_1 and A_2 be two bounded, measurable subsets of $\mathbb{R}^2$. There is a line in $\mathbb{R}^2$ that bisects both A_1 and A_2.*

Remark 8.8.11 This result is *not* elementary. Try finding such a line even in the simple case where A_1 and A_2 are both triangular regions.

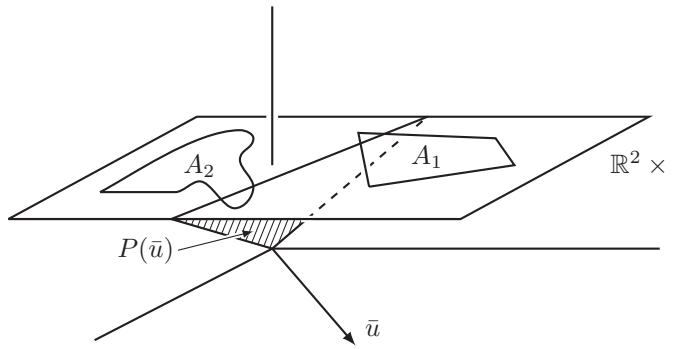


Figure 8.8.4

Proof of the Theorem: Suppose that A_1 and A_2 both lie in the plane $\mathbb{R}^2 \times \{1\}$ in $\mathbb{R}^3$. For each unit vector $\overrightarrow{u}$ in $\mathbb{R}^3$, consider the plane $P(\overrightarrow{u})$ through the origin perpendicular to $\overrightarrow{u}$. Let $f_i(\overrightarrow{u})$ equal the measure of the part of A_i that lies on the same side of $P(\overrightarrow{u})$ as does the vector $\overrightarrow{u}$. See Figure 8.8.4. Note that

$$f_i(\overrightarrow{u}) + f_i(-\overrightarrow{u}) = \text{measure}\, A_i\,.$$

Now the function

$$\overrightarrow{u} \mapsto (f_1(\overrightarrow{u}), f_2(\overrightarrow{u}))$$

is a continuous map of $\mathbb{S}^2$ into $\mathbb{R}^2$. By the Borsuk-Ulam theorem,

$$(f_1(\overrightarrow{a}), f_2(\overrightarrow{a})) = (f_1(-\overrightarrow{a}), f_2(-\overrightarrow{a}))$$

for some $\overrightarrow{a} \in \mathbb{S}^2$. Then

$$f_i(\overrightarrow{a}) = \frac{1}{2}(\text{measure}\, A_i)$$

for $i = 1, 2$. The line in which the plane $P(\overrightarrow{a})$ intersects the plane $\mathbb{R}^2 \times \{1\}$ is thus the desired line that bisects each of A_1 and A_2. □

The proof of this last theorem generalizes to prove the ham sandwich theorem, that is, to show that for n bounded measurable sets in $\mathbb{R}^n$, there is an $n-1$ plane in $\mathbb{R}^n$ that bisects them all. The motivation for the name of the ham sandwich theorem is the following: One thinks of a ham sandwich as consisting of two slices of bread and a slice of ham, so in case $n = 3$, the theorem says that one can bisect each slice of bread and the slice of ham with a single whack of the knife.

EXERCISES

1. Let T be as in Example 8.8.4. Let $g : \mathrm{sd}\, T \to T$ be a simplicial approximation to the identity. Check that if $\mathrm{sd}\, \gamma$ is the sum of all the n-simplices of $\mathrm{sd}\, T$, oriented counterclockwise, then $g^{\#}(w^1) \frown \mathrm{sd}\, \gamma$ is the cycle c_1 pictured in Figure 8.8.2.

2. **(a)** If $\phi : R \to R'$ is a homomorphism of commutative rings with unity, then show that the induced homomorphism

$$H^*(X; R) \to H^*(X; R')$$

is a ring homomorphism.

(b) Use the coefficient homomorphism $\mathbb{Z} \to \mathbb{Z}/2$ to compute the cohomology rings of $\mathbb{P}^{2k}$, of $\mathbb{P}^{2k+1}$, and of $\mathbb{P}^{\infty}$.

3. Show that at any given point in time there are antipodal points on the surface of the earth at which the temperature and barometric pressure are equal.

4. Compute the cohomology rings of the following:

(a) $\mathbb{S}^3 \times \mathbb{P}^5$ with $\mathbb{Z}$ and $\mathbb{Z}/2$ coefficients.

(b) $\mathbb{P}^2 \times \mathbb{P}^3$ with $\mathbb{Z}/2$ coefficients.

5. Assume that $\mathbb{C}P^n$ can be triangulated [and indeed it can be].

(a) Compute the cohomology rings $H^*(\mathbb{C}P^n)$ and $H^*(\mathbb{C}P^{\infty})$.

(b) Show that if $f : \mathbb{C}P^n \to \mathbb{C}P^n$, then the degree of f equals a^n for some integer a.

(c) Show that every map $f : \mathbb{C}P^{2n} \to \mathbb{C}P^{2n}$ has a fixed point.

(d) Show that there exists no map $f : \mathbb{C}P^{2n} \to \mathbb{C}P^{2n}$ of degree -1.

(e) If $f : \mathbb{C}P^{2n+1} \to \mathbb{C}P^{2n+1}$ has no fixed point, then what is the degree of f?

6.* Suppose we form a manifold M by taking two copies of $\mathbb{C}P^2$, deleting a small open 4-ball from each, and pasting the two remaining spaces together along their boundary 3-spheres. Discuss the cohomology ring of M. Compare with the cohomology ring of $\mathbb{S}^2 \times \mathbb{S}^2$.

7. Let X be a connected, triangulable homology 7-manifold. Suppose that

$$\begin{aligned} H_7(X) &\cong \mathbb{Z}, \\ H_6(X) &\cong \mathbb{Z}, \\ H_5(X) &\cong \mathbb{Z}/2, \\ H_4(X) &\cong \mathbb{Z} \oplus \mathbb{Z}/3. \end{aligned}$$

Give all the information you can about the cohomology groups of X and of the cohomology ring $H^*(X)$.

8. Let X be a compact, orientable, triangulable homology manifold of dimension $4n+2$. Show that it cannot be true that $H_{2n+1}(X) \cong \mathbb{Z}$.

9. Let $X_0 = \mathbb{S}^2$. Let X_n be the n-fold connected sum of the torus, T, with itself. Let Y_n be the n-fold connected sum of $\mathbb{P}^2$ with itself. Given f, consider the condition:

$$f_* \text{ is a non-trivial homomorphism of } H_2 \text{ with } \mathbb{Z}/2 \text{ coefficients.} \quad (*)$$

(a) Show that there is no map $f : Y_{2n} \to X_n$ satisfying $(*)$. [**Hint:** If $\phi : A \to B$ is a homomorphism, where A and B are free and $\operatorname{rank} A > \operatorname{rank} B$, then $\ker \phi$ has positive rank and is a direct summand in A. Choose a basis $\beta_1, \ldots, \beta_{2n}$ for $H^1(X_n)$ such that $f^*(\beta_1) = 0$. Pass to $\mathbb{Z}/2$ coefficients. Apply the proof of Theorem 8.8.3.]

(b) Show that there is a map $f : Y_{2n+1} \to X_n$ satisfying $(*)$. [**Hint:** First show that $Y_3 \approx T \# \mathbb{P}^2$.]

(c) Show that there is a map $f : Y_m \to X_n$ satisfying $(*)$ if and only if $m \geq 2n+1$.

10. Give an example of a continuous map $f : \mathbb{S}^2 \to \mathbb{S}^2$ that is not antipode preserving. Is there such an example in every dimension?

11. Formulate a suitable ham sandwich theorem in four-dimensional space. Can you prove it?

8.9 *Application: Homotopy Classification of Lens Spaces

In this section, we assume familiarity with lens spaces [Section 4.14].

Given the ring structure in cohomology, defined by the cup product, the Poincaré duality isomorphism

$$H^*(X; R) \xrightarrow{\frown \Gamma} H_*(X; R)$$

gives us an induced ring structure in homology called the *homology intersection ring*. This ring was known well before cohomology was discovered. The intersection product $\alpha \cdot \beta$ of two homology classes was defined, roughly speaking, as the geometric intersection of two cycles that represent α and β. Defining the homology intersection ring in this way, and deriving its properties, involved a good deal of work. Furthermore, although one proved that the multiplication operation was a topological invariant, one found that in general it was not preserved by continuous maps, even between manifolds of the same

dimension. As a consequence, it seems hardly surprising that all efforts to extend the intersection product from manifolds to arbitrary polyhedra were unsuccessful.

Nowadays we view the matter differently. We consider the cohomology ring as the natural object and view the existence of the homology intersection ring for manifolds as a happy accident resulting from the Poincaré duality isomorphism.

We are not going to study the homology intersection ring in any detail. We shall, however, prove one lemma interpreting the ring operation as a geometric intersection of cycles. This lemma will enable us to make computations in the cohomology rings of lens spaces.

Notation 8.9.1 Throughout this section, X will denote a compact, connected, triangulable n-manifold.

(a) In case X is orientable, we let Γ denote an orientation class for X, and we let R denote an arbitrary commutative ring with unity.

(b) In case X is not orientable, we let Γ, denote the class $\Gamma_{(2)}$, and we let R denote $\mathbb{Z}/2$.

Definition 8.9.2 We define the *intersection product* of two homology classes of X as follows. Given $\alpha_p \in H_p(X;R)$ and $\beta_q \in H_q(X;R)$, choose α^{n-p} and β^{n-q} so that

$$\alpha^{n-p} \frown \Gamma = \alpha_p \quad \text{and} \quad \beta^{n-q} \frown \Gamma = \beta_q\,.$$

Then define

$$\alpha_p \cdot \beta_q = (\alpha^{n-p} \cup \beta^{n-q}) \frown \Gamma\,.$$

Note that when X is orientable, the sign of the result depends on the orientation class Γ. Notice also that the product of α_p and β_q has dimension $p+q-n$. This fact makes the connection with geometric intersection plausible, for this is just the dimension of the intersection in $\mathbb{R}^n$ of a general plane of dimension p and a general plane of dimension q.

Lemma 8.9.3 *We use Notation 8.9.1. Let α_p and β_q be homology classes of X. Let $p+q=n$. Let α_p be represented by a cycle c_p of $\operatorname{sd} X$ that is carried by the dual p-skeleton of X. Let β_q be represented by a cycle of d_q of X.*

Suppose that the carriers of c_p and d_q meet in a single point, the barycenter of the q-simplex σ of X. If d_q has coefficient $\pm a$ on the simplex σ, and c_p has coefficient $\pm b$ on the dual cell $D(\sigma)$, then $\alpha_p \cdot \beta_q$ is represented [up to sign] by the 0-chain $ab\,\widehat{\sigma}$ of $\operatorname{sd} X$.

Proof: Let α^q and β^p be cohomology classes such that

$$\alpha^q \frown \Gamma = \alpha_p \quad \text{and} \quad \beta^p \frown \Gamma = \beta_q\,.$$

Then, by the definition of the homology intersection ring,

$$\begin{aligned}\alpha_p \cdot \beta_q &= (\alpha^q \smile \beta^p) \frown \Gamma \\ &= \alpha^q \frown (\beta^p \frown \Gamma) \\ &= \alpha^q \frown \beta_q \,.\end{aligned}$$

Because the homomorphism $\epsilon_* : H_0(X; R) \to R$ induced by the augmentation map ϵ is an isomorphism, to prove our result it suffices to show that

$$\pm ab = \epsilon_*(\alpha_p \cdot \beta_q) = \epsilon_*(\alpha^q \frown \beta_q) = \langle \alpha^q, \beta_q \rangle \,.$$

We evaluate this Kronecker index by finding a representative cocycle for α^q. Orient the simplices of X. Let γ be an n-cycle of X representing Γ. Let $\psi : C^q(X) \to D_p(X)$ be the isomorphism defined by the diagram

$$\begin{array}{ccc} C^q(X) & \xrightarrow{\psi} & D_p(X) \\ \big\downarrow g^{\#} & & \big\downarrow j \\ C^q(\mathrm{sd}\, X) & \xrightarrow{\frown \mathrm{sd}\, \gamma} & C_p(\mathrm{sd}\, X) \,, \end{array}$$

where g is a simplicial approximation to the identity. If σ_i is an oriented q-simplex of X and σ_i^* is the corresponding elementary cochain, then $\psi(\sigma_i^*)$ is a fundamental cycle for the p-block $\overline{D}(\sigma_i)$ dual to σ_i—which we denote by z_i.

Now the cycle representing α_p has the form

$$c_p = \sum b_j z_j = \sum b_j \psi(\sigma_j^*) \,.$$

We assert that α^q is represented by the cocycle defined by the equation

$$c^q = \sum b_j \sigma_j^* \,.$$

The fact that $\partial\psi = \pm\psi\delta$ implies that c^q is a cocycle. And the fact that ψ induces the isomorphism $\frown \Gamma$ means that c^q represents α^q.

Now the cycle d_q has the form

$$d_q = \sum a_i \sigma_i \,.$$

The dual cell $\overline{D}(\sigma_j)$ carrying the chain z_j intersects the simplex σ_i only if $i = j$. In this case they intersect in the barycenter $\widehat{\sigma}_i$ of σ_i. The carrier of c_p consists of those dual cells $\overline{D}(\sigma_j)$ for which $b_j \neq 0$. But the carrier of d_q consists of those simplices σ_i for which $a_i \neq 0$. Since these carriers intersect in only one point $\widehat{\sigma}$, it follows that the only index i for which both the coefficients b_i and a_i are non-zero is the index for which $\sigma_i = \sigma$. Our lemma follows:

$$\langle \alpha^q, \beta_q \rangle = \langle c^q, d_q \rangle = \sum b_j a_i \langle \sigma_j^*, \sigma_i \rangle = \sum b_i a_i = \pm ba \,. \qquad \square$$

To apply this result to the cohomology of lens spaces, we need a certain operation called the *Bockstein*, which we introduced for homology in Definition 3.3.10.

Definition 8.9.4 Let $\mathcal{C}$ be a free chain complex. Given an exact sequence of abelian groups

$$0 \to G \to G' \to G'' \to 0\,,$$

consider the associated short exact sequences

$$\begin{aligned} 0 &\to C_p \otimes G \to C_p \otimes G' \to C_p \otimes G'' \to 0\,, \\ 0 &\to \mathrm{Hom}(C_p, G) \to \mathrm{Hom}(C_p, G') \to \mathrm{Hom}(C_p, G'') \to 0\,. \end{aligned}$$

From the zig-zag lemma [Lemma 3.3.2], we obtain homomorphisms

$$\begin{aligned} \beta_* &: H_p(\mathcal{C}; G'') \to H_{p-1}(\mathcal{C}; G)\,, \\ \beta^* &: H^p(\mathcal{C}; G'') \to H^{p+1}(\mathcal{C}; G) \end{aligned}$$

that are natural with respect to homomorphisms induced by continuous maps. They are called the *Bockstein homomorphisms* associated with the coefficient sequence in question.

Lemma 8.9.5 *We use Notation 8.9.1, and we assume that X is orientable. Then the Bockstein homomorphisms commute with the Poincaré duality isomorphism up to sign.*

Proof: Let $\phi : C^k(X) \to D_{n-k}(X)$ be the Poincaré duality isomorphism of Theorem 8.5.2. Consider the commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C^k(X) \otimes G & \longrightarrow & C^k(X) \otimes G' & \longrightarrow & C^k(X) \otimes G'' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & D_{n-k}(X) \otimes G & \longrightarrow & D_{n-k}(X) \otimes G' & \longrightarrow & D_{n-k}(X) \otimes G'' & \longrightarrow & 0 \end{array}$$

where the vertical homomorphisms are induced by ϕ. Since $\partial\phi = \phi\delta$, one has an induced homomorphism of the corresponding exact homology sequences. In particular, the diagram

$$\begin{array}{ccc} H^k(X; G'') & \xrightarrow{\beta^*} & H^{k+1}(X; G) \\ \downarrow\cong & & \downarrow\cong \\ H_{n-k}(X; G'') & \xrightarrow{\beta_*} & H_{n-k-1}(X; G) \end{array}$$

commutes. Here we use the fact that one has a natural isomorphism

$$C^i(X) \otimes G \cong \mathrm{Hom}(C_i(X), G)$$

of cochain complexes. So we may use either to compute the cohomology of X. We also use the naturality of the zig-zag construction to conclude that the

cochain complex $\{C^i(X) \otimes G\}$ can be used to compute β^*. Also $\{D_i(X) \otimes G\}$ can be used to compute β_*.

The isomorphism induced by ϕ differs from that induced by $\frown \Gamma$ only by a sign, so that our lemma now follows. □

Now we apply these results to lens spaces. Recall that if n and k are relatively prime positive integers, then the lens space $X = L(n,k)$ is a compact, connected, orientable, triangulable 3-manifold. Furthermore, X has the structure of a CW complex. Its cellular chain complex $\mathcal{D}(X)$ is infinite cyclic in each dimension $i = 0, 1, 2, 3$. And the boundary operator is either 0 or multiplication by n. It follows that

$$H_i(X; \mathbb{Z}/n) \cong \mathbb{Z}/n \cong H^i(X; \mathbb{Z}/n)$$

for $i = 1, 2, 3$.

Theorem 8.9.6 *Let $X = L(n,k)$. Let Λ generate the infinite cyclic group $H^3(X)$. Let $\Lambda_{(n)}$ denote its image in $H^3(X; \mathbb{Z}/n)$ under the coefficient homomorphism induced by $\mathbb{Z} \to \mathbb{Z}/n$. Let*

$$\beta^* : H^1(X; \mathbb{Z}/n) \to H^2(X; \mathbb{Z}/n)$$

be the Bockstein homomorphism associated with the coefficient sequence

$$0 \longrightarrow \mathbb{Z}/n \xrightarrow{\phi_n} \mathbb{Z}/n^2 \longrightarrow \mathbb{Z}/n \longrightarrow 0\,,$$

where $\phi_n : \mathbb{Z}/n \to \mathbb{Z}/n^2$ is multiplication by n. Consider the ring $H^*(X; \mathbb{Z}/n)$. There is a generator u for $H^1(X; \mathbb{Z}/n)$ such that

$$u \smile \beta^*(u) = \pm \frac{1}{[k]} \Lambda_{(n)}\,,$$

where $1/[k]$ is the unique inverse to $[k]$ in the ring $\mathbb{Z}/n$.

Proof: Let

$$\beta_* : H_2(X; \mathbb{Z}/n) \to H_1(X; \mathbb{Z}/n)$$

be the homology Bockstein associated with the given coefficient sequence. Since the Bockstein homomorphisms commute with Poincaré duality up to sign, it suffices to show that in homology there exists a generator w for $H_2(X; \mathbb{Z}/n)$ such that

$$w \cdot \beta_*(w) = \pm(1/[k])\Lambda_{(n)} \frown \Gamma\,.$$

We know that $\Lambda_{(n)} \frown \Gamma$ is a generator of $H_0(X; \mathbb{Z}/n)$. Also it equals the homology class of $v_0 \otimes [1]$, where v_0 is a vertex of X and $[1] \in \mathbb{Z}/n$.

First, we define w. Let v, e, and c denote fundamental cycles for the cells of X in dimensions 0, 1, 2, respectively—as indicated in Figure 8.9.1. As we showed in the proof of Theorem 4.14.14, $\partial c = ne$. When c is considered as a chain with $\mathbb{Z}/n$ coefficients, then c is a cycle. Let w be the homology class of c. It generates $H_2(X; \mathbb{Z}/n)$.

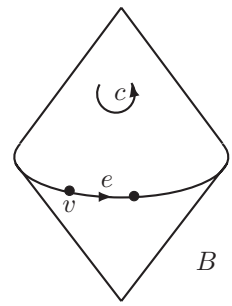


Figure 8.9.1

We now compute $\beta_*(w)$. We may use the cellular chain complex of X to compute β_*. Applying the zig-zag lemma, we begin with c, which is a cycle with $\mathbb{Z}/n$ coefficients. Then we consider it as a chain with $\mathbb{Z}/n^2$ coeffcients. Next we take its boundary [which is still ne] and finally we divide by n. Thus $\beta_*(w) = \{e\}$, where e is considered to be a cycle with $\mathbb{Z}/n$ coefficients.

Finally, we compute the intersection product $w \cdot \beta_*(w) = \{c\} \cdot \{e\}$. Unfortunately, we are not in a position to compute this product directly, for the carrier of e intersects the carrier of c in much more than a single point. We shall thus replace c and e by homologous cycles in order to carry out the computation.

Note first that c is homologous to the cycle d pictured in Figure 8.9.2. Indeed, the boundary of the solid upper half of the figure, suitably oriented, equals the chain $c - d$.

Second, let z be the cycle that runs from the top of the polyhedron B to the bottom, as pictured in Figure 8.9.3. We assert that, when considered in X, it is homologous to the cycle $-ke$. For there are k copies of the 1-cell e lying between the vertex v and its image $r_\theta(v)$ under rotation through the angle $\theta = 2\pi k/n$. Then, computed in B, the boundary of the shaded 2-chain in the figure equals the chain $ke + z + s_1 + s_2$, where s_1 and s_2 are the two slanted 1-chains pictured. These slanted 1-chains cancel each other after the identifications made in forming X. Thus the boundary of the shaded 2-chain equals $ke + z$ when computed in $L(n, k)$.

Now the intersection product of $\{d\}$ and $\{z\}$ is easy to compute, because their carriers intersect in a single point. Actually, we must triangulate X and push z off onto the dual 1-skeleton in order for the preceding lemma to

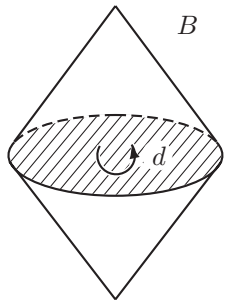


Figure 8.9.2

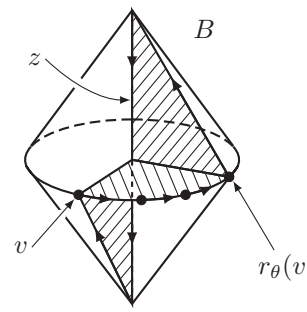


Figure 8.9.3

apply. But this is not hard to do in such a way that the resulting carrier still intersects d in only one point. It then follows from Lemma 8.9.3 that $\{d\} \cdot \{z\}$ equals ± 1 times our specified generator of H_0. Since $\{d\} = w$ and $\{z\} = \{-ke\} = -k\beta^*(w)$, our result is proved. □

Theorem 8.9.7 *For $X = L(n,k)$ and $Y = L(n,\ell)$ to have the same homotopy type, it is necessary that*

$$k = \pm a^2 \ell \pmod{n}$$

for some a.

Proof: Let $f : X \to Y$ be a homotopy equivalence. Let Λ generate $H^3(X)$ and let Λ' generate $H^3(Y)$. Let $u \in H^1(X;\mathbb{Z}/n)$ and $v \in H^1(Y;\mathbb{Z}/n)$ be chosen as in the preceding lemma in such a way that

$$u \smile \beta^*(u) = \pm\frac{1}{[k]}\Lambda_{(n)} \quad \text{and} \quad v \smile \beta^*(v) = \pm\frac{1}{[\ell]}\Lambda'_{(n)}\,.$$

Now $f^*(v) = au$ for some integer a. Thus

$$\begin{aligned} \pm\frac{1}{[\ell]} f^*(\Lambda'_{(n)}) &= f^*(v \smile \beta^*(v)) \\ &= f^*(v) \smile \beta^*(f^*(v)) \\ &= (au) \smile \beta^*(au) \\ &= a^2(u \smile \beta^*(u)) \;=\; \pm\frac{a^2}{[k]}\Lambda_{(n)}\,. \end{aligned}$$

Now f is a homotopy equivalence and therefore f^* maps Λ' to $\pm\Lambda$. So it maps $\Lambda'_{(n)}$ to $\pm\Lambda_{(n)}$. We conclude that

$$\frac{1}{[\ell]} = \pm\frac{a^2}{[k]}\,.$$

Thus the theorem is proved. □

The condition stated in this theorem is in fact both necessary and sufficient for $L(n,k)$ and $L(n,\ell)$ to have the same homotopy type. The sufficiency is a theorem of J. H. C. Whitehead. Thus the lens spaces are a rare class of spaces for which both the homeomorphism and the homotopy type classification problems have been solved. [See Theorem 4.14.15 and Remark 4.14.16.]

At one time, there was a classical conjecture, due to Hurewicz, to the effect that two compact manifolds having the same homotopy type should be homeomorphic as well. This conjecture is certainly true for 2-manifolds. But the lens spaces $L(7,k)$ provide a counterexample in dimension 3. The spaces $L(7,1)$ and $L(7,2)$ have the same homotopy type, since $2 \equiv (3)^2 \cdot 1 \,(\mathrm{mod}\, 7)$. But they are *not* homeomorphic [see Theorem 4.14.15 and Remark 4.14.16], since

$$2 \not\equiv \pm 1 \,(\mathrm{mod}\, 7) \quad \text{and} \quad 2 \cdot 1 \not\equiv \pm 1 \,(\mathrm{mod}\, 7)\,.$$

Exercise 3 below shows that the cohomology ring $H^*(L(n,k);\mathbb{Z}/n)$ is not by itself adequate to classify the lens spaces up to homotopy type. One needs also the Bockstein operation β^*. The Bockstein operation is a particular example of what is called a "cohomology operation."

Definition 8.9.8 A *cohomology operation* θ of type $(p,q;G,G')$ is a set map

$$\theta : H^p(X,A;G) \to H^q(X,A;G;)\,,$$

defined for all topological pairs (X,A) that is natural with respect to homomorphisms induced by continuous maps.

The existence of cohomology operations provides the cohomology of a space with a much richer structure than that of a ring. Frequently, as was the case with lens spaces, these operations give one valuable information about the spaces in question.

EXERCISES

1. Show that if $f : L(n,k) \to L(n,\ell)$ is a map of degree d, then

$$kd \equiv \pm a^2 \ell \,(\mathrm{mod}\, n)$$

for some integer a.

2. Using the results stated here, Theorem 4.14.15, and Remark 4.14.16, classify up to homeomorphism and up to homotopy type the manifolds $L(7,k)$ for $k = 1,2,\ldots,6$ and the manifolds $L(10,k)$ for k relatively prime to 10.

3. Let $X = L(n,k)$.

(a) Show that if u_i generates $H^i(X;\mathbb{Z}/n)$ for $i = 1, 2$, then $u_1 \cup u_2$ generates $H^3(X;\mathbb{Z}/n)$. [**Hint:** First show that there exist generators u_1, u_2 such that $u_1 \cup u_2$ generates H^3 by showing that $\beta^* : H^1 \to H^2$ is an isomorphism.]

(b) Show that $u_1 \cup u_2$ is either 0 or of order 2.

4. Give two examples of Bockstein homomorphisms for two different free chain complexes.

8.10 Lefschetz Duality

As one might expect, the Poincaré duality theorem generalizes to relative homology manifolds. The generalization is due to Lefschetz. It takes a particularly nice form in the case of manifolds.

First we need a lemma. Recall that A is said to be a full subcomplex of the complex X if every simplex of X whose vertices are in A is itself in A.

In general, a subcomplex A of X need not be full. But if we pass to the first barycentric subdivision, then we can show easily that $\operatorname{sd} A$ is a full subcomplex of $\operatorname{sd} X$. Let $s = \widehat{\sigma}_1\widehat{\sigma}_2 \dots \widehat{\sigma}_k$ be a simplex of $\operatorname{sd} X$, where $\sigma_1 \succ \sigma_2 \succ \cdots \succ \sigma_k$. Then $s \subset \sigma_1$. If each vertex of s belongs to $\operatorname{sd} A$, then in particular $\widehat{\sigma}_1 \in \operatorname{sd} A$ so that $\operatorname{Int}\sigma_1$ intersects $|A|$. It follows that $\sigma_1 \in A$ so that $s \subset \sigma_1 \subset |A|$ as claimed.

Lemma 8.10.1 *Let A be a full subcomplex of the finite simplicial complex X. Let C consist of all simplices of X that are disjoint from $|A|$. Then $|A|$ is a deformation retract of $|X| \setminus |C|$, and $|C|$ is a deformation retract of $|X| \setminus |A|$.*

Proof: First we note that each vertex of X belongs to either A or C. Second, we note that C is a full subcomplex of X. If the vertices of σ are in C, then the simplex σ cannot intersect $|A|$, so it is in C. Third, we note that A consists of all simplices X disjoint from $|C|$. Every simplex of A is disjoint from $|C|$. Conversely, if σ is disjoint from $|C|$, then each vertex of σ is in A, hence σ belongs to A.

By symmetry, it suffices to show that $|A|$ is a deformation retract of $|X| \setminus |C|$. Let σ be a simplex of X that belongs to neither A nor C. Then $\sigma = s * t$, where s is spanned by the vertices of σ that belong to A and t is spanned by the vertices that belong to C. Then $s \in A$ and $t \in C$. Each point x of $(\sigma \setminus s) \setminus t$ lies on a unique line segment joining a point of s with a point of t. Let us denote the endpoint of this line segment that lies in $|A|$ by $f_\sigma(X)$. We extend f_σ to the simplex s by letting it equal the identity on s. Then f_σ is continuous on $\sigma \setminus t$. [Indeed,if $s = v_0v_1 \dots v_p$ and $t = v_{p+1}v_{p+2} \dots v_n$, and if x

lies in $\sigma \setminus t$, then

$$x = \sum_{i=0}^{n} \alpha_i v_i \quad \text{implies} \quad f_\sigma(x) = \sum_{i=0}^{p} \frac{\alpha_i}{\lambda} v_i \,,$$

where $\lambda = \sum_{i=0}^{p} \alpha_i$. See the proof of Lemma 8.2.2.]

It is immediate that any two of the functions f_σ agree on the common part of their domains. Therefore we can define a continuous function

$$f : |X| \setminus |C| \to |A|$$

that retracts $|X| \setminus |C|$ onto $|A|$ by the equations

$$f(x) = \begin{cases} x & \text{if } \ x \in |A| \\ f_\sigma(x) & \text{if } \ x \in \sigma \setminus |C| \text{ and } \sigma \notin A \,. \end{cases}$$

The function $F(x,t) = (1-t)x + tf(x)$ is then the required deformation retract. □

The preceding lemma holds without assuming X is finite, but in the infinite case it requires some care to show that F is continuous. The problem does not arise in the finite case.

Definition 8.10.2 Let (X, A) be a compact triangulated relative homology n-manifold. We say that (X, A) is *orientable* if it is possible to orient all the n-simplices σ_i of X not in A so that their sum $\gamma = \sum \sigma_i$ is a cycle of (X, A). Such a cycle γ will be called an *orientation cycle* for (X, A).

Theorem 8.10.3 (Lefschetz duality) *Let (X, A) be a compact, triangulated, relative homology n-manifold. If (X, A) is orientable, then there are isomorphisms*

$$\begin{aligned} H^k(X, A; G) &\cong H_{n-k}(|X| \setminus |A|; G) \,, \\ H_k(X, A; G) &\cong H^{n-k}(|X| \setminus |A|; G) \end{aligned}$$

for all G. If (X, A) is not orientable, then these isomorphisms exist for $g = \mathbb{Z}/2$.

Remark 8.10.4 In the statement of the theorem, the homology and cohomology groups of $|X| \setminus |A|$ should be interpreted as singular groups, since $|X| \setminus |A|$ is not the polytope of a subcomplex of X. [It is triangulable, but that we have not yet proved.]

Proof of the Theorem: Let X^* denote the subcomplex of the first barycentric subdivision of X consisting of all simplices of $\operatorname{sd} X$ that are disjoint from $|A|$. Now $|A|$ is the polytope of a full subcomplex of $\operatorname{sd} X$. By the preceding lemma, $|X^*|$ is a deformation retract of $|X| \setminus |A|$. Therefore we may replace $|X| \setminus |A|$ by X^* in the statement of the theorem.

We consider, as in the proof of Poincaré duality, the collection of blocks $D(\sigma)$ dual to the simplices of X. We shall prove the following:

The space $|X^*|$ equals the union of all those blocks $D(\sigma)$ dual to simplices σ of X that are not in A.

To prove this fact, let $s = \widehat{\sigma}_1\widehat{\sigma}_2 \dots \widehat{\sigma}_k$ be a simplex of $\operatorname{sd} X$, where $\sigma_1 \succ \sigma_2 \succ \cdots \succ \sigma_k$. Then s lies in the dual block $\overline{D}(\sigma_k)$. If s is disjoint from $|A|$, then the vertex $\widehat{\sigma}_k$ is, in particular, not in $|A|$. Hence the simplex σ_k does not belong to A. Conversely, if σ_k does not belong to A, then neither do the simplices $\sigma_1, \sigma_2, \dots, \sigma_{k-1}$, since they have σ_k as a face. Hence s does not intersect $|A|$. See Figure 8.10.1.

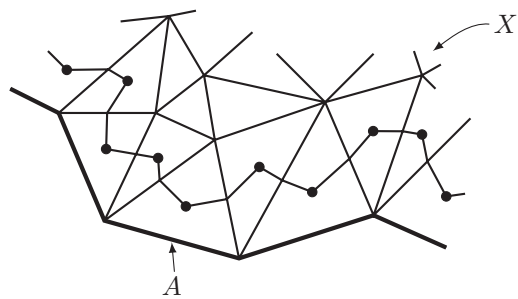


Figure 8.10.1

Now for each block $D(\sigma)$ for which σ is not in A, the point $\widehat{\sigma}$ is in $|X| \setminus |A|$, so that $H_i(|X|, |X| \setminus \widehat{\sigma})$ is infinite cyclic for $i = n$ and vanishes otherwise. It follows that $(\overline{D}(\sigma), \dot{D}(\sigma))$ is a homological cell of dimension $n - \dim \sigma$. Let $\mathcal{D}(X^*)$ denote the dual chain complex of X^*. The group $D_p(X^*)$ is generated by fundamental cycles for the homological cells $(\overline{D}(\sigma), \dot{D}(\sigma))$ of dimension p. Just as before, the inclusion $\mathcal{D}(X^*) \to \mathcal{C}(X^*)$ induces an isomorphism in cohomology and homology. [See Theorem 8.4.8.]

Now we consider the case where (X, A) is orientable. Recall that the relative cohomology group $C^k(X, A)$ can be naturally considered as the subgroup of $C^k(X)$ consisting of all cochains of X that vanish on simplices of A. It is therefore free abelian with a basis consisting of the cochains σ^*, as σ ranges over all k-simplices of X not in A. The argument used in the first proof of Poincaré duality goes through with no change to give us an isomorphism

$$C^k(X, A) \xrightarrow{\phi} D_{n-k}(X^*)\,,$$

having the property that $\phi\delta = \partial\phi$. The existence of an isomorphism

$$H^k(X, A) \cong H_{n-k}(X^*) \cong H_{n-k}(|X| \setminus |A|)$$

follows at once. This is the first of our duality isomorphisms.

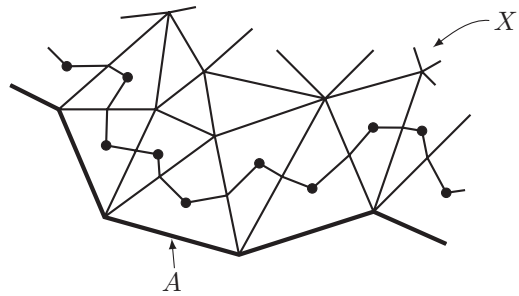


Figure 8.10.1

To obtain the second duality isomorphism, we recall that the group $\mathrm{Hom}(C^k(X,A),\mathbb{Z})$ is naturally isomorphic to the group $C_k(X,A)$. [See Step 4 of Theorem 7.6.1.] Since ϕ is an isomorphism, so is its dual

$$\mathrm{Hom}(C^k(X,A),\mathbb{Z}) \xleftarrow{\widetilde{\phi}} \mathrm{Hom}(D_{n-k}(X^*),\mathbb{Z})\,.$$

Therefore one has an isomorphism

$$H_k(X,A) \underset{\cong}{\xleftarrow{\phi^*}} H^{n-k}(X^*) \cong H^{n-k}(|X| \setminus |A|)\,.$$

The proof for arbitrary coefficients if (X,A) is orientable, or for $\mathbb{Z}/2$ coefficients if (X,A) is not orientable, goes through without difficulty, just as in the proof of Poincaré duality. □

Corollary 8.10.5 *Let (X,A) be a compact, triangulated relative homology n-manifold with $|X| \setminus |A|$ connected. If σ and σ' are two n-simplices of X not in A, then there is a sequence*

$$\sigma = \sigma_0, \sigma_1, \ldots, \sigma_k = \sigma'$$

of n-simplices not in A such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$-simplex not in A for each i.

Proof: We define two n-simplices of X not in A to be equivalent if there is such a sequence joining them. The sum of the members of any one equivalence class is a relative cycle of (X,A) with $\mathbb{Z}/2$ coefficients. We conclude that there is only one equivalence class, since

$$H_n(X,A;\mathbb{Z}/2) \cong H^0(|X| \setminus |A|;\mathbb{Z}/2) \cong \mathbb{Z}/2\,.$$ □

Corollary 8.10.6 *Let (X,A) be a compact, triangulated relative homology n-manifold. Assume that $|X| \setminus |A|$ is connected. Then $H_n(X,A) \cong \mathbb{Z}$ if (X,A) is orientable and $H_n(X,A) = 0$ if (X,A) is non-orientable.*

Proof: The proof follows the pattern of Corollary 8.5.4. □

Corollary 8.10.7 *Let (X, A) be a compact, triangulated relative homology n manifold. Then (X, A) is orientable if and only if, for each component X of $|X| \setminus |A|$, one has $H_n(\overline{X}_i, \overline{X}_i \cap A) \cong \mathbb{Z}$.*

Remark 8.10.8 It follows that orientability of (X, A) does not depend on the triangulation of (X, A).

Definition 8.10.9 Let (X, A) be a compact, triangulable relative homology n-manifold. Let $X_1, X_2, \ldots, X_k$ be the components of $X \setminus A$. Let $A_i = \overline{X}_i \cap A$. If (X, A) is orientable, then $H_n(\overline{X}_i, A_i)$ is infinite cyclic. A generator $\Gamma^{(i)}$ of this group is called an *orientation class* for $(\overline{X}_i, A_i)$. The image of the classes $\Gamma^{(i)}$ under the isomorphism

$$\bigoplus H_n(\overline{X}_i, A_i) \cong H_n(X, A)$$

induced by inclusion is called an *orientation class* for (X, A) and is denoted by Γ. Similarly, if (X, A) is not necessarily orientable, and if $\Gamma^{(i)}_{(2)}$ is the unique non-trivial element of $H_n(\overline{X}_i, A_i; \mathbb{Z}/2)$, then the image of these classes in $H_n(X, A; \mathbb{Z}/2)$ is denoted $\Gamma_{(2)}$ and is called an *orientation class for* (X, A) *over* $\mathbb{Z}/2$.

If (X, A) is given a specific triangulation, then Γ is represented by the sum γ of all n-simplices of X not in A, suitably oriented. And $\Gamma_{(2)}$ is represented by the sum of all these n-simplices, each with coefficient $[1] \in \mathbb{Z}/2$.

A natural question to ask at this point is whether the second version of the proof of Poincaré duality, which involves cap products, generalizes to the relative case. The answer is that it does not. Let us examine where the difficulty lies.

We begin as in the proof of Theorem 8.7.2. Choose a map

$$g : (\operatorname{sd} X, \operatorname{sd} A) \to (X, A)\,,$$

which is a simplicial approximation to the identity. Let γ be an orientation cycle for (X, A). Consider the diagram

$$\begin{array}{ccc} C^k(X, A) & \overset{\psi}{\dashrightarrow} & D_{n-k}(X^*) \\ \Big\downarrow{\scriptstyle g^\#} & & \Big\downarrow{\scriptstyle j} \\ C^k(\operatorname{sd} X, \operatorname{sd} X) & \xrightarrow{\frown \operatorname{sd}\gamma} & C_{n-k}(\operatorname{sd} X)\,, \end{array}$$

where j is inclusion as before. The proof of Theorem 8.7.2 goes through to show that the composite of $g^\#$ and $\frown \operatorname{sd}\gamma$ carries $C^k(X, A)$ isomorphically onto the subgroup of $C_{n-k}(\operatorname{sd} X)$ which is the image of $D_{n-k}(X^*)$ under j. Up to

sign, the resulting isomorphism ψ is a suitable candidate for our isomorphism ϕ just as before. [Here we are of course using the relative cap product

$$C^k(\operatorname{sd} X, \operatorname{sd} A) \otimes C_n(\operatorname{sd} X, \operatorname{sd} A) \to C_{n-k}(\operatorname{sd} X)$$

in defining the map $\frown \operatorname{sd} \gamma$.]

So far, no difficulties have appeared. The problem arises when we pass to the cohomology-homology level. On this level, we have a homomorphism

$$H^k(X, A) \xrightarrow{\frown \Gamma} H_{n-k}(X)\,.$$

The range of this map is *not* the group $H_{n-k}(|X| \setminus |A|)$ that we wish it to be.

Thus Lefschetz duality cannot in general be expressed by the cap product with an orientation class for (X, A). All that one can say is the following:

Theorem 8.10.10 *Let (X, A) be a compact, triangulable relative homology n-manifold. If (X, A) is orientable, then let Γ denote an orientation class for (X, A). Then the following diagram commutes, up to a sign that depends on k and n:*

$$\begin{array}{ccc} H^n(X, A; G) & \xrightarrow[\cong]{\phi_*} & H^{n-k}(X \setminus A; G) \\ & \searrow^{\frown \Gamma} & \downarrow j_* \\ & & H_{n-k}(X; G) \end{array}$$

[Here ϕ_* is the Lefschetz duality isomorphism and j is inclusion.] The same result holds if (X, A) is not orientable as long as one replaces Γ by $\Gamma_{(2)}$ and replaces G by $\mathbb{Z}/2$.

Proof: Choose a triangulation of (X, A). Then consider the diagram

$$\begin{array}{ccccc} C^k(X, A) & \xrightarrow[\cong]{\phi} & D_{n-k}(X^*) & \xrightarrow{\eta} & S_{n-k}(|X^*|) \\ \downarrow{\scriptstyle g^\#} & & \downarrow & & \downarrow \\ & & & & S_{n-k}(|X| \setminus |A|) \\ & & & & \downarrow \\ C^k(\operatorname{sd} X, \operatorname{sd} A) & \xrightarrow{\frown \operatorname{sd} \gamma} & C_{n-k}(\operatorname{sd} X) & \xrightarrow{\eta} & S_{n-k}(|X|) \end{array}$$

Here η is the chain map carrying simplicial chains to singular chains. The unlabeled maps are induced by inclusion. The proof of Theorem 8.7.2 applies to show that the first square commutes up to sign. The rest of the diagram clearly commutes. □

One situation in which the Lefschetz duality isomorphism is actually given by the cap product is the case of a manifold with boundary. We consider that situation now.

Definition 8.10.11 Let M be an n-manifold with boundary. We say that $\operatorname{Bd} M$ has a *product neighborhood* in M if there is a homeomorphism

$$h : \operatorname{Bd} M \times [0,1) \to U$$

whose image is an open set in M and such that $h(x,0) = x$ for each $x \in \operatorname{Bd} M$.

It is in fact true that such a product neighborhood always exists, but the proof is decidedly non-trivial [see [Bro62]].

Theorem 8.10.12 (Poincaré–Lefschetz duality) *Let M be a compact, triangulable n-manifold with boundary such that $\operatorname{Bd} M$ has a product neighborhood in M.*

If $(M, \operatorname{Bd} M)$ is orientable, let $\Gamma \in H_n(M, \operatorname{Bd} M)$ be an orientation class for $(M, \operatorname{Bd} M)$. Then there are isomorphisms

$$H^k(M, \operatorname{Bd} M; G) \xrightarrow{\frown \Gamma} H_{n-k}(M; G)\,,$$

$$H^k(M; G) \xrightarrow{\frown \Gamma} H_{n-k}(M, \operatorname{Bd} M; G)$$

for arbitrary G. If $(M, \operatorname{Bd} M)$ is not orientable, then the same result holds if G is replaced by $\mathbb{Z}/2$ and Γ is replaced by $\Gamma_{(2)}$.

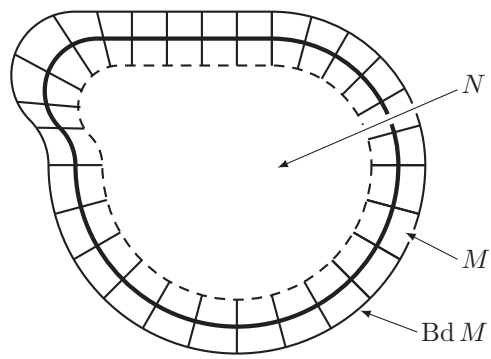


Figure 8.10.2

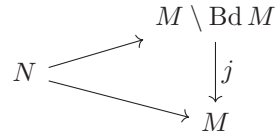
Proof: We first prove that inclusion induces an isomorphism

$$j_* : H_{n-k}(M \setminus \operatorname{Bd} M; G) \to H_{n-k}(M; G)\,.$$

Let $h : \operatorname{Bd} M \times [0,1) \to M$ be a product neighborhood of the boundary. Let

$$N = M \setminus h(\operatorname{Bd} M \times [0, 1/2))\,.$$

See Figure 8.10.2. Now $\operatorname{Bd} M \times [1/2, 1)$ is a deformation retract of both $\operatorname{Bd} M \times [0,1)$ and of $\operatorname{Bd} M \times (0,1)$. Hence N is a deformation retract of both M and $M \setminus \operatorname{Bd} M$. Thus the left-hand inclusions in the diagram



are homotopy equivalences. Hence so is j.

The first of the two isomorphisms stated in our theorem now follows from the naturality diagram of Theorem 8.10.10, setting $A = \operatorname{Bd} M$.

We now derive the other isomorphism. Let us triangulate $(M, \operatorname{Bd} M)$. The space $\operatorname{Bd} M$ is a union of simplices of M [by Theorem 4.8.11]. Being an $n-1$ manifold, it is the union of $(n-1)$-simplices. Now we claim that each $(n-1)$-simplex s of $\operatorname{Bd} M$ is a face of exactly one n-simplex σ of M. It must be a face of at least one such simplex, because m is a union of n-simplices. And it cannot be a face of more than one, since by Lemma 8.3.2,

$$H_n(M, M \setminus \widehat{s}\,) \cong \widetilde{H}_0(\operatorname{Lk}(s, M))\,,$$

and the former group vanishes because $\widehat{s}$ is a point of $\operatorname{Bd} M$.

Suppose now that γ is an orientation cycle for $(M, \operatorname{Bd} M)$. Then γ is a sum of *all* the n-simplices of M. Since each $(n-1)$-simplex of $\operatorname{Bd} M$ is the face of exactly one n-simplex of M, the chain $\partial\gamma$ has coefficient ± 1 on each $(n-1)$-simplex of $\operatorname{Bd} M$. Since $\partial\gamma$ is a cycle, it must be an orientation cycle for $\operatorname{Bd} M$.

Let $\Gamma = \{\gamma\}$. Then the class $\partial_*\Gamma = \{\partial\gamma\}$ is an orientation class for $\operatorname{Bd} M$. Consider the following diagram:

$$\begin{array}{ccccccccc}
H^{k-1}(\mathrm{Bd}M) & \to & H^k(M, \mathrm{Bd}M) & \to & H^k(M) & \to & H^k(\operatorname{Bd} M) & \to & H^{k+1}(M, \operatorname{Bd} M) \\
\downarrow \frown \partial_*\Gamma & & \downarrow \frown \Gamma & & \downarrow \frown \Gamma & & \downarrow \frown \partial_*\Gamma & & \downarrow \frown \Gamma \\
H_{n-k}(\mathrm{Bd}M) & \to & H_{n-k}(M) & \to & H_{n-k}(M, \mathrm{Bd}M) & \to & H_{n-k-1}(\mathrm{Bd}M) & \to & H_{n-k-1}(M)
\end{array}$$

where all the groups are assumed to have G coefficients. It is easy to check that this diagram commutes up to sign. The first and fourth vertical maps are isomorphisms by Poincaré duality, and the second and fifth are isomorphisms by Poincaré–Lefschetz duality. Therefore the third map is an isomorphism as well.

The same argument applies if M is non-orientable provided one replaces G by $\mathbb{Z}/2$ and Γ by $\Gamma_{(2)}$. □

As an application of Poincaré–Lefschetz duality, we consider the following question: Given a compact n-manifold, under what conditions is it the boundary of a compact $n+1$ manifold M? There is an entire theory that deals with this question. It is called *cobordism theory*. Here we prove just one elementary result.

Theorem 8.10.13 *Let M be a compact, triangulable manifold with boundary. Suppose that* $\dim M = 2m+1$. *Assume that* $\operatorname{Bd} M$ *is nonempty and has a product neighborhood in M. Then the vector space $H_m(\operatorname{Bd} M; \mathbb{Z}/2)$ has even dimension.*

Proof: Step 1: Consider an exact sequence of vector spaces and linear transformations

$$\cdots \longrightarrow A_{k-1} \xrightarrow{\phi_{k-1}} A_k \xrightarrow{\phi_k} A_{k+1} \longrightarrow \cdots ,$$

where A_k and ϕ_k are defined for all integers k and $\dim A_k = 0$ for $|k|$ sufficiently large. We assert that

$$\begin{aligned} \operatorname{rank} \phi_k &= \dim A_k - \dim A_{k-1} + \dim A_{k-2} - \cdots \qquad \text{and} \\ \operatorname{rank} \phi_{k-1} &= \dim A_k - \dim A_{k+1} + \dim A_{k+2} - \cdots . \end{aligned}$$

Exactness at A_i tells us that

$$\dim A_i = \operatorname{rank} \phi_{i-1} + \operatorname{rank} \phi_i .$$

Summation gives us the desired equations.

Step 2: Using $\mathbb{Z}/2$ coefficients, we have an exact sequence of vector spaces

$$\cdots \to H^k(M; \mathbb{Z}/2) \xrightarrow{\mu_k} H^k(\operatorname{Bd} M; \mathbb{Z}/2) \xrightarrow{\nu_k} H^{k+1}(M, \operatorname{Bd} M; \mathbb{Z}/2) \longrightarrow \cdots .$$

Let

$$\begin{aligned} \beta_k &= \dim H^k(M; \mathbb{Z}/2) , \\ \gamma_k &= \dim H^k(\operatorname{Bd} M; \mathbb{Z}/2) , \\ \alpha_{k+1} &= \dim H^{k+1}(M, \operatorname{Bd} M; \mathbb{Z}/2) . \end{aligned}$$

We apply **Step 1** to this sequence. Beginning at the term $H^k(\operatorname{Bd} M)$ and summing first to the left and then to the right, we have

$$\operatorname{rank} \nu_k = \sum_{i=0}^{\infty} (-1)^i (\gamma_{k-i} - \beta_{k-i} + \alpha_{k-i}) , \qquad (*)$$

$$\operatorname{rank} \mu_k = \sum_{i=0}^{\infty} (-1)^i (\gamma_{k+i} - \alpha_{k+i+1} + \beta_{k+i+1}) . \qquad (**)$$

Now Lefschetz duality and algebraic duality give us isomorphisms

$$H^j(M, \operatorname{Bd} M; \mathbb{Z}/2) \cong H_{n-j}(M; \mathbb{Z}/2) \cong H^{n-j}(M; \mathbb{Z}/2) ,$$

$$H^j(\operatorname{Bd} M; \mathbb{Z}/2) \cong H_{n-j-1}(\operatorname{Bd} M; \mathbb{Z}/2) \cong H^{n-j-1}(\operatorname{Bd} M; \mathbb{Z}/2) .$$

Therefore $\alpha_j = \beta_{n-j}$ and $\gamma_j = \gamma_{n-j-1}$. Substituting these results into $(**)$ and comparing with $(*)$, we find that

$$\operatorname{rank} \mu_k = \operatorname{rank} \nu_{n-k-1} .$$

In particular, since $n = 2m + 1$, $\operatorname{rank} \mu_m = \operatorname{rank} \nu_m$. Then

$$\begin{aligned} \dim H_m(\operatorname{Bd} M; \mathbb{Z}/2) &= \dim H^m(\operatorname{Bd} M; \mathbb{Z}/2) \\ &= \gamma_m \\ &= \operatorname{rank} \mu_m + \operatorname{rank} \nu_m \\ &= 2(\operatorname{rank} \nu_m) , \end{aligned}$$

which is even. □

Corollary 8.10.14 *The manifold $\mathbb{P}^{2m}$ is not the boundary of a compact, triangulable $2m + 1$ manifold.*

EXERCISES

1. Let (X, A) be a compact, triangulable relative homology n-manifold. Assume that $X \setminus A$ is connected. Show that if X is orientable, then

$$H_n(X, A; G) \cong G \cong H^n(X, A; G) ,$$

while if X is non-orientable, then

$$H_n(X, A; G) \cong \ker(\phi_2) ,$$

$$H^n(X, A; G) \cong G/2G .$$

As usual, $\phi_2 : G \to G$ is multiplication by 2.

2. Let M be a compact, triangulable orientable n-manifold with boundary. Suppose that $\operatorname{Bd} M$ is the disjoint union of two $n - 1$ manifolds V_0 and V_1 and that $\operatorname{Bd} M$ has a product neighborhood in M. Let Γ be an orientation cycle for $(M, \operatorname{Bd} M)$. Show that

$$H^k(M, V_0) \xrightarrow{\frown \Gamma} H_{n-k}(M, V_1)$$

is an isomorphism. [**Hint:** Consider the exact sequence of the triple $(M, \operatorname{Bd} M, V_1)$.]

3. Using the fact that $\mathbb{C}P^n$ can be triangulated, show that the manifold $\mathbb{C}P^{2m}$ is not the boundary of any manifold.

4. Which compact 2-manifolds bound?

5. Let M be a compact, triangulable, orientable n-manifold with boundary. Show that the cup product defines a dual pairing

$$\frac{H^k(M, \operatorname{Bd} M)}{T^k(M, \operatorname{Bd} M)} \otimes \frac{H^{n-k}(M)}{T^{n-k}(M)} \to H^n(M, \operatorname{Bd} M)\,,$$

where T^i denotes the torsion subgroup of H^i.

6. Let M be the torus with two open discs removed. See Figure 8.10.3.

(a) Compute $H^1(M)$ and $H^1(M, \operatorname{Bd} M)$. Draw generating cocycles.

(b) Verify the existence of the dual pairing of Exercise 5.

(c) Show that the cup product does not define a dual pairing:

$$\frac{H^1(M, \operatorname{Bd} M)}{T^1(M, \operatorname{Bd} M)} \otimes \frac{H^1(M, \operatorname{Bd} M)}{T^1(M, \operatorname{Bd} M)} \to H^2(M, \operatorname{Bd} M)\,.$$

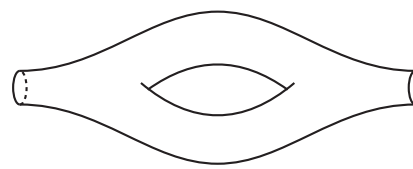


Figure 8.10.3

7. Let X be a compact, triangulable homology n-manifold. Let A and B be polytopes of subcomplexes of a triangulation of X, with $B \subset A$. Let G be arbitrary if X is orientable. Let $G = \mathbb{Z}/2$ otherwise. Show that there are isomorphisms

$$H^k(A, B; G) \cong H_{n-k}(X \setminus B, X \setminus A; G)\,,$$
$$H_k(A, B; G) \cong H^{n-k}(X \setminus B, X \setminus A; G)\,.$$

[**Hint:** Triangulate X so that A and B are polytopes of full subcomplexes. If C is a full subcomplex of X, let X_C denote the collection of all simplices of $\operatorname{sd} X$ disjoint from C. Consider the following diagram and its dual:

$$\begin{array}{ccccccccc}
0 & \longleftarrow & C^k(A, B) & \longleftarrow & C^k(X, B) & \longleftarrow & C^k(X, A) & \longleftarrow & 0 \\
 & & & & \phi\downarrow & & \phi\downarrow & & \\
0 & \longleftarrow & D_{n-k}(X_B, X_A) & \longleftarrow & D_{n-k}(X_B) & \longleftarrow & D_{n-k}(X_A) & \longleftarrow & 0\,.]
\end{array}$$

8.* Generalize Lefschetz duality to the non-compact case.

9. Let (X, A) be a triangulated, relative homology n-manifold, not compact. Show that if $|X| \setminus A$ is connected, then (X, A) is a relative pseudo n-manifold.

10. Give two examples of compact, triangulable, relative homology n-manifolds in dimension 2, and specify orientation classes for each one.

8.11 Alexander Duality

The Alexander duality theorem is almost as old as the Poincaré duality theorem. It predates Lefschetz duality by some years. In its original form it dealt with the relationship between the betti numbers and torsion coefficients of a subcomplex A of the n-sphere $\mathbb{S}^n$, and the betti numbers and torsion coefficients of the complement $\mathbb{S}^n \setminus A$. Nowadays the Alexander duality theorem is formulated in terms of cohomology and it is proved using Lefschetz duality.

Theorem 8.11.1 *Let A be a proper, nonempty subset of $\mathbb{S}^n$. Suppose that $(\mathbb{S}^n, A)$ is triangulable. Then there is an isomorphism*

$$\widetilde{H}^k(A) \cong \widetilde{H}_{n-k-1}(\mathbb{S}^n \setminus A)\,.$$

Proof: Assume that $n > 0$ to avoid triviality. Triangulate the pair $(\mathbb{S}^n, A)$.

Step 1: We prove the theorem first in the case $k \neq n, n-1$. Consider the exact sequence

$$\widetilde{H}^{k+1}(\mathbb{S}^n) \longleftarrow H^{k+1}(\mathbb{S}^n, A) \xleftarrow{\delta^*} \widetilde{H}^k(A) \longleftarrow \widetilde{H}^k(\mathbb{S}^n)\,.$$

Because the end groups vanish, δ^* is an isomorphism. Now since $(\mathbb{S}^n, A)$ is a relative n-manifold, we can apply Lefschetz duality to conclude that

$$H^{k+1}(\mathbb{S}^n, A) \cong H_{n-k-1}(\mathbb{S}^n \setminus A) = \widetilde{H}_{n-k-1}(\mathbb{S}^n \setminus A)\,.$$

Combining this isomorphism with that given by δ^* gives us our desired isomorphism.

Step 2: We now show that the theorem holds in the case $k = n-1$. The preceding argument will need some modifications. As before, one has the exact sequence

$$H^n(\mathbb{S}^n) \xleftarrow{j^*} H^n(\mathbb{S}^n, A) \xleftarrow{\delta^*} \widetilde{H}^{n-1}(A) \longleftarrow 0\,,$$

where $j : \mathbb{S}^n \to (\mathbb{S}^n, A)$ is inclusion. We conclude that

$$\widetilde{H}^{n-1}(A) \cong \ker j^* .$$

Let Γ be an orientation class for $\mathbb{S}^n$ and let $k : (\mathbb{S}^n \setminus A) \to \mathbb{S}^n$ be inclusion. We apply Poincaré and Lefschetz duality to obtain the isomorphisms in the following diagram:

$$\begin{array}{ccc} H^n(\mathbb{S}^n, A) & \xrightarrow{\quad j_* \quad} & H^n(\mathbb{S}^n) \\ \cong \big\downarrow \phi_* & \frown\, j_*\Gamma & \cong \big\downarrow \frown\, \Gamma \\ H_0(\mathbb{S}^n \setminus A) & \xrightarrow[\quad k_* \quad]{} & H_0(\mathbb{S}^n) \end{array}$$

Since this diagram commutes up to sign, it follows that

$$\ker j^* \cong \ker k_* .$$

It now follows from exactness of the sequence

$$0 \to \widetilde{H}_0(\mathbb{S}^n \setminus A) \longrightarrow H_0(\mathbb{S}^n \setminus A) \xrightarrow{k_*} H_0(\mathbb{S}^n) \to 0 ,$$

whose proof we leave as an exercise for the reader, that

$$\ker k_* \cong \widetilde{H}_0(\mathbb{S}^n \setminus A) .$$

Combining these isomorphisms gives us our desired isomorphism.

Step 3: It remains to show that the theorem holds for $k = n$. We prove first that the inclusion $i : A \to \mathbb{S}^n$ induces the zero homomorphism

$$H^n(A) \xleftarrow{i^*} H^n(\mathbb{S}^n) .$$

The group $H^n(\mathbb{S}^n)$ is infinite cyclic, and is generated by the cohomology class of σ^*, where σ is any oriented simplex of $\mathbb{S}^n$. Since A is a *proper* subcomplex of $\mathbb{S}^n$, we can choose σ to be outside A. Then $i^\#(\sigma^*)$ is the zero cochain.

Now consider the exact sequence

$$0 = H^{n+1}(\mathbb{S}^n, A) \xleftarrow{\delta^*} H^n(A) \xleftarrow{i^*} H^n(\mathbb{S}^n) .$$

Since i^* is the zero homomorphism, it follows that $H^n(A) = 0$. Therefore the groups

$$\widetilde{H}^n(A) = H^n(A) \quad \text{and} \quad \widetilde{H}_{-1}(\mathbb{S}^{n-1} \setminus A)$$

are isomorphic—because both vanish. □

Corollary 8.11.2 (The polyhedral Jordan curve theorem) *Let $n > 0$. If A is a subset of $\mathbb{S}^n$ that is homeomorphic to $\mathbb{S}^{n-1}$, and if $(\mathbb{S}^n, A)$ is triangulable, then $\mathbb{S}^n \setminus A$ has precisely two path components, of which A is the common boundary.*

Proof: A is a proper subset of $\mathbb{S}^n$ since $\mathbb{S}^{n-1}$ is not homeomorphic to $\mathbb{S}^n$. The fact that $\mathbb{S}^n \setminus A$ has exactly two path components follows from the fact that $\widetilde{H}^{n-1}(A)$ is infinite cyclic because $A \approx \mathbb{S}^{n-1}$ and from the fact that

$$\widetilde{H}^{n-1}(A) \cong \widetilde{H}_0(\mathbb{S}^n \setminus A)\,.$$

The fact that A is the common boundary of the two path components C_1 and C_2 follows, as in the proof of Theorem 4.9.5, from the fact that if s is an $(n-1)$-simplex in a [very fine] subdivision of A, then letting $B = A \setminus \operatorname{Int} s$, we have

$$0 = \widetilde{H}^{n-1}(B) \cong \widetilde{H}_0(\mathbb{S}^n \setminus B)\,.$$

Thus one can connect a point of C_1 to a point of C_2 by a path that intersects A in points of $\operatorname{Int} s$. □

EXERCISES

1. Let A be a nonempty subset of X. Show that if X is path connected, then

$$0 \to \widetilde{H}_0(A) \to H_0(A) \to H_0(X) \to 0$$

is exact. This fact completes the proof of Alexander duality.

2. Suppose that A is a proper nonempty subset of $\mathbb{S}^n$ and $(\mathbb{S}^n, A)$ is triangulable. Show that if A is acyclic, then $\mathbb{S}^n \setminus A$ is acyclic.

3. Let A be a homology $n-1$ manifold. Show that if A is homeomorphic to a subset B of $\mathbb{S}^n$ and $(\mathbb{S}^n, B)$ is triangulable, then A is orientable.

4. Prove the following version of Alexander duality: Let A be a proper nonempty subset of $\mathbb{S}^n$. If $(\mathbb{S}^n, A)$ is triangulable, then

$$\widetilde{H}_k(A) \cong \widetilde{H}^{n-k-1}(\mathbb{S}^n \setminus A)\,.$$

[**Hint:** Find a complex C that is a deformation retract of $\mathbb{S}^n \setminus A$.]

5. Prove Alexander duality for the n-ball: Let A be a proper nonempty subset of $\mathbb{B}^n$. Let $\partial A = A \cap \operatorname{Bd} \mathbb{B}^n$. If $(\mathbb{B}^n, A)$ is triangulable, show that

$$H^k(A, \partial A) \cong \widetilde{H}_{n-k-1}(\mathbb{B}^n \setminus A)\,.$$

[**Hint:** Consider the subset $(A \times \{1\}) \cup (\partial A \times I) \cup (\mathbb{B}^n \times \{0\})$ of $\operatorname{Bd}(\mathbb{B}^n \times I)$.]

8.12 "Natural" Versions of Lefschetz and Alexander Duality

The naturality property of the Lefschetz duality isomorphism

$$H^k(X, A) \xrightarrow{\phi_*} H_{n-k}(X \setminus A)$$

stated in Theorem 8.10.10 is rather unsatisfactory in general. It is not clear as it stands, for instance, that the isomorphism ϕ_* is independent of the triangulation of (X, A), or even that it is invariant under subdivision of X.

This lack of naturality carries over to Alexander duality. It is reflected in the fact that we were able in the last section to prove only the polyhedral version of the Jordan curve theorem.

In this section, we obtain more natural versions of the Lefschetz and Alexander isomorphisms. These will enable us, after we have introduced another version of cohomology due to Čěch, to construct the Alexander duality isomorphism in the situation where A is an arbitrary closed subset of $\mathbb{S}^n$.

First, a definition and a lemma:

Definition 8.12.1 Let A be the polytope of a subcomplex of the finite complex X. Define $\mathrm{St}(A, X)$ to be the union of all the sets $\mathrm{St}(\sigma, X)$ as σ ranges over all the simplices of X lying in A. It is called the *star of A in X*.

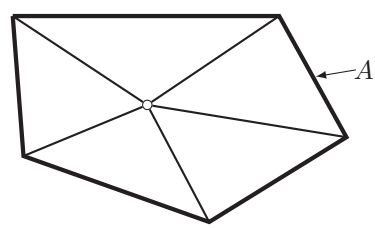


Figure 8.12.1

If C is the collection of all simplices of X disjoint from A, then it is immediate that

$$\mathrm{St}(A, X) = |X| \setminus |C|\,.$$

If A is the polytope of a *full* subcomplex of X, then it follows from Lemma 8.10.1 that A is a deformation retract of $\mathrm{St}(A, X)$. Note, however, that A need not be a deformation retract of the closed star $\overline{\mathrm{St}}(A, X)$. See Figure 8.12.1.

Lemma 8.12.2 *Let X be a finite complex. Let A be the polytope of a full subcomplex of X. Let C be the union of all simplices of X disjoint from A. Let* $\operatorname{sd} X$ *be the first barycentric subdivision of X.*

(a) *The space $|X|$ is the disjoint union of the three sets*

$$\begin{aligned} U_A &= \operatorname{St}(A, \operatorname{sd} X)\,, \\ U_C &= \operatorname{St}(C, \operatorname{sd} X)\,, \\ B &= \overline{U}_A \setminus U_A = \overline{U}_C \setminus U_C\,. \end{aligned}$$

(b) *The following inclusions are homotopy equivalences:*

$$A \to U_A \to \overline{U}_A \to \operatorname{St}(A, X)\,,$$

$$C \to U_C \to \overline{U}_C \to \operatorname{St}(C, X)\,.$$

Proof: The sets U_A and U_C are pictured in Figures 8.12.2 and 8.12.3. Let

$$s = \widehat{\sigma}_1, \ldots, \widehat{\sigma}_k$$

denote the general simplex of $\operatorname{sd} X$, where $\sigma_1 \succ \cdots \succ \sigma_k$.

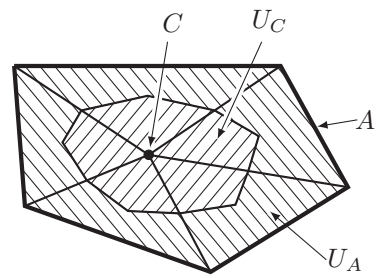


Figure 8.12.2

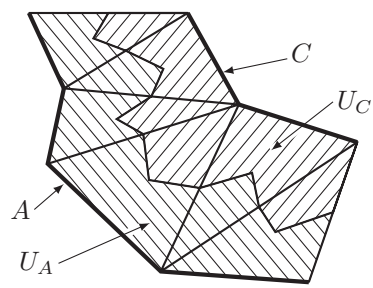


Figure 8.12.3

Step 1: We show that U_A is the union of the interiors of those simplices $s = \widehat{\sigma}_1 \ldots \widehat{\sigma}_k$ for which $\sigma_k \subset A$. If $\sigma_k \subset A$, then $\widehat{\sigma}_k \in A$, so that by definition we have

$$\operatorname{Int} s \subset \operatorname{St}(\widehat{\sigma}_k, \operatorname{sd} X) \subset U_A\,.$$

Conversely, if $\operatorname{Int} s \subset U_A$, then some vertex $\widehat{\sigma}_j$ of s is in A. It follows that σ_j lies in A and so does its face σ_k.

It follows by symmetry that U_C is the union of the interiors of those simplices s for which $\sigma_k \subset C$. Therefore U_A and U_C are disjoint open sets in $|X|$.

Step 2: We show now that if σ_k lies in neither A nor C, then $s = \widehat{\sigma}_1 \dots \widehat{\sigma}_k$ lies in $\overline{U}_A \setminus U_A$ and in $\overline{U}_C \setminus U_C$. Thus **(a)** is proved.

The simplex σ_k must have a vertex v in A and a vertex w in C. Then

$$\widehat{\sigma}_1 \dots \widehat{\sigma}_k \widehat{v} \quad \text{and} \quad \widehat{\sigma}_1 \dots \widehat{\sigma}_k \widehat{w}$$

are simplices of $\operatorname{sd} X$ whose interiors lie in U_A and U_C, respectively. Their common face s thus lies in $\overline{U}_A \setminus U_A$ and in $\overline{U} \setminus U_C$.

Step 3: Let $D = \overline{U}_A$. By **(a)**, $D = X \setminus U_C$. We now show that D is the polytope of a full subcomplex of $\operatorname{sd} X$ and that

$$\operatorname{St}(D, \operatorname{sd} X) = \operatorname{St}(A, X)\,.$$

First we show that D is full. If the interior of $s = \widehat{\sigma}_1 \dots \widehat{\sigma}_k$ lies in U_C, then $\sigma_k \subset C$ so that $\widehat{\sigma}_k \notin D$. It follows that if all the vertices of s lie in D, then s must lie in D.

We show now that the vertex $\widehat{\sigma}$ of $\operatorname{sd} X$ lies in D if and only if σ intersects A. Now a vertex $\widehat{\sigma}$ of $\operatorname{sd} X$ lies in U_C if and only if it lies in C, and this occurs if and only if $\sigma \subset C$. Therefore $\widehat{\sigma}$ lies in D if and only if σ does not lie in C, that is, if and only if σ intersects A.

Now we prove that $\operatorname{St}(D, \operatorname{sd} X) \subset \operatorname{St}(A, X)$. Suppose that $s = \widehat{\sigma}_1 \dots \widehat{\sigma}_k$ has a vertex $\widehat{\sigma}_i$ in D. Then, as just proved, σ_i must intersect A so that σ_i has a vertex v in A. Since $\sigma_1 \succ \sigma_i \succ v$, we must have

$$\operatorname{Int} s \subset \operatorname{Int} \sigma_1 \subset \operatorname{St}(v, X) \subset \operatorname{St}(A, X)\,.$$

To prove the reverse inclusion, let σ be a simplex of X having a vertex v in A. Then $\widehat{\sigma}$ is in D. Now $\operatorname{Int} \sigma$ is the union of the interiors of those simplices of $\operatorname{sd} X$ whose largest vertex [recall Definition 8.4.1] is $\widehat{\sigma}$. Thus

$$\operatorname{Int} \sigma \subset \operatorname{St}(\widehat{\sigma}, \operatorname{sd} X) \subset \operatorname{St}(D, \operatorname{sd} X)\,.$$

Step 4: We now prove **(b)**. Consider the inclusions

$$A \xrightarrow{i} U_A \xrightarrow{j} D \xrightarrow{k} \operatorname{St}(D, \operatorname{sd} X)\,.$$

Since A and D are polytopes of full subcomplexes of $\operatorname{sd} X$, the maps i and k are homotopy equivalences. Because $\operatorname{St}(D, \operatorname{sd} X) = \operatorname{St}(A, X)$ and A is a full

subcomplex of X, the map $k \circ j \circ i$ is a homotopy equivalence. It follows that j is a homotopy equivalence.

By symmetry, the inclusions

$$C \to U_C \to \overline{U}_C \to \mathrm{St}(C, X)$$

are also homotopy equivalences. □

Definition 8.12.3 Let (X, A) be a triangulable pair. If $D \subset X$, then we say that D is a *polyhedron in* (X, A) if there is some triangulation of the pair (X, A) relative to which D is the polytope of a subcomplex. If $A = \emptyset$, then we say simply that D is a polyhedron in X.

Note that if C and D are polyhedra in X, it does not follow that there is a single triangulation of X relative to which both C and D are polytopes of subcomplexes. For example, both the x-axis C and the set

$$D = (0, 0) \cup \{(x, x \sin(1/x)) : x \neq 0\}$$

are polyhedra in $\mathbb{R}^2$, but there is no triangulation of $\mathbb{R}^2$ relative to which both are polytopes of subcomplexes.

Lemma 8.12.4 *Let D be a polyhedron in the compact triangulable space X. Then there are arbitrarily small neighborhoods U of D such that*

(a) *$\overline{U}$ and $X \setminus U$ are polyhedra in X.*

(b) *The following inclusions are homotopy equivalences:*

$$D \to U \to \overline{U} \quad \text{and} \quad (X \setminus \overline{U}) \to (X \setminus U) \to (X \setminus D)\,.$$

Remark 8.12.5 In fact, if X is triangulated so that D is the polytope of a full subcomplex of X, then $U = \mathrm{St}(D, \mathrm{sd}^N X)$ satisfies these conditions for all $n \geq 1$.

Proof: It suffices to show that if D is the polytope of a full subcomplex of X, then $U = \mathrm{St}(D, \mathrm{sd}\, X)$ satisfies the requirements of the lemma.

The preceding lemma shows that the inclusions $D \to U \to \overline{U}$ are homotopy equivalences. To consider the other inclusions, let A be the union of all simplices of X disjoint from D. Then one has the following diagram:

$$\begin{array}{ccccccc}
A & \longrightarrow & \mathrm{St}(A, \mathrm{sd}\, X) & \longrightarrow & \mathrm{St}(A, \mathrm{sd}\, X) & \longrightarrow & \mathrm{St}(A, X) \\
 & & \| & & \| & & \| \\
 & & (X \setminus \overline{U}) & \longrightarrow & (X \setminus U) & \longrightarrow & (X \setminus D)
\end{array}$$

where the first two equalities follow from **(a)** of the preceding lemma, and the third by definition. By the preceding lemma, the inclusion maps of this diagram are homotopy equivalences. □

Theorem 8.12.6 (Lefschetz duality, natural version) *Let (X, A) be a compact, triangulable relative homology n-manifold. There is a function assigning to each polyhedron D in (X, A) that contains A an isomorphism*

$$\lambda_D : H^k(X, D; G) \to H_{n-k}(X \setminus D; G)\,.$$

This assignment is natural with respect to inclusions of polyhedra. The group G is arbitrary if (X, A) is orientable, and $G = \mathbb{Z}/2$ otherwise.

Proof: Let Γ be an orientation class for (X, A) in the orientable case. Let it denote an orientation class over $\mathbb{Z}/2$ otherwise.

Step 1: If U is any neighborhood of A such that $\overline{U}$ is a polyhedron in (X, A), let $X_U = X \setminus U$. Then X_U is also a polyhedron in (X, A). Let Γ_U denote the image of Γ under the homomorphism

$$H_n(X, A) \xrightarrow{m_*} H_n(X, \overline{U}) \xrightarrow[\cong]{k_*^{-1}} H_n(X_U, \operatorname{Bd} U)\,.$$

[Here m and k are inclusion maps, and $\operatorname{Bd} U = \overline{U} \setminus U$.] We show that Γ_U is an orientation class for the relative homology manifold $(X_U, \operatorname{Bd} U)$ in the orientable case, and an orientation class over $\mathbb{Z}/2$ in the non-orientable case.

Triangulate (X, A) so that $\overline{U}$ is the polytope of a subcomplex. Then Γ is represented by the sum of all n-simplices of X not in A, suitably oriented. Its image under $k_*^{-1} \circ m_*$ is represented by the sum of those simplices not in $\overline{U}$. It is automatically a cycle of $(X_U, \operatorname{Bd} U)$.

We think of Γ_U as the "restriction" of Γ to X_U.

Step 2: Let D be a polyhedron in (X, A) containing A. Let U be any neighborhood of D satisfying the conditions of the preceding lemma, such that A, D, and $\overline{U}$ are all polytopes of subcomplexes of a triangulation of X. See Figure 8.12.4. The following diagram commutes up to sign:

$$\begin{array}{ccc} H^k(X_U, \operatorname{Bd} U) & \xrightarrow[\cong]{\phi_*} & H_{n-k}(X_U \setminus \operatorname{Bd} U) \\ & \searrow^{\frown\, \Gamma_U} & \downarrow j_* \\ & & H_{n-k}(X_U)\,. \end{array}$$

Here ϕ_* is the Lefschetz duality isomorphism for the triangulated relative homology manifold $(X_U, \operatorname{Bd} U)$. We omit G from the notation for simplicity. Now $X_U = X \setminus U$ and $X_U \setminus \operatorname{Bd} U = X \setminus \overline{U}$. Therefore the inclusion j is a homotopy equivalence. It follows that $\frown\, \Gamma_U$ is an isomorphism in simplicial theory. Thus it is also an isomorphism in singular theory.

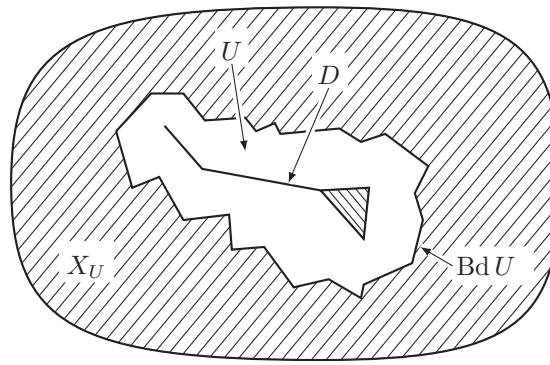


Figure 8.12.4

Consider now the following diagram in singular theory:

$$H^k(X,D) \xleftarrow{i^*} H^k(X,\overline{U}) \xrightarrow{k^*} H^k(X_U, \mathrm{Bd}\, U) \xrightarrow{\frown \Gamma_U} H_{n-k}(X_U) \xrightarrow{\ell_*} H_{n-k}(X \setminus D)\,,$$

where i, k, and ℓ are inclusions. The homomorphism k^* is an isomorphism because k is an excision map. The homomorphisms i^* and ℓ_* are isomorphisms because $i : D \to \overline{U}$ and $\ell : (X \setminus U) \to (X \setminus D)$ are homotopy equivalences.

We denote the composite isomorphism for the present by

$$\lambda_{D,U} : H^k(X,D) \to H_{n-k}(X \setminus D)\,.$$

Note that it does not depend on the specific triangulation involved, as did the Lefschetz isomorphism ϕ_*.

Step 3: We check a version of naturality for the isomorphism $\lambda_{D,U}$A. Suppose that E is a polyhedron in (X, A) such that $A \subset E \subset D$, and suppose that V is a neighborhood of E chosen as in **Step 2** such that $\overline{V} \subset U$. [Such a neighborhood of E always exists by the preceding lemma.] See Figure 8.12.5. Set $X_V = X \setminus V$. We show that $\lambda_{D,U}$ and $\lambda_{E,V}$ commute with the homomorphisms induced by inclusion.

To prove this fact, we consider the following formidable diagram:

$$\begin{array}{ccccccccc}
H^k(X,D) & \leftarrow & H^k(X,\overline{U}) & \rightarrow & H^k(X_U,\operatorname{Bd} U) & \xrightarrow{\frown\,\Gamma_U} & H_{n-k}(X_U) & \rightarrow & H_{n-k}(X\setminus D) \\
\downarrow & & \downarrow & \nearrow & \uparrow & & \downarrow & & \downarrow \\
 & & & & H^k(X_V,\overline{U}\setminus V) & \overset{\frown\,\Gamma'}{\searrow} & & & \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
H^k(X,E) & \leftarrow & H^k(X,\overline{V}) & \rightarrow & H^k(X_V,\operatorname{Bd} V) & \xrightarrow{\frown\,\Gamma_V} & H_{n-k}(X_V) & \rightarrow & H_{n-k}(X\setminus E).
\end{array}$$

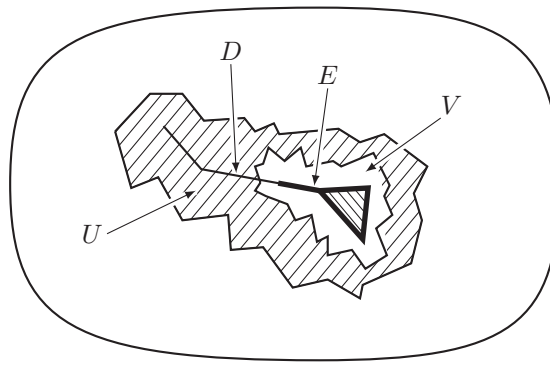


Figure 8.12.5

Here we must interpret these groups as singular groups because the pair $(X_V, \overline{U} \setminus V)$ is not necessarily triangulable. This arises from the fact that $\overline{U}$ and $\overline{V}$ may not be polytopes of subcomplexes of the *same* triangulation of X. The top row of maps in this last diagram defines $\lambda_{D,U}$ and the bottom row defines $\lambda_{E,V}$. All unlabeled maps are induced by inclusion. The class Γ' is the appropriate "restriction" of Γ—that is, the image of Γ under

$$H_n(X,A) \longrightarrow H_n(X,\overline{U}) \xleftarrow[\cong]{} H_n(X_V,\overline{U}\setminus V)\,.$$

All squares and triangles in the diagram commute, by naturality of the cap product. Our result follows.

Step 4: We now show that $\lambda_{D,U}$ is independent of the choice of U. We apply the preceding step to the case $D = E$. If U and U' are two neighborhoods of D satisfying the conditions of Lemma 8.12.4, then we can choose V [also satisfying these conditions] so that $D \subset \overline{V} \subset U \cap U'$. Then, by **Step 3**,

$$\lambda_{D,U} = \lambda_{D,V} = \lambda_{D,U'}\,.$$

We define $\lambda_D = \lambda_{D,U}$. Naturality of λ_D with respect to inclusions $E \to D$ now follows from **Step 3**. □

Corollary 8.12.7 (Alexander duality) *Let n be fixed. There is a function assigning to each proper nonempty polyhedron A in $\mathbb{S}^n$ an isomorphism*

$$\alpha_A : \widetilde{H}^k(A) \to \widetilde{H}_{n-k-1}(\mathbb{S}^n \setminus A)\,.$$

This assignment is natural with respect to inclusions.

Proof: The case $k < n-1$ follows by noting that α_A may be defined as the composite of the isomorphisms

$$\widetilde{H}^k(a) \xrightarrow{\delta^*} H^{k+1}(\mathbb{S}^n, A) \xrightarrow{\lambda_A} H_{n-k-1}(\mathbb{S}^n \setminus A)\,,$$

both of which are natural. The case $k = n-1$ follows similarly from naturality of the diagrams used in defining the Alexander isomorphism. The case $k = n$ is trivial. □

EXERCISES

1. Let D be a polyhedron in $\mathbb{S}^n$ that is the union of two disjoint sets A and B, with

$$A \approx \mathbb{S}^k \quad \text{and} \quad B \approx \mathbb{S}^{n-k-1}\,,$$

where $0 \le k \le n-1$. Inclusion maps induce homomorphisms

$$\phi : \widetilde{H}_k(A) \to \widetilde{H}_k(\mathbb{S}^n \setminus B)\,,$$

$$\psi : \widetilde{H}_{n-k-1}(B) \to \widetilde{H}_{n-k-1}(\mathbb{S}^n \setminus A)\,.$$

Alexander duality implies that these groups are all infinite cyclic. Show that, up to sign, ϕ and ψ both equal multiplication by the same integer m. [See Exercise 3 of Section 4.9.] This integer, the *linking number*, measures how many times A links B. [**Hint:** Triangulate $\mathbb{S}^n$ so that A and B are polytopes of full subcomplex, and no simplex intersects both A and B. Let C be the union of all simplices that are disjoint from A. Begin with the following diagram:

$$\begin{array}{ccc} \widetilde{H}^{n-k-1}(C) & \longrightarrow & \widetilde{H}^{n-k-1}(B) \\ \downarrow \alpha_C & & \downarrow \alpha_B \\ \widetilde{H}_k(\mathbb{S}^n \setminus C) & \longrightarrow & \widetilde{H}_k(\mathbb{S}^n \setminus B)] \end{array}\,.$$

2. Explain why Figures 8.12.2 and 8.12.3 are each actually the star of some ordered pair.

8.13 Čech Cohomology

Until now we have studied two homology and cohomology theories—namely, the simplicial theory and the singular theory. There are a number of others, most notably the Čech theory. This theory turns out to be particularly satisfactory in the case of cohomology. This is what we discuss now.

We shall construct the Čech cohomology groups of a topological space and show that they agree with the simplicial groups when both are defined. We also construct a topological space for which the Čech and singular theories disagree. We shall use Čech cohomology in the next section to prove a generalized version of Alexander duality.

We begin with the concept of a directed set.

Definition 8.13.1 A *directed set* J is a set with a relation $<$ such that

(1) $\alpha < \alpha$ for all $\alpha \in J$.

(2) $\alpha < \beta$ and $\beta < \gamma$ implies that $\alpha < \gamma$.

(3) Given α and β, there exists a δ such that $\alpha < \delta$ and $\beta < \delta$.

The element δ is called an *upper bound* for α and β.

EXAMPLE 8.13.2 Any simply-ordered set is a directed set under the relation $\leq$.

EXAMPLE 8.13.3 Consider the family whose elements are open coverings $\mathcal{A}$ of a topological space X. We make this family into a directed set by declaring $\mathcal{A} < \mathcal{B}$ if $\mathcal{B}$ is a *refinement* of $\mathcal{A}$. This means that for each element B of $\mathcal{B}$, there is at least one element A of $\mathcal{A}$ containing it. Properties **(1)** and **(2)** are now immediate. To check **(3)**, we note that given open coverings $\mathcal{A}$ and $\mathcal{B}$ of X, the collection

$$\mathcal{D} = \{A \cap B : A \in \mathcal{A} \text{ and } B \in \mathcal{B}\}$$

is an open covering of X that refines both $\mathcal{A}$ and $\mathcal{B}$.

Definition 8.13.4 A *direct system* of abelian groups and homomorphisms, corresponding to the directed set J, is an indexed family $\{G_\alpha\}_{\alpha \in J}$ of abelian groups, along with a family of homomorphisms

$$f_{\alpha\beta} : G_\alpha \to G_\beta \,,$$

defined for every pair of indices such that $\alpha < \beta$ and such that:

(1) $f_{\alpha\alpha} : G_\alpha \to G_\alpha$ is the identity.

(2) If $\alpha < \beta < \gamma$, then $f_{\beta\gamma} \circ f_{\alpha\beta} = f_{\alpha\gamma}$. That is,

the following diagram commutes:

$$\begin{array}{ccc} G_\alpha & \xrightarrow{f_{\alpha\gamma}} & G_\gamma \\ & {\scriptstyle f_{\alpha\beta}}\searrow \quad \nearrow{\scriptstyle f_{\beta\gamma}} & \\ & G_\beta & \end{array}$$

Definition 8.13.5 Given a direct system of abelian groups and homomorphisms, we define a group called the *direct limit* of this system as follows. Take the disjoint union of the groups G_α, and introduce an equivalence relation by declaring $g_\alpha \sim g_\beta$ [for $g_\alpha \in G_\alpha$ and $g_\beta \in G_\beta$] if, for some upper bound δ of α and β, we have

$$f_{\alpha\delta}(g_\alpha) = f_{\beta\delta}(g_\beta)\,.$$

The direct limit is the set of equivalence classes of $\sim$. It is denoted by

$$\operatorname{dir\,lim}{}_{\alpha\in J} G_\alpha\,.$$

We make it into an abelian group by defining

$$\{g_\alpha\} + \{g_\beta\} = \{f_{\alpha\delta}(g_\alpha) + f_{\beta\delta}(g_\beta)\}\,,$$

where δ is some upper bound for α and β.

It is easy to see that this operation is well defined and makes the direct limit into an abelian group. We note the following elementary facts:

(1) If all the maps $f_{\alpha\beta}$ are isomorphisms, then $\operatorname{dir\,lim} G_\alpha$ is isomorphic to any one of the groups G_α.

(2) If all the maps $f_{\alpha\beta}$ are zero-homomorphisms, then $\operatorname{dir\,lim} G_\alpha$ is the trivial group. More generally, if for each α there is a β such that $\alpha < \beta$ and $f_{\alpha\beta}$ is the zero homomorphism, then $\operatorname{dir\,lim} G_\alpha$ is the trivial group.

Definition 8.13.6 Let J and K be two directed sets. Let $\{G_\alpha, f_{\alpha\beta}\}$ and $\{H_\gamma, g_{\gamma\delta}\}$ be associated direct systems of abelian groups and homomorphisms. A *map* Φ *of direct systems* is first, a set map $\phi : J \to K$ that preserves the order relation, and second, for each $\alpha \in J$, a homomorphism

$$\phi_\alpha : G_\alpha \to H_{\phi(\alpha)}\,,$$

such that commutativity holds in the following diagram, where $\gamma = \phi(\alpha)$ and $\delta = \phi(\beta)$ and $\alpha < \beta$:

$$\begin{array}{ccc} G_\alpha & \xrightarrow{f_{\alpha\beta}} & G_\beta \\ {\scriptstyle \phi_\alpha}\downarrow & & \downarrow{\scriptstyle \phi_\beta} \\ H_\gamma & \xrightarrow{g_{\gamma\delta}} & H_\delta \end{array}$$

Such a map Φ induces a homomorphism, called the *direct limit* of the homomorphisms ϕ_α:

$$\underrightarrow{\Phi} : \operatorname{dir\,lim}_{\alpha\in J} G_\alpha \longrightarrow \operatorname{dir\,lim}_{\gamma\in K} H_\gamma \,.$$

It maps the equivalence class of $g_\alpha \in G_\alpha$ to the equivalence class of $\phi_\alpha(g_\alpha)$.

EXAMPLE 8.13.7 Let $\{G_\alpha, f_{\alpha\beta}\}$ be a direct system of abelian groups and homomorphisms. Let H be an abelian group. If, for each α, one has a homomorphism $\phi_\alpha : G_\alpha \to H$ and if $\phi_\beta \circ f_{\alpha\beta} = \phi_\alpha$ whenever $\alpha < \beta$, then one has an induced homomorphism

$$\underrightarrow{\Phi} : \operatorname{dir\,lim}_{\alpha\in J} G_\alpha \longrightarrow H \,.$$

It is a special case of the preceding construction, in which the second direct system consists of the single group H.

EXAMPLE 8.13.8 Suppose that one has a sequence of abelian groups and homomorphisms

$$G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} G_3 \longrightarrow \cdots \,.$$

It becomes a direct system with index set $J = \mathbb{Z}_+$, the positive integers, if we define

$$f_{mn} = f_{n-1} \circ f_{n-2} \circ \cdots \circ f_m$$

whenever $m < n$.

For example, let each group G_i equal the integers. Let $\mu : \mathbb{Z} \to \mathbb{Z}$ be multiplication by 2. If each map f_i equals μ, then one has the direct system

$$\mathbb{Z} \xrightarrow{\mu} \mathbb{Z} \xrightarrow{\mu} \cdots \,.$$

Its direct limit is readily seen to be isomorphic to the group H of dyadic rationals [which is the additive group of all rationals of the form $m/2^n$ for m and n integers]. Define

$$\phi_n : G_n \to H$$

by the equation $\phi_n(m) = m/2^n$. Then one checks that $\phi_n \circ f_{n-1} = \phi_{n-1}$. It is easy to check that $\underrightarrow{\Phi}$ is both injective and surjective.

Definition 8.13.9 If J is a directed set, then a subset J_0 of J is said to be *cofinal* in J if for each $\alpha \in J$ there exists a $\delta \in J_0$ with $\alpha < \delta$.

Lemma 8.13.10 *Suppose that one is given a direct system $\{G_\alpha, f_{\alpha\beta}\}$ of abelian groups and homomorphisms, indexed by the directed set J. If J_0 is cofinal in J, then J_0 is a directed set, and inclusion induces an isomorphism*

$$\operatorname{dir\,lim}_{\alpha\in J_0} G_\alpha \cong \operatorname{dir\,lim}_{\alpha\in J} G_\alpha \,.$$

Proof: The axioms for a directed set are easy to check. Given α, β in J_0, they have an upper bound in J, and hence an upper bound in J_0. Let $\phi : J_0 \to J$ be the inclusion map, and for each $\alpha \in J_0$, let $\phi_\alpha : G_\alpha \to G_{\phi(\alpha)}$ be the identity. Then one has an induced homomorphism

$$\underrightarrow{\Phi} : \operatorname{dir\,lim}_{\alpha \in J_0} G_\alpha \longrightarrow \operatorname{dir\,lim}_{\alpha \in J} G_\alpha \,.$$

It is surjective, since given g_α, it is equivalent to some g_δ for $\delta \in J_0$. To show that its kernel vanishes, we suppose that $\alpha \in J_0$ and $g_\alpha \sim 0_\beta$ for some $\beta \in J$. Then there is some element δ of J such that

$$f_{\alpha\delta}(g_\alpha) = f_{\beta\delta}(0_\beta) = 0_\delta \,.$$

Choose $\epsilon \in J_0$ with $\delta < \epsilon$. Then

$$f_{\alpha\epsilon}(g_\alpha) = f_{\delta\epsilon}(0_\delta) = 0_\epsilon \,,$$

so $g_\alpha \sim 0_\epsilon$, where $\epsilon \in J_0$. □

Now we define the Čech cohomology groups.

Definition 8.13.11 Let $\mathcal{A}$ be a collection of subsets of the space X. We define an abstract simplicial complex called the *nerve* of $\mathcal{A}$, denoted by $N(\mathcal{A})$. Its vertices are the elements of $\mathcal{A}$ and its simplices are the finite subcollections $\{A_1, A_2, \ldots, A_n\}$ of $\mathcal{A}$ such that

$$A_1 \cap A_2 \cap \cdots \cap A_n \neq \emptyset \,.$$

Now if $\mathcal{B}$ is a collection of sets refining $\mathcal{A}$, then we can define a map $g : \mathcal{B} \to \mathcal{A}$ by choosing $g(B)$ to be an element of $\mathcal{A}$ that contains B. If $\{B_1, B_2, \ldots, B_n\}$ is a simplex of $N(\mathcal{B})$, then $\{g(B_1), g(B_2), \ldots, g(B_n)\}$ is a simplex of $N(\mathcal{A})$, just because $\cap B_i$ is nonempty and contained in $\cap g(B_i)$. Thus the vertex map g induces a simplicial map

$$g : N(\mathcal{B}) \to N(\mathcal{A}) \,.$$

There is some arbitrariness in the choice of g in the last definition, but any other choice g' for g is contiguous to g, since

$$\cap B_i \subset \cap (g(B_i) \cap g'(B_i)) \,.$$

Thus we can make the following definition:

Definition 8.13.12 If $\mathcal{B}$ is a refinement of $\mathcal{A}$, then we have uniquely defined homomorphisms

$$g_* : H_k(N(\mathcal{B}); G) \to H_k(N(\mathcal{A}); G) \,,$$
$$g^* : H^k(N(\mathcal{A}); G) \to H^k(N(\mathcal{B}); G) \,,$$

induced by the simplicial map g satisfying the condition $g(B) \supset B$ for all $B \in \mathcal{B}$. We call them the *homomorphisms induced by refinement.*

Definition 8.13.13 Let J be the directed set consisting of all open coverings of the space X, directed by letting $\mathcal{A} < \mathcal{B}$ if $\mathcal{B}$ is a refinement of $\mathcal{A}$. Construct a direct system by assigning to the element $\mathcal{A}$ of J the group

$$H^k(N(\mathcal{A}); G)\,,$$

and by assigning to the pair $\mathcal{A} < \mathcal{B}$ the homomorphism

$$f_{\mathcal{A}\mathcal{B}} : H^k(N(\mathcal{A}); G) \to H^k(N(\mathcal{B}); G)\,,$$

induced by refinement. We define the *Čech cohomology group* of X in dimension k, with coefficients in G, by the equation

$$\check{H}^k(X; G) = \operatorname{dir\,lim}_{\mathcal{A}\in J} H^k(N(\mathcal{A}); G)\,.$$

We define the reduced Čech cohomology of X, $\check{\tilde{H}}^k(A)$, in a similar manner, as the direct limit of the reduced groups $\widetilde{H}^k(N(\mathcal{A}); G)$.

Since these groups depend only on the collection of open coverings of X, they are obviously topological invariants of X.

In spite of the abstractness of this definition, it gives us nothing new in the case of a triangulable space, as we now show.

Theorem 8.13.14 *Let K be a simplicial complex. Then*

$$\check{H}^k(|K|; G) \cong H^k(K; G)\,.$$

The same is true for reduced cohomology.

Proof: For any complex K, let $\mathcal{A}(K)$ be the covering of $|K|$ by the open stars of its vertices. The vertex correspondence f_K that assigns to the vertex v of K the vertex $\operatorname{St} v$ of $N(\mathcal{A}(K))$ defines an isomorphism between the complex K and the abstract complex $N(\mathcal{A}(K))$. For $v_0, v_1, \ldots, v_n$ is a simplex of K if and only if

$$\operatorname{St} v_0 \cap \operatorname{St} v_1 \cap \cdots \cap \operatorname{St} v_n \neq \emptyset\,.$$

This is equivalent to the statement that $\{\operatorname{St} v_0, \operatorname{St} v_1, \ldots, \operatorname{St} v_n\}$ is a simplex of $N(\mathcal{A}(K))$.

If K' is a subdivision of K, then there is a simplicial approximation to the identity $h : K' \to K$. It is induced by a vertex map h specified by the condition

$$\operatorname{St}(w, K') \subset \operatorname{St}(h(w), K)\,.$$

The same vertex correspondence can be used to define a simplicial map $g : N(\mathcal{A}(K')) \to N(\mathcal{A}(K))$. It assigns to the vertex $\operatorname{St}(w, K')$ of $N(\mathcal{A}(K'))$ the vertex $\operatorname{St}(h(w), K)$ of $N(\mathcal{A}(K))$. Then the following diagram commutes:

$$\begin{array}{ccc} K' & \xrightarrow[\cong]{f_{K'}} & N(\mathcal{A}(K')) \\ \downarrow h & & \downarrow g \\ K & \xrightarrow[\cong]{f_K} & N(\mathcal{A}(K)) \end{array}$$

All four of the induced maps in cohomology are isomorphisms.

Now consider the family J_0 of open coverings of $|K|$ of the form $\mathcal{A}(K')$ as K' ranges over all subdivisions of K. The general simplicial approximation theorem implies that this family is cofinal in the family J of all open coverings of $|K|$. [If K is finite, then the finite simplicial approximation theorem will suffice.] Therefore

$$\check{H}(|K|;G) \cong \operatorname{dir\,lim} H^k(K';G)\,,$$

where the direct system on the right is indexed by the family of all subdivisions of K. Each of the maps h^* in the latter direct system is an isomorphism, so the direct limit is isomorphic to $H^k(K;G)$ as desired. □

This theorem shows that Čech cohomology agrees with simplicial cohomology when both are defined. But what does Čech cohomology look like for more general spaces? In order to make the required computations, we shall prove a basic "continuity property" of Čech cohomology. First we need a lemma.

Lemma 8.13.15 *Let Y be a compact subspace of the normal space X. Consider the directed set L consisting of coverings $\mathcal{A}$ of Y by sets that are open in X [rather than in Y]. Then*

$$\check{H}^k(Y;G) \cong \operatorname{dir\,lim}_{\mathcal{A}\in L} H^k(N(\mathcal{A});G)\,.$$

The same is true in reduced cohomology.

Proof: Step 1: Let $U_1, U_2, \ldots, U_n$ be an open covering of X. We show that there exists an open covering $V_1, V_2, \ldots, V_n$ of X such that $\overline{V}_i \subset U_i$ for each i.

The set $X \setminus (U_2 \cup \cdots \cup U_n)$ is a closed subset of U_1. By normality, we can choose an open set V_1 containing it so that $\overline{V}_1 \subset U_1$. Then $V_1, U_2, \ldots, U_n$ covers X. Apply the same construction to choose V_2 containing $X \setminus (V_1 \cup U_3 \cup \cdots \cup U_n)$ such that $\overline{V}_2 \subset U_2$. Then $V_1, V_2, U_3, \ldots, U_n$ covers X. Continue this argument in a similar fashion.

Step 2: Let $\mathcal{C} = \{C_1, C_2, \ldots, C_n\}$ and $\mathcal{D} = \{D_1, D_2, \ldots, D_n\}$ be two indexed families having the same index set. We say that their nerves are *naturally isomorphic* provided that

$$C_{i_1} \cap \cdots \cap C_{i_p} \neq \emptyset \iff D_{i_1} \cap \cdots \cap D_{i_p} \neq \emptyset\,.$$

This means that the vertex map $C_i \to D_i$ induces a simplicial isomorphism $N(\mathcal{C}) \to N(\mathcal{D})$.

We prove the following. Let $\{C_1, C_2, \ldots, C_n\}$ be a collection of closed sets in X, and let $\{U_1, U_2, \ldots, U_n\}$ be a collection of open sets in X, with $C_i \subset U_i$ for each i. Then there exists a collection $\{W_1, W_2, \ldots, W_n\}$ of open subsets of X such that $C_i \subset W_i$ and $\overline{W}_i \subset U_i$ for all i. And also the nerves of $\{C_1, \ldots, C_n\}$ and $\{\overline{W}_1, \ldots, \overline{W}_n\}$ are naturally isomorphic.

Consider the collection $\mathcal{E}$ of all subsets of X of the form

$$E_{i_1,i_2,\ldots,i_p} = C_{i_1} \cap C_{i_2} \cap \cdots \cap C_{i_p}\,.$$

Let E be the union of all such sets that do not intersect C_1. Then E is closed in X and disjoint from C_1. Choose W_1 to be an open subset of X containing C_1 whose closure is disjoint from E and lies in U_1. We show that the nerves of $\{\overline{W}_1, C_2, \ldots, C_n\}$ and $\{C_1, C_2, \ldots, C_n\}$ are naturally isomorphic. For this purpose, it suffices to show that

$$\overline{W}_1 \cap C_{i_1} \cap C_{i_2} \cap \cdots \cap C_{i_p} \neq \emptyset \iff C_1 \cap C_{i_1} \cap C_{i_2} \cdots \cap c_{i_p} \neq \emptyset\,.$$

The implication $\Leftarrow$ is trivial since $C_1 \subset \overline{W}_1$. We prove the other implication. If the right-hand set is empty, then the set $C_{i_1} \cap C_{i_2} \cap \cdots \cap C_{i_p}$ is disjoint from C_1. Thus it is contained in E by definition. Then it is disjoint from $\overline{W}_1$ by definition of W_1. Hence the left-hand set is empty as well.

Apply this construction a second time to choose W_2 so that $C_2 \subset W_2$ and $\overline{W}_2 \subset U_2$ and the nerves of

$$\{\overline{W}_1, \overline{W}_2, C_3, C_4, \ldots, C_n\} \quad \text{and} \quad \{\overline{W}_1, C_2, C_3, C_4, \ldots, C_n\}$$

are naturally isomorphic. Continue this argument in a similar fashion.

Step 3: A covering $\mathcal{W} = \{W_1, W_2, \ldots, W_n\}$ of Y by sets open in X is said to be *adapted to* Y if the nerves of $\{W_1, W_2, \ldots, W_n\}$ and $\{W_1 \cap Y, W_2 \cap Y, \ldots, W_n \cap Y\}$ are naturally isomorphic. Let L be the family of all coverings of Y by sets open in X. Let L_0 be the subfamily consisting of those coverings that are adapted to Y. We show that L_0 is cofinal in L.

Let $\mathcal{A}$ be a covering of Y by sets open in X. Pass to a finite subcollection $\{U_1, U_2, \ldots, U_n\}$ covering Y. Consider the covering

$$\{U_1 \cap Y, U_2 \cap Y, \ldots, U_n \cap Y\}$$

of Y by sets open in Y. Applying **Step 1** to the normal space Y, choose a collection $\{V_1, V_2, \ldots, V_n\}$ of sets open in Y that cover Y and such that $\overline{V}_i \subset (U_i \cap Y)$ for each i. Let $C_i = \overline{V}_i$ and apply **Step 2** to find a collection of open sets $\mathcal{W} = \{W_1, W_2 \ldots, W_n\}$ of X such that $C_i \subset W_i$ and $\overline{W}_i \subset U_i$ and such that the nerves of $\{C_1, C_2, \ldots, C_n\}$ and $\{\overline{W}_1, \overline{W_2}, \ldots, \overline{W}_n\}$ are naturally isomorphic.

Now $\mathcal{W}$ refines $\mathcal{A}$ and $\mathcal{W}$ covers Y. The fact that $\mathcal{W}$ is adapted to Y follows from the implication

$$\begin{array}{ccc} C_{i_1} \cap C_{i_2} \cap \cdots \cap C_{i_p} \neq \emptyset & \iff & \overline{W}_{i_1} \cap \overline{W}_{i_2} \cap \cdots \cap \overline{W}_{i_p} \neq \emptyset \\ \Downarrow & & \Uparrow \\ W_{i_1} \cap W_{i_2} \cap \cdots \cap W_{i_p} \cap Y \neq \emptyset & \implies & W_{i_1} \cap W_{i_2} \cap \cdots \cap W_{i_p} \neq \emptyset \end{array}$$

Step 4: In general, if $\mathcal{A} = \{A_1, A_2, \ldots, A_n\}$ is a covering of Y by sets open in X, we let $\mathcal{A} \cap Y$ denote the covering $\{A_1 \cap Y, A_2 \cap Y, \ldots, A_n \cap Y\}$ of Y by sets open in Y. Consider the following four directed sets:

$$\begin{aligned} L &= \{\mathcal{A} : \mathcal{A} \text{ is a covering of } Y \text{ by sets open in } X\}, \\ L_0 &= \{\mathcal{W} : \mathcal{W} \in L \text{ and } \mathcal{W} \text{ is adapted to } Y\}, \\ J &= \{\mathcal{B} : \mathcal{B} \text{ is a covering of } Y \text{ by sets open in } Y\}, \\ J_0 &= \{\mathcal{W} \cap Y : \mathcal{W} \in L_0\}. \end{aligned}$$

Note that every element $\mathcal{B}$ of J equals $\mathcal{A} \cap Y$ for some $\mathcal{A}$ in L.

We have shown that L_0 is cofinal in L. It follows at once that J_0 is cofinal in J. Given $\mathcal{B} \in J$, it equals $\mathcal{A} \cap Y$ for some $\mathcal{A} \in L$. Then if $\mathcal{W}$ is an element of L_0 that refines $\mathcal{A}$, it follows that $\mathcal{W} \cap Y$ refines $\mathcal{A} \cap Y = \mathcal{B}$.

The lemma follows, because

$$\begin{aligned} \check{H}(Y; G) &= \operatorname{dir\,lim}{}_J H^k(N(\mathcal{B}); G) && \text{by definition,} \\ &\cong \operatorname{dir\,lim}{}_{J_0} H^k(N(\mathcal{W} \cap Y); G) && \text{since } J_0 \text{ is cofinal in } J, \\ &\cong \operatorname{dir\,lim}{}_{L_0} H^k(N(\mathcal{W}); G) && \text{since } N(\mathcal{W}) \cong N(\mathcal{W} \cap Y), \\ &\cong \operatorname{dir\,lim}{}_{L} H^k(N(\mathcal{A}); G) && \text{since } L_0 \text{ is cofinal in } L. \end{aligned}$$

□

Theorem 8.13.16 *Let X be a compact, triangulable space. Let $D_1 \supset D_2 \supset \cdots$ be a sequence of polyhedra in X whose intersection is Y. Then*

$$\check{H}^k(Y; G) \cong \operatorname{dir\,lim} H^k(D_n; G).$$

The same result holds in reduced cohomology.

Proof: We shall omit G from the notation for simplicity. The direct limit in the conclusion of the theorem does not depend on the particular triangulations of the spaces D_n. Let X_n be a triangulation of X chosen so that D_n is the polytope of a full subcomplex of X_n, and so that the maximum diameter of a simplex of X_n is less than $1/n$. And so that the inclusion map $|X_{n+1}| \to |X_n|$ satisfies the star condition. Let us denote the subcomplex of X_n whose polytope is D_n by K_n. Then the isomorphism

$$\operatorname{dir\,lim} H^k(D_n) \cong \operatorname{dir\,lim} H^k(K_n)$$

holds trivially.

Step 1: Given n, consider the covering of D_n by sets open in D_n defined by

$$\mathcal{A}'_n = \{\mathrm{St}(v, K_n) : v \in K_n\}.$$

Then, as we noted in the proof of Theorem 8.13.14, one has a simplicial isomorphism $f_n : K_n \to N(\mathcal{A}'_n)$ defined by $f_n(v) = \mathrm{St}(v, K_n)$. The same argument as was used there can be applied to pass to an isomorphism of the cohomology direct limits, as we now show.

Let $h : K_{n+1} \to K_n$ be a simplicial approximation to the inclusion map $|K_{n+1}| \to |K_n|$. We then define $g : \mathcal{A}'_{n+1} \to \mathcal{A}'_n$ by the equation

$$g(\mathrm{St}(w, K_{n+1})) = \mathrm{St}(h(w), K_n)\,.$$

The map g induces a simplicial map $g : N(\mathcal{A}'_{n+1}) \to N(\mathcal{A}'_n)$ such that the following diagram commutes

$$\begin{array}{ccc} K_{n+1} & \xrightarrow[\cong]{f_{n+1}} & N(\mathcal{A}'_{n+1}) \\ \downarrow h & & \downarrow g \\ K_n & \xrightarrow[\cong]{f_n} & N(\mathcal{A}'_n) \end{array}$$

We conclude that

$$\mathrm{dir\,lim}\, H^k(K_n) \cong \mathrm{dir\,lim}\, H^k(N(\mathcal{A}'_n))\,,$$

just as before. Note that in this case we cannot identify the left-hand group with a particular simplicial cohomology group, since h^* need not be an isomorphism.

Step 2: Given n, consider the covering of D_n defined by

$$\mathcal{A}_n = \{\mathrm{St}(v, X_n) : v \in K_n\}\,.$$

This is a covering of D_n by sets open in X, whereas the collection $\mathcal{A}'_n$ was a covering of D_n by sets open in D_n. The collection $\mathcal{A}_{n+1}$ is a refinement of $\mathcal{A}_n$ because the inclusion $|X_{n+1}| \to |X_n|$ satisfies the star condition.

We now show that the nerves of $\mathcal{A}_n$ and $\mathcal{A}'_n$ are naturally isomorphic with the vertex map in question carrying $\mathrm{St}(v, K_n)$ to $\mathrm{St}(v, X_n)$ for each $v \in K_n$. Since $\mathrm{St}(v, K_n) \subset \mathrm{St}(v, X_n)$, it is immediate that

$$\cap\, \mathrm{St}(v_i, K_n) \neq \emptyset \Rightarrow \cap\, \mathrm{St}(v_i, X_n) \neq \emptyset\,.$$

We next prove the reverse implication. If $\cap\,\mathrm{St}\,(v_i, X_n) \neq \emptyset$, then the vertices v_i span a simplex σ of X_n. Because the v_i are vertices of K_n and K_n is full, σ must belong to K_n. Then $\cap\, \mathrm{St}(v_i, K_n) \neq \emptyset$ as desired.

It follows at once that

$$\mathrm{dir\,lim}\, H^k(N(\mathcal{A}'_n)) \cong \mathrm{dir\,lim}\, H^k(N(\mathcal{A}_n))\,.$$

Step 3: We now show that the family of coverings $\{\mathcal{A}_1, \mathcal{A}_2, \dots\}$ of Y is cofinal in the family L of all coverings $\mathcal{B}$ of Y by sets open in X.

First, we note the following: If D is a compact subset of the metric space X, and if $\mathcal{B}$ is a covering of D by sets open in X, then there is a $\delta > 0$ such that any set of diameter less than δ that intersects D necessarily lies in an element of $\mathcal{B}$. This is an extension of the Lebesgue number lemma of general topology [see Lemma 2.4.1]. Its proof is left as an exercise for the reader.

Now let $\mathcal{B}$ be an arbitrary covering of Y by sets open in X. First we show that $\mathcal{B}$ covers $D_n = |K_n|$ for some n, say $n = N$. For if not then we can choose for each n a point $x_n \in D_n$ such that x_n is not in any element of $\mathcal{B}$. Some subsequence x_{n_i} converges, necessarily to a point of Y (since $Y = \cap D_n$). But $\mathcal{B}$ covers a neighborhood of Y, so it must cover the point x_{n_i} for i sufficiently large.

We now apply the extended Lebesgue number lemma. Choose $\delta > 0$ so that any set of diameter less than δ that intersects D_N necessarily lies in an element of $\mathcal{B}$. Then choose $m > N$ large enough so that $1/m < \delta/2$. It follows that $\mathcal{A}_m$ refines $\mathcal{B}$. For, given a vertex v of K_m, we have

$$\text{diam} \ \ \text{St}(v, X_m) \leq 2/m < \delta \,.$$

Since $v \in |K_m| \subset |K_N|$, the set $\text{St}(v, K_m)$ intersects $|K_N|$ so that it lies in an element of $\mathcal{B}$.

Step 4: We now prove the theorem. We have

$$\begin{aligned} \text{dir lim}\, H^k(D_n) &\cong \text{dir lim}\, H^k(K_n) \\ &\cong \text{dir lim}\, H^k(N(\mathcal{A}'_n)) \qquad \text{by } \textbf{Step 1}\,, \\ &\cong \text{dir lim}\, H^k(N(\mathcal{A}_n)) \qquad \text{by } \textbf{Step 2}\,, \\ &\cong \text{dir lim}_{\mathcal{B}\in L} H^k(N(\mathcal{B})) \qquad \text{by } \textbf{Step 3}\,. \end{aligned}$$

The latter group is isomorphic to $\check{H}^k(Y)$ by the preceding lemma. □

We use this theorem to compute the Čech cohomology of the following non-triangulable space.

Corollary 8.13.17 *Let X be the closed topologist's sine curve [see Definition 4.6.5]. Then*

$$\begin{aligned} H^1(X) &= 0 \qquad \text{(singular cohomology)}\,, \\ \check{H}^1(X) &\cong \mathbb{Z} \qquad \text{(Čech cohomology)}\,. \end{aligned}$$

Proof: For convenience, we express X as the union of line segments in the plane, as shown in Figure 8.13.1.

Step 1: First we compute the singular homology and cohomology of X.

Let U be the intersection of X with the set of all points (x, y) of $\mathbb{R}^2$ for which $y > 1/8$, and let V be the intersection of X with those (x, y) for which $y < 3/8$.

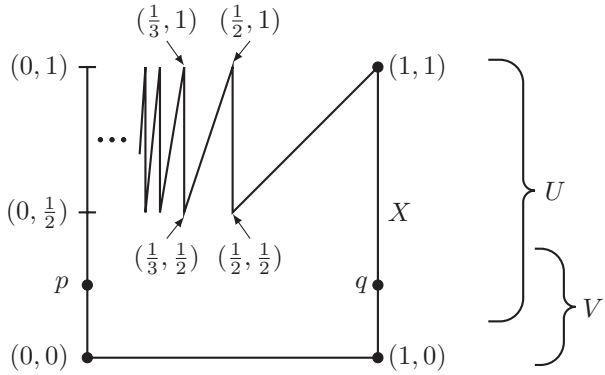


Figure 8.13.1

Because U and V are open, we have a Mayer–Vietoris sequence in singular homology:

$$H_1(U) \oplus H_1(V) \longrightarrow H_1(X) \longrightarrow \widetilde{H}_0(U \cap V) \xrightarrow{i_*} \widetilde{H}_0(U) \oplus \widetilde{H}(V)\,.$$

Now V is homeomorphic to an open interval. So $H_1(V) = \widetilde{H}_0(V) = 0$. The space U has two path components U_1 and U_2, one of which is homeomorphic to an open interval, and the other to a half-open interval. Therefore

$$H_1(U) \cong H_1(U_1) \oplus H_1(U_2) = 0\,.$$

The group $\widetilde{H}_0(U)$ is infinite cyclic and is generated by the 0-chain $q-p$, where $p = (0, 1/4) \in U_1$ and $q = (1, 1/4) \in U_2$. The space $U \cap V$ consists of two disjoint open line segments, so that $\widetilde{H}_0(U \cap V)$ is also infinite cyclic and is generated by $q - p$. It follows that i_* is an isomorphism, so that $H_1(X) = 0$. The universal coefficient theorem for cohomology now implies that $H^1(X) = 0$. [Alternatively, one can use a Mayer-Vietoris sequence in cohomology to prove this fact.]

Step 2: Now we compute the Čech cohomology of X.

We express X as an intersection of polyhedra as follows: For each n, let C_n be the closed rectangular region pictured in Figure 8.13.2. Let $D_n = X \cup C_n$. Then D_n is a polyhedron. By the preceding theorem,

$$\check{H}^1(X) \cong \operatorname{dir\,lim} H^1(D_n)\,.$$

Now D_n is a space having the homotopy type of a circle, so that $H^1(D_n) \cong \mathbb{Z}$.

Furthermore, D_{n+1} is readily seen to be a deformation retract of D_n so that the inclusion i induces an isomorphism

$$H^1(D_{n+1}) \xleftarrow{i^*} H^1(D_n)\,.$$

It follows that $\check{H}(X) \cong \mathbb{Z}$. □

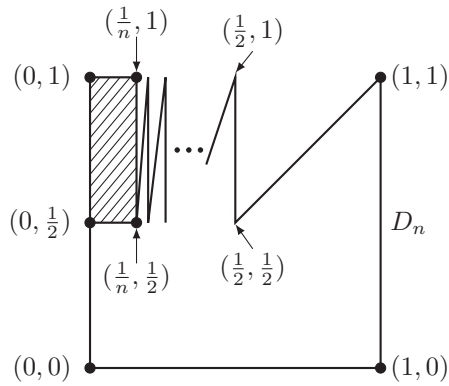


Figure 8.13.2

We have only touched on the subject of Čech cohomology. For a full-scale development one needs to define the homomorphism induced by a continuous map [which is easy], extend the definition to the cohomology of a pair (X, A), and verify the Eilenberg–Steenrod axioms [which requires some work]. The interested reader is referred to Chapters 9 and 10 of [ES52].

EXERCISES

1. Check the details of Example 8.13.8.

2. For each n, let $\mu_n : \mathbb{Z} \to \mathbb{Z}$ be multiplication by n. The direct limit of the system

$$\mathbb{Z} \xrightarrow{\phi_2} \mathbb{Z} \xrightarrow{\phi_3} \mathbb{Z} \xrightarrow{\phi_4} \mathbb{Z} \longrightarrow \cdots$$

is a familiar group. What is it?

3. Check that the system of groups and homomorphisms used in defining Čech cohomology is a direct system.

4. Verify the extended Lebesgue number lemma quoted in the proof of Theorem 8.13.16.

5. Let X denote the solid torus $\mathbb{B}^2 \times \mathbb{S}^1$ and let $f : X \to X$ be the imbedding pictured in Figure 8.13.3. Let $X_0 = X$ and let $X_1 = f(X_0)$. In general, let $X_{n+1} = f(X_n)$. Let $S = \cap X_n$. The space S is often called a *solenoid*.

(a) Describe the space X_2.

(b) Compute $\check{H}^1(S)$.

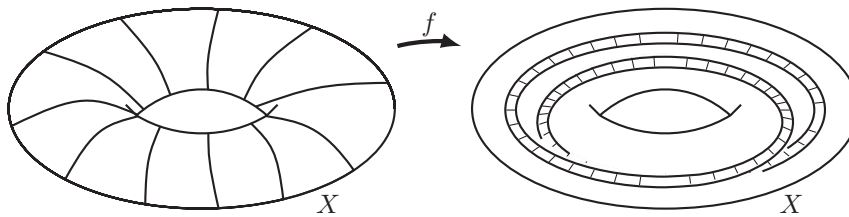


Figure 8.13.3

6. Given X, let $\mathcal{C}$ be the collection of compact subspaces of X, directed by letting $C < D$ if $C \subset D$. Show that $\mathcal{C}$ is a directed set. Define a direct system by the rules

$$G_C = H_i(C)\,,$$

$$F_{C,D} = i_* : H_i(C) \to H_i(D)\,,$$

where i is inclusion and H_i denotes singular homology. Show that

$$H_1(X) \cong \operatorname{dir\,lim} H_i(C)\,.$$

[This is just another way of expressing the compact support property of singular homology.]

7. Let X be any set. Let S be the collection of all subsets of X [that is, S is the power set of X]. Explain how to turn S into a directed set.

8.14 Alexander–Pontryagin Duality

Now we prove a generalized version of the Alexander duality theorem. This version applies to an arbitrary closed subspace of $\mathbb{S}^n$, not just to a polyhedron in $\mathbb{S}^n$. As an application, we derive a version of the Jordan curve theorem more general than any we have seen up until now.

Theorem 8.14.1 (Alexander–Pontryagin duality) *Let A be a proper, non-empty, closed subset of $\mathbb{S}^n$. Then*

$$\check{\widetilde{H}}^k(A) \cong \widetilde{H}_{n-k-1}(\mathbb{S}^n \setminus A)\,.$$

Here the left-hand space is Čech cohomology and the right-hand space is singular homology.

Proof: Choose a triangulation X of $\mathbb{S}^n$ fine enough that not every simplex of X intersects A. Consider the sequence $\operatorname{sd} X$, $\operatorname{sd}^2 X, \ldots$ of successively finer subdivisions of X.

Let A_m denote the subcomplex of $\operatorname{sd}^m X$ consisting of those simplices that intersect A, along with their faces. Then $|A_m|$ is a proper nonempty polyhedron in X and $|A_{m+1}| \subset |A_m|$ for all m. Furthermore, A is the intersection of the sets $|A_m|$.

The Alexander duality theorem gives us an isomorphism

$$\alpha_m : \widetilde{H}^k(A_m; G) \to \widetilde{H}_{n-k-1}(X \setminus A; G)\,.$$

Since this isomorphism is natural with respect to inclusions, it induces an isomorphism of the direct limits

$$\alpha : \operatorname{dir\,lim} \widetilde{H}^k(A_m; G) \to \operatorname{dir\,lim} \widetilde{H}_{n-k-1}(X \setminus A_m; G)\,.$$

The left-hand group is isomorphic to the reduced Čech cohomology group $\check{\widetilde{H}}^k(A; G)$ by Theorem 8.13.16. We show that the right-hand group is isomorphic to the singular homology group $\check{H}_{n-k-1}(X \setminus A; G)$. Then the theorem is proved.

Consider the homomorphisms induced by inclusion

$$\widetilde{H}_{n-k-1}(X \setminus A_m) \to \widetilde{H}_{n-k-1}(X \setminus A)\,.$$

They induce a homomorphism of the direct limit

$$\operatorname{dir\,lim} \widetilde{H}_{n-k-1}(X \setminus A_m) \to \widetilde{H}_{n-k-1}(X \setminus A)\,.$$

The fact that this homomorphism is an isomorphism follows from the compact support property of singular homology. Every singular simplex $T : \Delta_p \to X \setminus A$ lies in a compact set and hence lies in one of the open sets $X \setminus A_m$. Thus every singular cycle of $X \setminus A$ is carried by $X \setminus A_m$ for some m. Furthermore, any homology between two such singular cycles is carried by $X \setminus A_{m+p}$ for some p. □

Corollary 8.14.2 *Let $n > 1$. Let M be a compact, connected, triangulable $n-1$ manifold. Suppose that $h : M \to \mathbb{S}^n$ is an imbedding. Then M is orientable, and $\mathbb{S}^n \setminus h(M)$ has precisely two path components of which $h(M)$ is the common boundary.*

Proof: By Alexander duality, we know that $\check{H}^{n-1}(M) \cong \widetilde{H}_0(\mathbb{S}^n \setminus h(M))$. Since M is triangulable, its Čech and simplicial cohomology groups are isomorphic. If M were non-orientable, we would have $H^{n-1}(M) \cong \mathbb{Z}/2$ although the group $\widetilde{H}_0(\mathbb{S}^n \setminus h(M))$ is free abelian. We conclude that M is orientable. Hence $H^{n-1}(M) \cong \mathbb{Z}$ and $\mathbb{S}^n \setminus h(M)$ has precisely two path components U and V.

The proof that $h(M)$ is the common boundary of U and V proceeds just as in the proof of the Jordan curve theorem [Theorem 4.9.5]. All one needs is the fact that if we delete the interior of a small open $n-1$ simplex s from M, then $h(M \setminus \operatorname{Int} s)$ does not separate $\mathbb{S}^n$. And that follows from the fact that [letting $M_0 = h(M \setminus \operatorname{Int} s)$] we have

$$\widetilde{H}_0(\mathbb{S}^n \setminus M_0) \cong \check{H}^{n-1}(M_0) \cong H^{n-1}(M_0) = 0\,. \qquad \square$$



An Overhand Knot
Figure 8.14.1

EXERCISES

1. Let X be the closed topologist's sine curve [see Definition 4.6.5]. If $f : X \to \mathbb{S}^n$ is an imbedding, calculate $\widetilde{H}_i(S^n \setminus f(X))$.

2. Show that Alexander–Pontryagin duality does not hold if Čech cohomology is replaced by singular cohomology.

3. Show that neither the Klein bottle nor the projective plane can be imbedded in $\mathbb{R}^3$. But they can be imbedded in $\mathbb{R}^4$. [**Hint:** If one removes an open disc from $\mathbb{P}^2$, what remains is a Möbius band.]

4. Let A and B be closed sets in $\mathbb{S}^n$. Show that if $A \approx B$, then

$$H_i(\mathbb{S}^n \setminus A) \cong H_i(\mathbb{S}^n \setminus B)$$

for all i. [**Note:** It does not follow that $\mathbb{S}^n \setminus A$ and $\mathbb{S}^n \setminus B$ are themselves homeomorphic. Figure 8.14.1 pictures an imbedding of $\mathbb{S}^1$ in $\mathbb{S}^3$ called an overhand knot. Its complement C is not homeomorphic to the complement D of the standard imbedding of $\mathbb{S}^1$ in $\mathbb{S}^3$. For

although the homology groups of C and D are isomorphic, their fundamental groups are different. See [CF77], Chapter 6.]

5. Explain why Corollary 8.14.2 is a generalization of the Jordan curve theorem. Explain why this result is not inconsistent with the Alexander horned sphere. [See Section 4.9.]

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De Morgan, Augustus (27 June 1806–18 March 1871)

Edwards, Robert Duncan (born 1942)

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He published under the name Felix Klein.

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Almost all of his publications are under the name Henri Lebesgue.

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MacLane, Leslie Saunders (4 August 1909–14 April 2005)

Most of his publications are under the name Saunders Mac Lane, but for many his surname is spelled MacLane. A few just use the initial 'S.' He did not use the name 'Leslie.'

Mayer, Walther (11 March 1887–10 September 1948)

Mazur, Barry Charles (born 19 December 1937)

Most of his publications are under the name Barry Mazur

Möbius, August Ferdinand (17 November 1790–26 September 1868)

His articles in the *Journal für die Reine und Angewandte Mathematik* are under the name A. F. Mobius.

Moise, Edwin Evariste (22 December 1918–18 December 1998)

Most of his publications are under the name Edwin E. Moise.

Morse, Harold Calvin Marston (24 March 1892–22 June 1977)

Almost all of his publications are under the name Marston Morse.

Noether, Amalie Emmy (23 March 1882–14 April 1935)

Almost all of her publications are under the name Emmy Noether.

Peano, Giuseppe (27 August 1858–20 April 1932)

Poincaré, Jules Henri (29 April 1854–17 July 1912)

Almost all of his publications are under either the name Henri Poincaré or H. Poincaré.

Pontryagin, Lev Semenovich (3 September 1908–3 May 1988)

Most of his publications are under either the name L. S. Pontryagin or L. S. Pontrjagin.

Reidemeister, Kurt Werner Friedrich (13 October 1893–8 July 1971)

Riemann, Georg Friedrich Bernhard (17 September 1826–20 July 1866)

He published under the name Bernhard Riemann.

Schoenflies, Artur Moritz (17 April 1853–27 May 1928)

He published under the names Artur Schoenflies, A. Schoenflies, and A. Schönflies.

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Most of his publications are under the name N. E. Steenrod.

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Most of his publications are under the name M. H. Stone.

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Almost all of his publications are under the name Heinrich Tietze.

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Urysohn, Pavel Samuilovich (3 February, 1898–17 August, 1924)

Most of his publications are under the name S. M. Ulam or S. Ulam.

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Weyl, Hermann Klaus Hugo (9 November 1885–8 December 1955)

He published under the name Hermann Weyl.

Whitehead, John Henry Constantine (11 November 1904–8 May 1960)

Almost all of his publications are under the name J. H. C. Whitehead.

Whitney, Hassler (23 March 1907–10 May 1989)

Zilber, Joseph Abraham (27 July 1923–4 October 2009)

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Index of Notation

Notation	Name	Page
$\mathbb{R}^N$	euclidean space	2
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$\operatorname{Int}\sigma$	interior of simplex σ	5
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$\mathbb{S}^n$	unit sphere in $\mathbb{R}^n$	5
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E_-^{n-1}	lower unit hemisphere in $\mathbb{R}^n$	5
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$\operatorname{Bd} A$	topological boundary	6
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$\|K\|$	polytope of simplicial complex K	9
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$\overline{\operatorname{St}}\, v$	closed star of vertex v	12
$\operatorname{Lk} v$	link of vertex v	12
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$\mathbb{Q}$	rationals	23
$\mathbb{C}$	complex numbers	23
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$\operatorname{im} f$	image of homomorphism f	24
$\operatorname{cok} f$	cokernel of homomorphism f	24
$\cong$	isomorphic	24

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$\prod_{\alpha\in J} G_\alpha$	direct product of groups G_α	26
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